

Towards a Brave New Variational Geometry

Foundations

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List of Notation

Notation	Reference	Notation	Reference
<i>Ambient spectra and Postnikov notation</i>			
$\mathrm{Sp}, \mathrm{Sp}^{\otimes}$	1.2.1, p. 22	$\mathbb{S}$	1.2.1, p. 22
$\mathrm{Map}_{\mathcal{C}}(X, Y)$	1.1.1, p. 20	$\mathrm{map}_{\mathcal{C}}(X, Y)$	1.1.1, p. 20
$\underline{\mathrm{Hom}}_{\mathrm{Sp}}(X, -)$	1.2.2, p. 22	$\mathrm{Sp}_{\geq n}, \mathrm{Sp}_{\leq n}$	1.2.1, p. 23
$\tau_{\geq n}^{\mathrm{Sp}}, \tau_{\leq n}^{\mathrm{Sp}}$	1.2.1, p. 23	$\mathrm{Sp}^{\heartsuit}$	1.2.2, p. 24
$H_{\mathrm{Ab}} : \mathrm{Ab} \xrightarrow{\sim} \mathrm{Sp}^{\heartsuit}$	1.2.5, p. 24	$\mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L)$	1.1.5, p. 21
<i>Connective commutative spectral algebra</i>			
$\mathrm{CAlg}(\mathrm{Sp})$	1.3.1, p. 26	$\mathrm{Sym} : \mathrm{Sp} \rightarrow \mathrm{CAlg}(\mathrm{Sp})$	1.3.2, p. 27
$\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}}$	1.3.3, p. 28	$\tau_{\geq 0}^{\mathrm{CAlg}}$	1.3.4, p. 28
$\pi_0 \dashv H_{\mathrm{CRing}}$	1.3.6, p. 30		
<i>Affine spectral geometry and modules</i>			
$\mathrm{Aff}_{\mathrm{Sp}} = (\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}})^{\mathrm{op}}$	1.4.1, p. 32	$\mathrm{Spec}(A)$	1.4.1, p. 32
$\mathrm{Spec}(B) \times_{\mathrm{Spec}(A)} \mathrm{Spec}(C) \simeq \mathrm{Spec}(B \otimes_A C)$	1.4.2, p. 32	$X_{\mathrm{cl}} = \mathrm{Spec} \pi_0(A)$	1.14.1, p. 117
$ X = \mathrm{Spec} \pi_0(A) $	1.4.2, p. 33	Mod_A	1.5.1, p. 33
$- \otimes_A -$	1.5.1, p. 34	$\underline{\mathrm{Hom}}_A(M, -)$	1.5.1, p. 34
$f^* = - \otimes_A B \dashv f_*$	1.5.2, p. 34	$\mathbf{Mod}_{\mathrm{Sp}} : \mathrm{CAlg}(\mathrm{Sp}) \rightarrow \mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L)$	1.5.3, p. 35
$\mathrm{CAlg}(\mathrm{Mod}_A) \simeq \mathrm{CAlg}(\mathrm{Sp})_A /$	1.5.5, p. 36	$(\mathrm{Mod}_A)_{\geq 0}, (\mathrm{Mod}_A)_{\leq 0}$	1.5.4, p. 37
$\mathrm{Mod}_A^{\heartsuit} \simeq \mathrm{Mod}_{\pi_0(A)}$	1.5.8, p. 38	Sym_A	1.6.4, p. 45
<i>Prestacks and quasi-coherent coefficients</i>			
$\mathrm{PSh}(\mathrm{Aff}_{\mathrm{Sp}})$	1.7.1, p. 50	h_A	1.7.2, p. 50
$h : \mathrm{Aff}_{\mathrm{Sp}} \hookrightarrow \mathrm{PSh}(\mathrm{Aff}_{\mathrm{Sp}})$	1.7.1, p. 51	$\delta : \mathrm{AffSch} \hookrightarrow \mathrm{Aff}_{\mathrm{Sp}}$	1.14.1, p. 117
$\delta_! \dashv \delta^*$	1.14.1, p. 118	$X_{\mathrm{cl}} = \delta^* X$	1.14.1, p. 118
$\mathrm{QCoh}_{\mathrm{aff}}(\mathrm{Spec} A) = \mathrm{Mod}_A$	1.5.3, p. 36	$\mathrm{AffPt}(X)$	1.7.3, p. 51
$\mathrm{QCoh}(X) =$	1.7.4, p. 51	$\mathcal{O}_X$	1.7.5, p. 57
$\varprojlim_{(\mathrm{Spec} A \rightarrow X) \in \mathrm{AffPt}(X)^{\mathrm{op}}} \mathrm{Mod}_A$			
$\Gamma(X, \mathcal{F}) = \mathrm{map}_{\mathrm{QCoh}(X)}(\mathcal{O}_X, \mathcal{F})$	1.7.6, p. 57		
<i>Spectral Zariski geometry and global schemes</i>			
$A[f^{-1}]$	1.8.1, p. 60	$D_A^{\mathrm{sp}}(f) = \mathrm{Spec}(A[f^{-1}])$	1.8.4, p. 61
τ_{Zar}	1.8.3, p. 65	$\mathrm{Stk}_{\mathrm{Zar}}^{\mathrm{sp}}$	1.9.1, p. 75
$a_{\mathrm{Zar}}^{\mathrm{sp}}$	1.9.1, p. 75	$\mathrm{Sch}^{\mathrm{sp}}$	1.9.3, p. 76
$d = a_{\mathrm{Zar}}^{\mathrm{sp}} \circ \delta_!, \quad d \dashv \delta^*$	1.14.1, p. 118	$(-)^{\mathrm{disc}} : \mathrm{Sch} \hookrightarrow \mathrm{Sch}^{\mathrm{sp}}$	1.14.1, p. 118
$(-)_{\mathrm{cl}} : \mathrm{Sch}^{\mathrm{sp}} \rightarrow \mathrm{Sch}$	1.14.1, p. 118	$i_X : X_{\mathrm{cl}}^{\mathrm{disc}} \rightarrow X$	1.14.1, p. 121
$ X = X_{\mathrm{cl}} $	1.9.4, p. 81		
<i>Global coefficients and relative affine geometry</i>			
$\mathrm{AffOp}(X)$	1.10.1, p. 83	$\mathrm{QCoh}(X)_{\geq 0}, \mathrm{QCoh}(X)_{\leq 0}$	1.10.2, p. 86
$\mathrm{QCoh}(X)^{\heartsuit} \simeq \mathrm{QCoh}_{\mathrm{ab}}(X_{\mathrm{cl}})$	1.10.2, p. 86	$\mathrm{QCAlg}(X) = \mathrm{CAlg}(\mathrm{QCoh}(X))$	1.10.3, p. 89

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List of Notation, continued

Notation	Reference	Notation	Reference
$\mathrm{QCalg}(X)^{\mathrm{cn}}$	1.10.3, p. 89	$\mathrm{Spec}_X : \mathrm{QCalg}(X)^{\mathrm{cn}, \mathrm{op}} \rightarrow \mathrm{Sch}_{/X}^{\mathrm{sp}}$	1.11.1, p. 92
<i>Closed and linear geometry</i>			
$X \setminus Z$	1.12.1, p. 100	$\mathrm{Sym}_{\mathcal{O}_X}$	1.13.3, p. 108
$\mathbb{L}_X(\mathcal{E}) = \mathrm{Spec}_X \mathrm{Sym}_{\mathcal{O}_X}(\mathcal{E})$	1.13.3, p. 109	$\mathbf{V}_X(\mathcal{E}) = \mathbb{L}_X(\mathcal{E}^\vee)$	1.13.4, p. 111
$\mathbb{A}_X^n$	1.13.5, p. 113	$\mathbb{A}_X^1$	1.13.5, p. 113
$0, 1 : X \rightrightarrows \mathbb{A}_X^1$	1.13.2, p. 113	$\mathbb{G}_{m,X} = \mathbb{A}_X^1 \setminus 0(X)$	1.13.6, p. 114
$A\{t\} = \mathrm{Sym}_A(A)$	1.13.11, p. 114	$R\{t\} = \mathrm{Sym}_{HR}(HR)$	1.13.13, p. 116
<i>Tangent and cotangent differentiation</i>			
$L_{B/A}, \quad L_B = L_{B/\mathbb{S}}$	2.1.1, p. 126	$L_{X/S}, \quad L_X = L_{X/\mathrm{Spec}(\mathbb{S})}$	2.1.1, p. 126
$\mathrm{Der}_A(B, M)$	2.3.1, p. 129	$\mathrm{Der}_S(X; M) =$ $\mathrm{Map}_{\mathrm{QCoh}(X)}(L_{X/S}, M)$	2.5.2, p. 141
$\mathcal{C}_{A//B} = (\mathcal{C}_A)/_B$	2.1.2, p. 126	$\mathrm{Stab}(\mathrm{CAlg}(\mathrm{Sp})_{/B}) \simeq \mathrm{Mod}_B$	2.2.1, p. 127
$B \oplus M$	2.2.1, p. 128	$\epsilon_{B/A} : B \otimes_A \mathrm{cofib}(A \rightarrow B) \rightarrow L_{B/A}$	2.4.1, p. 134
<i>Differential-linear coefficient fibres</i>			
$\mathcal{L}_X := \mathrm{QCoh}(X)$	2.6.1, p. 144	$\mathrm{CoCom}(\mathcal{L}_X) = \mathrm{CAlg}(\mathcal{L}_X^{\mathrm{op}})^{\mathrm{op}}$	2.6.1, p. 144
$U_X : \mathrm{CoCom}(\mathcal{L}_X) \rightarrow \mathcal{L}_X, \quad U_X \dashv$ C_X	2.6.1, p. 144	$!_X = U_X C_X$	2.6.1, p. 144
$\Delta_M : !_X M \rightarrow !_X M \otimes !_X M, \quad e_M :$ $!_X M \rightarrow \mathcal{O}_X$	2.6.1, p. 144	$\chi_{M,N} : !_X(M \oplus N) \xrightarrow{\sim}$ $!_X M \otimes !_X N, \quad \chi_0 : !_X 0 \xrightarrow{\sim} \mathcal{O}_X$	2.6.2, p. 147
$P_X(M), \quad U_X P_X(M) \simeq \mathcal{O}_X \oplus M$	2.6.1, p. 150	$H_M^{\mathrm{inf}} : \mathcal{O}_X \oplus M \rightarrow !_X(\mathcal{O}_X \oplus M)$	2.6.2, p. 151
$\eta_M : M \rightarrow !_X M$	2.6.4, p. 153	$\nabla_M : !_X M \otimes !_X M \rightarrow !_X M, \quad u_M :$ $\mathcal{O}_X \rightarrow !_X M$	2.6.5, p. 154
$\tilde{\mathrm{d}}_M : C_X(M) \times P_X(M) \rightarrow C_X(M)$	2.6.6, p. 156	$\mathrm{d}_M : !_X M \otimes M \rightarrow !_X M, \quad Df =$ $f \circ \mathrm{d}_M$	2.6.7, p. 157
$\mathcal{O}_!(M) = \underline{\mathrm{Hom}}_B(!_B M, B)$	2.6.2, p. 159	$j_M : \mathrm{Sym}_B(M^\vee) \rightarrow \mathcal{O}_!(M)$	2.6.2, p. 159
$\beta_{f,M} : f^*(!_X M) \rightarrow !_Y(f^* M)$	2.6.9, p. 164		
<i>Structured and finite infinitesimal geometry</i>			
$B_\eta = B \times_{B \oplus M[1]} B, \quad q_\eta : B_\eta \rightarrow B$	2.7.1, p. 168	$\mathrm{SqZ}_A^{\mathrm{str}}(B; M)$	2.7.1, p. 170
$X_\eta, \quad i_\eta : X \rightarrow X_\eta$	2.7.2, p. 180	$\mathrm{SqZ}_S^{\mathrm{str}}(X; M)$	2.7.1, p. 183
$\mathrm{Inf}_{\mathrm{Sp}}^{\mathrm{fin}}$	2.8.3, p. 188	$\mathrm{Aff}_{\mathrm{Sp}}^{\mathrm{inf}} \subseteq \mathrm{Fun}(\Delta^1, \mathrm{Aff}_{\mathrm{Sp}})$	2.9.1, p. 192
$\mathrm{dom}, \mathrm{cod} : \mathrm{Aff}_{\mathrm{Sp}}^{\mathrm{inf}} \rightarrow \mathrm{Aff}_{\mathrm{Sp}}, \quad i(T) =$ id_T	2.9.1, p. 192	$\tau_{\mathrm{Zar}}^{\mathrm{inf}}$	2.9.3, p. 194
$\mathcal{H}_{\mathrm{Sp}} := \mathrm{Stk}_{\mathrm{Zar}}^{\mathrm{sp}}$	2.9.3, p. 197	$\mathcal{H}_{\mathrm{inf}} := \mathrm{Stk}_{\mathrm{inf}, \mathrm{Zar}}^{\mathrm{sp}}$	2.9.3, p. 197
$\mathrm{Inf}^{\mathrm{fin}}(X/S)$	2.9.4, p. 199		
<i>Differential-cohesive endpoint semantics</i>			
$i_! \dashv i^* \dashv i_* \dashv i^!$	2.10.2, p. 203	$i_! = \mathrm{cod}^*, \quad i_* = \mathrm{dom}^*$	2.10.2, p. 201
$\mathfrak{R} = i_! i^*, \quad \mathfrak{S} = i_* i^*, \quad \& = i_* i^!$	2.10.1, p. 204	$\mathfrak{R} \dashv \mathfrak{S} \dashv \&$	2.10.4, p. 205
$\mathfrak{S}_B E = B \times_{\mathfrak{S} B} \mathfrak{S} E$	2.10.1, p. 207	$\Theta_f : i_! X \rightarrow \mathfrak{S}_{i_! S}(i_! X)$	2.10.2, p. 208
$T_{\mathrm{inf}}(X/S) = i_! X \times_{\mathfrak{S}_{i_! S}(i_! X)} i_! X$	2.10.4, p. 209	$T_{\mathrm{inf}}^{[\bullet]}(X/S) = \check{C}(\Theta_f)$	2.10.4, p. 209
$\mathbf{h}_j^{\mathrm{inf}}$	2.10.3, p. 210	$\Theta_f^{\mathrm{rep}}(j)$	2.10.4, p. 211
$D_{X/S}^{(1)}, \quad i_! : X \rightarrow D_{X/S}^{(1)}, \quad h_0, h_1 :$	2.11.1, p. 219		
$D_{X/S}^{(1)} \rightrightarrows X$			
<i>First-order deformation and equation calculus</i>			
$\mathrm{Def}_{A'/A}(B)$	2.12.1, p. 224	$\mathrm{Def}_{S'/S}(X)$	2.12.2, p. 226
$\mathrm{ob}(B/A')$	2.12.2, p. 225	$\mathrm{ob}(X/S')$	2.12.4, p. 230
$N_{Z/X}^* = L_{Z/X}[-1]$	2.13.2, p. 232	$N_{Z/X} = (N_{Z/X}^*)^\vee$	2.13.3, p. 232
$Z(s)$	2.13.1, p. 233		
<i>Stable endpoint semantics and BNV-type reconstruction</i>			
$\mathcal{D}_{\mathrm{Sp}} = \mathrm{Sp}((\mathcal{H}_{\mathrm{Sp}})_*)$	2.14.1, p. 240	$\mathcal{D}_{\mathrm{inf}} = \mathrm{Sp}((\mathcal{H}_{\mathrm{inf}})_*)$	2.14.1, p. 240

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List of Notation, continued

Notation	Reference	Notation	Reference
$i_!^{\text{st}} \dashv i^{*,\text{st}} \dashv i_*^{\text{st}} \dashv i^{!,\text{st}}$	2.14.1, p. 240	$\mathcal{Z}(E) = \text{cofib}(\&^{\text{st}} E \rightarrow E)$	2.14.3, p. 245
$\mathfrak{R}^{\text{st}} = i_!^{\text{st}} i^{*,\text{st}}, \quad \mathfrak{S}^{\text{st}} =$	2.14.2, p. 241	$\chi_E : i^{!,\text{st}} E \rightarrow i^{*,\text{st}} E$	2.14.1, p. 245
$i_*^{\text{st}}, i^{*,\text{st}}, \quad \&^{\text{st}} = i_*^{\text{st}}, i^{!,\text{st}}$		$\widehat{\mathcal{T}}_{\text{inf}}$	2.14.5, p. 247
$\mathcal{A}(E) = \text{fib}(E \rightarrow \mathfrak{S}^{\text{st}} E)$	2.14.3, p. 245	$\widehat{E}^n(K) = \pi_{-n} \text{map}_{\mathcal{D}_{\text{inf}}}(K, E)$	2.14.1, p. 247
$\mathcal{D}_{\text{inf}}^{\text{BNV}} = \ker(i^{!,\text{st}})$	2.14.4, p. 245		
$\phi_E : i^{*,\text{st}} E \rightarrow i^{*,\text{st}} \mathcal{Z}(E)$	2.14.3, p. 245		
$\text{Diff}(U, P, \phi) = P \times_{\mathfrak{S}^{\text{st}} P} i_*^{\text{st}} U$	2.14.1, p. 247		
$E \simeq \mathcal{Z}(E) \times_{i_*^{\text{st}} i^{*,\text{st}} \mathcal{Z}(E)} i_*^{\text{st}} i^{*,\text{st}} E$	2.14.4, p. 246		
<i>Postnikov reconstruction and filtered formal order</i>			
$\text{adic}(A \rightarrow B)$	3.3.3, p. 265	$A_B^\wedge = F^0 \widehat{\text{adic}}(A \rightarrow B)$	3.3.3, p. 265
<i>Anchored cotangent and comonadic reconstruction</i>			
$\text{AnchCot}_{B/R}^{\text{sp}} = (\text{Mod}_B)_{L_{B/R}[-1]/}$	3.6.1, p. 297	$\text{AnchCot}_{B/R, \geq 0}^{\text{aperf}}$	3.6.1, p. 297
$\text{cot}_{B/R}(A) = (L_{B/R}[-1] \rightarrow$	3.6.2, p. 298	$\text{sqz}_{B/R}(M, \alpha) = (B_\alpha \rightarrow B)$	3.6.2, p. 298
$L_{B/A}[-1])$			
$\mathbb{C}_{B/R}^{\text{sqz}} = \text{cot}_{B/R} \text{sqz}_{B/R}$	3.6.3, p. 303	$\text{Coalg}_{\mathbb{C}_{B/R}^{\text{sqz}}}(\text{AnchCot}_{B/R, \geq 0}^{\text{aperf}})$	3.6.1, p. 303
<i>Artinian tests and formal-moduli semantics</i>			
$\text{Art}_{B/R}^{\text{sp}}$	3.7.1, p. 305	$\mathcal{P}_{B/R}^{\text{Art,sp}} = \text{Fun}(\text{Art}_{B/R}^{\text{sp}}, \mathcal{S})$	3.8.1, p. 324
$W_{B/R}^{\text{Sch,Art}}$	3.8.2, p. 325	$\text{FMP}_{B/R}^{\text{sp}}$	3.8.3, p. 326
<i>Centre-free reconstruction semantics</i>			
$L_S^{\text{Recon}} \dashv \iota_S$	3.9.1, p. 332	$\text{Recon}_S = \iota_S L_S^{\text{Recon}}$	3.9.4, p. 336
<i>Centred tangent recovery and stable reconstruction</i>			
$\mathbf{T}_{Y/V, u}^{\text{cn}} : (\text{Mod}_B)_{\geq 0} \rightarrow \text{Sp}$	3.10.4, p. 341	$\text{Mod}_B^+ = \bigcup_{r \in \mathbb{Z}} (\text{Mod}_B)_{\geq r}$	3.10.3, p. 345
$\mathbf{T}_{Y/V, u}^{\text{sp}, +} : \text{Mod}_B^+ \rightarrow \text{Sp}$	3.10.1, p. 346	$\mathbf{T}_{U/Y; V}^{\text{sp}, +} : \text{Mod}_B^+ \rightarrow \text{Sp}$	3.10.4, p. 348
$\mathcal{D}_{\text{Sp}/S} = \text{Sp}(((\mathcal{H}_{\text{Sp}})_{/S})^*)$	3.11.1, p. 350	$\mathcal{D}_{\text{Sp}/S}^{\text{Recon}} = \text{Sp}(((\mathcal{H}_{\text{Sp}})_{/S})^{\text{Recon}})^*$	3.11.1, p. 350
$L_S^{\text{Recon, st}} \dashv \iota_S^{\text{st}}$	3.11.1, p. 350	$\text{Recon}_S^{\text{st}} = \iota_S^{\text{st}} L_S^{\text{Recon, st}}$	3.11.1, p. 350
<i>Spectral de Rham reflection and effective descent</i>			
$L^{\text{dR}} \dashv \iota^{\text{dR}}, \quad \text{dR} = \iota^{\text{dR}} L^{\text{dR}}$	4.2.1, p. 364	$X_{\text{dR}}, \quad \eta_X^{\text{dR}} : X \rightarrow X_{\text{dR}}$	4.2.1, p. 364
$X_{\text{dR}/S} = S \times_{S_{\text{dR}}} X_{\text{dR}}$	4.2.4, p. 370	dR_S	4.2.4, p. 370
$X \xrightarrow{e_{X/S}} Q_{X/S} \xrightarrow{m_{X/S}} X_{\text{dR}/S}$	4.3.1, p. 374	$\mathcal{R}_{X/S, \bullet}^{\text{dR}} \simeq \check{C}_\bullet(e_{X/S})$	4.3.6, p. 383
$\text{Nbh}_{Z/S}^{\text{dR}}(Y) = Y \times_{Y_{\text{dR}/S}} Z_{\text{dR}/S}$	4.3.5, p. 385		
<i>Stable de Rham scalar calculus</i>			
$\Sigma_f \dashv f^* \dashv \Pi_f$	4.4.1, p. 388	$\text{Real}_S^{\text{st}} : \text{QCoh}(S) \rightarrow \mathcal{D}_{\text{Sp}/S}$	4.4.4, p. 397
$\mathcal{O}_S^{\text{add}} = \text{Real}_S^{\text{st}}(\mathcal{O}_S)$	4.4.9, p. 400	$C_{\text{dR}}^\bullet(X/S)$	4.5.1, p. 403
$\text{DR}_{X/S} = \text{Tot } C_{\text{dR}}^\bullet(X/S)$	4.5.5, p. 408	$\text{DR}_{X/S}^{\text{full}}$	4.5.5, p. 408
$\Omega_{\text{dR}}^q(X/S)$	4.6.1, p. 410	$d_{\text{dR}} : \Omega_{\text{dR}}^q(X/S) \rightarrow \Omega_{\text{dR}}^{q+1}(X/S)$	4.6.2, p. 412
$\lambda_{X/S}^1 : \Omega_{\text{dR}}^1(X/S) \rightarrow$	4.8.1, p. 446	$\kappa : \text{dR} \rightarrow \text{Red}_{\text{cl}}$	4.9.1, p. 452
$\Pi_p \text{Real}_X^{\text{st}}(L_{X/S})$			
<i>Crystalline disks, jets, and intrinsic PDEs</i>			
$\text{Disk}_{X/S}^{\text{dR}, \infty} = e_{X/S}^* \Sigma_{e_{X/S}}$	5.2.1, p. 473	$J_{X/S}^\infty = e_{X/S}^* \Pi_{e_{X/S}}$	5.2.5, p. 471
$\text{Disk}_{X/S}^{\text{dR}, \infty} \dashv J_{X/S}^\infty$	5.2.1, p. 473	$\text{PDE}_{X/S}^{\text{fi}} = (\mathcal{H}_{\text{Sp}})_{/Q_{X/S}}$	5.3.2, p. 482
$\mathbb{F} \dashv U_J \dashv \mathbb{K}$	5.4.1, p. 492	$j^\infty \phi = J^\infty(\phi) \rho_X$	5.4.2, p. 494
<i>Differential operators, equations, and prolongation</i>			
$\text{DiffOp}_{X/S} \simeq \text{coKl}(J_{X/S}^\infty)$	5.4.2, p. 495	$\text{Eqn}_{X/S} = (\text{id}_{(\mathcal{H}_{\text{Sp}})_{/X}} \downarrow J_{X/S}^\infty)$	5.4.5, p. 498
$\text{Sol}(i)$	5.4.5, p. 503	$\mathbf{Sol}_{X/S}(i)$	5.4.5, p. 503
$\text{FSol}_i(Z)$	5.5.3, p. 508	$\text{Prol}^{\text{Cart}}(i), \quad \text{Prol}(i)$	5.5.2, p. 507
$c_i : \text{Prol}(i) \rightarrow E$	5.5.5, p. 511	$\rho_i : E \rightarrow J^\infty E$	5.5.9, p. 515

Continued on the following page

List of Notation, continued

Notation	Reference	Notation	Reference
$J_{X/S}^{\infty,\text{red},\text{cl}}$	5.7.1, p. 540		

Introduction: Spectral Contexts and the Infinitesimal Foundations of Variational Geometry

1 From variational geometry to the spectral problem

Variational geometry studies fields through their dependence on parameters, their infinitesimal variation, and the differential equations that constrain them. In classical geometry this leads to jet spaces, formal integrability, variational sequences, and the variational bicomplex [28, 38, 44, 20, 42]. In mathematical physics these structures meet symplectic and multisymplectic geometry, gauge symmetry, cohomological information, and higher geometric structures [43, 41, 13, 16, 15, 1]. A homotopical formulation of variational geometry should therefore retain the deformation and coherence data carried by these interactions while still supporting the local and infinitesimal constructions on which differential geometry depends.

Spectral algebraic geometry provides a setting in which such deformation data are intrinsic, but it also introduces the structural problem that organizes this dissertation. Its nonlinear affine geometry is built from connective commutative ring spectra, whereas its natural coefficient categories are fully stable module categories [34, 37]. The connective layer governs spectral schemes, relative affine spaces, and nonlinear context extension. The stable layer carries cotangent complexes, mapping spectra, fibres, cofibres, suspensions, and totalizations. A spectral approach to variational geometry must use both layers without identifying them, and it must make their interaction coherent under change of geometric base.

This dissertation is the first stage of a longer programme aimed at developing such a framework. The eventual goal is a spectral-algebraic formulation of variational geometry suitable for foundational questions in mathematical physics, with higher gauge theory and derived field theory as principal sources of motivation and intended applications. The present work develops the geometric and infinitesimal infrastructure needed before a genuinely variational theory can be formulated. It reaches spectral infinitesimal variation, formal reconstruction, de Rham descent, crystalline transport, infinite jets, and a formal differential-equation calculus. Lagrangian structures, Euler–Lagrange equations, variational complexes, and a spectral theory of higher gauge fields are not constructed here and belong to later stages of the programme.

The first problem is organizational: the connective nonlinear geometry and the stable linear coefficients must be assembled into one indexed system. We seek a geometry in which change of base transports coefficients, local data satisfy descent, relative algebra determines geometry over a fixed base, and infinitesimal extensions encode variation. The language of categorical logic is useful for expressing this organization. A spectral scheme is read as a *context*, pullback as *substitution*, and

relative spectrum as *comprehension* or context extension [25]. These terms describe the indexed role of the constructions; the constructions themselves remain spectral-geometric and higher-categorical.

The guiding question of the dissertation is:

How much of the infinitesimal and jet-theoretic foundation of variational geometry can be constructed from connective spectral contexts and their fully stable quasi-coherent coefficients?

The answer developed here is that these two layers already support a coherent infinitesimal and infinite-jet calculus. The construction keeps fibrewise stable differentiation distinct from nonlinear infinitesimal geometry, connects them through the cotangent classifier, and then passes from finite infinitesimal variation to reconstruction, de Rham localization, effective descent, and crystalline jet transport. On specified complete fixed-centre domains, the nonlinear augmentation is reconstructed from shifted-cotangent data together with the coalgebra coherence generated by structured square-zero formation; separately, Postnikov continuity and nilpotent excision define a centre-free reconstruction localization. Making finite infinitesimal variation invisible produces a de Rham reflector and an effective quotient. Dependent adjoints along that quotient then produce the formal-disk and infinite-jet operators from which crystalline transport and formal differential equations are obtained.

2 The connective–stable architecture of infinitesimal variation

The programme requires nonlinear spectral geometry and stable linear coefficients to vary together without being collapsed into a single kind of object. We therefore organize spectral contexts, quasi-coherent coefficients, and change of base into one indexed geometry, with relative spectrum providing the return from connective coefficient algebra to geometric context extension.

The fixed affine background is

$$\mathrm{Aff}_{\mathrm{Sp}} = (\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}})^{\mathrm{op}}.$$

For an affine context $X = \mathrm{Spec}(A)$, the coefficient fibre is

$$\mathrm{QCoh}_{\mathrm{aff}}(X) = \mathrm{Mod}_A,$$

and a morphism

$$f : \mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A)$$

represented by $A \rightarrow B$ acts on coefficients by extension of scalars,

$$f^* = - \otimes_A B : \mathrm{Mod}_A \longrightarrow \mathrm{Mod}_B.$$

Spectral Zariski descent globalizes this assignment to

$$\mathrm{QCoh} : (\mathrm{Sch}^{\mathrm{sp}})^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L),$$

so every spectral context carries a stable symmetric-monoidal category of quasi-coherent coefficients, and every change of context carries a coherent strong symmetric-monoidal substitution functor [34, 37].

This coefficient doctrine remains connected to nonlinear geometry through relative spectrum. For a spectral scheme X , relative spectrum identifies connective quasi-coherent commutative algebras with affine morphisms over X :

$$\mathrm{QCAlg}(X)^{\mathrm{cn}, \mathrm{op}} \simeq (\mathrm{Sch}_{/X}^{\mathrm{sp}})_{\mathrm{aff}}.$$

Thus, for $\mathcal{A} \in \mathrm{QCAlg}(X)^{\mathrm{cn}}$, the context extension

$$\mathrm{Spec}_X(\mathcal{A}) \longrightarrow X$$

represents algebra terms after substitution. This is the comprehension principle used throughout the dissertation. Crucially, comprehension returns *connective* coefficient algebra to geometry; it does not identify the geometric and stable layers.

The connective–stable distinction remains active after comprehension. Geometry is built from connective commutative algebra, whereas cotangent complexes, mapping spectra, stabilizations, fibres, cofibres, and totalizations live in fully stable coefficient categories. When cocommutative coalgebras appear below, $\mathrm{CoCom}(\mathcal{L}_X)$ denotes the infinity-category of cocommutative coalgebra objects internal to the stable symmetric-monoidal fibre $\mathcal{L}_X$; stability is asserted for the ambient coefficient fibre, not for the coalgebra category itself.

With this indexed background in place, differentiation presents two different problems. The first asks how coefficients vary *inside* a stable fibre. The second asks how the nonlinear *context itself* varies infinitesimally. The dissertation constructs these two mechanisms independently and compares them only through a controlled cotangent interface.

For the fibrewise linear problem, one has affinely the tangent equivalence

$$\mathrm{Stab}(\mathrm{CAlg}(\mathrm{Sp})_{/B}) \simeq \mathrm{Mod}_B.$$

For a general spectral scheme X , however, the descended stable fibre $\mathrm{QCoh}(X)$ is used directly; no global identification of this fibre with a stabilization of a global nonlinear category is required or asserted. Writing $\mathcal{L}_X := \mathrm{QCoh}(X)$, the forgetful functor from cocommutative coalgebras admits a cofree right adjoint,

$$U_X : \mathrm{CoCom}(\mathcal{L}_X) \rightleftarrows \mathcal{L}_X : C_X, \quad U_X \dashv C_X.$$

The composite

$$!_X = U_X C_X$$

is the fibrewise differential exponential. Its canonical coalgebra, primitive, and bialgebra structures produce the codereliction and deriving transformation. After passage to the homotopy category, these maps recover the corresponding differential-storage structure in the sense of differential categories [6, 8, 14, 29]. The existence and behaviour of the spectral cofree coalgebra construction is closely related to recent work on spectral $\mathbb{E}_\infty$ -coalgebras [4].

Change of geometric base does not in general preserve this exponential by an equivalence. Instead, for $f : Y \rightarrow X$, cofreedom and pullback construct the canonical oplax comparison

$$\beta_{f,M} : f^*(!_X M) \longrightarrow !_Y(f^* M).$$

The resulting fibrewise calculus is coherently compatible with substitution in the oplax sense actually supplied by the construction; no general equivalence is asserted.

The geometric differentiation problem is controlled by the cotangent classifier. For a relative context $X \rightarrow S$, affine cotangent complexes glue by spectral Zariski descent to a canonically determined object

$$L_{X/S} \in \mathrm{QCoh}(X).$$

Global derivation terms are represented by

$$\mathrm{Der}_S(X; M) := \mathrm{Map}_{\mathrm{QCoh}(X)}(L_{X/S}, M),$$

recovering on compatible affine charts the usual spectral cotangent representation of derivations. Cotangent base change and transitivity supply the substitution law and chain rule. Degree-one cotangent classes then construct structured square-zero extensions, and finite composites of their underlying arrows produce the finite spectral infinitesimal substitutions used later in the dissertation [30, 34].

The cofree exponential differentiates inside the stable coefficient fibre, whereas cotangent classes generate nonlinear infinitesimal geometry. Their common first-order interface is the cotangent classifier: on free finite-projective models, the convolution derivative obtained from the cofree calculus compares with the universal spectral Kähler derivation. The same cotangent object then controls square-zero variation and reappears in the reconstruction theory.

Finite infinitesimal substitutions provide the next structure needed by the programme: they admit a Zariski semantics of their own. The resulting endpoint adjunctions are constructed first and only then recognized as a differential-cohesive configuration. Stabilizing that endpoint semantics produces an exact-adjoint fracture pattern. Its formal shape agrees with familiar pullback constructions in differential cohomology, but the comparison imports no independent curvature, period, cycle, or transfer package [24, 11]. At this stage the dissertation has produced two differential mechanisms, their cotangent interface, and a finite-infinitesimal geometry on which the later reconstruction and descent problems can be posed.

3 From infinitesimal data to reconstruction, descent, and jets

Chapter 3 reverses the preceding direction and asks when first-order and finite-infinitesimal data determine the nonlinear geometry from which they arose. It addresses this reconstruction problem through two approximation coordinates, one measuring homotopical height and the other formal order at a chosen centre.

Postnikov height resolves a connective spectral algebra into canonical square-zero stages and recovers it as the inverse limit of its truncations. Formal order is supplied instead by the canonical adic filtration of an augmentation $A \rightarrow B$. Completion preserves the associated graded, and the resulting linear layers are controlled by the shifted relative cotangent complex:

$$\widehat{\mathrm{Gr\,adic}}(A \rightarrow B) \simeq \mathrm{Sym}_B^{\mathrm{gr}}(L_{B/A}[-1]\langle 1 \rangle).$$

The two coordinates answer different questions: Postnikov height measures how much homotopical information remains, while the adic filtration measures order of vanishing at the chosen centre. This part of the dissertation is developed in close contact with derived completion and formal integration [7, 2, 9].

On a specified fixed-centre domain, the formal-order construction becomes a reconstruction theorem. Let B be a connective centre over a connective base R . On connective almost finitely presented augmentations $A \rightarrow B$ that are surjective on π_0 and fixed by canonical adic completion, the shifted-cotangent/structured-square-zero adjunction is comonadic. Complete augmentations on this domain are reconstructed from anchored shifted-cotangent data together with the coalgebra coherence generated by structured square-zero formation. The bounded coherent sector recovers the spectral Artinian test objects used in formal-moduli theory.

This fixed-centre result is not the only reconstruction mechanism in the chapter. A separate centre-free construction imposes Postnikov continuity and nilpotent excision and produces a left-exact reconstruction localization. Returning to a centre, split square-zero lifting yields an exact tangent functor on eventually connective coefficients; for represented targets it is represented by the cotangent complex, and the corresponding mapping-spectrum functor extends canonically to all modules. The crucial interface is not an identification of the reconstruction comonad with the fibrewise differential exponential. Rather, the cotangent classifier appears again as the common boundary object through which the constructions can be compared.

Reconstruction asks when infinitesimal data determine nonlinear geometry. Chapter 4 asks a different question: what remains when finite infinitesimal variation is declared invisible? Square-zero substitutions generate an accessible left-exact de Rham localization, in the spirit of infinitesimal and crystalline geometry [40, 19, 5]. For a relative context $X \rightarrow S$, this produces the universal relative de Rham reflection

$$X \longrightarrow X_{\mathrm{dR}/S}.$$

Universal localization alone is not enough for the later descent theory, because its unit need not be an effective epimorphism. The unit is therefore factored through its effective image,

$$X \xrightarrow{e_{X/S}} Q_{X/S} \xrightarrow{m_{X/S}} X_{\mathrm{dR}/S}.$$

The two targets play different roles. The object $X_{\mathrm{dR}/S}$ is the universal de Rham-local reflection, whereas $Q_{X/S}$ is the effective quotient actually presented by the generalized points of X . The Čech nerve of $e_{X/S}$ is the effective de Rham relation, and this effective relation, rather than the universal reflection by itself, is the object that supports the subsequent descent and jet constructions.

After stabilizing the slice semantics, a distinguished additive scalar object provides stable scalar functions on the simplices of this relation. These functions form a cosimplicial commutative algebra

$$C_{\mathrm{dR}}^\bullet(X/S),$$

whose totalization computes stable scalar functions on the effective quotient. The complete totalization filtration exposes intrinsic de Rham forms as its normalized layers; the connecting morphisms supply a homotopy-coherent differential, while the remaining filtration data encode multiplication, higher closedness, exact-form lifts, and primitive data. Restriction to the first structured infinitesimal neighbourhood recovers the stable realization of the cotangent classifier together with the universal derivation. A separate reduced-classical comparison probes a different boundary: it measures convergence toward the reduced classical sector rather than first-order variation.

Once the effective relation has been constructed, the remaining jet-theoretic problem is to make that relation act coherently on objects varying over X . Pullback along

$$e_{X/S} : X \longrightarrow Q_{X/S}$$

has both dependent adjoints,

$$\Sigma_{e_{X/S}} \dashv e_{X/S}^* \dashv \Pi_{e_{X/S}}.$$

The two composites encode the same effective infinitesimal relation in monadic and comonadic form:

$$\mathrm{Disk}_{X/S}^{\mathrm{dR}, \infty} = e_{X/S}^* \Sigma_{e_{X/S}}, \quad J_{X/S}^\infty = e_{X/S}^* \Pi_{e_{X/S}}.$$

The first is the formal-disk monad and the second the infinite-jet comonad. This is an intrinsically infinite-order construction; no global finite-order jet tower is used in its definition.

Effective descent then identifies quotient descent, formal-disk action, and jet coaction as three presentations of the same infinitesimal compatibility:

$$(\mathcal{H}_{\text{Sp}})_{/Q_{X/S}} \simeq \text{Alg}_{\text{Disk}_{X/S}^{\text{dR}, \infty}}((\mathcal{H}_{\text{Sp}})_{/X}) \simeq \text{Coalg}_{J_{X/S}^{\infty}}((\mathcal{H}_{\text{Sp}})_{/X}).$$

Crystalline transport is the relation-theoretic form of this same effective descent datum; it is not an independently chosen field on an object. This places the spectral construction in direct conversation with synthetic and categorical jet formalisms [28, 38, 27, 21].

The jet comonad now supplies the formal equation calculus. Differential operators are co-Kleisli morphisms out of $J_{X/S}^{\infty}$, generalized equations are morphisms into infinite-jet objects, and Cartan prolongation is the universal repair that equips an ambient equation with compatible crystalline transport. A prolongation-complete generalized equation canonically acquires the resulting jet-coalgebra and effective-descent structure and hence determines an intrinsic formally integrable PDE. Here formal integrability means infinitesimal descent: it does not assert analytic existence, regularity, ellipticity, or functional-analytic convergence of solutions.

The global jet construction is deliberately infinite-order. On represented affine diagonal augmentations, the adic filtration of Chapter 3 provides genuine finite formal-order stages with cotangent-controlled associated graded. A global tower $J_{X/S}^k$, however, would require a base-change-coherent simplicial globalization of those affine stages. No such tower is assumed in the construction of J^{∞} or in the formal-integrability results. The effective quotient $Q_{X/S}$ is thus the common geometric source of de Rham descent, formal-disk action, jet coaction, and crystalline transport, which present the same underlying infinitesimal compatibility in different forms.

4 Main results and relation to existing work

The thesis of the dissertation is that the indexed geometry of connective spectral contexts and fully stable quasi-coherent coefficients contains enough structure to support an infinitesimal and jet-theoretic foundation for spectral variational geometry. The principal results can be summarized as follows.

1. **Spectral contexts and stable coefficients form a base-change-compatible indexed doctrine.** Connective spectral schemes, stable quasi-coherent coefficient fibres, substitution, spectral Zariski descent, and relative-spectrum comprehension are organized in one indexed geometry. This supplies the fixed semantic background in which nonlinear context extension and stable coefficient transport can be used together while remaining distinct.
2. **Two distinct differential mechanisms are constructed and compared through cotangent data.** Each stable coefficient fibre carries a cofree cocommutative-coalgebra differential calculus, while the global cotangent classifier represents nonlinear first-order variation and constructs structured square-zero extensions. The two mechanisms are not identified: they meet on the stated free finite-projective comparison domain through the universal spectral Kähler derivation. Closing square-zero extensions under finite composition produces the infinitesimal test geometry, whose endpoint semantics yields differential cohesion and, after stabilization, an exact-adjoint fracture theorem and its subsequent BNV-type recognition.
3. **Complete fixed-centre geometry is reconstructed comonadically from shifted-cotangent data on its theorem domain.** Postnikov stages relate homotopical approximation to finite infinitesimal substitutions, while the canonical adic filtration supplies formal order and its cotangent-controlled associated graded. For a connective centre, connective almost finitely

presented augmentations that are surjective on π_0 and fixed by canonical adic completion are recovered comonadically from anchored shifted-cotangent coalgebra data. A separate centre-free construction uses Postnikov continuity and nilpotent excision to define a left-exact reconstruction localization. Tangent recovery returns these reconstruction mechanisms to the cotangent interface.

4. **Finite-infinitesimal invariance produces a left-exact de Rham reflector and an effective descent relation.** Effective-image factorization separates the universal de Rham-local reflection from the quotient actually presented by generalized points. Stable scalar descent on the resulting relation produces intrinsic forms, a homotopy-coherent de Rham differential, and the associated multiplicative and higher descent data. Formal étale descent and affine density then provide internal comprehension objects for the resulting form judgments.
5. **Effective descent generates formal disks, infinite jets, crystalline transport, and a formal PDE calculus.** Dependent adjoints along the effective quotient produce the formal-disk monad and infinite-jet comonad. Effective descent identifies quotient objects, disk actions, jet coactions, and crystalline transport as presentations of one infinitesimal compatibility structure. The jet comonad then organizes differential operators, generalized equations, solution objects, Cartan prolongation, and the intrinsic formal-integrability criterion.

These results use several established mathematical traditions, but at different interfaces of the construction. Spectral and derived algebraic geometry provide the connective geometric setting, quasi-coherent coefficients, and cotangent deformation theory [30, 31, 34, 37]. Categorical logic and linear/nonlinear semantics provide language for indexed substitution and context extension [25, 6], while differential categories, differential linear logic, and spectral cofree coalgebras provide comparison targets for the fibrewise differential branch [8, 14, 29, 4]. Formal integration and derived deformation theory supply the nearest comparison for the fixed-centre reconstruction problem [32, 9]. Infinitesimal sites, crystals, and derived formal geometry provide the surrounding language for de Rham localization and crystalline descent [40, 19, 17], and synthetic and categorical approaches to jets and PDEs provide the principal comparison for the final jet-theoretic calculus [28, 38, 27, 21].

The novelty is not that each of these mathematical ingredients is new in isolation. It is that the relevant operations are constructed and connected inside a single connective–stable indexed architecture, with the passages between nonlinear geometry, stable coefficients, infinitesimal variation, reconstruction, effective descent, and jets supplied by explicit universal constructions rather than by adding independent compatibility data.

The comparison with differential cohomology is correspondingly limited. The stabilized endpoint fracture theorem has the same formal exact-adjunction shape as sheaf-of-spectra models of differential cohomology [24, 11], but the dissertation does not identify its stable infinitesimal coefficients with a geometric differential-cohomology theory. No independent curvature, period, cycle, differential-form, or transfer package is supplied by this comparison. The phrase *BNV-type* is used only for this formal recognition, after the stabilized endpoint fracture theorem has already been established.

5 Foundational conventions, scope, and organization

The main ambient convention is the distinction between connective geometry and stable coefficients. Coordinate algebras lie in

$$\mathbf{CAlg}(\mathbf{Sp})^{\mathbf{cn}},$$

while module and quasi-coherent coefficient categories retain their full stable structure. All categories are infinity-categories unless a truncation is specified. Mapping spaces are written $\mathrm{Map}_{\mathcal{C}}(X, Y)$, and mapping spectra in stable infinity-categories are written $\mathrm{map}_{\mathcal{C}}(X, Y)$.

A second convention concerns higher coherence. Structure supplied functorially by a construction is retained through that construction: adjunctions provide units, counits, and mates; Cartesian and coCartesian transport provide substitution coherence; Kan extensions and descent provide comparison and gluing maps; and effective descent provides the simplicial coherence used in crystalline transport. Later notation suppresses these higher coherences once their source has been fixed. By contrast, affine charts, local trivializations, representatives of homotopy classes, finite square-zero factorizations, local equations, and auxiliary centres introduced inside proofs are proof witnesses and are not retained as additional structure.

This distinction is particularly important for finite infinitesimal arrows. Membership in the finite-composition closure of square-zero substitutions is an intrinsic property, whereas the length of a chosen factorization is not an intrinsic infinitesimal order. Formal order is supplied separately by the canonical adic filtration on the domain where that construction applies. Conditions such as almost finite presentation, coherence, completeness, Artinianity, formal smoothness, and formal étaleness likewise specify theorem domains rather than additional fields of structure.

The differential equations constructed here are formal and spectral-geometric. Intrinsic formal integrability means effective descent along the finite-infinitesimal de Rham relation, equivalently an object over $Q_{X/S}$. It does not imply analytic existence or regularity of solutions. The explicitly physical part of the broader programme lies beyond the present scope: Lagrangian field theory, Euler–Lagrange constructions, variational bicomplexes, higher gauge fields, and their physical interpretation are not claimed in this dissertation.

The five technical chapters follow the dependency order established above. Chapters 1–2 construct and differentiate the indexed spectral doctrine. Chapter 3 asks when the resulting infinitesimal data reconstruct nonlinear geometry. Chapters 4–5 instead pass to finite-infinitesimal invariance, effective de Rham descent, and the resulting crystalline jet calculus. Together they provide the infinitesimal and jet-theoretic foundation on which the later variational programme is intended to build.

Note on use of AI

Generative AI systems were used during the research and development of this dissertation as tools for mathematical investigation, discussion, and editorial refinement. Responsibility for the mathematical content, arguments, and conclusions rests with the author.

Chapter 1

Spectral Algebraic Geometry: Theories, Contexts, and Comprehension

1.1 Orientation: theories, models, contexts, and comprehension

Geometry and logic are both organized by change of context. In geometry, objects and coefficients are transported along maps of bases, local data are reconstructed by descent, and relative algebra can determine new spaces over a given base. In categorical semantics, the corresponding operations appear as substitution and context extension. The question of this chapter is whether these operations can be organized together in spectral algebraic geometry: can geometric change of base, stable coefficients, locality, and context extension be made into one coherent indexed geometry? We use the words **context**, **substitution**, and **comprehension** for these roles. The terminology is drawn from indexed categories in categorical logic and semantics, while the constructions themselves are spectral-geometric; compare [25].

The obstruction is that the geometric and linear layers have different homotopical character. Affine coordinates are connective commutative ring spectra, whereas the natural coefficient categories are stable module categories. Mapping spectra, shifts, cofibres, cotangent complexes, and other linearized constructions require the full stable layer. The chapter therefore keeps connective geometry and fully stable coefficient theory distinct while making their changes of context coherent and their local data reconstructible.

The core construction has three stages. First, connective commutative ring spectra are obtained as models of the commutative operadic theory and then read contravariantly as affine spectral contexts. In symbols,

$$\mathrm{Comm}^{\otimes} \longrightarrow \mathrm{Sp}_{\geq 0}^{\otimes} \longrightarrow \mathrm{Alg}_{\mathrm{Comm}^{\otimes}}(\mathrm{Sp}_{\geq 0}^{\otimes}) \simeq \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}} \longrightarrow \mathrm{Aff}_{\mathrm{Sp}} := (\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}})^{\mathrm{op}}.$$

Here $\mathrm{Comm}^{\otimes}$ is the commutative ∞ -operad and $\mathrm{Sp}_{\geq 0}^{\otimes}$ is the symmetric-monoidal universe in which its coordinate models are interpreted. The word **theory** is used in this operadic sense: this is the symmetric-monoidal analogue of interpreting an algebraic theory in a semantic universe, not an identification of an ∞ -operad with a finite-product Lawvere theory. Thus the first stage is the

pattern **theory** $\rightarrow$ **models** $\rightarrow$ **affines**, specialized to connective spectral algebra; compare [34, 37, 26].

Second, each affine context is equipped with a stable coefficient fibre. If $X = \mathrm{Spec}(A)$, then

$$\mathrm{QCoh}_{\mathrm{aff}}(X) := \mathrm{Mod}_A,$$

and a context morphism

$$f : \mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A)$$

represented by $A \rightarrow B$ acts on coefficients by extension of scalars,

$$f^* = - \otimes_A B : \mathrm{Mod}_A \longrightarrow \mathrm{Mod}_B.$$

We read this pullback geometrically as substitution. The module construction makes these substitutions functorial and strong symmetric monoidal, with restriction of scalars as right adjoint. The affine coefficient system is then extended to generalized contexts and, after Zariski descent, restricts to

$$\mathrm{QCoh} : (\mathrm{Sch}^{\mathrm{sp}})^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L), \quad X \longmapsto \mathrm{QCoh}(X).$$

The detailed Kan-extension and adjoint formalism will be introduced only when it becomes mathematically active. At this stage the point is that stable coefficients vary coherently with geometric context.

Third, locality selects the geometric contexts and relative algebra returns coefficient data to geometry. Spectral localizations generate the Zariski topology, quasi-coherent coefficients satisfy descent, and affine contexts glue to spectral schemes. Over a spectral scheme X , relative spectrum then identifies connective quasi-coherent commutative algebras with affine morphisms over X :

$$\mathrm{QCAlg}(X)^{\mathrm{cn}, \mathrm{op}} \xrightarrow{\simeq} (\mathrm{Sch}_{/X}^{\mathrm{sp}})_{\mathrm{aff}}.$$

Its universal property identifies maps into the corresponding affine extension with algebra terms after substitution. This is the comprehension mechanism constructed in the chapter: suitable algebraic data over a context are represented by a geometric extension of that context, compatibly with change of base.

These three stages form the structural spine of the chapter:

theory $\rightarrow$ models $\rightarrow$ contexts $\rightarrow$ substitution $\rightarrow$ descent $\rightarrow$ comprehension.

The distinction between connective coordinates and stable coefficients remains active throughout. Coordinate algebras lie in $\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}}$, while their coefficient categories retain the full stable structure required by linear and infinitesimal constructions. Once this indexed geometry has been established, the later sections develop its consequences: closed subcontexts and gluing, linear and section representations, distinguished affine contexts, and the recovery of the ordinary 0-truncated sector. The purpose of the intervening theory is to make all of these constructions interact without conflating the connective and stable levels.

The foundational spectral-algebraic and spectral-geometric results used in this construction are drawn from [34, 37, 30, 26]. The following conventions fix the technical background once, so that the main argument can subsequently refer to it without reopening coherence, size, or variance issues.

Convention 1.1.1. All categories are ∞ -categories unless explicitly stated otherwise. Mapping spaces are denoted by

$$\mathrm{Map}_{\mathcal{C}}(X, Y).$$

Mapping spectra in a stable ∞ -category are denoted by

$$\mathrm{map}_{\mathcal{C}}(X, Y).$$

Equivalences are denoted by

$$\simeq.$$

Convention 1.1.2 (Higher coherence). Whenever a structure is produced by a functor, an adjunction, coCartesian transport, an operadic construction, a Kan extension, a limit, or a descent totalization, we use the coherent higher structure supplied by that construction. No independent coherence conditions are imposed. After the source of the coherent structure has been specified, its higher homotopies are suppressed from the notation.

Convention 1.1.3 (Constructed structure and proof witnesses). Choices of representatives, atlases, trivializations, cell expressions, and presentations made inside proofs are not retained as additional structure unless explicitly stated. Likewise, comparison, exchange, base-change, projection, and coherence morphisms are those constructed from the displayed functors, adjunctions, Kan extensions, or descent limits. Their invertibility is a theorem to be proved or cited where it is used. Terminology belonging to an external structured-geometric presentation is not transported into the native category unless it is explicitly defined here.

Convention 1.1.4 (Size). Fix Grothendieck universes

$$\mathbb{U} \subset \mathbb{V}.$$

Spectra, ring spectra, modules, and geometric objects are taken to have $\mathbb{U}$ -small underlying data. The ∞ -categories collecting all such objects are regarded at the $\mathbb{V}$ -level, and presheaf, sheaf, functor, and presentable-category constructions are formed in that enlarged ambient universe. Enlargement functors are suppressed when they induce the canonical fully faithful identification of the construction in question.

Convention 1.1.5 (Monoidal structures on categories). The notation

$$\mathrm{CAlg}(\mathrm{Cat}_{\infty})$$

uses the Cartesian symmetric monoidal structure on Cat_{∞} . The notation

$$\mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L)$$

uses the tensor product of presentable stable ∞ -categories. Its morphisms are colimit-preserving strong symmetric monoidal functors. We write

$$\mathrm{Pr}_{\mathrm{st}}^R$$

for presentable stable ∞ -categories with accessible right-adjoint functors. Passage to right adjoints gives the canonical equivalence

$$(\mathrm{Pr}_{\mathrm{st}}^L)^{\mathrm{op}} \xrightarrow[\simeq]{\mathrm{RA} \mathrm{dj}} \mathrm{Pr}_{\mathrm{st}}^R.$$

Its functoriality supplies the coherent composition of right adjoints used below.

1.2 Spectral semantic universes

The construction begins by separating the two spectral roles identified in the orientation. The full category $\mathbf{Sp}$ supplies stable coefficients and linear constructions, whereas $\mathbf{Sp}_{\geq 0}$ supplies the symmetric-monoidal universe for connective coordinates. This section records only the structural features of spectra needed to keep those roles distinct and to compare them through the Postnikov t -structure.

1.2.1 The stable symmetric monoidal category of spectra

Proposition 1.2.1 (Spectral semantic category). *The ∞ -category of spectra admits a canonical structure*

$$\mathbf{Sp}^{\otimes} \in \mathrm{CAlg}(\mathbf{Pr}_{\mathrm{st}}^L).$$

Its tensor product is the smash product

$$- \otimes -: \mathbf{Sp} \times \mathbf{Sp} \longrightarrow \mathbf{Sp},$$

its tensor unit is the sphere spectrum $\mathbb{S}$, and its tensor product preserves small colimits separately in each variable.

Proof.

- (1) **Stabilize pointed spaces.** The stabilization of pointed spaces is a presentable stable ∞ -category.
- (2) **Stabilize the smash product.** The smash product on pointed spaces stabilizes to a symmetric monoidal product on spectra.
- (3) **Identify the tensor unit.** The stabilization universal property identifies its tensor unit with the sphere spectrum.
- (4) **Preserve colimits separately.** Separate preservation of colimits follows from the presentably symmetric monoidal construction; see [34].

□

Proposition 1.2.2 (Closedness of spectra). *For every spectrum X , the functor*

$$- \otimes X : \mathbf{Sp} \longrightarrow \mathbf{Sp}$$

admits a right adjoint

$$\underline{\mathrm{Hom}}_{\mathbf{Sp}}(X, -) : \mathbf{Sp} \longrightarrow \mathbf{Sp}.$$

Thus there are natural equivalences

$$\mathrm{Map}_{\mathbf{Sp}}(Z \otimes X, Y) \simeq \mathrm{Map}_{\mathbf{Sp}}(Z, \underline{\mathrm{Hom}}_{\mathbf{Sp}}(X, Y)).$$

Proof.

- (1) **Preserve small colimits.** The functor $- \otimes X$ preserves small colimits by 1.2.1.
- (2) **Apply the adjoint functor theorem.** Since $\mathbf{Sp}$ is presentable, the adjoint functor theorem supplies the right adjoint.

□

The closed structure will later be used through its adjunction, which we fix in the following form:

$$\mathrm{Sp} \begin{array}{c} \xrightarrow{-\otimes X} \\ \xleftarrow{\underline{\mathrm{Hom}}_{\mathrm{Sp}}(X, -)} \end{array} \mathrm{Sp}.$$

1.2.2 The canonical t -structure

Definition 1.2.1 (Postnikov t -structure). For every integer n , let

$$\mathrm{Sp}_{\geq n}$$

be the full subcategory of spectra X for which

$$\pi_i(X) = 0 \quad (i < n),$$

and let

$$\mathrm{Sp}_{\leq n}$$

be the full subcategory of spectra X for which

$$\pi_i(X) = 0 \quad (i > n).$$

The pair

$$(\mathrm{Sp}_{\geq 0}, \mathrm{Sp}_{\leq 0})$$

is the **canonical Postnikov t -structure** on Sp . Its truncation functors are denoted

$$\tau_{\geq n}^{\mathrm{Sp}} : \mathrm{Sp} \longrightarrow \mathrm{Sp}_{\geq n}$$

and

$$\tau_{\leq n}^{\mathrm{Sp}} : \mathrm{Sp} \longrightarrow \mathrm{Sp}_{\leq n}.$$

Proposition 1.2.3 (Postnikov truncation triangle). *For every spectrum X and every integer n , there is a natural fibre and cofibre sequence*

$$\tau_{\geq n+1}^{\mathrm{Sp}} X \longrightarrow X \longrightarrow \tau_{\leq n}^{\mathrm{Sp}} X.$$

The functors $\tau_{\geq n}^{\mathrm{Sp}}$ and $\tau_{\leq n}^{\mathrm{Sp}}$ are respectively right and left adjoint to the inclusions of the corresponding full subcategories.

Proof.

- (1) **Identify the truncation sequence.** This is the defining truncation sequence of the Postnikov t -structure.
- (2) **Use stability.** The equality of fibre and cofibre sequences follows from stability.

□

Proposition 1.2.4 (Completeness and detection). *The Postnikov t -structure is accessible, compatible with filtered colimits, left complete, and right complete. In particular, there are natural equivalences*

$$X \simeq \varprojlim_n \tau_{\leq n}^{\mathrm{Sp}} X$$

and

$$X \simeq \varinjlim_n \tau_{\geq -n}^{\mathrm{Sp}} X.$$

A morphism of spectra is an equivalence if and only if it induces an isomorphism on every homotopy group.

Proof.

- (1) **Invoke the Postnikov completeness theorem.** These are the established completeness and accessibility properties of the Postnikov t -structure; see [34].

□

Definition 1.2.2 (Discrete spectrum). A spectrum X is **discrete** if

$$X \in \mathrm{Sp}_{\geq 0} \cap \mathrm{Sp}_{\leq 0}.$$

The heart of the Postnikov t -structure is denoted

$$\mathrm{Sp}^{\heartsuit}.$$

Proposition 1.2.5 (Eilenberg–Mac Lane equivalence). *The Eilenberg–Mac Lane functor induces an equivalence*

$$H_{\mathrm{Ab}}: \mathrm{Ab} \xrightarrow{\sim} \mathrm{Sp}^{\heartsuit}.$$

Under this equivalence, the inverse functor is π_0 .

Proof.

- (1) **Recover a discrete spectrum from degree zero.** A discrete spectrum is determined by its degree-zero homotopy group.
- (2) **Realize every abelian group.** Every abelian group is realized by an Eilenberg–Mac Lane spectrum.

□

1.2.3 Compatibility with the smash product

Proposition 1.2.6 (Connectivity of the smash product). *If*

$$X \in \mathrm{Sp}_{\geq m} \quad \text{and} \quad Y \in \mathrm{Sp}_{\geq n},$$

then

$$X \otimes Y \in \mathrm{Sp}_{\geq m+n}.$$

In particular, $\mathrm{Sp}_{\geq 0}$ is closed under the smash product and contains the tensor unit $\mathbb{S}$.

Proof.

- (1) **Reduce to the connective case.** After suspension, it is enough to consider $m = n = 0$.
- (2) **Choose connective generators.** Connective spectra are generated under colimits and extensions by suspension spectra of pointed spaces in nonnegative degrees.
- (3) **Tensor the generators.** The smash product preserves colimits separately and carries the corresponding generators to connective spectra.
- (4) **Invoke monoidal compatibility.** Equivalently, the compatibility follows from the monoidal compatibility of the Postnikov t -structure; see [34].

□

Corollary 1.2.1 (Right t -exactness of connective tensoring). *For every connective spectrum X , the functor*

$$- \otimes X : \mathrm{Sp} \longrightarrow \mathrm{Sp}$$

is right t -exact.

Proof.

- (1) **Apply connectivity of the smash product.** If Y is connective, then $Y \otimes X$ is connective by 1.2.6.

□

1.2.4 The connective coordinate universe

Proposition 1.2.7 (Connective spectral semantic universe). *The full subcategory*

$$\mathrm{Sp}_{\geq 0} \subseteq \mathrm{Sp}$$

inherits a presentably symmetric monoidal structure $\mathrm{Sp}_{\geq 0}^{\otimes}$. Its tensor product is the restriction of the smash product, its tensor unit is $\mathbb{S}$, and the inclusion

$$\mathrm{Sp}_{\geq 0}^{\otimes} \hookrightarrow \mathrm{Sp}^{\otimes}$$

is strong symmetric monoidal. Tensoring in $\mathrm{Sp}_{\geq 0}$ preserves small colimits separately in each variable.

Proof.

- (1) **Closure under the monoidal product.** The tensor unit is connective, and 1.2.6 shows that the smash product of connective spectra is connective. Hence the monoidal structure on Sp restricts to $\mathrm{Sp}_{\geq 0}$.
- (2) **Presentability.** The connective aisle of the accessible Postnikov t -structure is a presentable full subcategory closed under small colimits.
- (3) **Compute connective colimits.** Those colimits are computed in Sp .
- (4) **Preserve colimits under tensoring.** Since the smash product on Sp preserves colimits separately, its restriction has the same property on connective spectra. See [34].

□

The preceding results isolate the two universes needed later: $\mathrm{Sp}_{\geq 0}^{\otimes}$ for connective coordinates and $\mathrm{Sp}^{\otimes}$ for stable coefficients. The next section uses the connective universe to form the coordinate models.

1.3 The commutative ∞ -operadic theory and its spectral models

With the two spectral roles separated, the next step is to construct the coordinate models. Interpreting the commutative ∞ -operad in $\mathrm{Sp}^\otimes$ gives commutative ring spectra; restricting the semantic universe to $\mathrm{Sp}_{\geq 0}^\otimes$ selects the connective models that will become affine contexts. The results below record exactly the free, reflective, discrete, and Postnikov structures used in that passage.

Let

$$\mathrm{Comm}^\otimes \longrightarrow N(\mathrm{Fin}_*)$$

denote the commutative ∞ -operad.

Definition 1.3.1 (Commutative ring spectrum). A **commutative ring spectrum** is an algebra over the commutative ∞ -operad in spectra:

$$A \in \mathrm{Alg}_{\mathrm{Comm}^\otimes}(\mathrm{Sp}^\otimes).$$

The resulting ∞ -category is denoted

$$\mathrm{CAlg}(\mathrm{Sp}) := \mathrm{Alg}_{\mathrm{Comm}^\otimes}(\mathrm{Sp}^\otimes).$$

Its objects are also called $\mathbb{E}_\infty$ -rings.

Remark 1.3.1 (Symmetric monoidal envelope). The operadic definition above has an equivalent symmetric-monoidal presentation. Let

$$\mathrm{Env}(\mathrm{Comm})^\otimes$$

denote the symmetric monoidal envelope of the commutative ∞ -operad. Its universal property gives a natural equivalence

$$\mathrm{CAlg}(\mathrm{Sp}) \simeq \mathrm{Fun}^\otimes(\mathrm{Env}(\mathrm{Comm})^\otimes, \mathrm{Sp}^\otimes).$$

Thus a commutative ring spectrum may be read either as an operadic algebra or as a symmetric-monoidal interpretation of the envelope; in either form, the commutative operations and their higher coherences are retained. See [34, Proposition 2.2.4.9].

For later comparison, the two presentations are related by the canonical equivalence

$$\mathrm{Alg}_{\mathrm{Comm}^\otimes}(\mathrm{Sp}^\otimes) \xrightarrow{\simeq} \mathrm{Fun}^\otimes(\mathrm{Env}(\mathrm{Comm})^\otimes, \mathrm{Sp}^\otimes).$$

Remark 1.3.2 (Theory, semantic universe, and relative affine geometry). The preceding equivalence isolates the semantic role of the commutative ∞ -operad. A commutative ring spectrum is an interpretation of the commutative operadic theory in a symmetric-monoidal spectral universe. In the connective coordinate universe, the theory-to-model passage is

$$\mathrm{Comm}^\otimes \longrightarrow \mathrm{Sp}_{\geq 0}^\otimes \longrightarrow \mathrm{Alg}_{\mathrm{Comm}^\otimes}(\mathrm{Sp}_{\geq 0}^\otimes) \simeq \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}}.$$

The variance reversal of 1.4.1 then reads these models as affine contexts. This theory–models–affines pattern is the absolute instance of the relative-algebraic-geometric construction that will reappear inside $\mathrm{QCoh}(X)$ under relative spectrum. See [34, 37, 26].

Proposition 1.3.1 (Presentability of commutative ring spectra). *The ∞ -category*

$$\mathrm{CAlg}(\mathrm{Sp})$$

is presentable. Its forgetful functor

$$\mathrm{oblv}_{\mathrm{CAlg}}: \mathrm{CAlg}(\mathrm{Sp}) \longrightarrow \mathrm{Sp}$$

is accessible, creates limits and sifted colimits, and admits a left adjoint.

Proof.

- (1) **Apply presentability for operadic algebras.** This is the presentability theorem for algebras over an accessible ∞ -operad in a presentably symmetric monoidal ∞ -category whose tensor product preserves small colimits separately; see [34].

□

1.3.1 Free commutative ring spectra

Definition 1.3.2 (Free commutative algebra). The left adjoint to the forgetful functor is denoted

$$\mathrm{Sym}: \mathrm{Sp} \longrightarrow \mathrm{CAlg}(\mathrm{Sp}).$$

It is the **free commutative algebra functor** associated to the commutative ∞ -operad.

The free–forgetful adjunction will be used repeatedly:

$$\mathrm{Sp} \begin{array}{c} \xrightarrow{\mathrm{Sym}} \\ \xleftarrow{\mathrm{oblv}_{\mathrm{CAlg}}} \end{array} \mathrm{CAlg}(\mathrm{Sp}).$$

Proposition 1.3.2 (Underlying free algebra). *For every spectrum M , there is a natural equivalence of underlying spectra*

$$\mathrm{oblv}_{\mathrm{CAlg}} \mathrm{Sym}(M) \simeq \bigoplus_{n \geq 0} (M^{\otimes n})_{h\Sigma_n},$$

where the summand for $n = 0$ is the sphere spectrum.

Proof.

- (1) **Apply the free-algebra formula.** This is the explicit free-algebra formula for the commutative ∞ -operad in a presentably symmetric monoidal ∞ -category; see [34].

□

Corollary 1.3.1 (Connectivity of free algebras). *If M is connective, then $\mathrm{Sym}(M)$ is connective.*

Proof.

- (1) **Check tensor powers.** Every tensor power $M^{\otimes n}$ is connective by 1.2.6.
- (2) **Check the remaining colimits.** Homotopy orbits and small coproducts of connective spectra are connective.
- (3) **Apply the free-algebra formula.** Apply 1.3.2.

□

1.3.2 Connective commutative ring spectra

Definition 1.3.3 (Connective $\mathbb{E}_\infty$ -ring). A commutative ring spectrum A is **connective** if its underlying spectrum is connective. The full subcategory of connective commutative ring spectra is denoted

$$\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}} \subseteq \mathrm{CAlg}(\mathrm{Sp}).$$

Proposition 1.3.3 (Connective models as algebras in connective spectra). *The strong symmetric monoidal inclusion*

$$\mathrm{Sp}_{\geq 0}^{\otimes} \hookrightarrow \mathrm{Sp}^{\otimes}$$

induces a fully faithful functor

$$\mathrm{Alg}_{\mathrm{Comm}^{\otimes}}(\mathrm{Sp}_{\geq 0}^{\otimes}) \longrightarrow \mathrm{Alg}_{\mathrm{Comm}^{\otimes}}(\mathrm{Sp}^{\otimes}) = \mathrm{CAlg}(\mathrm{Sp}),$$

whose essential image is $\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}}$. Hence there is a canonical equivalence

$$\mathrm{Alg}_{\mathrm{Comm}^{\otimes}}(\mathrm{Sp}_{\geq 0}^{\otimes}) \xrightarrow{\simeq} \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}}.$$

Proof.

- (1) **Restrict the operadic structure.** By 1.2.7, connective spectra form a full symmetric monoidal subcategory of spectra. An algebra object in this subcategory is therefore a commutative ring spectrum whose underlying spectrum is connective.
- (2) **Identify the essential image.** Conversely, if a commutative ring spectrum has connective underlying spectrum, all of its operadic structure maps already lie in the full subcategory $\mathrm{Sp}_{\geq 0}$. Full faithfulness and the displayed equivalence follow. Compare [34, 37].

□

Proposition 1.3.4 (Presentability, connective cover, and limits). *The ∞ -category*

$$\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}}$$

is presentable. The inclusion

$$\iota: \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}} \hookrightarrow \mathrm{CAlg}(\mathrm{Sp})$$

preserves small colimits and admits a right adjoint

$$\tau_{\geq 0}^{\mathrm{CAlg}}: \mathrm{CAlg}(\mathrm{Sp}) \longrightarrow \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}},$$

with

$$\mathrm{obv}_{\mathrm{CAlg}}(\tau_{\geq 0}^{\mathrm{CAlg}} A) \simeq \tau_{\geq 0}^{\mathrm{Sp}} \mathrm{obv}_{\mathrm{CAlg}}(A).$$

Thus

$$\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}} \begin{array}{c} \xrightarrow{\iota} \\ \xleftarrow{\tau_{\geq 0}^{\mathrm{CAlg}}} \end{array} \mathrm{CAlg}(\mathrm{Sp}) \quad \iota \dashv \tau_{\geq 0}^{\mathrm{CAlg}}.$$

For every small diagram

$$D: K \longrightarrow \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}},$$

its limit is

$$\varprojlim_K^{\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}}} D \simeq \tau_{\geq 0}^{\mathrm{CAlg}} \left(\varprojlim_K^{\mathrm{CAlg}(\mathrm{Sp})} \iota D \right).$$

The inclusion ι fails to preserve finite limits in general.

Proof.

- (1) **Connective cover.** The connective-cover theorem for structured ring spectra identifies $\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}}$ as a presentable full subcategory closed under small colimits, and identifies the right adjoint of its inclusion with connective cover on underlying spectra; see [34, 37]. Hence

$$\iota \dashv \tau_{\geq 0}^{\mathrm{CAlg}} \quad \text{and} \quad \mathrm{oblv}_{\mathrm{CAlg}} \tau_{\geq 0}^{\mathrm{CAlg}} \simeq \tau_{\geq 0}^{\mathrm{Sp}} \mathrm{oblv}_{\mathrm{CAlg}}.$$

- (2) **Limit formula.** Put

$$L := \varprojlim_K^{\mathrm{CAlg}(\mathrm{Sp})} \iota D.$$

For every connective commutative ring spectrum C , adjunction and the universal property of L give

$$\begin{aligned} \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}}}(C, \tau_{\geq 0}^{\mathrm{CAlg}} L) &\simeq \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})}(\iota C, L) \\ &\simeq \varprojlim_{k \in K} \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})}(\iota C, \iota D(k)) \\ &\simeq \varprojlim_{k \in K} \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}}}(C, D(k)). \end{aligned}$$

Yoneda identifies $\tau_{\geq 0}^{\mathrm{CAlg}} L$ with the limit of D .

- (3) **Failure of finite-limit preservation.** Consider

$$P := H\mathbb{Z} \times_{H(\mathbb{Z}[x])} H\mathbb{Z},$$

where both maps are induced by inclusion of constant polynomials. The forgetful functor creates limits, so the underlying spectra fit into

$$P \longrightarrow H\mathbb{Z} \oplus H\mathbb{Z} \xrightarrow{\mathrm{diff}} H(\mathbb{Z}[x]).$$

The long exact homotopy sequence gives

$$\pi_{-1}(P) \simeq \mathrm{coker}(\mathbb{Z} \oplus \mathbb{Z} \rightarrow \mathbb{Z}[x]) \simeq \mathbb{Z}[x]/\mathbb{Z},$$

so P is nonconnective. The connective pullback is therefore $\tau_{\geq 0}^{\mathrm{CAlg}} P$.

□

Corollary 1.3.2 (Connective pushouts). *If*

$$B \longleftarrow A \longrightarrow C$$

is a span in $\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}}$, then the pushout

$$B \otimes_A C$$

computed in $\mathrm{CAlg}(\mathrm{Sp})$ is connective.

Proof.

- (1) **Identify the pushout as a colimit.** It is a colimit of connective commutative ring spectra.
 (2) **Apply closure under colimits.** Apply 1.3.4.

□

1.3.3 The 0-truncated algebraic sector

Proposition 1.3.5 (Discrete commutative ring spectra). *Operadic Postnikov truncation induces an equivalence*

$$\mathrm{CAlg}(\mathrm{Sp})_{\leq 0}^{\mathrm{cn}} \xrightarrow{\pi_0} \mathrm{CRing}$$

whose inverse is the Eilenberg–Mac Lane construction

$$H_{\mathrm{CRing}}: \mathrm{CRing} \hookrightarrow \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}}.$$

Consequently, H_{CRing} is fully faithful and its essential image consists of the commutative ring spectra with discrete underlying spectrum.

Proof.

- (1) **Operadic truncation.** The Postnikov truncation functors on spectra lift to commutative algebra objects, and the full subcategory of connective 0-truncated commutative ring spectra is equivalent to ordinary commutative rings; see [34, 37].
- (2) **Identification of the inverse.** Under this equivalence, degree-zero homotopy is the functor to ordinary commutative rings, while the inverse sends R to HR . The asserted full faithfulness and description of the essential image follow.

□

Proposition 1.3.6 (Degree-zero reflection adjunction). *The functor*

$$\pi_0: \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}} \longrightarrow \mathrm{CRing}$$

is left adjoint to the Eilenberg–Mac Lane functor. Thus

$$\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}} \begin{array}{c} \xrightarrow{\pi_0} \\ \xleftarrow{H_{\mathrm{CRing}}} \end{array} \mathrm{CRing} \quad \pi_0 \dashv H_{\mathrm{CRing}},$$

and there are natural equivalences

$$\mathrm{Map}_{\mathrm{CRing}}(\pi_0(A), R) \simeq \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})}(A, HR).$$

The adjunction unit is the natural morphism

$$A \longrightarrow H\pi_0(A).$$

Proof.

- (1) **Reflection to discrete rings.** By 1.3.5, ordinary commutative rings identify with the full subcategory of connective 0-truncated commutative ring spectra.
- (2) **Mapping-space computation.** Operadic 0-truncation is left adjoint to the inclusion of this full subcategory. Under the equivalence with CRing , the reflector is π_0 and the inclusion is H_{CRing} . This gives the displayed adjunction and its unit; see [34].

□

1.3.4 Operadic Postnikov stages

Proposition 1.3.7 (Operadic Postnikov truncations). *For every integer $n \geq 0$, the truncation functor on spectra lifts to a functor*

$$\tau_{\leq n}^{\text{CAlg}}: \text{CAlg}(\text{Sp})^{\text{cn}} \longrightarrow \text{CAlg}(\text{Sp})^{\text{cn}}$$

whose underlying spectrum is the ordinary Postnikov truncation. For every connective commutative ring spectrum A , there is a natural equivalence

$$A \simeq \varprojlim_n \tau_{\leq n}^{\text{CAlg}} A$$

in $\text{CAlg}(\text{Sp})$.

Proof.

- (1) **Use reflectivity of truncated algebras.** The full subcategory of n -truncated objects in commutative ring spectra is reflective.
- (2) **Identify the reflector on connective objects.** The forgetful functor identifies the reflector on connective objects with the spectral Postnikov truncation.
- (3) **Transfer Postnikov convergence.** Since the forgetful functor creates limits, Postnikov convergence in spectra from 1.2.4 implies convergence in commutative ring spectra. See [34, 30].

□

Remark 1.3.3 (Postnikov notation on algebra objects). The notation

$$\text{CAlg}(\text{Sp})^{\text{cn}}, \quad \text{CAlg}(\text{Sp})_{\leq n}^{\text{cn}}, \quad \tau_{\leq n}^{\text{CAlg}}$$

is inherited from the Postnikov structure on underlying spectra: connectivity and truncation are detected after forgetting the commutative algebra structure. No t -structure on the nonstable ∞ -category $\text{CAlg}(\text{Sp})$ is being asserted. The same convention will apply to connective quasi-coherent commutative algebra objects, whose connectivity is measured in the underlying stable coefficient category.

The coordinate category $\text{CAlg}(\text{Sp})^{\text{cn}}$ is now equipped with the free, reflective, discrete, and Postnikov structures needed later. Its next role is geometric rather than algebraic: variance reversal turns these models into affine spectral contexts.

1.4 Affine spectral contexts

The connective coordinate models now become geometry. Reversing variance turns a connective commutative ring spectrum into an affine spectral context; algebraic pushouts consequently become geometric fibre products, while π_0 provides the topological support on which Zariski locality will be built. This section records that affine geometric interface before stable coefficients are attached to it.

Definition 1.4.1 (Affine spectral scheme). The ∞ -category of **affine spectral schemes** is

$$\mathrm{Aff}_{\mathrm{Sp}} := (\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}})^{\mathrm{op}}.$$

For

$$A \in \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}},$$

the corresponding affine spectral scheme is denoted

$$\mathrm{Spec}(A).$$

Proposition 1.4.1 (Mapping spaces of affine spectral schemes). *For connective commutative ring spectra A and B , there is a natural equivalence*

$$\mathrm{Map}_{\mathrm{Aff}_{\mathrm{Sp}}}(\mathrm{Spec}(B), \mathrm{Spec}(A)) \simeq \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})}(A, B).$$

Proof.

- (1) **Unfold the opposite category.** The mapping-space equivalence is the definition of the opposite ∞ -category.

□

Thus an affine context morphism reverses the direction of its coordinate map:

$$A \xrightarrow{\varphi} B \iff \mathrm{Spec}(B) \xrightarrow{\mathrm{Spec}(\varphi)} \mathrm{Spec}(A).$$

1.4.1 Finite limits of affine spectral schemes

Proposition 1.4.2 (Affine spectral fibre products). *For a diagram of affine spectral schemes*

$$\mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A) \longleftarrow \mathrm{Spec}(C),$$

there is a canonical equivalence

$$\mathrm{Spec}(B) \times_{\mathrm{Spec}(A)} \mathrm{Spec}(C) \simeq \mathrm{Spec}(B \otimes_A C).$$

The commutative ring spectrum $B \otimes_A C$ is connective.

Proof.

- (1) **Reverse the pushout.** The fibre product in the opposite category is represented by the pushout of commutative ring spectra.
- (2) **Check connectivity.** Connectivity follows from 1.3.2.

□

This variance reversal is summarized by the correspondence between algebraic pushouts and affine fibre products:

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & B \otimes_A C \end{array} \iff \begin{array}{ccc} \mathrm{Spec}(B \otimes_A C) & \longrightarrow & \mathrm{Spec}(B) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(C) & \longrightarrow & \mathrm{Spec}(A). \end{array}$$

Corollary 1.4.1 (Finite-limit closure). *The ∞ -category $\mathrm{Aff}_{\mathrm{Sp}}$ admits finite limits.*

Proof.

- (1) **Identify the terminal object.** The terminal affine spectral scheme is $\mathrm{Spec}(\mathbb{S})$.
- (2) **Construct binary fibre products.** Binary fibre products exist by 1.4.2.

□

1.4.2 Underlying topological support

Definition 1.4.2 (Underlying topological space). The underlying topological space of $X = \mathrm{Spec}(A)$ is

$$|X| := |\mathrm{Spec} \pi_0(A)|.$$

We have therefore fixed the affine geometric base: connective coordinate algebras determine affine contexts, their pushouts determine fibre products, and $|\mathrm{Spec} \pi_0(A)|$ supplies the support for locality. The next step is to attach the stable coefficient fibre carried by each such context.

1.5 Stable coefficients in affine context

The stable linear information needed later lives in Mod_A for $X = \mathrm{Spec}(A)$, and an affine context morphism acts on it by extension of scalars. This section constructs the affine coefficient system: substitution is extension of scalars, restriction of scalars is its right adjoint, and relative commutative algebras already encode affine extensions over a fixed base.

Fix a connective commutative ring spectrum

$$A \in \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}}.$$

Definition 1.5.1 (Module spectrum). The ∞ -category of A -module spectra is denoted

$$\mathrm{Mod}_A(\mathrm{Sp})$$

or, when no ambiguity is possible, simply

$$\mathrm{Mod}_A.$$

Theorem 1.5.1 (Spectral module category). *The category*

$$\mathrm{Mod}_A$$

is a presentably stable symmetric monoidal ∞ -category. Its tensor product is the relative tensor product

$$- \otimes_A -,$$

its tensor unit is the regular module A , and its tensor product preserves small colimits separately in each variable.

Proof.

- (1) **Apply the operadic module construction.** This is the operadic module construction for a commutative algebra object in a presentably symmetric monoidal stable ∞ -category; see [34, Theorem 4.5.2.1].

- (2) **Create limits and colimits in spectra.** Limits and colimits are created in spectra.
- (3) **Deduce stability.** Stability follows from stability of Sp .

□

Construction 1.5.1 (Relative tensor product). For a right A -module M and a left A -module N , the relative tensor product is represented by the geometric realization of the two-sided bar construction

$$M \otimes_A N \simeq \left| [n] \mapsto M \otimes A^{\otimes n} \otimes N \right|.$$

Since A is commutative, its left and right module theories are canonically identified.

Proposition 1.5.1 (Closedness of module spectra). *For every A -module M , the functor*

$$- \otimes_A M : \mathrm{Mod}_A \longrightarrow \mathrm{Mod}_A$$

admits a right adjoint

$$\underline{\mathrm{Hom}}_A(M, -).$$

Proof.

- (1) **Preserve small colimits.** The functor preserves small colimits by 1.5.1.
- (2) **Apply the adjoint functor theorem.** Presentability of Mod_A and the adjoint functor theorem supply the right adjoint.

□

1.5.1 Extension and restriction of scalars

Definition 1.5.2 (Extension and restriction of scalars). Let

$$\varphi : A \longrightarrow B$$

be a morphism of commutative ring spectra. The extension-of-scalars functor is

$$\mathrm{Ext}_\varphi := - \otimes_A B : \mathrm{Mod}_A \longrightarrow \mathrm{Mod}_B.$$

Its right adjoint is the restriction-of-scalars functor

$$\mathrm{Res}_\varphi : \mathrm{Mod}_B \longrightarrow \mathrm{Mod}_A.$$

The substitution/right-adjoint pair is therefore

$$\mathrm{Mod}_A \begin{array}{c} \xrightarrow{\mathrm{Ext}_\varphi} \\ \xleftarrow{\mathrm{Res}_\varphi} \end{array} \mathrm{Mod}_B.$$

For composable coordinate maps $A \rightarrow B \rightarrow C$, coherent composition of substitution has the form

$$\begin{array}{ccc} \mathrm{Mod}_A & \xrightarrow{- \otimes_A B} & \mathrm{Mod}_B \\ & \searrow - \otimes_A C & \downarrow - \otimes_B C \\ & & \mathrm{Mod}_C. \end{array}$$

Proposition 1.5.2 (Monoidality of spectral base change). *The functor*

$$\mathrm{Ext}_\varphi = - \otimes_A B$$

is colimit-preserving and strong symmetric monoidal. In particular, there is a natural equivalence

$$(M \otimes_A N) \otimes_A B \simeq (M \otimes_A B) \otimes_B (N \otimes_A B).$$

Proof.

- (1) **Identify extension of scalars.** The coCartesian operadic module construction identifies the transition functor associated to φ with extension of scalars.
- (2) **Supply the monoidal structure.** The same construction equips it with its strong symmetric monoidal structure; see [34, Theorem 4.5.3.1 and Remark 4.5.3.2].

□

Strong symmetric monoidality is the compatibility needed to transport the coefficient tensor product through substitution:

$$\begin{array}{ccc} \mathrm{Mod}_A \times \mathrm{Mod}_A & \xrightarrow{- \otimes_A -} & \mathrm{Mod}_A \\ \mathrm{Ext}_\varphi \times \mathrm{Ext}_\varphi \downarrow & & \downarrow \mathrm{Ext}_\varphi \\ \mathrm{Mod}_B \times \mathrm{Mod}_B & \xrightarrow{- \otimes_B -} & \mathrm{Mod}_B. \end{array}$$

Proposition 1.5.3 (Functoriality of module categories). *The assignments*

$$A \mapsto \mathrm{Mod}_A$$

and

$$(A \rightarrow B) \mapsto (- \otimes_A B)$$

define a functor

$$\mathbf{Mod}_{\mathrm{Sp}}: \mathrm{CAlg}(\mathrm{Sp}) \longrightarrow \mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L).$$

For composable maps $A \rightarrow B \rightarrow C$, there is a canonical coherent equivalence

$$(- \otimes_A B) \otimes_B C \simeq - \otimes_A C.$$

Proof.

- (1) **Straighten the module family.** Straightening the coCartesian family of module categories over $\mathrm{CAlg}(\mathrm{Sp})$ gives the asserted functor.
- (2) **Use coCartesian composition.** The coCartesian composition law supplies the coherent comparison for composable algebra maps; see [34].

□

1.5.2 The affine coefficient system

Definition 1.5.3 (Affine quasi-coherent modules). For an affine spectral scheme $\mathrm{Spec}(A)$, define

$$\mathrm{QCoh}_{\mathrm{aff}}(\mathrm{Spec}(A)) := \mathrm{Mod}_A.$$

For a morphism

$$\mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A),$$

the pullback functor is extension of scalars

$$- \otimes_A B : \mathrm{Mod}_A \longrightarrow \mathrm{Mod}_B.$$

Proposition 1.5.4 (Affine quasi-coherent coefficient system). *The assignments of 1.5.3 define a functor*

$$\mathrm{QCoh}_{\mathrm{aff}} : (\mathrm{Aff}_{\mathrm{Sp}})^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(\mathrm{Pr})^L_{\mathrm{st}}.$$

Proof.

- (1) **Restrict the module functor.** This is the restriction of the functor $\mathbf{Mod}_{\mathrm{Sp}}$ from 1.5.3 to connective commutative ring spectra.

□

Remark 1.5.1 (Affine indexed doctrine). The variance reversal from coordinate algebras to affine geometry also reverses the way the module functor is read. The covariant algebraic functor of 1.5.3 becomes

$$\mathrm{QCoh}_{\mathrm{aff}} : (\mathrm{Aff}_{\mathrm{Sp}})^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(\mathrm{Pr})^L_{\mathrm{st}}.$$

Accordingly, extension of scalars is substitution along an affine context morphism, and restriction of scalars is its right adjoint. This is the complete affine form of the indexed coefficient doctrine; the generalized and global forms will be obtained from it by extension and descent.

1.5.3 Relative commutative algebra objects

Proposition 1.5.5 (Relative commutative spectral algebras). *There is a natural equivalence*

$$\mathrm{CAlg}(\mathrm{Mod}_A) \simeq \mathrm{CAlg}(\mathrm{Sp})_{A/}.$$

Under this equivalence, a commutative algebra object B in A -modules is identified with its structural morphism

$$A \longrightarrow B.$$

Proof.

- (1) **Specialize the operadic equivalence.** This is the commutative specialization of the operadic equivalence between algebras in a module category and algebras under the base algebra; see [34, Corollary 3.4.1.7].

□

Proposition 1.5.6 (Connective relative algebras). *The equivalence of 1.5.5 restricts to an equivalence*

$$\mathrm{CAlg}(\mathrm{Mod}_A)^{\mathrm{cn}} \simeq \mathrm{CAlg}(\mathrm{Sp})_A^{\mathrm{cn}},$$

where connectivity on the left is detected on the underlying A -module. Consequently,

$$\mathrm{Aff}_{\mathrm{Sp}} / \mathrm{Spec}(A) \simeq (\mathrm{CAlg}(\mathrm{Mod}_A)^{\mathrm{cn}})^{\mathrm{op}}.$$

Proof.

- (1) **Preserve connective objects.** The equivalence preserves underlying spectra and therefore preserves connective objects.
- (2) **Pass to opposites.** Passing to opposite categories gives the affine statement.

□

1.5.4 The induced module t -structure

Definition 1.5.4 (Module Postnikov t -structure). Let

$$(\mathrm{Mod}_A)_{\geq 0}$$

be the full subcategory of A -modules whose underlying spectra are connective, and let

$$(\mathrm{Mod}_A)_{\leq 0}$$

be the full subcategory whose underlying spectra are coconnective.

Proposition 1.5.7 (Induced module t -structure). *The pair*

$$((\mathrm{Mod}_A)_{\geq 0}, (\mathrm{Mod}_A)_{\leq 0})$$

is an accessible, left complete, and right complete t -structure on Mod_A . The forgetful functor

$$\mathrm{oblv}_A: \mathrm{Mod}_A \longrightarrow \mathrm{Sp}$$

is t -exact and creates Postnikov truncations.

Proof.

- (1) **Create limits and colimits in spectra.** Limits and colimits of A -modules are created in spectra.
- (2) **Induce the truncation module structure.** The module structure on a truncation is induced by the connective multiplication of A .
- (3) **Detect the t -structure axioms.** The orthogonality and truncation axioms are detected after forgetting to spectra.
- (4) **Transfer completeness.** Completeness follows from completeness of the Postnikov t -structure and creation of limits and colimits. See [34, 37].

□

Proposition 1.5.8 (Heart of the module category). *There is a natural equivalence of abelian categories*

$$\mathrm{Mod}_A^\heartsuit \simeq \mathrm{Mod}_{\pi_0(A)}.$$

Proof.

- (1) **Recover the underlying discrete module.** A discrete A -module is determined by its degree-zero homotopy group.
- (2) **Factor the action through degree zero.** The A -action factors uniquely through the discrete commutative ring spectrum $H\pi_0(A)$.
- (3) **Identify the ordinary module structure.** It is therefore equivalent to an ordinary $\pi_0(A)$ -module structure.

□

Proposition 1.5.9 (Exactness and right t -exactness of base change). *Let $\varphi : A \rightarrow B$ be a morphism of connective commutative ring spectra. Then:*

(i) *the extension-of-scalars functor*

$$\mathrm{Ext}_\varphi : \mathrm{Mod}_A \longrightarrow \mathrm{Mod}_B$$

is exact as a functor of stable ∞ -categories;

(ii) *the functor Ext_φ is right t -exact;*

(iii) *the restriction-of-scalars functor Res_φ is t -exact.*

Proof.

- (1) **Stable exactness.** The functor $\mathrm{Ext}_\varphi = - \otimes_A B$ preserves all colimits. A colimit-preserving functor between stable ∞ -categories preserves finite colimits, and finite colimits agree with finite limits. Hence Ext_φ is exact.
- (2) **Right t -exactness.** The A -module B is connective. The relative tensor product of two connective A -modules is connective, so Ext_φ preserves connective objects.
- (3) **Restriction of scalars.** Restriction of scalars preserves the underlying spectrum of a module. Therefore it preserves both connective and coconnective objects and is t -exact.

□

Remark 1.5.2 (Stable exactness and geometric flatness). Stable exactness and geometric flatness measure different properties. Extension of scalars is exact between stable ∞ -categories for every morphism of commutative ring spectra. Flatness adds compatibility with the Postnikov t -structures, as formalized in 1.6.1.

The affine coefficient system is now in place: every affine context carries a stable symmetric-monoidal fibre, substitution is coherent extension of scalars, and relative commutative algebras already describe affine extensions over that context. The remaining local algebraic issue is to distinguish geometrically well-behaved substitutions from the exactness that is automatic in every stable fibre.

1.6 Substitution, geometric exactness, and finiteness

Geometric regularity is detected by compatibility with the Postnikov t -structures: flatness will mean that substitution is t -exact. Extension of scalars remains exact for every morphism of commutative ring spectra. This section develops that condition together with the finiteness, perfectness, and projectivity notions required later for Zariski locality.

1.6.1 Flat module spectra

Definition 1.6.1 (Flat module spectrum). Let A be a connective commutative ring spectrum. A connective A -module M is **flat** if the functor

$$- \otimes_A M : \text{Mod}_A \longrightarrow \text{Mod}_A$$

is t -exact.

For a connective module, tensoring is already right t -exact. Thus the additional content of flatness is left t -exactness: tensoring with M must also preserve coconnective modules.

Theorem 1.6.1 (Spectral Lazard theorem). *Let A be a connective commutative ring spectrum and let $M \in (\text{Mod}_A)_{\geq 0}$. The following conditions are equivalent.*

(i) *The module M is flat.*

(ii) *The $\pi_0(A)$ -module $\pi_0(M)$ is flat and, for every integer n , the canonical morphism*

$$\pi_n(A) \otimes_{\pi_0(A)} \pi_0(M) \longrightarrow \pi_n(M)$$

is an isomorphism.

(iii) *For every A -module N and every integer n , the canonical morphism*

$$\pi_n(N) \otimes_{\pi_0(A)} \pi_0(M) \longrightarrow \pi_n(N \otimes_A M)$$

is an isomorphism.

(iv) *The module M is a filtered colimit of finite free A -modules.*

Proof.

(1) **Invoke the spectral flatness criterion.** For a connective base ring, the spectral flatness criterion and Lazard theorem identify the homotopy-group condition in (ii), left t -exactness of $- \otimes_A M$, and expression of M as a filtered colimit of finite free modules; see [34, Definition 7.2.2.10 and Theorem 7.2.2.15].

(2) **Use right t -exactness.** Since M is connective, tensoring with M is already right t -exact.

(3) **Identify three equivalent conditions.** Thus the cited theorem gives

$$(i) \iff (ii) \iff (iv).$$

(4) **Choose a filtered free presentation.** Assume (iv) and choose

$$M \simeq \varinjlim_{\alpha} F_{\alpha}$$

with each F_{α} finite free.

- (5) **Compute homotopy groups after tensoring.** For every A -module N ,

$$\begin{aligned}\pi_n(N \otimes_A M) &\simeq \varinjlim_{\alpha} \pi_n(N \otimes_A F_{\alpha}) \\ &\simeq \varinjlim_{\alpha} (\pi_n(N) \otimes_{\pi_0(A)} \pi_0(F_{\alpha})) \\ &\simeq \pi_n(N) \otimes_{\pi_0(A)} \pi_0(M).\end{aligned}$$

- (6) **Pass through filtered colimits.** Here tensor products and homotopy groups commute with filtered colimits, and the formula for a finite free module is immediate. This proves (iv) $\Rightarrow$ (iii).
 (7) **Recover the coefficient formula.** Assume (iii). Taking $N = A$ gives the homotopy-group formula in (ii). It remains to prove that $\pi_0(M)$ is flat over $\pi_0(A)$.
 (8) **Choose a short exact sequence.** Let

$$0 \longrightarrow P' \longrightarrow P \longrightarrow P'' \longrightarrow 0$$

be a short exact sequence of ordinary $\pi_0(A)$ -modules, regarded as discrete A -modules. It determines a fibre sequence in Mod_A .

- (9) **Tensor the fibre sequence.** Since tensoring with M is exact as a functor of stable ∞ -categories, there is a fibre sequence

$$P' \otimes_A M \longrightarrow P \otimes_A M \longrightarrow P'' \otimes_A M.$$

- (10) **Apply condition (iii).** Condition (iii) shows that each of these three modules is discrete and identifies its degree-zero homotopy group with the ordinary tensor product by $\pi_0(M)$.
 (11) **Take degree-zero homotopy.** Taking π_0 therefore gives a short exact sequence

$$0 \longrightarrow P' \otimes_{\pi_0(A)} \pi_0(M) \longrightarrow P \otimes_{\pi_0(A)} \pi_0(M) \longrightarrow P'' \otimes_{\pi_0(A)} \pi_0(M) \longrightarrow 0.$$

- (12) **Deduce ordinary flatness.** Thus $- \otimes_{\pi_0(A)} \pi_0(M)$ is exact, so $\pi_0(M)$ is flat.
 (13) **Complete the equivalence proof.** This proves (iii) $\Rightarrow$ (ii), and all four conditions are equivalent.

□

Corollary 1.6.1 (Homotopy groups of a flat module). *If M is a flat A -module, then its graded homotopy groups are obtained from $\pi_0(M)$ by base change:*

$$\pi_n(M) \simeq \pi_n(A) \otimes_{\pi_0(A)} \pi_0(M)$$

for every integer n .

Proof.

- (1) **Apply the Lazard criterion.** Apply condition (ii) of 1.6.1.

□

Proposition 1.6.1 (Permanence of flat modules). *Let A be a connective commutative ring spectrum.*

(i) *Finite free and finite projective A -modules are flat.*

(ii) *Filtered colimits of flat A -modules are flat.*

(iii) *If M and N are flat A -modules, then*

$$M \otimes_A N$$

is flat over A .

(iv) *Let $A \rightarrow B$ be a flat morphism of connective commutative ring spectra. If N is flat as a B -module, then N , regarded by restriction of scalars as an A -module, is flat over A .*

(v) *Let $A \rightarrow B$ be any morphism of connective commutative ring spectra. If M is flat over A , then*

$$M \otimes_A B$$

is flat over B .

Proof.

(1) **Check finite free modules.** For a finite free module $F \simeq A^{\oplus r}$, tensoring with F is a finite direct sum of the identity functor and is therefore t -exact.

(2) **Pass to retracts.** If M is a retract of a flat module F , then $- \otimes_A M$ is a retract of $- \otimes_A F$. The connective and coconnective subcategories are closed under retracts, so finite projective modules are flat.

(3) **Commute tensoring with the filtered colimit.** For a filtered diagram $\{M_\alpha\}$ of flat modules and every $K \in \text{Mod}_A$,

$$K \otimes_A \varinjlim_\alpha M_\alpha \simeq \varinjlim_\alpha (K \otimes_A M_\alpha).$$

(4) **Preserve both halves of the t -structure.** Filtered colimits preserve both halves of the Postnikov t -structure, since homotopy groups commute with filtered colimits.

(5) **Conclude flatness of the colimit.** Hence the colimit module is flat.

(6) **Rewrite tensoring by the tensor product.** If M and N are flat, associativity gives

$$- \otimes_A (M \otimes_A N) \simeq (- \otimes_A M) \otimes_A N.$$

(7) **Use t -exact composition.** The right side is a composite of t -exact functors, proving (iii).

(8) **Rewrite tensoring after restriction.** Let $A \rightarrow B$ be flat and let N be flat over B . For every A -module K ,

$$K \otimes_A N \simeq (K \otimes_A B) \otimes_B N$$

as underlying A -modules.

(9) **Compose the t -exact functors.** Extension to B , tensoring with N , and restriction back to A are all t -exact. Their composite is therefore t -exact.

- (10) **Rewrite the base-changed tensor product.** Let $A \rightarrow B$ be arbitrary and let M be flat over A . For a B -module K ,

$$K \otimes_B (M \otimes_A B) \simeq K \otimes_A M,$$

where K on the right is restricted to A .

- (11) **Use restriction and flat tensoring.** Restriction preserves the underlying spectrum and hence the Postnikov t -structure, while tensoring with M is t -exact.
- (12) **Conclude flatness after base change.** Thus $M \otimes_A B$ is flat over B .

□

1.6.2 Flat morphisms of connective commutative ring spectra

Definition 1.6.2 (Flat spectral algebra morphism). A morphism

$$\varphi: A \longrightarrow B$$

in $\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}}$ is **flat** if B , regarded as an A -module, is flat.

Proposition 1.6.2 (Homotopy groups of a flat algebra). *If $A \rightarrow B$ is flat, then $\pi_0(B)$ is flat over $\pi_0(A)$ and there are natural isomorphisms*

$$\pi_n(B) \simeq \pi_n(A) \otimes_{\pi_0(A)} \pi_0(B)$$

for every integer n .

Proof.

- (1) **Apply the Lazard criterion to the algebra.** Apply condition (ii) of 1.6.1 to the A -module B .

□

Proposition 1.6.3 (Permanence of flat algebra morphisms). *Flat morphisms in $\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}}$ are stable under equivalence, composition, base change, and filtered colimits in the arrow category.*

Proof.

- (1) **Handle equivalences.** Stability under equivalence follows from the definition.
- (2) **Factor extension of scalars under composition.** For flat maps $A \rightarrow B \rightarrow C$, the extension-of-scalars functor factors as

$$\mathrm{Mod}_A \xrightarrow{- \otimes_A B} \mathrm{Mod}_B \xrightarrow{- \otimes_B C} \mathrm{Mod}_C.$$

- (3) **Conclude composition stability.** Both factors are t -exact, so their composite is t -exact.
- (4) **Form the base-change square.** Given $A \rightarrow A'$, form

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ A' & \longrightarrow & B' := B \otimes_A A'. \end{array}$$

- (5) **Rewrite the base-changed tensor product.** For every A' -module M , associativity gives

$$M \otimes_{A'} B' \simeq M \otimes_A B,$$

where the right side uses restriction of scalars along $A \rightarrow A'$.

- (6) **Use t -exactness.** Restriction is t -exact and tensoring with B is t -exact; hence $A' \rightarrow B'$ is flat.

- (7) **Form the filtered colimits.** Let $\{A_\alpha \rightarrow B_\alpha\}_\alpha$ be filtered and flat, and put

$$A := \varinjlim_\alpha A_\alpha, \quad B := \varinjlim_\alpha B_\alpha.$$

- (8) **Base change each target to the colimit base.** Define

$$B'_\alpha := A \otimes_{A_\alpha} B_\alpha.$$

- (9) **Preserve flatness under base change.** Base-change stability makes every $A \rightarrow B'_\alpha$ flat.

- (10) **Identify the filtered colimit in the undercategory.** The canonical morphisms $B'_\alpha \rightarrow B$ exhibit B as the filtered colimit of the B'_α in $\mathrm{CAlg}(\mathrm{Sp})_{A/}$: mapping into an A -algebra converts both sides into the same limit of mapping spaces.

- (11) **Identify the underlying module colimit.** The underlying A -module of B is therefore a filtered colimit of flat A -modules.

- (12) **Apply permanence of flat modules.** By 1.6.1, it is flat, so $A \rightarrow B$ is flat.

□

Proposition 1.6.4 (Degree-zero homotopy of connective pushouts). *Let*

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ A' & \longrightarrow & B' := B \otimes_A A' \end{array}$$

be a pushout square in connective commutative ring spectra. Then the canonical morphism

$$\pi_0(B) \otimes_{\pi_0(A)} \pi_0(A') \longrightarrow \pi_0(B')$$

is an isomorphism.

Proof.

- (1) **Identify degree zero as a left adjoint.** The adjunction of 1.3.6 shows that

$$\pi_0 : \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}} \longrightarrow \mathrm{CRing}$$

is a left adjoint.

- (2) **Preserve pushouts.** It therefore preserves all small colimits, in particular pushouts.

- (3) **Compute degree zero of the pushout.** Hence

$$\pi_0(B') \simeq \pi_0(B) \otimes_{\pi_0(A)} \pi_0(A').$$

□

1.6.3 Local finite presentation and finite cell algebras

Definition 1.6.3 (Local finite presentation). A morphism

$$A \longrightarrow B$$

of connective commutative ring spectra is **locally of finite presentation** if B is compact in

$$\mathrm{CAlg}(\mathrm{Sp})_{A/}.$$

For an affine morphism

$$\mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A),$$

we also say **of finite presentation**; affineness supplies the global quasi-compactness condition. This terminology agrees with the spectral geometric convention of [37, 26].

Proposition 1.6.5 (Compactness may be checked connectively). *Let A and B be connective commutative ring spectra and regard B as an A -algebra. Then B is compact in $\mathrm{CAlg}(\mathrm{Sp})_{A/}$ if and only if it is compact in $\mathrm{CAlg}(\mathrm{Sp})_{A/}^{\mathrm{cn}}$.*

Proof.

- (1) **Construct the connective reflection adjunction.** Since A is connective, the connective-cover adjunction of 1.3.4 induces an adjunction

$$\mathrm{CAlg}(\mathrm{Sp})_{A/}^{\mathrm{cn}} \begin{array}{c} \xrightarrow{\iota_A} \\ \xleftarrow{\tau_{\geq 0, A}^{\mathrm{CAlg}}} \end{array} \mathrm{CAlg}(\mathrm{Sp})_{A/}.$$

- (2) **Identify its right adjoint.** The right adjoint is obtained by applying connective cover to the target algebra; the structural map from A factors uniquely through that connective cover by adjunction.
- (3) **Commute spectral connective cover with filtered colimits.** The functor $\tau_{\geq 0}^{\mathrm{Sp}}$ on spectra commutes with filtered colimits, and the forgetful functor from commutative ring spectra creates sifted colimits.
- (4) **Transfer filtered-colimit preservation to the undercategory.** Hence $\tau_{\geq 0, A}^{\mathrm{CAlg}}$ commutes with filtered colimits.
- (5) **Compute the mapping space against a filtered colimit.** If B is compact in the connective undercategory and $\{C_\alpha\}$ is filtered in $\mathrm{CAlg}(\mathrm{Sp})_{A/}$, then

$$\begin{aligned} \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})_{A/}} \left(B, \varinjlim_{\alpha} C_{\alpha} \right) &\simeq \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})_{A/}^{\mathrm{cn}}} \left(B, \tau_{\geq 0, A}^{\mathrm{CAlg}} \varinjlim_{\alpha} C_{\alpha} \right) \\ &\simeq \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})_{A/}^{\mathrm{cn}}} \left(B, \varinjlim_{\alpha} \tau_{\geq 0, A}^{\mathrm{CAlg}} C_{\alpha} \right) \\ &\simeq \varinjlim_{\alpha} \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})_{A/}} (B, C_{\alpha}). \end{aligned}$$

- (6) **Conclude compactness in the unrestricted undercategory.** Thus B is compact in the unrestricted undercategory.

- (7) **Prove the converse by restriction.** The converse follows by restricting the same compactness condition to filtered diagrams of connective A -algebras.

□

Definition 1.6.4 (Finite cell algebra). Write

$$\mathrm{Sym}_A: \mathrm{Mod}_A \longrightarrow \mathrm{CAlg}(\mathrm{Mod}_A) \simeq \mathrm{CAlg}(\mathrm{Sp})_{A/}$$

for the free commutative A -algebra functor. A connective commutative A -algebra is a **finite cell algebra** if it belongs to the smallest full subcategory of $\mathrm{CAlg}(\mathrm{Sp})_{A/}^{\mathrm{cn}}$ which contains

$$\mathrm{Sym}_A(A^{\oplus n}), \quad n \geq 0,$$

and is closed under finite colimits. Equivalently, it is obtained by a finite iteration of finite colimits starting from finitely generated free commutative A -algebras. No separate presentation datum is part of the definition.

Proposition 1.6.6 (Degree zero of relative free commutative algebras). *Let A be a connective commutative ring spectrum and let $M \in (\mathrm{Mod}_A)_{\geq 0}$. Then the canonical map of ordinary commutative $\pi_0(A)$ -algebras is an isomorphism*

$$\mathrm{Sym}_{\pi_0(A)}(\pi_0(M)) \xrightarrow{\cong} \pi_0 \mathrm{Sym}_A(M).$$

In particular,

$$\pi_0 \mathrm{Sym}_A(A^{\oplus n}) \simeq \pi_0(A)[t_1, \dots, t_n].$$

Proof.

- (1) **Compute degree zero of a relative tensor product.** For connective A -modules M_1 and M_2 , the bar construction of 1.5.1 is a simplicial connective spectrum. On connective spectra, π_0 preserves colimits, and the monoidal structure induced on the heart is the ordinary tensor product. Hence geometric realization of the bar construction gives

$$\pi_0(M_1 \otimes_A M_2) \simeq \pi_0(M_1) \otimes_{\pi_0(A)} \pi_0(M_2).$$

- (2) **Iterate the tensor-product formula.** Iterating yields

$$\pi_0(M^{\otimes_A r}) \simeq \pi_0(M)^{\otimes_{\pi_0(A)} r}.$$

- (3) **Write the relative free-algebra formula.** The relative free-algebra formula gives

$$\mathrm{oblv}_A \mathrm{Sym}_A(M) \simeq \bigoplus_{r \geq 0} (M^{\otimes_A r})_{h\Sigma_r}.$$

- (4) **Apply degree zero to the colimits.** Since every tensor power is connective, π_0 carries the homotopy orbit colimit to ordinary coinvariants and carries the coproduct to the coproduct.

- (5) **Compute the symmetric algebra.** Therefore

$$\begin{aligned} \pi_0 \mathrm{Sym}_A(M) &\simeq \bigoplus_{r \geq 0} (\pi_0(M)^{\otimes_{\pi_0(A)} r})_{\Sigma_r} \\ &\simeq \mathrm{Sym}_{\pi_0(A)}(\pi_0(M)). \end{aligned}$$

- (6) **Identify the multiplication.** The multiplication is the one induced by concatenation of tensor powers, hence agrees with ordinary symmetric-algebra multiplication.
- (7) **Specialize to a finite free module.** Taking $M = A^{\oplus n}$ gives the polynomial-algebra formula. Compare the relative free-algebra construction of [34].

□

Remark 1.6.1 (Finite-cell and finite-presentation terminology). Two different finiteness notions are being kept distinct. The term **finite cell** refers to the explicit finite-colimit closure of finitely generated free commutative A -algebras. The phrase **locally of finite presentation** refers instead to the compactness condition of 1.6.3. In [34] the former class is called the subcategory of algebras of finite presentation, while the latter objects are called locally of finite presentation. The terminology here is chosen to keep construction separate from compactness. Neither notion retains a choice of generators, relations, or presentation data.

Proposition 1.6.7 (Finite-cell criterion). *A connective commutative A -algebra is locally of finite presentation if and only if it is a retract of a finite cell commutative A -algebra.*

Proof.

- (1) **Define the finite free algebra.** For every integer $n \geq 0$, put

$$F_n := \mathrm{Sym}_A(A^{\oplus n}).$$

- (2) **Compute mapping spaces against a filtered colimit.** Let $\{B_\alpha\}$ be a filtered diagram in $\mathrm{CAlg}(\mathrm{Sp})_{A/}^{\mathrm{cn}}$. Since the forgetful functor to A -modules creates filtered colimits, the free-algebra adjunction gives

$$\begin{aligned} \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})_{A/}^{\mathrm{cn}}} \left(F_n, \varinjlim_{\alpha} B_{\alpha} \right) &\simeq \mathrm{Map}_{\mathrm{Mod}_A} \left(A^{\oplus n}, \varinjlim_{\alpha} B_{\alpha} \right) \\ &\simeq \varinjlim_{\alpha} \mathrm{Map}_{\mathrm{Mod}_A} (A^{\oplus n}, B_{\alpha}) \\ &\simeq \varinjlim_{\alpha} \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})_{A/}^{\mathrm{cn}}} (F_n, B_{\alpha}). \end{aligned}$$

- (3) **Conclude compactness of finite free algebras.** Thus every finitely generated free commutative A -algebra is compact.
- (4) **Pass compactness through finite colimits.** Finite colimits of compact objects are compact because mapping out of a finite colimit is a finite limit of mapping spaces and finite limits of spaces commute with filtered colimits.
- (5) **Pass compactness through retracts.** Hence every finite cell A -algebra is compact, and so is every retract of a finite cell A -algebra.
- (6) **Invoke the Ind-completion theorem.** The finite-colimit closure of the finitely generated free commutative A -algebras has Ind-completion

$$\mathrm{Ind}(\mathrm{CAlg}(\mathrm{Sp})_{A/}^{\mathrm{fc}}) \simeq \mathrm{CAlg}(\mathrm{Sp})_{A/}^{\mathrm{cn}},$$

where $\mathrm{CAlg}(\mathrm{Sp})_{A/}^{\mathrm{fc}}$ denotes the full finite-cell subcategory of 1.6.4. This is the connective commutative specialization of [34, Definition 7.2.4.26 and Proposition 7.2.4.27].

- (7) **Rewrite the theorem as a filtered-colimit presentation.** Equivalently, every connective commutative A -algebra is a filtered colimit of finite cell A -algebras.
- (8) **Choose a finite-cell presentation of the compact algebra.** Let B be compact and choose

$$B \simeq \varinjlim_{\alpha} F_{\alpha}$$

with every F_{α} finite cell.

- (9) **Apply compactness to the identity mapping space.** Compactness gives

$$\mathrm{Map}(B, B) \simeq \varinjlim_{\alpha} \mathrm{Map}(B, F_{\alpha}).$$

- (10) **Represent the identity at a finite-cell stage.** Hence the point id_B is represented at some stage by a morphism $s : B \rightarrow F_{\alpha}$.
- (11) **Construct the retraction.** If $r : F_{\alpha} \rightarrow B$ is the colimit structure morphism, then

$$r \circ s \simeq \mathrm{id}_B.$$

- (12) **Conclude the retract statement.** Thus B is a retract of the finite cell algebra F_{α} .
- (13) **Return to the stated definition.** By 1.6.5, compactness in the connective undercategory is equivalent to compactness in $\mathrm{CAlg}(\mathrm{Sp})_{A/}$, which is the local finite-presentation condition of 1.6.3.

□

Proposition 1.6.8 (Permanence of finite presentation). *Morphisms locally of finite presentation are stable under equivalence, composition, and base change.*

Proof.

- (1) **Equivalences.** Compactness is invariant under equivalence, giving the first permanence property.
- (2) **Base change.** Base change

$$\mathrm{CAlg}(\mathrm{Sp})_{A/} \longrightarrow \mathrm{CAlg}(\mathrm{Sp})_{A'/}, \quad B \longmapsto B \otimes_A A',$$

is left adjoint to restriction of scalars. Restriction preserves filtered colimits, so the base-change functor preserves compact objects.

- (3) **Composition.** Local finite presentation is stable under composition for morphisms of connective commutative ring spectra; this is the compactness permanence theorem for spectral algebra morphisms. See [34, 37]. Applying it to $A \rightarrow B \rightarrow C$ gives local finite presentation of $A \rightarrow C$.

□

1.6.4 Perfect and projective modules

Definition 1.6.5 (Perfect module). An A -module M is **perfect** if it belongs to the thick subcategory of Mod_A generated by the tensor unit A .

Proposition 1.6.9 (Perfect equals dualizable). *An A -module is perfect if and only if it is dualizable in the symmetric monoidal ∞ -category Mod_A .*

Proof.

- (1) **Place the unit in both classes.** The unit A is compact and dualizable.
- (2) **Close dualizable objects thickly.** The dualizable objects form a thick subcategory.
- (3) **Show a dualizable module is compact.** Conversely, a dualizable A -module is compact because its mapping functor is represented after tensoring with its dual.
- (4) **Identify the compact objects.** The compact objects of Mod_A form the thick subcategory generated by A . See [34].

□

Definition 1.6.6 (Tor-amplitude). An A -module M has **Tor-amplitude in $[a, b]$** if, for every discrete A -module N , the module

$$M \otimes_A N$$

has homotopy groups only in degrees a through b .

Definition 1.6.7 (Finite projective module). An A -module M is **finite projective** if it is a retract of a finite free A -module.

Proposition 1.6.10 (Characterization of finite projectives). *For an A -module M , the following are equivalent.*

- (i) *The module M is finite projective.*
- (ii) *The module M is perfect and has Tor-amplitude contained in $[0, 0]$.*
- (iii) *The module M is perfect, connective, and flat.*

Proof.

- (1) **Check finite free modules.** A finite free A -module is perfect, connective, flat, and has Tor-amplitude in $[0, 0]$.
- (2) **Pass the properties to retracts.** These properties are stable under retracts. Hence (i) implies both (ii) and (iii).
- (3) **Pass to the discrete base.** Assume (iii) and put

$$M_0 := M \otimes_A H\pi_0(A).$$

- (4) **Compute the homotopy of the base change.** Base change preserves perfectness, and flatness gives

$$\pi_n(M_0) \simeq \pi_n(H\pi_0(A)) \otimes_{\pi_0(A)} \pi_0(M).$$

- (5) **Identify $\pi_0(M)$ as finitely generated projective.** Thus M_0 is discrete with $\pi_0(M_0) \simeq \pi_0(M)$. A discrete perfect module over $H\pi_0(A)$ corresponds to a perfect ordinary $\pi_0(A)$ -module, so $\pi_0(M)$ is finitely presented. It is also flat by 1.6.1; hence it is finitely generated projective.

- (6) **Choose an idempotent.** Choose an idempotent

$$e \in M_r(\pi_0(A))$$

with image $\pi_0(M)$.

- (7) **Split the idempotent.** The corresponding idempotent endomorphism of $A^{\oplus r}$ splits in Mod_A , producing a finite projective module P with

$$\pi_0(P) \simeq \pi_0(M).$$

- (8) **Lift the degree-zero isomorphism.** Because P is a retract of a finite free module, the map

$$\pi_0 \text{Map}_{\text{Mod}_A}(P, M) \longrightarrow \text{Hom}_{\pi_0(A)}(\pi_0(P), \pi_0(M))$$

is surjective. Lift a chosen isomorphism on π_0 to $u : P \rightarrow M$.

- (9) **Compare all homotopy groups.** Both modules are flat, so

$$\pi_n(u) \simeq \pi_n(A) \otimes_{\pi_0(A)} \pi_0(u)$$

is an isomorphism for every n .

- (10) **Conclude finite projectivity.** Hence u is an equivalence and M is finite projective.

- (11) **Pass again to the discrete base.** Assume (ii) and again put

$$M_0 := M \otimes_A H\pi_0(A).$$

The Tor-amplitude condition makes M_0 discrete. Perfectness over a connective ring spectrum makes M bounded below.

- (12) **Choose the first nonzero homotopy degree.** Let

$$I := \text{fib}(A \rightarrow H\pi_0(A)),$$

so $I \in (\text{Mod}_A)_{\geq 1}$. If $M \neq 0$, let r be the least integer with $\pi_r(M) \neq 0$. Then

$$I \otimes_A M \in (\text{Mod}_A)_{\geq r+1}.$$

- (13) **Tensor the defining fibre sequence.** Tensoring the fibre sequence

$$I \longrightarrow A \longrightarrow H\pi_0(A)$$

with M gives

$$I \otimes_A M \longrightarrow M \longrightarrow M_0.$$

- (14) **Compare the first nonzero homotopy group.** Hence $\pi_r(M) \rightarrow \pi_r(M_0)$ is an isomorphism.

- (15) **Deduce connectivity.** Since M_0 is discrete, $r = 0$. Therefore M is connective.
- (16) **Define the tensor functor.** Let $F = - \otimes_A M$. Connectivity of M makes F right t -exact, while Tor-amplitude $[0, 0]$ makes F carry discrete modules to discrete modules.
- (17) **Approximate a coconnective module by bounded truncations.** For $N \in (\text{Mod}_A)_{\leq 0}$, right completeness gives

$$N \simeq \varinjlim_k \tau_{\geq -k}^A N.$$

- (18) **Build each bounded truncation from discrete pieces.** Each bounded truncation is obtained by finitely many extensions from the shifted discrete modules $H\pi_j(N)[j]$, $-k \leq j \leq 0$.
- (19) **Apply exactness to the bounded truncations.** Exactness of F therefore gives

$$F(\tau_{\geq -k}^A N) \in (\text{Mod}_A)_{\leq 0}.$$

- (20) **Pass to the filtered colimit.** Filtered colimits preserve coconnective objects, so $F(N) \in (\text{Mod}_A)_{\leq 0}$.
- (21) **Conclude t -exactness.** Thus F is t -exact and M is flat.
- (22) **Return to finite projectivity.** Step (10) now gives (ii) $\Rightarrow$ (i).

□

At the affine level, the geometric conditions needed for locality now have precise algebraic forms: flatness is Postnikov-exact substitution, local finite presentation is compactness of the coordinate algebra, and finite projectivity is perfect connective flatness. The next problem is scope: the coefficient system must be extended beyond affines before locality can select the global geometric contexts.

1.7 Generalized contexts and extension of the coefficient system

The coefficient system has so far been constructed on affines. The next step is therefore to enlarge the base before imposing any geometric locality condition, so that locality can be formulated globally. We pass from affine contexts to space-valued prestacks and extend QCoh_{aff} by right Kan extension along opposite Yoneda. The resulting coefficient fibres exist on arbitrary generalized contexts, while unstraightening records the same substitution data as Cartesian transport.

1.7.1 Spectral prestacks and Yoneda

Definition 1.7.1 (Spectral prestacks). The ∞ -category of space-valued presheaves on affine spectral schemes is

$$\text{PSh}(\text{Aff}_{\text{Sp}}) := \text{Fun}((\text{Aff}_{\text{Sp}})^{\text{op}}, \mathcal{S}).$$

Its objects are called **spectral prestacks**.

Definition 1.7.2 (Functor of points). For $A \in \text{CAlg}(\text{Sp})^{\text{cn}}$, the functor of points of $\text{Spec}(A)$ is

$$h_A: (\text{Aff}_{\text{Sp}})^{\text{op}} \longrightarrow \mathcal{S},$$

defined by

$$h_A(\text{Spec}(B)) := \text{Map}_{\text{CAlg}(\text{Sp})}(A, B).$$

Proposition 1.7.1 (Spectral Yoneda embedding). *The assignment*

$$\mathrm{Spec}(A) \mapsto h_A$$

defines a fully faithful functor

$$h: \mathrm{Aff}_{\mathrm{Sp}} \hookrightarrow \mathrm{PSh}(\mathrm{Aff}_{\mathrm{Sp}}).$$

Proof.

(1) **Apply Yoneda.** The ∞ -categorical Yoneda lemma gives the asserted full faithfulness. □

1.7.2 Affine points and right Kan extension

Definition 1.7.3 (Affine points of a prestack). For a spectral prestack

$$X \in \mathrm{PSh}(\mathrm{Aff}_{\mathrm{Sp}}),$$

let

$$\mathrm{AffPt}(X)$$

denote the ∞ -category of pairs

$$(\mathrm{Spec}(A), x), \quad x \in X(A),$$

equivalently the category of objects of $\mathrm{Aff}_{\mathrm{Sp}}$ mapping to X through the Yoneda embedding. This notation is used instead of $\mathrm{Aff}_{\mathrm{Sp}}/X$, since an arbitrary prestack X is not an object of $\mathrm{Aff}_{\mathrm{Sp}}$.

Definition 1.7.4 (Quasi-coherent modules on a spectral prestack). For a spectral prestack X , define

$$\mathrm{QCoh}(X) := \varprojlim_{(\mathrm{Spec}(A) \rightarrow X) \in \mathrm{AffPt}(X)^{\mathrm{op}}} \mathrm{Mod}_A.$$

The limit is formed in

$$\mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L).$$

The affine-point formula makes the generalized coefficient fibre explicit:

$$\mathrm{AffPt}(X)^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L)$$

$$(\mathrm{Spec}(A) \rightarrow X) \longmapsto \mathrm{Mod}_A.$$

A morphism of generalized contexts $X \rightarrow Y$ acts on affine points by composition. The induced comparison of indexing diagrams is

$$\begin{array}{ccc} \mathrm{AffPt}(X)^{\mathrm{op}} & \xrightarrow{\quad} & \mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L) \\ \text{postcomposition} \downarrow & \nearrow & \\ \mathrm{AffPt}(Y)^{\mathrm{op}} & & \end{array}$$

and the resulting map on limits is the substitution functor $\mathrm{QCoh}(Y) \rightarrow \mathrm{QCoh}(X)$.

Theorem 1.7.1 (Affine right Kan extension). *Let*

$$h: \text{Aff}_{\text{Sp}} \longrightarrow \text{PSh}(\text{Aff}_{\text{Sp}})$$

be the Yoneda embedding. The construction of 1.7.4 defines a functor

$$\text{QCoh}: \text{PSh}(\text{Aff}_{\text{Sp}})^{\text{op}} \longrightarrow \text{CAlg}(\text{Pr}_{\text{st}}^L)$$

which is the right Kan extension of

$$\text{QCoh}_{\text{aff}}: (\text{Aff}_{\text{Sp}})^{\text{op}} \longrightarrow \text{CAlg}(\text{Pr}_{\text{st}}^L)$$

along the opposite Yoneda embedding

$$h^{\text{op}}: (\text{Aff}_{\text{Sp}})^{\text{op}} \longrightarrow \text{PSh}(\text{Aff}_{\text{Sp}})^{\text{op}}.$$

Equivalently, QCoh is the unique small-limit-preserving functor on $\text{PSh}(\text{Aff}_{\text{Sp}})^{\text{op}}$ whose restriction along h^{op} is QCoh_{aff} .

Proof.

- (1) **Pointwise right Kan extension.** For a spectral prestack X , the pointwise formula along h^{op} is

$$\text{Ran}_{h^{\text{op}}}(\text{QCoh}_{\text{aff}})(X) \simeq \varprojlim_{(\text{Spec}(A) \rightarrow X) \in \text{AffPt}(X)^{\text{op}}} \text{Mod}_A.$$

This is exactly 1.7.4.

- (2) **Functoriality.** A morphism $X \rightarrow Y$ induces

$$\text{AffPt}(X) \longrightarrow \text{AffPt}(Y)$$

by composition. Passing to opposites and using the universal property of the two limits produces

$$\text{QCoh}(Y) \longrightarrow \text{QCoh}(X).$$

- (3) **Universal property.** The Yoneda embedding exhibits $\text{PSh}(\text{Aff}_{\text{Sp}})$ as the free cocompletion of Aff_{Sp} ; after passage to opposites, $\text{PSh}(\text{Aff}_{\text{Sp}})^{\text{op}}$ is generated under limits by the image of h^{op} . Hence a small-limit-preserving extension is uniquely determined by its restriction to representables, and the right Kan extension has the asserted universal property.

□

Corollary 1.7.1 (Recovery on affine spectral schemes). *For every connective commutative ring spectrum A , there is a canonical symmetric monoidal equivalence*

$$\text{QCoh}(\text{Spec}(A)) \simeq \text{Mod}_A.$$

Proof.

- (1) **Identify a terminal affine point.** For the representable prestack

$$X = \mathrm{Spec}(A),$$

the identity map

$$\mathrm{id}_X : X \longrightarrow X$$

is terminal in the affine-point category

$$\mathrm{AffPt}(X).$$

- (2) **Pass to the opposite indexing category.** It is therefore initial in

$$\mathrm{AffPt}(X)^{\mathrm{op}}.$$

- (3) **Evaluate the limit at the initial object.** A limit of a diagram indexed by an ∞ -category with an initial object is canonically its value at that initial object. Hence

$$\mathrm{QCoh}(\mathrm{Spec}(A)) \simeq \mathrm{Mod}_A.$$

- (4) **Preserve the symmetric monoidal structure.** The equivalence is symmetric monoidal because the defining limit is formed in

$$\mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L).$$

□

Corollary 1.7.2 (Colimits of prestacks). *For every small diagram of spectral prestacks X_α , there is a canonical equivalence*

$$\mathrm{QCoh}\left(\varinjlim_{\alpha} X_{\alpha}\right) \simeq \varprojlim_{\alpha} \mathrm{QCoh}(X_{\alpha}).$$

Proof.

- (1) **Apply limit preservation.** Apply the limit-preservation assertion of 1.7.1 to the opposite functor.

□

1.7.3 The coefficient fibration

Construction 1.7.1 (Grothendieck construction of quasi-coherent coefficients). Forget the monoidal structure of the right-Kan-extended functor and regard

$$\mathrm{QCoh} : \mathrm{PSh}(\mathrm{Aff}_{\mathrm{Sp}})^{\mathrm{op}} \longrightarrow \mathrm{Cat}_{\infty}.$$

Unstraightening produces a Cartesian fibration

$$\pi_{\mathrm{QCoh}} : \int \mathrm{QCoh} \longrightarrow \mathrm{PSh}(\mathrm{Aff}_{\mathrm{Sp}}).$$

The fibre over a generalized context X is $\mathrm{QCoh}(X)$. For a morphism of contexts $f : Y \rightarrow X$ and an object $\mathcal{F} \in \mathrm{QCoh}(X)$, Cartesian transport along f has domain

$$(Y, f^*\mathcal{F})$$

and codomain $(X, \mathcal{F})$. The coherence of iterated substitution is therefore the Cartesian-composition coherence of this fibration, rather than additional structure imposed on individual pullback functors.

$$\begin{array}{ccc} (Y, f^*\mathcal{F}) & \longrightarrow & (X, \mathcal{F}) \\ \downarrow & & \downarrow \\ Y & \xrightarrow{f} & X. \end{array}$$

Remark 1.7.1 (Lawvere-style indexed semantics). The Cartesian fibration π_{QCoh} is the indexed-categorical object encoded by the language of coefficients in context: its base consists of contexts, its fibres are coefficient categories, and its Cartesian transport is substitution. The right adjoints to substitution will be obtained functorially by passage to right adjoints, after which the canonical mate transformations are constructed from the resulting adjunctions. We call this indexed organization a **Lawvere-style doctrine**. The term **hyperdoctrine** is reserved for settings in which the additional dependent adjoints exist and the relevant exchange transformations have been proved invertible. Compare [25].

1.7.4 Pullback, direct image, and the structure sheaf

Proposition 1.7.2 (Quasi-coherent pullback). *Every morphism of spectral prestacks*

$$f : X \longrightarrow Y$$

induces a colimit-preserving strong symmetric monoidal functor

$$f^* : \mathrm{QCoh}(Y) \longrightarrow \mathrm{QCoh}(X).$$

For affine spectral schemes it agrees with extension of scalars.

Proof.

- (1) **Obtain the transition functor.** Functoriality of the right Kan extension gives the transition functor.
- (2) **Read off its monoidal properties.** The target of the right Kan extension is $\mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L)$, so the transition is colimit-preserving and strong symmetric monoidal.
- (3) **Recover the affine case.** The affine identification follows from 1.7.1 and the defining affine coefficient system.

□

Corollary 1.7.3 (Quasi-coherent direct image). *The pullback functor of 1.7.2 admits a right adjoint*

$$f_* : \mathrm{QCoh}(X) \longrightarrow \mathrm{QCoh}(Y).$$

Proof.

- (1) **Preserve small colimits.** The functor f^* preserves small colimits between presentable ∞ -categories.
- (2) **Apply the adjoint functor theorem.** The adjoint functor theorem supplies a right adjoint.

□

Proposition 1.7.3 (Coherent direct-image coefficient system). *Forget the symmetric monoidal enhancement of quasi-coherent pullback and write*

$$\mathrm{QCoh}^* : \mathrm{PSh}(\mathrm{Aff}_{\mathrm{Sp}})^{\mathrm{op}} \longrightarrow \mathrm{Pr}_{\mathrm{st}}^L.$$

Passage to right adjoints defines a functor

$$\mathrm{QCoh}_* : \mathrm{PSh}(\mathrm{Aff}_{\mathrm{Sp}}) \longrightarrow \mathrm{Pr}_{\mathrm{st}}^R$$

whose value on a morphism $f : X \rightarrow Y$ is the right adjoint

$$f_* : \mathrm{QCoh}(X) \longrightarrow \mathrm{QCoh}(Y)$$

of f^ . In particular, the identity and composition equivalences*

$$(\mathrm{id}_X)_* \simeq \mathrm{id}_{\mathrm{QCoh}(X)}, \quad (g \circ f)_* \simeq g_* f_*$$

and all their higher associativity coherences are supplied by the functor QCoh_ ; no coherent family of adjoints is chosen independently.*

Proof.

- (1) **Pass functorially to right adjoints.** Taking opposites gives

$$\mathrm{PSh}(\mathrm{Aff}_{\mathrm{Sp}}) \xrightarrow{(\mathrm{QCoh}^*)^{\mathrm{op}}} (\mathrm{Pr}_{\mathrm{st}}^L)^{\mathrm{op}} \xrightarrow[\simeq]{\mathrm{RAdj}} \mathrm{Pr}_{\mathrm{st}}^R.$$

Define the composite to be QCoh_* . The value of RAdj on a colimit-preserving functor is its right adjoint, so the transition attached to $f : X \rightarrow Y$ is exactly the right adjoint of

$$f^* : \mathrm{QCoh}(Y) \longrightarrow \mathrm{QCoh}(X).$$

Existence of this right adjoint is 1.7.3.

- (2) **Read off higher coherence.** For composable morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z$, the two adjunction directions may be displayed simultaneously as

$$\mathrm{QCoh}(Z) \xrightleftharpoons[g_*]{g^*} \mathrm{QCoh}(Y) \xrightleftharpoons[f_*]{f^*} \mathrm{QCoh}(X).$$

Functoriality of QCoh_* supplies $(g \circ f)_* \simeq g_* f_*$, while its higher functoriality supplies the unit and associativity homotopies. This is the coherent structure used below. Compare the standard passage-to-adjoints formalism in [34].

□

Proposition 1.7.4 (Canonical composition, base-change, and projection morphisms). *The functorial right adjoints of 1.7.3 carry the following canonical mate transformations.*

(i) *For composable morphisms*

$$X \xrightarrow{f} Y \xrightarrow{g} Z,$$

there is a canonical coherent equivalence

$$(g \circ f)_* \simeq g_* f_*.$$

(ii) *For every Cartesian square of spectral prestacks*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ f' \downarrow & & \downarrow f \\ Y' & \xrightarrow{g} & Y, \end{array}$$

there is a canonical base-change morphism

$$\mathrm{BC}_{f,g} : g^* f_* \longrightarrow f'_* g'^*.$$

(iii) *For $\mathcal{E} \in \mathrm{QCoh}(Y)$ and $\mathcal{F} \in \mathrm{QCoh}(X)$, there is a canonical projection morphism*

$$\mathrm{PF}_f(\mathcal{E}, \mathcal{F}) : \mathcal{E} \otimes f_* \mathcal{F} \longrightarrow f_*(f^* \mathcal{E} \otimes \mathcal{F}).$$

No equivalence assertion is part of the construction in (ii) or (iii).

Proof.

(1) **Composition from the direct-image functor.** The equivalence

$$(g \circ f)_* \simeq g_* f_*$$

and all of its higher associativity coherence are the composition data of the functor QCoh_* constructed in 1.7.3. Thus no pointwise choice of coherent right adjoints is required.

(2) **Base change as an adjoint mate.** The Cartesian square supplies the canonical pullback equivalence

$$g'^* f^* \simeq f'^* g^*.$$

For $\mathcal{F} \in \mathrm{QCoh}(X)$, define $\mathrm{BC}_{f,g}(\mathcal{F})$ to be the composite

$$\begin{aligned} g^* f_* \mathcal{F} &\longrightarrow f'_* f'^* g^* f_* \mathcal{F} \\ &\simeq f'_* g'^* f^* f_* \mathcal{F} \\ &\longrightarrow f'_* g'^* \mathcal{F}, \end{aligned}$$

where the first arrow is the unit of $f'^* \dashv f'_*$, the middle equivalence is the pullback comparison, and the final arrow is induced by the counit of $f^* \dashv f_*$. This is precisely the adjoint mate of the pullback equivalence.

- (3) **Projection morphism as an adjoint mate.** Define the displayed projection morphism to be adjoint to

$$\begin{aligned} f^*(\mathcal{E} \otimes f_*\mathcal{F}) &\simeq f^*\mathcal{E} \otimes f^*f_*\mathcal{F} \\ &\longrightarrow f^*\mathcal{E} \otimes \mathcal{F}, \end{aligned}$$

where the final morphism uses the counit. Naturality and all coherence follow from the calculus of adjoint mates and the coherent monoidality of pullback.

□

Remark 1.7.2 (No general dependent left adjoint). The indexed coefficient doctrine does not supply a left adjoint to every substitution functor. A concrete obstruction already appears for extension of scalars

$$- \otimes_{H\mathbb{Z}} H\mathbb{Q} : \text{Mod}_{H\mathbb{Z}} \longrightarrow \text{Mod}_{H\mathbb{Q}}.$$

This functor does not preserve infinite products. On degree-zero homotopy, the canonical map

$$\left(\prod_{n \geq 1} \mathbb{Z} \right) \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \prod_{n \geq 1} \mathbb{Q}$$

is not surjective: elements on the left have a common denominator, whereas

$$\left(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right)$$

does not. Thus this substitution functor is not a right adjoint and cannot itself admit a left adjoint. Any dependent left adjoints considered in a more specialized geometric setting must therefore come from additional geometry rather than from the indexed doctrine alone.

Definition 1.7.5 (Structure sheaf). The **structure sheaf** of a spectral prestack X is the tensor unit

$$\mathcal{O}_X \in \text{QCoh}(X).$$

Definition 1.7.6 (Global sections). For $\mathcal{F} \in \text{QCoh}(X)$, define its spectrum of global sections by

$$\Gamma(X, \mathcal{F}) := \text{map}_{\text{QCoh}(X)}(\mathcal{O}_X, \mathcal{F}).$$

Proposition 1.7.5 (Pullback of the structure sheaf). *For every morphism $f : X \rightarrow Y$, there is a canonical equivalence*

$$f^*\mathcal{O}_Y \simeq \mathcal{O}_X.$$

Proof.

- (1) **Preserve the tensor unit.** The functor f^* is strong symmetric monoidal and therefore preserves the tensor unit.

□

1.7.5 Connectivity on prestacks

Definition 1.7.7 (Objectwise connective quasi-coherent module). A quasi-coherent module $\mathcal{F} \in \mathbf{QCoh}(X)$ is **objectwise connective** if, for every affine point

$$x: \mathrm{Spec}(A) \longrightarrow X,$$

its pullback

$$x^* \mathcal{F}$$

belongs to $(\mathrm{Mod}_A)_{\geq 0}$. Objectwise coconnective modules are defined similarly.

The affine coefficient system has now been extended to every spectral prestack, with coherent substitution encoded either functorially or by the associated Cartesian fibration. The remaining step toward geometry is locality: we must identify which generalized contexts can be reconstructed from affine opens, and show that the coefficient system reconstructs with them.

1.8 Zariski locality and coherent descent

The generalized coefficient system is available before geometry has been selected. Zariski locality now performs that selection. We first characterize affine opens and principal spectral localizations, then define covering families by joint conservativity of substitution. Descendability and the Amitsur complex convert this local detection criterion into reconstruction of representables, modules, and commutative algebras.

Convention 1.8.1 (Sheaves and hyperdescent). Unless explicitly stated otherwise,

$$\mathrm{Sh}_\tau(\mathcal{C})$$

denotes the ∞ -category of τ -sheaves of spaces without hypercompletion. When hyperdescent is required, the hypercomplete ∞ -topos is denoted by

$$\mathrm{Sh}_\tau(\mathcal{C})^\wedge.$$

Convention 1.8.2 (Effective epimorphisms and descent data). In an ∞ -topos, a morphism

$$p: U \longrightarrow X$$

is an **effective epimorphism** if the augmentation from its Čech nerve $U_\bullet \rightarrow X$ exhibits X as the geometric realization

$$X \simeq \mathrm{colim}_{[n] \in \Delta^{\mathrm{op}}} U_n.$$

Whenever descent data for a cosimplicial diagram of ∞ -categories $\mathcal{C}^\bullet$ are mentioned, they mean an object of the totalization

$$\mathrm{Tot}_{[n] \in \Delta} \mathcal{C}^n.$$

Thus the compatibility morphisms and all higher cocycle coherences are part of the displayed limit rather than additional axiomatic data. We also use the standard local-surjectivity characterization of effective epimorphisms in a sheaf ∞ -topos: sections lift after passage to a covering family; see [37].

Lemma 1.8.1 (Mapping descent along an effective epimorphism). *Let $\mathcal{X}$ be an ∞ -topos, let*

$$p : U \longrightarrow X$$

be an effective epimorphism with Čech nerve $U_\bullet$, and let

$$Y \longrightarrow X, \quad Z \longrightarrow X$$

be objects of the slice $\mathcal{X}_{/X}$. Then there is a natural equivalence

$$\mathrm{Map}_X(Y, Z) \simeq \mathrm{Tot} \, \mathrm{Map}_{U_n}(Y \times_X U_n, Z \times_X U_n).$$

In particular, morphisms and all homotopies between them may be reconstructed from their restrictions to an effective epimorphic cover and its iterated intersections.

Proof.

- (1) **Base change the effective epimorphism.** Effective epimorphisms are stable under base change in an ∞ -topos. Hence

$$Y \times_X U \longrightarrow Y$$

is an effective epimorphism in the slice $\mathcal{X}_{/X}$, with Čech nerve

$$[n] \longmapsto Y \times_X U_n.$$

Therefore

$$Y \simeq \mathrm{colim}_{[n] \in \Delta^{\mathrm{op}}} Y \times_X U_n$$

in $\mathcal{X}_{/X}$.

- (2) **Map out of the geometric realization.** Mapping from the preceding colimit into Z gives

$$\mathrm{Map}_X(Y, Z) \simeq \mathrm{Tot} \, \mathrm{Map}_X(Y \times_X U_n, Z).$$

Since $Y \times_X U_n$ lies over U_n , the universal property of the slice identifies

$$\mathrm{Map}_X(Y \times_X U_n, Z) \simeq \mathrm{Map}_{U_n}(Y \times_X U_n, Z \times_X U_n).$$

Combining the displayed equivalences proves the assertion. □

1.8.1 Affine monomorphisms and open immersions

Proposition 1.8.1 (Affine monomorphism criterion). *Let $A \rightarrow B$ be a morphism of connective commutative ring spectra. The following are equivalent.*

- (i) *The morphism*

$$\mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A)$$

is a monomorphism of spectral prestacks.

(ii) *The multiplication morphism*

$$B \otimes_A B \longrightarrow B$$

is an equivalence.

Proof.

- (1) **Use the diagonal criterion.** A morphism $X \rightarrow Y$ is a monomorphism if and only if its diagonal is an equivalence.
- (2) **Identify the affine diagonal.** By 1.4.2, the diagonal of $\mathrm{Spec}(B) \rightarrow \mathrm{Spec}(A)$ is represented by the multiplication map $B \otimes_A B \rightarrow B$.

□

Definition 1.8.1 (Affine open immersion). A morphism

$$\mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A)$$

of affine spectral schemes is an **affine open immersion** if the algebra map $A \rightarrow B$ is flat, locally of finite presentation, and satisfies

$$B \otimes_A B \xrightarrow{\sim} B.$$

Proposition 1.8.2 (Permanence of affine open immersions). *Affine open immersions are stable under equivalence, composition, and base change.*

Proof.

- (1) **Preserve flatness and finite presentation.** Flatness and finite presentation have the required permanence properties by 1.6.3 and 1.6.8.
- (2) **Preserve monomorphisms.** Monomorphisms are stable under equivalence, composition, and base change.
- (3) **Apply the definition.** Apply 1.8.1.

□

1.8.2 Principal opens

Construction 1.8.1 (Spectral localization). Let A be connective and let

$$f \in \pi_0(A).$$

The **localization of A at f** is the initial commutative A -algebra

$$A[f^{-1}]$$

in which the image of f is invertible.

Proposition 1.8.3 (Homotopy groups of a localization). *For every integer n , there is a natural isomorphism*

$$\pi_n(A[f^{-1}]) \simeq \pi_n(A)[f^{-1}].$$

In particular, $A \rightarrow A[f^{-1}]$ is flat.

Proof.

- (1) **Underlying module of the localization.** Choose a representative of the homotopy class of multiplication by f in $\text{Map}_{\text{Mod}_A}(A, A)$. The spectral localization theorem identifies the underlying A -module of $A[f^{-1}]$ with the telescope

$$A \xrightarrow{f} A \xrightarrow{f} A \xrightarrow{f} \cdots.$$

Any representative of the same degree-zero class computes the same localization object by its universal property, so this proof choice is not part of the construction. See [34].

- (2) **Homotopy groups.** Homotopy groups commute with filtered colimits, so

$$\begin{aligned} \pi_n(A[f^{-1}]) &\simeq \varinjlim (\pi_n(A) \xrightarrow{f} \pi_n(A) \xrightarrow{f} \cdots) \\ &\simeq \pi_n(A)[f^{-1}]. \end{aligned}$$

- (3) **Flatness.** In degree zero this gives $\pi_0(A[f^{-1}]) \simeq \pi_0(A)[f^{-1}]$, a flat $\pi_0(A)$ -algebra. Together with the displayed homotopy formula, 1.6.1 implies that $A \rightarrow A[f^{-1}]$ is flat. □

Proposition 1.8.4 (Principal spectral opens). *The morphism*

$$D_A^{\text{sp}}(f) := \text{Spec}(A[f^{-1}]) \longrightarrow \text{Spec}(A)$$

is an affine open immersion.

Proof.

- (1) **Flatness.** The localization map is flat by 1.8.3.
 (2) **Choose the universal polynomial algebra.** Let

$$P := \text{Sym}_{\mathbb{S}}(\mathbb{S})$$

with universal generator t .

- (3) **Choose a representative of the class.** Choose a point $\tilde{f} \in \Omega^\infty A$ representing the component $f \in \pi_0(A)$, and use $t \mapsto \tilde{f}$ to regard A as a P -algebra for this proof.
 (4) **Introduce the two free P -algebras.** In $\text{CAlg}(\text{Mod}_P)$, let

$$R := \text{Sym}_P(P), \quad Q := \text{Sym}_P(P),$$

with generators u and z , respectively.

- (5) **Form the pushout.** Form

$$\begin{array}{ccc} Q & \xrightarrow{z \mapsto tu-1} & R \\ \downarrow z \mapsto 0 & & \downarrow \\ P & \longrightarrow & P[t^{-1}]. \end{array}$$

- (6) **Describe maps out of the pushout.** A map from this pushout to a commutative P -algebra C is a choice of $u \in \Omega^\infty C$ together with a path $tu \simeq 1$.
- (7) **Use invertibility of t .** If the image of t is a unit, multiplication by t is an equivalence of C -modules, so

$$\Omega^\infty C \xrightarrow{t^-} \Omega^\infty C$$

is an equivalence of spaces and the homotopy fibre over 1 is contractible.

- (8) **Identify the universal localization.** The pushout therefore has the universal property of $P[t^{-1}]$.
- (9) **Identify the finite-cell generators.** The algebras Q and R are finitely generated free commutative P -algebras, while P is the free P -algebra on the zero module.
- (10) **Conclude finite-cell structure.** The displayed pushout is therefore a finite colimit of the generators appearing in 1.6.4.
- (11) **Apply the finite-cell criterion.** Hence $P[t^{-1}]$ is a finite cell P -algebra and is locally of finite presentation by 1.6.7.
- (12) **Base change the universal localization.** For the chosen representative $\tilde{f}$, the base change $A \otimes_P P[t^{-1}]$ has the universal property of an A -algebra in which the degree-zero class f is invertible.
- (13) **Identify it with $A[f^{-1}]$.** Hence the localization universal property gives a canonical equivalence

$$A[f^{-1}] \simeq A \otimes_P P[t^{-1}].$$

- (14) **Remove dependence on the representative.** In particular, the object on the left and the conclusion below are independent of the chosen representative of f .
- (15) **Preserve local finite presentation.** Local finite presentation is preserved by base change, so $A \rightarrow A[f^{-1}]$ is locally of finite presentation.
- (16) **Verify the multiplication equivalence.** The multiplication map

$$A[f^{-1}] \otimes_A A[f^{-1}] \longrightarrow A[f^{-1}]$$

is an equivalence by the same universal property.

- (17) **Conclude that the principal localization is open.** Thus $D_A^{\text{sp}}(f) \rightarrow \text{Spec } A$ is an affine open immersion.

□

Proposition 1.8.5 (Principal opens form a basis). *Every affine open immersion*

$$\text{Spec}(B) \longrightarrow \text{Spec}(A)$$

is covered by principal spectral opens $D_A^{\text{sp}}(f)$ with $f \in \pi_0(A)$. More precisely, there is a family of elements $\{f_\alpha\} \subseteq \pi_0(A)$ such that every $D_A^{\text{sp}}(f_\alpha) \rightarrow \text{Spec}(A)$ factors through $\text{Spec}(B)$, and after pullback to $\text{Spec}(B)$ these principal opens are jointly conservative on Mod_B .

$$\begin{array}{ccc}
 D_A^{\text{sp}}(f_\alpha f_\beta) & \longrightarrow & D_A^{\text{sp}}(f_\beta) \\
 \downarrow & & \downarrow \\
 D_A^{\text{sp}}(f_\alpha) & \longrightarrow & \text{Spec}(A).
 \end{array}$$

Proof.

- (1) **Pass flatness to degree zero.** Let $A \rightarrow B$ represent the affine open immersion. Flatness gives a flat ring map $\pi_0(A) \rightarrow \pi_0(B)$.
- (2) **Pass the diagonal equivalence to degree zero.** Applying π_0 to the diagonal equivalence

$$B \otimes_A B \xrightarrow{\sim} B$$

and using 1.6.4 shows that $\pi_0(A) \rightarrow \pi_0(B)$ is a ring epimorphism.

- (3) **Set up finite presentation in degree zero.** Local finite presentation of $A \rightarrow B$ also descends to degree zero. Let $\{R_\lambda\}$ be a filtered diagram of $\pi_0(A)$ -algebras.
- (4) **Commute Eilenberg–Mac Lane with the filtered colimit.** Filtered colimits of discrete connective commutative ring spectra are created on underlying spectra and remain discrete, so there is a canonical equivalence

$$H\left(\varinjlim_\lambda R_\lambda\right) \simeq \varinjlim_\lambda H R_\lambda.$$

- (5) **Use compactness to compare mapping spaces.** Compactness of B and 1.3.6 therefore give

$$\begin{aligned}
 \text{Map}_{\text{CRing}_{\pi_0(A)/}}\left(\pi_0(B), \varinjlim_\lambda R_\lambda\right) &\simeq \text{Map}_{\text{CAlg}(\text{Sp})_{A/}}\left(B, H\left(\varinjlim_\lambda R_\lambda\right)\right) \\
 &\simeq \text{Map}_{\text{CAlg}(\text{Sp})_{A/}}\left(B, \varinjlim_\lambda H R_\lambda\right) \\
 &\simeq \varinjlim_\lambda \text{Map}_{\text{CAlg}(\text{Sp})_{A/}}(B, H R_\lambda) \\
 &\simeq \varinjlim_\lambda \text{Map}_{\text{CRing}_{\pi_0(A)/}}(\pi_0(B), R_\lambda).
 \end{aligned}$$

- (6) **Conclude finite presentation in degree zero.** Hence $\pi_0(B)$ is finitely presented over $\pi_0(A)$.
- (7) **Identify the resulting ordinary open immersion.** A flat finitely presented ring epimorphism determines an open immersion on prime spectra; equivalently, its image is locally principal. See [37].
- (8) **Choose a principal neighbourhood in degree zero.** Let $\mathfrak{q} \in \text{Spec } \pi_0(B)$ and let $\mathfrak{p} \in \text{Spec } \pi_0(A)$ be its image. Choose $f \in \pi_0(A)$ such that $\mathfrak{p} \in D(f)$ and the principal open $D(f)$ is contained in the image of $\text{Spec } \pi_0(B) \rightarrow \text{Spec } \pi_0(A)$.

- (9) **Form the spectral pullback.** Form

$$\begin{array}{ccc} P := \mathrm{Spec}(B) \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A[f^{-1}]) & \longrightarrow & \mathrm{Spec}(B) \\ \downarrow p & & \downarrow \\ \mathrm{Spec}(A[f^{-1}]) & \longrightarrow & \mathrm{Spec}(A). \end{array}$$

- (10) **Identify its affine coordinate algebra.** The object P is affine and p is an affine open immersion by base change. Its coordinate algebra is $B \otimes_A A[f^{-1}]$.
- (11) **Compute its degree-zero ring.** By 1.6.4, its degree-zero ring is

$$\pi_0(B) \otimes_{\pi_0(A)} \pi_0(A)[f^{-1}].$$

- (12) **Use the chosen principal neighbourhood.** By the choice of the principal neighbourhood, this ring is canonically $\pi_0(A)[f^{-1}]$.

- (13) **Analyze the comparison algebra map.** The map

$$A[f^{-1}] \longrightarrow B \otimes_A A[f^{-1}]$$

is flat and induces an isomorphism on π_0 .

- (14) **Detect its equivalence on homotopy groups.** The flat homotopy-group formula therefore makes it an isomorphism on every homotopy group, hence an equivalence.

- (15) **Obtain the spectral factorization.** Thus

$$P \xrightarrow{\sim} \mathrm{Spec}(A[f^{-1}]),$$

and the inverse exhibits $D_A^{\mathrm{sp}}(f)$ as an open spectral subscheme factoring through $\mathrm{Spec}(B)$.

- (16) **Repeat the construction over every prime.** Repeat the construction for every prime of $\pi_0(B)$. After pullback to $\mathrm{Spec}(B)$, the resulting principal localizations cover the prime spectrum of $\pi_0(B)$.
- (17) **Detect vanishing on the principal cover.** If a B -module restricts to zero on all of them, every homotopy group localizes to zero on this principal cover and hence vanishes.
- (18) **Conclude joint conservativity.** Homotopy groups detect equivalences, so the pullback functors are jointly conservative on Mod_B .

□

1.8.3 The affine spectral Zariski topology

Definition 1.8.2 (Affine Zariski covering family). A small family of affine open immersions

$$\{\mathrm{Spec}(B_i) \rightarrow \mathrm{Spec}(A)\}_{i \in I}$$

is an **affine spectral Zariski covering family** if the substitution functors

$$\{- \otimes_A B_i : \mathrm{Mod}_A \rightarrow \mathrm{Mod}_{B_i}\}_{i \in I}$$

are jointly conservative.

The definition makes locality part of the coefficient system itself: an affine family covers exactly when substitution along those context changes jointly detects coefficients.

Proposition 1.8.6 (The spectral Zariski pretopology). *Affine spectral Zariski covering families define a Grothendieck pretopology on Aff_{Sp} .*

Proof.

- (1) **Identity covers.** The identity substitution on Mod_A is conservative, so the identity family covers.
- (2) **Base change.** Let $\{A \rightarrow B_i\}$ be covering and let $A \rightarrow A'$ be any morphism. Put $B'_i := B_i \otimes_A A'$. If $M \in \text{Mod}_{A'}$ satisfies $M \otimes_{A'} B'_i \simeq 0$ for every i , then, after restriction to A ,

$$M \otimes_A B_i \simeq M \otimes_{A'} B'_i \simeq 0.$$

Joint conservativity over A gives $M \simeq 0$. The base-changed family therefore covers. Its members remain affine immersions by 1.8.2.

- (3) **Transitivity.** Suppose $\{A \rightarrow B_i\}$ is covering and, for every i , $\{B_i \rightarrow C_{ij}\}_j$ is covering. If $M \in \text{Mod}_A$ restricts to zero over every C_{ij} , then $M \otimes_A B_i \simeq 0$ for every i by joint conservativity over B_i . Joint conservativity over A then gives $M \simeq 0$. Thus the composite family covers.

□

Definition 1.8.3 (Spectral Zariski topology). The **spectral Zariski topology**

$$\tau_{\text{Zar}}$$

on Aff_{Sp} is the Grothendieck topology generated by the pretopology of 1.8.6.

Proposition 1.8.7 (Finite principal refinements). *Every affine spectral Zariski covering family of $\text{Spec}(A)$ admits a finite refinement by principal spectral opens*

$$\{D_A^{\text{sp}}(f_j) \rightarrow \text{Spec}(A)\}_{j=1}^r$$

with $f_1, \dots, f_r \in \pi_0(A)$ generating the unit ideal. The refinement may be displayed as

$$\coprod_{j=1}^r D_A^{\text{sp}}(f_j) \longrightarrow \coprod_{i \in I} \text{Spec}(B_i) \longrightarrow \text{Spec}(A).$$

Consequently, finite principal covers generate the spectral Zariski topology on affines.

Proof.

- (1) **Refine each member by principal opens.** Let $\{\text{Spec}(B_i) \rightarrow \text{Spec}(A)\}_{i \in I}$ be a covering family. By 1.8.5, each $\text{Spec}(B_i)$ is covered by principal spectral opens of $\text{Spec}(A)$ which factor through $\text{Spec}(B_i)$.
- (2) **Form the combined principal family.** Let $\{D_A^{\text{sp}}(f_\alpha)\}_{\alpha \in J}$ be the combined family.
- (3) **Assume vanishing on the combined family.** If $M \in \text{Mod}_A$ restricts to zero on every $D_A^{\text{sp}}(f_\alpha)$, then for each i the module $M \otimes_A B_i$ restricts to zero on a jointly conservative principal cover of $\text{Spec}(B_i)$.

- (4) **Descend vanishing to each original member.** Hence $M \otimes_A B_i \simeq 0$ for every i .
- (5) **Use the original cover.** Since the original family is covering, $M \simeq 0$.
- (6) **Conclude conservativity of the combined family.** Thus the combined principal family is jointly conservative.
- (7) **Define the coefficient-detecting module.** Let $J_0 \subseteq \pi_0(A)$ be the ideal generated by all f_α , and regard

$$M := H(\pi_0(A)/J_0)$$

as an A -module through

$$A \longrightarrow H\pi_0(A) \longrightarrow H(\pi_0(A)/J_0).$$

- (8) **Apply the localization tensor formula.** For every α , the map $A \rightarrow A[f_\alpha^{-1}]$ is flat by 1.8.3. Applying the tensor formula of 1.6.1 to the flat A -module $A[f_\alpha^{-1}]$ gives

$$\pi_n(M \otimes_A A[f_\alpha^{-1}]) \simeq \pi_n(M)[f_\alpha^{-1}] = 0$$

for all n , because M is discrete and f_α acts by zero on $\pi_0(M)$.

- (9) **Deduce vanishing after every principal localization.** Hence

$$M \otimes_A A[f_\alpha^{-1}] \simeq 0$$

for every α .

- (10) **Use joint conservativity to identify the unit ideal.** Joint conservativity of the combined principal family gives $M \simeq 0$, so

$$\pi_0(A)/J_0 = 0 \quad \text{and therefore} \quad J_0 = (1).$$

- (11) **Extract finitely many generators.** By the definition of the generated ideal, the relation $1 \in J_0$ involves only finitely many generators: there are $f_1, \dots, f_r$ among the f_α and $a_1, \dots, a_r \in \pi_0(A)$ with

$$1 = a_1 f_1 + \dots + a_r f_r.$$

- (12) **Obtain the finite principal refinement.** Thus $(f_1, \dots, f_r) = (1)$, and the corresponding finite principal family is the required refinement.

□

Proposition 1.8.8 (Coefficient criterion for Zariski covers). *For a family of affine open immersions*

$$\{\mathrm{Spec}(B_i) \rightarrow \mathrm{Spec}(A)\}_{i \in I},$$

the following conditions are equivalent.

- (i) *The family is an affine spectral Zariski covering family.*
- (ii) *The pullback functors $\{- \otimes_A B_i\}_{i \in I}$ are jointly conservative on Mod_A .*
- (iii) *The family admits a finite refinement by principal spectral opens $D_A^{\mathrm{sp}}(f_j)$ with $(f_1, \dots, f_r) = (1)$ in $\pi_0(A)$.*

Proof.

- (1) **Identify covering with conservative substitution.** Conditions (i) and (ii) are equivalent by 1.8.2. The implication (i) $\Rightarrow$ (iii) is 1.8.7.
- (2) **Principal unit-ideal families are conservative.** Assume (iii) and let $M \in \text{Mod}_A$ restrict to zero over every member of the original family. It then restricts to zero over every refining principal open, so by 1.8.3

$$\pi_n(M)[f_j^{-1}] = 0$$

for every n and j . Since the f_j generate the unit ideal, an ordinary $\pi_0(A)$ -module which vanishes after all these localizations is zero. Thus every $\pi_n(M)$ vanishes, and 1.2.4 gives $M \simeq 0$. Hence the original family is jointly conservative. $\square$

For a finite covering family, let

$$U := \coprod_i \text{Spec}(B_i)$$

in $\text{PSh}(\text{Aff}_{\text{Sp}})$. Its Čech nerve packages all iterated overlaps; the first stage is

$$U \times_{\text{Spec}(A)} U \rightrightarrows U \longrightarrow \text{Spec}(A),$$

and each affine overlap is represented by the corresponding tensor product $B_i \otimes_A B_j$. The next results show that these overlap diagrams detect covers and reconstruct the objects being covered.

$$\begin{array}{ccc} \text{Spec}(B_i \otimes_A B_j) & \longrightarrow & \text{Spec}(B_j) \\ \downarrow & & \downarrow \\ \text{Spec}(B_i) & \longrightarrow & \text{Spec}(A). \end{array}$$

1.8.4 Descendability and Zariski descent

Definition 1.8.4 (Thick tensor ideal). Let $\mathcal{C}^\otimes$ be a stable symmetric monoidal ∞ -category. A full subcategory

$$\mathcal{I} \subseteq \mathcal{C}$$

is a **thick tensor ideal** if it is closed under finite limits, finite colimits, retracts, and tensoring with arbitrary objects of $\mathcal{C}$.

Definition 1.8.5 (Descendable morphism). A morphism

$$A \longrightarrow B$$

of commutative ring spectra is **descendable** if B generates Mod_A as a thick tensor ideal.

Proposition 1.8.9 (Permanence of descendability). *Descendable morphisms are stable under equivalence, base change, and composition. If $A \rightarrow B \rightarrow C$ is a factorization and $A \rightarrow C$ is descendable, then $A \rightarrow B$ is descendable.*

Proof.

- (1) **Handle base change.** Base change follows because extension of scalars is exact and strong symmetric monoidal.
- (2) **Invoke the remaining permanence results.** The remaining assertions are the permanence results of [39, Proposition 3.24].

□

Lemma 1.8.2 (Modules over finite product algebras). *For a finite family of commutative ring spectra $\{B_i\}_{i \in I}$, there is a canonical symmetric monoidal equivalence*

$$\mathrm{Mod}_{\prod_i B_i} \simeq \prod_i \mathrm{Mod}_{B_i}.$$

Proof.

- (1) **Identify the underlying biproduct.** The finite product of commutative ring spectra has underlying finite biproduct of spectra.
- (2) **Split modules by idempotents.** Its orthogonal idempotents split every module into its components.
- (3) **Construct the inverse functors.** Extension of scalars along the projections and finite direct sum of restrictions of scalars define mutually inverse symmetric monoidal functors, exactly as for ordinary product rings.

□

Lemma 1.8.3 (Finite products and base change). *Let $A \rightarrow D$ be a morphism of commutative ring spectra and let $\{B_i\}_{i \in I}$ be a finite family of commutative A -algebras. The canonical morphism*

$$D \otimes_A \left(\prod_{i \in I} B_i \right) \longrightarrow \prod_{i \in I} (D \otimes_A B_i)$$

is an equivalence. Consequently,

$$\left(\prod_i B_i \right)^{\otimes_A (n+1)} \simeq \prod_{(i_0, \dots, i_n)} B_{i_0} \otimes_A \cdots \otimes_A B_{i_n}.$$

Proof.

- (1) **Underlying modules.** Finite products and finite coproducts agree in the stable category Mod_A , so

$$\prod_i B_i \simeq \bigoplus_i B_i$$

as A -modules.

- (2) **Extension of scalars.** Tensoring with D preserves coproducts, giving

$$D \otimes_A \prod_i B_i \simeq \bigoplus_i (D \otimes_A B_i) \simeq \prod_i (D \otimes_A B_i)$$

on underlying D -modules. The canonical algebra morphism is therefore an equivalence. Iteration gives the second formula.

□

Proposition 1.8.10 (Finite Zariski covers yield descendable product algebras). *Let*

$$\{\mathrm{Spec}(B_i) \rightarrow \mathrm{Spec}(A)\}_{i \in I}$$

be a finite affine spectral Zariski covering family and put

$$B := \prod_{i \in I} B_i.$$

Then $A \rightarrow B$ is descendable.

Proof.

(1) **Reduce to a finite principal cover.** By 1.8.7, choose a finite principal refinement

$$\{D_A^{\mathrm{sp}}(f_j) \rightarrow \mathrm{Spec}(A)\}_{j=1}^r$$

subordinate to the given cover. Put

$$A_j := A[f_j^{-1}], \quad C := \prod_{j=1}^r A_j.$$

The refinement gives a factorization

$$A \longrightarrow B \longrightarrow C.$$

By 1.8.9, it is enough to prove that $A \rightarrow C$ is descendable.

(2) **Fibres of the localization maps.** For each j , let

$$K_j := \mathrm{fib}(A \longrightarrow A_j) \in \mathrm{Mod}_A.$$

Localization is idempotent, so

$$A_j \otimes_A A_j \simeq A_j.$$

Exactness of extension of scalars therefore gives

$$K_j \otimes_A A_j \simeq \mathrm{fib}(A_j \rightarrow A_j \otimes_A A_j) \simeq 0.$$

(3) **The product of the fibres vanishes.** Put

$$K := K_1 \otimes_A \cdots \otimes_A K_r.$$

For every j , one tensor factor becomes zero after extension to A_j , hence

$$K \otimes_A A_j \simeq 0.$$

The principal spectral opens $D_A^{\mathrm{sp}}(f_j)$ cover $\mathrm{Spec}(A)$, so the functors $- \otimes_A A_j$ are jointly conservative by 1.8.8. Consequently

$$K \simeq 0.$$

- (4) **Place the localizations in the thick tensor ideal.** Let $\mathcal{I}$ be the thick tensor ideal of Mod_A generated by C . Every A_j is a retract of the underlying finite product C , so $A_j \in \mathcal{I}$.

- (5) **Define the inductive objects and target.** Define

$$L_m := K_1 \otimes_A \cdots \otimes_A K_m, \quad L_0 := A,$$

and let $u_m : L_m \rightarrow A$ be the tensor product of the maps $K_j \rightarrow A$, followed by multiplication. We prove inductively that

$$\text{cofib}(u_m) \in \mathcal{I}.$$

- (6) **Verify the base case.** For $m = 1$, the defining fibre sequence gives

$$\text{cofib}(K_1 \rightarrow A) \simeq A_1 \in \mathcal{I}.$$

- (7) **Factor the inductive map.** For $m > 1$, factor

$$L_m = L_{m-1} \otimes_A K_m \longrightarrow L_{m-1} \xrightarrow{u_{m-1}} A.$$

- (8) **Identify the first cofiber.** The cofiber of the first map is

$$L_{m-1} \otimes_A A_m,$$

which lies in $\mathcal{I}$ because $A_m \in \mathcal{I}$ and $\mathcal{I}$ is a tensor ideal.

- (9) **Apply the composite cofiber sequence.** The cofiber sequence associated to a composite in a stable ∞ -category then expresses $\text{cofib}(u_m)$ as an extension of $L_{m-1} \otimes_A A_m$ by $\text{cofib}(u_{m-1})$. Hence $\text{cofib}(u_m) \in \mathcal{I}$.

- (10) **Use vanishing of the total fibre product.** For $m = r$, Step (3) gives $L_r = K \simeq 0$. Therefore

$$A \simeq \text{cofib}(0 \rightarrow A) \simeq \text{cofib}(u_r) \in \mathcal{I}.$$

- (11) **Generate the tensor unit.** Thus C generates the tensor unit and $A \rightarrow C$ is descendable.

- (12) **Return to the original cover.** The factorization $A \rightarrow B \rightarrow C$ and the intermediate-algebra property of descendability imply that $A \rightarrow B$ is descendable; see [39, Proposition 3.24].

□

Theorem 1.8.1 (Amitsur convergence). *If $A \rightarrow B$ is descendable, then the augmented Amitsur object*

$$A \longrightarrow B \rightrightarrows B \otimes_A B \cdots$$

is a limit diagram in $\text{CAlg}(\text{Sp})$. In particular,

$$A \simeq \text{Tot}_{[n] \in \Delta} B^{\otimes_A (n+1)}.$$

Proof.

- (1) **Regard the cover algebra in the module category.** Regard B as a commutative algebra object of Mod_A .

- (2) **Translate descendability.** Descendability means that B generates the tensor unit.
- (3) **Apply the descendability theorem.** The descendability theorem identifies the augmented cobar object as a limit diagram in Mod_A ; see [39, Proposition 3.20].
- (4) **Create the same limit in commutative algebras.** The forgetful functor from commutative A -algebras to A -modules creates limits, so the same object is a limit in commutative A -algebras.
- (5) **Pass to commutative ring spectra.** It is therefore a limit in $\text{CAlg}(\text{Sp})$.

□

Theorem 1.8.2 (Affine module descent). *If $A \rightarrow B$ is descendable, then extension of scalars induces an equivalence in $\text{CAlg}(\text{Cat}_\infty)$*

$$\text{Mod}_A \xrightarrow{\cong} \text{Tot}_{[n] \in \Delta} \text{Mod}_{B^{\otimes_A(n+1)}}.$$

Proof.

- (1) **Apply comonadicity.** The extension–restriction adjunction is comonadic for a descendable map.
- (2) **Identify the cobar resolution.** Its cobar resolution is the displayed Amitsur diagram; see [39, Proposition 3.22].
- (3) **Retain the symmetric monoidal structure.** Every transition functor is strong symmetric monoidal, so the comparison is an equivalence of symmetric monoidal ∞ -categories.

□

Theorem 1.8.3 (Affine algebra descent). *If $A \rightarrow B$ is descendable, then base change induces an equivalence*

$$\text{CAlg}(\text{Sp})_{A/} \xrightarrow{\cong} \text{Tot}_{[n] \in \Delta} \text{CAlg}(\text{Sp})_{B^{\otimes_A(n+1)}/}.$$

$$\begin{array}{ccc} \text{Mod}_A & \xrightarrow{\cong} & \text{Tot}_{[n] \in \Delta} \text{Mod}_{B^{\otimes_A(n+1)}} \\ \text{CAlg} \downarrow & & \downarrow \text{CAlg} \\ \text{CAlg}(\text{Mod}_A) & \xrightarrow{\cong} & \text{CAlg}\left(\text{Tot}_{[n] \in \Delta} \text{Mod}_{B^{\otimes_A(n+1)}}\right) \xrightarrow{\cong} \text{Tot}_{[n] \in \Delta} \text{CAlg}(\text{Mod}_{B^{\otimes_A(n+1)}}). \end{array}$$

Proof.

- (1) **Apply affine module descent.** By 1.8.2, extension of scalars gives a symmetric monoidal equivalence

$$\text{Mod}_A \xrightarrow{\cong} \text{Tot}_{[n] \in \Delta} \text{Mod}_{B^{\otimes_A(n+1)}}.$$

- (2) **Use limit preservation of commutative algebra objects.** The functor assigning to a symmetric monoidal ∞ -category its ∞ -category of commutative algebra objects preserves limits.

(3) **Apply commutative algebra objects.** Applying it to the displayed equivalence gives

$$\mathrm{CAlg}(\mathrm{Mod}_A) \xrightarrow{\simeq} \mathrm{Tot}_{[n] \in \Delta} \mathrm{CAlg}(\mathrm{Mod}_{B^{\otimes_A(n+1)}}).$$

(4) **Identify relative commutative algebras.** Using 1.5.5 in every cosimplicial degree identifies this equivalence with

$$\mathrm{CAlg}(\mathrm{Sp})_{A/} \xrightarrow{\simeq} \mathrm{Tot}_{[n] \in \Delta} \mathrm{CAlg}(\mathrm{Sp})_{B^{\otimes_A(n+1)}/}.$$

□

Corollary 1.8.1 (Connective algebra descent for finite Zariski covers). *Let*

$$\{\mathrm{Spec}(B_i) \rightarrow \mathrm{Spec}(A)\}_{i \in I}$$

be a finite affine spectral Zariski covering family and put

$$B := \prod_{i \in I} B_i.$$

Then the equivalence of 1.8.3 restricts to an equivalence

$$\mathrm{CAlg}(\mathrm{Sp})_{A/}^{\mathrm{cn}} \xrightarrow{\simeq} \mathrm{Tot}_{[n] \in \Delta} \mathrm{CAlg}(\mathrm{Sp})_{B^{\otimes_A(n+1)}/}^{\mathrm{cn}}.$$

Proof.

(1) **Flat descendable cover algebra.** The morphism $A \rightarrow B$ is descendable by 1.8.10. Each B_i is flat over A , and the finite product B is the finite direct sum of the B_i as an A -module. Hence $A \rightarrow B$ is flat. Every Amitsur term $B^{\otimes_A(n+1)}$ is therefore connective, and base change along $A \rightarrow B$ is t -exact and conservative.

(2) **Descend an object of the connective totalization.** Let an object of the right-hand totalization be given whose component in every cosimplicial degree is connective. By 1.8.3, it descends to a commutative A -algebra C . Its base change $C \otimes_A B$ is connective, so

$$\tau_{\leq -1}^A(C) \otimes_A B \simeq \tau_{\leq -1}^B(C \otimes_A B) \simeq 0,$$

where the algebras are regarded as their underlying modules. Conservativity of $- \otimes_A B$ gives

$$\tau_{\leq -1}^A(C) \simeq 0,$$

hence the underlying A -module of C , and therefore C , is connective.

(3) **Restrict the descent equivalence.** Essential surjectivity on connective objects follows from the preceding step, while full faithfulness is inherited from the unrestricted algebra descent equivalence. This gives the asserted totalization equivalence.

□

Theorem 1.8.4 (Subcanonicity of the spectral Zariski topology). *Every representable spectral prestack is a sheaf for τ_{Zar} .*

Proof.

- (1) **Reduction to finite principal covers.** By 1.8.7, finite principal covers generate the affine spectral Zariski topology. It therefore suffices to prove Čech descent for a finite affine cover $\{\mathrm{Spec}(B_i) \rightarrow \mathrm{Spec}(A)\}$.
- (2) **Passage to one descendable algebra.** Put $B := \prod_i B_i$. By 1.8.10, $A \rightarrow B$ is descendable, so 1.8.1 gives

$$A \simeq \mathrm{Tot}_{[n] \in \Delta} B^{\otimes_A (n+1)}.$$

- (3) **Mapping into the Amitsur totalization.** For every connective commutative ring spectrum C , mapping out of C converts this limit into

$$\mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})}(C, A) \simeq \mathrm{Tot}_{[n] \in \Delta} \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})}(C, B^{\otimes_A (n+1)}).$$

By 1.8.3, the Amitsur terms decompose into the iterated intersections of the original covering family. This is the Čech descent condition for the representable h_C .

□

Theorem 1.8.5 (Zariski descent for affine quasi-coherent modules). *Let*

$$U := \coprod_i \mathrm{Spec}(B_i) \longrightarrow \mathrm{Spec}(A)$$

be a finite affine spectral Zariski cover, and let $U_\bullet$ be its Čech nerve. Then the comparison functor is an equivalence

$$\mathrm{Mod}_A \xrightarrow{\simeq} \mathrm{Tot}_{[n] \in \Delta} \mathrm{QCoh}(U_n).$$

Proof.

- (1) **Identify the prestack Čech terms.** Put $B := \prod_i B_i$. By 1.8.3, the degree- n Amitsur algebra decomposes as

$$B^{\otimes_A (n+1)} \simeq \prod_{(i_0, \dots, i_n)} B_{i_0} \otimes_A \cdots \otimes_A B_{i_n}.$$

The corresponding affine components are exactly the iterated fibre products occurring in the prestack Čech nerve of $U \rightarrow \mathrm{Spec}(A)$.

- (2) **Compute quasi-coherent modules on the coproducts.** Since QCoh carries prestack colimits to limits by 1.7.2, and modules over a finite product split by 1.8.2,

$$\mathrm{QCoh}(U_n) \simeq \prod_{(i_0, \dots, i_n)} \mathrm{Mod}_{B_{i_0} \otimes_A \cdots \otimes_A B_{i_n}} \simeq \mathrm{Mod}_{B^{\otimes_A (n+1)}}.$$

- (3) **Apply descendable module descent.** The map $A \rightarrow B$ is descendable by 1.8.10. Therefore 1.8.2 gives

$$\mathrm{Mod}_A \xrightarrow{\simeq} \mathrm{Tot}_{[n] \in \Delta} \mathrm{Mod}_{B^{\otimes_A (n+1)}}.$$

Under the identifications of the preceding steps this is the asserted Čech descent equivalence.

□

Proposition 1.8.11 (Affine open covers and effective epimorphisms). *Let*

$$\{U_i = \operatorname{Spec}(B_i) \longrightarrow X = \operatorname{Spec}(A)\}_{i \in I}$$

be a small family of affine open immersions. The following conditions are equivalent.

- (i) *The family is an affine spectral Zariski covering family.*
- (ii) *The induced morphism of spectral Zariski stacks*

$$\coprod_{i \in I} U_i \longrightarrow X$$

is an effective epimorphism.

For every affine base change $T \rightarrow X$, the covering morphism is carried to

$$\begin{array}{ccc} \coprod_i (T \times_X U_i) & \longrightarrow & \coprod_i U_i \\ \downarrow & & \downarrow \\ T & \longrightarrow & X, \end{array}$$

and the left vertical morphism is again effective epimorphic.

Proof.

- (1) **A covering family is locally surjective.** Assume (i). For every affine test $T \rightarrow X$, base change of the family is again a Zariski covering family by 1.8.6. Hence the section of X over T determined by $T \rightarrow X$ lifts locally to one of the pullbacks $T \times_X U_i$. The local-surjectivity characterization of effective epimorphisms from 1.8.2 therefore makes the coproduct map an effective epimorphism.
- (2) **An effective affine-open family is conservative.** Assume (ii). Apply local surjectivity to the identity section $\operatorname{id}_X \in X(X)$. The resulting covering sieve contains, because the topology is generated by the pretopology of 1.8.6, an affine spectral Zariski covering family

$$\{V_j \longrightarrow X\}_{j \in J}$$

such that each $V_j \rightarrow X$ factors through some U_i . If $M \in \operatorname{Mod}_A$ restricts to zero on every U_i , it restricts to zero on every V_j . Joint conservativity of the latter covering family gives $M \simeq 0$. Thus the original family is jointly conservative and is a covering family by 1.8.2.

□

The affine locality problem is now solved in the form needed for gluing. Principal spectral opens generate the topology, joint conservativity detects covers, and the same covering data reconstruct representables, modules, and connective commutative algebras. The next section uses these reconstruction results to pass from affine contexts to global spectral schemes.

1.9 Global spectral contexts

The affine site is now subcanonical, and its covering families reconstruct both geometric and coefficient data. We can therefore pass from local contexts to global ones by gluing affine opens. This section defines spectral schemes as Zariski sheaves with effective affine open atlases, establishes the closure properties needed later, and finally identifies this functor-of-points model with the standard structured spectral-scheme model.

Definition 1.9.1 (Spectral Zariski stack). A **spectral Zariski stack** is a sheaf of spaces on Aff_{Sp} for the spectral Zariski topology. The ∞ -category of spectral Zariski stacks is denoted

$$\text{Stk}_{\text{Zar}}^{\text{sp}} := \text{Sh}_{\tau_{\text{Zar}}}(\text{Aff}_{\text{Sp}}).$$

Construction 1.9.1 (Spectral Zariski sheafification). The inclusion

$$\text{Stk}_{\text{Zar}}^{\text{sp}} \hookrightarrow \text{PSh}(\text{Aff}_{\text{Sp}})$$

is an accessible left-exact localization. Its left adjoint is denoted

$$a_{\text{Zar}}^{\text{sp}} : \text{PSh}(\text{Aff}_{\text{Sp}}) \longrightarrow \text{Stk}_{\text{Zar}}^{\text{sp}}$$

and is called **spectral Zariski sheafification**. It preserves all small colimits and finite limits. See [37].

Subcanonicity allows affine contexts to be regarded without ambiguity as objects of the Zariski topos:

$$\text{Aff}_{\text{Sp}} \xhookrightarrow{h} \text{Stk}_{\text{Zar}}^{\text{sp}} \hookrightarrow \text{PSh}(\text{Aff}_{\text{Sp}}).$$

1.9.1 Representable open immersions

Definition 1.9.2 (Open immersion of spectral stacks). Let

$$j : U \longrightarrow X$$

be a morphism of spectral Zariski stacks.

Suppose first that X is an affine spectral scheme. Then j is an **open immersion** if:

- (i) j is a monomorphism;
- (ii) there exists a small family of affine spectral schemes $\{U_\alpha\}_{\alpha \in I}$ and morphisms

$$U_\alpha \longrightarrow U$$

such that

$$\coprod_{\alpha \in I} U_\alpha \longrightarrow U$$

is an effective epimorphism and every composite

$$U_\alpha \longrightarrow U \longrightarrow X$$

is an affine open immersion in the sense of 1.8.1.

For a general spectral Zariski stack X , the morphism j is an open immersion if, for every affine spectral scheme T and every morphism $T \rightarrow X$, the pullback

$$U \times_X T \longrightarrow T$$

is an open immersion in the preceding sense.

Proposition 1.9.1 (Permanence of open immersions). *Open immersions of spectral Zariski stacks are stable under equivalence, composition, and base change.*

Proof.

- (1) **Handle equivalences.** Equivalences preserve open immersions.
- (2) **Handle base change.** Stability under base change is part of the representable definition.
- (3) **Reduce composition to an affine target.** For composable open immersions

$$U \xrightarrow{j} V \xrightarrow{k} X,$$

base change along an affine point $T \rightarrow X$ reduces the assertion to the case $X = T$ affine. Choose an affine open atlas $\coprod_{\alpha} V_{\alpha} \rightarrow V$ whose composites to T are affine open immersions. Pull U back to each V_{α} and choose an affine open atlas $\coprod_{\beta} U_{\alpha\beta} \rightarrow U \times_V V_{\alpha}$.

- (4) **Compose the affine charts.** Every $U_{\alpha\beta} \rightarrow V_{\alpha} \rightarrow T$ is an affine open immersion by 1.8.2. The coproduct

$$\coprod_{\alpha,\beta} U_{\alpha\beta} \longrightarrow U$$

is an effective epimorphism, and $U \rightarrow T$ is a monomorphism. Hence $U \rightarrow T$ satisfies the definition of an open immersion. This proves composition stability. □

1.9.2 Spectral schemes

Definition 1.9.3 (Spectral scheme). A **spectral scheme** is a spectral Zariski stack X for which there exists a small family of affine spectral schemes $\{U_i\}_{i \in I}$ and a morphism

$$\coprod_{i \in I} U_i \longrightarrow X$$

such that:

- (i) the displayed morphism is an effective epimorphism;
- (ii) every component $U_i \rightarrow X$ is an open immersion.

Such a family is an **affine spectral Zariski atlas** of X .

The full subcategory of

$$\mathrm{Stk}_{\mathrm{Zar}}^{\mathrm{sp}}$$

spanned by spectral schemes is denoted

$$\mathrm{Sch}^{\mathrm{sp}}.$$

Proposition 1.9.2 (Affine spectral schemes are spectral schemes). *Every affine spectral scheme is a spectral scheme.*

Proof.

- (1) **Use the identity atlas.** Use the singleton atlas given by the identity map.

□

Proposition 1.9.3 (Coproducts of spectral schemes). *Small coproducts of spectral schemes exist and are spectral schemes.*

Proof.

- (1) **Form the disjoint union in the Zariski topos.** Coproducts exist in the ambient ∞ -topos of spectral Zariski stacks. The inclusion of a coproduct summand is an open immersion. This is the usual disjoint-union construction for spectral schemes: after affine base change, the summand pullbacks are open-and-closed affine pieces, equivalently the pieces determined by idempotents in degree zero; see [37].
- (2) **Coproduct the affine atlases.** Choose affine open atlases $\coprod_{\alpha} U_{i\alpha} \rightarrow X_i$. The composite

$$\coprod_{i,\alpha} U_{i\alpha} \longrightarrow \coprod_i X_i$$

is an effective epimorphism, and each component is an open immersion by composition with the open summand inclusion. Hence the ambient coproduct is a spectral scheme and is the coproduct in Sch^{sp} .

□

Lemma 1.9.1 (Open substacks of spectral schemes). *Let X be a spectral scheme and let*

$$U \longrightarrow X$$

be an open immersion of spectral Zariski stacks. Then U is a spectral scheme.

Proof.

- (1) **Pull back an affine atlas.** Choose an affine spectral Zariski atlas

$$\coprod_i X_i \longrightarrow X.$$

Put $U_i := U \times_X X_i$. Each $U_i \rightarrow X_i$ is an open immersion by base change.

- (2) **Refine over each affine chart.** By the definition of an open immersion with affine target, every U_i admits an effective epimorphism

$$\coprod_{\alpha} U_{i\alpha} \longrightarrow U_i$$

by affine spectral schemes whose composites $U_{i\alpha} \rightarrow X_i$ are affine open immersions.

- (3) **Assemble the atlas.** Effective epimorphisms are stable under base change and composition in an ∞ -topos. Hence

$$\coprod_{i,\alpha} U_{i\alpha} \longrightarrow U$$

is an effective epimorphism. Every component is an open immersion by composition and base change. Thus it is an affine spectral Zariski atlas of U .

□

Proposition 1.9.4 (Small unions of open substacks). *Let X be a spectral Zariski stack and let*

$$\{j_\alpha : U_\alpha \rightarrow X\}_{\alpha \in I}$$

be a small family of open immersions. Factor the induced morphism in the spectral Zariski ∞ -topos as an effective epimorphism followed by a monomorphism:

$$\coprod_{\alpha \in I} U_\alpha \longrightarrow \twoheadrightarrow U \xhookrightarrow{j} X.$$

Then $j : U \rightarrow X$ is an open immersion. This construction is stable under arbitrary base change. If X is a spectral scheme, then U is a spectral scheme.

Proof.

- (1) **Base change the image factorization.** Image factorizations in an ∞ -topos are stable under pullback. For every affine point $T \rightarrow X$, the pullback $U_T := U \times_X T$ is therefore the image in

$$\coprod_{\alpha} (U_\alpha \times_X T) \longrightarrow \twoheadrightarrow U_T \xhookrightarrow{\quad} T.$$

Each $U_\alpha \times_X T \rightarrow T$ is open by base-change stability of open immersions.

- (2) **Produce an affine open atlas of the image.** For every α , choose an affine open atlas

$$\coprod_{\beta} V_{\alpha\beta} \longrightarrow U_\alpha \times_X T.$$

Every composite $V_{\alpha\beta} \rightarrow T$ is an affine open immersion. Since effective epimorphisms are stable under coproduct and composition, the induced map

$$\coprod_{\alpha,\beta} V_{\alpha\beta} \longrightarrow U_T$$

is an effective epimorphism. The morphism $U_T \rightarrow T$ is a monomorphism by construction. Hence $U_T \rightarrow T$ is an open immersion in the sense of 1.9.2.

- (3) **Globalize the affine-point argument.** Since the preceding argument applies to every affine point of X , the morphism $U \rightarrow X$ is an open immersion.
- (4) **Record base-change stability.** The construction commutes with arbitrary base change because it was defined by the universal image factorization.

- (5) **Recognize the open as a spectral scheme.** If X is a spectral scheme, then U is a spectral scheme by 1.9.1.

□

Theorem 1.9.1 (Fibre products of spectral schemes). *Let*

$$X \longrightarrow Z \longleftarrow Y$$

be morphisms of spectral schemes. Then the fibre product

$$X \times_Z Y$$

is a spectral scheme.

Proof.

- (1) **Choose one affine atlas of the base.** Choose an affine spectral Zariski atlas

$$\coprod_k W_k \longrightarrow Z.$$

Put

$$X_k := X \times_Z W_k, \quad Y_k := Y \times_Z W_k.$$

Since $W_k \rightarrow Z$ is open, the projections $X_k \rightarrow X$ and $Y_k \rightarrow Y$ are open. Hence X_k and Y_k are spectral schemes by 1.9.1.

- (2) **Choose affine atlases over the same base chart.** For every k , choose affine open atlases

$$\coprod_{\alpha} U_{k\alpha} \longrightarrow X_k, \quad \coprod_{\beta} V_{k\beta} \longrightarrow Y_k.$$

By construction, every $U_{k\alpha}$ and $V_{k\beta}$ maps to the same affine spectral scheme W_k .

- (3) **Form affine fibre products.** Put

$$P_{k\alpha\beta} := U_{k\alpha} \times_{W_k} V_{k\beta}.$$

Each $P_{k\alpha\beta}$ is affine by 1.4.2. The canonical map

$$P_{k\alpha\beta} \longrightarrow X \times_Z Y$$

is an open immersion, since it is obtained by composing base changes of the open immersions $U_{k\alpha} \rightarrow X_k$, $V_{k\beta} \rightarrow Y_k$, and $W_k \rightarrow Z$.

- (4) **Verify that the affine charts cover.** The pullbacks

$$(X \times_Z Y) \times_Z W_k \simeq X_k \times_{W_k} Y_k$$

cover $X \times_Z Y$. For fixed k , base changing the atlas of X_k to $X_k \times_{W_k} Y_k$, and then base changing the atlas of Y_k , shows that

$$\coprod_{\alpha, \beta} P_{k\alpha\beta} \longrightarrow X_k \times_{W_k} Y_k$$

is an effective epimorphism. Universality and composition of effective epimorphisms therefore give an effective epimorphism

$$\coprod_{k, \alpha, \beta} P_{k\alpha\beta} \longrightarrow X \times_Z Y.$$

Thus these affine open immersions form an affine spectral Zariski atlas.

□

Corollary 1.9.1 (Finite limits of spectral schemes). *The ∞ -category of spectral schemes admits finite limits, computed in spectral Zariski stacks.*

Proof.

- (1) **Identify the terminal object.** The terminal object is $\mathrm{Spec}(\mathbb{S})$.
- (2) **Construct binary fibre products.** Binary fibre products exist by 1.9.1.

□

1.9.3 Comparison with structured spectral schemes

Theorem 1.9.2 (Recognition of the structured spectral model). *Let*

$$\mathrm{Sch}_{\mathrm{str}}^{\mathrm{sp}}$$

denote the ∞ -category of connective 0-localic spectral schemes in the structured-space sense of spectral algebraic geometry. The standard functor of points

$$\Phi(\mathfrak{X})(\mathrm{Spec} A) := \mathrm{Map}(\mathrm{Spec} A, \mathfrak{X})$$

induces an equivalence

$$\mathrm{Sch}_{\mathrm{str}}^{\mathrm{sp}} \xrightarrow[\simeq]{\Phi} \mathrm{Sch}^{\mathrm{sp}} \hookrightarrow \mathrm{Stk}_{\mathrm{Zar}}^{\mathrm{sp}}.$$

Under this equivalence:

- (i) $\Phi(\mathrm{Spec} A) \simeq h_A$ for every connective commutative ring spectrum A ;
- (ii) affine morphisms, open immersions, and finite fibre products agree in the two models.

Proof.

- (1) **Verify that the native affine geometry is the standard connective spectral Zariski geometry.** Affines are the opposites of connective commutative ring spectra by 1.4.1. Principal localizations are affine open immersions by 1.8.4, principal opens form a basis by 1.8.5, and 1.8.4 identifies representables as Zariski sheaves. Thus the affine objects, local morphisms, and covering families used in the native construction are the standard connective spectral Zariski ones.
- (2) **Invoke the structured-space recognition theorem.** The comparison theorem between connective 0-localic structured spectral schemes and their functors of points identifies precisely the Zariski stacks admitting affine open atlases with the essential image of the structured model; see [37, 26]. It identifies an affine structured spectral scheme with h_A , and representable properties such as affineness and open immersion are preserved under the comparison. Since functors of points preserve finite limits objectwise, finite fibre products agree as well. This proves all assertions.

□

Remark 1.9.1 (The structured model is used only for recognition). The comparison theorem identifies the spectral schemes constructed above with the structured-space model of [37] while leaving the data carried by an object of Sch^{sp} unchanged. Terminology such as **admissible** belongs to the external structured presentation. The open morphisms and covering families used here remain the spectral localizations and conservative families constructed before the comparison. The structured model serves to recognize this geometry as the standard connective spectral one.

1.9.4 Underlying topological support and qcqs geometry

Definition 1.9.4 (Underlying topological space). Let X be a spectral scheme and choose an affine spectral Zariski atlas

$$\coprod_i \text{Spec}(A_i) \longrightarrow X.$$

The spaces $|\text{Spec } \pi_0(A_i)|$ glue along the open subsets represented by the affine spectral intersections and their principal-open refinements. The resulting topological space is independent of the chosen atlas by common refinement and is denoted

$$|X|.$$

For $X = \text{Spec}(A)$ this agrees with 1.4.2. See [37] for the equivalent structured-space construction.

Definition 1.9.5 (Quasi-compact spectral scheme). A spectral scheme X is **quasi-compact** if it admits a finite affine open cover. A morphism $f : X \rightarrow Y$ is quasi-compact if the pullback over every affine open of Y is quasi-compact.

Definition 1.9.6 (Quasi-separated morphisms and spectral schemes). A morphism

$$f : X \longrightarrow Y$$

of spectral schemes is **quasi-separated** if its diagonal

$$\Delta_f : X \longrightarrow X \times_Y X$$

is quasi-compact.

A spectral scheme X is **quasi-separated** if its structural morphism

$$X \longrightarrow \text{Spec}(\mathbb{S})$$

is quasi-separated. Equivalently, the absolute diagonal

$$X \longrightarrow X \times X$$

is quasi-compact.

A spectral scheme is **qcqs** if it is quasi-compact and quasi-separated.

The global geometric base is now available: spectral schemes are obtained by Zariski gluing from the affine contexts, remain closed under the finite-limit and open-subobject operations needed below, and agree with the standard connective structured model. The coefficient system must now be restricted to this geometric base and shown to admit the corresponding global structures.

1.10 The global quasi-coherent coefficient system

For a global spectral context, the coefficient fibre should be recoverable from the affine fibres on any atlas. The affine descent theorem provides that reconstruction and thereby turns the right-Kan-extended prestack assignment into the coefficient system used on spectral schemes. This section computes $\mathrm{QCoh}(X)$ from affine charts and globalizes the Postnikov structure, flatness, affine direct image, and quasi-coherent commutative algebra objects. These are precisely the ingredients needed before coefficient algebra can be returned to geometry by relative spectrum.

1.10.1 Geometric descent of quasi-coherent coefficients

Theorem 1.10.1 (Atlas computation of quasi-coherent modules). *Let X be a spectral scheme and let*

$$U \longrightarrow X$$

be an affine Zariski atlas. Write $U_\bullet$ for its Čech nerve in spectral schemes. Then the canonical comparison functor is a symmetric monoidal equivalence

$$\mathrm{QCoh}(X) \xrightarrow{\cong} \mathrm{Tot}_{[n] \in \Delta} \mathrm{QCoh}(U_n).$$

Equivalently, the augmented cosimplicial coefficient diagram

$$\mathrm{QCoh}(X) \longrightarrow \mathrm{QCoh}(U) \begin{array}{c} \xrightarrow{\quad} \\ \xleftarrow{\quad} \end{array} \mathrm{QCoh}(U \times_X U) \begin{array}{c} \xrightarrow{\quad} \\ \xleftarrow{\quad} \end{array} \cdots$$

is a limit diagram in symmetric monoidal presentable stable ∞ -categories.

Proof.

- (1) **Restriction to the Čech nerve.** Restriction of an object of $\mathrm{QCoh}(X)$ to the atlas and all its iterated intersections produces an object of the totalization and therefore the comparison functor

$$\mathrm{QCoh}(X) \longrightarrow \mathrm{Tot}_{[n] \in \Delta} \mathrm{QCoh}(U_n).$$

- (2) **Pull the atlas back to an affine point.** Let $v : V = \mathrm{Spec}(A) \rightarrow X$ be an affine point. The pullback $U_V := U \times_X V \rightarrow V$ is an effective epimorphism by open subschemes.
- (3) **Refine by affine opens.** Refine its components by affine open atlases.
- (4) **Recognize an affine Zariski cover.** The resulting coproduct of affine opens is still an effective epimorphism over V , so 1.8.11 identifies the resulting affine family as an affine spectral Zariski covering family of V .
- (5) **Choose a finite principal refinement.** Hence 1.8.7 supplies a finite principal refinement

$$\coprod_{i=1}^r V_i \longrightarrow V, \quad V_i = D_V^{\mathrm{sp}}(f_i), \quad (f_1, \dots, f_r) = (1).$$

- (6) **Descend to an A -module.** Restrict the given object of the Čech totalization to this finite affine cover. By 1.8.5, it descends to an A -module M_V .

- (7) **Independence of principal refinement.** Suppose $\{D_V^{\text{sp}}(f_i)\}_i$ and $\{D_V^{\text{sp}}(g_j)\}_j$ are two finite principal refinements. Their pairwise intersections are

$$D_V^{\text{sp}}(f_i) \times_V D_V^{\text{sp}}(g_j) \simeq D_V^{\text{sp}}(f_i g_j).$$

Since $(f_i)_i = (1)$ and $(g_j)_j = (1)$, the products $\{f_i g_j\}_{i,j}$ also generate the unit ideal. Thus these intersections form a common finite principal refinement. Full faithfulness in affine module descent identifies the two descended modules by a canonical equivalence, and the space of such identifications is contractible.

- (8) **Compatibility under affine base change.** For a morphism $V' = \text{Spec}(A') \rightarrow V = \text{Spec}(A)$ over X , base change of the chosen principal refinement is the finite principal family

$$D_{V'}^{\text{sp}}(f_i) \longrightarrow V'.$$

The images of the f_i still generate the unit ideal, so this is again a cover. Both

$$M_V \otimes_A A' \quad \text{and} \quad M_{V'}$$

correspond to the same object of its affine descent totalization. Hence affine descent yields a canonical equivalence

$$M_V \otimes_A A' \simeq M_{V'}.$$

Full faithfulness of affine descent identifies these transition equivalences and all their higher homotopies with the corresponding morphisms in the totalization.

- (9) **Assemble the compatible affine-point family.** The compatible family $\{M_V\}$ defines an object of the affine-point limit

$$\text{QCoh}(X) = \varprojlim_{(\text{Spec } A \rightarrow X) \in \text{AffPt}(X)^{\text{op}}} \text{Mod}_A.$$

- (10) **Identify the inverse equivalence.** Affine descent shows that this construction and restriction to the Čech nerve are inverse.
- (11) **Preserve the symmetric monoidal structure.** All restriction functors are strong symmetric monoidal and the defining limits are formed in $\text{CAlg}(\text{Pr}_{\text{st}}^L)$, so the equivalence is symmetric monoidal.

□

Corollary 1.10.1 (Zariski locality of quasi-coherent modules). *A quasi-coherent module on a spectral scheme is equivalently a family of modules on the members of an affine Zariski atlas, together with coherent descent equivalences on all iterated intersections.*

Proof.

- (1) **Unfold the descent totalization.** Unfold the totalization in 1.10.1.

□

Lemma 1.10.1 (Computation on the affine-open basis). *Let X be a spectral scheme, and let*

$$\text{AffOp}(X)$$

denote the ∞ -category of affine open spectral subschemes of X . Restriction induces a symmetric monoidal equivalence

$$\text{QCoh}(X) \xrightarrow{\simeq} \varprojlim_{V \in \text{AffOp}(X)^{\text{op}}} \text{QCoh}(V).$$

Proof.

- (1) **Restriction to affine opens.** Restriction gives a symmetric monoidal functor

$$\rho : \mathrm{QCoh}(X) \longrightarrow \varprojlim_{V \in \mathrm{AffOp}(X)^{\mathrm{op}}} \mathrm{QCoh}(V).$$

- (2) **Choose an affine atlas and an affine cover of the intersection.** Let $\{M_V\}_V$ be an object of the limit on the right and choose an affine open atlas $\{U_i \rightarrow X\}$. The intersection

$$U_{ij} := U_i \times_X U_j$$

need not be affine. Choose an affine open cover $\{W_\alpha \rightarrow U_{ij}\}$.

- (3) **Compare the prescribed modules on the affine cover.** Every W_α is also an affine open of X , so the compatibility morphisms of the limit give

$$M_{U_i}|_{W_\alpha} \simeq M_{W_\alpha} \simeq M_{U_j}|_{W_\alpha}.$$

- (4) **Check compatibility on refinements.** On further affine refinements these equivalences agree because they are restrictions of one object of the categorical limit.

- (5) **Glue the overlap equivalence.** By 1.10.1 applied to U_{ij} , they glue to a canonical equivalence

$$M_{U_i}|_{U_{ij}} \xrightarrow{\simeq} M_{U_j}|_{U_{ij}}.$$

- (6) **Higher compatibility.** On triple and higher intersections, every required equality between composites may be checked after an affine open cover. There it is exactly the corresponding equality among the transition maps of the object $\{M_V\}_V$ in the limit. Full faithfulness of affine Zariski descent therefore supplies the global equality and all higher coherences. Thus the M_{U_i} define an object of

$$\mathrm{Tot}_{[n] \in \Delta} \mathrm{QCoh}(U_n).$$

- (7) **Glue the descent object.** By 1.10.1, the resulting descent object glues to $M \in \mathrm{QCoh}(X)$.

- (8) **Compare it with a prescribed affine-open value.** Let $V \subseteq X$ be any affine open. The opens $V \times_X U_i$ cover V ; after refining them by affine opens, both $M|_V$ and the prescribed object M_V restrict to the same members of the original compatible family.

- (9) **Apply affine descent on the affine open.** Affine descent on V gives a canonical equivalence

$$M|_V \simeq M_V.$$

- (10) **Conclude that the constructions are inverse.** Hence this construction is inverse to ρ .

- (11) **Monoidality.** All restriction and gluing functors used above arise from limits in $\mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L)$. The equivalence is therefore symmetric monoidal.

□

1.10.2 Global flatness and local detection

Definition 1.10.1 (Flat morphisms and flat quasi-coherent modules). A morphism

$$f: Y \longrightarrow X$$

of spectral schemes is **flat** if, for every affine spectral scheme

$$U = \operatorname{Spec}(A) \longrightarrow X$$

and every affine open

$$V = \operatorname{Spec}(B) \subseteq Y \times_X U,$$

the induced morphism $A \rightarrow B$ is flat.

A quasi-coherent module $\mathcal{F} \in \operatorname{QCoh}(X)$ is **flat** if, for every affine open $U = \operatorname{Spec}(A) \subseteq X$, the corresponding A -module $\mathcal{F}|_U$ is flat.

Proposition 1.10.1 (Zariski locality of flatness). *Flatness of morphisms of spectral schemes is local on the source and the target for the spectral Zariski topology. Consequently, in 1.10.1 it is enough to verify flatness on an affine Zariski atlas of the target and an affine Zariski atlas of each pullback to the source.*

Proof.

- (1) **Source locality in the affine case.** Let $A \rightarrow B$ be a map of connective commutative ring spectra and let $\{\operatorname{Spec}(B_i) \rightarrow \operatorname{Spec}(B)\}$ be a Zariski cover such that every composite $A \rightarrow B_i$ is flat. For $N \in (\operatorname{Mod}_A)_{\leq 0}$, put

$$M := N \otimes_A B.$$

Then

$$M \otimes_B B_i \simeq N \otimes_A B_i \in (\operatorname{Mod}_{B_i})_{\leq 0}.$$

- (2) **Detect coconnectivity after the cover.** The maps $B \rightarrow B_i$ are open and hence flat, so extension of scalars is t -exact. Therefore

$$\tau_{\geq 1}^B(M) \otimes_B B_i \simeq \tau_{\geq 1}^{B_i}(M \otimes_B B_i) \simeq 0.$$

Joint conservativity from 1.8.8 gives $\tau_{\geq 1}^B(M) \simeq 0$. Thus $- \otimes_A B$ preserves coconnective modules; right t -exactness is automatic, so $A \rightarrow B$ is flat.

- (3) **Establish target locality.** Locality on the target follows from base-change stability of flatness and the source-locality just proved.
- (4) **Globalize by affine atlases.** Refining source and target by affine open atlases yields the stated global locality criterion.

□

Proposition 1.10.2 (Zariski locality of flat modules). *Let A be a connective commutative ring spectrum, let*

$$\{\operatorname{Spec}(A_i) \rightarrow \operatorname{Spec}(A)\}_{i \in I}$$

be an affine spectral Zariski covering family, and let

$$M \in \operatorname{Mod}_A.$$

Then M is flat over A if and only if

$$M \otimes_A A_i$$

is flat over A_i for every i .

Proof.

- (1) **Base change of a flat module.** If M is flat over A , then each $M \otimes_A A_i$ is flat over A_i by 1.6.1.
- (2) **Recover connectivity from the cover.** Conversely, put $M_i := M \otimes_A A_i$ and assume every M_i is flat. The maps $A \rightarrow A_i$ are open and hence flat, so extension of scalars is t -exact. Since every M_i is connective,

$$\tau_{\leq -1}^A(M) \otimes_A A_i \simeq \tau_{\leq -1}^{A_i}(M_i) \simeq 0.$$

Joint conservativity gives $\tau_{\leq -1}^A(M) \simeq 0$, so M is connective.

- (3) **Recover left t -exactness.** Let $N \in (\text{Mod}_A)_{\leq 0}$. For every i ,

$$\begin{aligned} (N \otimes_A M) \otimes_A A_i &\simeq (N \otimes_A A_i) \otimes_{A_i} (M \otimes_A A_i) \\ &\simeq (N \otimes_A A_i) \otimes_{A_i} M_i. \end{aligned}$$

Flatness of $A \rightarrow A_i$ makes $N \otimes_A A_i$ coconnective, and flatness of M_i keeps the displayed tensor product coconnective. Hence

$$\tau_{\geq 1}^A(N \otimes_A M) \otimes_A A_i \simeq 0.$$

Joint conservativity gives $\tau_{\geq 1}^A(N \otimes_A M) \simeq 0$.

- (4) **Conclude flatness.** Tensoring with the connective module M is right t -exact, and the preceding step proves left t -exactness. Therefore M is flat.

□

1.10.3 The global Postnikov structure and affine direct image

Theorem 1.10.2 (Canonical t -structure on global quasi-coherent modules). *Let X be a spectral scheme. The full subcategories*

$$\text{QCoh}(X)_{\geq 0} \quad \text{and} \quad \text{QCoh}(X)_{\leq 0}$$

of quasi-coherent modules which are respectively connective and coconnective on every affine open define an accessible, left complete, and right complete t -structure on $\text{QCoh}(X)$. Pullback along every morphism of spectral schemes is right t -exact, and pullback along a flat morphism is t -exact.

Proof.

- (1) **Limit presentation.** For each affine open write $V = \text{Spec}(A_V)$. By 1.10.1,

$$\text{QCoh}(X) \simeq \varprojlim_{V = \text{Spec}(A_V) \in \text{AffOp}(X)^{\text{op}}} \text{Mod}_{A_V}.$$

Every transition functor is pullback along an open immersion and is t -exact because open immersions are flat.

- (2) **Construct global truncations componentwise.** For a compatible family $\mathcal{F} = \{\mathcal{F}_V\}_V$, apply affine Postnikov truncation componentwise.
- (3) **Assemble the truncation triangle.** The t -exact transition functors preserve the compatibility maps, so the affine truncation triangles assemble to

$$\tau_{\geq 0}^X \mathcal{F} \longrightarrow \mathcal{F} \longrightarrow \tau_{\leq -1}^X \mathcal{F}.$$

- (4) **Establish global orthogonality.** Mapping spectra in the categorical limit are limits of the affine mapping spectra, so affine orthogonality implies global orthogonality.
- (5) **Accessibility.** Choose a regular cardinal κ for which the small affine-open diagram, its transition functors, and the affine truncation functors are κ -accessible. The componentwise truncation functors on the limit are then κ -accessible.
- (6) **Left completeness.** For each affine open V , consider the tower

$$\cdots \longrightarrow (\mathrm{Mod}_{A_V})_{\leq n} \xrightarrow{\tau_{\leq n-1}^{A_V}} (\mathrm{Mod}_{A_V})_{\leq n-1} \longrightarrow \cdots$$

Affine left completeness gives an equivalence in Cat_∞

$$\mathrm{Mod}_{A_V} \simeq \varprojlim_n (\mathrm{Mod}_{A_V})_{\leq n}.$$

Naturality in V and commutation of limits yield

$$\begin{aligned} \mathrm{QCoh}(X) &\simeq \varprojlim_V \varprojlim_n (\mathrm{Mod}_{A_V})_{\leq n} \\ &\simeq \varprojlim_n \mathrm{QCoh}(X)_{\leq n}. \end{aligned}$$

- (7) **Right completeness.** The canonical morphism

$$\varinjlim_n \tau_{\geq -n}^X \mathcal{F} \longrightarrow \mathcal{F}$$

becomes an equivalence after restriction to every affine open by affine right completeness. Affine-open restrictions jointly detect equivalences, so it is an equivalence globally.

- (8) **Choose an affine open in the source and an affine atlas in the target.** Let $f : Y \rightarrow X$ and let $\mathcal{F} \in \mathrm{QCoh}(X)_{\geq 0}$. Choose an affine open $V = \mathrm{Spec}(B) \subseteq Y$ and an affine open atlas $\{U_i = \mathrm{Spec}(A_i) \rightarrow X\}$. The opens

$$V_i := V \times_X U_i$$

cover V .

- (9) **Refine the pullback cover by affine opens.** Refine each V_i by affine opens $W_{ij} = \mathrm{Spec}(B_{ij})$.
- (10) **Check local connectivity after extension of scalars.** On W_{ij} , the restriction of $f^* \mathcal{F}$ is extension of scalars along $A_i \rightarrow B_{ij}$, hence is connective by 1.5.9.
- (11) **Recognize the refinement as a Zariski cover.** Let $M \in \mathrm{Mod}_B$ represent $(f^* \mathcal{F})|_V$. The coproduct of the affine opens $W_{ij} \rightarrow V$ is an effective epimorphism, so 1.8.11 makes them an affine spectral Zariski covering family of V . Their structural maps $B \rightarrow B_{ij}$ are flat. Therefore

$$\tau_{\leq -1}^B(M) \otimes_B B_{ij} \simeq \tau_{\leq -1}^{B_{ij}}(M \otimes_B B_{ij}) \simeq 0.$$

- (12) **Detect global connectivity by joint conservativity.** Joint conservativity gives $\tau_{\leq -1}^B(M) \simeq 0$. Thus $f^*\mathcal{F}$ is connective on every affine open of Y , proving right t -exactness.
- (13) **t -exactness for flat pullback.** If f is flat, every affine map $A_i \rightarrow B_{ij}$ above is flat, so affine extension of scalars is t -exact. For $\mathcal{F} \in \mathrm{QCoh}(X)_{\leq 0}$, the same argument gives

$$\tau_{\geq 1}^B(M) \otimes_B B_{ij} \simeq \tau_{\geq 1}^{B_{ij}}(M \otimes_B B_{ij}) \simeq 0,$$

and joint conservativity gives $\tau_{\geq 1}^B(M) \simeq 0$. Hence flat pullback preserves both halves of the global Postnikov t -structure.

□

Definition 1.10.2 (Affine morphism of spectral schemes). A morphism $p : Y \rightarrow X$ of spectral schemes is **affine** if, for every affine open $U \rightarrow X$, the pullback

$$Y \times_X U$$

is an affine spectral scheme.

Proposition 1.10.3 (Affine direct image is restriction of scalars). *Let*

$$p : Y \longrightarrow X$$

be an affine morphism of spectral schemes. For every affine open $U = \mathrm{Spec}(A) \subseteq X$, write

$$Y_U := Y \times_X U \simeq \mathrm{Spec}(B).$$

Then for every $\mathcal{F} \in \mathrm{QCoh}(Y)$ there is a canonical equivalence of A -modules

$$(p_*\mathcal{F})|_U \simeq \mathrm{Res}_A^B(\mathcal{F}|_{Y_U}).$$

These equivalences are compatible with restriction to smaller affine opens.

Proof.

- (1) **Compatibility under restriction.** Let $U' = \mathrm{Spec}(A') \rightarrow U = \mathrm{Spec}(A)$ be an affine open. Then

$$Y_{U'} \simeq \mathrm{Spec}(B'), \quad B' := B \otimes_A A'.$$

For every B -module M , associativity of relative tensor products gives a canonical equivalence of A' -modules

$$\begin{aligned} \mathrm{Res}_A^B(M) \otimes_A A' &\simeq M \otimes_A A' \\ &\simeq M \otimes_B (B \otimes_A A') \\ &\simeq \mathrm{Res}_{A'}^{B'}(M \otimes_B B'). \end{aligned}$$

The associativity coherence supplies compatibility for chains of affine open restrictions.

- (2) **Construct the candidate direct image.** The preceding compatibilities show that the modules

$$U = \mathrm{Spec}(A) \longmapsto \mathrm{Res}_A^B(\mathcal{F}|_{Y_U})$$

define an object $G \in \mathrm{QCoh}(X)$ by 1.10.1.

- (3) **Apply the affine adjunction chartwise.** Let $\mathcal{E} \in \mathrm{QCoh}(X)$. Choose an affine open atlas of X and affine refinements of all iterated intersections. On every affine chart the affine extension–restriction adjunction gives

$$\mathrm{Map}_{\mathrm{Mod}_A}(\mathcal{E}|_U, \mathrm{Res}_A^B(\mathcal{F}|_{Y_U})) \simeq \mathrm{Map}_{\mathrm{Mod}_B}(\mathcal{E}|_U \otimes_A B, \mathcal{F}|_{Y_U}).$$

- (4) **Check compatibility with restriction.** These equivalences commute with restriction by Step (1).
 (5) **Pass to descent totalizations.** Taking the descent totalizations of 1.10.1 yields

$$\mathrm{Map}_{\mathrm{QCoh}(X)}(\mathcal{E}, G) \simeq \mathrm{Map}_{\mathrm{QCoh}(Y)}(p^*\mathcal{E}, \mathcal{F}).$$

- (6) **Identify the right adjoint.** Hence G is right adjoint to p^* . Uniqueness of right adjoints identifies $G \simeq p_*\mathcal{F}$, proving the formula.

$$\begin{array}{ccc} \mathrm{Mod}_B & \xrightarrow{-\otimes_B B'} & \mathrm{Mod}_{B'} \\ \mathrm{Res}_A^B \downarrow & \simeq & \downarrow \mathrm{Res}_{A'}^{B'} \\ \mathrm{Mod}_A & \xrightarrow{-\otimes_A A'} & \mathrm{Mod}_{A'}. \end{array}$$

□

1.10.4 Quasi-coherent commutative algebras

Definition 1.10.3 (Quasi-coherent commutative algebra). For a spectral scheme X , define

$$\mathrm{QCAlg}(X) := \mathrm{CAlg}(\mathrm{QCoh}(X)).$$

Let

$$\mathrm{oblv}_X^{\mathrm{alg}}: \mathrm{QCAlg}(X) \longrightarrow \mathrm{QCoh}(X)$$

denote the forgetful functor. The full subcategory of algebra objects whose underlying quasi-coherent modules are connective is denoted

$$\mathrm{QCAlg}(X)^{\mathrm{cn}} := \left\{ \mathcal{A} \in \mathrm{QCAlg}(X) \mid \mathrm{oblv}_X^{\mathrm{alg}}(\mathcal{A}) \in \mathrm{QCoh}(X)_{\geq 0} \right\}.$$

Remark 1.10.1 (Connectivity of quasi-coherent algebras). Connectivity of a quasi-coherent commutative algebra is measured in the underlying stable coefficient category. Thus the superscript cn in $\mathrm{QCAlg}(X)^{\mathrm{cn}}$ means that the underlying object of $\mathrm{QCoh}(X)$ is connective. It does not assert a t -structure on the generally nonstable ∞ -category $\mathrm{QCAlg}(X)$; compare 1.3.3.

Proposition 1.10.4 (Pullback of quasi-coherent algebras). *Every morphism $f: X \rightarrow Y$ of spectral schemes induces a functor*

$$f^*: \mathrm{QCAlg}(Y) \longrightarrow \mathrm{QCAlg}(X).$$

It preserves connective algebra objects, and for composable morphisms there are canonical coherent equivalences

$$(g \circ f)^* \simeq f^* g^*.$$

Proof.

- (1) **Pass to commutative algebra objects.** The quasi-coherent pullback functor is strong symmetric monoidal by 1.7.2, hence carries commutative algebra objects to commutative algebra objects.
- (2) **Preserve connectivity.** Right t -exactness gives preservation of connectivity.
- (3) **Inherit coherence.** Coherence is inherited from the functoriality of quasi-coherent pullback.

□

Proposition 1.10.5 (Zariski descent for quasi-coherent algebras). *Let X be a spectral scheme and let $U \rightarrow X$ be an affine Zariski atlas with Čech nerve $U_\bullet$. Then restriction induces an equivalence*

$$\mathrm{QCAlg}(X) \xrightarrow{\simeq} \mathrm{Tot}_{[n] \in \Delta} \mathrm{QCAlg}(U_n).$$

It restricts to an equivalence on connective algebra objects:

$$\mathrm{QCAlg}(X)^{\mathrm{cn}} \xrightarrow{\simeq} \mathrm{Tot}_{[n] \in \Delta} \mathrm{QCAlg}(U_n)^{\mathrm{cn}}.$$

$$\begin{array}{ccc} \mathrm{QCoh}(X) & \xrightarrow{\simeq} & \mathrm{Tot}_{[n] \in \Delta} \mathrm{QCoh}(U_n) \\ \mathrm{CAlg} \downarrow & & \downarrow \mathrm{CAlg} \\ \mathrm{QCAlg}(X) & \xrightarrow{\simeq} & \mathrm{CAlg}\left(\mathrm{Tot}_{[n] \in \Delta} \mathrm{QCoh}(U_n)\right) \xrightarrow{\simeq} \mathrm{Tot}_{[n] \in \Delta} \mathrm{QCAlg}(U_n). \end{array}$$

Proof.

- (1) **Apply commutative algebra objects to QCoh descent.** By 1.10.1,

$$\mathrm{QCoh}(X) \xrightarrow{\simeq} \mathrm{Tot}_{[n] \in \Delta} \mathrm{QCoh}(U_n)$$

is an equivalence of symmetric monoidal ∞ -categories. The functor assigning commutative algebra objects preserves limits of symmetric monoidal ∞ -categories; see [34]. Hence

$$\mathrm{CAlg}(\mathrm{QCoh}(X)) \xrightarrow{\simeq} \mathrm{Tot}_{[n] \in \Delta} \mathrm{CAlg}(\mathrm{QCoh}(U_n)).$$

This is the first assertion.

- (2) **Restrict to connective objects.** Connectivity of a quasi-coherent module is detected on affine opens by 1.10.2. Therefore an algebra object is connective exactly when all its restrictions to the affine atlas are connective, and the equivalence restricts as asserted.

□

Proposition 1.10.6 (Lax monoidality of direct image). *For every morphism $f : X \rightarrow Y$ of spectral schemes, the right adjoint*

$$f_* : \mathrm{QCoh}(X) \longrightarrow \mathrm{QCoh}(Y)$$

carries a canonical lax symmetric monoidal structure. Consequently there is an adjunction on commutative algebra objects

$$\mathrm{QCAlg}(Y) \begin{array}{c} \xrightarrow{f^*} \\ \xleftarrow{f_*} \end{array} \mathrm{QCAlg}(X).$$

If $f = p$ is affine and $U = \mathrm{Spec}(A) \subseteq Y$ has pullback $X \times_Y U \simeq \mathrm{Spec}(B)$, then the commutative algebra $p_*\mathcal{O}_X$ restricts to B as a commutative A -algebra.

Proof.

- (1) **Construct the lax unit.** The morphism

$$\mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X$$

is the adjoint of the canonical equivalence

$$f^*\mathcal{O}_Y \simeq \mathcal{O}_X.$$

- (2) **Construct the lax tensor product.** For $\mathcal{F}, \mathcal{G} \in \mathrm{QCoh}(X)$, define

$$f_*\mathcal{F} \otimes f_*\mathcal{G} \longrightarrow f_*(\mathcal{F} \otimes \mathcal{G})$$

to be adjoint to the composite

$$\begin{aligned} f^*(f_*\mathcal{F} \otimes f_*\mathcal{G}) &\simeq f^*f_*\mathcal{F} \otimes f^*f_*\mathcal{G} \\ &\longrightarrow \mathcal{F} \otimes \mathcal{G}, \end{aligned}$$

where the final morphism is the tensor product of the two adjunction counits.

- (3) **Obtain coherence by adjoint mates.** Associativity, commutativity, and unitality are the mates under the adjunction of the corresponding identities for the strong symmetric monoidal functor f^* .
- (4) **Identify the canonical lax monoidal structure.** This is the canonical lax symmetric monoidal structure on the right adjoint of a strong symmetric monoidal left adjoint; see [34]. Thus no independent coherence datum is chosen.
- (5) **Lift to commutative algebra objects.** A lax symmetric monoidal functor carries commutative algebra objects to commutative algebra objects, and the original adjunction lifts to the displayed adjunction on QCAlg .
- (6) **Affine identification on the structure sheaf.** Suppose $f = p$ is affine and, over $U = \mathrm{Spec}(A) \subseteq Y$, write $X \times_Y U \simeq \mathrm{Spec}(B)$. By 1.10.3, the underlying A -module of $(p_*\mathcal{O}_X)|_U$ is B . Under the affine extension–restriction adjunction, the lax multiplication constructed in Step (2) is the ordinary multiplication

$$B \otimes_A B \longrightarrow B,$$

and the lax unit is $A \rightarrow B$. Hence the induced commutative A -algebra is canonically the original algebra B .

□

The global coefficient system can now be computed from any affine atlas and carries the structures needed for geometric re-entry: a global Postnikov structure, local flatness, affine direct image, and quasi-coherent commutative algebras satisfying descent. Relative spectrum can therefore be introduced as the construction that represents connective algebra in a fibre by affine geometry over the base.

1.11 Affine comprehension

The indexed geometry now has all three ingredients needed for representability: a geometric base of spectral schemes, stable coefficient fibres varying by substitution, and Zariski descent. What remains is the return from algebra in a coefficient fibre to geometry over its context. Relative spectrum provides that return. For a spectral scheme X , connective objects of $\mathrm{QCAlg}(X)$ determine affine extensions of X , and their algebra terms are represented after arbitrary substitution.

Remark 1.11.1 (Relative algebraic geometry in context). Affine comprehension repeats the absolute algebra-to-geometry construction inside each context. The absolute passage

$$\mathrm{Sp}_{\geq 0}^{\otimes} \longrightarrow \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}} \longrightarrow \mathrm{Aff}_{\mathrm{Sp}}$$

becomes fibrewise

$$\mathrm{QCoh}(X)^{\otimes} \longrightarrow \mathrm{QCAlg}(X)^{\mathrm{cn}} \longrightarrow \mathrm{QCAlg}(X)^{\mathrm{cn}, \mathrm{op}} \simeq (\mathrm{Sch}_{/X}^{\mathrm{sp}})_{\mathrm{aff}}.$$

The relative spectrum theorem identifies the final term with the actual affine slice geometry over X . The displayed parallel is therefore both an analogy and a construction.

Theorem 1.11.1 (Relative spectrum construction). *Let X be a spectral scheme. There is a functor*

$$\mathrm{Spec}_X : \mathrm{QCAlg}(X)^{\mathrm{cn}, \mathrm{op}} \longrightarrow \mathrm{Sch}_{/X}^{\mathrm{sp}}$$

characterized by the following properties.

(i) *If $X = \mathrm{Spec}(A)$ and $\mathcal{A}$ corresponds to a connective commutative A -algebra B , then*

$$\mathrm{Spec}_X(\mathcal{A}) \simeq \mathrm{Spec}(B).$$

(ii) *For every morphism $f : Y \rightarrow X$, there is a natural equivalence*

$$\mathrm{Map}_X(Y, \mathrm{Spec}_X(\mathcal{A})) \simeq \mathrm{Map}_{\mathrm{QCAlg}(Y)}(f^* \mathcal{A}, \mathcal{O}_Y).$$

Proof.

(1) **Define pullback on algebra objects.** Let T be an affine spectral scheme. Pullback defines a functor

$$X(T) \longrightarrow \mathrm{QCAlg}(T)^{\simeq}, \quad f \longmapsto f^* \mathcal{A}.$$

(2) **Define the contravariant mapping-space functor.** The functor

$$\mathrm{QCAlg}(T)^{\simeq, \mathrm{op}} \longrightarrow \mathcal{S}, \quad \mathcal{B} \longmapsto \mathrm{Map}_{\mathrm{QCAlg}(T)}(\mathcal{B}, \mathcal{O}_T),$$

is contravariant in the algebra variable.

(3) **Use inversion to obtain the fibre functor.** Since $X(T)$ is an ∞ -groupoid, every path $e : f \rightarrow f'$ induces an equivalence

$$e^* : f^* \mathcal{A} \xrightarrow{\simeq} f'^* \mathcal{A},$$

and transport from the fibre over f to the fibre over f' is precomposition with $(e^*)^{-1}$. Equivalently, compose the contravariant mapping-space functor with the inversion equivalence $X(T) \simeq X(T)^{\mathrm{op}}$. This gives a functor

$$\Psi_{\mathcal{A}, T} : X(T) \longrightarrow \mathcal{S}, \quad f \longmapsto \mathrm{Map}_{\mathrm{QCAlg}(T)}(f^* \mathcal{A}, \mathcal{O}_T).$$

- (4) **Unstraighten the fibre functor.** Unstraighten $\Psi_{\mathcal{A},T}$. Let

$$p_T : F_{\mathcal{A}}(T) \longrightarrow X(T)$$

be the left fibration classified by this functor.

- (5) **Identify the total space as a colimit.** Its total space is canonically the colimit

$$F_{\mathcal{A}}(T) \simeq \operatorname{colim}_{f \in X(T)} \Psi_{\mathcal{A},T}(f).$$

- (6) **Describe its objects and coherence.** Thus an object of the total space is precisely the Grothendieck-construction object represented by a pair

$$(f, \alpha), \quad f : T \rightarrow X, \quad \alpha : f^* \mathcal{A} \rightarrow \mathcal{O}_T,$$

with all path and higher-coherence information supplied by unstraightening rather than imposed separately.

- (7) **Define pullback on pairs.** For $g : T' \rightarrow T$, pullback sends

$$(f, \alpha) \longmapsto (f \circ g, g^* \alpha).$$

- (8) **Use the coherent pullback equivalences.** This uses the coherent equivalences

$$g^* f^* \mathcal{A} \simeq (f \circ g)^* \mathcal{A}, \quad g^* \mathcal{O}_T \simeq \mathcal{O}_{T'}.$$

- (9) **Assemble a prestack over X .** Naturality of unstraightening therefore assembles the spaces $F_{\mathcal{A}}(T)$ into a spectral prestack

$$F_{\mathcal{A}} \longrightarrow X.$$

- (10) **Establish contravariant functoriality in the algebra.** A morphism $\mathcal{A} \rightarrow \mathcal{B}$ acts by precomposition on the mapping-space fibres, so $\mathcal{A} \mapsto F_{\mathcal{A}}$ is contravariantly functorial.
- (11) **Specify descent data on a finite affine cover.** Let $T_{\bullet} \rightarrow T$ be the Čech nerve of a finite affine spectral Zariski cover. An object of

$$\operatorname{Tot}_{[n] \in \Delta} F_{\mathcal{A}}(T_n)$$

consists of a coherent family of maps $f_n : T_n \rightarrow X$ and compatible algebra maps

$$\alpha_n : f_n^* \mathcal{A} \longrightarrow \mathcal{O}_{T_n}.$$

- (12) **Glue the maps to X .** Since X is a spectral Zariski sheaf, the f_n glue uniquely to $f : T \rightarrow X$.

- (13) **Apply algebra descent.** By 1.10.5,

$$\operatorname{QCAlg}(T) \simeq \operatorname{Tot}_{[n] \in \Delta} \operatorname{QCAlg}(T_n).$$

- (14) **Glue the algebra maps.** Mapping spaces in a limit of ∞ -categories are limits of the corresponding mapping spaces, so the α_n glue uniquely to

$$\alpha : f^* \mathcal{A} \longrightarrow \mathcal{O}_T.$$

(15) **Obtain descent for the total space.** Hence

$$F_{\mathcal{A}}(T) \xrightarrow{\cong} \operatorname{Tot}_{[n] \in \Delta} F_{\mathcal{A}}(T_n).$$

(16) **Conclude the prestack is a Zariski stack.** Finite principal covers generate the topology by 1.8.7; therefore $F_{\mathcal{A}}$ is a spectral Zariski stack.

(17) **Choose an affine base and algebra.** Let $U = \operatorname{Spec}(A) \rightarrow X$ be affine and let $\mathcal{A}|_U$ correspond to a connective commutative A -algebra B .

(18) **Compute the fibre on an affine test.** For $T = \operatorname{Spec}(C) \rightarrow U$, the fibre over the chosen map $T \rightarrow U \rightarrow X$ is

$$\begin{aligned} \operatorname{Map}_{\operatorname{QCAlg}(T)}((\mathcal{A}|_U)|_T, \mathcal{O}_T) &\simeq \operatorname{Map}_{\operatorname{CAlg}(\operatorname{Sp})_{C/}}(B \otimes_A C, C) \\ &\simeq \operatorname{Map}_{\operatorname{CAlg}(\operatorname{Sp})_{A/}}(B, C) \\ &\simeq \operatorname{Map}_U(T, \operatorname{Spec} B). \end{aligned}$$

(19) **Apply Yoneda over the affine base.** Naturality in $T \rightarrow U$ and Yoneda give a canonical equivalence

$$\begin{array}{ccc} F_{\mathcal{A}} \times_X U & \xrightarrow{\cong} & \operatorname{Spec}(B) \\ \downarrow & & \downarrow \\ U & \xlongequal{\quad} & U. \end{array}$$

(20) **Pull back an affine atlas.** Choose an affine spectral Zariski atlas $\coprod_i U_i \rightarrow X$. Pulling it back gives an effective epimorphism

$$\coprod_i (F_{\mathcal{A}} \times_X U_i) \longrightarrow F_{\mathcal{A}}.$$

(21) **Identify the pullback charts as affine opens.** Every source component is affine by Step (19), and its map to $F_{\mathcal{A}}$ is an open immersion by base change. Hence $F_{\mathcal{A}}$ is a spectral scheme over X .

(22) **Define the relative spectrum.** Define

$$\operatorname{Spec}_X(\mathcal{A}) := F_{\mathcal{A}}.$$

(23) **Define the two presheaves on open subschemes.** Fix $f : Y \rightarrow X$. On the Zariski site of open spectral subschemes of Y , define

$$\mathscr{P}(V) := \operatorname{Map}_X(V, \operatorname{Spec}_X \mathcal{A})$$

and

$$\mathscr{Q}(V) := \operatorname{Map}_{\operatorname{QCAlg}(V)}((f|_V)^* \mathcal{A}, \mathcal{O}_V).$$

(24) **Verify that $\mathscr{P}$ is a sheaf.** The functor $\mathscr{P}$ is a Zariski sheaf because $\operatorname{Spec}_X \mathcal{A}$ is a spectral Zariski stack.

(25) **Verify that $\mathcal{Q}$ is a sheaf.** To verify that $\mathcal{Q}$ is a Zariski sheaf, it is enough to check descent on affine open covers, since affine opens form a basis. For such a cover, 1.10.5 identifies the algebra category on V with the corresponding totalization, and mapping spaces in that totalization are the totalizations of the local mapping spaces. Hence $\mathcal{Q}$ satisfies Zariski descent.

(26) **Compare the two sheaves on affine opens.** If $V \subseteq Y$ is affine, Step (19) gives a natural equivalence

$$\mathcal{P}(V) \simeq \mathcal{Q}(V).$$

(27) **Extend the comparison from the affine basis.** Affine opens form a basis of the Zariski topology, so these affine equivalences determine an equivalence of sheaves $\mathcal{P} \simeq \mathcal{Q}$.

(28) **Evaluate on Y .** Evaluating on Y gives

$$\mathrm{Map}_X(Y, \mathrm{Spec}_X(\mathcal{A})) \simeq \mathrm{Map}_{\mathrm{QCAlg}(Y)}(f^*\mathcal{A}, \mathcal{O}_Y).$$

(29) **Record naturality.** Naturality in Y , f , and $\mathcal{A}$ is inherited from the construction.

(30) **Record uniqueness.** Yoneda gives uniqueness of the represented object.

□

Proposition 1.11.1 (Universal variable and term abstraction). *Let $\mathcal{A} \in \mathrm{QCAlg}(X)^{\mathrm{cn}}$ and write*

$$\pi_{\mathcal{A}} : X.\mathcal{A} := \mathrm{Spec}_X(\mathcal{A}) \longrightarrow X.$$

There is a canonical universal algebra term

$$\mathrm{var}_{\mathcal{A}} : \pi_{\mathcal{A}}^*\mathcal{A} \longrightarrow \mathcal{O}_{X.\mathcal{A}}.$$

For every $f : T \rightarrow X$, composition with the relative-spectrum universal property gives a natural equivalence between

(i) *algebra terms $a : f^*\mathcal{A} \rightarrow \mathcal{O}_T$, and*

(ii) *morphisms $\langle f, a \rangle : T \rightarrow X.\mathcal{A}$ over X ,*

characterized by

$$\langle f, a \rangle^* \mathrm{var}_{\mathcal{A}} \simeq a.$$

*Under substitution $g : T' \rightarrow T$, the term a is carried to g^*a , and there is a canonical coherent equivalence*

$$\langle f \circ g, g^*a \rangle \simeq \langle f, a \rangle \circ g.$$

Proof.

(1) **Construct the universal variable.** Apply the equivalence of 1.11.1 with $Y = X.\mathcal{A}$ and $f = \pi_{\mathcal{A}}$. The identity map

$$\mathrm{id}_{X.\mathcal{A}} \in \mathrm{Map}_X(X.\mathcal{A}, X.\mathcal{A})$$

corresponds to a canonical map

$$\pi_{\mathcal{A}}^*\mathcal{A} \longrightarrow \mathcal{O}_{X.\mathcal{A}},$$

which is $\mathrm{var}_{\mathcal{A}}$.

- (2) **Abstract a term.** For $a : f^* \mathcal{A} \rightarrow \mathcal{O}_T$, let $\langle f, a \rangle$ be its image under the inverse of the same representing equivalence. Naturality of the representing equivalence identifies pullback of $\mathbf{var}_{\mathcal{A}}$ with a , and also gives the displayed compatibility under $g : T' \rightarrow T$. All higher coherence is the naturality coherence of the relative-spectrum equivalence.

□

Remark 1.11.2 (Context extension and terms). The universal property of relative spectrum has the expected comprehension form. For $\mathcal{A} \in \mathbf{QCAlg}(X)^{\text{cn}}$, write

$$\pi_{\mathcal{A}} : \text{Spec}_X(\mathcal{A}) \longrightarrow X.$$

After substitution along $f : Y \rightarrow X$, maps into this context extension are precisely algebra terms:

$$Y \longrightarrow \text{Spec}_X(\mathcal{A}) \quad \text{over } X \quad \Longleftrightarrow \quad f^* \mathcal{A} \longrightarrow \mathcal{O}_Y.$$

For a term $a : f^* \mathcal{A} \rightarrow \mathcal{O}_T$, the corresponding term-abstraction map is represented by

$$\begin{array}{ccc} T & \xrightarrow{\langle f, a \rangle} & X.\mathcal{A} \\ & \searrow f & \swarrow \pi_{\mathcal{A}} \\ & X & \end{array}$$

together with $\langle f, a \rangle^* \mathbf{var}_{\mathcal{A}} \simeq a$. In particular, sections of $\pi_{\mathcal{A}}$ correspond to algebra maps $\mathcal{A} \rightarrow \mathcal{O}_X$, and the identity of the comprehending object corresponds to the universal variable of 1.11.1. The word **comprehension** refers to this representability statement; it does not identify $\mathbf{QCAlg}(X)$ with a literal Lawvere predicate fibre.

Proposition 1.11.2 (Base change for relative spectra). *Let $f : Y \rightarrow X$ be a morphism of spectral schemes and let $\mathcal{A} \in \mathbf{QCAlg}(X)^{\text{cn}}$. There is a canonical equivalence over Y*

$$\text{Spec}_X(\mathcal{A}) \times_X Y \simeq \text{Spec}_Y(f^* \mathcal{A}).$$

Equivalently, the canonical square

$$\begin{array}{ccc} \text{Spec}_Y(f^* \mathcal{A}) & \longrightarrow & \text{Spec}_X(\mathcal{A}) \\ \downarrow & & \downarrow \pi_{\mathcal{A}} \\ Y & \xrightarrow{f} & X \end{array}$$

is Cartesian.

Proof.

- (1) **Compare the two functors of points.** For every $T \rightarrow Y$,

$$\begin{aligned} \text{Map}_Y(T, \text{Spec}_X(\mathcal{A}) \times_X Y) &\simeq \text{Map}_X(T, \text{Spec}_X(\mathcal{A})) \\ &\simeq \text{Map}_{\mathbf{QCAlg}(T)}(\mathcal{A}|_T, \mathcal{O}_T) \\ &\simeq \text{Map}_Y(T, \text{Spec}_Y(f^* \mathcal{A})). \end{aligned}$$

The middle equivalence is the relative-spectrum universal property.

- (2) **Apply Yoneda.** The comparison is natural in T , so Yoneda gives the canonical equivalence over Y .

□

Theorem 1.11.2 (Affine comprehension and classification of affine morphisms). *For every spectral scheme X , relative spectrum induces an equivalence*

$$\mathrm{QCAlg}(X)^{\mathrm{cn}, \mathrm{op}} \xrightarrow{\simeq} (\mathrm{Sch}_{/X}^{\mathrm{sp}})_{\mathrm{aff}}$$

onto the full subcategory of affine morphisms over X . Under this equivalence, for every morphism $f : Y \rightarrow X$, pullback of affine morphisms corresponds to reindexing of quasi-coherent algebras:

$$\mathrm{Spec}_X(\mathcal{A}) \times_X Y \simeq \mathrm{Spec}_Y(f^* \mathcal{A}).$$

Thus affine comprehension is stable under substitution in the Cartesian square

$$\begin{array}{ccc} \mathrm{Spec}_Y(f^* \mathcal{A}) & \longrightarrow & \mathrm{Spec}_X(\mathcal{A}) \\ \downarrow & & \downarrow \\ Y & \xrightarrow{f} & X. \end{array}$$

$$\begin{array}{ccc} \mathrm{QCAlg}(X)^{\mathrm{cn}, \mathrm{op}} & \xrightarrow{\mathrm{Spec}_X} & (\mathrm{Sch}_{/X}^{\mathrm{sp}})_{\mathrm{aff}} \\ (f^*)^{\mathrm{op}} \downarrow & & \downarrow - \times_X Y \\ \mathrm{QCAlg}(Y)^{\mathrm{cn}, \mathrm{op}} & \xrightarrow{\mathrm{Spec}_Y} & (\mathrm{Sch}_{/Y}^{\mathrm{sp}})_{\mathrm{aff}}. \end{array}$$

Proof.

- (1) **Attach the direct-image algebra.** Let $p : Y \rightarrow X$ be affine. By 1.10.6, the structure sheaf determines a commutative algebra

$$\mathcal{A}_p := p_* \mathcal{O}_Y \in \mathrm{QCAlg}(X).$$

- (2) **Identify it on an affine open.** For an affine open $U = \mathrm{Spec}(A) \subseteq X$, write

$$Y_U \simeq \mathrm{Spec}(B).$$

By 1.10.6,

$$\mathcal{A}_p|_U \simeq B$$

as a commutative A -algebra.

- (3) **Detect connectivity locally.** Since B is connective, connectivity of $\mathcal{A}_p$ is detected on affine opens, and

$$\mathcal{A}_p \in \mathrm{QCAlg}(X)^{\mathrm{cn}}.$$

- (4) **Apply the counit on algebra objects.** The counit of the lifted algebra adjunction gives

$$p^* p_* \mathcal{O}_Y \longrightarrow \mathcal{O}_Y$$

in $\mathrm{QCAlg}(Y)$.

- (5) **Use the relative-spectrum universal property.** The universal property of relative spectrum therefore determines a morphism

$$\eta_p : Y \longrightarrow \mathrm{Spec}_X(\mathcal{A}_p)$$

over X .

- (6) **Identify the comparison after affine base change.** After base change to $U = \mathrm{Spec}(A) \subseteq X$, the preceding morphism is the map

$$\mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(B)$$

corresponding contravariantly to the identity of B .

- (7) **Conclude local equivalence.** Hence it is an equivalence after base change to an affine open atlas of X .
- (8) **Match the local inverses on overlaps.** The local inverses are uniquely characterized as inverses to the restrictions of η_p , so their restrictions to all iterated intersections agree.
- (9) **Glue the local inverses.** Mapping descent, 1.8.1, glues them to a global inverse. Thus:

$$Y \xrightarrow{\simeq} \mathrm{Spec}_X(p_*\mathcal{O}_Y).$$

- (10) **Conclude essential surjectivity.** This proves essential surjectivity.
- (11) **Identify the direct-image algebra locally.** Let

$$q_{\mathcal{B}} : \mathrm{Spec}_X(\mathcal{B}) \longrightarrow X$$

with $\mathcal{B} \in \mathrm{QCAlg}(X)^{\mathrm{cn}}$. On every affine open $U = \mathrm{Spec}(A)$, if $\mathcal{B}|_U$ corresponds to the connective A -algebra B , then 1.10.6 gives an equivalence of commutative A -algebras

$$(q_{\mathcal{B},*}\mathcal{O}_{\mathrm{Spec}_X \mathcal{B}})|_U \simeq B \simeq \mathcal{B}|_U.$$

- (12) **Check compatibility under restriction.** The equivalences are compatible under restriction.
- (13) **Glue the local algebra equivalences.** The affine-open basis theorem therefore gives a canonical algebra equivalence

$$q_{\mathcal{B},*}\mathcal{O}_{\mathrm{Spec}_X \mathcal{B}} \simeq \mathcal{B}.$$

- (14) **Compute the mapping-space equivalence.** For connective algebras $\mathcal{A}, \mathcal{B}$, the relative-spectrum universal property and the algebra adjunction give

$$\begin{aligned} \mathrm{Map}_X(\mathrm{Spec}_X \mathcal{B}, \mathrm{Spec}_X \mathcal{A}) &\simeq \mathrm{Map}_{\mathrm{QCAlg}(\mathrm{Spec}_X \mathcal{B})}(q_{\mathcal{B}}^*\mathcal{A}, \mathcal{O}_{\mathrm{Spec}_X \mathcal{B}}) \\ &\simeq \mathrm{Map}_{\mathrm{QCAlg}(X)}(\mathcal{A}, q_{\mathcal{B},*}\mathcal{O}_{\mathrm{Spec}_X \mathcal{B}}) \\ &\simeq \mathrm{Map}_{\mathrm{QCAlg}(X)}(\mathcal{A}, \mathcal{B}). \end{aligned}$$

- (15) **Conclude full faithfulness.** Thus relative spectrum is fully faithful.

- (16) **Combine with essential surjectivity.** Together with Step (10), this proves the asserted anti-equivalence, with explicit inverse

$$p : Y \rightarrow X \longmapsto p_* \mathcal{O}_Y.$$

□

Affine comprehension is the main structural return from coefficients to geometry. Connective quasi-coherent commutative algebras are exactly affine extensions of the base, and this identification commutes with substitution. The remaining sections exploit this representability rather than adding a new layer to the core construction.

1.12 Closed context geometry, complements, and gluing

Affine comprehension supplies geometric extensions of a context. To manipulate those extensions internally, we next record the complementary geometric operations used later: closed subcontexts, their open complements, and gluing along closed immersions. Each is constructed from the spectral geometry already developed, with π_0 used locally to detect the relevant closed condition but without introducing a global classical truncation.

1.12.1 Closed immersions

Definition 1.12.1 (Affine closed immersion). A morphism

$$\mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A)$$

of affine spectral schemes is a **closed immersion** if the induced ordinary ring homomorphism

$$\pi_0(A) \longrightarrow \pi_0(B)$$

is surjective.

Definition 1.12.2 (Closed immersion of spectral schemes). A morphism

$$i : Z \longrightarrow X$$

of spectral schemes is a **closed immersion** if, after base change along every affine open $U \rightarrow X$, the pullback

$$Z \times_X U \longrightarrow U$$

is an affine closed immersion.

Proposition 1.12.1 (Permanence of closed immersions). *Closed immersions of spectral schemes are stable under equivalence, composition, and base change.*

Proof.

- (1) **Reduce to the affine target.** The assertion is local on the target.
- (2) **Handle equivalences.** On affines, equivalences preserve surjectivity on π_0 .
- (3) **Handle composition.** A composite of surjective ring maps is surjective.

- (4) **Compute degree zero after base change.** Let $A \rightarrow B$ be surjective on π_0 and let $A \rightarrow A'$ be any morphism of connective commutative ring spectra. By 1.6.4,

$$\pi_0(B \otimes_A A') \simeq \pi_0(B) \otimes_{\pi_0(A)} \pi_0(A').$$

- (5) **Preserve surjectivity under base change.** The induced map $\pi_0(A') \rightarrow \pi_0(B \otimes_A A')$ is the base change of the surjection $\pi_0(A) \rightarrow \pi_0(B)$, hence is surjective.
- (6) **Globalize from the affine-local definition.** Thus affine closed immersions are stable under base change, and the global assertion follows from the affine-local definition. $\square$

1.12.2 Open complements

Lemma 1.12.1 (Vanishing criterion for spectral schemes). *Let A be a connective commutative ring spectrum. If $\pi_0(A) = 0$, then $A \simeq 0$ and $\mathrm{Spec}(A) \simeq \emptyset$. More generally, a spectral scheme is empty if an affine spectral Zariski atlas consists entirely of empty affine schemes.*

Proof.

- (1) **Affine vanishing.** Every $\pi_n(A)$ is naturally a module over $\pi_0(A)$. If the latter is the zero ring, all homotopy groups of A vanish. By 1.2.4, $A \simeq 0$, and its representable functor of points is initial.
- (2) **Global vanishing.** If an affine atlas of X has only empty members, its coproduct is the initial object. The atlas map is an effective epimorphism. In an ∞ -topos an effective epimorphism from the initial object has initial target, so $X \simeq \emptyset$. $\square$

Theorem 1.12.1 (Complement of a closed immersion). *Let*

$$i : Z \longrightarrow X$$

be a closed immersion of spectral schemes. There exists an open immersion

$$j : X \setminus Z \hookrightarrow X$$

characterized by the universal property that a morphism $T \rightarrow X$ factors through $X \setminus Z$ if and only if

$$T \times_X Z \simeq \emptyset.$$

Proof.

- (1) **Construct the complement over an affine context.** Suppose first that $X = \mathrm{Spec}(A)$ and $Z = \mathrm{Spec}(B)$, with $\pi_0(A) \rightarrow \pi_0(B)$ surjective. Put

$$I := \ker(\pi_0(A) \rightarrow \pi_0(B)).$$

In the spectral Zariski ∞ -topos, form the image

$$\coprod_{f \in I} D_A^{\mathrm{sp}}(f) \longrightarrow U \hookrightarrow X.$$

By 1.9.4, this image is the open union of the displayed principal spectral opens and is stable under arbitrary base change. Since X is a spectral scheme, U is a spectral scheme.

- (2) **The constructed open is disjoint from the closed part.** For $f \in I$, the affine intersection

$$D_A^{\text{sp}}(f) \times_X Z \simeq \text{Spec}(B \otimes_A A[f^{-1}])$$

has degree-zero ring

$$\pi_0(B) \otimes_{\pi_0(A)} \pi_0(A)[f^{-1}].$$

The image of f is zero in $\pi_0(B)$ and invertible in the localization, so this ring is zero. By 1.12.1, the intersection is empty. Since the principal opens cover U , mapping descent gives

$$U \times_X Z \simeq \emptyset.$$

- (3) **Translate emptiness to degree zero.** Let $T = \text{Spec}(C) \rightarrow X$ satisfy $T \times_X Z \simeq \emptyset$. Then

$$0 = \pi_0(C \otimes_A B) \simeq \pi_0(C) \otimes_{\pi_0(A)} \pi_0(B) \simeq \pi_0(C)/I\pi_0(C).$$

- (4) **Generate the unit ideal.** Thus the image of I generates the unit ideal in $\pi_0(C)$.
 (5) **Choose finitely many generators.** Choose finitely many $f_1, \dots, f_r \in I$ whose images generate the unit ideal.
 (6) **Construct the principal cover.** The principal opens

$$\{D_C^{\text{sp}}(f_j) \rightarrow T\}_{j=1}^r$$

form a Zariski cover,

- (7) **Produce the local factorizations.** Each composite to X factors through the corresponding $D_A^{\text{sp}}(f_j) \subseteq U$.
 (8) **Check agreement on intersections.** Since $U \rightarrow X$ is a monomorphism, these local factorizations agree on all intersections.
 (9) **Glue the local factorizations.** Mapping descent along the effective epimorphic cover glues them to a unique factorization $T \rightarrow U \rightarrow X$.
 (10) **Verify the converse.** The converse follows from Step (2).
 (11) **Choose an affine atlas.** For general X , choose an affine open atlas $\{X_a \rightarrow X\}$.
 (12) **Construct the local complements.** The pullbacks $Z_a := Z \times_X X_a$ are affine closed immersions, and the preceding affine steps construct their open complements $U_a \subseteq X_a$.
 (13) **Compare the local complements on overlaps.** The affine universal property is preserved by arbitrary base change, so the U_a agree canonically on all iterated intersections.
 (14) **Glue the global open subscheme.** They therefore glue to an open spectral subscheme $U \hookrightarrow X$.
 (15) **Verify the global universal property.** The same mapping-descent argument, now on an affine cover of an arbitrary $T \rightarrow X$, shows that

$$T \rightarrow U \iff T \times_X Z \simeq \emptyset.$$

(16) **Define the complement.** Define $X \setminus Z := U$.

(17) **Record uniqueness.** The universal property gives uniqueness.

□

Proposition 1.12.2 (Base change of complements). *Let $Z \rightarrow X$ be closed and let $Y \rightarrow X$ be any morphism. Then there is a canonical equivalence*

$$Y \times_X (X \setminus Z) \simeq Y \setminus (Y \times_X Z).$$

$$\begin{array}{ccc} Y \setminus (Y \times_X Z) & \longrightarrow & X \setminus Z \\ \downarrow & & \downarrow \\ Y & \longrightarrow & X. \end{array}$$

Proof.

- (1) **Translate maps into the pullback.** For every $T \rightarrow Y$, a map $T \rightarrow Y \times_X (X \setminus Z)$ over Y is equivalently a map $T \rightarrow Y$ whose composite with X factors through $X \setminus Z$.
- (2) **Apply the complement criterion.** By 1.12.1, this is equivalent to

$$T \times_X Z \simeq \emptyset.$$

- (3) **Compare the two intersections.** Since

$$T \times_Y (Y \times_X Z) \simeq T \times_X Z,$$

the same condition represents maps into the complement of $Y \times_X Z$ in Y .

- (4) **Apply Yoneda.** Yoneda gives the asserted equivalence.

□

1.12.3 Pushouts along closed immersions

Theorem 1.12.2 (Closed pushouts). *Suppose*

$$Z \longrightarrow X \quad \text{and} \quad Z \longrightarrow Y$$

are closed immersions of spectral schemes. Then the pushout

$$X \coprod_Z Y$$

exists in spectral schemes. The canonical maps

$$X \longrightarrow X \coprod_Z Y \quad \text{and} \quad Y \longrightarrow X \coprod_Z Y$$

are closed immersions. The construction is compatible with arbitrary base change. The resulting cocartesian square is displayed as

$$\begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow & \lrcorner & \downarrow \\ Y & \longrightarrow & X \amalg_Z Y \end{array}$$

Proof.

- (1) **Affine pullback algebra.** First suppose

$$X = \operatorname{Spec}(A), \quad Y = \operatorname{Spec}(B), \quad Z = \operatorname{Spec}(C),$$

with $A \rightarrow C \leftarrow B$ surjective on π_0 . Put

$$D := A \times_C B$$

in $\operatorname{CAlg}(\operatorname{Sp})$. Since the forgetful functor creates limits, the underlying spectra fit into

$$D \longrightarrow A \oplus B \xrightarrow{\operatorname{diff}} C.$$

- (2) **Compute $\pi_{-1}(D)$.** The long exact homotopy sequence gives

$$\pi_{-1}(D) \simeq \operatorname{coker}(\pi_0(A) \oplus \pi_0(B) \rightarrow \pi_0(C)) = 0,$$

because either structural map onto $\pi_0(C)$ is surjective.

- (3) **Deduce connectivity of D .** Lower negative homotopy groups vanish automatically, so D is connective.

- (4) **Compute $\pi_0(D)$.** The same sequence identifies

$$\pi_0(D) \simeq \pi_0(A) \times_{\pi_0(C)} \pi_0(B),$$

whose projections to $\pi_0(A)$ and $\pi_0(B)$ are surjective.

- (5) **Deduce closedness of the affine maps.** Hence

$$\begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y & \longrightarrow & \operatorname{Spec}(D) \end{array}$$

is a pushout square of affine spectral schemes with closed horizontal and vertical maps.

- (6) **Affine base change.** For any map $D \rightarrow D'$, exactness of extension of scalars sends the fibre sequence above to

$$D' \longrightarrow (A \otimes_D D') \oplus (B \otimes_D D') \longrightarrow C \otimes_D D'.$$

The forgetful functor from commutative ring spectra creates limits, so the induced algebra map

$$D' \longrightarrow (A \otimes_D D') \times_{C \otimes_D D'} (B \otimes_D D')$$

is an equivalence. Thus the affine pushout is preserved by arbitrary base change.

- (7) **Pass to the structured spectral model.** By 1.9.2, the functor-of-points spectral schemes used here identify with connective 0-localic structured spectral schemes.
- (8) **Match the affine closed-immersion condition.** On affines, the closed-immersion condition in the structured closed-gluing theorem is exactly surjectivity on π_0 , agreeing with 1.12.1; see [31, Theorem 4.4 and Remark 4.5].
- (9) **Apply the closed-gluing theorem.** Hence the spectral-scheme closed-gluing theorem applies and gives a pushout

$$\begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y & \longrightarrow & P \end{array}$$

with the two canonical maps into P closed; see [31, Theorem 6.1 and Remark 6.2].

- (10) **Identify the pushout.** Hence $P \simeq X \amalg_Z Y$.
- (11) **Global base change.** The closed-gluing base-change theorem identifies, for every $T \rightarrow P$, the square

$$\begin{array}{ccc} T \times_P Z & \longrightarrow & T \times_P X \\ \downarrow & & \downarrow \\ T \times_P Y & \longrightarrow & T \end{array}$$

as a pushout. Its proof is affine-local and applies to spectral schemes; see [31, Proposition 9.2]. This proves compatibility with arbitrary base change.

□

Remark 1.12.1 (Closed gluing carries no presentation data). The construction of the closed pushout uses several proof witnesses: the affine pullback algebra $A \times_C B$, affine charts used to verify connectivity, and the structured-space comparison used for the global gluing theorem. None is retained as structure on the result. The spectral scheme $X \amalg_Z Y$ is determined by its pushout universal property, with no chosen affine presentation, gluing presentation, or admissibility datum.

Closed immersions therefore admit the two operations that will remain active below: intrinsic open complements and base-change-stable pushouts. Together with affine comprehension, this gives a geometric calculus in which the linear constructions of the next section can be represented and manipulated.

1.13 Linear comprehension and distinguished affine contexts

Affine comprehension represents connective commutative algebra in a coefficient fibre. Applying it to free commutative algebras produces the linear geometric objects needed here. There are two related representation problems: relative linear spaces represent linear functionals, whereas section stacks represent sections. Finite locally free coefficients are the sector in which duality identifies the latter with affine geometry. The constructions below organize these two variances and then recover affine spaces, the brave new affine line, and the spectral multiplicative group as distinguished examples.

1.13.1 Affine and global local freeness

Definition 1.13.1 (Locally free module). Let A be connective. An A -module M is **locally free of rank n** if there exists a Zariski covering family

$$\{A \rightarrow A_i\}_{i \in I}$$

as defined in 1.8.2, such that

$$M \otimes_A A_i \simeq A_i^{\oplus n}$$

for every i .

Definition 1.13.2 (Locally free quasi-coherent module). Let X be a spectral scheme. A quasi-coherent module $\mathcal{E} \in \mathrm{QCoh}(X)$ is **locally free of rank n** if there is a Zariski open cover $\{U_i \rightarrow X\}$ such that

$$\mathcal{E}|_{U_i} \simeq \mathcal{O}_{U_i}^{\oplus n}$$

for every i .

A **vector bundle of rank n** on X means a locally free quasi-coherent module of rank n .

Proposition 1.13.1 (Locally free modules are finite projective). *Let*

$$\mathcal{E} \in \mathrm{QCoh}(X)$$

be a quasi-coherent module. Suppose there is a Zariski open cover $\{U_i \rightarrow X\}$ and nonnegative integers n_i such that

$$\mathcal{E}|_{U_i} \simeq \mathcal{O}_{U_i}^{\oplus n_i}$$

for every i . Then $\mathcal{E}$ is dualizable, connective, and flat. On every affine open it corresponds to a finite projective module.

Proof.

- (1) **Choose an affine trivializing atlas.** Refine the given Zariski cover by affine open subschemes and choose an affine Zariski atlas $\{U_i \rightarrow X\}$ with

$$E_i := \mathcal{E}|_{U_i} \simeq \mathcal{O}_{U_i}^{\oplus n_i}.$$

- (2) **Dualize the transition equivalences.** Each E_i is dualizable. On an overlap U_{ij} , the descent equivalence

$$g_{ij} : E_i|_{U_{ij}} \xrightarrow{\sim} E_j|_{U_{ij}}$$

induces functorially under dualization an equivalence

$$(g_{ij}^{-1})^\vee : E_i^\vee|_{U_{ij}} \xrightarrow{\sim} E_j^\vee|_{U_{ij}}.$$

- (3) **Transfer the cocycle coherences.** Dualization is a functor on the maximal ∞ -groupoid of dualizable objects, so the cocycle and all higher compatibilities are the images of those already carried by the descent object $\mathcal{E}$.
- (4) **Glue the local duals.** By 1.10.1, the $E_i^\vee$ therefore glue to $\mathcal{E}^\vee \in \mathrm{QCoh}(X)$.

- (5) **Glue evaluation and coevaluation.** The local evaluation and coevaluation maps are natural under the transition equivalences and hence glue to

$$\mathcal{E}^\vee \otimes \mathcal{E} \longrightarrow \mathcal{O}_X, \quad \mathcal{O}_X \longrightarrow \mathcal{E} \otimes \mathcal{E}^\vee.$$

- (6) **Check the triangle identities.** Their triangle identities hold after restriction to the covering family, and affine-open restrictions jointly detect equivalences and homotopies by descent.
- (7) **Conclude global dualizability.** Thus $\mathcal{E}$ is globally dualizable.
- (8) **Connectivity.** Connectivity is detected on affine opens by 1.10.2. Every local trivialization is connective, so $\mathcal{E}$ is connective.
- (9) **Choose an affine open.** Let $U = \operatorname{Spec}(A) \subseteq X$ be affine and let $M \in \operatorname{Mod}_A$ represent $\mathcal{E}|_U$.
- (10) **Pull back the trivializing cover.** Pull back the trivializing cover to U and refine it by affine opens.
- (11) **Recognize the affine Zariski cover.** The resulting affine-open coproduct is effective epimorphic, hence is an affine spectral Zariski cover by 1.8.11.
- (12) **Record the local finite-free forms.** Thus there are affine opens $\{\operatorname{Spec}(A_j) \rightarrow \operatorname{Spec}(A)\}$ and nonnegative integers r_j with

$$M \otimes_A A_j \simeq A_j^{\oplus r_j}.$$

- (13) **Apply locality of flatness.** The local modules are flat, so 1.10.2 makes M flat.
- (14) **Conclude global flatness.** Thus $\mathcal{E}$ is flat on every affine open.
- (15) **Finite projectivity on affine opens.** On an affine open $U = \operatorname{Spec}(A)$, dualizability gives perfectness of the corresponding module by 1.6.9. Together with connectivity and flatness, 1.6.10 implies that the module is finite projective.

□

Theorem 1.13.1 (Finite projective modules and vector bundles). *Let A be a connective commutative ring spectrum and let $P \in \operatorname{Mod}_A$. The following are equivalent.*

- (i) P is finite projective.
- (ii) P is Zariski-locally finite free of finite rank.
- (iii) P is perfect, connective, and flat.

Consequently, a quasi-coherent module on a spectral scheme is Zariski-locally free of locally constant finite rank exactly when its restriction to every affine open is finite projective. The local ranks then define a locally constant function

$$|X| \longrightarrow \mathbb{N}.$$

It is a vector bundle of rank n in the sense of 1.13.2 exactly when this rank function is identically equal to n .

Proof.

- (1) **Algebraic characterization.** The equivalence (i) $\Leftrightarrow$ (iii) is 1.6.10. The implication (ii) $\Rightarrow$ (iii) follows from 1.13.1.

- (2) **Degree-zero local triviality.** Assume P finite projective. Then P is flat, and

$$\pi_n(P) \simeq \pi_n(A) \otimes_{\pi_0(A)} \pi_0(P).$$

The module $\pi_0(P)$ is finitely generated projective over $\pi_0(A)$. Hence there are finitely many elements $f_i \in \pi_0(A)$ generating the unit ideal such that

$$\pi_0(P)[f_i^{-1}] \simeq \pi_0(A)[f_i^{-1}]^{\oplus r_i}.$$

- (3) **Lift the local bases.** Put

$$A_i := A[f_i^{-1}], \quad P_i := P \otimes_A A_i.$$

A chosen basis of $\pi_0(P_i)$ defines a homomorphism

$$\pi_0(A_i)^{\oplus r_i} \xrightarrow{\simeq} \pi_0(P_i).$$

Since maps from a finite free A_i -module are finite tuples of elements, this basis lifts to an A_i -linear morphism

$$u_i : A_i^{\oplus r_i} \longrightarrow P_i.$$

- (4) **Detect the lifted basis on homotopy groups.** Both sides are flat. Hence, for every integer n ,

$$\pi_n(u_i) \simeq \pi_n(A_i) \otimes_{\pi_0(A_i)} \pi_0(u_i),$$

which is an isomorphism. Homotopy groups detect equivalences, so

$$A_i^{\oplus r_i} \xrightarrow{\simeq} P_i.$$

Thus P is Zariski-locally finite free.

- (5) **Globalize affine local freeness.** Apply the affine equivalence on an affine Zariski atlas of a spectral scheme. If a quasi-coherent module restricts to a finite projective module on every affine open, then each affine restriction is Zariski-locally finite free by the preceding steps.
- (6) **Glue the local trivializations.** These local trivializations together give a Zariski cover on which the quasi-coherent module is finite free.
- (7) **Construct the locally constant rank function.** The ranks agree after restriction to overlaps, since the rank of a finitely generated projective module is preserved by base change. They therefore determine a locally constant function

$$|X| \longrightarrow \mathbb{N}.$$

- (8) **Prove the converse local criterion.** Conversely, a Zariski-locally finite free quasi-coherent module restricts on every affine open to a locally free module, hence to a finite projective module by 1.13.1.

- (9) **Characterize fixed rank.** Finally, the module is locally free of the fixed rank n precisely when the resulting locally constant rank function is identically n . This is exactly the notion of a vector bundle of rank n from 1.13.2.

□

Local freeness is preserved by substitution; a trivializing cover pulls back through

$$\begin{array}{ccc} Y \times_X U_i & \longrightarrow & U_i \\ \downarrow & & \downarrow \\ Y & \xrightarrow{f} & X. \end{array}$$

Proposition 1.13.2 (Base change of locally free modules). *If $f : Y \rightarrow X$ is any morphism and $\mathcal{E}$ is locally free of rank n , then $f^*\mathcal{E}$ is locally free of rank n .*

Proof.

- (1) **Pull back the trivializing cover.** Pull back a trivializing Zariski cover of X .
- (2) **Preserve the local data.** Open covers and finite free modules are preserved under base change.

□

1.13.2 The relative free commutative algebra

Proposition 1.13.3 (Relative free commutative algebra). *For every spectral scheme X , the forgetful functor*

$$\mathrm{oblv}_X^{\mathrm{alg}} : \mathrm{QCAlg}(X) \longrightarrow \mathrm{QCoh}(X)$$

admits a left adjoint

$$\mathrm{Sym}_{\mathcal{O}_X} : \mathrm{QCoh}(X) \longrightarrow \mathrm{QCAlg}(X).$$

The construction is compatible with pullback: for every $f : Y \rightarrow X$ there is a canonical equivalence

$$f^* \mathrm{Sym}_{\mathcal{O}_X}(\mathcal{E}) \simeq \mathrm{Sym}_{\mathcal{O}_Y}(f^*\mathcal{E}).$$

$$\begin{array}{ccc} \mathrm{QCoh}(X) & \xrightarrow{\mathrm{Sym}_{\mathcal{O}_X}} & \mathrm{QCAlg}(X) \\ f^* \downarrow & & \downarrow f^* \\ \mathrm{QCoh}(Y) & \xrightarrow{\mathrm{Sym}_{\mathcal{O}_Y}} & \mathrm{QCAlg}(Y). \end{array}$$

Proof.

- (1) **Construct the free-algebra adjunction.** The category $\mathrm{QCoh}(X)$ is presentably symmetric monoidal, so the operadic free-algebra theorem supplies the adjunction.
- (2) **Commute the construction with pullback.** Since pullback is colimit-preserving and strong symmetric monoidal, it commutes with the operadic free commutative algebra construction.

□

Proposition 1.13.4 (Connectivity of the relative free algebra). *If*

$$\mathcal{E} \in \mathrm{QCoh}(X)_{\geq 0},$$

then

$$\mathrm{Sym}_{\mathcal{O}_X}(\mathcal{E}) \in \mathrm{QCalg}(X)^{\mathrm{cn}}.$$

Proof.

- (1) **Reduce to an affine open.** The assertion is affine-local. Let $U = \mathrm{Spec}(A) \subseteq X$, and let $M \in (\mathrm{Mod}_A)_{\geq 0}$ correspond to $\mathcal{E}|_U$.
- (2) **Write the underlying free algebra.** The underlying A -module of the free commutative A -algebra is

$$\bigoplus_{k \geq 0} (M^{\otimes_A k})_{h\Sigma_k}.$$

- (3) **Check tensor-power connectivity.** Every tensor power $M^{\otimes_A k}$ is connective.
- (4) **Check colimit connectivity.** Homotopy orbits and small coproducts preserve connective objects, so the displayed A -module is connective.
- (5) **Detect connectivity globally.** Connectivity of a quasi-coherent algebra is detected on affine opens, and the result follows.

□

1.13.3 Total spaces and affine spaces

Definition 1.13.3 (Relative linear space). For a connective quasi-coherent module $\mathcal{E} \in \mathrm{QCoh}(X)_{\geq 0}$, define its **relative linear space** by

$$\mathbb{L}_X(\mathcal{E}) := \mathrm{Spec}_X \mathrm{Sym}_{\mathcal{O}_X}(\mathcal{E}).$$

$$\begin{array}{ccc} \mathrm{QCoh}(X)_{\geq 0}^{\mathrm{op}} & \xrightarrow{(\mathrm{Sym}_{\mathcal{O}_X})^{\mathrm{op}}} & \mathrm{QCalg}(X)^{\mathrm{cn}, \mathrm{op}} \\ & \searrow \mathbb{L}_X & \downarrow \mathrm{Spec}_X \\ & & (\mathrm{Sch}_X^{\mathrm{sp}})_{\mathrm{aff}}. \end{array}$$

Proposition 1.13.5 (Functor of points of a relative linear space). *For every morphism $f : T \rightarrow X$, there is a natural equivalence*

$$\mathrm{Map}_X(T, \mathbb{L}_X(\mathcal{E})) \simeq \mathrm{Map}_{\mathrm{QCoh}(T)}(f^* \mathcal{E}, \mathcal{O}_T).$$

The variance is summarized by

$$\begin{array}{ccc} T & \xrightarrow{\ell} & \mathbb{L}_X(\mathcal{E}) \\ & \searrow f & \swarrow \\ & X & \end{array} \quad \longleftrightarrow \quad f^* \mathcal{E} \xrightarrow{\ell^\sharp} \mathcal{O}_T.$$

Proof.

- (1) **Apply relative spectrum.** Apply the relative spectrum universal property of 1.11.1.
- (2) **Apply the free-algebra adjunction.** Apply the free-algebra adjunction of 1.13.3.

□

Construction 1.13.1 (Section stack of a quasi-coherent coefficient). Let X be a spectral scheme and let $\mathcal{E} \in \mathrm{QCoh}(X)$. Define a spectral prestack over X , denoted

$$\mathrm{Sect}_X(\mathcal{E}) \longrightarrow X,$$

by the following functor of affine points. For an affine spectral scheme T , the map to $X(T)$ has fibre over $f : T \rightarrow X$

$$\mathrm{Map}_{\mathrm{QCoh}(T)}(\mathcal{O}_T, f^*\mathcal{E}).$$

Equivalently, its total space is the Grothendieck construction of the functor

$$X(T) \longrightarrow \mathcal{S}, \quad f \longmapsto \mathrm{Map}_{\mathrm{QCoh}(T)}(\mathcal{O}_T, f^*\mathcal{E}).$$

Proposition 1.13.6 (Descent and base change for section stacks). *For every $\mathcal{E} \in \mathrm{QCoh}(X)$, the prestack $\mathrm{Sect}_X(\mathcal{E})$ is a spectral Zariski stack over X . For every morphism $g : Y \rightarrow X$, there is a canonical equivalence over Y*

$$\mathrm{Sect}_X(\mathcal{E}) \times_X Y \simeq \mathrm{Sect}_Y(g^*\mathcal{E}).$$

Thus the square

$$\begin{array}{ccc} \mathrm{Sect}_Y(g^*\mathcal{E}) & \longrightarrow & \mathrm{Sect}_X(\mathcal{E}) \\ \downarrow & & \downarrow \\ Y & \xrightarrow{g} & X \end{array}$$

is Cartesian.

Proof.

- (1) **Specify descent data.** Let $T_\bullet \rightarrow T$ be the Čech nerve of an affine spectral Zariski cover. An object of

$$\mathrm{Tot}_{[n] \in \Delta} \mathrm{Sect}_X(\mathcal{E})(T_n)$$

consists of a coherent family of maps $f_n : T_n \rightarrow X$ and compatible sections

$$s_n : \mathcal{O}_{T_n} \longrightarrow f_n^*\mathcal{E}.$$

- (2) **Glue the maps to X .** Since X is a spectral Zariski sheaf, the f_n glue uniquely to a map $f : T \rightarrow X$.
- (3) **Apply quasi-coherent descent.** By 1.10.1,

$$\mathrm{QCoh}(T) \simeq \mathrm{Tot}_{[n] \in \Delta} \mathrm{QCoh}(T_n),$$

and under this equivalence $\mathcal{O}_T$ and $f^*\mathcal{E}$ correspond to their restrictions.

(4) **Identify the mapping-space totalization.** Therefore

$$\mathrm{Map}_{\mathrm{QCoh}(T)}(\mathcal{O}_T, f^*\mathcal{E}) \simeq \mathrm{Tot}_{[n] \in \Delta} \mathrm{Map}_{\mathrm{QCoh}(T_n)}(\mathcal{O}_{T_n}, f_n^*\mathcal{E}).$$

(5) **Glue the local sections.** The s_n therefore glue uniquely, together with all higher homotopies, to a section $s : \mathcal{O}_T \rightarrow f^*\mathcal{E}$.

(6) **Conclude descent of the total prestack.** This constructs the inverse to the restriction map and proves descent for the total prestack.

(7) **Base change.** A T -point of the left-hand side over a map $h : T \rightarrow Y$ is a section

$$\mathcal{O}_T \longrightarrow (g \circ h)^*\mathcal{E} \simeq h^*g^*\mathcal{E}.$$

This is exactly a T -point of $\mathrm{Sect}_Y(g^*\mathcal{E})$ over h . Naturality in T and Yoneda give the claimed equivalence. □

Definition 1.13.4 (Total space representing sections). Let $\mathcal{E}$ be a finite locally free quasi-coherent module on X . Define its **section total space** by

$$\mathbf{V}_X(\mathcal{E}) := \mathbb{L}_X(\mathcal{E}^\vee).$$

Proposition 1.13.7 (Affine representability and base change for section total spaces). *Let $\mathcal{E}$ be finite locally free on X . Then the section stack of 1.13.1 is represented by the affine spectral scheme $\mathbf{V}_X(\mathcal{E})$:*

$$\mathrm{Sect}_X(\mathcal{E}) \simeq \mathbf{V}_X(\mathcal{E})$$

over X . Equivalently, for every morphism $f : T \rightarrow X$, there is a natural equivalence

$$\mathrm{Map}_X(T, \mathbf{V}_X(\mathcal{E})) \simeq \mathrm{Map}_{\mathrm{QCoh}(T)}(\mathcal{O}_T, f^*\mathcal{E}).$$

For every morphism $g : Y \rightarrow X$, there is a canonical equivalence

$$\mathbf{V}_X(\mathcal{E}) \times_X Y \simeq \mathbf{V}_Y(g^*\mathcal{E}).$$

For the finite free module $\mathcal{O}_X^{\oplus n}$, the canonical self-duality identifies

$$\mathbf{V}_X(\mathcal{O}_X^{\oplus n}) \simeq \mathbb{L}_X(\mathcal{O}_X^{\oplus n}).$$

$$\begin{array}{ccc} \mathbf{V}_Y(g^*\mathcal{E}) & \longrightarrow & \mathbf{V}_X(\mathcal{E}) \\ \downarrow & & \downarrow \\ Y & \xrightarrow{g} & X. \end{array}$$

Proof.

- (1) **Represent sections by duality.** Strong symmetric monoidal pullback preserves duals, so

$$f^*(\mathcal{E}^\vee) \simeq (f^*\mathcal{E})^\vee.$$

By 1.13.5 and the tensor–Hom adjunction for the dualizable module $f^*\mathcal{E}$,

$$\begin{aligned} \mathrm{Map}_X(T, \mathbf{V}_X(\mathcal{E})) &\simeq \mathrm{Map}_{\mathrm{QCoh}(T)}((f^*\mathcal{E})^\vee, \mathcal{O}_T) \\ &\simeq \mathrm{Map}_{\mathrm{QCoh}(T)}(\mathcal{O}_T, f^*\mathcal{E}). \end{aligned}$$

- (2) **Base change.** Relative linear spaces commute with base change by 1.11.2 and 1.13.3. Together with preservation of duals by pullback this gives

$$\mathbf{V}_X(\mathcal{E}) \times_X Y \simeq \mathbb{L}_Y((g^*\mathcal{E})^\vee) = \mathbf{V}_Y(g^*\mathcal{E}).$$

- (3) **Finite free self-duality.** The direct-sum decomposition gives the canonical duality

$$(\mathcal{O}_X^{\oplus n})^\vee \simeq \mathcal{O}_X^{\oplus n},$$

yielding the final equivalence. □

Remark 1.13.1 (Linear comprehension). The two linear representation problems now fit into a single picture. For any coefficient $\mathcal{E} \in \mathrm{QCoh}(X)$, 1.13.1 constructs the stack representing sections

$$\mathcal{O}_T \longrightarrow f^*\mathcal{E}.$$

When $\mathcal{E}$ is finite locally free, 1.13.7 identifies this stack with the affine context $\mathbf{V}_X(\mathcal{E})$. Thus section comprehension is available as a stack for arbitrary quasi-coherent coefficients and becomes affine representable in the finite locally free sector. In that sector, duality gives the comparison

$$\mathbb{L}_X(\mathcal{E}) \simeq \mathbf{V}_X(\mathcal{E}^\vee),$$

so linear-functional and section representations differ exactly by duality.

Remark 1.13.2 (Coordinate-module and section conventions). The relative linear space and the section total space represent opposite variances, and duality is required to identify them. A map

$$T \longrightarrow \mathbb{L}_X(\mathcal{E})$$

over X represents a linear functional

$$f^*\mathcal{E} \longrightarrow \mathcal{O}_T,$$

whereas, for finite locally free $\mathcal{F}$, a map

$$T \longrightarrow \mathbf{V}_X(\mathcal{F})$$

represents a section

$$\mathcal{O}_T \longrightarrow f^*\mathcal{F}.$$

For finite locally free $\mathcal{E}$, duality relates the two conventions by

$$\mathbb{L}_X(\mathcal{E}) \simeq \mathbf{V}_X(\mathcal{E}^\vee).$$

The distinction is therefore one of variance, not an implicit choice of a self-duality $\mathcal{E} \simeq \mathcal{E}^\vee$.

Definition 1.13.5 (Brave new affine space). For an integer $n \geq 0$, define

$$\mathbb{A}_X^n := \mathbb{L}_X(\mathcal{O}_X^{\oplus n}) = \operatorname{Spec}_X \operatorname{Sym}_{\mathcal{O}_X}(\mathcal{O}_X^{\oplus n}).$$

The case $n = 1$ is the **brave new affine line**

$$\mathbb{A}_X^1 = \operatorname{Spec}_X \operatorname{Sym}_{\mathcal{O}_X}(\mathcal{O}_X).$$

Proposition 1.13.8 (Base change of affine space). *For every morphism $f : Y \rightarrow X$, there is a canonical equivalence*

$$\mathbb{A}_X^n \times_X Y \simeq \mathbb{A}_Y^n.$$

$$\begin{array}{ccc} \mathbb{A}_Y^n & \longrightarrow & \mathbb{A}_X^n \\ \downarrow & & \downarrow \\ Y & \xrightarrow{f} & X. \end{array}$$

Proof.

- (1) **Commute the free algebra with pullback.** Use the pullback compatibility of the free algebra from 1.13.3.
- (2) **Apply relative-spectrum base change.** Use relative spectrum base change from 1.11.2.

□

Construction 1.13.2 (Zero and unit sections). The zero endomorphism and identity endomorphism of $\mathcal{O}_X$ determine maps of quasi-coherent modules

$$\mathcal{O}_X \rightrightarrows \mathcal{O}_X.$$

By the free-algebra and relative-spectrum adjunctions, they induce sections

$$0, 1 : X \rightrightarrows \mathbb{A}_X^1.$$

Proposition 1.13.9 (Sections of brave new affine space are closed). *Let $n \geq 0$. Every section*

$$\begin{array}{ccc} X & \xrightarrow{s} & \mathbb{A}_X^n \\ & \searrow & \swarrow \\ & X & \end{array}$$

is a closed immersion.

Proof.

- (1) **Reduce to an affine context.** Closed immersions are local on the target, so take $X = \operatorname{Spec}(A)$. The section s corresponds contravariantly to an A -algebra morphism

$$\operatorname{Sym}_A(A^{\oplus n}) \longrightarrow A.$$

By the free-algebra adjunction it is determined by an n -tuple of elements of $\Omega^\infty A$.

(2) **Pass to degree zero.** By 1.6.6, the induced ordinary ring morphism is an evaluation map

$$\pi_0(A)[t_1, \dots, t_n] \longrightarrow \pi_0(A), \quad t_i \longmapsto a_i.$$

It is surjective because it restricts to the identity on the constant subring $\pi_0(A)$. Therefore the corresponding affine morphism is a closed immersion by 1.12.1.

□

Proposition 1.13.10 (The zero section is closed). *The zero section*

$$0: X \longrightarrow \mathbb{A}_X^1$$

is a closed immersion.

Proof.

(1) **Recognize the zero map as a section.** The zero section is a section of $\mathbb{A}_X^1 \rightarrow X$.

(2) **Apply closedness of sections.** Apply 1.13.9.

□

Definition 1.13.6 (Spectral multiplicative group). The **spectral multiplicative group** over X is the open complement

$$\mathbb{G}_{m,X} := \mathbb{A}_X^1 \setminus 0(X).$$

Proposition 1.13.11 (Group structure on the spectral multiplicative group). *The spectral scheme*

$$\mathbb{G}_{m,X}$$

is canonically a commutative group object in spectral schemes over X . For every morphism $T \rightarrow X$, there is a natural equivalence

$$\mathrm{Map}_X(T, \mathbb{G}_{m,X}) \simeq \mathrm{Aut}_{\mathrm{QCoh}(T)}(\mathcal{O}_T).$$

Proof.

(1) **Affine coordinate algebra.** Work first over $X = \mathrm{Spec}(A)$ and write

$$A\{t\} := \mathrm{Sym}_A(A).$$

By the complement construction and 1.8.4,

$$\mathbb{G}_{m,X} \simeq \mathrm{Spec}(A\{t\}[t^{-1}]).$$

(2) **Coordinate algebra of the product.** There is a canonical equivalence

$$A\{t_1, t_2\}[(t_1 t_2)^{-1}] \simeq A\{t_1, t_2\}[t_1^{-1}, t_2^{-1}],$$

because an inverse of $t_1 t_2$ gives

$$t_1^{-1} = t_2(t_1 t_2)^{-1}, \quad t_2^{-1} = t_1(t_1 t_2)^{-1}.$$

The right side is the coordinate algebra of $\mathbb{G}_{m,X} \times_X \mathbb{G}_{m,X}$.

- (3) **Define the Hopf algebra maps.** The universal property of localization gives

$$\begin{aligned}\Delta(t) &= t_1 t_2, \\ \epsilon(t) &= 1, \\ \text{inv}(t) &= t^{-1}.\end{aligned}$$

Thus

$$\begin{array}{ccccc} & & A\{t\}[t^{-1}] & & \\ & \swarrow \epsilon & \downarrow \text{inv} & \searrow \Delta & \\ A & & A\{t\}[t^{-1}] & & A\{t_1, t_2\}[t_1^{-1}, t_2^{-1}]. \end{array}$$

- (4) **Verify the group identities.** The commutative group identities are the identities satisfied by the universal invertible element.
- (5) **Pass to relative spectra.** Passing contravariantly to relative spectra produces multiplication, unit, and inversion.
- (6) **Identify points of affine line with endomorphisms.** A map $T \rightarrow \mathbb{A}_X^1$ corresponds by 1.13.5 to an $\mathcal{O}_T$ -linear endomorphism of $\mathcal{O}_T$.
- (7) **Reduce invertibility to an affine test.** The assertion that such an endomorphism is invertible is Zariski-local on T , so take $T = \text{Spec}(C)$.
- (8) **Identify an endomorphism with multiplication by a point.** Then

$$\text{Map}_{\text{Mod}_C}(C, C) \simeq \Omega^\infty C,$$

and a point $c \in \Omega^\infty C$ acts by multiplication $m_c : C \rightarrow C$.

- (9) **Show an equivalence gives a unit in degree zero.** If m_c is an equivalence, multiplication by $[c]$ on $\pi_0(C)$ is an isomorphism, so $[c] \in \pi_0(C)^\times$.
- (10) **Show a unit in degree zero gives an equivalence.** Conversely, if $[c]$ is a unit, choose the inverse component $[d]$. The products cd and dc lie in the component of the unit, so multiplication by c and multiplication by d are inverse equivalences of C -modules. Thus m_c is invertible exactly when $[c]$ is a unit.
- (11) **Characterize factorization through the localization.** By the universal property of localization, the corresponding map $T \rightarrow \mathbb{A}_X^1$ factors through the principal open $D_{A\{t\}}^{\text{sp}}(t) = \mathbb{G}_{m,X}$ exactly when the image of t is a unit in $\pi_0(C)$.
- (12) **Identify the functor of points.** Therefore

$$\text{Map}_X(T, \mathbb{G}_{m,X}) \simeq \text{Aut}_{\text{QCoh}(T)}(\mathcal{O}_T).$$

- (13) **Globalize by Zariski descent.** Zariski descent extends this equivalence to arbitrary spectral schemes T over X .
- (14) **Globalization.** The affine constructions commute with restriction to affine opens and therefore glue over an arbitrary spectral scheme X .

□

Proposition 1.13.12 (Base change of the multiplicative group). *For every morphism $Y \rightarrow X$, there is a canonical equivalence*

$$\mathbb{G}_{m,X} \times_X Y \simeq \mathbb{G}_{m,Y}.$$

$$\begin{array}{ccc} \mathbb{G}_{m,Y} & \longrightarrow & \mathbb{G}_{m,X} \\ \downarrow & & \downarrow \\ Y & \longrightarrow & X. \end{array}$$

Proof.

- (1) **Apply affine-space base change.** Use affine-space base change.
- (2) **Apply base change of open complements.** Use base change of open complements from 1.12.2.

□

Proposition 1.13.13 (Homotopy of the brave new polynomial algebra). *Let A be a connective commutative ring spectrum and put*

$$A\{t\} := \mathrm{Sym}_A(A).$$

Then, as an underlying A -module,

$$A\{t\} \simeq \bigoplus_{n \geq 0} A \otimes \Sigma_+^\infty B\Sigma_n.$$

Consequently, if

$$A_m(K) := \pi_m(A \otimes \Sigma_+^\infty K),$$

then

$$\pi_m(A\{t\}) \simeq \bigoplus_{n \geq 0} A_m(B\Sigma_n)$$

for every $m \geq 0$, while

$$\pi_0(A\{t\}) \simeq \pi_0(A)[t].$$

In particular, for $A = HR$ with R a commutative ring,

$$\pi_m(HR\{t\}) \simeq \bigoplus_{n \geq 0} H_m(B\Sigma_n; R),$$

recovering the degree-wise calculation over a discrete spectral base. These positive-degree groups can be nonzero: for example,

$$H_1(B\Sigma_2; \mathbb{Z}) \simeq \mathbb{Z}/2,$$

so $\pi_1(H\mathbb{Z}\{t\})$ contains a $\mathbb{Z}/2$ -summand. They need not be nonzero for every base: if R is a $\mathbb{Q}$ -algebra, transfer for the finite groups Σ_n gives

$$H_m(B\Sigma_n; R) = 0 \quad (m > 0),$$

and the rank-one free commutative algebra is discrete. The formula therefore displays exactly when the brave new affine line carries additional positive spectral homotopy.

Proof.

- (1) **Apply the relative free-algebra formula.** By the relative version of 1.3.2,

$$A\{t\} \simeq \bigoplus_{n \geq 0} (A^{\otimes A^n})_{h\Sigma_n} \simeq \bigoplus_{n \geq 0} (A)_{h\Sigma_n}.$$

The Σ_n -action is trivial on the tensor power of the rank-one free A -module, so

$$(A)_{h\Sigma_n} \simeq A \otimes \Sigma_+^\infty B\Sigma_n.$$

- (2) **Take homotopy groups.** Homotopy groups commute with the displayed coproduct, giving

$$\pi_m(A\{t\}) \simeq \bigoplus_{n \geq 0} \pi_m(A)_{h\Sigma_n}.$$

The degree-zero algebra identification is already 1.6.6 applied to the connective A -module A :

$$\pi_0(A\{t\}) \simeq \pi_0(A)[t].$$

When $A = HR$, the generalized homology groups above are ordinary homology groups $H_m(B\Sigma_n; R)$. Compare [34, 26].

□

The two linear representation conventions are now separated and related precisely: relative linear spaces represent functionals, section stacks represent sections, and finite locally free coefficients make the latter affine by duality. The final substantive step is retrospective rather than foundational: having built the spectral geometry natively, we recover its ordinary 0-truncated sector as an interface with classical algebraic geometry.

1.14 Reconstructing the 0-truncated comparison

The spectral geometry above was defined natively. Ordinary schemes can now be recovered from it through the degree-zero reflection already constructed for connective commutative ring spectra. This section extracts that 0-truncated interface: first on affines, then globally by Zariski gluing. The completed spectral construction therefore yields the comparison as a consequence.

Let

$$\delta : \text{AffSch} \hookrightarrow \text{Aff}_{\text{Sp}}, \quad \text{Spec}(R) \mapsto \text{Spec}(HR),$$

be the fully faithful embedding of ordinary affine schemes obtained from the Eilenberg–Mac Lane functor.

Proposition 1.14.1 (Restriction to 0-truncated affine tests). *For every connective commutative ring spectrum A , restriction of the spectral functor of points along δ is represented by $\text{Spec } \pi_0(A)$: for every commutative ring R ,*

$$\begin{aligned} h_A(\text{Spec}(HR)) &\simeq \text{Map}_{\text{CAlg}(\text{Sp})}(A, HR) \\ &\simeq \text{Map}_{\text{CRing}}(\pi_0(A), R). \end{aligned}$$

Moreover, for $r \in R$, there is a canonical equivalence

$$HR[r^{-1}] \simeq H(R[r^{-1}]).$$

Consequently, restriction along δ carries the spectral principal Zariski basis to the usual principal Zariski basis.

$$\begin{array}{ccc}
 t_0 D_A^{\text{sp}}(f) & \longrightarrow & t_0 \text{Spec}(A) \\
 \simeq \downarrow & & \downarrow \simeq \\
 D_{\pi_0(A)}(f) & \longrightarrow & \text{Spec } \pi_0(A).
 \end{array}$$

Proof.

- (1) **Restrict the affine functor of points.** The mapping-space equivalence is exactly the adjunction $\pi_0 \dashv H_{\text{CRing}}$ of 1.3.6. Ordinary affine Yoneda therefore identifies the restricted functor with $\text{Spec } \pi_0(A)$.
- (2) **Compare principal localization.** By 1.8.3, the localization $HR[r^{-1}]$ is connective, has degree-zero homotopy $R[r^{-1}]$, and has no positive homotopy groups. Hence it is canonically $H(R[r^{-1}])$. Thus the principal covering families are preserved under restriction.

□

Theorem 1.14.1 (Global 0-truncated reconstruction and adjunction). *The affine enhancement δ extends by Zariski gluing to a functor*

$$(-)^{\text{disc}} : \text{Sch} \longrightarrow \text{Sch}^{\text{sp}},$$

characterized on affines by

$$(\text{Spec } R)^{\text{disc}} \simeq \text{Spec}(HR).$$

Restriction of the functor of points along δ carries every spectral scheme X to a Zariski sheaf represented by an ordinary scheme $t_0 X$, and the two constructions form an adjunction

$$\text{Sch} \begin{array}{c} \xrightarrow{(-)^{\text{disc}}} \\ \xleftarrow{t_0} \end{array} \text{Sch}^{\text{sp}}. \quad (-)^{\text{disc}} \dashv t_0.$$

It has the following properties.

- (i) *For every connective commutative ring spectrum A ,*

$$t_0 \text{Spec}(A) \simeq \text{Spec } \pi_0(A).$$

- (ii) *The adjunction unit*

$$S \longrightarrow t_0(S^{\text{disc}})$$

is an equivalence for every ordinary scheme S . Hence $(-)^{\text{disc}}$ is fully faithful.

- (iii) *The right adjoint t_0 preserves finite limits.*

- (iv) *Principal opens correspond under t_0 :*

$$t_0 D_A^{\text{sp}}(f) \simeq D_{\pi_0(A)}(f).$$

Consequently the posets of open subschemes of X and of $t_0 X$ are canonically equivalent.

(v) The adjunction counit is a natural morphism

$$\iota_X : (t_0 X)^{\text{disc}} \longrightarrow X,$$

whose restriction to $\text{Spec}(A) \subseteq X$ is the morphism represented by

$$A \longrightarrow H\pi_0(A).$$

(vi) Under the standard connective spectral-scheme recognition theorem, affineness, quasi-compactness, quasi-separatedness, and the qcqs condition are detected by t_0 .

The construction in this theorem is an interface with the 0-truncated sector; none of its detection statements is used in the native construction of spectral schemes, coefficients, locality, or comprehension above.

$$\begin{array}{ccc} \text{AffSch} & \hookrightarrow & \text{Sch} \\ \delta \downarrow & & \downarrow (-)^{\text{disc}} \\ \text{Aff}_{\text{Sp}} & \hookrightarrow & \text{Sch}^{\text{Sp}}. \end{array}$$

Proof.

- (1) **Choose an affine atlas of the ordinary scheme.** Let an ordinary scheme S have an affine Zariski atlas $\coprod_i \text{Spec}(R_i) \rightarrow S$.
- (2) **Enhance principal intersections spectrally.** By 1.14.1, Eilenberg–Mac Lane enhancement carries principal intersections and their restriction maps to the corresponding spectral principal intersections.
- (3) **Glue the discrete enhancement.** Spectral Zariski descent therefore glues the affines $\text{Spec}(HR_i)$ to a spectral scheme S^{disc} .
- (4) **Remove dependence on the atlas.** Common refinement and mapping descent make the construction independent of the chosen atlas and functorial in S .
- (5) **Restrict a spectral affine atlas to degree zero.** Conversely, let

$$\coprod_i \text{Spec}(A_i) \longrightarrow X$$

be an affine spectral Zariski atlas. Restriction along δ carries each chart to $\text{Spec} \pi_0(A_i)$ and every principal spectral cover to the corresponding ordinary principal cover.

- (6) **Recognize the ordinary scheme.** Since principal covers generate both Zariski topologies, the restricted functor is an ordinary Zariski sheaf with an effective affine open atlas and is therefore represented by an ordinary scheme.
- (7) **Define $t_0 X$.** Define that scheme to be $t_0 X$.
- (8) **Record the affine formula.** On affines this gives $t_0 \text{Spec}(A) \simeq \text{Spec} \pi_0(A)$.

- (9) **Establish the affine mapping equivalence.** Let $T = \operatorname{Spec} R$ be an ordinary affine scheme and let X be an arbitrary spectral scheme. By the functor-of-points definition and the construction of $t_0 X$, there are natural equivalences

$$\begin{aligned} \operatorname{Map}_{\operatorname{Sch}^{\operatorname{sp}}}(T^{\operatorname{disc}}, X) &\simeq X(\operatorname{Spec}(HR)) \\ &\simeq (t_0 X)(\operatorname{Spec} R) \\ &\simeq \operatorname{Map}_{\operatorname{Sch}}(T, t_0 X). \end{aligned}$$

- (10) **Identify its first two components.** The first equivalence is spectral Yoneda and the second is the defining restriction of the functor of points along δ .
- (11) **Choose an affine atlas of a general ordinary scheme.** For a general ordinary scheme S , choose an affine Zariski atlas $U \rightarrow S$.
- (12) **Apply mapping descent.** Its discrete enhancement is an effective affine open atlas $U^{\operatorname{disc}} \rightarrow S^{\operatorname{disc}}$, and mapping descent on the two sides gives

$$\operatorname{Map}_{\operatorname{Sch}^{\operatorname{sp}}}(S^{\operatorname{disc}}, X) \xrightarrow{\simeq} \operatorname{Map}_{\operatorname{Sch}}(S, t_0 X).$$

- (13) **Establish naturality.** Naturality of the affine equivalence on the Čech nerve makes this equivalence natural in both variables.
- (14) **Conclude the global adjunction.** This is the asserted adjunction.
- (15) **Identify the unit on affines.** For an ordinary affine scheme $\operatorname{Spec} R$, the unit is the identity after the canonical identification

$$t_0 \operatorname{Spec}(HR) \simeq \operatorname{Spec} R.$$

- (16) **Globalize the unit.** Zariski descent therefore makes the unit an equivalence for every ordinary scheme.
- (17) **Deduce full faithfulness.** Thus discrete enhancement is fully faithful.
- (18) **Identify the counit on affines.** The counit on $\operatorname{Spec} A$ is contravariantly represented by the unit of the algebraic reflection

$$A \longrightarrow H\pi_0(A),$$

- (19) **Glue the counit.** These affine maps glue by naturality.
- (20) **Use the right-adjoint property to preserve finite limits.** Finally, as a right adjoint, t_0 preserves every limit that exists in spectral schemes, in particular the finite limits constructed in 1.9.1.
- (21) **Identify principal open subobjects.** The equivalence

$$HR[r^{-1}] \simeq H(R[r^{-1}])$$

of 1.14.1 identifies principal spectral opens with ordinary principal opens.

- (22) **Pass to small unions.** Small unions of open substacks are constructed by 1.9.4 and commute with base change.

- (23) **Glue the open-subobject correspondence.** Since principal opens form a basis on both sides, the principal correspondence glues to a canonical equivalence of open-subobject posets.
- (24) **Invoke only the remaining recognition properties.** The standard connective spectral-scheme comparison identifies the functor just constructed with the usual 0-truncation of spectral algebraic geometry, under which affineness, quasi-compactness, and quasi-separatedness are detected; see [37]. This invocation is used only for these recognition properties and imports no additional structure on X .

□

Corollary 1.14.1 (The truncation counit is a closed immersion). *For every spectral scheme X , the counit*

$$\iota_X : (t_0 X)^{\text{disc}} \longrightarrow X$$

of 1.14.1 is a closed immersion. It induces a homeomorphism on underlying topological spaces

$$|(t_0 X)^{\text{disc}}| \xrightarrow{\cong} |X|,$$

and its 0-truncation

$$t_0(\iota_X) : t_0((t_0 X)^{\text{disc}}) \longrightarrow t_0 X$$

is an equivalence.

Proof.

- (1) **Reduce closedness to an affine target.** The assertion that ι_X is a closed immersion is local on the target. Let

$$X = \text{Spec}(A)$$

be affine.

- (2) **Identify the affine form of the counit.** By 1.14.1,

$$t_0 X \simeq \text{Spec } \pi_0(A),$$

and the restriction of the counit is

$$\text{Spec}(H\pi_0(A)) \longrightarrow \text{Spec}(A),$$

represented contravariantly by the canonical morphism

$$A \longrightarrow H\pi_0(A).$$

- (3) **Compute degree-zero homotopy.** On degree-zero homotopy this induces the canonical isomorphism

$$\pi_0(A) \xrightarrow{\cong} \pi_0(H\pi_0(A)) \simeq \pi_0(A).$$

- (4) **Deduce affine closedness.** In particular, it is surjective. Hence the displayed affine morphism is a closed immersion by 1.12.1.
- (5) **Globalize closedness.** Since closed immersions are defined affine-locally on the target, ι_X is a closed immersion for every spectral scheme X .

(6) **Underlying topological support.** On the same affine open,

$$|\mathrm{Spec}(H\pi_0(A))| = |\mathrm{Spec}\pi_0(A)| = |\mathrm{Spec}(A)|.$$

Under these identifications the map induced by ι_X is the identity. These affine identifications are compatible on intersections and therefore glue to a homeomorphism

$$|(t_0X)^{\mathrm{disc}}| \xrightarrow{\cong} |X|.$$

(7) **Apply the 0-truncation functor.** On an affine open $X = \mathrm{Spec}(A)$, applying t_0 to the counit gives

$$\mathrm{Spec}\pi_0(H\pi_0(A)) \longrightarrow \mathrm{Spec}\pi_0(A).$$

Since

$$\pi_0(H\pi_0(A)) \simeq \pi_0(A),$$

this morphism is an equivalence. The affine equivalences agree on overlaps, so they glue to a global equivalence

$$t_0((t_0X)^{\mathrm{disc}}) \xrightarrow{\cong} t_0X.$$

□

Convention 1.14.1 (Classical comparison notation). For compatibility with later chapters, when the 0-truncated comparison is used we write

$$X^{\mathrm{cl}} := t_0X, \quad f^{\mathrm{cl}} := t_0(f).$$

The notation is an abbreviation for the right-adjoint reconstruction above; it does not reintroduce classical geometry into the native construction.

The 0-truncated comparison therefore closes the circle without altering the native construction: ordinary schemes embed discretely, t_0 recovers their degree-zero geometry, and the truncation counit identifies the ordinary support inside the spectral scheme. We can now summarize the chapter in terms of the structures that remain available for later use.

1.15 Chapter summary

The chapter has produced a coherent spectral geometry whose persistent structure can be compressed into four interacting outputs.

First, the geometric contexts are spectral schemes built from connective coordinates. The connective symmetric-monoidal category $\mathrm{Sp}_{\geq 0}^{\otimes}$ supplies the semantic universe for the commutative operadic theory, giving

$$\mathrm{Alg}_{\mathrm{Comm}^{\otimes}}(\mathrm{Sp}_{\geq 0}^{\otimes}) \simeq \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}}.$$

Variance reversal produces the affine contexts

$$\mathrm{Aff}_{\mathrm{Sp}} = (\mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}})^{\mathrm{op}},$$

principal spectral localizations generate their Zariski topology, and Zariski gluing produces $\mathrm{Sch}^{\mathrm{sp}}$. The resulting category has the finite-limit and open-subobject operations used throughout the later geometry and agrees with the standard connective 0-localic structured model.

Second, every spectral context carries a fully stable coefficient fibre. On an affine context $\mathrm{Spec}(A)$ the fibre is Mod_A , and substitution along $\mathrm{Spec}(B) \rightarrow \mathrm{Spec}(A)$ is extension of scalars. Right Kan extension and Zariski descent globalize these fibres to the coefficient system

$$\mathrm{QCoh} : (\mathrm{Sch}^{\mathrm{sp}})^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L).$$

Its pullbacks are coherent and strong symmetric monoidal, its right adjoints are functorial, and the global fibres inherit the Postnikov structure needed to formulate flatness and connective algebra without truncating the stable coefficient theory.

Third, locality is detected through those coefficients and reconstructs both geometry and algebra. Affine open immersions are flat, locally finitely presented monomorphisms, and an affine family covers exactly when the associated substitutions are jointly conservative. Finite principal covers are descendable, so Amitsur and comonadic descent reconstruct representables, modules, and commutative algebras from their restrictions. The geometric site and the coefficient system are therefore linked by one reconstruction mechanism rather than by two independent locality theories.

Fourth, relative spectrum returns connective coefficient algebra to geometry. For every spectral scheme X ,

$$\mathrm{QCAlg}(X)^{\mathrm{cn}, \mathrm{op}} \xrightarrow{\simeq} (\mathrm{Sch}_{/X}^{\mathrm{sp}})_{\mathrm{aff}},$$

and this equivalence is compatible with arbitrary substitution. Maps into the resulting affine extension represent algebra terms $f^* \mathcal{A} \rightarrow \mathcal{O}_Y$, with the identity of the relative spectrum corresponding to the universal variable. This is the chapter's comprehension principle: connective relative algebra is represented by affine geometry over the base.

The subsequent constructions are consequences and specializations of this core. Closed immersions admit intrinsic open complements and base-change-stable pushouts. Free commutative algebras give relative linear spaces representing linear functionals; arbitrary coefficients have section stacks; and finite locally free coefficients have affine section total spaces by duality. The brave new affine spaces and $\mathbb{G}_m$ are distinguished instances. Finally, the adjunction

$$(-)^{\mathrm{disc}} \dashv t_0$$

recovers ordinary algebraic geometry as the 0-truncated interface, with $t_0 \mathrm{Spec}(A) \simeq \mathrm{Spec} \pi_0(A)$ and the same underlying topological support.

The dependency structure of the chapter is therefore

operadic theory $\longrightarrow$ connective models $\longrightarrow$ spectral contexts $\longrightarrow$ stable substitution $\longrightarrow$ Zariski descent $\longrightarrow$ affine an

What remains active is the coexistence of two levels. Geometry is built from connective commutative ring spectra, while coefficients remain fully stable. Substitution transports the stable fibres coherently, locality detects and reconstructs their geometric base, and relative spectrum returns connective coefficient algebras to geometry. The later closed, linear, and classical comparison constructions all operate inside this same indexed structure.

Chapter 2

Differentiation of the Spectral Doctrine

2.1 Orientation: how does the spectral doctrine differentiate?

Chapter 1 constructed the native spectral contexts used throughout the manuscript and organized their stable coefficients into the indexed symmetric-monoidal doctrine

$$\mathrm{QCoh} : (\mathrm{Sch}^{\mathrm{sp}})^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(\mathrm{Pr}_{\mathrm{st}}^L),$$

and established the substitution, Zariski descent, relative-spectrum comprehension, and open-closed and linear context calculus needed below. Those structures are fixed input. The present chapter asks what differential structure is already carried by this spectral doctrine:

How does the spectral doctrine differentiate?

The chapter answers this question through two complementary constructions. The first is fibrewise and linear: the stable coefficient fibre is identified affinely with the tangent stabilization of commutative spectral algebra and is then equipped with the cofree cocommutative-coalgebra differential exponential. The second is geometric: the cotangent complex represents first-order variation, its degree-one classes construct square-zero thickenings, and finite composites of those thickenings generate an infinitesimal test geometry. These constructions are independent in origin. They meet on free finite-projective models, where the derivative obtained from cofree infinitesimal translation is compared with the universal spectral Kähler derivation. The geometric branch then produces the endpoint adjunctions, infinitesimal lifting and deformation calculus, and, after stabilization, an exact fracture reconstruction.

Both branches use the same preparatory differential core:

spectral doctrine $\longrightarrow$ affine tangent linearization $\longrightarrow$ cotangent-classified variation and chain rule.

The tangent identification supplies the stable semantic environment. The relative and global cotangent classifiers then organize variation by substitution and transitivity. This cotangent calculus

directly drives the geometric branch and later supplies the comparison interface with the linear branch; it does not construct the cofree exponential.

The fibrewise linear branch is tracked by

$$\mathrm{QCoh}(X), \quad !_X, \quad \eta, \quad \mathbf{d}.$$

Cofree cocommutative coalgebras construct the exponential and its canonical monoidal structure. Primitive infinitesimal coalgebras then produce the infinitesimal augmentation, codereliction, and deriving transformation. Only after these maps have been constructed do finite-projective models compare their convolution derivative with the spectral universal derivation.

The geometric branch is tracked by

$$\mathcal{H}_{\mathrm{inf}}, \quad \mathfrak{R} \dashv \mathfrak{S} \dashv \&.$$

Degree-one cotangent classes first produce structured square-zero contexts. Forgetting their presentations and closing under finite composition gives the infinitesimal test category and its Zariski topos. Once the endpoint adjunctions have been constructed, their modal form is

$$\mathfrak{R} \dashv \mathfrak{S} \dashv \&.$$

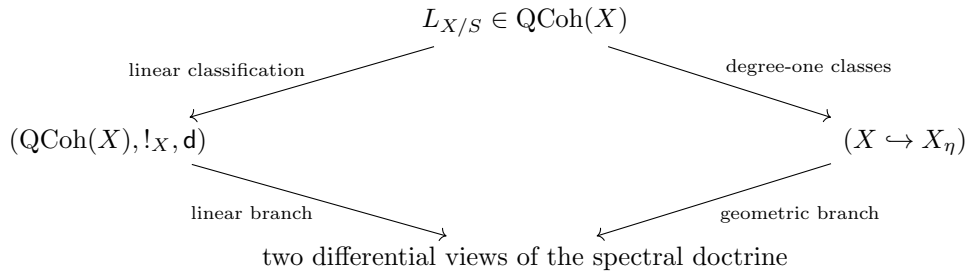
Cotangent mapping spectra then compute the fibres of the associated lifting comparison and control the first-order deformation and local-equation problems.

For navigation, the order of the two constructions is:

linear branch:	cofree cocommutative coalgebra exponential → primitive infinitesimal augmentation → codereliction and deriving transformation → comparison with the spectral universal derivation,
geometric branch:	structured square-zero context extension → finite infinitesimal contexts and their Zariski topos → differential cohesion ($\mathfrak{R} \dashv \mathfrak{S} \dashv \&$) → formal étalification and infinitesimal disks → cotangent lifting, deformation, and equation calculus → stable endpoint semantics and BNV reconstruction.

The display records construction order and later recognition statements. Neither branch presents the other, and the BNV terminology in the final line is only a forward reference to a comparison made after the stable fracture theorem has been proved internally.

The cotangent classifier interfaces with both differential views:



This diagram is mnemonic rather than a dependency graph. In particular, its left-hand arrow marks the later classification and comparison interface; it does not construct the cofree coalgebra modality from the cotangent complex.

The chapter therefore proceeds from affine tangent linearization to the relative and global cotangent calculus, completes the cofree linear branch, and then returns to degree-one cotangent classes to construct the geometric branch. Only after the infinitesimal site, endpoint semantics, and cotangent lifting calculus are in place do we stabilize the endpoint adjunctions. Construction precedes recognition throughout.

All conventions of Chapter 1 remain in force. Coordinate algebras and spectral schemes remain connective, whereas module categories, cotangent complexes, mapping spectra, suspensions, fibres, cofibres, and stabilizations are formed in their full stable ambient categories. Coherence supplied functorially is suppressed from notation. Affine charts, refinements, local equations, and factorizations chosen only inside proofs remain proof witnesses rather than additional structure.

Convention 2.1.1 (Relative and absolute notation). For a morphism of commutative ring spectra $A \rightarrow B$, its relative cotangent complex is written

$$L_{B/A} \in \text{Mod}_B.$$

For a morphism of spectral schemes $f : X \rightarrow S$, its relative cotangent complex is written

$$L_{X/S} \in \text{QCoh}(X).$$

When the base is the sphere spectrum or $\text{Spec}(\mathbb{S})$, respectively, we abbreviate

$$L_B := L_{B/\mathbb{S}}, \quad L_X := L_{X/\text{Spec}(\mathbb{S})}.$$

The space of A -linear derivations from B to a B -module M is written

$$\text{Der}_A(B, M).$$

No notation introduced in this chapter changes the variance conventions of Chapter 1.

Convention 2.1.2 (Under-over categories). Let $\mathcal{C}$ be an ∞ -category and fix a morphism $A \rightarrow B$ in $\mathcal{C}$. We write

$$\mathcal{C}_{A//B} := (\mathcal{C}_{A/})_{/B}.$$

Thus an object is a factorization

$$A \longrightarrow C \longrightarrow B$$

of the fixed morphism, with its canonical coherent comparison. In particular, $\text{CAlg}(\text{Sp})_{A//B}$ denotes commutative spectral algebras under A and over B , and $\text{CAlg}(\text{Sp})_{B//B}$ denotes retractive commutative B -algebras. This notation is an ordinary slice construction and introduces no additional structure.

Convention 2.1.3 (Theorem domains are properties). Connectivity, finite local freeness, local finite presentation, nilpotence, and representability by spectral schemes are predicates specifying theorem domains in this chapter. They are not auxiliary fields of structure attached to an object.

Convention 2.1.4 (Hooked arrows and geometric inclusions). A hooked arrow is used graphically for a specified geometric inclusion, in particular for open immersions and, when convenient, for closed or infinitesimal inclusions. Unless monomorphism is stated separately, the glyph $\hookrightarrow$ does not assert that the morphism is a monomorphism in the ambient ∞ -category. Thus a derived closed or square-zero immersion may be displayed with a hook without suppressing its higher derived self-intersection information.

Foundational tangent and cotangent input is taken from [34, 30, 37], while cofree spectral co-commutative coalgebras are used as in [4]. The differential-storage recognition uses [8, 29], with the linear/nonlinear and differential-lambda-calculus background of [6, 14]. The endpoint configuration is compared with differential cohesion only after its adjoint quadruple has been constructed; at that point we use the notation

$$\mathfrak{R} \dashv \mathfrak{S} \dashv \mathfrak{L}$$

for reduction, infinitesimal shape, and infinitesimal flat. Likewise, the stabilized endpoint semantics is first analyzed by the exact-adjunction fracture theorem proved below and only afterward compared with [12, Sections 2–3] and [24].

2.2 Stabilization and the linear semantics of a model

Chapter 1 attached the stable coefficient fibre Mod_B to an affine context $\mathrm{Spec}(B)$. To differentiate the doctrine, we first identify this fibre intrinsically as the linearization of the nonlinear coordinate category at B . This section establishes that tangent identification and records the split square-zero object supplied by the zeroth-space functor, which will make derivation terms visible in the next section.

2.2.1 The tangent category of commutative spectral algebra

Theorem 2.2.1 (Tangent category at a commutative ring spectrum). *Let $B \in \mathrm{CAlg}(\mathrm{Sp})$. There is a canonical equivalence of stable ∞ -categories*

$$\mathrm{Stab}(\mathrm{CAlg}(\mathrm{Sp})_{/B}) \simeq \mathrm{Mod}_B.$$

Here $\mathrm{Stab}(\mathrm{CAlg}(\mathrm{Sp})_{/B})$ denotes spectrum objects in the pointed slice, equivalently the stabilization whose zeroth-space functor lands in $(\mathrm{CAlg}(\mathrm{Sp})_{/B})_*$. Under this equivalence, the zeroth-space functor

$$\Omega^\infty : \mathrm{Stab}(\mathrm{CAlg}(\mathrm{Sp})_{/B}) \longrightarrow (\mathrm{CAlg}(\mathrm{Sp})_{/B})_*$$

identifies a B -module M with the split square-zero augmented algebra

$$B \longrightarrow B \oplus M \longrightarrow B.$$

Proof.

- (1) **Identify the resulting object.** The tangent theorem for commutative $\mathbb{E}_\infty$ -algebras identifies the stabilization of the slice of commutative algebras at B with the category of modules over B ; see [34, 30].
- (2) **Use terminality.** A pointed object of $\mathrm{CAlg}(\mathrm{Sp})_{/B}$ is a commutative algebra $C \rightarrow B$ together with a section from the terminal object $B \rightarrow B$, hence a diagram

$$B \longrightarrow C \longrightarrow B$$

whose composite is equivalent to id_B .

- (3) **Transport the structure.** Under the tangent equivalence, the zeroth-space functor sends M to the split square-zero algebra whose underlying B -module is $B \oplus M$ and whose multiplication on the M -summand is coherently null.

□

Remark 2.2.1 (Linear semantics at a model). The theorem identifies the stable coefficient fibre of an affine spectral context with the linearization of the nonlinear model category at its coordinate algebra:

$$\mathrm{CAlg}(\mathrm{Sp})/B \longrightarrow \mathrm{Stab}(\mathrm{CAlg}(\mathrm{Sp})/B) \simeq \mathrm{Mod}_B.$$

Thus the module doctrine of Chapter 1 simultaneously supplies the tangent semantic fibres needed for deformation theory.

2.2.2 Split square-zero algebra

Construction 2.2.1 (Split square-zero algebra). For $B \in \mathrm{CAlg}(\mathrm{Sp})$ and $M \in \mathrm{Mod}_B$, write

$$B \oplus M$$

for the commutative ring spectrum supplied by 2.2.1, with section and augmentation

$$B \xrightarrow{s} B \oplus M \xrightarrow{p} B, \quad p \circ s \simeq \mathrm{id}_B.$$

The underlying B -module is the direct sum $B \oplus M$, and the product of two elements in the coefficient summand is coherently zero.

Proposition 2.2.1 (Functoriality of split square-zero formation). *For every commutative ring spectrum B , split square-zero formation defines a limit-preserving functor*

$$\mathrm{Mod}_B \longrightarrow \mathrm{CAlg}(\mathrm{Sp})_{B//B}, \quad M \longmapsto (B \rightarrow B \oplus M \rightarrow B).$$

If B and M are connective, then $B \oplus M$ is connective.

Proof.

- (1) **Transport the structure.** Under 2.2.1, the displayed functor is the zeroth-space functor of a stabilization.
- (2) **Use the adjunction.** It is a right adjoint and therefore preserves limits.
- (3) **Verify connectivity.** The underlying spectrum of the split square-zero algebra is $B \oplus M$, so connectivity of both summands implies connectivity of the algebra.

□

The stable coefficient fibre has now been recovered from tangent stabilization, and every coefficient determines a split square-zero augmented algebra. What is still missing is the universal coefficient that classifies sections of these augmentations.

2.3 Derivation terms and cotangent classifiers

Relative derivations are the terms represented by split square-zero augmentations. This section first formulates those terms and then constructs their representing object, the relative cotangent complex. The free-algebra calculation at the end gives the basic model used repeatedly later.

2.3.1 Relative derivation terms

Definition 2.3.1 (Spectral derivation). Let $A \rightarrow B$ be a morphism of commutative ring spectra and let $M \in \text{Mod}_B$. The space of A -linear derivations from B to M is

$$\text{Der}_A(B, M) := \text{Map}_{\text{CAlg}(\text{Sp})_{A//B}}(B, B \oplus M).$$

Thus an A -linear derivation is a section of the augmentation $B \oplus M \rightarrow B$ in commutative A -algebras.

Proposition 2.3.1 (Elementary functoriality of derivations). *Let $A \rightarrow B \rightarrow C$ be morphisms of commutative ring spectra.*

(i) *A morphism of B -modules $M \rightarrow M'$ induces*

$$\text{Der}_A(B, M) \longrightarrow \text{Der}_A(B, M').$$

(ii) *For every C -module N , restriction along $B \rightarrow C$ induces*

$$\text{Der}_A(C, N) \longrightarrow \text{Der}_A(B, N),$$

where N is regarded on the right as a B -module.

Proof.

- (1) **Handle the first component.** The first map is postcomposition with the morphism of split square-zero algebras induced by $M \rightarrow M'$.
- (2) **Identify the resulting object.** For the second, compose an A -algebra section $C \rightarrow C \oplus N$ with $B \rightarrow C$ and use the canonical pullback identification

$$B \times_C (C \oplus N) \simeq B \oplus N.$$

- (3) **Verify naturality.** Naturality follows from functoriality of pullback and split square-zero formation.

□

Derivations are now identified as sections of split square-zero augmentations. The next step replaces this family of mapping spaces by one universal coefficient in the tangent fibre.

2.3.2 The cotangent classifier

Theorem 2.3.1 (Existence of the relative cotangent classifier). *Let $A \rightarrow B$ be a morphism of commutative ring spectra. There exists a B -module*

$$L_{B/A} \in \text{Mod}_B$$

and a universal A -linear derivation such that, for every B -module M , there is a natural equivalence

$$\text{Map}_{\text{Mod}_B}(L_{B/A}, M) \simeq \text{Der}_A(B, M).$$

The construction is functorial in commutative triangles of commutative ring spectra.

Proof.

- (1) **relative cotangent object of over.** The relative cotangent object of B over A is defined in the tangent bundle of $\mathbf{CAlg}(\mathbf{Sp})$.
- (2) **Use the adjunction.** The tangent equivalence 2.2.1 identifies its fibre at B with $\mathbf{Mod}_B$, and the stabilization adjunction gives the displayed representing equivalence.
- (3) **Verify functoriality.** Functoriality is the functoriality of the relative cotangent object in the tangent bundle; see [34, 30].

□

Corollary 2.3.1 (Cotangent classifier of an equivalence). *If $A \rightarrow B$ is an equivalence, then*

$$L_{B/A} \simeq 0.$$

Proof.

- (1) **Identify contractibility.** Relative derivation spaces are contractible for every coefficient module, so their representing object is zero.

□

Remark 2.3.1 (Classifier terminology). The phrase **cotangent classifier** records the role of $L_{B/A}$ in the indexed semantics. It does not introduce a second object: throughout the chapter the cotangent classifier is exactly the usual relative cotangent complex.

The cotangent classifier is defined abstractly by its representing property. Its first concrete calculation is on free spectral coordinates, where the classifier exposes the linear generators directly.

2.3.3 Free models

Proposition 2.3.2 (Cotangent classifier of a free commutative algebra). *Let $A \in \mathbf{CAlg}(\mathbf{Sp})$, let $M \in \mathbf{Mod}_A$, and put*

$$B := \mathbf{Sym}_A(M).$$

Then there is a natural equivalence of B -modules

$$L_{B/A} \simeq B \otimes_A M.$$

Proof.

- (1) **Use the adjunction.** For every B -module N , the free-algebra and extension-of-scalars adjunctions give

$$\begin{aligned} \mathbf{Der}_A(B, N) &\simeq \mathbf{Map}_{\mathbf{Mod}_A}(M, N) \\ &\simeq \mathbf{Map}_{\mathbf{Mod}_B}(B \otimes_A M, N). \end{aligned}$$

- (2) **Apply Yoneda.** Compare with 2.3.1 and apply Yoneda.

□

Corollary 2.3.2 (Cotangent classifier of brave-new affine space). *For*

$$A\{t_1, \dots, t_n\} := \mathbf{Sym}_A(A^{\oplus n}),$$

one has

$$L_{A\{t_1, \dots, t_n\}/A} \simeq A\{t_1, \dots, t_n\}^{\oplus n}.$$

Equivalently, the affine context $\mathbb{A}_{\mathbf{Spec} A}^n$ of 1.13.5 has free rank- n relative cotangent classifier.

We now have the relative cotangent classifier and its free-model calculation. For it to serve as the differential structure of an indexed geometry, it must also behave coherently under change of context and under composition.

2.4 Substitution, the chain rule, and intrinsic cotangent estimates

This section supplies that calculus. Cotangent base change is the substitution law, while transitivity is the chain rule. The subsequent connectivity, Hurewicz, rigidity, and finiteness results record the intrinsic estimates needed when cotangent data are later converted into geometric lifting and equation problems.

2.4.1 Substitution and transitivity

Theorem 2.4.1 (Substitution law for cotangent classifiers). *Let*

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ A' & \longrightarrow & B' := B \otimes_A A' \end{array}$$

be a pushout square of commutative ring spectra. Then there is a natural equivalence of B' -modules

$$L_{B/A} \otimes_B B' \simeq L_{B'/A'}.$$

Proof.

(1) **For every -module.** For every B' -module N ,

$$\begin{aligned} \mathrm{Map}_{\mathrm{Mod}_{B'}}(L_{B/A} \otimes_B B', N) &\simeq \mathrm{Map}_{\mathrm{Mod}_B}(L_{B/A}, N) \\ &\simeq \mathrm{Der}_A(B, N) \\ &\simeq \mathrm{Der}_{A'}(B', N) \\ &\simeq \mathrm{Map}_{\mathrm{Mod}_{B'}}(L_{B'/A'}, N). \end{aligned}$$

(2) **Apply the universal property.** The middle equivalence is the pushout universal property.

(3) **Apply Yoneda.** Yoneda gives the result. □

Corollary 2.4.1 (Substitution of derivation terms). *In the situation of 2.4.1, every B' -module N admits a natural equivalence*

$$\mathrm{Der}_{A'}(B', N) \simeq \mathrm{Der}_A(B, N).$$

Theorem 2.4.2 (Chain rule for cotangent classifiers). *For composable morphisms*

$$A \longrightarrow B \longrightarrow C$$

of commutative ring spectra, there is a natural cofibre sequence of C -modules

$$L_{B/A} \otimes_B C \longrightarrow L_{C/A} \longrightarrow L_{C/B}.$$

Proof.

- (1) **Fix the notation.** Let $N \in \text{Mod}_C$.
- (2) **Compute the fibre.** Restriction of an A -linear derivation of C along $B \rightarrow C$ gives a natural fibre sequence of derivation spaces

$$\text{Der}_B(C, N) \longrightarrow \text{Der}_A(C, N) \longrightarrow \text{Der}_A(B, N),$$

where N is regarded in the last term as a B -module by restriction of scalars.

- (3) **Compute the fibre.** The homotopy fibre over the zero restricted derivation consists exactly of those A -linear derivations of C which vanish on B , that is, the B -linear derivations.
- (4) **Use the adjunction.** By the cotangent representing property and extension–restriction adjunction, the three spaces are naturally

$$\begin{aligned} \text{Der}_B(C, N) &\simeq \text{Map}_{\text{Mod}_C}(L_{C/B}, N), \\ \text{Der}_A(C, N) &\simeq \text{Map}_{\text{Mod}_C}(L_{C/A}, N), \\ \text{Der}_A(B, N) &\simeq \text{Map}_{\text{Mod}_C}(L_{B/A} \otimes_B C, N). \end{aligned}$$

- (5) **Transport the same comparison.** The same equivalences hold naturally after replacing N by every shift $N[k]$.
- (6) **Compute the mapping object.** Since the mapping spectrum in a stable ∞ -category is determined by these shifted mapping spaces, the preceding fibre sequences assemble to a natural fibre sequence of mapping spectra

$$\text{map}_{\text{Mod}_C}(L_{C/B}, N) \longrightarrow \text{map}_{\text{Mod}_C}(L_{C/A}, N) \longrightarrow \text{map}_{\text{Mod}_C}(L_{B/A} \otimes_B C, N).$$

- (7) **Identify the resulting comparison.** This holds naturally for every C -module N .
- (8) **Apply Yoneda.** Stable Yoneda therefore identifies

$$L_{B/A} \otimes_B C \longrightarrow L_{C/A} \longrightarrow L_{C/B}$$

as a cofibre sequence in Mod_C .

□

Lemma 2.4.1 (Spectral localizations have trivial relative cotangent). *Let $A \in \text{CAlg}(\text{Sp})$ and let $f \in \pi_0(A)$. Then*

$$L_{A[f^{-1}]/A} \simeq 0.$$

The equivalence is natural under further localization.

Proof.

- (1) **Fix the notation.** Put $A_f := A[f^{-1}]$.
- (2) **Identify the resulting object.** For every A_f -module N , the derivation classifier identifies

$$\text{Map}_{\text{Mod}_{A_f}}(L_{A_f/A}, N) \simeq \text{Der}_A(A_f, N).$$

- (3) **Compute the fibre.** The right-hand side is the homotopy fibre over id_{A_f} of

$$\text{Map}_{\text{CAlg}(\text{Sp})_{A_f}}(A_f, A_f \oplus N) \longrightarrow \text{Map}_{\text{CAlg}(\text{Sp})_{A_f}}(A_f, A_f),$$

where the map is induced by the split augmentation $A_f \oplus N \rightarrow A_f$.

- (4) **Verify invertibility.** In both target algebras the image of f is invertible.
- (5) **Apply the universal property.** The universal property of spectral localization therefore makes both displayed mapping spaces contractible, and the augmentation-induced map between them is an equivalence.
- (6) **Conclude the step.** Hence
- $$\text{Der}_A(A_f, N) \simeq *$$
- for every N .
- (7) **Apply Yoneda.** The representing property and Yoneda give $L_{A_f/A} \simeq 0$.
- (8) **Apply the universal property.** Naturality is inherited from the functorial universal property of localization.

□

Proposition 2.4.1 (Affine open immersions have trivial relative cotangent). *Let*

$$j : \text{Spec}(B) \hookrightarrow \text{Spec}(A)$$

be an affine spectral Zariski open immersion. Then

$$L_{B/A} \simeq 0.$$

Proof.

- (1) **Construct the factorization.** Choose a finite family of principal spectral opens

$$D_A^{\text{sp}}(f_\alpha) = \text{Spec}(A[f_\alpha^{-1}]) \longrightarrow \text{Spec}(A)$$

which factor through $\text{Spec}(B)$ and whose pullbacks cover $\text{Spec}(B)$; such a family exists by the principal-basis theorem of Chapter 1.

- (2) **Fix the notation.** Put

$$B_\alpha := B \otimes_A A[f_\alpha^{-1}].$$

- (3) **Construct the factorization.** Because the principal open factors through $\text{Spec}(B)$, the canonical map

$$B_\alpha \longrightarrow A[f_\alpha^{-1}]$$

is an equivalence.

- (4) **Apply cotangent transitivity.** By 2.4.1, principal spectral localization has vanishing relative cotangent complex, so transitivity for $A \rightarrow B \rightarrow B_\alpha$ gives

$$L_{B/A} \otimes_B B_\alpha \simeq 0.$$

- (5) **Use joint conservativity.** The substitution functors $- \otimes_B B_\alpha$ are jointly conservative for a finite affine Zariski cover by Chapter 1.
- (6) **Conclude the step.** Therefore $L_{B/A} \simeq 0$.

□

Remark 2.4.1 (Substitution and composition). Theorem 2.4.1 is the substitution law for the differentiated spectral doctrine. Theorem 2.4.2 is its chain rule: variation along a composite context map is assembled from variation in the first stage and variation relative to that stage. No further exchange law is imposed; these morphisms and equivalences are the canonical ones constructed by the representing property.

Substitution and the chain rule describe how cotangent classifiers interact with context change. We now compare the classifier with the underlying relative homotopy of a map; this is the bridge to conormal and rigidity arguments.

2.4.2 Connectivity and relative Hurewicz comparison

Proposition 2.4.2 (Connectivity of the relative cotangent classifier). *Let $A \rightarrow B$ be a morphism of connective commutative ring spectra. Then*

$$L_{B/A} \in (\text{Mod}_B)_{\geq 0}.$$

Proof.

- (1) **Verify connectivity.** This is the connective relative cotangent theorem for commutative ring spectra; see [34].
- (2) **Verify connectivity.** Equivalently, resolve B over A by connective free commutative algebras, apply 2.3.2, and use compatibility with geometric realization.

□

Construction 2.4.1 (Canonical relative Hurewicz morphism). Let

$$A \longrightarrow B$$

be a morphism of commutative ring spectra and put

$$Q := \text{cofib}(A \rightarrow B)$$

in Mod_A . For a B -module N , let

$$d : B \longrightarrow B \oplus N$$

be an A -linear derivation and let

$$d_0 : B \longrightarrow B \oplus N$$

be the zero derivation. On underlying A -modules, the difference $d - d_0$ has zero component in the B -summand and therefore determines an A -linear morphism

$$\delta_d : B \longrightarrow N.$$

Because d and d_0 agree after restriction to A , the composite $A \rightarrow B \rightarrow N$ is canonically null and δ_d factors through a canonical morphism

$$\bar{\delta}_d : Q \longrightarrow N.$$

Adjunction gives a B -linear morphism

$$B \otimes_A Q \longrightarrow N.$$

The construction is natural in N . Using

$$\mathrm{Der}_A(B, N) \simeq \mathrm{Map}_{\mathrm{Mod}_B}(L_{B/A}, N),$$

Yoneda therefore determines a canonical natural morphism

$$\epsilon_{B/A} : B \otimes_A Q \longrightarrow L_{B/A}.$$

We call $\epsilon_{B/A}$ the **relative Hurewicz morphism**. It is characterized by the property that, for every classifying map $\eta : L_{B/A} \rightarrow N$, the adjoint of $\eta \epsilon_{B/A}$ is the factor $Q \rightarrow N$ of the underlying relative derivation.

To compare with the standard construction, let

$$L_B \longrightarrow L_{B/A}$$

be the transitivity morphism. Its associated square-zero extension has fibre $L_{B/A}[-1]$, and the induced boundary $Q \rightarrow L_{B/A}$ is, by the preceding characterization, the adjoint of $\epsilon_{B/A}$. This is exactly the construction of [34, Construction 7.4.3.10]. Thus the morphism above is the standard canonical Hurewicz morphism.

Theorem 2.4.3 (Relative Hurewicz–conormal comparison). *Let $f : A \rightarrow B$ be a morphism of connective commutative ring spectra which is surjective on π_0 , and put*

$$K := \mathrm{fib}(A \rightarrow B).$$

Then K is connective, $L_{B/A}$ is 1-connective, and shifting the Hurewicz morphism of 2.4.1 gives a canonical natural morphism

$$B \otimes_A K \longrightarrow L_{B/A}[-1]$$

which is an isomorphism on π_0 . More generally, if

$$Q := \mathrm{cofib}(A \rightarrow B)$$

is n -connective for $n \geq 1$, then the fibre of

$$\epsilon_{B/A} : B \otimes_A Q \longrightarrow L_{B/A}$$

is $2n$ -connective. In particular, the Hurewicz morphism induces an isomorphism on the first possible nonzero homotopy group.

Proof.

- (1) **Verify connectivity.** Surjectivity of $\pi_0(A) \rightarrow \pi_0(B)$ makes the fibre K connective and gives a canonical equivalence

$$Q = \mathrm{cofib}(A \rightarrow B) \simeq K[1].$$

- (2) **Verify connectivity.** The connectivity estimate for the canonical Hurewicz morphism is [34, Theorem 7.4.3.12].
- (3) **Verify connectivity.** Applied with $n = 1$, it makes $L_{B/A}$ 1-connective and the fibre of $B \otimes_A Q \rightarrow L_{B/A}$ 2-connective.
- (4) **Derive the comparison.** Shifting by $[-1]$ gives the displayed conormal comparison and its π_0 -isomorphism.
- (5) **Transport the same comparison.** The same theorem for arbitrary $n \geq 1$ gives the final statement.

□

Lemma 2.4.2 (Degree-zero quotient of the derived conormal module). *In the situation of 2.4.3, put*

$$R := \pi_0(A), \quad S := \pi_0(B), \quad I := \ker(R \rightarrow S).$$

Then the relative fibre determines a canonical surjection

$$\pi_0(B \otimes_A K) \twoheadrightarrow I/I^2.$$

Under the Hurewicz isomorphism of 2.4.3, this is a canonical quotient of $\pi_0(L_{B/A}[-1])$ onto the ordinary conormal module I/I^2 .

Proof.

- (1) **Verify surjectivity.** The long exact homotopy sequence of $K \rightarrow A \rightarrow B$ shows that the image of $\pi_0(K) \rightarrow R$ is exactly I , so there is a surjection

$$\pi_0(K) \twoheadrightarrow I.$$

- (2) **Verify connectivity.** Because B and K are connective, the connective tensor-product spectral sequence gives

$$\pi_0(B \otimes_A K) \simeq S \otimes_R \pi_0(K).$$

- (3) **Verify surjectivity.** Tensoring the preceding surjection with $S = R/I$ gives

$$S \otimes_R \pi_0(K) \twoheadrightarrow S \otimes_R I \simeq I/I^2.$$

- (4) **Verify naturality.** Naturality follows from naturality of the fibre sequence and scalar extension.

□

Corollary 2.4.2 (Cotangent rigidity). *Let $A \rightarrow B$ be a morphism of connective commutative ring spectra. If*

$$\pi_0(A) \xrightarrow{\cong} \pi_0(B) \quad \text{and} \quad L_{B/A} \simeq 0,$$

then $A \rightarrow B$ is an equivalence.

Proof.

- (1) **Identify the resulting comparison.** This is the cotangent rigidity consequence of the relative Hurewicz theorem; see [34, Corollary 7.4.3.4].
- (2) **Identify the map.** The proof there applies the first-nonzero-relative-homotopy comparison supplied by the Hurewicz morphism.
- (3) **Verify connectivity.** The hypotheses displayed here are exactly its connective theorem domain.

□

The Hurewicz comparison controls the first nonzero relative homotopy and yields cotangent rigidity. A second intrinsic use of the same classifier is to detect local finite presentation.

2.4.3 Finiteness detected by cotangent data

Theorem 2.4.4 (Cotangent criterion for local finite presentation). *Let $A \rightarrow B$ be a morphism of connective commutative ring spectra.*

- (i) *If B is locally of finite presentation over A , then $L_{B/A}$ is a perfect B -module.*
- (ii) *Conversely, if $L_{B/A}$ is perfect and $\pi_0(B)$ is finitely presented as a $\pi_0(A)$ -algebra, then B is locally of finite presentation over A .*

Proof.

- (1) **Verify connectivity.** This is the connective spectral cotangent criterion for local finite presentation; see [30, 34, 37].
- (2) **theorem is invoked here as foundational spectral.** The theorem is invoked here as foundational spectral algebra rather than reproved by a parallel cell theory.

□

These results give the affine cotangent interface needed later: substitution and the chain rule control variation, the Hurewicz comparison controls the first relative homotopy, and cotangent data detect the stated finiteness property. The remaining problem is globalization, because the subsequent geometric constructions require one cotangent object on an arbitrary spectral scheme rather than a collection of affine choices.

2.5 The intrinsic global cotangent doctrine

We globalize the affine classifier using only structures already constructed in Chapter 1. Compatible affine cotangent complexes are transported by substitution and glued by spectral Zariski descent in QCoh . The structured ringed-topos cotangent complex is used only afterward as a recognition of the resulting intrinsic object.

In the globalizations below, once affine data and comparison maps have been shown compatible on common refinements, full faithfulness of spectral Zariski descent is used to glue them uniquely. The affine covers and refinements used for this verification remain proof witnesses.

2.5.1 Construction by affine coefficient descent

Definition 2.5.1 (Compatible affine pair for a relative context). Let $f : X \rightarrow S$ be a morphism of spectral schemes. A **compatible affine pair** for f is a diagram

$$\begin{array}{ccc} U = \operatorname{Spec}(B) & \hookrightarrow & X \\ \downarrow & & \downarrow f \\ V = \operatorname{Spec}(A) & \hookrightarrow & S \end{array}$$

in which the horizontal arrows are affine open immersions. A refinement is a commutative diagram of such pairs induced by affine open restriction in the source and target.

Lemma 2.5.1 (Compatibility of affine cotangent classifiers). *Let (U, V) and (U', V') be compatible affine pairs for $f : X \rightarrow S$, with*

$$U' = \operatorname{Spec}(B') \rightarrow U = \operatorname{Spec}(B), \quad V' = \operatorname{Spec}(A') \rightarrow V = \operatorname{Spec}(A)$$

a refinement. Then there is a canonical equivalence

$$L_{B/A} \otimes_B B' \xrightarrow{\simeq} L_{B'/A'}.$$

These equivalences are coherent under identities, composition, and iterated refinement.

Proof.

(1) **square is a pushout.** The square

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ A' & \longrightarrow & B \otimes_A A' \end{array}$$

is a pushout.

(2) **Apply cotangent base change.** Cotangent base change gives

$$L_{B/A} \otimes_B (B \otimes_A A') \simeq L_{(B \otimes_A A')/A'}.$$

(3) **Identify the map.** The further morphism $B \otimes_A A' \rightarrow B'$ represents an affine open immersion, so its relative cotangent complex vanishes.

(4) **Apply cotangent transitivity.** Transitivity therefore identifies

$$L_{(B \otimes_A A')/A'} \otimes_{B \otimes_A A'} B' \simeq L_{B'/A'}.$$

(5) **Derive the comparison.** Composing yields the displayed equivalence.

(6) **Apply cotangent base change.** All comparison maps are obtained from functorial cotangent base change and transitivity, so their higher compatibilities are inherited from those constructions.

□

Construction 2.5.1 (Native global relative cotangent classifier). Let $f : X \rightarrow S$ be a morphism of spectral schemes. On every compatible affine pair

$$U = \operatorname{Spec}(B) \longrightarrow V = \operatorname{Spec}(A)$$

place the quasi-coherent coefficient corresponding to $L_{B/A}$. By 2.5.1, these coefficients agree canonically on every common affine refinement, with all higher compatibility. These source charts form a Zariski basis of X : given an affine open of X , pull back an affine open cover of S and refine the resulting cover by affine principal opens of the source. The quasi-coherent descent theorem of Chapter 1 produces a canonically determined object

$$L_{X/S} \in \operatorname{QCoh}(X).$$

We call it the **global relative cotangent classifier**.

Theorem 2.5.1 (Affine characterization and uniqueness). *For every morphism $f : X \rightarrow S$, the object of 2.5.1 is characterized by the following property: whenever*

$$U = \operatorname{Spec}(B) \longrightarrow X$$

is an affine open whose composite with S factors through an affine open $V = \operatorname{Spec}(A) \rightarrow S$, there is a canonical equivalence

$$L_{X/S}|_U \simeq L_{B/A},$$

compatible with every further affine restriction. The space of descent gluings of this specified compatible affine cotangent datum is contractible. Moreover

$$L_{X/S} \in \operatorname{QCoh}(X)_{\geq 0}.$$

Proof.

- (1) **Set up the argument.** Existence is the descent construction above.
- (2) **Use full faithfulness.** Full faithfulness of spectral Zariski descent makes the space of descent gluings of the fixed affine datum contractible and proves uniqueness up to its canonical contractible comparison space.
- (3) **Verify connectivity.** Each affine restriction $L_{B/A}$ is connective by 2.4.2; connectivity in $\operatorname{QCoh}(X)$ is detected on affine opens, so $L_{X/S}$ is connective.

□

Remark 2.5.1 (Recognition in the structured spectral model). Under the comparison of Chapter 1 with structured spectral schemes, the intrinsically descended object $L_{X/S}$ identifies with the sheaf-level cotangent complex

$$L_{\mathcal{O}_X/f^{-1}\mathcal{O}_S}$$

of [37]. This comparison recognizes an object already constructed in the native coefficient doctrine. No small ringed topos, affine atlas, refinement, or independent descent datum is retained in the definition of $L_{X/S}$.

The global object has now been constructed by affine coefficient descent. Its first structural test is whether the affine substitution law survives that gluing and hence defines genuine change of context globally.

2.5.2 Global substitution and transitivity

Theorem 2.5.2 (Global cotangent substitution law). *Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ f' \downarrow & & \downarrow f \\ S' & \xrightarrow{g} & S \end{array}$$

be Cartesian in spectral schemes. Then there is a canonical equivalence

$$L_{X'/S'} \simeq (g')^* L_{X/S}.$$

The equivalence is coherent under identities, composition, and iterated Cartesian base change.

Proof.

- (1) **Choose the local data.** Choose a compatible affine pair

$$U = \operatorname{Spec}(B) \longrightarrow V = \operatorname{Spec}(A)$$

for $X \rightarrow S$.

- (2) **After pullback to , choose an affine.** After pullback to S' , choose an affine open $V' = \operatorname{Spec}(A') \rightarrow S'$ mapping to V , form

$$\bar{B} := B \otimes_A A',$$

and refine the corresponding source open by an affine open $U' = \operatorname{Spec}(B') \rightarrow \operatorname{Spec}(\bar{B})$.

- (3) **Identify the map.** The latter map is an affine spectral Zariski open immersion.
 (4) **affine comparison on this refinement.** The affine comparison on this refinement is therefore

$$\begin{aligned} L_{B/A} \otimes_B B' &\simeq (L_{B/A} \otimes_B \bar{B}) \otimes_{\bar{B}} B' \\ &\simeq L_{\bar{B}/A'} \otimes_{\bar{B}} B' \\ &\simeq L_{B'/A'}. \end{aligned}$$

- (5) **Apply cotangent base change.** The middle equivalence is cotangent base change and the last is transitivity together with $L_{B'/\bar{B}} \simeq 0$.
 (6) **this is precisely the compatible-pair comparison of.** Equivalently, this is precisely the compatible-pair comparison of 2.5.1.
 (7) **Verify compatibility.** These affine comparisons commute with every further compatible refinement by the coherence established in that lemma.
 (8) **Use full faithfulness.** Full faithfulness of quasi-coherent Zariski descent therefore glues them uniquely to

$$(g')^* L_{X/S} \xrightarrow{\simeq} L_{X'/S'}.$$

- (9) **Apply base change.** Identity, composition, and iterated-base-change coherence are inherited from the affine cotangent base-change and transitivity constructions and are detected on the same affine basis.

□

Theorem 2.5.3 (Global cotangent chain rule). *For composable morphisms of spectral schemes*

$$X \xrightarrow{f} Y \xrightarrow{g} S,$$

there is a canonical cofibre sequence in $\mathrm{QCoh}(X)$

$$f^*L_{Y/S} \longrightarrow L_{X/S} \longrightarrow L_{X/Y}.$$

Proof.

- (1) **Restrict the construction.** Restrict to an affine open of X , refine its image in Y and S by compatible affine opens, and apply 2.4.2.
- (2) **Compute the fibre.** The affine cofibre sequences and their maps are compatible under further restriction.
- (3) **Apply Zariski descent.** Since finite limits and colimits in the stable category $\mathrm{QCoh}(X)$ are detected by affine Zariski descent, the local sequences glue to the asserted global cofibre sequence.

□

Corollary 2.5.1 (Open context extensions have trivial relative tangent). *If $j : U \hookrightarrow X$ is an open immersion of spectral schemes, then*

$$L_{U/X} \simeq 0.$$

Proof.

- (1) **Set up the argument.** The assertion is local on source and target.
- (2) **After restricting to compatible affine opens it.** After restricting to compatible affine opens it is exactly 2.4.1.
- (3) **Glue the local data.** The affine vanishings glue uniquely by 2.5.1.

□

With global substitution and the chain rule in place, the global cotangent object can be used exactly as in the affine case: mapping out of it defines relative derivation terms on a spectral context.

2.5.3 Global derivation terms

Definition 2.5.2 (Relative derivation term on a spectral context). Let $f : X \rightarrow S$ be a spectral scheme and let $M \in \mathrm{QCoh}(X)$. Define the space of relative derivation terms with coefficient M by

$$\mathrm{Der}_S(X; M) := \mathrm{Map}_{\mathrm{QCoh}(X)}(L_{X/S}, M).$$

The universal element corresponding to $\mathrm{id}_{L_{X/S}}$ is called the universal relative S -derivation term.

Proposition 2.5.1 (Affine computation of global derivation terms). *Let $X \rightarrow S$ and M be as above. Choose an affine spectral Zariski atlas $U \rightarrow X$, let $U_\bullet \rightarrow X$ be its Čech nerve, and on each U_n choose an affine-open basis subordinate to compatible affine opens of S . Then*

$$\mathrm{Der}_S(X; M) \simeq \mathrm{Tot}_{[n] \in \Delta} \mathrm{Map}_{\mathrm{QCoh}(U_n)}(L_{X/S}|_{U_n}, M|_{U_n}).$$

Each mapping space on the right is in turn computed on the chosen compatible affine-open basis by 2.3.1. The resulting equivalence is independent of the computational atlas and all affine refinements.

Proof.

- (1) **Identify the resulting object.** Chapter 1 identifies $\mathrm{QCoh}(X)$ with the limit of its values on the ordinary Čech nerve of an affine Zariski atlas; no hypercompletion is used.
- (2) **Compute the mapping object.** Mapping spaces in a limit of ∞ -categories are limits of mapping spaces.
- (3) **Conclude the step.** Therefore

$$\mathrm{Map}_{\mathrm{QCoh}(X)}(L_{X/S}, M) \simeq \mathrm{Tot}_{[n] \in \Delta} \mathrm{Map}_{\mathrm{QCoh}(U_n)}(L_{X/S}|_{U_n}, M|_{U_n}).$$

- (4) **Compute the mapping object.** On every compatible affine open $\mathrm{Spec}(B) \rightarrow \mathrm{Spec}(A)$, 2.5.1 identifies the restricted cotangent object with $L_{B/A}$, and the affine mapping space is computed by 2.3.1.
- (5) **Use full faithfulness.** Common affine refinements and full faithfulness of Zariski descent prove independence of all proof choices.

□

The preceding results can be compressed into an indexed statement. Rather than treating each base-change equivalence separately, we regard the global cotangent classifier as a section over the category of relative contexts.

2.5.4 The cotangent classifier as a Cartesian section

Let

$$\mathrm{Rel}_{\mathrm{Cart}}^{\mathrm{sp}} \subseteq \mathrm{Fun}(\Delta^1, \mathrm{Sch}^{\mathrm{sp}})$$

be the wide subcategory whose objects are relative spectral contexts $X \rightarrow S$ and whose morphisms are Cartesian squares. Write

$$s : \mathrm{Rel}_{\mathrm{Cart}}^{\mathrm{sp}} \longrightarrow \mathrm{Sch}^{\mathrm{sp}}, \quad s(X \rightarrow S) = X$$

for source evaluation. Pulling the Cartesian fibration $\int \mathrm{QCoh} \rightarrow \mathrm{Sch}^{\mathrm{sp}}$ of Chapter 1 back along s gives the coefficient fibration over relative contexts. The theorem below says that the cotangent classifier is a Cartesian section of precisely this fibration.

Theorem 2.5.4 (Cartesian cotangent section of the differentiated doctrine). *The assignment*

$$(X \rightarrow S) \longmapsto L_{X/S} \in \mathrm{QCoh}(X)$$

defines a canonical Cartesian section of the pulled-back coefficient fibration over $\mathrm{Rel}_{\mathrm{Cart}}^{\mathrm{SP}}$. Equivalently, for every Cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow & & \downarrow \\ S' & \xrightarrow{g} & S, \end{array}$$

the Cartesian transport of $L_{X/S}$ is canonically

$$(g')^* L_{X/S} \simeq L_{X'/S'}.$$

The identity, composition, and higher-pasting coherences are those of the Cartesian transport and of global cotangent substitution.

Proof.

- (1) **Restrict the construction.** The global cotangent object is defined by its compatible affine restrictions.
- (2) **Identify the resulting object.** For every Cartesian square the equivalence of 2.5.2 identifies the pullback of those affine classifiers with the classifiers of the pulled-back relative context.
- (3) **These equivalences.** These equivalences are coherent under identities and iterated Cartesian squares by that theorem.
- (4) **They are therefore exactly the data asserting.** They are therefore exactly the data asserting that the section is Cartesian in the pulled-back coefficient fibration.

□

Remark 2.5.2 (Differentiated substitution and chain rule). The Cartesian-section theorem is the substitution law of the differentiated spectral doctrine. The global cofibre sequence

$$f^* L_{Y/S} \longrightarrow L_{X/S} \longrightarrow L_{X/Y}$$

of 2.5.3 is its chain rule. Both are constructed consequences of the cotangent representing property and spectral Zariski descent; no independent differential coherence is imposed.

The resulting interface is now fixed: affinely the stable coefficient fibre is the tangent linearization, while globally $L_{X/S}$ classifies variation, is transported Cartesianly by substitution, and satisfies the chain rule. The first branch can therefore remain entirely inside the stable coefficient fibre. The cotangent classifier will re-enter only at the comparison with the universal spectral derivation and then as the input to the geometric branch.

2.6 Differential-linear structure of the stable coefficient fibres

The first branch remains entirely inside the stable coefficient fibre. Affinely this fibre is literally the tangent stabilization identified in 2.2.1; for a general spectral scheme we use the Zariski-descended fibre $\mathrm{QCoh}(X)$ and make no additional global stabilization claim. We now construct on every such stable coefficient fibre the cofree cocommutative-coalgebra exponential, its primitive infinitesimal structure, and the resulting codereliction and deriving transformation. The branch concludes by comparing that derivative with the spectral universal derivation and by recording its actual oplax behavior under substitution.

2.6.1 Cofree cocommutative coalgebras

Definition 2.6.1 (Cocommutative coalgebras in a stable coefficient fibre). For a spectral context X , put

$$\mathcal{L}_X := \mathrm{QCoh}(X)$$

and define

$$\mathrm{CoCom}(\mathcal{L}_X) := \mathrm{CAlg}(\mathcal{L}_X^{\mathrm{op}})^{\mathrm{op}}.$$

Write

$$U_X : \mathrm{CoCom}(\mathcal{L}_X) \longrightarrow \mathcal{L}_X$$

for the forgetful functor.

Theorem 2.6.1 (Cofree coalgebra and the linear exponential). *For every spectral context X , the forgetful functor U_X admits a cofree right adjoint*

$$C_X : \mathcal{L}_X \longrightarrow \mathrm{CoCom}(\mathcal{L}_X).$$

Define

$$!_X := U_X C_X : \mathcal{L}_X \longrightarrow \mathcal{L}_X.$$

Then $!_X$ carries the canonical comonad structure induced by $U_X \dashv C_X$. For every $M \in \mathcal{L}_X$, the object $!_X M$ carries canonical cocommutative comonoid maps

$$\Delta_M : !_X M \longrightarrow !_X M \otimes !_X M, \quad e_M : !_X M \longrightarrow \mathcal{O}_X.$$

Proof.

- (1) **Use preservation.** The category $\mathcal{L}_X = \mathrm{QCoh}(X)$ is presentable, stable, and symmetric monoidal, and its tensor product is accessible and preserves colimits separately.
- (2) **existence theorem for cofree spectral cocommutative coalgebras.** The existence theorem for cofree spectral cocommutative coalgebras therefore applies; see [4].
- (3) **Use the product structure.** The same source identifies $\mathrm{CoCom}(\mathcal{L}_X)$ as Cartesian symmetric monoidal and the forgetful functor U_X as symmetric monoidal, so the underlying object of a product of coalgebras is the tensor product of the underlying coefficients.
- (4) **Use the adjunction.** The adjunction $U_X \dashv C_X$ produces the comonad $U_X C_X$.
- (5) **Use the stated property.** Since $C_X(M)$ is a cocommutative coalgebra, its underlying object carries the stated diagonal and counit.

□

Throughout this section we repeatedly compare morphisms into a cofree coalgebra. We use the following consequence of the adjunction without reintroducing it each time: two such morphisms are canonically compared once their cofree adjuncts are canonically compared, and the corresponding higher coherence is transported through the same mapping-space equivalence. Thus later proofs will expose the adjunct that controls a comparison rather than repeat the general uniqueness argument.

Proposition 2.6.1 (Canonical monoidal structure of the cofree exponential). *For $M, N \in \mathcal{L}_X$, there is a canonical coalgebra morphism*

$$\tilde{m}_{M,N} : C_X(M) \times C_X(N) \longrightarrow C_X(M \otimes N)$$

whose cofree adjunct is

$$!_X M \otimes !_X N \xrightarrow{\varepsilon_M \otimes \varepsilon_N} M \otimes N.$$

Write

$$m_{M,N} := U_X \tilde{m}_{M,N} : !_X M \otimes !_X N \longrightarrow !_X (M \otimes N).$$

The identity of the tensor unit similarly has a canonical cofree lift whose underlying map is

$$m_{\mathcal{O}_X} : \mathcal{O}_X \longrightarrow !_X \mathcal{O}_X.$$

These maps give C_X its canonical lax symmetric-monoidal structure and make $!_X = U_X C_X$ a lax symmetric-monoidal comonad. The comonad counit and comultiplication are monoidal, and every $m_{M,N}$ and $m_{\mathcal{O}_X}$ is a morphism of cocommutative coalgebras. Consequently, after passage to the homotopy category, the maps

$$(!_X, \delta, \varepsilon, \Delta, e, m, m_{\mathcal{O}})$$

form the ordinary monoidal coalgebra-modality structure associated to the cofree cocommutative coalgebra.

Proof.

- (1) **product coalgebra.** The product coalgebra

$$C_X(M) \times C_X(N)$$

has underlying coefficient

$$!_X M \otimes !_X N.$$

- (2) **Use the cofree universal property.** The cofree mapping-space equivalence therefore gives a canonical coalgebra morphism

$$\tilde{m}_{M,N}$$

corresponding to the specified map

$$\varepsilon_M \otimes \varepsilon_N : !_X M \otimes !_X N \longrightarrow M \otimes N.$$

- (3) **Establish the next comparison.** Equivalently,

$$\varepsilon_{M \otimes N} m_{M,N} \simeq \varepsilon_M \otimes \varepsilon_N.$$

- (4) **Use the adjunction.** For the tensor unit, the terminal cocommutative coalgebra has underlying object $\mathcal{O}_X$; applying the same adjunction to $\text{id}_{\mathcal{O}_X}$ gives $m_{\mathcal{O}_X}$.
- (5) **Associativity, unitality, and symmetry require no additional.** Associativity, unitality, and symmetry require no additional choices.

- (6) **Use the cofree universal property.** For example, the two coalgebra maps

$$C_X(M) \times C_X(N) \times C_X(P) \rightrightarrows C_X(M \otimes N \otimes P)$$

obtained from the two bracketings are carried by the cofree mapping-space equivalence to the same specified adjunct

$$\varepsilon_M \otimes \varepsilon_N \otimes \varepsilon_P.$$

- (7) **Use the cofree universal property.** By the cofree comparison convention above, the common adjunct determines the canonical comparison between the two coalgebra maps and its higher coherence.
- (8) **unit and symmetry comparisons.** The unit and symmetry comparisons are obtained identically.
- (9) **Use the adjunction.** This is precisely the canonical lax symmetric-monoidal structure on the right adjoint of the symmetric monoidal functor U_X .
- (10) **counit is monoidal by the defining adjunct.** The counit is monoidal by the defining adjunct equation above.
- (11) **For the comultiplication, recall that where.** For the comultiplication, recall that

$$\delta_M = U_X \alpha_{C_X(M)},$$

where α is the unit of $U_X \dashv C_X$.

- (12) **Compare the two coalgebra.** Compare the two coalgebra maps from

$$C_X(M) \times C_X(N)$$

to

$$C_X(!_X(M \otimes N))$$

which express monoidality of δ .

- (13) **Use the cofree universal property.** After composition with the cofree counit, both correspond to the same specified adjunct $m_{M,N}$, using the triangle identity

$$\varepsilon_{U_X D} U_X \alpha_D \simeq \text{id}_{U_X D}.$$

- (14) **Transport the same comparison.** The same cofree comparison principle gives the canonical monoidality comparison for δ , together with its inherited higher coherence.
- (15) **Conclude the argument.** Finally,

$$\tilde{m}_{M,N}$$

and the unit lift are morphisms in

$$\text{CoCom}(\mathcal{L}_X)$$

by construction.

- (16) **Use preservation.** Their underlying maps therefore preserve Δ and e .

- (17) **These are exactly the remaining monoidal coalgebra-modality.** These are exactly the remaining monoidal coalgebra-modality compatibilities after passage to the homotopy category. $\square$

Remark 2.6.1 (No naive symmetric-power formula). The notation $!_X M$ denotes the underlying object of the cofree cocommutative coalgebra. Its definition throughout this chapter is the universal property of C_X . In spectra the cosifted-limit and Tate phenomena governing cofree coalgebras obstruct a naive infinite-product formula in general; see [4].

Proposition 2.6.2 (Seely equivalences in the stable coefficient fibre). *For $M, N \in \mathcal{L}_X$, there are canonical equivalences*

$$\chi_{M,N} : !_X(M \oplus N) \xrightarrow{\simeq} !_X M \otimes !_X N, \quad \chi_0 : !_X 0 \xrightarrow{\simeq} \mathcal{O}_X.$$

They are coherent under identities, associativity, symmetry, and finite iteration. Under the product description of $\text{CoCom}(\mathcal{L}_X)$, the two components of the inverse comparison

$$\chi_{M,N}^{-1} : !_X M \otimes !_X N \longrightarrow !_X(M \oplus N)$$

after composition with the cofree counit are

$$\varepsilon_M \otimes e_N : !_X M \otimes !_X N \longrightarrow M, \quad e_M \otimes \varepsilon_N : !_X M \otimes !_X N \longrightarrow N.$$

After passage to the homotopy category these are the Seely isomorphisms of the storage modality associated to the cofree cocommutative coalgebra.

Proof.

- (1) **Use the product structure.** The category of cocommutative coalgebras is Cartesian on the theorem domain of 2.6.1, and its Cartesian product is carried by U_X to the tensor product of underlying objects; see [4].
- (2) **Use the adjunction.** Since C_X is a right adjoint, it preserves products.
- (3) **Use the product structure.** Finite products in the stable category $\mathcal{L}_X$ are biproducts, so there is a canonical equivalence

$$C_X(M \oplus N) \simeq C_X(M) \times C_X(N).$$

- (4) **Use terminality.** Applying U_X gives $\chi_{M,N}$, and preservation of the terminal object gives χ_0 .
- (5) **Identify the resulting object.** To identify the comparison intrinsically, recall that a product projection

$$C_X(M) \times C_X(N) \longrightarrow C_X(M)$$

has underlying map

$$!_X M \otimes !_X N \xrightarrow{1 \otimes e_N} !_X M \otimes \mathcal{O}_X \simeq !_X M,$$

and similarly for the second projection.

- (6) **Use the product structure.** Hence the product comparison $!_X(M \oplus N) \rightarrow !_X M \otimes !_X N$ is exactly the canonical Seely comparison obtained from the coalgebra diagonal followed by the two biproduct projections.

- (7) **Use the cofree universal property.** Conversely, the inverse product equivalence is the unique coalgebra map into $C_X(M \oplus N)$ whose cofree adjunct has components

$$\varepsilon_M \otimes e_N \quad \text{and} \quad e_M \otimes \varepsilon_N.$$

- (8) **Use the adjunction.** The identity, associativity, symmetry, unit, and all higher-pasting coherences are inherited from product preservation by the right adjoint C_X and the Cartesian coherence of $\text{CoCom}(\mathcal{L}_X)$.

□

Proposition 2.6.3 (Canonical identification of the monoidal storage maps). *The lax symmetric-monoidal maps*

$$m_{M,N} : !_X M \otimes !_X N \longrightarrow !_X(M \otimes N), \quad m_{\mathcal{O}_X} : \mathcal{O}_X \longrightarrow !_X \mathcal{O}_X$$

constructed in 2.6.1 are canonically the monoidal maps induced from the Seely equivalences of 2.6.2. Explicitly, if $m_{M,N}^{\text{Seely}}$ denotes the ordinary storage-modality composite recalled in [29, Section 2]

$$\begin{aligned} !_X M \otimes !_X N &\xrightarrow{\chi_{M,N}^{-1}} !_X(M \oplus N) \xrightarrow{\delta_{M \oplus N}} !_X !_X(M \oplus N) \\ &\xrightarrow{!_X(\chi_{M,N})} !_X(!_X M \otimes !_X N) \xrightarrow{!_X(\varepsilon_M \otimes \varepsilon_N)} !_X(M \otimes N), \end{aligned}$$

then there is a canonical equivalence

$$m_{M,N}^{\text{Seely}} \simeq m_{M,N}.$$

The corresponding tensor-unit maps are canonically equivalent as well.

Proof.

- (1) **Compare the two constructions.** Both $m_{M,N}^{\text{Seely}}$ and $m_{M,N}$ underlie coalgebra maps

$$C_X(M) \times C_X(N) \longrightarrow C_X(M \otimes N).$$

- (2) **Use the cofree universal property.** By the cofree mapping-space equivalence it is enough to compare their composites with the cofree counit.
- (3) **Verify naturality.** Naturality of ε , followed by the comonad counit identity, gives

$$\begin{aligned} \varepsilon_{M \otimes N} m_{M,N}^{\text{Seely}} &\simeq (\varepsilon_M \otimes \varepsilon_N) \varepsilon_{!_X M \otimes !_X N} !_X(\chi_{M,N}) \delta_{M \oplus N} \chi_{M,N}^{-1} \\ &\simeq (\varepsilon_M \otimes \varepsilon_N) \chi_{M,N} \varepsilon_{!_X(M \oplus N)} \delta_{M \oplus N} \chi_{M,N}^{-1} \\ &\simeq \varepsilon_M \otimes \varepsilon_N. \end{aligned}$$

- (4) **Use the cofree universal property.** This is exactly the specified cofree adjunct of $m_{M,N}$.
- (5) **Derive the comparison.** The cofree comparison principle therefore gives the claimed canonical comparison, with its higher coherence.
- (6) **Use terminality.** The tensor-unit statement is the same argument applied to the terminal coalgebra.

□

Remark 2.6.2 (The storage structure is constructed). The exponential modality used below is the canonical storage package induced by the adjunction $U_X \dashv C_X$ and product preservation of the cofree right adjoint. The comonad maps come from the adjunction, the Seely equivalences come from product preservation, and 2.6.3 proves that the directly constructed monoidal maps $m_{M,N}$ and $m_{\mathcal{O}_X}$ are exactly the canonical monoidal storage maps induced from those Seely equivalences. The corresponding identification of the additive bialgebra maps is proved below after those maps have been constructed.

Differentiation now requires a canonical primitive infinitesimal element that can be transported into the cofree coalgebra.

2.6.2 Primitive infinitesimal coalgebras and coherent infinitesimal augmentation

Lemma 2.6.1 (Opposite split square-zero construction). *Let $\mathcal{C}^\otimes$ be a stable symmetric-monoidal ∞ -category whose tensor product is exact separately in each variable. Then $(\mathcal{C}^{\text{op}})^\otimes$ is again stable symmetric monoidal with tensor product exact separately, and every $M \in \mathcal{C}$, regarded as an object of $\mathcal{C}^{\text{op}}$, determines functorially a split square-zero commutative algebra*

$$\mathbb{K}_{\mathcal{C}} \oplus M \in \text{CAlg}(\mathcal{C}^{\text{op}})$$

whose augmentation ideal is M . Its underlying object is the biproduct $\mathbb{K}_{\mathcal{C}} \oplus M$, the unit acts in the evident way, and the product on the M -summand is coherently null.

Proof.

- (1) **Use exactness.** Opposition preserves stability, and an exact functor between stable ∞ -categories remains exact after passage to opposites.
- (2) **Use exactness.** Hence the tensor product on $\mathcal{C}^{\text{op}}$ is exact separately.
- (3) **Identify the resulting object.** In a stable symmetric-monoidal ∞ -category, augmentation ideal and unitalization identify augmented commutative algebra objects with nonunital commutative algebra objects; see [34, Proposition 5.4.4.10].
- (4) **Equip functorially with the trivial nonunital commutative algebra.** Equip M functorially with the trivial nonunital commutative algebra structure: every multiplication involving at least two inputs from M is the zero morphism, with the coherence supplied by the zero object of the stable category.
- (5) **Use the product structure.** Its unitalization has underlying biproduct

$$\mathbb{K}_{\mathcal{C}} \oplus M,$$

the tensor unit acts by the canonical unit constraints, and the product on $M \otimes M$ is coherently null.

- (6) **Verify compatibility.** Applying this construction in $\mathcal{C}^{\text{op}}$ gives the stated split square-zero commutative algebra.

- (7) **Verify functoriality.** Functoriality is inherited from trivial nonunital algebra formation and unitalization.
- (8) **Identify the resulting comparison.** This finite algebraic construction does not require presentability of $\mathcal{C}^{\text{op}}$.

□

Construction 2.6.1 (Primitive infinitesimal coalgebra). Let

$$\mathcal{L}_X := \text{QCoh}(X).$$

The tensor product on $\mathcal{L}_X$ is exact separately, so 2.6.1 applies. For $M \in \mathcal{L}_X$, form in $\mathcal{L}_X^{\text{op}}$ the functorial split square-zero commutative algebra with underlying object

$$\mathcal{O}_X \oplus M$$

and square-zero augmentation ideal M . Passing to the opposite category defines a cocommutative $\mathbb{E}_\infty$ -coalgebra

$$P_X(M) \in \text{CoCom}(\mathcal{L}_X), \quad U_X P_X(M) \simeq \mathcal{O}_X \oplus M.$$

Write

$$\text{pr}_0 : U_X P_X(M) \longrightarrow \mathcal{O}_X, \quad \text{pr}_1 : U_X P_X(M) \longrightarrow M$$

for the counit and coefficient projection on underlying objects, and write ι_0, ι_1 for the two biproduct inclusions. The unit summand is group-like and the coefficient summand is primitive. In particular, the binary comultiplication has the familiar underlying form

$$\Delta(1) = 1 \otimes 1, \quad \Delta(m) = m \otimes 1 + 1 \otimes m.$$

The full $\mathbb{E}_\infty$ -coalgebra coherence is the one supplied by the opposite split square-zero algebra construction; the displayed binary formula is only its underlying first operation.

Proposition 2.6.4 (Square-zero and primitive infinitesimal structures). *The underlying stable object $\mathcal{O}_X \oplus M$ supports two canonical and complementary infinitesimal structures:*

<i>split square-zero algebra</i> $M \cdot M \simeq 0$	<i>primitive $\mathbb{E}_\infty$-coalgebra</i> $\Delta(m) = m \otimes 1 + 1 \otimes m.$
--	---

Both structures are functorial in M . The first is the zeroth-space image of tangent linearization, while the second is obtained by applying the same split square-zero construction in the opposite stable symmetric-monoidal ∞ -category.

Proof.

- (1) **algebra structure.** The algebra structure is 2.2.1.
- (2) **coalgebra structure.** The coalgebra structure is 2.6.1.
- (3) **Compare the two constructions.** Both constructions are functorial split square-zero constructions, respectively in $\mathcal{L}_X$ and $\mathcal{L}_X^{\text{op}}$, so no separate collection of arity-wise coherences is chosen.

□

Construction 2.6.2 (Canonical coherent infinitesimal augmentation). Let

$$\alpha_D : D \longrightarrow C_X U_X D$$

denote the unit of the adjunction $U_X \dashv C_X$ at a cocommutative coalgebra D . For $M \in \mathcal{L}_X$, let

$$\hat{H}_M^{\text{inf}} := \alpha_{P_X(M)} : P_X(M) \longrightarrow C_X U_X P_X(M)$$

and write

$$H_M^{\text{inf}} := U_X \hat{H}_M^{\text{inf}} : \mathcal{O}_X \oplus M \longrightarrow !_X(\mathcal{O}_X \oplus M).$$

Thus $\hat{H}_M^{\text{inf}}$ is a morphism of cocommutative $\mathbb{E}_\infty$ -coalgebras, while its underlying map H_M^{inf} is the canonical coalgebra structure for the comonad $!_X = U_X C_X$:

$$\varepsilon_{\mathcal{O}_X \oplus M} H_M^{\text{inf}} \simeq \text{id}_{\mathcal{O}_X \oplus M},$$

and

$$\delta_{\mathcal{O}_X \oplus M} H_M^{\text{inf}} \simeq !_X(H_M^{\text{inf}}) H_M^{\text{inf}}.$$

All these equivalences and their higher compatibility are the unit and triangle coherences of the adjunction.

Construction 2.6.3 (Primitive tensor comparison). Let $D \in \text{CoCom}(\mathcal{L}_X)$ and $N \in \mathcal{L}_X$. There is a canonical coalgebra morphism

$$\vartheta_{D,N} : D \times P_X(N) \longrightarrow P_X(U_X D \otimes N).$$

To construct it, pass to $\mathcal{L}_X^{\text{op}}$. The product on the left becomes the tensor product of the commutative algebra corresponding to D with the split square-zero algebra $\mathcal{O}_X \oplus N$. Its summand $U_X D \otimes N$ is a square-zero ideal because the N -summand is square-zero. Hence the split square-zero universal construction gives a canonical commutative-algebra morphism

$$\mathcal{O}_X \oplus (U_X D \otimes N) \longrightarrow A_D \otimes (\mathcal{O}_X \oplus N),$$

where $A_D \in \text{CAlg}(\mathcal{L}_X^{\text{op}})$ is the commutative algebra corresponding to D . Passing to opposites gives $\vartheta_{D,N}$. Under the symmetric-monoidal equivalence

$$U_X(D \times P_X(N)) \simeq U_X D \otimes (\mathcal{O}_X \oplus N),$$

its underlying map has the two components

$$e_D \otimes \text{pr}_0 : U_X D \otimes (\mathcal{O}_X \oplus N) \longrightarrow \mathcal{O}_X$$

and

$$1 \otimes \text{pr}_1 : U_X D \otimes (\mathcal{O}_X \oplus N) \longrightarrow U_X D \otimes N.$$

Naturality of the adjunction unit makes $U_X \vartheta_{D,N}$ a morphism between the canonical $!_X$ -coalgebra structures induced from $D \times P_X(N)$ and $P_X(U_X D \otimes N)$.

Proposition 2.6.5 (The adjunction unit is an infinitesimal augmentation). *The maps H_M^{inf} of 2.6.2 satisfy the three infinitesimal-augmentation identities coherently. Explicitly:*

(i) H_M^{inf} is a $!_X$ -coalgebra structure:

$$\varepsilon H_M^{\text{inf}} \simeq \text{id}, \quad \delta H_M^{\text{inf}} \simeq !_X(H_M^{\text{inf}}) H_M^{\text{inf}};$$

(ii) it is compatible with the primitive cocommutative coalgebra structure: if Λ_M denotes the co-multiplication of $P_X(M)$, then

$$\Delta_{\mathcal{O}_X \oplus M} H_M^{\text{inf}} \simeq (H_M^{\text{inf}} \otimes H_M^{\text{inf}}) \Lambda_M, \quad e_{\mathcal{O}_X \oplus M} H_M^{\text{inf}} \simeq \text{pr}_0;$$

(iii) for $M, N \in \mathcal{L}_X$, the primitive tensor comparison

$$U_X \vartheta_{C_X(M), N} : !_X M \otimes (\mathcal{O}_X \oplus N) \longrightarrow \mathcal{O}_X \oplus (!_X M \otimes N)$$

is a morphism of $!_X$ -coalgebras, where the source coalgebra structure is the canonical tensor product structure

$$\begin{aligned} !_X M \otimes (\mathcal{O}_X \oplus N) &\xrightarrow{\delta_M \otimes H_N^{\text{inf}}} !_X(!_X M) \otimes !_X(\mathcal{O}_X \oplus N) \\ &\xrightarrow{m_{!_X M, \mathcal{O}_X \oplus N}} !_X(!_X M \otimes (\mathcal{O}_X \oplus N)), \end{aligned}$$

and the target coalgebra structure is $H_{!_X M \otimes N}^{\text{inf}}$.

After passage to the homotopy category these are precisely the conditions IA.1–IA.3 of the infinitesimal augmentation of [29, Definition 10].

Proof.

(1) **Use the adjunction.** Part (i) is the comonad-coalgebra equation for the underlying adjunction-unit map

$$U_X \alpha_{P_X(M)} = H_M^{\text{inf}}.$$

(2) **Derive the comparison.** Part (ii) follows because $\alpha_{P_X(M)} : P_X(M) \rightarrow C_X U_X P_X(M)$ is a morphism in $\text{CoCom}(\mathcal{L}_X)$; applying U_X gives the two displayed coalgebra compatibilities.

(3) **For (iii), first note that for arbitrary coalgebras there.** For (iii), first note that for arbitrary coalgebras D, E there is a canonical equivalence

$$U_X \alpha_{D \times E} \simeq m_{U_X D, U_X E} (U_X \alpha_D \otimes U_X \alpha_E).$$

(4) **compare the two coalgebra.** Indeed, compare the two coalgebra maps

$$D \times E \rightrightarrows C_X(U_X D \otimes U_X E).$$

(5) **Use the cofree universal property.** After composition with the cofree counit, both have adjunct $\text{id}_{U_X D \otimes U_X E}$, by the triangle identities for $U_X \dashv C_X$.

(6) **cofree comparison principle therefore.** The cofree comparison principle therefore supplies the displayed comparison and its higher coherence.

(7) **Apply this with and .** Apply this with $D = C_X(M)$ and $E = P_X(N)$.

(8) **Then so the canonical -coalgebra structure on the underlying product.** Then

$$U_X \alpha_{C_X(M)} = \delta_M, \quad U_X \alpha_{P_X(N)} = H_N^{\text{inf}},$$

so the canonical $!_X$ -coalgebra structure on the underlying product $!_X M \otimes (\mathcal{O}_X \oplus N)$ is exactly the source structure displayed in (iii).

(9) **Verify naturality.** Naturality of the adjunction unit for the coalgebra morphism

$$\vartheta_{C_X(M),N} : C_X(M) \times P_X(N) \longrightarrow P_X(!_X M \otimes N)$$

gives

$$!_X(U_X \vartheta_{C_X(M),N}) U_X \alpha_{C_X(M) \times P_X(N)} \simeq H_{!_X M \otimes N}^{\text{inf}} U_X \vartheta_{C_X(M),N}.$$

(10) **Identify the resulting comparison.** This is exactly the required $!_X$ -coalgebra-morphism comparison.

(11) **Verify naturality.** All higher compatibility comes from naturality of α , the monoidal coherences of 2.6.1, and functoriality of products in $\text{CoCom}(\mathcal{L}_X)$.

□

Construction 2.6.4 (Codereliction). Let

$$\text{pr}_1 : U_X P_X(M) = \mathcal{O}_X \oplus M \longrightarrow M$$

be the coefficient projection. Its cofree adjunct is the canonical coalgebra morphism

$$\begin{aligned} H_M : P_X(M) &\xrightarrow{\alpha_{P_X(M)}} C_X(\mathcal{O}_X \oplus M) \\ &\xrightarrow{C_X(\text{pr}_1)} C_X(M). \end{aligned}$$

Equivalently, H_M is characterized by

$$\varepsilon_M U_X H_M \simeq \text{pr}_1,$$

and functoriality of the cofree adjunction identifies its underlying map as

$$U_X H_M \simeq !_X(\text{pr}_1) H_M^{\text{inf}}.$$

Restricting along the coefficient inclusion therefore gives

$$\eta_M : M \xrightarrow{\iota_1} \mathcal{O}_X \oplus M \xrightarrow{U_X H_M} !_X M.$$

We call η_M the **codereliction map**. Its recognition as a codereliction in the ordinary differential-storage sense is made after passage to the homotopy category in 2.6.2.

Proposition 2.6.6 (Basic coherent codereliction identities). *Let*

$$\varepsilon_M : !_X M \longrightarrow M$$

be the comonad counit, let $e_M : !_X M \rightarrow \mathcal{O}_X$ be the coalgebra counit, and let

$$u_M : \mathcal{O}_X \longrightarrow !_X M$$

be the group-like morphism obtained by composing $\iota_0 : \mathcal{O}_X \rightarrow P_X(M)$ with $U_X H_M$. Then there are canonical coherent equivalences

$$\varepsilon_M \eta_M \simeq \text{id}_M, \quad e_M \eta_M \simeq 0,$$

and

$$\Delta_M \eta_M \simeq \eta_M \otimes u_M + u_M \otimes \eta_M.$$

These are the linear, constant, and primitive Leibniz identities for the codereliction.

Proof.

- (1) **Restrict the construction.** The first identity follows from the defining adjunct equation $\varepsilon_M U_X H_M \simeq \text{pr}_1$ after restriction along ι_1 .
- (2) **Identify the map.** The map H_M is a morphism of cocommutative $\mathbb{E}_\infty$ -coalgebras.
- (3) **Use preservation.** It therefore preserves the coalgebra counit and the full coalgebra comultiplication.
- (4) **On the coefficient summand of , the counit.** On the coefficient summand of $P_X(M)$, the counit is zero and the binary comultiplication is primitive.
- (5) **Restrict the construction.** Restriction along ι_1 gives the second and third displayed equivalences.
- (6) **Their higher compatibility.** Their higher compatibility is inherited from the single coalgebra morphism H_M , rather than imposed on the three formulas separately.

□

The primitive infinitesimal input can now be combined with the bialgebra multiplication to produce infinitesimal translation and hence the deriving transformation.

2.6.3 Coalgebra-level infinitesimal translation and deriving transformation

Construction 2.6.5 (Bialgebra multiplication of the cofree exponential). The product coalgebra

$$C_X(M) \times C_X(M)$$

has underlying object $!_X M \otimes !_X M$. Define

$$b_M : !_X M \otimes !_X M \longrightarrow M$$

by

$$b_M := \varepsilon_M \otimes e_M + e_M \otimes \varepsilon_M.$$

By the cofree adjunction there is a canonical coalgebra morphism

$$\tilde{\nabla}_M : C_X(M) \times C_X(M) \longrightarrow C_X(M)$$

whose adjunct is b_M . Write

$$\nabla_M := U_X \tilde{\nabla}_M : !_X M \otimes !_X M \longrightarrow !_X M.$$

Similarly, $P_X(0)$, whose underlying object is $\mathcal{O}_X$, is the terminal cocommutative coalgebra. The zero morphism $\mathcal{O}_X \rightarrow M$ therefore has a cofree lift

$$\tilde{u}_M : P_X(0) \longrightarrow C_X(M),$$

whose underlying map is $u_M : \mathcal{O}_X \rightarrow !_X M$. This agrees with the map u_M defined from the group-like summand above, because both coalgebra maps have zero as their cofree adjunct.

Proposition 2.6.7 (Canonical identification of the additive storage maps). *Under the Seely equivalence*

$$\chi_{M,M} : !_X(M \oplus M) \xrightarrow{\simeq} !_X M \otimes !_X M,$$

the coalgebra morphism obtained from $C_X(+): C_X(M \oplus M) \rightarrow C_X(M)$ has underlying map

$$\nabla_M : !_X M \otimes !_X M \longrightarrow !_X M$$

of 2.6.5. Likewise the coalgebra map induced by $0 \rightarrow M$ has underlying map

$$u_M : \mathcal{O}_X \longrightarrow !_X M.$$

Hence ∇_M and u_M are exactly the additive bialgebra maps canonically induced by the Seely storage structure.

Proof.

- (1) **Transport across.** Transport $C_X(+)$ across

$$C_X(M \oplus M) \simeq C_X(M) \times C_X(M).$$

- (2) **Use the cofree universal property.** The resulting coalgebra map into $C_X(M)$ has cofree adjunct

$$(+)\varepsilon_{M \oplus M} \chi_{M,M}^{-1}.$$

- (3) **Invoke the cited result.** By the component calculation of 2.6.2, its two summands are

$$\varepsilon_M \otimes e_M \quad \text{and} \quad e_M \otimes \varepsilon_M.$$

- (4) **Conclude the step.** Thus the adjunct is

$$\varepsilon_M \otimes e_M + e_M \otimes \varepsilon_M,$$

which is exactly the map b_M defining $\tilde{\nabla}_M$ in 2.6.5.

- (5) **Derive the comparison.** Cofreeness therefore gives a canonical equivalence between the transported addition map and $\tilde{\nabla}_M$, hence between their underlying maps.
- (6) **Use the cofree universal property.** The zero-induced unit is treated identically: both coalgebra maps have zero as their cofree adjunct.
- (7) **Verify functoriality.** Coherence is inherited from the coherent commutative-group structure of biproducts and functoriality of C_X .

□

Proposition 2.6.8 (Coherent bicommutative bialgebra structure). *The maps $\nabla_M, u_M, \Delta_M, e_M$ make $!_X M$ a coherently bicommutative bialgebra object in $\mathcal{L}_X$. Under the Seely equivalence*

$$!_X(M \oplus M) \simeq !_X M \otimes !_X M,$$

the multiplication ∇_M is the morphism induced by addition $M \oplus M \rightarrow M$, and u_M is induced by $0 \rightarrow M$.

Proof.

- (1) **Use the product structure.** The addition and zero maps make every object of the stable ∞ -category $\mathcal{L}_X$ a coherently commutative group object for the biproduct monoidal structure.
- (2) **Applying and transporting through the product equivalences of.** Applying C_X and transporting through the product equivalences of 2.6.2 produces coalgebra maps

$$C_X(M) \times C_X(M) \longrightarrow C_X(M), \quad P_X(0) \longrightarrow C_X(M).$$

- (3) **Invoke the cited result.** By 2.6.7, their underlying maps are precisely ∇_M and u_M .
- (4) **Associativity, commutativity, and unitality.** Associativity, commutativity, and unitality are therefore the functorial images of the coherent addition and zero laws.
- (5) **Use the stated property.** Since the maps are morphisms in $\text{CoCom}(\mathcal{L}_X)$, multiplication and unit preserve Δ_M and e_M .
- (6) **Use the product structure.** These are the bicommutative bialgebra compatibilities, with all higher coherence inherited from the functorial product construction and the stable biproduct structure.

□

Construction 2.6.6 (Coalgebra-level infinitesimal translation). Define

$$a_M : !_X M \otimes (\mathcal{O}_X \oplus M) \longrightarrow M$$

by

$$a_M := e_M \otimes \text{pr}_1 + \varepsilon_M \otimes \text{pr}_0.$$

The source is the underlying object of the product coalgebra $C_X(M) \times P_X(M)$. The cofree adjunction therefore supplies a canonical coalgebra morphism

$$\tilde{\mathbf{d}}_M : C_X(M) \times P_X(M) \longrightarrow C_X(M)$$

characterized by

$$\varepsilon_M U_X \tilde{\mathbf{d}}_M \simeq a_M.$$

We call $\tilde{\mathbf{d}}_M$ **coalgebra-level infinitesimal translation**.

Proposition 2.6.9 (Infinitesimal translation from codereliction). *There is a canonical equivalence of coalgebra morphisms*

$$\tilde{\mathbf{d}}_M \simeq \tilde{\nabla}_M \circ (1 \times H_M).$$

Consequently, after restricting the underlying map along

$$!_X M \otimes M \xrightarrow{1 \otimes \iota_1} !_X M \otimes (\mathcal{O}_X \oplus M),$$

one obtains

$$\mathbf{d}_M : !_X M \otimes M \xrightarrow{1 \otimes \eta_M} !_X M \otimes !_X M \xrightarrow{\nabla_M} !_X M.$$

Proof.

- (1) **Compare the two constructions.** Both sides of the first displayed equivalence are coalgebra morphisms

$$C_X(M) \times P_X(M) \longrightarrow C_X(M).$$

- (2) **Use the cofree universal property.** By the cofree comparison principle, it is enough to compare their specified adjoints.
- (3) **Use the stated property.** Since H_M is a coalgebra morphism,

$$e_M U_X H_M \simeq \text{pr}_0, \quad \varepsilon_M U_X H_M \simeq \text{pr}_1.$$

- (4) **adjunct of.** The adjunct of

$$\tilde{\nabla}_M(1 \times H_M)$$

is consequently

$$\begin{aligned} & (\varepsilon_M \otimes e_M + e_M \otimes \varepsilon_M)(1 \otimes U_X H_M) \\ & \simeq \varepsilon_M \otimes \text{pr}_0 + e_M \otimes \text{pr}_1 \\ & = a_M. \end{aligned}$$

- (5) **Invoke the cited result.** By definition, a_M is also the specified adjunct of $\tilde{\mathbf{d}}_M$.
- (6) **Identify the equivalence.** The cofree comparison principle therefore gives the canonical equivalence

$$\tilde{\mathbf{d}}_M \simeq \tilde{\nabla}_M(1 \times H_M).$$

- (7) **Use the adjunction.** Its higher compatibility is inherited from the adjunction and the coalgebra maps entering the construction.
- (8) **Restrict the construction.** Restricting the resulting underlying map along the primitive inclusion

$$!_X M \otimes M \xrightarrow{1 \otimes \iota_1} !_X M \otimes (\mathcal{O}_X \oplus M)$$

gives

$$!_X M \otimes M \xrightarrow{1 \otimes \eta_M} !_X M \otimes !_X M \xrightarrow{\nabla_M} !_X M,$$

which is the asserted formula for $\mathbf{d}_M$.

□

Construction 2.6.7 (Deriving transformation). The map $\mathbf{d}_M$ of 2.6.9 is the **deriving-transformation map**; its ordinary deriving-transformation recognition is 2.6.2. For a coKleisli morphism

$$f : !_X M \longrightarrow N,$$

define its differential by

$$Df := f \circ \mathbf{d}_M : !_X M \otimes M \longrightarrow N.$$

The coalgebra morphism $\tilde{\mathbf{d}}_M$, rather than merely its restriction $\mathbf{d}_M$, is retained as the higher-categorical source of this operation.

Theorem 2.6.2 (Coherent differential-exponential structure of the stable coefficient fibre). *For every spectral context X , the preceding constructions canonically realize the differential-exponential structure associated to the cofree cocommutative coalgebra at the ordinary homotopy-category recognition interface. More precisely:*

(i) the homotopy category $h\mathcal{L}_X$, with

$$(!_X, \delta, \varepsilon, \Delta, e, m, m_{\mathcal{O}})$$

and the Seely isomorphisms of 2.6.2, is an additive storage category;

(ii) the image of H^{inf} in $h\mathcal{L}_X$ is an infinitesimal augmentation;

(iii) the image of the morphism

$$\eta_M : M \xrightarrow{\iota_1} \mathcal{O}_X \oplus M \xrightarrow{H_M^{\text{inf}}} !_X(\mathcal{O}_X \oplus M) \xrightarrow{!_X(\text{pr}_1)} !_X M$$

is therefore a codereliction in $h\mathcal{L}_X$;

(iv) the deriving transformation associated to this codereliction is the image of the already constructed morphism

$$\mathbf{d}_M = !_X M \otimes M \xrightarrow{1 \otimes \eta_M} !_X M \otimes !_X M \xrightarrow{\nabla_M} !_X M.$$

Thus $h\mathcal{L}_X$ is a differential storage category, hence in particular a differential category in the sense of Blute–Cockett–Seely. Before passage to the homotopy category, the primitive coalgebra, H^{inf} , the codereliction map, the monoidal and bialgebra maps, and the coalgebra-level infinitesimal translation are genuine morphisms in the stated ∞ -categories. The complete ordinary differential-storage equations are invoked only at the homotopy-category recognition interface; no separate ∞ -categorical algebraic theory of differential storage categories is assumed.

Proof.

- (1) **Use the cofree universal property.** The cofree adjunction gives the monoidal coalgebra modality of 2.6.1, and 2.6.2 gives canonical Seely isomorphisms.
- (2) **Identify the map.** The identification theorems 2.6.3 and 2.6.7 prove that the monoidal and additive bialgebra maps already constructed from cofreeness are exactly the storage-modality maps induced by those Seely isomorphisms.
- (3) **Use the product structure.** Since $\mathcal{L}_X$ is stable, $h\mathcal{L}_X$ is additive with finite biproducts.
- (4) **Conclude the step.** Hence the first assertion is the ordinary additive storage structure determined by these displayed spectral maps; no additional storage datum is chosen.
- (5) **coherent identities of the cited result descend.** The coherent identities of 2.6.5 descend exactly to IA.1–IA.3 of [29, Definition 10].
- (6) **Identify the resulting object.** Proposition 13 of that source therefore identifies

$$M \xrightarrow{\iota_1} \mathcal{O}_X \oplus M \xrightarrow{H_M^{\text{inf}}} !_X(\mathcal{O}_X \oplus M) \xrightarrow{!_X(\text{pr}_1)} !_X M$$

as a codereliction.

- (7) **Use the cofree universal property.** By 2.6.4, this is precisely the image of the map η_M already constructed as the cofree adjunct of the primitive projection; the recognition theorem introduces no second codereliction.

- (8) **Conclude the argument.** Finally, [29, Theorem 8] identifies coderelictions with deriving transformations in an additive storage category and gives

$$\mathbf{d}_M = \nabla_M(1 \otimes \eta_M).$$

- (9) **Restrict the construction.** This is the image of 2.6.7, since 2.6.9 constructs the same map as the primitive restriction of the coalgebra morphism $\tilde{\mathbf{d}}_M$.
- (10) **Conclude the step.** Thus the ordinary constant, Leibniz, linear, chain, interchange, and monoidal equations follow from the cited recognition results.
- (11) **Identify the map.** The specific higher maps and canonical homotopies constructed before passage to $h\mathcal{L}_X$ remain the higher-categorical source of that ordinary structure.

□

Remark 2.6.3 (Relation to differential linear syntax). The primary differential structure of this chapter lives in the stable symmetric-monoidal ∞ -category $\mathbf{QCoh}(X)$: the primitive $\mathbb{E}_\infty$ -coalgebra, infinitesimal augmentation, codereliction, and coalgebra-level infinitesimal translation are constructed before any passage to homotopy classes. The homotopy category is only a recognition interface with the differential-category terminology of [8, 29]. This places the stable coefficient fibres in the linear/nonlinear tradition of [6] and relates their coKleisli differential to the differential lambda-calculus of [14], without discarding the higher categorical information of the spectral doctrine.

It remains to compare this cofree derivative with cotangent-theoretic differentiation on free finite-linear spectral models.

2.6.4 Comparison with the spectral universal derivation

Definition 2.6.2 (Cofree function algebra). Let B be connective and let M be a finite projective B -module. Define

$$\mathcal{O}_!(M) := \underline{\mathrm{Hom}}_B(!_B M, B).$$

The cocommutative $\mathbb{E}_\infty$ -coalgebra structure of $C_B(M)$ gives $\mathcal{O}_!(M)$ its convolution commutative B -algebra structure. The comonad counit $\varepsilon_M : !_B M \rightarrow M$ induces a B -linear map

$$M^\vee \longrightarrow \mathcal{O}_!(M),$$

and hence a canonical commutative B -algebra morphism

$$j_M : \mathrm{Sym}_B(M^\vee) \longrightarrow \mathcal{O}_!(M).$$

Remark 2.6.4 (Stable algebraic nature of the cofree function algebra). The convolution algebra $\mathcal{O}_!(M)$ is formed as a commutative algebra object in the unrestricted stable coefficient category Mod_B ; equivalently,

$$\mathcal{O}_!(M) \in \mathrm{CAlg}(\mathrm{Mod}_B) \simeq \mathrm{CAlg}(\mathrm{Sp})_{B/},$$

with its underlying B -module allowed to range over the unrestricted stable coefficient category. The stability belongs to Mod_B ; its ∞ -category of commutative algebra objects is generally nonstable.

The comparison in this subsection is therefore formulated in the unrestricted commutative spectral-algebraic universe. A connective interpretation of $\mathcal{O}_!(M)$ as a spectral coordinate algebra or affine spectral context requires a separate connectivity theorem. The comparison map j_M below is an algebraic map in that unrestricted universe, in accordance with the connective-coordinate versus stable-coefficient convention of this manuscript.

Lemma 2.6.2 (Dual of the primitive infinitesimal coalgebra). *Let M be finite projective over B . Then $P_B(M) = B \oplus M$ is dualizable, and dualization carries its primitive cocommutative coalgebra structure to the split square-zero commutative B -algebra*

$$P_B(M)^\vee \simeq B \oplus M^\vee.$$

For every B -module K , there is a canonical equivalence

$$\underline{\mathrm{Hom}}_B(K \otimes_B P_B(M), B) \simeq \underline{\mathrm{Hom}}_B(K, B) \otimes_B P_B(M)^\vee.$$

When K underlies a cocommutative coalgebra, this equivalence identifies the convolution algebra on the left with the tensor-product commutative algebra on the right.

Proof.

- (1) **Finite projectivity.** Finite projectivity makes M , and hence $B \oplus M$, dualizable.
- (2) **contravariant symmetric-monoidal duality functor on dualizable objects.** The contravariant symmetric-monoidal duality functor on dualizable objects carries cocommutative coalgebras to commutative algebras.
- (3) **Derive the comparison.** Dualizing the primitive comultiplication gives multiplication which is the ordinary unit action on the $M^\vee$ -summand and vanishes on $M^\vee \otimes_B M^\vee$; this is exactly the split square-zero algebra.
- (4) **For the internal-Hom formula, for every , duality and closedness.** For the internal-Hom formula, for every $Z \in \mathrm{Mod}_B$, duality and closedness give

$$\begin{aligned} \mathrm{Map}_{\mathrm{Mod}_B}(Z, \underline{\mathrm{Hom}}_B(K, B) \otimes_B P_B(M)^\vee) &\simeq \mathrm{Map}_{\mathrm{Mod}_B}(Z \otimes_B P_B(M), \underline{\mathrm{Hom}}_B(K, B)) \\ &\simeq \mathrm{Map}_{\mathrm{Mod}_B}(Z \otimes_B P_B(M) \otimes_B K, B) \\ &\simeq \mathrm{Map}_{\mathrm{Mod}_B}(Z, \underline{\mathrm{Hom}}_B(K \otimes_B P_B(M), B)). \end{aligned}$$

- (5) **Apply Yoneda.** Yoneda gives the equivalence.
- (6) **Compatibility with convolution.** Compatibility with convolution is the symmetric-monoidal compatibility of duality: tensor products of coalgebras are sent to tensor products of their dual convolution algebras whenever the relevant factor is dualizable.

□

Construction 2.6.8 (Spectral convolution derivation). Let

$$R := \mathcal{O}_!(M).$$

Precomposition with the coalgebra morphism

$$\tilde{\mathrm{d}}_M : C_B(M) \times P_B(M) \longrightarrow C_B(M)$$

gives a morphism of commutative B -algebras

$$R \longrightarrow \underline{\mathrm{Hom}}_B(!_B M \otimes_B P_B(M), B).$$

By 2.6.2, identify the target canonically with

$$R \otimes_B (B \oplus M^\vee) \simeq R \oplus (R \otimes_B M^\vee).$$

Denote the resulting commutative B -algebra morphism by

$$\tilde{D} : R \longrightarrow R \oplus (R \otimes_B M^\vee).$$

It is a section of the split square-zero augmentation

$$R \oplus (R \otimes_B M^\vee) \longrightarrow R.$$

Indeed, restriction of $\tilde{\mathbf{d}}_M$ along the group-like summand

$$C_B(M) \simeq C_B(M) \times B \xrightarrow{1 \times \iota_0} C_B(M) \times P_B(M)$$

is the identity coalgebra morphism: after composing with ε_M , its adjunct is ε_M .

Thus

$$\tilde{D} \in \mathrm{Der}_B(R, R \otimes_B M^\vee)$$

in the sense of 2.3.1. Write

$$D : R \longrightarrow R \otimes_B M^\vee$$

for its coefficient component under the canonical decomposition

$$R \oplus (R \otimes_B M^\vee).$$

Equivalently,

$$\tilde{D} = (\mathrm{id}_R, D)$$

on underlying B -modules.

Under the internal-Hom adjunction, the coefficient component D is precisely precomposition with the deriving transformation

$$\mathbf{d}_M : !_B M \otimes_B M \longrightarrow !_B M.$$

Theorem 2.6.3 (Compatibility with the spectral Kähler derivative). *Let M be finite projective over a connective commutative ring spectrum B , and put*

$$A := \mathrm{Sym}_B(M^\vee).$$

Then

$$L_{A/B} \simeq A \otimes_B M^\vee.$$

Let

$$\tilde{d}_{\mathrm{univ}} : A \longrightarrow A \oplus (A \otimes_B M^\vee)$$

be the universal B -linear derivation section, and write

$$d_{\mathrm{univ}} : A \longrightarrow A \otimes_B M^\vee$$

for its coefficient component.

The coalgebra-level infinitesimal translation of 2.6.6 induces the spectral convolution derivation section

$$\tilde{D} : \mathcal{O}_!(M) \longrightarrow \mathcal{O}_!(M) \oplus (\mathcal{O}_!(M) \otimes_B M^\vee)$$

of 2.6.8. Write

$$D : \mathcal{O}_!(M) \longrightarrow \mathcal{O}_!(M) \otimes_B M^\vee$$

for its coefficient component. Then the square

$$\begin{array}{ccc} A & \xrightarrow{j_M} & \mathcal{O}_!(M) \\ d_{\text{univ}} \downarrow & & \downarrow D \\ A \otimes_B M^\vee & \xrightarrow{j_M \otimes 1} & \mathcal{O}_!(M) \otimes_B M^\vee \end{array}$$

commutes canonically.

Equivalently, the corresponding square of spectral derivation sections

$$\begin{array}{ccc} A & \xrightarrow{j_M} & \mathcal{O}_!(M) \\ \tilde{d}_{\text{univ}} \downarrow & & \downarrow \tilde{D} \\ A \oplus (A \otimes_B M^\vee) & \xrightarrow{j_M \oplus (j_M \otimes 1)} & \mathcal{O}_!(M) \oplus (\mathcal{O}_!(M) \otimes_B M^\vee) \end{array}$$

commutes canonically. Thus, on free finite-linear spectral models, differential-linear differentiation is canonically compatible along j_M with the universal spectral Kähler derivation.

Proof.

(1) **Apply the cotangent calculation.** The cotangent calculation is 2.3.2, applied to $M^\vee$.

(2) **Fix the notation.** Put

$$R := \mathcal{O}_!(M) = \underline{\text{Hom}}_B(!_B M, B).$$

(3) **Derive the comparison.** Finite projectivity of M gives a canonical duality equivalence

$$R \otimes_B M^\vee \simeq \underline{\text{Hom}}_B(!_B M \otimes_B M, B).$$

(4) **Transport the structure.** Under this equivalence, the coefficient component

$$D : R \longrightarrow R \otimes_B M^\vee$$

of the convolution derivation section is precomposition with the deriving transformation

$$\mathbf{d}_M : !_B M \otimes_B M \longrightarrow !_B M$$

by 2.6.8.

(5) **Fix the notation.** Let

$$g : M^\vee \longrightarrow R$$

be the generator map underlying

$$j_M : \text{Sym}_B(M^\vee) \longrightarrow R;$$

it is induced by the comonad counit

$$\varepsilon_M : !_B M \longrightarrow M.$$

- (6) **Use the adjunction.** Under the displayed internal-Hom equivalence, the composite

$$M^\vee \xrightarrow{g} R \xrightarrow{D} R \otimes_B M^\vee$$

is adjoint to

$$\begin{aligned} M^\vee \otimes_B !_B M \otimes_B M &\xrightarrow{1 \otimes d_M} M^\vee \otimes_B !_B M \\ &\xrightarrow{1 \otimes \varepsilon_M} M^\vee \otimes_B M \xrightarrow{\text{ev}} B. \end{aligned}$$

- (7) **Derive the comparison.** The defining adjoint equation for coalgebra-level infinitesimal translation, restricted to the primitive summand, gives the canonical homotopy

$$\varepsilon_M d_M \simeq e_M \otimes \text{id}_M : !_B M \otimes_B M \longrightarrow M.$$

- (8) **Use the adjunction.** Consequently the map induced on generators by Dg is canonically adjoint to

$$!_B M \otimes_B M \xrightarrow{e_M \otimes \text{id}_M} B \otimes_B M \simeq M.$$

- (9) **Use the adjunction.** Since the convolution unit

$$1_R : B \longrightarrow R$$

is adjoint to the coalgebra counit

$$e_M : !_B M \longrightarrow B,$$

this is exactly the internal-Hom representative of the canonical map

$$M^\vee \longrightarrow R \otimes_B M^\vee, \quad \lambda \longmapsto 1_R \otimes \lambda.$$

- (10) **Identify the map.** The coefficient component of the universal derivation section

$$\tilde{d}_{\text{univ}} : A \longrightarrow A \oplus (A \otimes_B M^\vee)$$

is characterized by the generator map

$$M^\vee \longrightarrow A \otimes_B M^\vee, \quad \lambda \longmapsto 1_A \otimes \lambda.$$

- (11) **After extension of coefficients along this.** After extension of coefficients along

$$j_M : A \longrightarrow R,$$

this becomes

$$M^\vee \longrightarrow R \otimes_B M^\vee, \quad \lambda \longmapsto 1_R \otimes \lambda.$$

- (12) **Now regard as an A -module through .** Now regard $R \otimes_B M^\vee$ as an A -module through j_M .

- (13) **Apply the cotangent calculation.** The free cotangent calculation gives

$$L_{A/B} \simeq A \otimes_B M^\vee,$$

and therefore the cotangent classifier and extension–restriction adjunction give a natural equivalence

$$\begin{aligned} \text{Der}_B(A, R \otimes_B M^\vee) &\simeq \text{Map}_{\text{Mod}_A}(A \otimes_B M^\vee, R \otimes_B M^\vee) \\ &\simeq \text{Map}_{\text{Mod}_B}(M^\vee, R \otimes_B M^\vee). \end{aligned}$$

Under this equivalence, a B -linear derivation from the free algebra $A = \text{Sym}_B(M^\vee)$ is determined precisely by its value on the free generator $M^\vee$.

- (14) **Identify the map.** The two B -linear derivations

$$D \circ j_M \quad \text{and} \quad (j_M \otimes 1) \circ d_{\text{univ}}$$

have just been shown to correspond to the same generator map

$$M^\vee \longrightarrow R \otimes_B M^\vee, \quad \lambda \longmapsto 1_R \otimes \lambda.$$

- (15) **Identify the equivalence.** The identity path of this generator map therefore transports through the preceding equivalence of derivation spaces to a canonical homotopy

$$D \circ j_M \simeq (j_M \otimes 1) \circ d_{\text{univ}}.$$

- (16) **Verify compatibility.** This is the canonical commuting square of coefficient components displayed in the theorem.
- (17) **Conclude the argument.** Finally, both square-zero derivation sections have the corresponding algebra map as their first component and the preceding derivation as their coefficient component.
- (18) **Verify compatibility.** Hence the same canonical homotopy gives the commuting square

$$\begin{array}{ccc} A & \xrightarrow{j_M} & R \\ \tilde{d}_{\text{univ}} \downarrow & & \downarrow \tilde{D} \\ A \oplus (A \otimes_B M^\vee) & \xrightarrow{j_M \oplus (j_M \otimes 1)} & R \oplus (R \otimes_B M^\vee). \end{array}$$

- (19) **No finite free retract, basis, or other presentation of.** No finite free retract, basis, or other presentation of M is used.

□

The final issue for the linear branch is substitution: pullback transports the cofree differential structure through a canonical comparison, but that comparison need not be invertible.

2.6.5 Substitution compatibility of the linear exponential

Construction 2.6.9 (Oplax substitution comparison for cofree coalgebras). Let $f : Y \rightarrow X$ be a morphism of spectral schemes. Strong symmetric monoidality of

$$f^* : \text{QCoh}(X) \longrightarrow \text{QCoh}(Y)$$

induces a functor on cocommutative coalgebras. For $M \in \text{QCoh}(X)$, the morphism

$$f^*(!_X M) \xrightarrow{f^* \varepsilon_M} f^* M$$

therefore classifies, by the cofree universal property in $\text{QCoh}(Y)$, a canonical coalgebra morphism

$$f^* C_X(M) \longrightarrow C_Y(f^* M).$$

On underlying coefficients this is a natural comparison

$$\beta_{f,M} : f^*(!_X M) \longrightarrow !_Y(f^* M).$$

Proposition 2.6.10 (Oplax substitution compatibility of the differential exponential). *The comparisons $\beta_{f,M}$ are coherent under identities, composition, and higher homotopies of substitution. They satisfy the following canonical compatibilities, with the strong symmetric-monoidal identifications for f^* suppressed from notation. Superscripts X and Y distinguish the corresponding comonad, monoidal, coalgebra, Seely, and bialgebra maps in the two coefficient fibres.*

(i) **Comonad.**

$$\varepsilon_{f^*M}^Y \beta_{f,M} \simeq f^*(\varepsilon_M^X),$$

and

$$\delta_{f^*M}^Y \beta_{f,M} \simeq !_Y(\beta_{f,M}) \beta_{f,!_X M} f^*(\delta_M^X).$$

(ii) **Monoidal structure.** For $M, N \in \mathbf{QCoh}(X)$,

$$\beta_{f,M \otimes N} f^*(m_{M,N}^X) \simeq m_{f^*M, f^*N}^Y (\beta_{f,M} \otimes \beta_{f,N}),$$

and the analogous unit comparison holds for $m_{\mathcal{O}_X}$ and $m_{\mathcal{O}_Y}$.

(iii) **Cocommutative coalgebra structure.** Since $\beta_{f,M}$ underlies a coalgebra morphism,

$$\Delta_{f^*M}^Y \beta_{f,M} \simeq (\beta_{f,M} \otimes \beta_{f,M}) f^*(\Delta_M^X), \quad e_{f^*M}^Y \beta_{f,M} \simeq f^*(e_M^X).$$

(iv) **Seely and additive bialgebra structure.** The Seely comparisons satisfy

$$\chi_{f^*M, f^*N}^Y \beta_{f,M \oplus N} \simeq (\beta_{f,M} \otimes \beta_{f,N}) f^*(\chi_{M,N}^X),$$

and the corresponding unit comparison holds. Moreover

$$\beta_{f,M} f^*(\nabla_M^X) \simeq \nabla_{f^*M}^Y (\beta_{f,M} \otimes \beta_{f,M}), \quad \beta_{f,M} f^*(u_M^X) \simeq u_{f^*M}^Y.$$

(v) **Primitive infinitesimals and codereliction.** Strong symmetric monoidality and exactness identify

$$f^*P_X(M) \simeq P_Y(f^*M),$$

and under this identification

$$\beta_{f,\mathcal{O}_X \oplus M} f^*(H_M^{\inf}) \simeq H_{f^*M}^{\inf}.$$

Consequently

$$\beta_{f,M} f^*(\eta_M) \simeq \eta_{f^*M}.$$

(vi) **Infinitesimal translation and differentiation.** The coalgebra-level infinitesimal translations satisfy

$$\beta_{f,M} \circ f^*(U_X \tilde{\mathbf{d}}_M) \simeq U_Y \tilde{\mathbf{d}}_{f^*M} \circ (\beta_{f,M} \otimes 1),$$

as maps

$$f^*(!_X M) \otimes P_Y(f^*M) \longrightarrow !_Y(f^*M).$$

Restricting to the primitive summand gives

$$\beta_{f,M} f^*(\mathbf{d}_M) \simeq \mathbf{d}_{f^*M} (\beta_{f,M} \otimes 1_{f^*M}).$$

Consequently the stable coefficient fibres carry an oplax indexed family of the coherent differential-exponential structures constructed above; no invertibility of β is used.

Proof.

- (1) **Use the adjunction.** The comparison $\beta_{f,M}$ is the underlying map of the coalgebra morphism adjoint to $f^*\varepsilon_M^X$.
- (2) **Its defining adjunct equation.** Its defining adjunct equation is exactly the first formula in (i).
- (3) **For comultiplication, both sides of the second formula.** For comultiplication, both sides of the second formula underlie coalgebra maps

$$f^*C_X(M) \longrightarrow C_Y(!_Y f^*M).$$

- (4) **Verify naturality.** After composition with the cofree counit in the target, the left side reduces to $\beta_{f,M}$ by the comonad counit law, while the right side reduces to the same map by naturality of ε_Y^Y , the defining adjunct equation for $\beta_{f,!_X M}$, and the comonad counit law for $!_X$.
- (5) **Verify naturality.** Cofreeness therefore gives the comparison; identity, composition, and higher-pasting coherences follow from the same adjunction naturality.
- (6) **For (ii), both sides.** For (ii), both sides are coalgebra maps from $f^*C_X(M) \times f^*C_X(N)$ to the cofree coalgebra $C_Y(f^*M \otimes f^*N)$.
- (7) **Derive the comparison.** Composing with the target counit gives on both sides

$$f^*(\varepsilon_M^X) \otimes f^*(\varepsilon_N^X).$$

- (8) **Use the cofree universal property.** The cofree mapping-space equivalence therefore gives the monoidal comparison.
- (9) **unit case.** The unit case is identical.
- (10) **Part (iii).** Part (iii) is immediate because the map defining β is a morphism in $\text{CoCom}(\text{QCoh}(Y))$.
- (11) **Use the product structure.** For (iv), the Seely comparison is induced by the Cartesian product comparison

$$C_X(M \oplus N) \xrightarrow{\sim} C_X(M) \times C_X(N).$$

- (12) **Apply and compare with the corresponding product comparison in the.** Apply f^* and compare with the corresponding product comparison in the target fibre.
- (13) **Apply the universal property.** Both maps to each product factor have the same cofree adjunct, so the product universal property gives the displayed compatibility with χ .
- (14) **Use exactness.** Exactness of f^* preserves biproduct addition and zero.
- (15) **Use the cofree universal property.** Since $\widetilde{\nabla}_M$ and $\widetilde{u}_M$ are the cofree coalgebra lifts of the maps induced by addition and zero, the same cofree-adjunct comparison gives compatibility with ∇ and u .
- (16) **For (v), the primitive coalgebra construction.** For (v), the primitive coalgebra construction is the opposite split square-zero algebra construction.

- (17) **Use the product structure.** Strong symmetric monoidality and preservation of finite biproducts by f^* therefore give $f^*P_X(M) \simeq P_Y(f^*M)$.

- (18) **Compare the two coalgebra.** Compare the two coalgebra maps

$$P_Y(f^*M) \rightrightarrows C_Y(\mathcal{O}_Y \oplus f^*M)$$

given by $\alpha_{P_Y(f^*M)}$ and by the pullback of $\alpha_{P_X(M)}$ followed by the cofree comparison β .

- (19) **Use the cofree universal property.** After composition with the cofree counit, both have adjunct $\text{id}_{\mathcal{O}_Y \oplus f^*M}$.

- (20) **Conclude the step.** Hence

$$\beta_{f, \mathcal{O}_X \oplus M} f^*(H_M^{\text{inf}}) \simeq H_{f^*M}^{\text{inf}}.$$

- (21) **Next compare the two coalgebra.** Next compare the two coalgebra maps

$$P_Y(f^*M) \rightrightarrows C_Y(f^*M)$$

given respectively by H_{f^*M} and by

$$f^*P_X(M) \xrightarrow{f^*H_M} f^*C_X(M) \longrightarrow C_Y(f^*M).$$

- (22) **Use the cofree universal property.** After composition with the cofree counit both have adjunct pr_1 .

- (23) **Conclude the step.** Hence they agree canonically.

- (24) **Restrict the construction.** Restriction along the primitive inclusion gives

$$\beta_{f, M} f^*(\eta_M) \simeq \eta_{f^*M}.$$

- (25) **For (vi), both displayed composites.** For (vi), both displayed composites are coalgebra maps into the cofree coalgebra $C_Y(f^*M)$.

- (26) **Compose them with .** Compose them with $\varepsilon_{f^*M}^Y$.

- (27) **For the first composite, the defining properties of.** For the first composite, the defining properties of β and $\tilde{d}_M$ give the pullback of

$$e_M^X \otimes \text{pr}_1 + \varepsilon_M^X \otimes \text{pr}_0.$$

- (28) **Derive the comparison.** For the second, part (iii) gives preservation of the coalgebra counit, while part (i) gives $\varepsilon_{f^*M}^Y \beta_{f, M} \simeq f^* \varepsilon_M^X$.

- (29) **Its adjunct.** Its adjunct is therefore the same map.

- (30) **Derive the comparison.** Cofreeness gives the canonical comparison, coherently in f and M .

- (31) **Restrict the construction.** Restriction along the primitive summand proves the final formula.

□

Remark 2.6.5 (No indexed equivalence is asserted). The canonical substitution structure proved here is oplax indexed, with the comparison

$$\beta_{f,M} : f^*(!_X M) \longrightarrow !_Y(f^* M).$$

An equivalence for this comparison would require an additional base-change theorem for cofree spectral coalgebras. Cofree formation is controlled by cosifted limits, and spectral Tate phenomena obstruct the naive interchange of those limits with tensor products in general; see [4]. The differential compatibility above is formulated with the actual comparison β .

The linear branch therefore exports three facts: every coefficient fibre carries the cofree differential-exponential structure, its derivative agrees with the spectral universal derivation on the stated finite-projective models, and substitution acts through the canonical oplax comparison. The geometric branch does not depend on this construction; it returns to the cotangent classifier and uses degree-one classes to build nonlinear contexts.

2.7 Structured infinitesimal context extension

A degree-one derivation term is converted functorially into a square-zero thickening whose centre is the original context. The conceptual passage is

$$L_{X/S} \xrightarrow{\eta} M[1]$$

classifying coeff square-zero construction

$$X \xhookrightarrow{i_\eta} X_\eta.$$

This is analogous in purpose to context formation, but it has different variance from the affine comprehension of Chapter 1. The section first constructs the affine structured extension and distinguishes its three base-change operations; it then proves the Zariski compatibility required to glue the construction intrinsically over an arbitrary spectral context.

Convention 2.7.1 (Structured and underlying infinitesimal arrows). A **structured** square-zero extension is determined by its degree-one derivation together with the coefficient occurring in that derivation; its square-zero algebra, augmentation, and pullback comparison are constructed functorially. An **underlying** square-zero extension is an arrow in the equivalence-closed essential image of that construction. Passing to the underlying arrow forgets the coefficient and classifying derivation. A coefficient, factorization, filtration, pullback model, local equation, relative refinement, or representative chosen later inside a proof is a proof witness and is not retained on the underlying arrow.

2.7.1 Affine structured square-zero extensions

Construction 2.7.1 (Structured square-zero extension classified by a derivation). Let $A \rightarrow B$ be a morphism of commutative ring spectra, let $M \in \text{Mod}_B$, and let

$$\eta : L_{B/A} \longrightarrow M[1].$$

Under 2.3.1, η represents an A -linear derivation

$$d_\eta : B \longrightarrow B \oplus M[1].$$

Let

$$d_0 : B \longrightarrow B \oplus M[1]$$

be the zero derivation and let

$$p : B \oplus M[1] \longrightarrow B$$

be the split augmentation. Put

$$\mathcal{A} := \mathrm{CAlg}(\mathrm{Sp})_{A/}.$$

Because

$$p d_\eta \simeq \mathrm{id}_B \simeq p d_0,$$

the maps d_η and d_0 are morphisms

$$\mathrm{id}_B \rightrightarrows p$$

in the slice $\mathcal{A}_{/B}$. Define the structured extension arrow by the pullback in that slice

$$(q_\eta : B_\eta \rightarrow B) := \mathrm{id}_B \times_p \mathrm{id}_B.$$

Since limits in a slice are created in the ambient category, its underlying commutative A -algebra is

$$B_\eta \simeq B \times_{B \oplus M[1]} B,$$

with specified pullback square

$$\begin{array}{ccc} B_\eta & \xrightarrow{q_\eta} & B \\ r_\eta \downarrow & & \downarrow d_\eta \\ B & \xrightarrow{d_0} & B \oplus M[1]. \end{array}$$

Applying the augmentation p to the pullback comparison gives a canonical homotopy

$$q_\eta \simeq r_\eta.$$

The classified square-zero arrow is $q_\eta : B_\eta \rightarrow B$; the second projection and the displayed homotopy are functorially determined by the pullback. The derivation η , its coefficient M , and this functorially constructed arrow form the **structured square-zero extension of B by M over A** classified by η .

Proposition 2.7.1 (Fibre and connectivity of a structured extension). *In 2.7.1, the homotopy fibre of*

$$q_\eta : B_\eta \longrightarrow B$$

is canonically equivalent to M , regarded as a B_η -module by restriction of scalars. The induced multiplication on that fibre is coherently null. If B and M are connective, then B_η is connective, $\pi_0(q_\eta)$ is surjective, and its kernel is square-zero.

Proof.

- (1) **forgetful functor creates limits, so the defining square.** The forgetful functor $\mathrm{CAlg}(\mathrm{Sp}) \rightarrow \mathrm{Sp}$ creates limits, so the defining square is Cartesian on underlying spectra.

- (2) **Transport the structure.** Under the canonical decomposition

$$B \oplus M[1],$$

the two sections have the form

$$d_\eta = (\text{id}_B, \delta_\eta), \quad d_0 = (\text{id}_B, 0),$$

where $\delta_\eta : B \rightarrow M[1]$ is the underlying relative derivation.

- (3) **Compute the fibre.** Consequently the pullback is canonically equivalent, compatibly with q_η , to the homotopy fibre of δ_η :

$$B_\eta \simeq \text{fib}(\delta_\eta) \longrightarrow B.$$

- (4) **Derive the comparison.** Equivalently, pullback invariance of fibres applied to the horizontal maps in the defining square gives

$$\text{fib}(q_\eta) \simeq \text{fib}(d_0) \simeq \Omega M[1] \simeq M.$$

- (5) **Compute the fibre.** Thus there is a canonical fibre sequence

$$M \longrightarrow B_\eta \xrightarrow{q_\eta} B \xrightarrow{\delta_\eta} M[1].$$

- (6) **tangent-theoretic derivation-to-extension construction equips the fibre with zero.** The tangent-theoretic derivation-to-extension construction equips the fibre with zero nonunital commutative multiplication; equivalently

$$M \otimes_{B_\eta} M \longrightarrow M$$

is coherently null.

- (7) **Compute the fibre.** If B and M are connective, the displayed fibre sequence makes B_η connective.
- (8) **Verify surjectivity.** Its long exact homotopy sequence gives surjectivity on π_0 , and the kernel is the image of $\pi_0(M)$.
- (9) **null multiplication on the fibre.** The null multiplication on the fibre makes the product of any two elements of that image zero.

□

Theorem 2.7.1 (Structured square-zero parameterization). *Let $A \rightarrow B$ be a morphism of commutative ring spectra and let $M \in \text{Mod}_B$. Let*

$$\text{SqZ}_A^{\text{str}}(B; M)$$

be the ∞ -groupoid whose objects are derivation terms $\eta : L_{B/A} \rightarrow M[1]$, read through the associated square-zero arrows of 2.7.1, and whose morphisms are induced by homotopies of classifying derivations. The square-zero algebra, augmentation, and pullback identification are outputs of the construction rather than further fields of data. Then there are natural equivalences

$$\begin{aligned} \text{SqZ}_A^{\text{str}}(B; M) &\simeq \text{Der}_A(B, M[1]) \\ &\simeq \text{Map}_{\text{Mod}_B}(L_{B/A}, M[1]). \end{aligned}$$

Proof.

- (1) **Handle the second component.** The second equivalence is 2.3.1.
- (2) **Handle the first component.** The first is the functorial pullback construction of 2.7.1.
- (3) **Identify contractibility.** For a fixed derivation, the space of pullback objects equipped with an identification with the specified pullback is contractible.
- (4) **Conclude the step.** Hence adjoining the constructed square-zero arrow adds no independent structure.

□

Lemma 2.7.1 (Square-zero boundary and the relative Hurewicz morphism). *Let $A \rightarrow B$ be a morphism of commutative ring spectra, let $M \in \text{Mod}_B$, and let*

$$\eta : L_{B/A} \longrightarrow M[1]$$

classify the structured square-zero extension

$$q_\eta : B_\eta \longrightarrow B.$$

Put

$$Q_\eta := \text{cofib}(A \rightarrow B_\eta), \quad Q := \text{cofib}(A \rightarrow B).$$

Octahedrality for $A \rightarrow B_\eta \rightarrow B$ supplies a cofibre sequence

$$Q_\eta \longrightarrow Q \xrightarrow{\beta_\eta} M[1].$$

Then β_η is canonically the composite

$$Q \longrightarrow B \otimes_A Q \xrightarrow{\epsilon_{B/A}} L_{B/A} \xrightarrow{\eta} M[1].$$

The identification is natural in the relative algebra, the coefficient, and the classifying derivation.

Proof.

- (1) **Fix the notation.** Let

$$\delta_\eta : B \longrightarrow M[1]$$

be the underlying A -linear morphism obtained by subtracting the zero derivation from the derivation section:

$$d_\eta - d_0 = (0, \delta_\eta).$$

- (2) **Invoke the cited result.** By 2.7.1, the slice pullback construction is canonically the fibre construction for δ_η , compatibly with the classified projection q_η .
- (3) **Compute the fibre.** Hence there is a canonical fibre–cofibre sequence

$$M \longrightarrow B_\eta \xrightarrow{q_\eta} B \xrightarrow{\delta_\eta} M[1].$$

- (4) **Restrict the construction.** The map δ_η is canonically null after restriction to A , because d_η is an A -linear derivation.

- (5) **Construct the factorization.** It therefore factors through

$$Q = \text{cofib}(A \rightarrow B) \longrightarrow M[1].$$

- (6) **Compute the fibre.** Octahedrality for $A \rightarrow B_\eta \xrightarrow{q_\eta} B$ identifies this factor with the boundary

$$\beta_\eta : Q \rightarrow M[1]$$

in the displayed cofibre sequence of the statement.

- (7) **Construct the factorization.** By the characterization of $\epsilon_{B/A}$ in 2.4.1, the factor of the underlying relative derivation δ_η through Q is exactly the composite

$$Q \longrightarrow B \otimes_A Q \xrightarrow{\epsilon_{B/A}} L_{B/A} \xrightarrow{\eta} M[1].$$

- (8) **Conclude the step.** Thus this composite is β_η .

- (9) **Verify naturality.** Naturality is inherited from the functorial square-zero pullback, cofibre, and Hurewicz constructions.

□

Proposition 2.7.2 (Relative-base change of structured square-zero extensions). *Let $A \rightarrow B$ and $A \rightarrow A'$ be morphisms of commutative ring spectra, put*

$$B' := B \otimes_A A', \quad M' := M \otimes_B B',$$

and let

$$\eta : L_{B/A} \longrightarrow M[1]$$

classify a structured square-zero extension $B_\eta \rightarrow B$. Let

$$\eta' : L_{B'/A'} \simeq L_{B/A} \otimes_B B' \longrightarrow M'[1]$$

be its cotangent base change. Then there is a canonical equivalence of augmented A' -algebras

$$B_\eta \otimes_A A' \xrightarrow{\sim} B'_{\eta'}$$

compatible with the maps to B' . These equivalences are coherent under iterated base change.

Proof.

- (1) **Fix the notation.** Write the defining structured square-zero pullback as

$$\begin{array}{ccc} B_\eta & \longrightarrow & B \\ \downarrow & & \downarrow d_\eta \\ B & \xrightarrow{d_0} & B \oplus M[1]. \end{array}$$

- (2) **Extension of scalars from to.** Extension of scalars from A to A' is symmetric monoidal.

- (3) **Apply cotangent base change.** Hence it carries the split square-zero algebra to the split square-zero algebra

$$(B \oplus M[1]) \otimes_A A' \simeq B' \oplus M'[1],$$

and cotangent base change identifies the scalar extension of the section d_η with the section $d_{\eta'}$.

- (4) **zero section.** The zero section is carried to the zero section.

- (5) **Apply base change.** Thus the base-changed square supplies a canonical morphism

$$B_\eta \otimes_A A' \longrightarrow B' \times_{B' \oplus M'[1]} B' = B'_{\eta'}.$$

- (6) **It remains to.** It remains to prove that this morphism is an equivalence; no preservation of arbitrary pullbacks by algebraic extension of scalars is being assumed.

- (7) **forgetful functor from commutative algebras to module.** The forgetful functor from commutative algebras to module spectra creates limits.

- (8) **Compute the fibre.** Consequently the defining pullback has an underlying fibre sequence

$$B_\eta \longrightarrow B \oplus B \xrightarrow{d_\eta - d_0} B \oplus M[1].$$

- (9) **Compute the fibre.** Tensoring this fibre sequence over A with A' is exact in module spectra and gives

$$B_\eta \otimes_A A' \longrightarrow B' \oplus B' \xrightarrow{d_{\eta'} - d_0} B' \oplus M'[1].$$

- (10) **fibre of the final arrow.** The fibre of the final arrow is exactly the underlying spectrum of $B'_{\eta'}$.

- (11) **Conclude the step.** Therefore the canonical algebra morphism above is an equivalence on underlying spectra and hence an equivalence of commutative A' -algebras.

- (12) **Apply cotangent base change.** Every map used in the construction is the canonical scalar-extension, cotangent-base-change, split-square-zero, or finite-limit comparison.

- (13) **Apply base change.** Their identity, composition, and higher-pasting coherences therefore give the stated coherence under iterated base change.

□

Lemma 2.7.2 (Algebraic pullback of structured square-zero extensions). *Let $A \rightarrow D$ be a morphism of commutative ring spectra and let*

$$q_\eta : E \longrightarrow D$$

be the structured square-zero extension classified by

$$\eta : L_{D/A} \longrightarrow M[1], \quad M \in \text{Mod}_D.$$

For every morphism of commutative A -algebras

$$u : D' \longrightarrow D,$$

put

$$E' := E \times_D D'.$$

Then $E' \rightarrow D'$ is canonically a structured square-zero extension with coefficient M regarded as a D' -module by restriction of scalars. The construction is coherent under identities, composition, and iterated algebraic pullback. If D' and the restricted coefficient are connective, then E' is connective.

Proof.

(1) **Fix the notation.** Put $\mathcal{A} := \mathrm{CAlg}(\mathrm{Sp})_{A/}$.

(2) **Identify the map.** The structured extension is the finite limit

$$q_\eta \simeq \mathrm{id}_D \times_p \mathrm{id}_D$$

in the slice $\mathcal{A}_{/D}$, where

$$p : D \oplus M[1] \longrightarrow D$$

is the split augmentation and the two maps $\mathrm{id}_D \rightrightarrows p$ are the derivation section d_η and the zero section.

(3) **Apply base change.** Base change along u is the pullback functor

$$u^* : \mathcal{A}_{/D} \longrightarrow \mathcal{A}_{/D'}, \quad (C \rightarrow D) \longmapsto (C \times_D D' \rightarrow D').$$

(4) **Use the adjunction.** It is right adjoint to postcomposition along u , hence preserves limits.

(5) **Conclude the step.** Therefore

$$\begin{aligned} u^* q_\eta &\simeq u^*(\mathrm{id}_D \times_p \mathrm{id}_D) \\ &\simeq u^* \mathrm{id}_D \times_{u^* p} u^* \mathrm{id}_D \\ &\simeq \mathrm{id}_{D'} \times_{u^* p} \mathrm{id}_{D'}. \end{aligned}$$

(6) **pullback of the split augmentation.** The pullback of the split augmentation is canonically

$$u^* p : D' \oplus \mathrm{Res}_{D'}^D M[1] \longrightarrow D'.$$

(7) **Verify compatibility.** Indeed, limits of commutative ring spectra are created on underlying spectra, and the pullback of $D \oplus M[1] \rightarrow D$ has underlying D' -module $D' \oplus \mathrm{Res}_{D'}^D M[1]$, with the functorially pulled-back square-zero multiplication.

(8) **two pulled-back sections.** The two pulled-back sections are respectively the pulled-back derivation and the zero derivation.

(9) **Conclude the step.** Consequently

$$E' \simeq D' \times_{D' \oplus \mathrm{Res}_{D'}^D M[1]} D',$$

which is the structured square-zero extension classified by that pulled-back derivation.

(10) **coefficient fibre.** The coefficient fibre is the restricted D' -module M .

(11) **Compute the fibre.** Connectivity follows from the resulting fibre sequence.

- (12) **Apply base change.** Identity, composition, and all higher-pasting coherences are those of functorial slice base change and therefore require no additional compatibility datum. $\square$

Theorem 2.7.2 (Geometric Cartesian base change of structured square-zero extensions). *Let*

$$q : E \longrightarrow B$$

be a structured square-zero extension of connective commutative ring spectra with connective coefficient $M \in \text{Mod}_B$. For every morphism of connective commutative ring spectra

$$E \longrightarrow C,$$

put

$$D := B \otimes_E C, \quad M_D := M \otimes_B D.$$

Then the pushout arrow

$$C \longrightarrow D$$

is canonically a structured square-zero extension with coefficient M_D . Equivalently, the Cartesian base change of $\text{Spec}(B) \rightarrow \text{Spec}(E)$ along $\text{Spec}(C) \rightarrow \text{Spec}(E)$ is a structured square-zero infinitesimal substitution. These transported structures are coherent under identities, composition, and iterated geometric Cartesian base change.

Proof.

- (1) **Forget any relative base that may.** Forget any relative base that may have been used to construct q .
- (2) **Verify connectivity.** A structured square-zero extension of connective commutative ring spectra is a point of the connective derivation category of [30].
- (3) **Verify connectivity.** In the notation of that source, the functor from connective derivations to connective algebra arrows factors through a left fibration over the category whose morphisms are pushout squares; see [30, Theorem 3.25].
- (4) **Verify connectivity.** For the displayed pushout

$$\begin{array}{ccc} E & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D = B \otimes_E C, \end{array}$$

the ring spectrum D is connective.

- (5) **Verify connectivity.** Moreover connective pushout gives

$$\pi_0(D) \simeq \pi_0(B) \otimes_{\pi_0(E)} \pi_0(C),$$

and $\pi_0(E) \rightarrow \pi_0(B)$ is surjective by 2.7.1.

- (6) **Verify surjectivity.** Hence $\pi_0(C) \rightarrow \pi_0(D)$ is surjective.

- (7) **Verify connectivity.** Thus the target arrow belongs to the connective, surjective-on- π_0 arrow category and the displayed square is a morphism in the pushout-square subcategory used in [30, Theorem 3.25].
- (8) **Left-fibration transport along this morphism therefore produces.** Left-fibration transport along this morphism therefore produces a structured square-zero presentation of $C \rightarrow D$.
- (9) **transported tangent coefficient.** The transported tangent coefficient is extension of scalars and is canonically

$$M \otimes_B D = M_D.$$

- (10) **left-fibration transport.** The left-fibration transport supplies the identity, composition, and iterated-pushout coherences.
- (11) **If the square.** If the square is equipped with a map from an additional base algebra, that map is retained on the pushed-out arrow; when a relative classifying derivation is required, it is constructed from this base structure by 2.8.2 after passing to the underlying square-zero arrow.

□

Remark 2.7.1 (Three distinct square-zero base-change operations). The preceding results distinguish three operations which will be used below. Relative-base change changes the base A of a derivation; algebraic pullback forms a limit $E \times_D D'$ in commutative algebras; geometric Cartesian base change forms a pushout $B \otimes_E C$ of coordinate algebras. Their variance is different, and no one of the three is used as a substitute for another.

Remark 2.7.2 (Structured data and underlying arrows). The parameterization theorem concerns structured extensions. Later we pass to the equivalence-closed class of their underlying arrows. At that point no chosen coefficient or classifying derivation is retained. Recognition of a structured presentation from an arbitrary underlying arrow is a separate theorem on whatever bounded domain it is available; it is not part of the definition of an infinitesimal substitution.

The affine construction is functorial in the coordinate algebra, but a global spectral thickening also requires compatibility with restriction to affine opens of the centre. We now prove that compatibility and use it to glue the affine square-zero pieces. The gluing data are supplied by the construction itself rather than retained independently.

2.7.2 Native global structured infinitesimal context extension

Lemma 2.7.3 (Zariski restriction of structured square-zero extensions). *Let $A \rightarrow B$ be connective, let $M \in (\text{Mod}_B)_{\geq 0}$, and let*

$$\eta : L_{B/A} \longrightarrow M[1]$$

classify the structured extension $B_\eta \rightarrow B$. Let

$$u : B \longrightarrow B'$$

represent an affine spectral Zariski open immersion and put

$$M' := B' \otimes_B M, \quad \eta' : L_{B'/A} \simeq B' \otimes_B L_{B/A} \longrightarrow M'[1],$$

where the first equivalence uses transitivity and $L_{B'/B} \simeq 0$. Functoriality of split square-zero formation and of the defining pullback produces a canonical morphism

$$u_\eta : B_\eta \longrightarrow B'_{\eta'}.$$

This morphism represents an affine spectral Zariski open immersion. Moreover the square of spectral schemes

$$\begin{array}{ccc} \mathrm{Spec}(B') & \longrightarrow & \mathrm{Spec}(B) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(B'_{\eta'}) & \longrightarrow & \mathrm{Spec}(B_\eta) \end{array}$$

is Cartesian. The construction and the Cartesian comparison are coherent under identities and composition of affine open restrictions.

Proof.

(1) **First suppose.** First suppose that $B' = B[f^{-1}]$ is principal.

(2) **Verify surjectivity.** The surjection

$$\pi_0(B_\eta) \twoheadrightarrow \pi_0(B)$$

allows a lift $\tilde{f} \in \pi_0(B_\eta)$ of f .

(3) **Localize the structured extension at.** Localize the structured extension at $\tilde{f}$:

$$B_\eta[\tilde{f}^{-1}] \longrightarrow B[f^{-1}].$$

(4) **Verify compatibility.** View the defining pullback square of B_η as a diagram of commutative B_η -algebras through its two projections to B and the common map to $B \oplus M[1]$.

(5) **Apply base change.** Base change this diagram along the principal localization

$$B_\eta \longrightarrow B_\eta[\tilde{f}^{-1}].$$

(6) **Apply base change.** On underlying module spectra principal localization is exact, so the base-changed diagram remains Cartesian; limits of commutative algebras are created on underlying spectra.

(7) **Apply base change.** The two base changes of B are canonically $B[f^{-1}]$.

(8) **Verify connectivity.** Indeed, because M is connective, $M[1]$ is 1-connective, so the derivation section and the zero section

$$B \rightrightarrows B \oplus M[1]$$

induce the same map on π_0 .

(9) **Conclude the step.** Hence the two projections $B_\eta \rightrightarrows B$ agree on π_0 , and both images of $\tilde{f}$ are f .

- (10) **Use spectral localization.** Likewise split square-zero formation commutes with localization, so the third vertex is

$$B[f^{-1}] \oplus M[f^{-1}][1].$$

- (11) **Use spectral localization.** Cotangent localization identifies the nonzero section with the derivation η' .

- (12) **Apply base change.** Hence the base-changed Cartesian square is precisely

$$B[f^{-1}] \times_{B[f^{-1}] \oplus M[f^{-1}][1]} B[f^{-1}],$$

and therefore

$$B_\eta[\tilde{f}^{-1}] \simeq B'_{\eta'}.$$

- (13) **Use spectral localization.** Thus u_η is a principal localization.
- (14) **Derive the comparison.** Base changing the corresponding open immersion along the centre $\mathrm{Spec}(B) \rightarrow \mathrm{Spec}(B_\eta)$ gives $\mathrm{Spec}(B[f^{-1}])$, proving the Cartesian assertion in the principal case.
- (15) **proof did not retain .** The proof did not retain $\tilde{f}$.
- (16) **Verify invertibility.** Indeed two lifts differ by an element of the square-zero kernel of $\pi_0(B_\eta) \rightarrow \pi_0(B)$; after either one is inverted, the other differs from it by a nilpotent perturbation and is therefore invertible as well.
- (17) **Conclude the step.** Thus the principal open subobject and the resulting comparison are independent of the proof choice.
- (18) **For a general affine open immersion.** For a general affine open immersion $\mathrm{Spec}(B') \rightarrow \mathrm{Spec}(B)$, choose a finite principal cover of its image.
- (19) **Use spectral localization.** On every principal member and on every finite intersection the preceding construction identifies the corresponding restriction of B'_η with a principal localization of B_η .
- (20) **Verify functoriality.** These identifications agree on refinements by functoriality of cotangent base change and structured square-zero formation.
- (21) **Apply Zariski descent.** Spectral Zariski descent then shows that $\mathrm{Spec}(B'_{\eta'}) \rightarrow \mathrm{Spec}(B_\eta)$ is an open immersion and that the displayed square is Cartesian.
- (22) **Use the stated property.** Since the canonical morphism u_η existed before choosing the principal cover, the cover is only a proof witness.
- (23) **Identity and composition coherence.** Identity and composition coherence are inherited from the functorial square-zero construction and may be checked on the principal basis.

□

Lemma 2.7.4 (Compatible-pair restriction of structured square-zero extensions). *Let (U, V) and (U', V') be compatible affine pairs for a morphism of spectral schemes, with*

$$U = \mathrm{Spec}(B), \quad V = \mathrm{Spec}(A), \quad U' = \mathrm{Spec}(B'), \quad V' = \mathrm{Spec}(A')$$

and with $(U', V') \rightarrow (U, V)$ a refinement. Let $M \in (\text{Mod}_B)_{\geq 0}$ and let

$$\eta : L_{B/A} \longrightarrow M[1].$$

Write

$$\bar{B} := B \otimes_A A', \quad \bar{M} := M \otimes_B \bar{B}, \quad M' := M \otimes_B B'.$$

Cotangent base change gives

$$\bar{\eta} : L_{\bar{B}/A'} \simeq L_{B/A} \otimes_B \bar{B} \longrightarrow \bar{M}[1].$$

The map $\bar{B} \rightarrow B'$ represents an affine spectral Zariski open immersion, and transitivity gives

$$L_{B'/A'} \simeq L_{\bar{B}/A'} \otimes_{\bar{B}} B'.$$

Let

$$\eta' : L_{B'/A'} \longrightarrow M'[1]$$

be the resulting restriction. Then there is a canonical morphism

$$B_\eta \longrightarrow B'_{\eta'}$$

which represents an affine spectral Zariski open immersion, and the square

$$\begin{array}{ccc} \text{Spec}(B') & \longrightarrow & \text{Spec}(B) \\ \downarrow & & \downarrow \\ \text{Spec}(B'_{\eta'}) & \longrightarrow & \text{Spec}(B_\eta) \end{array}$$

is Cartesian. These morphisms and Cartesian comparisons are coherent under identities, composition, and iterated compatible-pair refinement.

Proof.

- (1) **Apply base change.** The affine open $V' \rightarrow V$ base changes to an affine open

$$\bar{U} := U \times_V V' = \text{Spec}(\bar{B}) \longrightarrow U.$$

- (2) **Construct the factorization.** Since $U' \rightarrow U$ lies over V' , it factors canonically through an affine open immersion

$$U' \longrightarrow \bar{U},$$

so $\bar{B} \rightarrow B'$ represents an affine spectral Zariski open immersion.

- (3) **Apply base change.** Structured square-zero base change gives a canonical equivalence

$$B_\eta \otimes_A A' \xrightarrow{\sim} \bar{B}_{\bar{\eta}}.$$

- (4) **Apply base change.** Consequently

$$\text{Spec}(\bar{B}_{\bar{\eta}}) \longrightarrow \text{Spec}(B_\eta)$$

is the base change of the affine open $V' \rightarrow V$, and hence is an affine open immersion.

(5) **Apply the cited result, now over the fixed base .** Apply 2.7.3, now over the fixed base A' , to the affine open $\bar{B} \rightarrow B'$.

(6) **Derive the comparison.** It gives an affine open immersion

$$\mathrm{Spec}(B'_{\eta'}) \longrightarrow \mathrm{Spec}(\bar{B}_{\bar{\eta}})$$

whose pullback to the centre is $\mathrm{Spec}(B') \rightarrow \mathrm{Spec}(\bar{B})$.

(7) **Derive the comparison.** Composing the two open immersions gives the asserted map to $\mathrm{Spec}(B_{\eta})$.

(8) **centre square.** The centre square is Cartesian by pullback pasting.

(9) **on coordinate algebras.** Equivalently, on coordinate algebras,

$$\begin{aligned} B \otimes_{B_{\eta}} B'_{\eta'} &\simeq (B \otimes_{B_{\eta}} \bar{B}_{\bar{\eta}}) \otimes_{\bar{B}_{\bar{\eta}}} B'_{\eta'} \\ &\simeq \bar{B} \otimes_{\bar{B}_{\bar{\eta}}} B'_{\eta'} \\ &\simeq B'. \end{aligned}$$

(10) **Apply cotangent base change.** Every comparison used here is the canonical cotangent-base-change, structured-square-zero-base-change, or Zariski-restriction comparison.

(11) **Their identity, composition, and higher-pasting coherences therefore.** Their identity, composition, and higher-pasting coherences therefore give the last assertion.

□

Construction 2.7.2 (Global structured infinitesimal context extension). Let $f : X \rightarrow S$ be a morphism of spectral schemes, let $M \in \mathrm{QCoh}(X)_{\geq 0}$, and let

$$\eta : L_{X/S} \longrightarrow M[1].$$

For every compatible affine pair

$$U = \mathrm{Spec}(B) \longrightarrow V = \mathrm{Spec}(A)$$

for $X \rightarrow S$, restriction gives

$$\eta_U : L_{B/A} \longrightarrow M_U[1]$$

and hence the affine structured square-zero extension

$$B_{\eta_U} := B \times_{B \oplus M_U[1]} B.$$

For every compatible affine-pair refinement $(U', V') \rightarrow (U, V)$, the canonical morphism supplied by 2.7.4 identifies

$$\mathrm{Spec}(B'_{\eta_{U'}}) \longrightarrow \mathrm{Spec}(B_{\eta_U})$$

as the corresponding affine open restriction. These open maps are coherent under identities, composition, and further compatible refinement. Thus the constructed affine thickenings form a spectral Zariski gluing diagram. Define

$$X_{\eta}$$

to be the spectral scheme obtained by gluing this diagram, and let

$$i_\eta : X \longrightarrow X_\eta$$

be the morphism obtained by gluing the affine closed immersions

$$\mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(B_{\eta_U}).$$

Neither the affine cover nor any principal refinement used to verify the open transition maps is retained in X_η .

Proposition 2.7.3 (Geometric properties of global structured infinitesimal extension). *In 2.7.2:*

- (i) X_η is a spectral scheme;
- (ii) $i_\eta : X \rightarrow X_\eta$ is a closed immersion over S ;
- (iii) the underlying topological map is a homeomorphism;
- (iv) pullback along i_η induces an equivalence between the small spectral Zariski sites of X_η and X ;
- (v) on every compatible affine pair $U = \mathrm{Spec}(B) \rightarrow V = \mathrm{Spec}(A)$, the restriction is

$$\mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(B_{\eta_U});$$

- (vi) the construction commutes with arbitrary represented relative-base change: for every Cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S, \end{array}$$

if η' is the cotangent base change of η , then there is a canonical equivalence over S'

$$X'_{\eta'} \xrightarrow{\sim} X_\eta \times_S S',$$

compatible with the centres and coherent under iterated relative-base change.

Proof.

- (1) **Apply Zariski descent.** The compatible-pair open-transition theorem 2.7.4 makes the affine pieces $\mathrm{Spec}(B_{\eta_U})$ into a genuine Zariski gluing diagram, so spectral Zariski descent constructs a spectral scheme and proves (i) and (v).
- (2) **Verify surjectivity.** By 2.7.1, every affine coordinate map

$$B_{\eta_U} \longrightarrow B$$

is surjective on π_0 with square-zero kernel.

- (3) **Conclude the step.** Hence every local geometric map $\mathrm{Spec}(B) \rightarrow \mathrm{Spec}(B_{\eta_U})$ is a closed immersion, and closedness is local on the target.

(4) **Identify the resulting comparison.** This proves (ii).

(5) **Verify nilpotence.** A quotient of an ordinary ring by a nilpotent ideal induces a homeomorphism on prime spectra.

(6) **Conclude the step.** Thus

$$|\mathrm{Spec}(B)| \xrightarrow{\cong} |\mathrm{Spec}(B_{\eta_U})|$$

on every affine piece.

(7) **Glue the local data.** The homeomorphisms commute with the open transition maps of 2.7.4, so they glue to (iii).

(8) **Use spectral localization.** Principal spectral opens are detected by degree-zero localization.

(9) **Across a square-zero quotient, every principal open of the centre.** Across a square-zero quotient, every principal open of the centre has the unique corresponding principal open of the thickening, and conversely because the prime spectra agree.

(10) **Lemma the cited result.** Lemma 2.7.4 shows that these correspondences are compatible with intersections and covering families.

(11) **Principal opens form bases of both small.** Principal opens form bases of both small spectral Zariski sites, so we obtain the equivalence in (iv).

(12) **For (vi), restrict the Cartesian square to.** For (vi), restrict the Cartesian square to a compatible affine pair

$$U = \mathrm{Spec}(B) \longrightarrow V = \mathrm{Spec}(A)$$

for $X \rightarrow S$, and choose an affine open $V' = \mathrm{Spec}(A') \rightarrow S'$ mapping to V .

(13) **Fix the notation.** Put

$$\bar{B} := B \otimes_A A'.$$

(14) **Apply cotangent base change.** Cotangent base change transports η_U to a class

$$\bar{\eta} : L_{\bar{B}/A'} \longrightarrow (M_U \otimes_B \bar{B})[1],$$

and 2.7.2 gives the canonical equivalence

$$B_{\eta_U} \otimes_A A' \xrightarrow{\cong} \bar{B}_{\bar{\eta}}.$$

(15) **Now let be any compatible affine refinement.** Now let $U' = \mathrm{Spec}(B')$ be any compatible affine refinement of the pulled-back source over V' .

(16) **Then represents an affine spectral Zariski open.** Then $\bar{B} \rightarrow B'$ represents an affine spectral Zariski open immersion.

(17) **Restrict the construction.** By 2.7.4, the corresponding restriction of $\mathrm{Spec}(\bar{B}_{\bar{\eta}})$ is canonically $\mathrm{Spec}(B'_{\eta'})$, and its pullback to the centre is $\mathrm{Spec}(B')$.

(18) **Restrict the construction.** Hence, on the compatible affine basis of X' , the restrictions of $X_{\eta} \times_S S'$ are exactly the affine thickenings used to construct $X'_{\eta'}$.

- (19) **Apply Zariski descent.** The restriction comparisons commute with further affine refinement by the coherence of 2.7.4; spectral Zariski descent therefore glues them uniquely to

$$X'_{\eta'} \xrightarrow{\cong} X_{\eta} \times_S S'.$$

- (20) **Apply cotangent base change.** Identity, composition, and higher-pasting coherence are inherited from affine cotangent base change, structured square-zero scalar extension, open restriction, and descent.

□

Corollary 2.7.1 (Fixed-site ringed-topos recognition). *Through the site equivalence of 2.7.3(iv), the structured spectral-scheme recognition of X_{η} has the same underlying small Zariski ∞ -topos as X . On that identified topos its structure sheaf is the finite pullback*

$$\mathcal{O}_X^{\eta} \simeq \mathcal{O}_X \times_{\mathcal{O}_X \oplus M[1]} \mathcal{O}_X,$$

where the two maps are the sheaf-level derivation represented by η and the zero derivation.

Proof.

- (1) **Identify the equivalence.** The site equivalence is already a theorem of the native construction.
- (2) **Transport the structure.** Under the structured comparison, restrict to any affine open of the common site.
- (3) **structure sheaf of the constructed thickening.** The structure sheaf of the constructed thickening is represented there by B_{η_U} , which is exactly the displayed pullback of affine structure sheaves.
- (4) **Glue the local data.** These affine identifications commute with compatible affine restriction by 2.7.4 and hence glue.
- (5) **Conclude the step.** Thus the fixed-topos formula recognizes the native construction; it does not define it.

□

Definition 2.7.1 (Structured infinitesimal extension of a spectral context). Let $f : X \rightarrow S$ and $M \in \mathrm{QCoh}(X)_{\geq 0}$. A **structured infinitesimal extension of X over S by M** is a derivation term

$$\eta : L_{X/S} \longrightarrow M[1]$$

read together with the thickening $X \rightarrow X_{\eta}$ functorially constructed in 2.7.2. The thickening and its affine pullback comparisons are outputs of the construction, not additional fields of data. Write

$$\mathrm{Sq}Z_S^{\mathrm{str}}(X; M)$$

for the resulting ∞ -groupoid.

Theorem 2.7.3 (Global structured infinitesimal-extension parameterization). *For every $f : X \rightarrow S$ and $M \in \mathrm{QCoh}(X)_{\geq 0}$, the construction above gives a natural equivalence*

$$\mathrm{SqZ}_S^{\mathrm{str}}(X; M) \simeq \mathrm{Map}_{\mathrm{QCoh}(X)}(L_{X/S}, M[1]).$$

It is compatible with represented relative-base change.

Proof.

- (1) **right-hand side.** The right-hand side is, by definition, the space of global relative derivation terms.
- (2) **Apply Zariski descent.** Restriction to compatible affine pairs and quasi-coherent descent identify it with the limit of the affine derivation spaces

$$\mathrm{Map}_{\mathrm{Mod}_B}(L_{B/A}, M_U[1]).$$

- (3) **Invoke the cited result.** By 2.7.1, each affine space is the ∞ -groupoid of the corresponding constructed structured square-zero extension.
- (4) **Glue the local data.** Lemma 2.7.4 identifies the restriction maps on these groupoids with the open transition maps used to glue X_η .
- (5) **Apply Zariski descent.** Thus the descent limit is precisely the groupoid of global structured extensions constructed above.
- (6) **Apply base change.** Relative-base-change compatibility is 2.7.3(vi) together with 2.5.2.

□

The global structured construction therefore produces square-zero context extensions with the required base-change behavior. The infinitesimal test geometry, however, must depend only on the resulting arrows, not on chosen coefficients or classifying derivations. We therefore pass from structured outputs to an intrinsic class of underlying arrows.

2.8 Underlying square-zero substitutions and finite infinitesimal contexts

The first step forgets the presentation of a constructed square-zero arrow while retaining the property of belonging to its equivalence-closed essential image. The second closes this class under finite composition. This produces the presentation-free infinitesimal substitutions that will form the objects of the infinitesimal site.

2.8.1 Underlying square-zero substitutions

Definition 2.8.1 (Underlying spectral square-zero extension). A morphism

$$q : A' \longrightarrow A$$

of connective commutative ring spectra is a **spectral square-zero extension** if, as an object of $\mathrm{Fun}(\Delta^1, \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{cn}})$, it is equivalent to an arrow

$$q_\eta : A_\eta \longrightarrow A$$

constructed by 2.7.1 from a connective A -module M and an absolute degree-one derivation

$$\eta : L_A \longrightarrow M[1].$$

No structured presentation is retained on q .

Lemma 2.8.1 (Relative and absolute presentations have the same underlying arrow). *Let $A_0 \rightarrow A$ be a morphism of commutative ring spectra, let $M \in \text{Mod}_A$, and let*

$$\eta : L_{A/A_0} \longrightarrow M[1]$$

be a relative degree-one derivation. Put

$$\eta^{\text{abs}} : L_A \longrightarrow L_{A/A_0} \xrightarrow{\eta} M[1].$$

Then the structured square-zero extension classified by η over A_0 and the structured square-zero extension classified by η^{abs} over $\mathbb{S}$ have canonically the same underlying algebra arrow

$$A_\eta \longrightarrow A.$$

Consequently the equivalence-closed essential image in 2.8.1 is unchanged if relative structured presentations are allowed in its definition.

Proof.

- (1) **Transport the structure.** Under cotangent classification, precomposition $L_A \rightarrow L_{A/A_0}$ is exactly the operation of forgetting the A_0 -linearity of a relative derivation.
- (2) **Conclude the step.** Hence the relative class η and the absolute class η^{abs} determine the same underlying commutative-algebra section

$$d_\eta : A \longrightarrow A \oplus M[1];$$

the former additionally carries the specified compatibility with the A_0 -structure.

- (3) **Identify the map.** The zero section is likewise the same underlying map.
- (4) **Compare the two constructions.** Both square-zero algebras are therefore the same functorial pullback

$$A \times_{A \oplus M[1]} A,$$

and their projections to A agree.

- (5) **Only the relative base structure.** Only the relative base structure has been forgotten.

□

Definition 2.8.2 (Underlying square-zero infinitesimal substitution). A morphism of affine spectral contexts

$$\text{Spec}(A) \longrightarrow \text{Spec}(A')$$

is a **square-zero infinitesimal substitution** if its opposite coordinate map $A' \rightarrow A$ is a spectral square-zero extension. A morphism $j : U \rightarrow T$ of arbitrary spectral schemes is a square-zero infinitesimal substitution if it is affine and becomes one of the preceding affine arrows after pullback along every affine open of T .

Proposition 2.8.1 (Permanence of square-zero infinitesimal substitutions). *Square-zero infinitesimal substitutions are stable under equivalence and arbitrary base change. Every such substitution is a closed immersion, induces a homeomorphism on underlying topological spaces, and induces an equivalence of small spectral Zariski sites.*

Proof.

- (1) **Identify the equivalence.** The assertions are invariant under equivalence in the arrow category, so for an affine arrow choose, only inside the proof, a structured representative.
- (2) **Apply base change.** Geometric Cartesian base change of that representative is again structured by 2.7.2; forgetting the transported coefficient and derivation shows that the underlying class is stable under arbitrary base change.
- (3) **Verify connectivity.** The fibre and connectivity theorem for a structured representative gives surjectivity on π_0 with square-zero kernel, hence closedness and the homeomorphism on prime spectra.
- (4) **Restrict the construction.** The Zariski restriction theorem then identifies the small spectral Zariski sites.
- (5) **All claims descend through equivalence of arrows, and.** All claims descend through equivalence of arrows, and the global statement follows from the affine-local definition.

□

Remark 2.8.1 (No square-zero presentation datum). An underlying square-zero substitution is a property of an already specified morphism. A coefficient module or classifying derivation may be chosen inside a proof after replacing the arrow by an equivalent constructed arrow, but such a choice is not retained on the morphism. This distinction is mathematically necessary: [30, Warning 3.9] notes that an underlying square-zero extension need not determine its coefficient and classifying derivation uniquely, even up to equivalence. Recognition theorems such as [30, Theorem 3.17] recover structured presentations only on their stated bounded theorem domains; no such recognition is built into the present definition.

Lemma 2.8.2 (Relative refinement of a chosen square-zero presentation). *Let*

$$A \longrightarrow E \xrightarrow{q} B$$

be morphisms of connective commutative ring spectra and suppose that the underlying arrow q is a spectral square-zero extension. Choose, only inside the proof, an absolute structured representative supplied by 2.8.1, with connective B -module M and degree-one derivation

$$\eta : L_B \longrightarrow M[1].$$

Then the transported map $A \rightarrow E$ canonically determines a lift of η through transitivity to a relative derivation

$$\eta_A : L_{B/A} \longrightarrow M[1],$$

and the structured square-zero extension over A classified by η_A is equivalent to $E \rightarrow B$ as an arrow of commutative A -algebras. The relative derivation is a proof witness associated to the chosen structured presentation and is not retained on the underlying arrow.

Proof.

- (1) **Replace , only inside the proof, by the.** Replace q , only inside the proof, by the chosen structured presentation and transport the given map $A \rightarrow E$ across that equivalence.

- (2) **Conclude the step.** Thus

$$q : E \longrightarrow B$$

is literally the pullback in $\mathrm{CAlg}(\mathrm{Sp})/B$ of the two morphisms

$$d_\eta, d_0 : \mathrm{id}_B \rightrightarrows (B \oplus M[1] \rightarrow B).$$

- (3) **Identify the map.** The composite $A \rightarrow E \rightarrow B$ makes $A \rightarrow B$ an object of $\mathrm{CAlg}(\mathrm{Sp})/B$, and the transported map $A \rightarrow E$ is a point of

$$\mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})/B}(A, E).$$

- (4) **Compute the mapping object.** Mapping into the defining pullback identifies this mapping space with the space of paths between the two composites

$$A \longrightarrow B \xrightarrow{d_\eta, d_0} B \oplus M[1]$$

over B .

- (5) **Restrict the construction.** Hence the specified map $A \rightarrow E$ determines canonically a nullhomotopy of the restriction of the absolute derivation.

- (6) **Transport the structure.** Under cotangent classification this is a nullhomotopy of

$$L_A \otimes_A B \longrightarrow L_B \xrightarrow{\eta} M[1].$$

- (7) **Apply cotangent transitivity.** Apply $\mathrm{map}_{\mathrm{Mod}_B}(-, M[1])$ to the transitivity cofibre sequence

$$L_A \otimes_A B \longrightarrow L_B \longrightarrow L_{B/A}.$$

- (8) **Compute the fibre.** The chosen nullhomotopy is precisely the datum of a lift of η through this fibre sequence, and therefore determines a relative class

$$\eta_A : L_{B/A} \longrightarrow M[1].$$

- (9) **Restrict the construction.** Its associated relative derivation has the same underlying absolute derivation d_η and the same specified homotopy to the zero derivation after restriction along A .

- (10) **Identify contractibility.** Applying the slice pullback construction of 2.7.1 therefore recovers $E \rightarrow B$, together with its given commutative A -algebra structure, through a contractible space of comparisons.

- (11) **No coefficient or relative class.** No coefficient or relative class is retained after the proof.

□

The one-stage class is now intrinsic to the arrow. Finite infinitesimal contexts are obtained by closing this class under finite composition, without retaining a factorization as part of the resulting morphism.

2.8.2 Finite infinitesimal substitutions

Definition 2.8.3 (Finite spectral infinitesimal substitution). Let $\text{Inf}_{\text{Sp}}^{\text{fin}}$ be the smallest class of morphisms of spectral schemes which

- (i) contains all equivalences and all square-zero infinitesimal substitutions;
- (ii) is closed under finite composition.

A morphism in this class is a **finite spectral infinitesimal substitution**. No finite factorization is retained as structure on the morphism.

Proposition 2.8.2 (Permanence of finite infinitesimal substitutions). *Finite spectral infinitesimal substitutions are affine and are stable under equivalence, finite composition, and arbitrary base change.*

Proof.

- (1) **Set up the argument.** Equivalences are affine, square-zero infinitesimal substitutions are affine by 2.8.2, and affine morphisms are closed under finite composition.
- (2) **Conclude the step.** Hence every finite infinitesimal substitution is affine.
- (3) **Stability under equivalence and finite composition.** Stability under equivalence and finite composition is built into 2.8.3.
- (4) **Apply base change.** Let $\mathcal{B}$ be the class of morphisms whose arbitrary base changes belong to $\text{Inf}_{\text{Sp}}^{\text{fin}}$.
- (5) **Establish the next comparison.** Equivalences belong to $\mathcal{B}$, square-zero substitutions belong to $\mathcal{B}$ by 2.8.1, and $\mathcal{B}$ is closed under finite composition.
- (6) **Apply base change.** Minimality gives base-change stability.

□

Proposition 2.8.3 (Zariski support invariance). *Let*

$$j : U \longrightarrow T$$

be a finite spectral infinitesimal substitution. Then:

- (i) *the induced map $|U| \rightarrow |T|$ is a homeomorphism;*
- (ii) *pullback along j induces an equivalence between the posets of open spectral subschemes of T and U ;*
- (iii) *pullback induces an equivalence of their small spectral Zariski sites, preserving finite intersections, covering families, and Čech nerves.*

Proof.

- (1) **Each assertion holds for equivalences and for.** Each assertion holds for equivalences and for square-zero substitutions by 2.8.1.

- (2) **Identify the map.** The class of morphisms satisfying each assertion is closed under composition.
- (3) **Apply the minimality in the cited result.** Apply the minimality in 2.8.3.
- (4) **Use preservation.** Compatibility with Čech nerves follows because the site equivalence is induced by pullback and therefore preserves finite fibre products.

□

Remark 2.8.2 (Finite means generated by finite composition). The adjective **finite** refers only to generation by finite composition. No numerical order, length, filtration, or chosen sequence of square-zero stages is assigned to a finite infinitesimal substitution. A factorization may be chosen inside an induction, but the conclusion depends only on the underlying arrow.

Finite composition gives the formal class of infinitesimal substitutions. We next show that this terminology is reflected intrinsically on degree-zero coordinates: every such arrow is a nilpotent thickening there.

2.8.3 Nilpotence of finite infinitesimal substitutions

Lemma 2.8.3 (Nilpotence under composition of surjections). *Let*

$$R \xrightarrow{a} S \xrightarrow{b} T$$

be surjections of ordinary commutative rings. Put

$$I := \ker(a), \quad J := \ker(b), \quad K := \ker(ba).$$

If $I^m = 0$ and $J^n = 0$, then

$$K^n \subseteq I$$

and hence

$$K^{mn} = 0.$$

Proof.

- (1) **Apply the statement objectwise.** Every element of K maps into J under a .
- (2) **Conclude the step.** Therefore every product of n elements of K maps to zero in S , because $J^n = 0$.
- (3) **Conclude the step.** Hence $K^n \subseteq I$.
- (4) **Derive the comparison.** Raising to the m -th power gives $K^{mn} \subseteq I^m = 0$.

□

Proposition 2.8.4 (Finite infinitesimal substitutions are nilpotent thickenings on degree zero). *Let*

$$j : U \longrightarrow T$$

be a finite spectral infinitesimal substitution. Affine-locally on T , write

$$U = \operatorname{Spec}(B), \quad T = \operatorname{Spec}(A).$$

*Then $\pi_0(A) \rightarrow \pi_0(B)$ is surjective with nilpotent kernel. In particular, the adjective **infinitesimal** is intrinsic to the underlying arrow and does not depend on a chosen square-zero factorization.*

Proof.

- (1) **Construct the factorization.** Choose, only inside the proof, a finite factorization

$$U = U_0 \longrightarrow U_1 \longrightarrow \cdots \longrightarrow U_r = T$$

into underlying square-zero substitutions.

- (2) **Apply the statement objectwise.** Every stage is an affine morphism.

- (3) **Use the stated property.** Since $U_r = T$ is affine, U_{r-1} is affine; repeating this argument backwards shows inductively that every intermediate U_k is affine.

- (4) **Fix the notation.** Write

$$U_k = \operatorname{Spec}(A_k).$$

- (5) **Verify surjectivity.** For every stage the coordinate map

$$A_{k+1} \longrightarrow A_k$$

is surjective on π_0 with square-zero kernel by 2.8.1.

- (6) **Verify nilpotence.** Iterating 2.8.3 therefore shows that the kernel of

$$\pi_0(A_r) \longrightarrow \pi_0(A_0)$$

is nilpotent.

- (7) **Verify surjectivity.** Since nilpotence and surjectivity are properties of the composite ring map, the conclusion is independent of the chosen factorization.

- (8) **global statement.** The global statement is local on the affine target.

□

The nilpotence result makes the infinitesimal character of the underlying arrow independent of any chosen square-zero factorization. We can therefore compare the discrete sector with the familiar finite nilpotent pattern of ordinary algebraic geometry.

2.8.4 Discrete comparison with the classical finite infinitesimal pattern

Lemma 2.8.4 (Recognition of discrete square-zero quotients). *Let*

$$R' \twoheadrightarrow R$$

be a surjection of ordinary commutative rings with square-zero kernel J . Then the Eilenberg–Mac Lane morphism

$$HR' \longrightarrow HR$$

is a spectral square-zero extension in the sense of 2.8.1.

Proof.

- (1) **Verify surjectivity.** The morphism is a 0-small extension in the sense of [30, Definition 3.12 and Remark 3.16]: both rings are discrete, the map on degree zero is surjective, and the multiplication on its fibre HJ is null because $J^2 = 0$.

- (2) **Identify the resulting object.** The square-zero recognition theorem [30, Theorem 3.17] identifies every 0-small extension with a square-zero extension classified by a degree-one derivation.
- (3) **Conclude the step.** Hence $HR' \rightarrow HR$ belongs to the equivalence-closed essential image used in 2.8.1.
- (4) **No classifying derivation.** No classifying derivation is retained on the underlying arrow.

□

Proposition 2.8.5 (Nilpotent discrete thickenings are finite square-zero composites). *Let R be an ordinary commutative ring and let $I \subseteq R$ be a nilpotent ideal with $I^N = 0$. The discrete spectral thickening*

$$\mathrm{Spec} H(R/I) \longrightarrow \mathrm{Spec} HR$$

is a finite spectral infinitesimal substitution. More precisely, the coordinate quotient factors as

$$R = R/I^N \longrightarrow R/I^{N-1} \longrightarrow \cdots \longrightarrow R/I^2 \longrightarrow R/I,$$

and the kernel of

$$R/I^{k+1} \longrightarrow R/I^k$$

is I^k/I^{k+1} , which is square-zero.

Proof.

- (1) **For , one.** For $k \geq 1$, one has $2k \geq k + 1$, hence

$$I^{2k} \subseteq I^{k+1}.$$

- (2) **Conclude the step.** Thus multiplication on I^k/I^{k+1} is zero.
- (3) **Invoke the cited result.** By 2.8.4, Eilenberg–Mac Lane formation carries each quotient

$$R/I^{k+1} \longrightarrow R/I^k$$

to an underlying spectral square-zero extension.

- (4) **Derive the comparison.** Finite composition gives the asserted infinitesimal substitution.

□

Theorem 2.8.1 (Discrete finite-infinitesimal comparison). *Under the fully faithful discrete enhancement of Chapter 1, finite spectral infinitesimal substitutions between discrete affine contexts are exactly the ordinary nilpotent closed thickenings. Under this identification, codomain-pullback Zariski coverings agree with the usual Zariski coverings of nilpotent thickenings. Consequently the discrete affine sector of the spectral finite infinitesimal site has the classical finite infinitesimal-site pattern.*

Proof.

- (1) **One direction.** One direction is 2.8.5.
- (2) **Verify nilpotence.** Conversely, 2.8.4 makes the degree-zero kernel of a finite spectral infinitesimal substitution nilpotent.

- (3) **Identify the resulting object.** For discrete affine contexts the fully faithful enhancement identifies the spectral coordinate map with the corresponding ordinary quotient map.
- (4) **Use preservation.** Zariski opens are preserved and reflected by the discrete comparison of Chapter 1; codomain-pullback covers therefore agree with their classical counterparts.

□

We now have a presentation-free class of finite infinitesimal substitutions, stable under base change and invisible on Zariski support. The next task is to organize these arrows as contexts for a second indexed semantics and to put a locality theory on them.

2.9 The category of infinitesimal substitutions and its Zariski topology

Finite infinitesimal substitutions form a full subcategory of the arrow category. A morphism is therefore genuinely two-dimensional:

$$\begin{array}{ccc} U & \xrightarrow{\alpha} & U' \\ j \downarrow & & \downarrow j' \\ T & \xrightarrow{\beta} & T'. \end{array} \quad \text{dom}(j) = U, \quad \text{cod}(j) = T.$$

The identity-arrow embedding supplies ordinary affine contexts inside this category, while domain and codomain give its two endpoints. We first record that arrow-category structure and then transport the spectral Zariski topology through the codomain endpoint.

2.9.1 The infinitesimal arrow category

Definition 2.9.1 (Affine finite-infinitesimal category). Let

$$\text{Aff}_{\text{Sp}}^{\text{inf}} \subseteq \text{Fun}(\Delta^1, \text{Aff}_{\text{Sp}})$$

be the full subcategory spanned by finite spectral infinitesimal substitutions between affine spectral schemes. Define

$$\text{dom}, \text{cod} : \text{Aff}_{\text{Sp}}^{\text{inf}} \longrightarrow \text{Aff}_{\text{Sp}}$$

by

$$\text{dom}(U \rightarrow T) = U, \quad \text{cod}(U \rightarrow T) = T,$$

and let

$$i : \text{Aff}_{\text{Sp}} \longrightarrow \text{Aff}_{\text{Sp}}^{\text{inf}}, \quad i(T) = \text{id}_T.$$

Proposition 2.9.1 (Endpoint adjunctions). *There is a canonical adjoint triple*

$$\text{cod} \dashv i \dashv \text{dom}.$$

Proof.

- (1) **Fix the notation.** Let $j : U \rightarrow T$ be a finite infinitesimal substitution and $X \in \text{Aff}_{\text{Sp}}$.

- (2) **Verify compatibility.** A morphism

$$j \longrightarrow \text{id}_X$$

in the arrow category is a commutative square

$$\begin{array}{ccc} U & \longrightarrow & X \\ j \downarrow & & \parallel \\ T & \longrightarrow & X, \end{array}$$

and is determined by its lower map.

- (3) **Conclude the step.** Hence

$$\text{Map}_{\text{Aff}_{\text{Sp}}^{\text{inf}}}(j, iX) \simeq \text{Map}_{\text{Aff}_{\text{Sp}}}(j, X).$$

- (4) **a morphism.** Similarly, a morphism $\text{id}_X \rightarrow j$ is determined by its upper map $X \rightarrow U$, giving

$$\text{Map}_{\text{Aff}_{\text{Sp}}^{\text{inf}}}(iX, j) \simeq \text{Map}_{\text{Aff}_{\text{Sp}}}(X, \text{dom}(j)).$$

- (5) **Verify naturality.** Naturality gives the two adjunctions.

□

Proposition 2.9.2 (Finite products of infinitesimal substitutions). *The category $\text{Aff}_{\text{Sp}}^{\text{inf}}$ is closed under finite products in the arrow category, and*

$$i(T \times S) \simeq i(T) \times i(S).$$

Proof.

- (1) **Use terminality.** The terminal arrow is the identity of $\text{Spec}(\mathbb{S})$.
 (2) **For and , factor the product arrow.** For $U \rightarrow T$ and $V \rightarrow S$, factor the product arrow as

$$U \times V \longrightarrow T \times V \longrightarrow T \times S.$$

- (3) **Apply base change.** The first stage is a base change of $U \rightarrow T$ and the second a base change of $V \rightarrow S$.
 (4) **Compare the two constructions.** Both are finite infinitesimal substitutions by 2.8.2, hence so is their composite.

□

These endpoint functors determine the appropriate locality convention. Because finite infinitesimal substitutions preserve the small Zariski site, we cover the codomain and pull the cover back to the domain. This makes both endpoints continuous and later controls the Kan extensions.

2.9.2 The codomain-pullback spectral Zariski topology

Definition 2.9.2 (Codomain-pullback Zariski basis cover). Let $j : U \rightarrow T$ be an object of $\text{Aff}_{\text{Sp}}^{\text{inf}}$. A **codomain-pullback Zariski basis cover** of j is a finite family

$$\{j_\alpha : U_\alpha \rightarrow T_\alpha\}_{\alpha \in A} \longrightarrow j$$

such that $\{T_\alpha \rightarrow T\}_{\alpha \in A}$ is a finite affine spectral Zariski covering family and every square

$$\begin{array}{ccc} U_\alpha & \longrightarrow & U \\ j_\alpha \downarrow & & \downarrow j \\ T_\alpha & \longrightarrow & T \end{array}$$

is Cartesian. Thus

$$U_\alpha \simeq U \times_T T_\alpha.$$

Proposition 2.9.3 (The infinitesimal Zariski pretopology). *Codomain-pullback Zariski basis covers contain identities and are stable under pullback and finite composition. They therefore generate a Grothendieck topology*

$$\tau_{\text{Zar}}^{\text{inf}}$$

on $\text{Aff}_{\text{Sp}}^{\text{inf}}$.

Proof.

- (1) **Identity covers.** Identity covers are immediate.
- (2) **Pulling a codomain-pullback cover of back along.** Pulling a codomain-pullback cover of $j : U \rightarrow T$ back along a morphism $j' : U' \rightarrow T' \rightarrow j$ in the arrow category pulls the codomain cover back to a spectral Zariski cover of T' .
- (3) **Its domain.** Its domain is

$$U' \times_{T'} (T_\alpha \times_T T') \simeq U' \times_T T_\alpha,$$

so the resulting squares remain Cartesian.

- (4) **For composition, compose the codomain Zariski covers.** For composition, compose the codomain Zariski covers and use iterated pullback for the domains.
- (5) **spectral Zariski pretopology of Chapter the cited result.** The spectral Zariski pretopology of Chapter 1 is stable under both operations.

□

Proposition 2.9.4 (Endpoint continuity). *The functors dom, cod carry codomain-pullback Zariski basis covers to spectral Zariski covering families, and i carries finite affine spectral Zariski covers to codomain-pullback covers.*

Proof.

- (1) **For this.** For cod this is the definition.

- (2) **Apply base change.** For dom , the domain family is the base change

$$\{U \times_T T_\alpha \rightarrow U\}_\alpha$$

of a spectral Zariski cover.

- (3) **Derive the comparison.** For i , a cover $\{T_\alpha \rightarrow T\}$ gives the codomain-pullback family $\{\text{id}_{T_\alpha} \rightarrow \text{id}_T\}$.

□

Lemma 2.9.1 (Affine-open invariance under finite infinitesimal substitutions). *Let*

$$j : U \longrightarrow T$$

be a finite spectral infinitesimal substitution between affine spectral schemes. Pullback along j induces an equivalence

$$j^* : \text{AffOp}(T) \xrightarrow{\sim} \text{AffOp}(U)$$

between the categories of affine spectral Zariski open subobjects. Under this equivalence finite intersections and finite affine covering families are preserved.

Proof.

- (1) **First suppose.** First suppose that j is represented by a structured square-zero extension

$$\text{Spec}(B) \longrightarrow \text{Spec}(B_\eta).$$

- (2) **For an affine open, the cited result.** For an affine open $\text{Spec}(B') \rightarrow \text{Spec}(B)$, 2.7.3 constructs an affine open

$$\text{Spec}(B'_{\eta'}) \longrightarrow \text{Spec}(B_\eta)$$

whose pullback to $\text{Spec}(B)$ is $\text{Spec}(B')$.

- (3) **Restrict the construction.** The construction is compatible with identities, composition, and further affine restriction.
- (4) **Verify surjectivity.** Thus pullback is essentially surjective on affine opens for a structured square-zero stage.
- (5) **Identify the resulting object.** It is fully faithful because 2.8.1 identifies the small spectral Zariski sites of the source and target.
- (6) **Verify surjectivity.** In particular, it identifies the mapping posets of open subobjects, and hence the full subcategories of affine open subobjects once essential surjectivity has been established.
- (7) **Now let be an underlying square-zero substitution.** Now let j be an underlying square-zero substitution between affine spectral schemes.
- (8) **Its opposite affine coordinate.** Its opposite affine coordinate map is, by definition, an underlying spectral square-zero extension, hence is equivalent in the arrow category to a structured square-zero extension.

- (9) **Replace the arrow, only inside this proof.** Replace the arrow, only inside this proof, by such an equivalent structured representative.
- (10) **Affineness of an open subobject and the Cartesian pullback property.** Affineness of an open subobject and the Cartesian pullback property are invariant under equivalence in the arrow category.
- (11) **Conclude the step.** Therefore the preceding one-stage argument applies to every underlying square-zero substitution between affine spectral schemes.
- (12) **No structured presentation.** No structured presentation is retained.
- (13) **Conclude the argument.** Finally let

$$j : U \longrightarrow T$$

be a finite infinitesimal substitution between affine spectral schemes.

- (14) **Construct the factorization.** Choose, only inside the proof, a factorization

$$U = U_0 \longrightarrow U_1 \longrightarrow \cdots \longrightarrow U_r = T$$

into underlying square-zero substitutions.

- (15) **Apply the statement objectwise.** Every stage

$$U_i \longrightarrow U_{i+1}$$

is affine by the definition of a square-zero infinitesimal substitution.

- (16) **Use the stated property.** Since $U_r = T$ is affine, affineness of the morphism $U_{r-1} \rightarrow U_r$ implies that U_{r-1} is affine.
- (17) **Repeating the same argument backwards.** Repeating the same argument backwards shows inductively that every intermediate U_i is affine.
- (18) **Construct the factorization.** Hence the one-stage affine-open equivalence proved above may be iterated along the entire chosen factorization.
- (19) **Starting from an affine open of , this.** Starting from an affine open of $U = U_0$, this iteration produces an affine open of $T = U_r$ whose pullback is the original affine open.
- (20) **Verify surjectivity.** Thus

$$j^* : \text{AffOp}(T) \longrightarrow \text{AffOp}(U)$$

is essentially surjective.

- (21) **Use full faithfulness.** Full faithfulness follows intrinsically from 2.8.3, which identifies the small spectral Zariski sites of T and U .
- (22) **Construct the factorization.** The same intrinsic site equivalence shows that the iterated construction is independent of the chosen finite square-zero factorization: it is the unique open subobject of T corresponding to the specified open subobject of U .

- (23) **Use the stated property.** Since the site equivalence is induced by pullback, it preserves finite intersections and covering families.
- (24) **Conclude the step.** Therefore j^* is the asserted equivalence of affine-open categories with the stated compatibility.

□

Proposition 2.9.5 (Domain classification of infinitesimal Zariski covers). *For every $j : U \rightarrow T$ in $\mathbf{Aff}_{\mathbf{Sp}}^{\text{inf}}$, domain evaluation induces an equivalence between the category of codomain-pullback Zariski basis covers of j and the category of finite affine spectral Zariski covers of U , including refinement morphisms and finite intersections.*

Proof.

- (1) **codomain-pullback cover.** A codomain-pullback cover is determined by its codomain cover.
- (2) **Invoke the cited result.** By 2.9.1, pullback along $j : U \rightarrow T$ identifies finite affine Zariski covers of T with finite affine Zariski covers of U .
- (3) **Transport the same comparison.** The same lemma, together with the intrinsic site equivalence of 2.8.3, shows that refinement morphisms and finite intersections are carried to the corresponding domain refinements and intersections.
- (4) **Identify the resulting comparison.** This is exactly the claimed equivalence of cover categories.

□

The codomain-pullback pretopology is now established and is subcanonical on the resulting arrow category. Passing to sheaves produces the ambient infinitesimal topos in which the endpoint semantics will live.

2.9.3 The infinitesimal Zariski topos

Definition 2.9.3 (Finite-infinitesimal spectral Zariski topos). Define

$$\mathbf{Stk}_{\text{inf}, \text{Zar}}^{\text{sp}} := \mathbf{Sh}_{\tau_{\text{Zar}}^{\text{inf}}}(\mathbf{Aff}_{\mathbf{Sp}}^{\text{inf}}).$$

As in Chapter 1, no hypercompletion is understood unless explicitly stated. From this point onward write

$$\mathcal{H}_{\mathbf{Sp}} := \mathbf{Stk}_{\text{Zar}}^{\text{sp}}, \quad \mathcal{H}_{\text{inf}} := \mathbf{Stk}_{\text{inf}, \text{Zar}}^{\text{sp}}.$$

These symbols are fixed for the remainder of the chapter.

Proposition 2.9.6 (Subcanonicity of the infinitesimal Zariski topology). *The topology $\tau_{\text{Zar}}^{\text{inf}}$ is subcanonical.*

Proof.

- (1) **Fix the notation.** Let $k : V \rightarrow S$ be an object of $\mathbf{Aff}_{\mathbf{Sp}}^{\text{inf}}$.
- (2) **Compute the mapping object.** For $j : U \rightarrow T$, the mapping space in the arrow category is the pullback

$$\mathbf{Map}_{\mathbf{Aff}_{\mathbf{Sp}}^{\text{inf}}}(j, k) \simeq \mathbf{Map}_{\mathbf{Aff}_{\mathbf{Sp}}}(U, V) \times_{\mathbf{Map}_{\mathbf{Aff}_{\mathbf{Sp}}}(U, S)} \mathbf{Map}_{\mathbf{Aff}_{\mathbf{Sp}}}(T, S).$$

- (3) **For a codomain-pullback cover of , codomain evaluation.** For a codomain-pullback cover of j , codomain evaluation is a spectral Zariski cover of T , and domain evaluation is the corresponding spectral Zariski cover of U .
- (4) **Apply Zariski descent.** Representable functors on Aff_{Sp} satisfy Zariski descent by Chapter 1.
- (5) **Apply Zariski descent.** Limits commute with limits in spaces, so the displayed pullback of representable mapping functors satisfies the same descent condition.

□

The absolute infinitesimal topos supplies the global test geometry. For a fixed relative context $X \rightarrow S$, we also need the corresponding affine basis of infinitesimal thickenings incident with X and lying over S ; its covering families are inherited from the global topology rather than chosen anew.

2.9.4 The relative spectral infinitesimal site

Lemma 2.9.2 (Affineness across finite infinitesimal substitutions). *Let*

$$j : U \longrightarrow T$$

be a finite spectral infinitesimal substitution of spectral schemes. Then

$$U \text{ is affine} \iff T \text{ is affine.}$$

Proof.

- (1) **If U is affine, then T is affine because.** If T is affine, then U is affine because every finite infinitesimal substitution is affine.
- (2) **Prove the converse.** Conversely suppose that U is affine.
- (3) **Derive the comparison.** Applying the 0-truncated reconstruction of Chapter 1 gives a morphism of ordinary schemes

$$t_0 U \longrightarrow t_0 T.$$
- (4) **Verify nilpotence.** By 2.8.4, this morphism is Zariski-locally a surjective quotient by a nilpotent ideal; hence it is an ordinary nilpotent closed thickening.
- (5) **Affineness of ordinary schemes.** Affineness of ordinary schemes is invariant under nilpotent thickenings; we use this standard theorem only at the 0-truncated comparison interface (compare [37]).
- (6) **Use the stated property.** Since $t_0 U$ is affine, $t_0 T$ is affine.
- (7) **affineness-detection statement of the 0-truncated reconstruction theorem of.** The affineness-detection statement of the 0-truncated reconstruction theorem of Chapter 1 then implies that T is affine.
- (8) **Identify the resulting comparison.** This is the one use in the construction of the affine finite-infinitesimal basis where the downstream 0-truncated comparison interface of Chapter 1 is invoked.

- (9) **native definitions of finite infinitesimal substitutions and their.** The native definitions of finite infinitesimal substitutions and their Zariski topology are formulated intrinsically in spectral geometry.
- (10) **Construct the factorization.** The intrinsic nilpotence property of j supplies the required input, while any chosen finite square-zero factorization remains a proof witness.

□

Definition 2.9.4 (Relative finite spectral infinitesimal site). Let $X \rightarrow S$ be a morphism of spectral schemes. Define

$$\mathrm{Inf}^{\mathrm{fin}}(X/S)$$

to be the category whose objects are diagrams

$$\begin{array}{ccc} U & \hookrightarrow & X \\ \downarrow & & \downarrow \\ T & \longrightarrow & S \end{array}$$

in which $U \hookrightarrow X$ is an affine spectral Zariski open immersion and $U \hookrightarrow T$ is a finite spectral infinitesimal substitution. A morphism is a commutative morphism of such diagrams over $X \rightarrow S$. A covering family of $(U \hookrightarrow T)$ is a finite family obtained from a finite affine spectral Zariski cover $\{T_\alpha \rightarrow T\}$ by Cartesian pullback:

$$U_\alpha := U \times_T T_\alpha.$$

We call the resulting site the **relative finite spectral infinitesimal site of X/S** .

Proposition 2.9.7 (The relative site is inherited from the global infinitesimal geometry). *The covering families of $\mathrm{Inf}^{\mathrm{fin}}(X/S)$ are precisely the restrictions of the codomain-pullback Zariski covering families of $\mathrm{Aff}_{\mathrm{Sp}}^{\mathrm{inf}}$. They contain identities and are stable under pullback and composition inside the relative category $\mathrm{Inf}^{\mathrm{fin}}(X/S)$. No independent infinitesimal topology is chosen.*

Proof.

- (1) **For every object of the relative site.** For every object of the relative site, U is affine by definition and $U \rightarrow T$ is finite infinitesimal.
- (2) **Conclude the step.** Hence T is affine by 2.9.2.
- (3) **Conclude the step.** Thus every object has an underlying object $U \rightarrow T$ of $\mathrm{Aff}_{\mathrm{Sp}}^{\mathrm{inf}}$, and its covering definition is exactly 2.9.2.
- (4) **Fix the notation.** Let $(U' \rightarrow T') \rightarrow (U \rightarrow T)$ be a morphism in the relative category and let $\{T_\alpha \rightarrow T\}$ be a covering family.
- (5) **Restrict the construction.** The pullbacks

$$T'_\alpha := T' \times_T T_\alpha$$

form a finite affine spectral Zariski cover of the affine T' , and

$$U'_\alpha := U' \times_{T'} T'_\alpha$$

is the corresponding affine open restriction of U' .

- (6) **Apply base change.** Base-change stability of finite infinitesimal substitutions makes $U'_\alpha \rightarrow T'_\alpha$ an object of the same relative category.
- (7) **Identify the resulting comparison.** This is exactly pullback stability of the global codomain-pullback pretopology; composition is inherited in the same way.
- (8) **Identify the map.** The maps to $X \rightarrow S$ are incidence data and do not alter the covering condition.

□

Remark 2.9.1 (External base change and the affine basis). External base change for an arbitrary morphism of relative contexts $(X'/S') \rightarrow (X/S)$ is handled on the affine basis by refining the pullback of each affine open of X by affine spectral Zariski opens of X' . The construction is then performed on that affine refinement, as supplied by ordinary Zariski locality.

Remark 2.9.2 (Classical infinitesimal-site comparison). On discrete spectral contexts, 2.8.1 identifies the objects with nilpotent thickenings of affine open subcontexts and identifies the covering families with their ordinary Zariski coverings. Thus the construction is the spectral analogue of the finite infinitesimal site pattern of [40]; no ordinary scheme category is used in its native definition.

These constructions supply the infinitesimal test geometry: its arrow category, Zariski topology, sheaf topos, and relative affine-basis sites. The endpoint semantics can therefore be derived from this geometry rather than postulated separately.

2.10 Differential cohesion and the infinitesimal-shape calculus

The identity-arrow embedding has both domain and codomain adjoints at the level of affine infinitesimal contexts. This section promotes those endpoint relations to an adjoint quadruple on sheaf topoi. Only after the quadruple has been constructed do we recognize it as differential cohesion and package its composites as

$$\mathfrak{R} \dashv \mathfrak{S} \dashv \mathfrak{L}.$$

The resulting infinitesimal-shape reflector then supplies the formal lifting comparison and the infinitesimal disk relation used in the next section.

2.10.1 Kan-extension formulas

Let

$$i^* : \mathrm{PSh}(\mathrm{Aff}_{\mathrm{Sp}}^{\mathrm{inf}}) \longrightarrow \mathrm{PSh}(\mathrm{Aff}_{\mathrm{Sp}})$$

denote restriction along i^{op} . The endpoint adjunctions compute its left and right Kan extensions pointwise, which gives the first three sheaf adjoints.

Proposition 2.10.1 (Endpoint Kan extensions). *For every presheaf F on $\mathrm{Aff}_{\mathrm{Sp}}$, there are canonical equivalences*

$$\mathrm{Lan}_{i^{\mathrm{op}}} F \simeq \mathrm{cod}^* F, \quad \mathrm{Ran}_{i^{\mathrm{op}}} F \simeq \mathrm{dom}^* F.$$

Pointwise,

$$(\mathrm{cod}^* F)(U \rightarrow T) = F(T), \quad (\mathrm{dom}^* F)(U \rightarrow T) = F(U).$$

Proof.

- (1) **Use the adjunction.** The adjunctions $\text{cod} \dashv i \dashv \text{dom}$ of 2.9.1 become, after passage to opposite categories, the adjunctions computing left and right Kan extension along i^{op} .
- (2) **Use the adjunction.** A Kan extension along a functor possessing the relevant adjoint is computed by precomposition with that adjoint.

□

Proposition 2.10.2 (The first three sheaf adjoints). *The endpoint formulas restrict to sheaves and give adjunctions*

$$i_! \dashv i^* \dashv i_* : \text{Stk}_{\text{Zar}}^{\text{sp}} \rightleftarrows \text{Stk}_{\text{inf, Zar}}^{\text{sp}},$$

with

$$i_! = \text{cod}^*, \quad i_* = \text{dom}^*.$$

Both $i_!$ and i_* are fully faithful, and $i_!$ preserves all limits.

Proof.

- (1) **Apply Zariski descent.** On a codomain-pullback basis cover, the descent diagram for $\text{cod}^* F$ is exactly the Zariski descent diagram for the codomain cover.
- (2) **descent diagram for.** The descent diagram for $\text{dom}^* F$ is the one for the domain cover, which is Zariski by 2.9.4.
- (3) **Conclude the step.** Thus both precomposition functors carry Zariski sheaves to infinitesimal Zariski sheaves.
- (4) **Restrict the construction.** Conversely, restriction along i carries codomain-pullback covers of identity arrows to ordinary Zariski covers, so i^* preserves sheaves.
- (5) **Use the adjunction.** The presheaf adjunctions therefore restrict to sheaves.
- (6) **Use the stated property.** Since

$$\text{cod } i \simeq \text{id}_{\text{Aff}_{\mathbb{S}_p}} \simeq \text{dom } i,$$

we have

$$i^* i_! \simeq \text{id}, \quad i^* i_* \simeq \text{id},$$

so $i_!$ and i_* are fully faithful.

- (7) **Conclude the argument.** Finally $i_! = \text{cod}^*$ is precomposition and therefore preserves all limits.

□

Three endpoint adjoints have been obtained. The fourth exists once the domain embedding i_* is shown to preserve sheaf colimits, so the remaining task is to compare domain pullback with Zariski sheafification.

2.10.2 Domain sheafification and the fourth adjoint

Lemma 2.10.1 (Domain pullback preserves Zariski local equivalences). *Let*

$$u : P \longrightarrow Q$$

be a spectral Zariski local equivalence of presheaves on Aff_{Sp} . Then

$$\text{dom}^* u : \text{dom}^* P \longrightarrow \text{dom}^* Q$$

is a $\tau_{\text{Zar}}^{\text{inf}}$ -local equivalence.

Proof.

- (1) **Identify the equivalence.** The local-equivalence class of spectral Zariski sheafification is the strongly saturated class generated by covering-sieve monomorphisms $R \hookrightarrow h_V$.
- (2) **Fix the notation.** Let $\mathcal{K}$ be the class of morphisms whose domain pullback is infinitesimally Zariski local.
- (3) **Use preservation.** Precomposition with dom preserves colimits, equivalences, compositions, and retract diagrams, while the infinitesimal local-equivalence class is strongly saturated.
- (4) **It is enough to treat a covering-sieve.** It is enough to treat a covering-sieve generator.
- (5) **Fix , an object , and a.** Fix $R \hookrightarrow h_V$, an object $j : U \rightarrow T$, and a section

$$x \in (\text{dom}^* h_V)(j) = \text{Map}(U, V).$$

- (6) **Pull back the covering sieve along .** Pull back the covering sieve along x .
- (7) **It contains a finite affine Zariski basis.** It contains a finite affine Zariski basis cover $\{U_\alpha \rightarrow U\}$ by the finite-principal-refinement theorem of Chapter 1.
- (8) **Invoke the cited result.** By 2.9.5, this domain cover is the domain family of a codomain-pullback basis cover

$$\{j_\alpha : U_\alpha \rightarrow T_\alpha\} \longrightarrow j.$$
- (9) **On each , the restricted section.** On each j_α , the restricted section lies in the pulled-back sieve.
- (10) **Conclude the step.** Thus every local section of $\text{dom}^* h_V$ is infinitesimally Zariski-locally in the image of $\text{dom}^* R$.
- (11) **Establish the next comparison.** Sheafification makes the resulting monomorphism both a monomorphism and an effective epimorphism, hence an equivalence.
- (12) **Strong saturation.** Strong saturation proves the lemma.

□

Theorem 2.10.1 (Domain pullback commutes with sheafification). *Let*

$$a_{\text{Zar}}^{\text{sp}} : \text{PSh}(\text{Aff}_{\text{Sp}}) \longrightarrow \text{Stk}_{\text{Zar}}^{\text{sp}}$$

and

$$a_{\text{inf,Zar}}^{\text{sp}} : \text{PSh}(\text{Aff}_{\text{Sp}}^{\text{inf}}) \longrightarrow \text{Stk}_{\text{inf,Zar}}^{\text{sp}}$$

be the two sheafification functors. There is a canonical equivalence

$$a_{\text{inf,Zar}}^{\text{sp}} \text{dom}^* \simeq \text{dom}^* a_{\text{Zar}}^{\text{sp}}.$$

Consequently $i_* = \text{dom}^*$ preserves all small colimits.

Proof.

- (1) **For a presheaf, its sheafification unit.** For a presheaf P , its sheafification unit $P \rightarrow a_{\text{Zar}}^{\text{sp}} P$ is a Zariski local equivalence.
- (2) **Invoke the cited result.** By 2.10.1, its domain pullback is an infinitesimal Zariski local equivalence.
- (3) **target is already a sheaf by the.** The target $\text{dom}^* a_{\text{Zar}}^{\text{sp}} P$ is already a sheaf by 2.10.2; hence it realizes the infinitesimal sheafification of $\text{dom}^* P$.
- (4) **Colimits of sheaves.** Colimits of sheaves are obtained by sheafifying presheaf colimits.
- (5) **Use the stated property.** Since precomposition preserves presheaf colimits and domain pullback commutes with sheafification, dom^* preserves sheaf colimits.

□

Theorem 2.10.2 (The infinitesimal adjoint string). *The functor i_* admits a right adjoint $i^!$, giving an adjoint string*

$$i_! \dashv i^* \dashv i_* \dashv i^! : \text{Stk}_{\text{Zar}}^{\text{sp}} \rightleftarrows \text{Stk}_{\text{inf,Zar}}^{\text{sp}}.$$

For $E \in \text{Stk}_{\text{inf,Zar}}^{\text{sp}}$ and an affine spectral context T ,

$$(i^! E)(T) \simeq \text{Map}_{\text{Stk}_{\text{inf,Zar}}^{\text{sp}}} (i_* h_T, E).$$

Proof.

- (1) **Use the adjunction.** The first three adjoints are 2.10.2.
- (2) **Use preservation.** The domain embedding i_* preserves all small colimits by 2.10.1; the two sheaf categories are presentable.
- (3) **Use the adjunction.** The presentable adjoint functor theorem therefore supplies a right adjoint $i^!$.
- (4) **Apply Yoneda.** The pointwise formula follows from Yoneda and the adjunction $i_* \dashv i^!$.

□

The endpoint quadruple has now been constructed. Only at this point do we introduce the differential-cohesive terminology and package its composites as the three standard modalities.

2.10.3 Differential-cohesive recognition and modalities

Theorem 2.10.3 (Differential-cohesive recognition). *The canonically constructed adjoint quadruple*

$$i_! \dashv i^* \dashv i_* \dashv i^! : \mathcal{H}_{\text{Sp}} \rightleftarrows \mathcal{H}_{\text{inf}},$$

with $i_!$ and i_* fully faithful and $i_!$ left exact, is the ∞ -categorical adjoint configuration called **differential cohesion**; compare [27]. This is a recognition theorem for the endpoint semantics already constructed above and introduces no additional cohesion datum.

Proof.

(1) **Use full faithfulness.** The adjoint quadruple and full faithfulness are 2.10.2 and 2.10.2.

(2) **Use the stated property.** Since

$$i_! = \text{cod}^*,$$

it is precomposition and therefore preserves all limits, in particular finite limits.

(3) **Use the adjunction.** These are the structural properties used in the cited differential-cohesive adjoint pattern.

(4) **Use the adjunction.** Every functor and adjunction has already been constructed from the infinitesimal arrow category and its Zariski topology, so the terminology adds no data.

□

Remark 2.10.1 (Qualification of the cohesion terminology). We use **differential cohesion** only for the stated ∞ -categorical adjoint configuration constructed in this chapter, in the sense of the corresponding endpoint-adjunction definition in [27, Definition 2.15]. The terminology imports no additional synthetic differential-geometric axiom. In particular, no Kock–Lawvere principle, microlinearity hypothesis, specified infinitesimal object, formal-manifold condition, local-surjectivity axiom, or unqualified Lawvere–Menni cohesion over spaces is assumed unless it is separately constructed or stated as a theorem hypothesis. We also avoid the phrase **infinitesimally cohesive**, which has a distinct use in derived deformation theory.

Definition 2.10.1 (Differential-cohesive modalities). Define endofunctors of $\mathcal{H}_{\text{inf}}$ by the standard notation

$$\mathfrak{R} := i_! i^*, \quad \mathfrak{S} := i_* i^*, \quad \& := i_* i^!.$$

They are called, respectively, the **reduction modality**, the **infinitesimal shape modality**, and the **infinitesimal flat modality**. An object is **reduced** when the counit $\mathfrak{R}E \rightarrow E$ is an equivalence and **coreduced** when the unit $E \rightarrow \mathfrak{S}E$ is an equivalence.

Remark 2.10.2 (Reduction is relative to the infinitesimal endpoint). Here **reduction** means the differential-cohesive operation determined by the endpoint inclusion i : it removes the finite-infinitesimal direction encoded by an object of $\mathcal{H}_{\text{inf}}$. This operation is distinct from scheme-theoretic reduction $X \mapsto X_{\text{red}}$ and from the 0-truncation t_0 of Chapter 1. For a representable finite infinitesimal arrow $j : U \rightarrow T$, the calculation below gives

$$\mathfrak{R}(h_j) \simeq h_{\text{id}_U}.$$

Theorem 2.10.4 (Differential-cohesive modal triple). *There is a canonical adjoint triple*

$$\mathfrak{R} \dashv \mathfrak{S} \dashv \&.$$

The functors $\mathfrak{R}$ and $\&$ are idempotent comonads and $\mathfrak{S}$ is an idempotent monad. The modality $\mathfrak{R}$ preserves finite products, while $\mathfrak{S}$ is a left-exact reflector onto the essential image of i_ . There are canonical transformations*

$$\mathfrak{R}E \longrightarrow E \longrightarrow \mathfrak{S}E.$$

For a finite infinitesimal context $j : U \rightarrow T$,

$$(\mathfrak{R}E)(j) \simeq E(\mathrm{id}_T), \quad (\mathfrak{S}E)(j) \simeq E(\mathrm{id}_U),$$

and

$$(\&E)(j) \simeq \mathrm{Map}_{\mathcal{H}_{\mathrm{inf}}}(i_* h_U, E).$$

The composite $\mathfrak{R}E \rightarrow \mathfrak{S}E$ evaluates at j as restriction along the infinitesimal substitution

$$E(\mathrm{id}_T) \longrightarrow E(\mathrm{id}_U).$$

Proof.

- (1) **Use the adjunction.** For $E, F \in \mathcal{H}_{\mathrm{inf}}$, the adjoint quadruple gives

$$\begin{aligned} \mathrm{Map}(\mathfrak{R}E, F) &= \mathrm{Map}(i_! i^* E, F) \\ &\simeq \mathrm{Map}(i^* E, i^* F) \\ &\simeq \mathrm{Map}(E, i_* i^* F) = \mathrm{Map}(E, \mathfrak{S}F), \end{aligned}$$

so $\mathfrak{R} \dashv \mathfrak{S}$.

- (2) **Establish the next comparison.** Similarly $\mathfrak{S} \dashv \&$.

- (3) **Use full faithfulness.** Full faithfulness of $i_!$ and i_* gives the corresponding idempotence.

- (4) **Use preservation.** The functor $\mathfrak{R} = i_! i^*$ preserves finite products because both factors do; indeed $i_!$ preserves all limits here.

- (5) **Use the adjunction.** The unit $E \rightarrow i_* i^* E$ exhibits the essential image of the fully faithful right adjoint i_* as reflective, and both i^* and i_* preserve limits, so $\mathfrak{S}$ is left exact.

- (6) **pointwise formulas.** The pointwise formulas are the endpoint Kan-extension formulas of 2.10.1.

- (7) **last assertion.** The last assertion is evaluation of the canonical composite on

$$\mathrm{id}_U \longrightarrow j \longrightarrow \mathrm{id}_T.$$

□

Remark 2.10.3 (The modalities are derived, not imposed). The three modalities are composites of the constructed endpoint adjoints. Their adjunctions, idempotence, and natural transformations are consequences of that quadruple; none is input structure on an infinitesimal sheaf.

Proposition 2.10.3 (Reduction of a representable infinitesimal context). *Let $j : U \rightarrow T$ be an object of $\mathbf{Aff}_{\mathbf{Sp}}^{\text{inf}}$, and let h_j be its representable infinitesimal sheaf. Then*

$$i^*h_j \simeq h_U$$

and consequently

$$\mathfrak{R}(h_j) \simeq h_{\text{id}_U}.$$

Thus reduction sends a representable infinitesimal context to its source/reduction.

Proof.

(1) **Apply Yoneda.** For an affine spectral context V , Yoneda and the endpoint adjunction give

$$\begin{aligned} (i^*h_j)(V) &= \text{Map}_{\mathbf{Aff}_{\mathbf{Sp}}^{\text{inf}}}(\text{id}_V, j) \\ &\simeq \text{Map}_{\mathbf{Aff}_{\mathbf{Sp}}}(V, U) = h_U(V). \end{aligned}$$

(2) **Conclude the step.** Hence $i^*h_j \simeq h_U$.

(3) **Derive the comparison.** Applying $i_!$ gives the second equivalence.

□

Proposition 2.10.4 (Infinitesimal shape of an ordinary spectral stack). *For an ordinary spectral Zariski stack X , regarded in infinitesimal semantics through $i_!X$, there is a canonical equivalence*

$$\mathfrak{S}(i_!X) \simeq i_*X.$$

At a finite infinitesimal context $j : U \rightarrow T$,

$$(i_!X)(j) \simeq X(T), \quad (\mathfrak{S}i_!X)(j) \simeq X(U),$$

and the unit

$$i_!X \longrightarrow \mathfrak{S}(i_!X)$$

evaluates as restriction

$$X(T) \longrightarrow X(U).$$

Proof.

(1) **Use the stated property.** Since $i^*i_! \simeq \text{id}$, one has

$$\mathfrak{S}(i_!X) = i_*i^*\mathfrak{S}(i_!X) \simeq i_*X.$$

(2) **pointwise formulas.** The pointwise formulas are 2.10.1.

□

The modal triple supplies more than endofunctors: the left-exact reflector $\mathfrak{S}$ has a relative form in every slice. Its defining geometry is the pullback square

$$\begin{array}{ccc} \mathfrak{S}_B E & \longrightarrow & \mathfrak{S} E \\ \downarrow & & \downarrow \\ B & \longrightarrow & \mathfrak{S} B, \end{array}$$

which will be used as the formally étale reflection of a map over B .

2.10.4 Formal étalification and the lifting comparison

Definition 2.10.2 (Formally étale and formally smooth morphisms). Let $g : E \rightarrow B$ be a morphism in $\mathcal{H}_{\text{inf}}$. It is **formally étale** if the naturality square of the unit $\text{id} \rightarrow \mathfrak{S}$ is Cartesian, equivalently if the canonical comparison

$$E \longrightarrow B \times_{\mathfrak{S}B} \mathfrak{S}E$$

is an equivalence. It is **formally smooth** if the same comparison is an effective epimorphism.

Construction 2.10.1 (Relative infinitesimal shape and formal étalification). For $g : E \rightarrow B$ in $\mathcal{H}_{\text{inf}}$, define its **relative infinitesimal shape**, or **formal étalification**, by

$$\mathfrak{S}_B E := B \times_{\mathfrak{S}B} \mathfrak{S}E,$$

with its projection $\mathfrak{S}_B g : \mathfrak{S}_B E \rightarrow B$. The unit and g give the canonical factorization

$$E \xrightarrow{\theta_g} \mathfrak{S}_B E \xrightarrow{\mathfrak{S}_B g} B.$$

Proposition 2.10.5 (Relative infinitesimal-shape factorization). *The morphism θ_g becomes an equivalence after applying $\mathfrak{S}$, while $\mathfrak{S}_B g$ is formally étale. The construction is stable under base change in the slice over B . Moreover, $\mathfrak{S}_B g$ is the formally étale reflection of g : for every formally étale $Z \rightarrow B$, precomposition with θ_g induces a natural equivalence*

$$\text{Map}_{\mathcal{H}_{\text{inf}}/B}(\mathfrak{S}_B E, Z) \xrightarrow{\simeq} \text{Map}_{\mathcal{H}_{\text{inf}}/B}(E, Z).$$

Proof.

- (1) **Use exactness.** Left exactness and idempotence of $\mathfrak{S}$ give

$$\begin{aligned} \mathfrak{S}(\mathfrak{S}_B E) &\simeq \mathfrak{S}B \times_{\mathfrak{S}\mathfrak{S}B} \mathfrak{S}\mathfrak{S}E \\ &\simeq \mathfrak{S}E, \end{aligned}$$

so $\mathfrak{S}(\theta_g)$ is an equivalence.

- (2) **Verify naturality.** The defining square of $\mathfrak{S}_B E$ is exactly the unit-naturality square for $\mathfrak{S}_B g$, proving formal étaleness; left exactness gives base-change stability.
- (3) **Apply the universal property.** For the universal property, let $Z \rightarrow B$ be formally étale.
- (4) **Establish the next comparison.** Then

$$Z \simeq B \times_{\mathfrak{S}B} \mathfrak{S}Z.$$

- (5) **map over is therefore equivalently a map.** A map $E \rightarrow Z$ over B is therefore equivalently a map $E \rightarrow \mathfrak{S}Z$ whose composite to $\mathfrak{S}B$ is the prescribed one.
- (6) **Use the stated property.** Since $\mathfrak{S}Z$ is coreduced, the reflector gives

$$\text{Map}(E, \mathfrak{S}Z) \simeq \text{Map}(\mathfrak{S}E, \mathfrak{S}Z),$$

compatibly over $\mathfrak{S}B$.

(7) **Conclude the step.** Hence

$$\mathrm{Map}_B(E, Z) \simeq \mathrm{Map}_{\mathfrak{S}_B}(\mathfrak{S}E, \mathfrak{S}Z).$$

(8) **Compute the mapping object.** The same calculation for $\mathfrak{S}_B E$, using $\mathfrak{S}(\mathfrak{S}_B E) \simeq \mathfrak{S}E$, gives the identical mapping space.

(9) **Transport the structure.** Under these identifications precomposition with θ_g is the identity. $\square$

Construction 2.10.2 (The formal lifting comparison). Let $f : X \rightarrow S$ be a morphism in $\mathcal{H}_{\mathrm{Sp}}$. Define

$$\Theta_f := \theta_{i_! f} : i_! X \longrightarrow \mathfrak{S}_{i_! S}(i_! X).$$

Using $\mathfrak{S}(i_! X) \simeq i_* X$ and the corresponding equivalence for S , there is a canonical identification

$$\mathfrak{S}_{i_! S}(i_! X) \simeq i_! S \times_{i_* S} i_* X,$$

so Θ_f is the canonical comparison induced by $i_! \rightarrow i_*$.

Proposition 2.10.6 (The formal lifting comparison is the étalification unit). *The identification above exhibits Θ_f as the unit of the relative formally étale reflection of $i_! f$.*

Proof.

(1) **Identify the resulting comparison.** This is 2.10.4 substituted into the definition of $\mathfrak{S}_{i_! S}(i_! X)$. $\square$

Proposition 2.10.7 (Pointwise interpretation of the lifting comparison). *For every object*

$$j : U \longrightarrow T$$

of $\mathrm{Aff}_{\mathrm{Sp}}^{\mathrm{inf}}$,

$$\Theta_f(j) : X(T) \longrightarrow S(T) \times_{S(U)} X(U).$$

Its homotopy fibre over a compatible pair

$$T \longrightarrow S, \quad U \longrightarrow X$$

is the space of extensions $T \rightarrow X$ together with the specified compatibility homotopies.

Proof.

(1) **Identify the map.** The pointwise formulas for $i_!$ and i_* give the displayed map, and limits of sheaves are computed objectwise.

(2) **Apply the universal property.** The fibre statement is the universal property of the pullback of spaces. $\square$

Definition 2.10.3 (Formal smoothness and formal étaleness over ordinary spectral stacks). A morphism $f : X \rightarrow S$ in $\mathcal{H}_{\mathrm{Sp}}$ is **formally étale** if $i_! f$ is formally étale in $\mathcal{H}_{\mathrm{inf}}$, equivalently if Θ_f is an equivalence. It is **formally smooth** if $i_! f$ is formally smooth, equivalently if Θ_f is an effective epimorphism.

Theorem 2.10.5 (Formal etaleness is etaleness for infinitesimal shape). *For a morphism $f : X \rightarrow S$ of spectral Zariski stacks,*

$$f \text{ is formally étale} \iff i_!f \text{ is } \mathfrak{S}\text{-closed.}$$

Equivalently, the unit-naturality square

$$\begin{array}{ccc} i_!X & \longrightarrow & \mathfrak{S}(i_!X) \\ i_!f \downarrow & & \downarrow \mathfrak{S}(i_!f) \\ i_!S & \longrightarrow & \mathfrak{S}(i_!S) \end{array}$$

is Cartesian.

Proof.

- (1) **Identify the resulting comparison.** This is the definition of a formally étale morphism in $\mathcal{H}_{\text{inf}}$, together with 2.10.6. □

Proposition 2.10.8 (Base change of formal lifting properties). *Formally étale and formally smooth morphisms in $\mathcal{H}_{\text{inf}}$ are stable under base change. Consequently the same is true for formally étale and formally smooth morphisms in $\mathcal{H}_{\text{Sp}}$.*

Proof.

- (1) **Use exactness.** Left exactness of $\mathfrak{S}$ identifies the relative etalification comparison of a pullback with the pullback of the original comparison.
- (2) **Equivalences and effective epimorphisms in an ∞ -topos.** Equivalences and effective epimorphisms in an ∞ -topos are stable under pullback.
- (3) **Use preservation.** For morphisms in $\mathcal{H}_{\text{Sp}}$, the functor $i_!$ preserves finite limits, so the same argument applies after the fully faithful inclusion. □

Relative infinitesimal shape provides the canonical comparison from a map to its formally étale reflection. Taking its kernel pair turns that comparison into an intrinsic relation of infinitesimal equality.

2.10.5 Relative infinitesimal disks and infinitesimal equality

Definition 2.10.4 (Relative infinitesimal disk bundle and groupoid). For $f : X \rightarrow S$ in $\mathcal{H}_{\text{Sp}}$, define the **relative infinitesimal disk bundle**

$$T_{\text{inf}}(X/S) := i_!X \times_{\mathfrak{S}_{i_!S}(i_!X)} i_!X.$$

Its full **relative infinitesimal disk groupoid** is the Čech nerve

$$T_{\text{inf}}^{[\bullet]}(X/S) := \check{C}(\Theta_f).$$

Thus $T_{\text{inf}}(X/S)$ is the degree-one relation object of the Čech groupoid generated by the constant infinitesimal-path inclusion relative to S .

Proposition 2.10.9 (Pointwise infinitesimal equality). *For a finite infinitesimal context $j : U \rightarrow T$,*

$$T_{\text{inf}}(X/S)(j) \simeq X(T) \times_{S(T) \times_{S(U)} X(U)} X(T).$$

Thus two S -points of X over T are infinitesimally neighbouring exactly when their restrictions to the reduction U agree, with the specified compatibility over S .

Proof.

- (1) **Evaluate the defining kernel pair objectwise.** Evaluate the defining kernel pair objectwise and apply 2.10.7.

□

Remark 2.10.4 (Infinitesimal equality is generated by infinitesimal shape). The groupoid $T_{\text{inf}}^{[\bullet]}(X/S)$ is the Čech groupoid of the canonical map to relative infinitesimal shape. Its objects, disk pairs, and higher compositions are therefore generated functorially by the unit of $\mathfrak{S}$.

The modal formulas are pointwise on affine finite-infinitesimal arrows. Later lifting arguments also use represented finite infinitesimal arrows that need not themselves be affine objects of the test category. We therefore extend Yoneda only as far as needed to evaluate the same comparison on those represented arrows.

2.10.6 Infinitesimal Yoneda realization of represented finite arrows

Construction 2.10.3 (Infinitesimal Yoneda realization). Let

$$j : U \longrightarrow T$$

be a finite spectral infinitesimal substitution of spectral schemes, without an affineness requirement on U or T . Define a presheaf on $\text{Aff}_{\text{Sp}}^{\text{inf}}$ by

$$\mathbf{h}_j^{\text{inf}}(k : V \rightarrow W) := \text{Map}_{\text{Fun}(\Delta^1, \text{Sch}^{\text{sp}})}(k, j).$$

Since $\text{Aff}_{\text{Sp}}^{\text{inf}}$ is a full subcategory of the arrow category, this is the restriction of the ordinary arrow-category Yoneda functor along the affine finite-infinitesimal test category.

Proposition 2.10.10 (Descent and affine recovery for infinitesimal Yoneda realization). *The presheaf $\mathbf{h}_j^{\text{inf}}$ is a $\tau_{\text{Zar}}^{\text{inf}}$ -sheaf. Hence it determines an object of $\mathcal{H}_{\text{inf}}$. If j is an object of $\text{Aff}_{\text{Sp}}^{\text{inf}}$, then there is a canonical equivalence*

$$\mathbf{h}_j^{\text{inf}} \simeq h_j$$

with the representable infinitesimal sheaf.

Proof.

- (1) **Compute the mapping object.** For an affine finite-infinitesimal test arrow $k : V \rightarrow W$, the arrow-category mapping space is the pullback

$$\mathbf{h}_j^{\text{inf}}(k) \simeq \text{Map}_{\text{Sch}^{\text{sp}}}(V, U) \times_{\text{Map}_{\text{Sch}^{\text{sp}}}(V, T)} \text{Map}_{\text{Sch}^{\text{sp}}}(W, T).$$

For a codomain-pullback Zariski basis cover of k , the targets form a spectral Zariski cover of W and the sources form the corresponding spectral Zariski cover of V .

- (2) **Apply Zariski descent.** Represented spectral schemes satisfy spectral Zariski descent, and limits commute with limits in spaces.
- (3) **Identify the map.** The preceding pullback of mapping functors therefore satisfies descent.
- (4) **Use the stated property.** Since the basis covers generate $\tau_{\text{Zar}}^{\text{inf}}$, the presheaf is a sheaf.
- (5) **Compute the mapping object.** If j is an object of $\text{Aff}_{\text{Sp}}^{\text{inf}}$, both source and target lie in the affine infinitesimal arrow category and the displayed mapping space is exactly $\text{Map}_{\text{Aff}_{\text{Sp}}^{\text{inf}}}(k, j)$.
- (6) **Identify the resulting object.** Subcanonicity then identifies the sheaf with the representable h_j .

□

Construction 2.10.4 (Represented formal lifting comparison). Let

$$f : X \longrightarrow S$$

be a morphism in $\mathcal{H}_{\text{Sp}}$, and let

$$j : U \longrightarrow T$$

be a finite spectral infinitesimal substitution of spectral schemes, without an affineness requirement on U or T . Define the **represented formal lifting comparison** of f at j by

$$\begin{aligned} \Theta_f^{\text{rep}}(j) : \text{Map}_{\mathcal{H}_{\text{Sp}}}(T, X) &\longrightarrow \text{Map}_{\mathcal{H}_{\text{Sp}}}(T, S) \\ &\quad \times_{\text{Map}_{\mathcal{H}_{\text{Sp}}}(U, S)} \text{Map}_{\mathcal{H}_{\text{Sp}}}(U, X), \end{aligned}$$

where the map sends

$$b : T \longrightarrow X$$

to the pair

$$f \circ b : T \longrightarrow S, \quad b \circ j : U \longrightarrow X,$$

together with the canonical equality of their restrictions to U .

If j is an object of $\text{Aff}_{\text{Sp}}^{\text{inf}}$, then the endpoint formulas of 2.10.1 and the pointwise computation of 2.10.7 canonically identify

$$\Theta_f^{\text{rep}}(j) \simeq \Theta_f(j).$$

Thus Θ_f^{rep} extends the already constructed affine pointwise lifting comparison to represented finite infinitesimal arrows. It does not declare a nonaffine arrow to be an object of $\text{Aff}_{\text{Sp}}^{\text{inf}}$, nor does it introduce an additional infinitesimal-site structure.

This gives the endpoint interface needed for formal geometry: relative infinitesimal shape supplies the lifting comparison, infinitesimal disks its kernel-pair relation, and represented finite arrows the same pointwise interpretation. The remaining problem is to compute the fibres of the lifting comparison.

2.11 Cotangent analysis of infinitesimal lifting

For every morphism $f : X \rightarrow S$, differential cohesion has already constructed the relative formal-etalification unit

$$\Theta_f : i_! X \longrightarrow \mathfrak{S}_{i_! S}(i_! X).$$

Its fibres are nonlinear lifting spaces. On a structured square-zero stage, the cotangent classifier turns each such fibre into a nullhomotopy problem in a stable coefficient category. The geometric and linear descriptions fit into the pullback picture

$$\begin{array}{ccc} \text{Lift}_S(a; i) & \longrightarrow & X(T) \\ \downarrow & & \downarrow \Theta_f^{\text{rep}}(i) \\ * & \xrightarrow{(T \rightarrow S, a)} & S(T) \times_{S(U)} X(U). \end{array}$$

This section first computes that fibre and then uses one-stage detection to recognize formal étaleness and formal smoothness from cotangent data.

Proposition 2.11.1 (Formal lifting across a fixed affine square-zero extension). *Let*

$$A \longrightarrow B, \quad A \longrightarrow C$$

be morphisms of commutative ring spectra. Let

$$q : C' \longrightarrow C$$

be a structured square-zero extension in commutative A -algebras by a C -module J , classified by

$$\delta : L_{C/A} \longrightarrow J[1].$$

For an A -algebra morphism $\phi : B \rightarrow C$, define the obstruction term

$$o(\phi) : L_{B/A} \otimes_B C \longrightarrow L_{C/A} \xrightarrow{\delta} J[1],$$

where the first map is induced by cotangent functoriality. Then the space of lifts of ϕ to an A -algebra morphism $B \rightarrow C'$ is naturally equivalent to the space of nullhomotopies of $o(\phi)$.

Consequently:

(i) *a lift exists if and only if $[o(\phi)]$ is zero in*

$$\pi_0 \text{map}_{\text{Mod}_C}(L_{B/A} \otimes_B C, J[1]);$$

(ii) *when nonempty, the lift space is a torsor under*

$$\Omega^\infty \text{map}_{\text{Mod}_C}(L_{B/A} \otimes_B C, J);$$

(iii) *if $L_{B/A} \simeq 0$, the lift space is contractible.*

Proof.

(1) **Fix the notation.** Put

$$\mathcal{A} := \text{CAlg}(\text{Sp})_{A/}.$$

- (2) **Identify the map.** The structured extension $q : C' \rightarrow C$ is, by 2.7.1, the pullback in $\mathcal{A}/C$

$$q \simeq \text{id}_C \times_p \text{id}_C,$$

where

$$p : C \oplus J[1] \longrightarrow C$$

is the split augmentation and the two maps $\text{id}_C \rightrightarrows p$ are d_δ and d_0 .

- (3) **Identify the map.** The morphism $\phi : B \rightarrow C$ is an object of $\mathcal{A}/C$.
- (4) **space of lifts of through , including the specified homotopy over.** The space of lifts of ϕ through q , including the specified homotopy over C , is therefore

$$\text{Map}_{\mathcal{A}/C}(\phi, q).$$

- (5) **Compute the mapping object.** Mapping out of ϕ preserves the defining pullback, so

$$\begin{aligned} \text{Map}_{\mathcal{A}/C}(\phi, q) &\simeq \text{Map}_{\mathcal{A}/C}(\phi, \text{id}_C) \\ &\quad \times_{\text{Map}_{\mathcal{A}/C}(\phi, p)} \text{Map}_{\mathcal{A}/C}(\phi, \text{id}_C). \end{aligned}$$

- (6) **Compute the mapping object.** The object id_C is terminal in $\mathcal{A}/C$, hence each outer mapping space is contractible.

- (7) **Their two images in.** Their two images in $\text{Map}_{\mathcal{A}/C}(\phi, p)$ are the sections

$$d_\delta \phi \quad \text{and} \quad d_0 \phi.$$

- (8) **Conclude the step.** Thus the lift space is canonically the path space between these two derivations.

- (9) **Transport the structure.** Under the relative cotangent classifier, their difference is classified by the composite

$$L_{B/A} \otimes_B C \longrightarrow L_{C/A} \xrightarrow{\delta} J[1] = o(\phi).$$

- (10) **Conclude the step.** Consequently

$$\text{Lift}_A(\phi; q) \simeq \text{Null}(o(\phi)).$$

- (11) **Compute the mapping object.** A nullhomotopy exists exactly when the class of $o(\phi)$ vanishes in π_0 of the indicated mapping spectrum.

- (12) **Identify the resulting object.** When the nullhomotopy space is nonempty, translation by a chosen nullhomotopy identifies it as a torsor under the loop space

$$\Omega \Omega^\infty \text{map}_{\text{Mod}_C}(L_{B/A} \otimes_B C, J[1]) \simeq \Omega^\infty \text{map}_{\text{Mod}_C}(L_{B/A} \otimes_B C, J).$$

- (13) **Compute the mapping object.** If $L_{B/A} \simeq 0$, the relevant mapping spectrum is zero and the lift space is contractible.

□

Remark 2.11.1 (Lifting as a judgment). The proposition may be read as the equivalence

$$\{\text{extensions of a term across an infinitesimal context}\} \simeq \{\text{nullhomotopies of its canonical tangent boundary}\}.$$

The obstruction is the boundary term constructed from cotangent functoriality and the specified infinitesimal extension.

Theorem 2.11.1 (Global lifting judgment). *Let $f : X \rightarrow S$ be a morphism of spectral schemes and let*

$$i : U \longrightarrow T$$

be a structured infinitesimal extension over S by $J \in \mathbf{QCoh}(U)_{\geq 0}$, classified by

$$\delta : L_{U/S} \longrightarrow J[1].$$

For an S -morphism $a : U \rightarrow X$, define

$$o(a) : a^* L_{X/S} \longrightarrow L_{U/S} \xrightarrow{\delta} J[1].$$

The compatible pair

$$T \longrightarrow S, \quad a : U \longrightarrow X$$

defines a point of the target of the represented formal lifting comparison

$$\Theta_f^{\text{rep}}(i) : X(T) \longrightarrow S(T) \times_{S(U)} X(U)$$

of 2.10.4. Its homotopy fibre over that point is the space of S -morphisms $\tilde{a} : T \rightarrow X$ equipped with a specified homotopy

$$\tilde{a} \circ i \simeq a,$$

and there is a natural equivalence

$$\begin{aligned} \text{fib}_{(T \rightarrow S, a)}(\Theta_f^{\text{rep}}(i)) &\simeq \text{Lift}_S(a; i) \\ &\simeq \text{Null}_{\mathbf{QCoh}(U)}(o(a)). \end{aligned}$$

In particular, when nonempty, the lift space is a torsor under

$$\Omega^\infty \text{map}_{\mathbf{QCoh}(U)}(a^* L_{X/S}, J).$$

When i is affine, the canonical comparison

$$\Theta_f^{\text{rep}}(i) \simeq \Theta_f(i)$$

recovers the pointwise differential-cohesive lifting comparison of 2.10.7.

Proof.

- (1) **Choose the local data.** Choose an affine Zariski cover of T .
- (2) **Invoke the cited result.** By 2.7.3, its pullback to U is the corresponding affine cover, and the square-zero extension on every member is represented by a structured affine square-zero extension.
- (3) **Refine the image of each affine source.** Refine the image of each affine source in X and S by compatible affine opens.

- (4) **Restrict the construction.** On every resulting affine piece, 2.11.1 identifies the lift space with the nullhomotopy space of the restriction of $o(a)$.
- (5) **Verify naturality.** On finite intersections these equivalences agree by cotangent base change and naturality of the classifying derivation.
- (6) **Apply Zariski descent.** Mapping descent for the represented spectral scheme X identifies the global lift space with the totalization of the local lift spaces.
- (7) **Apply Zariski descent.** Quasi-coherent descent identifies

$$\mathrm{Map}_{\mathrm{QCoh}(U)}(a^*L_{X/S}, J[1])$$

with the totalization of the corresponding affine mapping spaces, and limits in spaces preserve homotopy fibres over the zero point.

- (8) **Identify the equivalence.** The affine nullhomotopy equivalences therefore totalize to the displayed global equivalence.
- (9) **Invoke the cited result.** By definition,

$$\Theta_f^{\mathrm{rep}}(i) : X(T) \longrightarrow S(T) \times_{S(U)} X(U)$$

has as its fibre over the compatible pair $(T \rightarrow S, a)$ precisely the space of maps

$$\tilde{a} : T \longrightarrow X$$

over S , equipped with the specified homotopy $\tilde{a}i \simeq a$.

- (10) **Conclude the step.** Hence this fibre is the same global lift space computed above.
- (11) **Compute the mapping object.** The torsor statement follows by looping the mapping spectrum.
- (12) **If is affine, the final assertion follows.** If i is affine, the final assertion follows from 2.10.4 and 2.10.7.

□

The structured one-stage calculation is the local building block. We now show that testing only square-zero stages already detects the lifting behavior on arbitrary finite infinitesimal substitutions, after which the cotangent criteria become intrinsic statements about represented spectral contexts.

2.11.1 One-stage detection and cotangent recognition

Proposition 2.11.2 (One-stage detection). *Let $f : X \rightarrow S$ be a morphism in $\mathcal{H}_{\mathrm{Sp}}$.*

Formal étaleness is equivalent to contractible lifting against every affine square-zero infinitesimal substitution, equivalently against every one-stage square-zero object

$$j : U \longrightarrow T$$

of $\mathrm{Aff}_{\mathrm{Sp}}^{\mathrm{inf}}$.

Formal smoothness is equivalent to Zariski-local existence of lifts against every such affine square-zero infinitesimal substitution.

Proof.

- (1) **forward implications.** The forward implications follow by evaluating Θ_f on one-stage objects of $\text{Aff}_{\text{Sp}}^{\text{inf}}$.
- (2) **Construct the factorization.** Conversely, let

$$j : U \longrightarrow T$$

be an arbitrary object of $\text{Aff}_{\text{Sp}}^{\text{inf}}$ and choose, only inside the proof, a finite factorization

$$U = U_0 \longrightarrow U_1 \longrightarrow \cdots \longrightarrow U_r = T$$

into underlying square-zero substitutions.

- (3) **Each intermediate.** Each intermediate U_k is affine: work backward from the affine target T , using affineness of every square-zero substitution.
- (4) **Identify contractibility.** For formal étaleness, the lift space across the first stage is contractible by the one-stage condition.
- (5) **Compute the fibre.** Applying the same statement to the resulting lift at the next stage and iterating expresses the lift space for the composite as a finite iterated homotopy fibre of contractible spaces; it is therefore contractible.
- (6) **Identify contractibility.** Hence every fibre of $\Theta_f(j)$ is contractible, so Θ_f is an equivalence.
- (7) **For formal smoothness, construct a lift successively.** For formal smoothness, construct a lift successively after Zariski refinement.
- (8) **Identify the map.** The square-zero map

$$U_k \longrightarrow U_{k+1}$$

identifies their small Zariski sites by 2.8.1, so a cover on one side transports uniquely to the other.

- (9) **Apply the one-stage local lifting property, refine finitely at each.** Apply the one-stage local lifting property, refine finitely at each step, and compose the resulting covers.
- (10) **After the finite number of stages a.** After the finite number of stages a lift exists over a Zariski cover of T .
- (11) **Verify surjectivity.** Thus Θ_f is locally surjective on sections and hence an effective epimorphism.
- (12) **Construct the factorization.** The factorization and intermediate covers are proof witnesses only.

□

Theorem 2.11.2 (Cotangent recognition of formal étaleness). *Let $f : X \rightarrow S$ be represented by spectral schemes. Then*

$$f \text{ is formally étale} \iff L_{X/S} \simeq 0.$$

Proof.

- (1) **Suppose first that .** Suppose first that $L_{X/S} \simeq 0$.
- (2) **Invoke the cited result.** By 2.11.2, it suffices to solve a one-stage underlying square-zero lifting problem.
- (3) **Fix the notation.** Let

$$j : U \longrightarrow T$$

be an affine square-zero infinitesimal substitution over S , together with $a : U \rightarrow X$.

- (4) **Choose the local data.** Choose, only inside the proof, a structured presentation of j .
- (5) **Construct the factorization.** Refine T by a finite affine Zariski cover whose composites with S factor through affine opens of S .
- (6) **On each member the chosen presentation may.** On each member the chosen presentation may be refined relative to that affine base by 2.8.2.
- (7) **Identify contractibility.** The global lifting calculation 2.11.1 then identifies the local lift space with the nullhomotopy space of a map out of the pullback of $L_{X/S}$, hence with a contractible space.
- (8) **Apply Zariski descent.** The same is true on all finite intersections, and mapping descent totalizes these contractible local lift spaces to a contractible global lift space.
- (9) **Identify contractibility.** Thus every one-stage lifting problem has a contractible solution space, and f is formally étale.
- (10) **Prove the converse.** Conversely suppose f is formally étale.
- (11) **Construct the factorization.** Let

$$U = \operatorname{Spec}(B) \hookrightarrow X$$

be an affine open whose composite with S factors through an affine open

$$V = \operatorname{Spec}(A) \hookrightarrow S.$$

- (12) **Verify connectivity.** For a connective B -module M , consider the split square-zero test

$$j : U = \operatorname{Spec}(B) \longrightarrow T := \operatorname{Spec}(B \oplus M).$$

- (13) **Identify the map.** The structural map $T \rightarrow V \rightarrow S$ and $U \rightarrow X$ define a one-stage lifting problem.
- (14) **Identify contractibility.** Formal étaleness makes its lift space contractible.
- (15) **Construct the factorization.** Every lift factors through $U \subseteq X$: the inverse image of U is an open subobject of T containing the centre, and $U \rightarrow T$ is a homeomorphism on underlying topological spaces.
- (16) **Identify contractibility.** Hence that open is all of T , and the factorization space is contractible because an open immersion is a monomorphism.

(17) **lift space.** The lift space is therefore the space of A -algebra sections

$$B \longrightarrow B \oplus M,$$

namely

$$\Omega^\infty \text{map}_{\text{Mod}_B}(L_{B/A}, M).$$

(18) **Verify connectivity.** Take $M = L_{B/A}$, which is connective.

(19) **Establish the next comparison.** Contractibility makes the identity of $L_{B/A}$ null, so $L_{B/A} \simeq 0$.

(20) **Glue the local data.** These affine vanishings are compatible under restriction and glue by 2.5.1, giving $L_{X/S} \simeq 0$.

□

Proposition 2.11.3 (Finite locally free cotangent complexes give formal smoothness). *Let $f : X \rightarrow S$ be a morphism of spectral schemes. If $L_{X/S}$ is finite locally free, then f is formally smooth.*

Proof.

- (1) **Invoke the cited result.** By 2.11.2, it suffices to solve a one-stage underlying square-zero lifting problem Zariski-locally.
- (2) **Fix the notation.** Let $U \rightarrow T$ be an affine square-zero infinitesimal substitution over S and let $U \rightarrow X$ be the given map.
- (3) **Cover its image in by affine opens.** Cover its image in X by affine opens, pull the cover back to the affine U , and choose a finite principal refinement.
- (4) **Zariski-site invariance across transports this to a finite.** Zariski-site invariance across $U \rightarrow T$ transports this to a finite affine cover $\{T_\alpha \rightarrow T\}$, with $U_\alpha \simeq U \times_T T_\alpha$, such that each restricted map lands in an affine open of X .
- (5) **On each member choose, only inside the.** On each member choose, only inside the proof, a structured presentation of the underlying square-zero arrow and refine it relative to an affine open of the base S using 2.8.2.
- (6) **Verify connectivity.** The obstruction of 2.11.1 then lies in

$$\pi_0 \text{map}_C(P, J[1]),$$

where P is the pullback of $L_{X/S}$ and J is the connective coefficient of that proof presentation.

- (7) **Finite local freeness.** Finite local freeness makes P finite projective, hence

$$\text{map}_C(P, J[1]) \simeq P^\vee \otimes_C J[1].$$

- (8) **Verify connectivity.** The right side is 1-connective, so its π_0 vanishes.
- (9) **Derive the comparison.** The obstruction is therefore zero and the fixed-extension lifting theorem gives a lift over each T_α .

(10) **These local lifts.** These local lifts are precisely the effective epimorphism condition for Θ_f .

□

The lifting comparison has now been identified cotangentially and its formal étale and smooth regimes have been recognized. The relative infinitesimal disk groupoid contains more than these lifting fibres, however: its diagonal relation has a canonical first-order piece whose difference map is itself universal.

2.11.2 The canonical first infinitesimal neighbourhood of the diagonal

We isolate that first-order piece as a structured thickening of the diagonal. Its two retractions to X agree on the centre, and their first-order difference recovers the universal relative derivation. This identifies the cotangent classifier as the linearization of infinitesimal equality without identifying all finite infinitesimal structure with first-order data.

Construction 2.11.1 (First structured infinitesimal neighbourhood). Let $p : X \rightarrow S$ be a morphism of spectral schemes and put

$$Y := X \times_S X.$$

Let $\Delta : X \rightarrow Y$ be the diagonal and regard Y over X through pr_1 . Global cotangent base change gives

$$L_{Y/X} \simeq \text{pr}_2^* L_{X/S}, \quad \Delta^* L_{Y/X} \simeq L_{X/S}.$$

Since

$$X \xrightarrow{\Delta} Y \xrightarrow{\text{pr}_1} X$$

is the identity, transitivity supplies a canonical connecting equivalence

$$\partial_\Delta : L_{X/Y} \xrightarrow{\sim} L_{X/S}[1].$$

Apply the global structured infinitesimal context-extension construction to ∂_Δ . This gives a canonical structured square-zero thickening

$$X \xrightarrow{i_1} D_{X/S}^{(1)} \xrightarrow{\nu_1} X \times_S X$$

with coefficient $L_{X/S}$. Define

$$h_0 := \text{pr}_1 \nu_1, \quad h_1 := \text{pr}_2 \nu_1 : D_{X/S}^{(1)} \rightrightarrows X.$$

Both are retractions of the centre:

$$h_0 i_1 \simeq \text{id}_X, \quad h_1 i_1 \simeq \text{id}_X.$$

Proposition 2.11.4 (Base change of the first neighbourhood). *The construction $D_{X/S}^{(1)}$, its centre, the two retractions, and the coefficient $L_{X/S}$ commute with arbitrary Cartesian base change of represented spectral contexts.*

Proof.

- (1) **Apply cotangent base change.** The diagonal, global cotangent base change, transitivity, the connecting morphism ∂_Δ , and global structured infinitesimal context extension all commute with represented Cartesian base change.

- (2) **two retractions.** The two retractions are obtained by composing with the two product projections.

□

Theorem 2.11.3 (The two diagonal retractions differ by the universal derivation). *Let $p : X \rightarrow S$. Split the first structured neighbourhood by the retraction h_0 . Under the resulting identification of its square-zero coefficient with $L_{X/S}$, the second retraction h_1 is the graph of the universal relative S -derivation. Equivalently, on every affine base test $S = \text{Spec}(R)$ and affine open $X = \text{Spec}(A)$, the induced derivation is*

$$d_{A/R} : A \longrightarrow L_{A/R}.$$

Thus the first-order difference of h_1 and h_0 is the universal relative derivation of the structure algebra; it is not an $\mathcal{O}_X$ -linear map from $\mathcal{O}_X$ to $L_{X/S}$.

Proof.

- (1) **Set up the argument.** The assertion is local on X , so take $S = \text{Spec}(R)$ and $X = \text{Spec}(A)$.

- (2) **Fix the notation.** Put

$$M := L_{A/R}, \quad B := A \otimes_R A.$$

- (3) **Fix the notation.** Let

$$j_0, j_1 : A \rightrightarrows B$$

be the two tensor-factor maps and let $\mu : B \rightarrow A$ be multiplication.

- (4) **Regard as an A -algebra through j_0 .** Regard B as an A -algebra through j_0 .

- (5) **Apply cotangent base change.** Cotangent base change gives

$$L_{B/A} \simeq B \otimes_{A, j_1} M,$$

and hence

$$P := A \otimes_B L_{B/A} \simeq M.$$

- (6) **Apply cotangent transitivity.** The composite $A \xrightarrow{j_0} B \xrightarrow{\mu} A$ is the identity, so transitivity gives

$$P \longrightarrow 0 \longrightarrow L_{A/B}$$

and therefore a connecting equivalence

$$\partial : L_{A/B} \xrightarrow{\simeq} P[1] \simeq M[1].$$

- (7) **Restrict the construction.** This is the affine restriction of ∂_Δ .

- (8) **Fix the notation.** Let $q : A' \rightarrow A$ be the structured square-zero B -algebra extension classified by ∂ , and let $\beta : B \rightarrow A'$ be its B -algebra structure.

- (9) **Identify the map.** The two tensor-factor maps give two sections of q ,

$$\alpha_i := \beta j_i : A \longrightarrow A', \quad i = 0, 1,$$

because $q\alpha_i = \mu j_i = \text{id}_A$.

- (10) **These algebra sections.** These algebra sections are contravariantly the geometric retractions h_i .
- (11) **Restrict the construction.** Restrict the structured extension along $j_0 : A \rightarrow B$.
- (12) **Identify contractibility.** Its relative cotangent complex is $L_{A/A} \simeq 0$, so square-zero classification identifies it, through a contractible space, with the split extension

$$A' \simeq A \oplus M$$

in such a way that

$$\alpha_0(a) = (a, 0).$$

- (13) **There is therefore a unique -derivation.** There is therefore a unique R -derivation

$$\delta : A \longrightarrow M$$

with

$$\alpha_1(a) = (a, \delta(a)).$$

- (14) **Transport the structure.** Under the same splitting, write

$$\beta(b) = (\mu(b), D(b)),$$

where $D : B \rightarrow M$ is a derivation relative to j_0 .

- (15) **Its classifying -linear morphism.** Its classifying A -linear morphism is

$$\bar{D} : P = A \otimes_B L_{B/A} \longrightarrow M.$$

- (16) **Apply cotangent transitivity.** Applying $\text{map}_{\text{Mod}_A}(-, M[1])$ to the transitivity sequence $P \rightarrow 0 \rightarrow L_{A/B}$ identifies maps $P \rightarrow M$ with maps $L_{A/B} \rightarrow M[1]$.
- (17) **Transport the structure.** Under this connecting equivalence, the classifying map ∂ corresponds to the identity of $P \simeq M$.
- (18) **Conclude the step.** Hence

$$\bar{D} = \text{id}_M.$$

- (19) **Restrict the construction.** The derivation δ is the restriction of D along j_1 .
- (20) **Apply base change.** Under the base-change identification $P \simeq L_{A/R} = M$, its classifying morphism is therefore id_M .
- (21) **Apply the universal property.** The universal property of the cotangent complex gives

$$\delta = d_{A/R}.$$

- (22) **Restrict the construction.** The affine identifications commute with restriction and therefore globalize.

□

Remark 2.11.2 (First order is not all infinitesimal order). The theorem identifies the first structured neighbourhood of the diagonal with the universal first-order variation. Higher finite infinitesimal relations may contain data beyond the classifier $L_{X/S}$, and higher infinitesimal functions may carry structure beyond exterior powers of the cotangent classifier. The first neighbourhood is the canonical linear quotient of the infinitesimal geometry.

Proposition 2.11.5 (The first diagonal neighbourhood is a represented first-order infinitesimal disk). *Let $p : X \rightarrow S$, and let*

$$i_1 : X \longrightarrow D_{X/S}^{(1)}$$

be the canonical first structured infinitesimal neighbourhood. The two retractions

$$h_0, h_1 : D_{X/S}^{(1)} \rightrightarrows X$$

determine a canonical morphism of infinitesimal sheaves

$$(h_0, h_1) : \mathbf{h}_{i_1}^{\text{inf}} \longrightarrow T_{\text{inf}}(X/S).$$

If X is affine—equivalently, by 2.9.2, if $D_{X/S}^{(1)}$ is affine—then i_1 is an object of $\text{Aff}_{\text{Sp}}^{\text{inf}}$, and the affine recovery of 2.10.10 identifies this morphism by Yoneda with a canonical point

$$(h_0, h_1) \in T_{\text{inf}}(X/S)(i_1).$$

Under the splitting by h_0 , the first-order difference of this represented neighbour is the universal relative derivation.

Proof.

- (1) **Fix the notation.** Let $k : V \rightarrow W$ be an affine finite-infinitesimal test arrow and let

$$k \longrightarrow i_1$$

be a point of $\mathbf{h}_{i_1}^{\text{inf}}(k)$.

- (2) **Verify compatibility.** Thus we have a commutative square

$$\begin{array}{ccc} V & \longrightarrow & X \\ k \downarrow & & \downarrow i_1 \\ W & \longrightarrow & D_{X/S}^{(1)} \end{array}$$

- (3) **Compose the lower.** Compose the lower map with h_0 and h_1 .

- (4) **Identify the resulting comparison.** This produces two maps

$$W \rightrightarrows X$$

over the same map $W \rightarrow S$.

- (5) **Restrict the construction.** Their restrictions to V agree with the given map $V \rightarrow X$, because

$$h_0 i_1 \simeq \text{id}_X \simeq h_1 i_1.$$

- (6) **Invoke the cited result.** By the pointwise formula of 2.10.9, this is a point of $T_{\text{inf}}(X/S)(k)$.
- (7) **Identify the map.** The construction is natural in both k and the map $k \rightarrow i_1$, hence defines the displayed morphism of sheaves.
- (8) **When is affine, the cited result makes.** When X is affine, 2.9.2 makes $D_{X/S}^{(1)}$ affine as well, so $\mathbf{h}_{i_1}^{\text{inf}} \simeq h_{i_1}$.
- (9) **Apply Yoneda.** Ordinary Yoneda in the infinitesimal topos gives the stated point.
- (10) **final assertion.** The final assertion is exactly 2.11.3.

□

Corollary 2.11.1 (Cotangent classification linearizes infinitesimal equality). *For every represented relative spectral context $X \rightarrow S$, the cotangent classifier $L_{X/S}$ is the universal linear coefficient measuring the first-order difference of infinitesimally neighbouring points along the canonical first neighbourhood of the diagonal.*

Proof.

- (1) **Handle the first component.** The first diagonal disk pair is universal by 2.11.3; derivation judgments are represented by $L_{X/S}$ by 2.5.2 and its affine classifier.
- (2) **Conclude the step.** Thus every first-order linear difference is obtained by applying a coefficient map out of $L_{X/S}$ to the universal one.

□

Two facts remain active from the lifting analysis: one-stage square-zero tests control finite infinitesimal lifting, and the first diagonal neighbourhood realizes the universal first-order difference geometrically. Applying the same mechanism to the base context gives the first-order deformation calculus.

2.12 First-order deformation spaces and the cotangent boundary calculus

A first-order deformation is a lifting problem in which the base is replaced by one structured square-zero thickening. After identifying the reduction, the geometry has the form

$$\begin{array}{ccc} X & \hookrightarrow & X' \\ \downarrow f & & \downarrow \\ S & \hookrightarrow & S'. \end{array}$$

Cotangent transitivity turns the existence of such a lift, its variation, and its automorphisms into one exact boundary calculus. Throughout this section **deformation** refers to this first-order square-zero situation.

2.12.1 Affine deformation spaces

Definition 2.12.1 (Affine deformation over an infinitesimal base). Let

$$A' \longrightarrow A$$

be a structured square-zero extension of connective commutative ring spectra by a connective A -module I . Let B be a connective commutative A -algebra. A **deformation of B over A'** is a pair (B', α) consisting of a connective commutative A' -algebra B' and an equivalence

$$\alpha : B' \otimes_{A'} A \xrightarrow{\simeq} B$$

in commutative A -algebras. Morphisms are equivalences over A' compatible with the specified reduction equivalences. The resulting ∞ -groupoid is denoted

$$\mathrm{Def}_{A'/A}(B).$$

Theorem 2.12.1 (Affine deformation as a homotopy fibre). *Suppose $A' \rightarrow A$ is classified by*

$$\eta_A : L_A \longrightarrow I[1],$$

and put

$$J := B \otimes_A I.$$

Base change gives

$$\eta_B : L_A \otimes_A B \longrightarrow J[1].$$

Then there is a natural equivalence

$$\mathrm{Def}_{A'/A}(B) \simeq \mathrm{fib}_{\eta_B} \left(\mathrm{Map}_{\mathrm{Mod}_B}(L_B, J[1]) \longrightarrow \mathrm{Map}_{\mathrm{Mod}_B}(L_A \otimes_A B, J[1]) \right).$$

Equivalently, a deformation is a morphism

$$\eta_e : L_B \longrightarrow J[1]$$

together with a specified homotopy between its restriction to $L_A \otimes_A B$ and η_B .

Proof.

- (1) **Verify connectivity.** The class $\eta_A : L_A \rightarrow I[1]$ is an object of the connective derivation category Der^+ of [30], because A and I are connective.
- (2) **Its associated algebra arrow.** Its associated algebra arrow is the structured square-zero extension $A' \rightarrow A$.
- (3) **Verify connectivity.** The connective deformation equivalence [30, Proposition 3.22 and Remark 3.23] identifies connective A' -algebras with connective derivations over η_A .
- (4) **Verify connectivity.** Restrict this equivalence to those objects whose reduction is the fixed connective A -algebra B .

- (5) **Verify compatibility.** The resulting fibre consists precisely of derivation classes

$$\eta_e : L_B \longrightarrow J[1]$$

together with a specified homotopy making the square

$$\begin{array}{ccc} L_A \otimes_A B & \xrightarrow{\eta_B} & J[1] \\ \downarrow & & \parallel \\ L_B & \xrightarrow{\eta_e} & J[1] \end{array}$$

commute.

- (6) **Apply the universal property.** By the universal property of a homotopy fibre, this is exactly

$$\mathrm{fib}_{\eta_B} (\mathrm{Map}_{\mathrm{Mod}_B}(L_B, J[1]) \longrightarrow \mathrm{Map}_{\mathrm{Mod}_B}(L_A \otimes_A B, J[1])) .$$

- (7) **Identify the equivalence.** The cited equivalence is an equivalence of ∞ -categories, so taking the indicated fibre gives an equivalence of the full deformation ∞ -groupoids, including all morphisms and higher homotopies.

□

Theorem 2.12.2 (Affine cotangent boundary calculus). *In the situation of 2.12.1, the transitivity cofibre sequence*

$$L_A \otimes_A B \longrightarrow L_B \longrightarrow L_{B/A}$$

determines a canonical boundary element

$$\mathrm{ob}(B/A') \in \pi_0 \mathrm{map}_{\mathrm{Mod}_B}(L_{B/A}, J[2]).$$

It satisfies:

- (i) $\mathrm{ob}(B/A') = 0$ if and only if $\mathrm{Def}_{A'/A}(B)$ is nonempty;
- (ii) when nonempty, the deformation space is a torsor under

$$\Omega^\infty \mathrm{map}_{\mathrm{Mod}_B}(L_{B/A}, J[1]);$$

- (iii) for a fixed deformation ξ , its automorphism space is naturally equivalent to

$$\Omega^\infty \mathrm{map}_{\mathrm{Mod}_B}(L_{B/A}, J).$$

Proof.

- (1) **Apply cotangent transitivity.** Apply the exact contravariant functor $\mathrm{map}_{\mathrm{Mod}_B}(-, J[1])$ to the transitivity sequence.
- (2) **Compute the fibre.** This gives a fibre sequence of spectra

$$\mathrm{map}(L_{B/A}, J[1]) \longrightarrow \mathrm{map}(L_B, J[1]) \longrightarrow \mathrm{map}(L_A \otimes_A B, J[1]) \xrightarrow{\partial} \mathrm{map}(L_{B/A}, J[2]).$$

(3) **Establish the next comparison.** Define

$$\mathrm{ob}(B/A') := \partial(\eta_B).$$

- (4) **Invoke the cited result.** By 2.12.1, a deformation is exactly a lift of the point η_B through the middle map.
- (5) **Such a lift exists exactly when its boundary** is null.
- (6) **Compute the fibre.** If the fibre is inhabited, translation by a chosen point identifies it with the homotopy fibre over zero, which is $\Omega^\infty \mathrm{map}(L_{B/A}, J[1])$.
- (7) **Identify the map.** The automorphism space is the based loop space of the deformation ∞ -groupoid at ξ , giving the final formula.

□

The affine deformation groupoid is now expressed as a homotopy fibre and its obstruction theory is the boundary map of cotangent transitivity. We next globalize this description using the invariance of the small Zariski site across the base thickening.

2.12.2 Global deformation spaces

Definition 2.12.2 (Deformation of a spectral context). Let

$$i : S \longrightarrow S'$$

be a structured infinitesimal extension of spectral schemes and let $f : X \rightarrow S$ be a spectral scheme. A **deformation of X over S'** is a pair (X', α) consisting of a spectral scheme $X' \rightarrow S'$ and an equivalence

$$\alpha : X' \times_{S'} S \xrightarrow{\sim} X$$

over S . Morphisms are equivalences over S' with the specified coherent compatibility with α . Write

$$\mathrm{Def}_{S'/S}(X)$$

for the resulting ∞ -groupoid.

Lemma 2.12.1 (Fixed Zariski-site description of a deformation). *Let $i : S \rightarrow S'$ be a structured infinitesimal extension and let $(X', \alpha) \in \mathrm{Def}_{S'/S}(X)$. Then the base-change map*

$$j : X \simeq X' \times_{S'} S \longrightarrow X'$$

is an underlying square-zero infinitesimal extension and induces an equivalence of small spectral Zariski sites. Through this equivalence, the deformation is equivalently a square-zero deformation of the native structure coefficients on the fixed spectral Zariski site of X , compatible with the corresponding base deformation. More precisely, restriction to affine opens identifies $\mathrm{Def}_{S'/S}(X)$ with the descent groupoid of compatible affine deformations over the corresponding affine restrictions of $S \rightarrow S'$.

Proof.

- (1) **Invoke the cited result.** By 2.7.3(ii),(v), the structured base extension $i : S \rightarrow S'$ is an underlying square-zero infinitesimal substitution.
- (2) **Apply base change.** The morphism j is its geometric Cartesian base change.
- (3) **Conclude the step.** Hence 2.8.1 makes j an underlying square-zero infinitesimal substitution and identifies the small spectral Zariski sites of X and X' .
- (4) **is affine.** In particular, j is affine.
- (5) **We first pass from a global deformation.** We first pass from a global deformation to affine coefficient data.
- (6) **Construct the factorization.** Choose an affine Zariski atlas

$$\{T'_\alpha = \operatorname{Spec}(C'_\alpha) \rightarrow X'\}_\alpha$$

and refine it so that each composite $T'_\alpha \rightarrow S'$ factors through an affine open

$$V'_\alpha = \operatorname{Spec}(A'_\alpha) \longrightarrow S'.$$

- (7) **Fix the notation.** Put

$$U_\alpha := T'_\alpha \times_{X'} X, \quad V_\alpha := V'_\alpha \times_{S'} S.$$

- (8) **Use the stated property.** Because both $j : X \rightarrow X'$ and $i : S \rightarrow S'$ are affine square-zero substitutions, U_α and V_α are affine.
- (9) **Fix the notation.** Write

$$U_\alpha = \operatorname{Spec}(C_\alpha), \quad V_\alpha = \operatorname{Spec}(A_\alpha).$$

- (10) **Verify connectivity.** The defining Cartesian squares give a commutative diagram of connective commutative ring spectra

$$\begin{array}{ccc} A'_\alpha & \longrightarrow & C'_\alpha \\ \downarrow & & \downarrow \\ A_\alpha & \longrightarrow & C_\alpha \end{array}$$

with a canonical equivalence

$$C'_\alpha \otimes_{A'_\alpha} A_\alpha \simeq C_\alpha.$$

- (11) **Conclude the step.** Thus every affine member is an affine deformation in the sense of 2.12.1.
- (12) **On overlaps and iterated overlaps, refine by.** On overlaps and iterated overlaps, refine by affine opens.
- (13) **Apply base change.** The resulting affine deformation data agree under restriction because they are obtained by Cartesian base change from the single global object X' .
- (14) **Transport the structure.** Under the small-site equivalence, the opens U_α form the corresponding affine basis of X .

- (15) **Prove the converse.** Conversely, suppose compatible affine deformation data are given on an affine basis of the fixed small Zariski site of X , subordinate to compatible affine pairs for $X \rightarrow S$.

- (16) **Derive the comparison.** For a pair

$$U = \mathrm{Spec}(B) \longrightarrow V = \mathrm{Spec}(A),$$

the site equivalence for the structured base extension $S \rightarrow S'$ gives the corresponding affine open

$$V' = \mathrm{Spec}(A') \longrightarrow S'.$$

- (17) **Verify connectivity.** The affine deformation datum is a connective commutative A' -algebra B' with

$$B' \otimes_{A'} A \simeq B.$$

- (18) **Apply base change.** The map $B' \rightarrow B$ is the coordinate pushout corresponding to the geometric Cartesian base change of the structured square-zero map $\mathrm{Spec}(A) \rightarrow \mathrm{Spec}(A')$.

- (19) **Conclude the step.** Hence it is itself a structured square-zero extension by 2.7.2.

- (20) **Restrict the construction.** If $(U_1, V_1) \rightarrow (U, V)$ is an affine refinement, the square-zero Zariski-restriction theorem 2.7.4 identifies the corresponding $\mathrm{Spec}(B'_1)$ with the affine open restriction of $\mathrm{Spec}(B')$.

- (21) **Transport the same comparison.** The same statement on all iterated refinements supplies the canonical higher pasting compatibilities.

- (22) **Glue the local data.** Hence the affine deformations form a spectral Zariski gluing diagram and glue to a spectral scheme

$$X' \longrightarrow S'.$$

- (23) **Glue the local data.** Their specified reductions glue to an equivalence

$$X' \times_{S'} S \xrightarrow{\sim} X.$$

- (24) **Use full faithfulness.** Full faithfulness of spectral Zariski descent shows that the space of such gluings is contractible and that morphisms and higher homotopies are obtained by the same descent.

- (25) **Identify the equivalence.** The two constructions are therefore inverse equivalences of ∞ -groupoids.

- (26) **No affine atlas or coefficient presentation.** No affine atlas or coefficient presentation is retained after the construction.

□

Theorem 2.12.3 (Global deformation as a homotopy fibre). *Let $i : S \rightarrow S'$ be a structured infinitesimal extension by $I \in \mathrm{QCoh}(S)_{\geq 0}$, classified by*

$$\eta_S : L_S \longrightarrow I[1].$$

Let $f : X \rightarrow S$, put

$$J := f^*I,$$

and let

$$\eta_X : f^*L_S \longrightarrow J[1]$$

be the pullback of η_S . Then

$$\mathrm{Def}_{S'/S}(X) \simeq \mathrm{fib}_{\eta_X} \left(\mathrm{Map}_{\mathrm{QCoh}(X)}(L_X, J[1]) \longrightarrow \mathrm{Map}_{\mathrm{QCoh}(X)}(f^*L_S, J[1]) \right).$$

Equivalently, a deformation is a morphism

$$\eta_e : L_X \longrightarrow J[1]$$

together with a specified homotopy

$$\eta_e|_{f^*L_S} \simeq \eta_X.$$

Proof.

- (1) **Choose the local data.** Choose an affine basis of compatible pairs

$$U_\alpha = \mathrm{Spec}(B_\alpha) \longrightarrow V_\alpha = \mathrm{Spec}(A_\alpha)$$

for $X \rightarrow S$, closed under passage to a chosen affine basis on finite intersections.

- (2) **structured extension transports each to its corresponding.** The structured extension $S \rightarrow S'$ transports each V_α to its corresponding affine square-zero thickening

$$V'_\alpha = \mathrm{Spec}(A'_\alpha) \longrightarrow S'.$$

- (3) **Apply Zariski descent.** By 2.12.1, restriction to this fixed basis identifies the global deformation groupoid with the descent limit of the affine deformation groupoids

$$\mathrm{Def}_{A'_\alpha/A_\alpha}(B_\alpha)$$

and their corresponding values on iterated affine intersections.

- (4) **Compute the fibre.** On every affine member, 2.12.1 identifies this deformation groupoid with the homotopy fibre over the restricted base class $\eta_{X,\alpha}$ of

$$\mathrm{Map}_{\mathrm{Mod}_{B_\alpha}}(L_{B_\alpha}, J_\alpha[1]) \longrightarrow \mathrm{Map}_{\mathrm{Mod}_{B_\alpha}}(L_{A_\alpha} \otimes_{A_\alpha} B_\alpha, J_\alpha[1]).$$

- (5) **Apply cotangent base change.** Cotangent base change makes these affine equivalences compatible with every restriction and every iterated intersection.

- (6) **Restrict the construction.** The affine characterization of the global cotangent complexes identifies the two resulting cosimplicial mapping diagrams with the restrictions of

$$\mathrm{Map}_{\mathrm{QCoh}(X)}(L_X, J[1]) \quad \text{and} \quad \mathrm{Map}_{\mathrm{QCoh}(X)}(f^*L_S, J[1]).$$

- (7) **Apply Zariski descent.** Quasi-coherent Zariski descent computes each global mapping space as the limit of its affine mapping diagram.

- (8) **Compute the fibre.** Since limits of spaces commute with homotopy fibres over compatible points, totalizing the affine deformation equivalences gives

$$\mathrm{Def}_{S'/S}(X) \simeq \mathrm{fib}_{\eta_X} \left(\mathrm{Map}_{\mathrm{QCoh}(X)}(L_X, J[1]) \longrightarrow \mathrm{Map}_{\mathrm{QCoh}(X)}(f^*L_S, J[1]) \right).$$

- (9) **Transport the same comparison.** The same descent calculation identifies morphisms and all higher homotopies, so this is an equivalence of ∞ -groupoids.
- (10) **Glue the local data.** Conversely, a point of the global homotopy fibre restricts to the compatible affine deformation data above, and the converse construction in 2.12.1 glues them to the corresponding spectral scheme over S' .
- (11) **Glue the local data.** Thus no independent global deformation or gluing datum is assumed. □

Theorem 2.12.4 (Global cotangent boundary calculus). *In the situation of 2.12.3, the global transitivity sequence*

$$f^*L_S \longrightarrow L_X \longrightarrow L_{X/S}$$

determines a canonical obstruction

$$\mathrm{ob}(X/S') \in \pi_0 \mathrm{map}_{\mathrm{QCoh}(X)}(L_{X/S}, J[2]).$$

It satisfies:

- (i) *the obstruction vanishes if and only if X admits a deformation over S' ;*
- (ii) *when it vanishes, $\mathrm{Def}_{S'/S}(X)$ is a torsor under*

$$\Omega^\infty \mathrm{map}_{\mathrm{QCoh}(X)}(L_{X/S}, J[1]);$$

- (iii) *for a chosen deformation ξ ,*

$$\mathrm{Aut}(\xi) \simeq \Omega^\infty \mathrm{map}_{\mathrm{QCoh}(X)}(L_{X/S}, J).$$

Proof.

- (1) **Apply to the cited result.** Apply $\mathrm{map}_{\mathrm{QCoh}(X)}(-, J[1])$ to 2.5.3.
- (2) **Compute the fibre.** The resulting fibre sequence has boundary

$$\partial : \mathrm{map}(f^*L_S, J[1]) \longrightarrow \mathrm{map}(L_{X/S}, J[2]).$$

- (3) **Establish the next comparison.** Define

$$\mathrm{ob}(X/S') := \partial(\eta_X).$$

- (4) **Now apply the homotopy-fibre description of the.** Now apply the homotopy-fibre description of 2.12.3 exactly as in the affine proof. □

Corollary 2.12.1 (Split first-order deformations). *If the base infinitesimal extension is split, so that its classifying map $L_S \rightarrow I[1]$ is zero, then*

$$\mathrm{Def}_{S'/S}(X) \simeq \mathrm{Map}_{\mathrm{QCoh}(X)}(L_{X/S}, f^*I[1]).$$

Proof.

- (1) **deformation space.** The deformation space is the fibre over zero in 2.12.3.
- (2) **Apply cotangent transitivity.** Apply the mapping-spectrum fibre sequence associated to transitivity.

□

The deformation calculus now gives, both affinely and globally, the obstruction, the deformation torsor, and the automorphism space from one cotangent exact triangle. A second geometric use of the same differential structure is to analyze contexts defined by equations.

2.13 Equations, conormal classifiers, and first-order variation

Linear comprehension from Chapter 1 turns a coefficient into a context of linear functionals. Equalizing such a term with the zero term gives a derived solution context. The relative cotangent complex of that closed context records the linear type of the imposed equations through its shifted conormal classifier.

2.13.1 Conormal classifiers and quasi-smooth closed contexts

Definition 2.13.1 (Local finite presentation for spectral schemes). A morphism $f : Y \rightarrow X$ of spectral schemes is **locally of finite presentation** if, for every affine open $U = \mathrm{Spec}(A) \subseteq X$ and every affine open $V = \mathrm{Spec}(B) \subseteq Y \times_X U$, the connective commutative ring spectrum B is locally of finite presentation over A in the sense of Chapter 1. This is a property of the morphism; no affine presentation is retained.

Proposition 2.13.1 (Locality and base change of local finite presentation). *Local finite presentation is local on source and target for the spectral Zariski topology and is stable under arbitrary base change and composition. For a closed immersion $Z \hookrightarrow X$, it is enough to test the property after pullback to affine opens of X .*

Proof.

- (1) **Apply base change.** The affine notion is stable under base change and composition and is local on affine spectral Zariski covers.
- (2) **Two affine computations admit common affine refinements by.** Two affine computations admit common affine refinements by the principal-open basis of Chapter 1; hence the definition is independent of charts.
- (3) **Apply base change.** The locality and base-change assertions follow after passing to such refinements.

- (4) **closed immersion.** A closed immersion is affine, so its pullback over an affine target is affine, giving the final assertion. □

Definition 2.13.2 (Conormal classifier). Let

$$i : Z \hookrightarrow X$$

be a closed immersion of spectral schemes. Its **conormal classifier** is

$$N_{Z/X}^* := L_{Z/X}[-1] \in \mathrm{QCoh}(Z).$$

Definition 2.13.3 (Quasi-smooth closed context extension). A closed immersion $i : Z \hookrightarrow X$ is **quasi-smooth** if it is locally of finite presentation and its conormal classifier $N_{Z/X}^*$ is finite locally free. For such an immersion define the normal coefficient

$$N_{Z/X} := (N_{Z/X}^*)^\vee.$$

Its associated normal linear context is the relative linear space of Chapter 1 determined by this finite locally free coefficient.

Proposition 2.13.2 (Base change of quasi-smooth closed contexts). *Quasi-smooth closed immersions are stable under arbitrary base change. If $Y \rightarrow X$ is any morphism and $Z_Y := Z \times_X Y$, then*

$$N_{Z_Y/Y}^* \simeq N_{Z/X}^*|_{Z_Y}.$$

Proof.

- (1) **Apply base change.** Closed immersions and local finite presentation are stable under base change.
- (2) **cotangent comparison.** The cotangent comparison is 2.5.2; shift by $[-1]$.
- (3) **Use preservation.** Finite local freeness is preserved by substitution in the coefficient doctrine of Chapter 1. □

Lemma 2.13.1 (Ambient lifting of principal opens along a closed immersion). *Let $A \rightarrow B$ be a morphism of connective commutative ring spectra which is surjective on π_0 . For every*

$$g \in \pi_0(B)$$

and every lift $f \in \pi_0(A)$, there is a canonical equivalence

$$B \otimes_A A[f^{-1}] \simeq B[g^{-1}].$$

Consequently the principal spectral open $D_B^{\mathrm{sp}}(g)$ is the Cartesian pullback of $D_A^{\mathrm{sp}}(f)$. In particular, every principal open neighbourhood in a closed affine subcontext is obtained by restricting an ambient principal open neighbourhood.

Proof.

- (1) **image of in.** The image of f in $\pi_0(B)$ is g .
- (2) **Verify invertibility.** Both displayed B -algebras represent the universal connective commutative B -algebra in which this class is invertible.
- (3) **Apply the universal property.** The universal property of spectral localization therefore supplies the canonical equivalence.
- (4) **Derive the comparison.** Passing to opposite affine contexts gives the Cartesian statement.
- (5) **No lift of.** No lift of g is retained: two lifts differ by an element of the kernel of $\pi_0(A) \rightarrow \pi_0(B)$, and the resulting principal opens have the same pullback to $\text{Spec}(B)$.

□

The conormal classifier provides the intrinsic first-order coefficient of a closed context. We now show that finite locally free linear equations produce exactly the quasi-smooth closed contexts locally, so that the equation presentation is a theorem rather than part of the structure.

2.13.2 Derived zero loci as equational context formation

Construction 2.13.1 (Derived solution context of a linear equation). Let $\mathcal{E} \in \text{QCoh}(X)$ be finite locally free and let

$$s : \mathcal{E} \longrightarrow \mathcal{O}_X$$

be a coefficient term. By the linear-functional representing property of 1.13.2, s determines a section

$$\sigma_s : X \longrightarrow \mathbb{L}_X(\mathcal{E}).$$

The zero morphism determines the zero section

$$0 : X \longrightarrow \mathbb{L}_X(\mathcal{E}).$$

Define the **derived solution context** or **derived zero locus** of s by the Cartesian square

$$\begin{array}{ccc} Z(s) & \longrightarrow & X \\ \downarrow & & \downarrow \sigma_s \\ X & \xrightarrow{0} & \mathbb{L}_X(\mathcal{E}). \end{array}$$

Thus $Z(s)$ is the context obtained by imposing the equation $s = 0$.

Theorem 2.13.1 (Derived linear equations are quasi-smooth). *In 2.13.1, the projection*

$$Z(s) \longrightarrow X$$

is a quasi-smooth closed immersion, and there is a canonical equivalence

$$N_{Z(s)/X}^* \simeq \mathcal{E}|_{Z(s)}.$$

Proof.

- (1) **Set up the argument.** The assertion is Zariski-local on X .

- (2) **Take an affine open on which.** Take an affine open $X = \operatorname{Spec}(A)$ on which

$$\mathcal{E} \simeq \mathcal{O}_X^{\oplus n}.$$

- (3) **Fix the notation.** Put

$$R := \operatorname{Sym}_A(A^{\oplus n}).$$

- (4) **Identify the map.** The section s and the zero section correspond contravariantly to two A -algebra maps

$$\epsilon_s, \epsilon_0 : R \rightrightarrows A.$$

- (5) **Fix the notation.** Write A_s and A_0 for the two copies of A equipped with these R -algebra structures.

- (6) **coordinate algebra of the zero locus.** The coordinate algebra of the zero locus is

$$C := A_s \otimes_R A_0.$$

- (7) **Apply cotangent transitivity.** Apply transitivity to

$$A \longrightarrow R \xrightarrow{\epsilon_0} A_0.$$

- (8) **Derive the comparison.** The composite is the identity, hence $L_{A_0/A} \simeq 0$, while 2.3.2 gives

$$L_{R/A} \simeq R \otimes A^{\oplus n}.$$

- (9) **Conclude the step.** Therefore

$$L_{A_0/R} \simeq A_0^{\oplus n}[1].$$

- (10) **zero section.** The zero section is a closed immersion by 1.13.9.

- (11) **It is locally of finite presentation by.** It is locally of finite presentation by 2.4.4: on degree zero its coordinate map is the quotient by the ideal generated by the n universal variables, hence is finitely presented, and its relative cotangent complex is finite free.

- (12) **Apply base change.** The projection $\operatorname{Spec}(C) \rightarrow \operatorname{Spec}(A_s)$ is its base change along σ_s , so it is a closed immersion locally of finite presentation.

- (13) **Apply cotangent base change.** Cotangent base change identifies

$$L_{C/A_s} \simeq L_{A_0/R} \otimes_{A_0} C \simeq C^{\oplus n}[1].$$

- (14) **Restrict the construction.** Thus

$$L_{C/A_s}[-1] \simeq C^{\oplus n},$$

which is the restriction of $\mathcal{E}$.

- (15) **Glue the local data.** The affine identifications are natural under further Zariski localization and therefore glue globally.

□

Lemma 2.13.2 (Universal equations and the conormal basis). *Let A be a connective commutative ring spectrum and put*

$$R := \mathrm{Sym}_A(A^{\oplus n}).$$

Let

$$\epsilon_0, \epsilon_f : R \rightrightarrows A$$

be respectively the zero augmentation and the map classified by points $f_1, \dots, f_n \in \Omega^\infty A$. Write A_{ϵ_f} and A_{ϵ_0} for the two copies of A with these R -algebra structures and put

$$C := A_{\epsilon_f} \otimes_R A_{\epsilon_0}.$$

Regard C as an A -algebra through $A = A_{\epsilon_f} \rightarrow C$, and put

$$K_C := \mathrm{fib}(A \rightarrow C).$$

For each $1 \leq i \leq n$, the i -th universal equation together with its canonical nullhomotopy in C determines a class

$$\lambda_i \in \pi_0(K_C).$$

Under the canonical equivalence

$$L_{C/A}[-1] \simeq C^{\oplus n},$$

the derived Hurewicz–conormal morphism

$$C \otimes_A K_C \longrightarrow L_{C/A}[-1]$$

carries the image of λ_i to the i -th standard basis vector of $\pi_0(C)^{\oplus n}$.

Proof.

(1) **Fix the notation.** Let $x_1, \dots, x_n$ denote the universal generators of $R = \mathrm{Sym}_A(A^{\oplus n})$.

(2) **Derive the comparison.** The universal derivation gives

$$L_{R/A} \simeq R \otimes_A A^{\oplus n},$$

under which dx_i is the i -th standard basis vector.

(3) **Apply cotangent transitivity.** Apply transitivity to

$$A \longrightarrow R \xrightarrow{\epsilon_0} A_{\epsilon_0}.$$

(4) **Apply cotangent transitivity.** The composite is the identity on the underlying A -algebra, so $L_{A_{\epsilon_0}/A} \simeq 0$, and transitivity gives

$$L_{A_{\epsilon_0}/R}[-1] \simeq L_{R/A} \otimes_R A_{\epsilon_0} \simeq A_{\epsilon_0}^{\oplus n}.$$

(5) **Apply cotangent transitivity.** The class determined by x_i and its nullhomotopy under the zero augmentation maps to the i -th basis vector; this is the boundary interpretation of the transitivity morphism applied to the universal derivation.

(6) **Apply cotangent base change.** The pushout square defining C and cotangent base change identify

$$L_{C/A} \simeq L_{A_{\epsilon_0}/R} \otimes_{A_{\epsilon_0}} C.$$

- (7) **Derive the comparison.** The canonical homotopy between the images of x_i through the two legs of the pushout gives $\lambda_i \in \pi_0(K_C)$.
- (8) **Verify naturality.** Naturality of the relative-cofibre-to-cotangent morphism identifies its image with the base change of the universal class for ϵ_0 , hence with the i -th standard basis vector.

□

Theorem 2.13.2 (Local equational characterization of quasi-smooth closed contexts). *Let $i : Z \hookrightarrow X$ be a closed immersion of spectral schemes. The following are equivalent.*

- (i) *i is quasi-smooth.*
- (ii) *Zariski-locally on X , the immersion is equivalent to the derived zero locus of a term*

$$\mathcal{E} \longrightarrow \mathcal{O}_X$$

for a finite locally free coefficient $\mathcal{E}$.

Under such an identification, the restricted coefficient $\mathcal{E}|_Z$ identifies canonically with $N_{Z/X}^$.*

Proof.

- (1) **Set up the argument.** The implication (ii) $\Rightarrow$ (i) is 2.13.1.
- (2) **Assume .** Assume (i).
- (3) **Establish the next comparison.** The assertion is local on X .
- (4) **On the complement of , the immersion.** On the complement of Z , the immersion is empty, and the empty immersion is the derived zero locus of the unit term $\mathcal{O}_X \rightarrow \mathcal{O}_X$: affine-locally its coordinate pushout has zero degree-zero ring and hence is the zero ring spectrum.
- (5) **It therefore suffices to work near a.** It therefore suffices to work near a point of Z .
- (6) **Verify connectivity.** Choose an affine neighbourhood and write

$$Z = \mathrm{Spec}(B) \longrightarrow X = \mathrm{Spec}(A),$$

where $A \rightarrow B$ is connective, surjective on π_0 , and locally of finite presentation.

- (7) **Shrink the ambient affine around the selected.** Shrink the ambient affine around the selected point so that

$$L_{B/A}[-1] \simeq B^{\oplus n}.$$

- (8) **Use spectral localization.** This shrinking may be performed in the ambient context by 2.13.1: choose a principal open of $\mathrm{Spec}(B)$ on which the conormal classifier is free, lift its defining class from $\pi_0(B)$ to $\pi_0(A)$, and replace the ambient affine by the corresponding principal localization.
- (9) **Fix the notation.** Put

$$K := \mathrm{fib}(A \rightarrow B), \quad R := \pi_0(A), \quad S := \pi_0(B), \quad I := \ker(R \rightarrow S).$$

- (10) **Derive the comparison.** The relative Hurewicz–conormal comparison gives

$$\pi_0(B \otimes_A K) \xrightarrow{\cong} \pi_0(L_{B/A}[-1]).$$

- (11) **Verify connectivity.** Because B and K are connective,

$$\pi_0(B \otimes_A K) \simeq S \otimes_R \pi_0(K) \simeq \pi_0(K)/I\pi_0(K).$$

- (12) **Verify surjectivity.** Hence $\pi_0(K) \rightarrow \pi_0(L_{B/A}[-1])$ is surjective.

- (13) **Fix the notation.** Let $e_1, \dots, e_n$ be the basis of $\pi_0(L_{B/A}[-1]) \simeq S^{\oplus n}$ determined by the chosen local trivialization.

- (14) **Choose the local data.** Choose points

$$\tilde{\kappa}_i \in \Omega^\infty K$$

whose components $\kappa_i \in \pi_0(K)$ map to the e_i .

- (15) **Their images in.** Their images in $\Omega^\infty A$ determine elements

$$f_1, \dots, f_n \in R$$

together with specified nullhomotopies of their images in B , so $f_i \in I$.

- (16) **Invoke the cited result.** By 2.4.2, the image of κ_i in I/I^2 is the class of f_i .

- (17) **Use the stated property.** Since the κ_i map to a basis of the derived conormal module, the classes of $f_1, \dots, f_n$ generate I/I^2 .

- (18) **We next prove that I is finitely generated.** We next prove that I is finitely generated.

- (19) **Verify connectivity.** Local finite presentation of $A \rightarrow B$ means that B is compact in connective commutative A -algebras.

- (20) **Use the adjunction.** If $\{T_\alpha\}$ is a filtered diagram of ordinary commutative R -algebras, Eilenberg–Mac Lane formation commutes with the filtered colimit and the $\pi_0 \dashv H$ adjunction gives

$$\begin{aligned} \mathrm{Map}_{\mathrm{CRing}_{R/}}(S, \mathrm{colim}_\alpha T_\alpha) &\simeq \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})_{A/}}(B, H \mathrm{colim}_\alpha T_\alpha) \\ &\simeq \mathrm{colim}_\alpha \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})_{A/}}(B, HT_\alpha) \\ &\simeq \mathrm{colim}_\alpha \mathrm{Map}_{\mathrm{CRing}_{R/}}(S, T_\alpha). \end{aligned}$$

- (21) **Conclude the step.** Thus S is finitely presented over R .

- (22) **Use the stated property.** Since $R \twoheadrightarrow S$, its kernel I is finitely generated.

- (23) **Establish the next comparison.** Set

$$Q := I/(f_1, \dots, f_n).$$

- (24) **Derive the comparison.** The module Q is finitely generated, and generation of I/I^2 gives

$$Q = IQ.$$

- (25) **Use spectral localization.** At the prime corresponding to the chosen point, I lies in the maximal ideal, so Nakayama's lemma gives $Q = 0$ after localization.

- (26) **Use the stated property.** Since Q is finitely generated, shrink once more so that

$$I = (f_1, \dots, f_n).$$

- (27) **Fix the notation.** Put

$$P := \operatorname{Sym}_A(A^{\oplus n}).$$

- (28) **Fix the notation.** Let $\epsilon_0 : P \rightarrow A$ be the zero augmentation and $\epsilon_f : P \rightarrow A$ the map classified by the selected points lifting $f_1, \dots, f_n$.

- (29) **Distinguish the two -algebra structures by writing and $\cdot$.** Distinguish the two P -algebra structures by writing A_{ϵ_f} and A_{ϵ_0} , and define

$$C := A_{\epsilon_f} \otimes_P A_{\epsilon_0}.$$

- (30) **Apply the universal property.** The specified nullhomotopies carried by the points $\tilde{\kappa}_i$ identify the two composites $P \rightrightarrows B$, so the pushout universal property gives an A -algebra morphism

$$C \longrightarrow B.$$

- (31) **Verify connectivity.** Connective pushouts give

$$\pi_0(C) \simeq R/(f_1, \dots, f_n) \simeq R/I \simeq S.$$

- (32) **Conclude the step.** Hence $C \rightarrow B$ is an isomorphism on π_0 .

- (33) **Invoke the cited result.** By 2.13.1,

$$L_{C/A}[-1] \simeq C^{\oplus n}.$$

- (34) **Fix the notation.** Put

$$K_C := \operatorname{fib}(A \rightarrow C), \quad K_B := \operatorname{fib}(A \rightarrow B) = K.$$

- (35) **For each universal equation, its canonical nullhomotopy in.** For each universal equation, its canonical nullhomotopy in C determines $\lambda_i \in \pi_0(K_C)$.

- (36) **Invoke the cited result.** By 2.13.2, the derived conormal map sends λ_i to the i -th standard basis vector.

- (37) **Transport the structure.** Under $K_C \rightarrow K_B$, the class λ_i maps to κ_i , because $C \rightarrow B$ was constructed from precisely those nullhomotopies.

- (38) **Verify naturality.** Naturality of the Hurewicz–conormal comparison gives a commutative square

$$\begin{array}{ccc} B \otimes_A K_C & \longrightarrow & (L_{C/A} \otimes_C B)[-1] \\ \downarrow & & \downarrow \\ B \otimes_A K_B & \longrightarrow & L_{B/A}[-1]. \end{array}$$

- (39) **Identify the map.** The right vertical map carries the standard basis to $e_1, \dots, e_n$ and therefore induces an isomorphism on π_0 .
- (40) **Compare the two constructions.** Both its source and target are finite projective B -modules.
- (41) **Use finite projectivity.** A morphism between finite projective B -modules which is an isomorphism on π_0 is an equivalence, by the flat homotopy-group formula for finite projectives.
- (42) **Conclude the step.** Therefore

$$L_{C/A} \otimes_C B \xrightarrow{\simeq} L_{B/A}.$$

- (43) **Apply cotangent transitivity.** Transitivity for $A \rightarrow C \rightarrow B$ gives $L_{B/C} \simeq 0$.
- (44) **Derive the comparison.** Together with $\pi_0(C) \cong \pi_0(B)$, cotangent rigidity yields $C \simeq B$.
- (45) **Conclude the step.** Thus the original closed immersion is, near the selected point, the derived zero locus of the constructed finite tuple of equations.

□

Remark 2.13.1 (No equation presentation is retained). The local tuple $(f_1, \dots, f_n)$, its chosen lifts from the relative fibre, and the trivialization of the conormal classifier occur only in the proof of 2.13.2. A quasi-smooth closed context is not equipped with local equations or a chosen zero-locus presentation.

The geometric branch has now reached its principal applications: cotangent data controls infinitesimal lifting, first-order deformations, and local equational presentations of quasi-smooth closed contexts. The final substantive construction is a stable consequence of the endpoint semantics itself.

2.14 Stable reconstruction from exact endpoint semantics

The geometric branch has produced the endpoint adjunction string before any stable differential-cohomological terminology was introduced. We now stabilize that string and isolate the exact categorical mechanism responsible for its fracture pattern. The order is deliberate: first prove an abstract fracture theorem for an exact adjoint triple with fully faithful middle functor, then specialize it to the stabilized infinitesimal semantics, and only afterward compare the resulting structure with established differential-cohomological fracture patterns.

2.14.1 Stabilization of the endpoint adjunction string

Definition 2.14.1 (Stable native and infinitesimal coefficient categories). Define

$$\mathcal{D}_{\text{Sp}} := \text{Sp}((\mathcal{H}_{\text{Sp}})_*), \quad \mathcal{D}_{\text{inf}} := \text{Sp}((\mathcal{H}_{\text{inf}})_*).$$

These are the stabilizations of the pointed native spectral Zariski topos and the pointed finite-infinitesimal spectral Zariski topos. We call objects of $\mathcal{D}_{\text{inf}}$ **stable infinitesimal coefficients**.

Theorem 2.14.1 (Stable endpoint adjunction string). *The endpoint adjunction string stabilizes canonically to an exact adjoint string*

$$i_!^{\text{st}} \dashv i^{*,\text{st}} \dashv i_*^{\text{st}} \dashv i^{!,\text{st}} : \mathcal{D}_{\text{Sp}} \rightleftarrows \mathcal{D}_{\text{inf}}.$$

The functors $i_!^{\text{st}}$ and i_*^{st} are fully faithful. Moreover i_*^{st} preserves all small limits and colimits. In particular,

$$i^{*,\text{st}} \dashv i_*^{\text{st}} \dashv i^{!,\text{st}}$$

is an exact adjoint triple with fully faithful middle functor.

Proof.

- (1) **Invoke the cited result.** By 2.10.2, the unstable endpoint semantics is

$$i_! \dashv i^* \dashv i_* \dashv i^!.$$

- (2) **Use the adjunction.** The functor $i_! = \text{cod}^*$ preserves all limits and, being a left adjoint, all colimits.
- (3) **Use the adjunction.** Restriction i^* preserves all limits and, as the left adjoint of i_* , all colimits.
- (4) **Use preservation.** The functor $i_* = \text{dom}^*$ preserves all limits and preserves all colimits by 2.10.1.
- (5) **Conclude the argument.** Finally $i^!$ preserves all limits.
- (6) **Use the adjunction.** Thus the left adjoint in each adjacent pair is left exact.
- (7) **Use the adjunction.** Passage to pointed objects and then to spectrum objects is therefore functorial for each adjacent adjunction and yields the displayed string.
- (8) **Apply the statement objectwise.** Every induced functor between the stable categories is exact.
- (9) **Use full faithfulness.** Stabilization preserves the unit and counit equivalences expressing full faithfulness of $i_!$ and i_* , so the two stabilized embeddings remain fully faithful.
- (10) **Use the adjunction.** Since i_*^{st} has both a left and a right adjoint, it preserves all small colimits and limits.

□

Definition 2.14.2 (Stable differential-cohesive modalities). Define exact endofunctors of $\mathcal{D}_{\text{inf}}$ by

$$\mathfrak{R}^{\text{st}} := i_!^{\text{st}} i_*^{\text{st}}, \quad \mathfrak{S}^{\text{st}} := i_*^{\text{st}} i_!^{\text{st}}, \quad \&^{\text{st}} := i_*^{\text{st}} i_!^{\text{st}}.$$

Proposition 2.14.1 (Stable modal triple). *There is a canonical exact adjoint triple*

$$\mathfrak{R}^{\text{st}} \dashv \mathfrak{S}^{\text{st}} \dashv \&^{\text{st}}.$$

All three endofunctors are idempotent. There are canonical transformations

$$\mathfrak{R}^{\text{st}} E \longrightarrow E \longrightarrow \mathfrak{S}^{\text{st}} E,$$

and the essential images of $\mathfrak{S}^{\text{st}}$ and $\&^{\text{st}}$ are the fully faithful copy of $\mathcal{D}_{\text{Sp}}$ determined by i_^{st} .*

Proof.

- (1) **Identify the resulting comparison.** This is the stabilized form of 2.10.4.
- (2) **Use full faithfulness.** The adjunctions follow by composing the stabilized endpoint adjunctions, and idempotence follows from full faithfulness.
- (3) **common essential image claim uses.** The common essential image claim uses

$$i_*^{\text{st}} i_!^{\text{st}} \simeq \text{id}, \quad i_!^{\text{st}} i_*^{\text{st}} \simeq \text{id}.$$

□

Stabilization produces the exact adjoint triple needed for the final reconstruction. Before naming any of its pieces, we isolate the general fracture mechanism that follows solely from exactness, adjunction, and full faithfulness.

2.14.2 An exact-adjoint fracture theorem

Theorem 2.14.2 (Exact-adjoint fracture from a fully faithful middle functor). *Let*

$$L \dashv j \dashv R : \mathcal{A} \rightleftarrows \mathcal{B}$$

be an exact adjoint triple of stable ∞ -categories with j fully faithful. For $E \in \mathcal{B}$, let

$$\eta_E : E \longrightarrow jLE, \quad \epsilon_E : jRE \longrightarrow E$$

be the unit and counit, and define

$$\mathbf{A}(E) := \text{fib}(\eta_E), \quad \mathbf{Z}(E) := \text{cofib}(\epsilon_E).$$

Put

$$\mathcal{B}^{\text{pure}} := \ker(R).$$

Then the following structures and equivalences are canonical.

- (i) *Under the equivalences $Lj \simeq \text{id}_{\mathcal{A}}$ and $Rj \simeq \text{id}_{\mathcal{A}}$, the two adjacent-mates morphisms*

$$L(\epsilon_E), \quad R(\eta_E) : RE \longrightarrow LE$$

are canonically equivalent. Write their common value as $\chi_E : RE \rightarrow LE$.

(ii) $Z(E) \in \mathcal{B}^{\text{pure}}$, and

$$Z : \mathcal{B} \longrightarrow \mathcal{B}^{\text{pure}}$$

is an exact reflector left adjoint to the inclusion. Moreover

$$LA(E) \simeq 0, \quad ZA(E) \simeq Z(E), \quad Z(jLE) \simeq 0.$$

(iii) Applying L to the cycle triangle gives the characteristic exact triangle

$$RE \xrightarrow{\chi_E} LE \xrightarrow{\phi_E} LZ(E).$$

Applying the cycle reflector to $A(E)$ gives the upper exact triangle

$$j\Omega LZ(E) \longrightarrow A(E) \longrightarrow Z(E).$$

(iv) The canonical square

$$\begin{array}{ccc} E & \longrightarrow & Z(E) \\ \eta_E \downarrow & & \downarrow \eta_{Z(E)} \\ jLE & \xrightarrow{j\phi_E} & jLZ(E) \end{array}$$

is Cartesian. Equivalently,

$$E \simeq Z(E) \times_{jLZ(E)} jLE.$$

(v) Let $\mathcal{T}_j$ be the pullback ∞ -category whose objects are triples

$$(U, P, \phi), \quad U \in \mathcal{A}, \quad P \in \mathcal{B}^{\text{pure}}, \quad \phi : U \rightarrow LP.$$

Then

$$\mathcal{B} \xrightarrow{\sim} \mathcal{T}_j, \quad E \longmapsto (LE, Z(E), \phi_E)$$

is an equivalence of stable ∞ -categories. An inverse is

$$\text{Diff}_j(U, P, \phi) := P \times_{jLP} jU.$$

If $\mathcal{A}$ and $\mathcal{B}$ are presentable and the adjoints are accessible, then $\mathcal{B}^{\text{pure}}$ is presentable and Z is an accessible exact localization.

Proof.

(1) **Use full faithfulness.** Full faithfulness of j gives canonical equivalences

$$Lj \simeq \text{id}_{\mathcal{A}}, \quad Rj \simeq \text{id}_{\mathcal{A}}.$$

(2) **Apply to the two candidate maps in (i).** Apply j to the two candidate maps in (i).

(3) **Verify naturality.** Naturality of the unit with respect to $\epsilon_E : jRE \rightarrow E$ gives

$$jL(\epsilon_E) \eta_{jRE} \simeq \eta_E \epsilon_E,$$

while naturality of the counit with respect to $\eta_E : E \rightarrow jLE$ gives

$$\epsilon_{jLE} jR(\eta_E) \simeq \eta_E \epsilon_E.$$

(4) **Identify the equivalence.** The maps η_{jRE} and ϵ_{jLE} are equivalences because their objects lie in the essential image of j .

(5) **Conclude the step.** Thus the two candidate maps have canonically equivalent images under the fully faithful functor j , proving (i).

(6) **Apply to.** Apply R to

$$jRE \xrightarrow{\epsilon_E} E \longrightarrow Z(E).$$

(7) **Handle the first component.** The first induced map is an equivalence, hence $RZ(E) \simeq 0$.

(8) **If , then.** If $P \in \ker(R)$, then

$$\mathrm{map}_{\mathcal{B}}(jRE, P) \simeq \mathrm{map}_{\mathcal{A}}(RE, RP) \simeq 0.$$

(9) **Compute the mapping object.** Mapping the cycle triangle into P gives

$$\mathrm{map}_{\mathcal{B}}(Z(E), P) \xrightarrow{\simeq} \mathrm{map}_{\mathcal{B}}(E, P),$$

so Z is the reflector onto $\ker(R)$.

(10) **Use the stated property.** Since jR and the identity functor are exact, their functorial cofiber Z is exact.

(11) **Applying to.** Applying L to

$$A(E) \longrightarrow E \xrightarrow{\eta_E} jLE$$

shows $LA(E) \simeq 0$.

(12) **Use exactness.** Since jLE lies in the image of j , its cycle quotient vanishes; exactness of Z applied to the same triangle therefore gives $ZA(E) \simeq Z(E)$.

(13) **Identify the resulting comparison.** This proves (ii).

(14) **Derive the comparison.** Applying L to the cycle triangle and using $Lj \simeq \mathrm{id}$ yields

$$RE \xrightarrow{\chi_E} LE \xrightarrow{\phi_E} LZ(E).$$

(15) **Use exactness.** Applying R to the deformation triangle gives

$$RA(E) \longrightarrow RE \xrightarrow{\chi_E} LE,$$

so exactness identifies

$$RA(E) \simeq \Omega LZ(E).$$

(16) **cycle triangle of , together.** The cycle triangle of $A(E)$, together with $ZA(E) \simeq Z(E)$, is therefore the upper triangle of (iii).

(17) **Verify naturality.** For (iv), naturality of η makes the displayed square commute.

(18) **Identify the map.** The fibre of its upper horizontal map is jRE .

(19) **Use exactness.** By the characteristic triangle and exactness of j , the fibre of the lower horizontal map is also canonically jRE .

- (20) **Transport the structure.** Under these identifications the induced map of fibres is the unit equivalence on an object in the image of j .
- (21) **square in a stable -category.** A square in a stable ∞ -category is Cartesian exactly when the induced map of horizontal fibres is an equivalence, proving the fracture formula.
- (22) **For (v), let.** For (v), let

$$E := P \times_{jLP} jU.$$
- (23) **Use exactness.** Exactness and $Lj \simeq \text{id}$ give

$$LE \simeq LP \times_{LP} U \simeq U.$$
- (24) **Use the stated property.** Since $RP \simeq 0$ and $Rj \simeq \text{id}$, applying R gives

$$RE \simeq 0 \times_{LP} U \simeq \text{fib}(\phi).$$
- (25) **projection therefore.** The projection $E \rightarrow P$ therefore has fibre

$$j \text{fib}(\phi) \simeq jRE.$$
- (26) **Use full faithfulness.** Its composite with the counit $jRE \rightarrow E$ becomes null after mapping to P because $RP = 0$; the resulting map from jRE to the fibre is an equivalence after applying R , hence is an equivalence by full faithfulness on the image of j .
- (27) **Conclude the step.** Thus $E \rightarrow P$ is the cycle quotient, so $Z(E) \simeq P$.
- (28) **Identify the resulting object.** Applying L to this quotient identifies the characteristic map with the given ϕ .
- (29) **Prove the converse.** Conversely, (iv) reconstructs every E from its characteristic triple.
- (30) **Identify the equivalence.** The two assignments are functorial and exact, hence give the asserted stable equivalence.
- (31) **and are accessible and so is the.** In the presentable case, R and j are accessible and so is the functorial cofiber Z .
- (32) **Use spectral localization.** An accessible reflective localization of a presentable ∞ -category has presentable target, proving the last statement.

□

The abstract reconstruction mechanism is now internal to the chapter: it uses only exactness, the adjacent adjunctions, and full faithfulness of the middle functor. We may therefore specialize it to the stabilized endpoint semantics. Only at this point do we attach BNV-type terminology to the resulting sectors and characteristic data; that terminology records a later comparison and is not part of their construction.

2.14.3 The stabilized spectral instance and BNV-type sectors

Put

$$L := i^{*,\text{st}}, \quad j := i_{*}^{\text{st}}, \quad R := i^{!,\text{st}}.$$

Lemma 2.14.1 (Adjacent-mates comparison). *For $E \in \mathcal{D}_{\text{inf}}$, the two maps*

$$i^{*,\text{st}}(\epsilon_E), \quad i^{!,\text{st}}(\eta_E) : i^{!,\text{st}}E \longrightarrow i^{*,\text{st}}E$$

are canonically equivalent. Their common value

$$\chi_E : i^{!,\text{st}}E \longrightarrow i^{*,\text{st}}E$$

is the adjacent-mates morphism of 2.14.2.

Proof.

- (1) **Apply the cited result(i) to the stabilized endpoint triple.** Apply 2.14.2(i) to the stabilized endpoint triple.

□

Definition 2.14.3 (Stable BNV-type sectors). For $E \in \mathcal{D}_{\text{inf}}$, define

$$\mathcal{A}(E) := \text{fib}(E \xrightarrow{\eta_E} \mathfrak{S}^{\text{st}}E), \quad \mathcal{Z}(E) := \text{cofib}(\&^{\text{st}}E \xrightarrow{\epsilon_E} E).$$

The endpoint values

$$i^{*,\text{st}}E, \quad i^{!,\text{st}}E$$

are respectively the underlying and secondary sectors. The map

$$\phi_E : i^{*,\text{st}}E \longrightarrow i^{*,\text{st}}\mathcal{Z}(E)$$

is the characteristic morphism obtained from the exact triangle

$$i^{!,\text{st}}E \xrightarrow{\chi_E} i^{*,\text{st}}E \xrightarrow{\phi_E} i^{*,\text{st}}\mathcal{Z}(E).$$

Definition 2.14.4 (BNV-pure cycle coefficient). Define

$$\mathcal{D}_{\text{inf}}^{\text{BNV}} := \ker(i^{!,\text{st}}) = \{P \in \mathcal{D}_{\text{inf}} \mid i^{!,\text{st}}P \simeq 0\}.$$

Objects of this full subcategory are called **BNV-pure cycle coefficients**. The terminology records the comparison made below; the subcategory itself is constructed solely from the endpoint right adjoint.

Proposition 2.14.2 (Cycle reflector and deformation identities). *The category $\mathcal{D}_{\text{inf}}^{\text{BNV}}$ is stable and*

$$\mathcal{Z} : \mathcal{D}_{\text{inf}} \longrightarrow \mathcal{D}_{\text{inf}}^{\text{BNV}}$$

is an exact reflector. Moreover

$$i^{!,\text{st}}\mathcal{Z}(E) \simeq 0, \quad \mathcal{Z}^2(E) \simeq \mathcal{Z}(E),$$

and

$$\boxed{i^{*,\text{st}}\mathcal{A}(E) \simeq 0, \quad \mathcal{Z}\mathcal{A}(E) \simeq \mathcal{Z}(E), \quad \mathcal{Z}\mathfrak{S}^{\text{st}}(E) \simeq 0.}$$

Proof.

- (1) **Identify the resulting comparison.** This is 2.14.2(ii) for the stabilized endpoint triple.

□

Proposition 2.14.3 (Presentability of the BNV-pure sector). *The stable category $\mathcal{D}_{\text{inf}}^{\text{BNV}}$ is pre-presentable and $\mathcal{Z}$ is an accessible exact localization. For every small diagram $k \mapsto P_k$ of BNV-pure coefficients,*

$$\mathcal{D}_{\text{inf}}^{\text{BNV}} \text{colim}_k P_k \simeq \mathcal{Z} \left(\mathcal{D}_{\text{inf}} \text{colim}_k P_k \right).$$

Proof.

- (1) **presentability statement.** The presentability statement is the last clause of 2.14.2.
- (2) **colimit formula.** The colimit formula is the standard formula for colimits in the target of a reflector.

□

Theorem 2.14.3 (Upper BNV-type triangle). *For every $E \in \mathcal{D}_{\text{inf}}$, there is a canonical exact triangle*

$$\boxed{i_*^{\text{st}} \Omega i^{*,\text{st}} \mathcal{Z}(E) \longrightarrow \mathcal{A}(E) \longrightarrow \mathcal{Z}(E).}$$

Proof.

- (1) **Identify the resulting comparison.** This is 2.14.2(iii).

□

The stable sectors and characteristic morphism are now defined. The remaining task is reconstruction: the fracture square must recover an arbitrary stable infinitesimal coefficient from its underlying value and pure cycle data.

2.14.4 Reconstruction and characteristic classification

Theorem 2.14.4 (Stable BNV-type reconstruction square). *For every $E \in \mathcal{D}_{\text{inf}}$, the canonical square*

$$\begin{array}{ccc} E & \longrightarrow & \mathcal{Z}(E) \\ \eta_E \downarrow & & \downarrow \eta_{\mathcal{Z}(E)} \\ i_*^{\text{st}} i^{*,\text{st}} E & \xrightarrow{i_*^{\text{st}} \phi_E} & i_*^{\text{st}} i^{*,\text{st}} \mathcal{Z}(E) \end{array}$$

is a pullback. Equivalently,

$$\boxed{E \simeq \mathcal{Z}(E) \times_{i_*^{\text{st}} i^{*,\text{st}} \mathcal{Z}(E)} i_*^{\text{st}} i^{*,\text{st}} E.}$$

Proof.

- (1) **Apply the cited result(iv).** Apply 2.14.2(iv).

□

Corollary 2.14.1 (BNV-type differential-cohomological pullback of mapping spectra). *For $K, E \in \mathcal{D}_{\text{inf}}$, mapping out of K gives a canonical homotopy pullback*

$$\begin{aligned} \text{map}_{\mathcal{D}_{\text{inf}}}(K, E) &\simeq \text{map}_{\mathcal{D}_{\text{inf}}}(K, \mathcal{Z}(E)) \\ &\quad \times_{\text{map}_{\mathcal{D}_{\text{inf}}}(K, i_*^{\text{st}} i^{*, \text{st}} \mathcal{Z}(E))} \text{map}_{\mathcal{D}_{\text{inf}}}(K, i_*^{\text{st}} i^{*, \text{st}} E). \end{aligned}$$

Thus the graded groups

$$\widehat{E}^n(K) := \pi_{-n} \text{map}_{\mathcal{D}_{\text{inf}}}(K, E)$$

carry the corresponding BNV-type differential-cohomological fracture. No identification with a geometric differential-cohomology theory is asserted without further cycle or curvature data.

Proof.

- (1) **Compute the mapping object.** Mapping spectra preserve limits in the second variable.
- (2) **Apply them to the cited result.** Apply them to 2.14.4.

□

Definition 2.14.5 (Characteristic pullback category). Let $\mathcal{T}_{\text{inf}}$ be the pullback ∞ -category whose objects are triples

$$(U, P, \phi), \quad U \in \mathcal{D}_{\text{Sp}}, \quad P \in \mathcal{D}_{\text{inf}}^{\text{BNV}}, \quad \phi : U \longrightarrow i_*^{\text{st}} P.$$

Equivalently it is the pullback

$$\begin{array}{ccc} \mathcal{T}_{\text{inf}} & \longrightarrow & \text{Fun}(\Delta^1, \mathcal{D}_{\text{Sp}}) \\ \downarrow & & \downarrow (\text{ev}_0, \text{ev}_1) \\ \mathcal{D}_{\text{Sp}} \times \mathcal{D}_{\text{inf}}^{\text{BNV}} & \xrightarrow{(\text{id}, i_*^{\text{st}})} & \mathcal{D}_{\text{Sp}} \times \mathcal{D}_{\text{Sp}}. \end{array}$$

All functors in this pullback are exact, so $\mathcal{T}_{\text{inf}}$ carries its canonical stable ∞ -category structure.

Construction 2.14.1 (Differential function object of characteristic data). For $(U, P, \phi) \in \mathcal{T}_{\text{inf}}$, define

$$\boxed{\text{Diff}(U, P, \phi) := P \times_{\mathfrak{S}^{\text{st}} P} i_*^{\text{st}} U,}$$

where $P \rightarrow \mathfrak{S}^{\text{st}} P$ is the stable infinitesimal-shape unit and

$$i_*^{\text{st}} U \xrightarrow{i_*^{\text{st}}(\phi)} i_*^{\text{st}} i^{*, \text{st}} P = \mathfrak{S}^{\text{st}} P$$

is induced by ϕ .

Theorem 2.14.5 (BNV-type characteristic classification). *There is a canonical equivalence of stable ∞ -categories*

$$\boxed{\mathcal{D}_{\text{inf}} \simeq \mathcal{T}_{\text{inf}}.}$$

It sends E to

$$(i_*^{\text{st}} E, \mathcal{Z}(E), \phi_E),$$

and its inverse is Diff . In particular,

$$i_*^{\text{st}} \text{Diff}(U, P, \phi) \simeq U, \quad \mathcal{Z} \text{Diff}(U, P, \phi) \simeq P,$$

and the recovered characteristic morphism is ϕ .

Proof.

- (1) **Identify the resulting comparison.** This is 2.14.2(v) for

$$i^{*,\text{st}} \dashv i_*^{\text{st}} \dashv i^{!,\text{st}}.$$

□

Proposition 2.14.4 (Colimits of characteristic triples). *Let $k \mapsto (U_k, P_k, \phi_k)$ be a small diagram in $\mathcal{T}_{\text{inf}}$. Put*

$$U := \text{colim}_k^{\mathcal{D}_{\text{Sp}}} U_k, \quad P := \text{colim}_k^{\mathcal{D}_{\text{inf}}^{\text{BNV}}} P_k.$$

The cocone maps induce

$$\lambda_P : \text{colim}_k i^{*,\text{st}} P_k \longrightarrow i^{*,\text{st}} P,$$

and the ϕ_k induce

$$\bar{\phi} : U \longrightarrow \text{colim}_k i^{*,\text{st}} P_k.$$

Then

$$\text{colim}_k^{\mathcal{T}_{\text{inf}}} (U_k, P_k, \phi_k) \simeq (U, P, \lambda_P \bar{\phi}).$$

Proof.

- (1) **Transport the diagram through the cited result.** Transport the diagram through 2.14.5.
- (2) **Use the adjunction.** The functor $i^{*,\text{st}}$ preserves colimits because it is a left adjoint, and $\mathcal{Z}$ preserves colimits into the pure subcategory because it is the reflector.
- (3) **Verify naturality.** Naturality of the cycle quotient identifies the resulting characteristic map with $\lambda_P \bar{\phi}$.

□

Remark 2.14.1 (Comparison with BNV and differential function spectra). The fracture square and characteristic classification above have the same formal exact-adjunction shape as the characteristic pullback decomposition of [12, Proposition 3.5], and as the homotopy-pullback pattern of differential function spectra in [24]. The comparison is made only after the fracture theorem has been proved internally. Consequently the name **stable BNV reconstruction** below is recognition terminology for this constructed pattern; it does not supply an additional cycle, form, curvature, period, or transfer datum.

Corollary 2.14.2 (Stable BNV reconstruction is intrinsic). *The stable BNV-type sectors, characteristic morphism, reconstruction square, and characteristic classification are functorial outputs of the stabilized endpoint adjunction string. No independent BNV package or differential function datum is part of the input of a stable infinitesimal coefficient.*

Proof.

- (1) **They are the specialization of the cited.** They are the specialization of 2.14.2 to the stabilized endpoint triple, followed only by the comparison terminology of the preceding remark.

□

The stable endpoint semantics is therefore determined by its characteristic data. We close the formal development by collecting the chapter's export interface. Before listing the exported structures, we fix the bookkeeping principle that distinguishes constructed structure from proof presentations.

2.15 Exported interface of the differentiated spectral doctrine

The first part of the interface distinguishes intrinsic constructed structure from auxiliary proof presentations. Constructions supplied functorially by universal properties, descent, adjunctions, and exact calculus are retained as structure; auxiliary presentations chosen only to prove their properties remain proof witnesses.

Theorem 2.15.1 (No differential or cohesive axiom disguised as presentation data). *None of the following is retained as additional input structure on a spectral context, an underlying infinitesimal substitution, or an infinitesimal sheaf:*

- (i) a chosen affine atlas, affine refinement, or independently retained descent datum used to construct $L_{X/S}$ or X_η ;
- (ii) a coefficient or derivation presenting an already specified underlying square-zero arrow;
- (iii) a finite square-zero factorization, nilpotent filtration, or numerical infinitesimal order attached to a finite infinitesimal substitution;
- (iv) a tuple of local equations presenting a quasi-smooth closed context;
- (v) a presentation of the cofree cocommutative coalgebra $C_X(M)$ by symmetric powers, homotopy fixed points, or any other chosen model;
- (vi) an independently specified monoidal exponential map, Seely comparison, primitive-coalgebra coherence, infinitesimal augmentation, coderelection, deriving transformation, or differential operator in a stable coefficient fibre, or a separate ordinary cofree-comonoid structure after passage to the homotopy category;
- (vii) an independently specified compatibility between substitution and the differential exponential: the actual comparison $\beta_{f,M}$ and its comonad, monoidal, coalgebra, coderelection, and deriving compatibilities are constructed from pullback and cofreeness;
- (viii) an independently specified cohesion structure, reduction operation, reduction modality, infinitesimal-shape modality, infinitesimal-flat modality, Kock–Lawvere principle, microlinearity datum, or synthetic infinitesimal-object package;
- (ix) an independently specified BNV underlying sector, secondary sector, deformation sector, cycle sector, characteristic morphism, differential-function reconstruction datum, curvature datum, period map, differential-form model, or transfer structure on a stable infinitesimal coefficient;
- (x) a Schlessinger-style lifting witness attached to formal smoothness or formal étaleness;
- (xi) a chosen representative for a relative infinitesimal-site object beyond the actual diagram $U \hookrightarrow T$ over $X \rightarrow S$.

The structures replacing these presentations are the cotangent representing property, Zariski descent, the cofree coalgebra adjunction and its canonical monoidal maps, the primitive coalgebra construction, the oplax substitution comparison, the endpoint adjunctions and their stabilizations, exact fibre and cofiber calculus, and their canonical composites.

Proof.

- (1) **Each item.** Each item is a direct consequence of the construction that introduced it.
- (2) **Apply Zariski descent.** Global cotangent objects and structured thickenings are determined by affine restriction and Zariski descent (2.5.1, 2.7.2); underlying square-zero and finite infinitesimal arrows are defined by essential image and finite closure rather than by retained presentations (2.8.1, 2.8.3); and local equations are used only in the recognition theorem 2.13.2.
- (3) **Verify functoriality.** In the linear branch, $!_X$, its monoidal and Seely maps, the primitive coalgebra, infinitesimal augmentation, codereliction, deriving transformation, and substitution comparison are all constructed from the cofree adjunction, product preservation, opposite split square-zero formation, and their functoriality (2.6.1– 2.6.10).
- (4) **In the geometric branch, the cohesive modalities.** In the geometric branch, the cohesive modalities are composites of the endpoint quadruple (2.10.2 and 2.10.1), while formal étalification and infinitesimal disks are obtained from the left-exact reflector $\mathfrak{S}$.
- (5) **Use the adjunction.** Finally, the stable sectors, characteristic morphism, reconstruction square, and characteristic classification are obtained by specializing the exact-adjoint fracture theorem 2.14.2 to the stabilized endpoint triple.
- (6) **listed presentations therefore remain either theorem-domain properties.** The listed presentations therefore remain either theorem-domain properties or temporary proof witnesses, exactly as stated.

□

Remark 2.15.1 (Genuine structured inputs). The audit preserves the structure that genuinely belongs to an object. A structured infinitesimal extension by M genuinely contains its derivation judgment $L_{X/S} \rightarrow M[1]$; a relative context genuinely contains the map $X \rightarrow S$; an object of the relative infinitesimal site genuinely contains the maps $U \hookrightarrow X$, $U \hookrightarrow T$, and $T \rightarrow S$. Presentations of already specified intrinsic objects and compatibility witnesses supplied functorially remain proof-level data.

With this bookkeeping convention fixed, the second part of the interface records the differential structures that later chapters may use. The theorem is deliberately a collecting statement; the conceptual retrospective is reserved for the chapter summary.

Theorem 2.15.2 (Differentiated spectral doctrine interface). *The constructions of this chapter canonically provide the following.*

- (i) **Affine tangent linearization.** *For every commutative ring spectrum B , the tangent stabilization of the ambient category of commutative spectral algebras at B is*

$$\mathrm{Stab}(\mathrm{CAlg}(\mathrm{Sp})/B) \simeq \mathrm{Mod}_B,$$

together with the split square-zero zeroth-space construction. Globally, $\mathrm{QCoh}(X)$ is the descended stable coefficient fibre; no additional global stabilization equivalence is asserted.

- (ii) **Cotangent-classified differentiation.** *Relative affine derivations are represented by $L_{B/A}$. Globally $(X \rightarrow S) \mapsto L_{X/S}$ is a connective Cartesian section under represented substitution and satisfies the global cotangent chain rule. On affine relative contexts, the cotangent classifier additionally satisfies the Hurewicz–conormal comparison, cotangent rigidity, and the stated criterion for local finite presentation.*

(iii) **Differential-linear stable coefficient structure.** Every stable coefficient fibre $\mathrm{QCoh}(X)$ carries the cofree cocommutative-coalgebra exponential $!_X$, its canonically constructed monoidal coalgebra-modality maps and Seelye equivalences, together with the canonical identification showing that these induce the same ordinary storage-modality and additive-bialgebra structure, primitive infinitesimal coalgebras, coherent infinitesimal augmentation, and canonical codereliction and deriving maps. Their images in the homotopy category give the ordinary differential-storage codereliction and deriving transformation. The actual comparison $\beta_{f,M} : f^*(!_X M) \rightarrow !_Y(f^* M)$ makes these structures oplaxly compatible with substitution, including the comonad, monoidal, coalgebra, codereliction, and deriving maps. On finite projective linear models the resulting convolution derivation is canonically compatible along j_M with the spectral universal derivation.

(iv) **Structured square-zero context extension.** Every connective coefficient and degree-one derivation

$$\eta : L_{X/S} \longrightarrow M[1]$$

constructs a spectral thickening $X \hookrightarrow X_\eta$, stable under represented relative-base change. Algebraic pullback, relative-base change, and geometric Cartesian base change are separately constructed, and geometric Cartesian base change preserves structured square-zero extensions.

(v) **Presentation-free finite infinitesimal contexts.** Underlying square-zero substitutions and their finite-composition closure are affine properties of arrows, stable under base change, nilpotent on degree-zero coordinate rings, and invariant on underlying Zariski support and small Zariski sites. A structured representative may be chosen only as a proof witness and can be refined relative to a displayed base when a lifting argument requires it.

(vi) **Infinitesimal Zariski semantics.** The category $\mathrm{Aff}_{\mathrm{Sp}}^{\mathrm{inf}}$, its codomain-pullback Zariski topology, its relative affine-basis sites, and its subcanonical sheaf category $\mathcal{H}_{\mathrm{inf}}$, together with the endpoint adjunctions

$$i_! \dashv i^* \dashv i_* \dashv i^!$$

(vii) **Differential cohesion.** The endpoint quadruple is recognized as differential cohesion and induces the standard modal triple

$$\mathbb{R} \dashv \mathbb{S} \dashv \mathbb{L},$$

where $\mathbb{R}$ is reduction, $\mathbb{S}$ is infinitesimal shape, and $\mathbb{L}$ is infinitesimal flat. The modality $\mathbb{S}$ is a left-exact reflector.

(viii) **Formal étalification and infinitesimal disks.** For $g : E \rightarrow B$ in $\mathcal{H}_{\mathrm{inf}}$, relative infinitesimal shape is

$$\mathbb{S}_B E = B \times_{\mathbb{S} B} \mathbb{S} E.$$

The unit $E \rightarrow \mathbb{S}_B E$ is the formally étale reflection in the slice. For $f : X \rightarrow S$, its kernel-pair Čech nerve gives the relative infinitesimal disk groupoid $T_{\mathrm{inf}}^{[\bullet]}(X/S)$.

(ix) **Cotangent lifting and formal geometry.** On an affine structured square-zero stage, the fibre of the modal lifting comparison Θ_f is the nullhomotopy space of the canonical cotangent boundary. For an arbitrary represented structured square-zero stage, the represented comparison Θ_f^{rep} has the same cotangent description. One-stage detection extends the affine modal criterion to finite infinitesimal contexts. For represented spectral contexts,

$$f \text{ formally étale} \iff L_{X/S} \simeq 0,$$

while finite local freeness of $L_{X/S}$ implies formal smoothness. The first structured neighbourhood of the diagonal maps canonically into $T_{\text{inf}}(X/S)$, and the difference of its two retractions is the universal relative derivation.

- (x) **First-order deformation boundary calculus.** First-order deformation spaces are homotopy fibres of restriction on derivations; cotangent transitivity constructs their obstruction, deformation torsor, and automorphism space.
- (xi) **Equational context formation.** Derived zero loci of finite locally free linear equations are quasi-smooth closed contexts, and every quasi-smooth closed immersion is locally of this form. The conormal classifier is the restricted coefficient of the local equation context.
- (xii) **Stable BNV differential-cohomological reconstruction.** Stabilization of the endpoint semantics gives stable presentable categories

$$\mathcal{D}_{\text{Sp}} = \text{Sp}((\mathcal{H}_{\text{Sp}})_*), \quad \mathcal{D}_{\text{inf}} = \text{Sp}((\mathcal{H}_{\text{inf}})_*),$$

and an exact adjoint triple

$$i^{*,\text{st}} \dashv i_*^{\text{st}} \dashv i^{!,\text{st}}.$$

The abstract exact-adjoint fracture theorem 2.14.2 applies to this fully faithful middle functor. Its stabilized modalities

$$\mathfrak{R}^{\text{st}}, \quad \mathfrak{S}^{\text{st}}, \quad \&^{\text{st}}$$

construct the deformation and cycle sectors

$$\mathcal{A}(E) = \text{fib}(E \rightarrow \mathfrak{S}^{\text{st}} E), \quad \mathcal{Z}(E) = \text{cofib}(\&^{\text{st}} E \rightarrow E),$$

while the already defined endpoint values

$$i^{*,\text{st}} E, \quad i^{!,\text{st}} E, \quad i^{*,\text{st}} \mathcal{Z}(E)$$

serve as the underlying, secondary, and underlying-cycle sectors without further renaming. The characteristic morphism is

$$\phi_E : i^{*,\text{st}} E \longrightarrow i^{*,\text{st}} \mathcal{Z}(E),$$

and the canonical reconstruction is

$$E \simeq \mathcal{Z}(E) \times_{i_*^{\text{st}} i^{*,\text{st}} \mathcal{Z}(E)} i_*^{\text{st}} i^{*,\text{st}} E.$$

Equivalently, stable infinitesimal coefficients are classified by characteristic triples

$$(U, P, \phi), \quad \phi : U \longrightarrow i^{*,\text{st}} P,$$

with P BNV-pure, meaning

$$i^{!,\text{st}} P \simeq 0,$$

and mapping spectra satisfy the corresponding BNV-type differential-cohomological pullback. No identification with a geometric differential-cohomology theory and no separate geometric interpretation of the pure cycle coefficient is included in this abstract reconstruction.

- (xiii) **Logical audit.** The constructions above distinguish genuine structured inputs from functorial outputs, theorem-domain properties, and temporary proof witnesses as in 2.15.1.

No additional differential, infinitesimal, cohesive, differential-cohomology, presentation, admissibility, or compatibility axiom is introduced by this collecting theorem.

Proof.

- (1) **Identify the resulting comparison.** This theorem only collects results already proved.
- (2) **Use the adjunction.** Items (i)–(iii) are the tangent, cotangent, and differential-linear constructions; items (iv)–(vi) are the square-zero, finite-infinitesimal, and infinitesimal-site constructions; items (vii)–(ix) are the endpoint modalities and cotangent lifting calculus; items (x) and (xi) are the deformation and equation calculi; item (xii) is the stabilized exact-adjoint fracture construction together with its subsequent BNV-type comparison; and item (xiii) is the bookkeeping audit immediately above.
- (3) **No new construction.** No new construction is made here.

□

2.16 Chapter summary

The chapter has exposed two complementary differential mechanisms already carried by the spectral doctrine. Affinely, the stable coefficient fibre is the tangent stabilization of commutative spectral algebra, and relative derivations in that tangent semantics are represented by the cotangent classifier with its substitution law and chain rule. Independently, the same stable coefficient fibres carry a cofree cocommutative-coalgebra differential-exponential calculus. On free finite-projective models the two descriptions meet: the convolution derivative determined by the cofree calculus is compatible with the universal spectral Kähler derivation.

The cotangent classifier then drives the nonlinear geometric branch. Degree-one cotangent classes produce structured square-zero extensions, whose presentation-free finite composites form the infinitesimal test geometry. Their Zariski topos yields the endpoint adjunctions and hence differential cohesion. The resulting formal lifting comparison is computed by the same cotangent classifier, so infinitesimal lifting, first-order deformation, and local equation formation appear as different instances of one cotangent boundary calculus.

Finally, stabilization converts the already constructed endpoint semantics into an exact adjoint triple, and the resulting fracture theorem reconstructs stable infinitesimal coefficients from characteristic data. Its BNV-type differential-cohomological form is recognized after this reconstruction has been obtained internally. The distinction inherited from Chapter 1 remains active throughout: geometry is connective, coefficients are fully stable, and every passage between the nonlinear, infinitesimal, and stable layers is supplied by a construction established in this chapter.

Chapter 3

Formal Reconstruction of the Spectral Doctrine

3.1 Orientation: from differentiation to reconstruction

Chapter 1 constructed the spectral doctrine by combining connective spectral contexts, stable co-efficient fibres, substitution, Zariski descent, and comprehension. Chapter 2 then differentiated that doctrine: cotangent complexes represent first-order variation, structured square-zero extensions realize it geometrically, finite composites generate an infinitesimal test geometry, and endpoint adjunctions produce differential cohesion. In parallel, each stable coefficient fibre carries the cofree cocommutative-coalgebra exponential and its differential-linear structure.

The unresolved problem is converse. The preceding chapter extracts first-order and infinitesimal data from a nonlinear context, but does not say when those data determine the context from which they arose. This chapter asks when differential data cease to be mere invariants and become reconstruction data.

The principal fixed-centre answer is comonadic. For a connective centre B over a connective base R , a complete almost finitely presented augmentation $A \rightarrow B$ is recovered from the anchored shifted-cotangent map

$$L_{B/R}[-1] \longrightarrow L_{B/A}[-1]$$

together with the coalgebra coherence generated by structured square-zero formation. No extra reconstruction structure is imposed: the relevant comonad is produced by the cotangent–square-zero adjunction constructed below.

The following adjunction records that destination. Its source and target notation will be defined only when those categories become active:

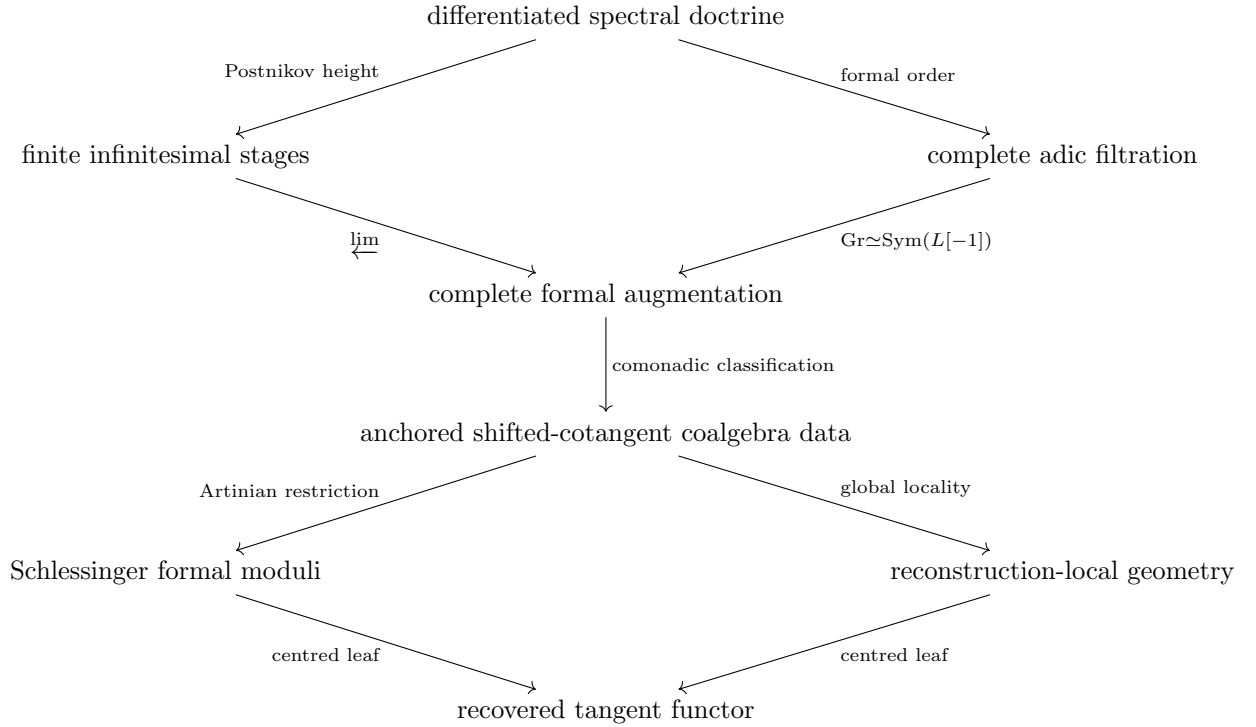
$$\mathrm{CAlg}(\mathrm{Sp})_{R//B}^{\mathrm{cn}, \mathrm{afp}, \wedge} \begin{array}{c} \xrightarrow{\mathrm{cot}_{B/R}} \\ \xleftarrow{\mathrm{sqz}_{B/R}} \end{array} \mathrm{AnchCot}_{B/R, \geq 0}^{\mathrm{aperf}}$$

For now the display is only a marker. The unrestricted adjunction will be constructed first and then restricted to the complete theorem domain.

Two approximation coordinates lead to this theorem. **Postnikov height** resolves a connective spectral algebra into canonical square-zero stages and recovers it by inverse limit. **Formal order at a centre** equips an augmentation $A \rightarrow B$ with a canonical decreasing filtration whose completion is

functorial and whose associated graded is freely generated by the shifted relative cotangent complex. The first coordinate measures homotopical height; the second measures order of vanishing at the chosen centre.

The dependency diagram records the full route of the chapter. At this stage only its upper spine is active: the two approximation mechanisms converge on complete formal augmentations, where comonadic reconstruction is proved. The lower labels name later exports of that theorem and need not yet be retained as definitions.



The argument has three phases. First we construct and control the two approximation mechanisms, culminating in the complete fixed-centre domain and the comonadic cotangent–square-zero theorem. Next we extract bounded Artinian tests and then remove the centre by imposing Postnikov continuity and nilpotent excision globally. Finally we return to a centre, stabilize split square-zero lifting, and compare the recovered tangent functor with the two differential branches of Chapter 2.

That final comparison is deliberately postponed. The cofree differential-linear structure remains in the stable coefficient fibre, whereas the square-zero branch supplies the nonlinear infinitesimal geometry used for reconstruction. They will be compared only after both reconstruction mechanisms are available, and only on the represented finite-projective domain where the universal derivation appears on both sides.

The conventions of Chapters 1 and 2 remain in force. Almost finite presentation, coherence, nilpotence, completeness, Artinianity, and reconstruction locality are properties specifying theorem domains. Finite square-zero factorizations, coherent refinements, Postnikov presentations, affine charts, and centre presentations used inside proofs remain proof witnesses rather than retained structure. Canonical filtrations, completion maps, cotangent maps, adjunction units and counits, and localization transformations are the functorially constructed maps used below.

Coordinate algebras remain connective. Filtered modules, cotangent complexes, totalizations, mapping spectra, and stabilizations are formed in their corresponding stable ambient categories.

3.2 Postnikov height and finite infinitesimal reconstruction

The first reconstruction coordinate is homotopical height. The preceding chapter defined finite spectral infinitesimal substitutions as finite composites of square-zero substitutions. The multiplicative Postnikov tower supplies canonical examples: each transition is square-zero, every finite stage lies over the same discrete centre $H\pi_0(B)$, and the original connective algebra is recovered from the entire tower.

For $n \geq 1$, the common centre is visible in the comparison square

$$\begin{array}{ccc} \tau_{\leq n}^{\text{CAlg}} B & \longrightarrow & \tau_{\leq n-1}^{\text{CAlg}} B \\ \downarrow & & \downarrow \\ H\pi_0(B) & \xlongequal{\quad} & H\pi_0(B). \end{array}$$

The upper arrow adds one homotopy layer, while the vertical maps keep every finite stage over the same discrete centre. The point is stronger than analogy with the infinitesimal site: the next theorem identifies each Postnikov transition with one of the structured square-zero extensions already constructed.

Theorem 3.2.1 (Postnikov stages are structured square-zero extensions). *Let B be a connective commutative ring spectrum. For every integer $n \geq 1$, the truncation morphism*

$$\tau_{\leq n}^{\text{CAlg}} B \longrightarrow \tau_{\leq n-1}^{\text{CAlg}} B$$

is canonically a structured square-zero extension with fibre

$$H\pi_n(B)[n].$$

The coefficient is regarded as a module over $\tau_{\leq n-1}^{\text{CAlg}} B$ through $\tau_{\leq n-1}^{\text{CAlg}} B \rightarrow H\pi_0(B)$, and the extension is classified by a canonical multiplicative Postnikov k -invariant

$$L_{\tau_{\leq n-1}^{\text{CAlg}} B} \longrightarrow H\pi_n(B)[n+1].$$

Proof.

- (1) **Identify the truncation range.** The truncation map has the displayed fibre and lies in the bounded small extension range of the connective square-zero recognition theorem for commutative ring spectra.
- (2) **Apply square-zero recognition.** That theorem identifies the truncation arrow with the structured square-zero extension classified by its multiplicative Postnikov k -invariant; see [34, Corollary 7.4.1.28].
- (3) **Record canonicity.** The classifying k -invariant is canonical because the Postnikov truncation functors and their transition maps are functorial.

□

Corollary 3.2.1 (Relative Postnikov extensions). *Let $A \rightarrow B$ be a morphism of connective commutative ring spectra. For $n \geq 1$, the map*

$$\tau_{\leq n}^{\text{CAlg}} B \longrightarrow \tau_{\leq n-1}^{\text{CAlg}} B$$

is canonically a structured square-zero extension in commutative A -algebras by $H\pi_n(B)[n]$, classified by

$$L_{\tau_{\leq n-1}^{\text{CAlg}} B/A} \longrightarrow H\pi_n(B)[n+1].$$

Proof.

- (1) **Lift the structural map.** The structural map

$$A \longrightarrow \tau_{\leq n}^{\text{CAlg}} B$$

lifts the map $A \rightarrow \tau_{\leq n-1}^{\text{CAlg}} B$ through the absolute Postnikov extension of 3.2.1.

- (2) **Obtain the absolute nullhomotopy.** The absolute classifying derivation therefore acquires a canonical nullhomotopy after restriction along

$$L_A \otimes_A \tau_{\leq n-1}^{\text{CAlg}} B \longrightarrow L_{\tau_{\leq n-1}^{\text{CAlg}} B}.$$

- (3) **Apply cotangent transitivity.** Cotangent transitivity gives the cofiber sequence

$$L_A \otimes_A \tau_{\leq n-1}^{\text{CAlg}} B \longrightarrow L_{\tau_{\leq n-1}^{\text{CAlg}} B} \longrightarrow L_{\tau_{\leq n-1}^{\text{CAlg}} B/A}.$$

- (4) **Factor through the relative cotangent complex.** The compatible absolute derivation and nullhomotopy determine a factorization through $L_{\tau_{\leq n-1}^{\text{CAlg}} B/A}$, through a contractible space of choices,

$$L_{\tau_{\leq n-1}^{\text{CAlg}} B/A} \longrightarrow H\pi_n(B)[n+1].$$

- (5) **Classify the relative extension.** Structured square-zero classification in commutative A -algebras identifies the Postnikov transition with the extension classified by this relative map.

□

Corollary 3.2.2 (Infinitesimal Postnikov reconstruction). *Every connective commutative ring spectrum is reconstructed by the canonical inverse tower*

$$B \simeq \varprojlim_n \tau_{\leq n}^{\text{CAlg}} B, \quad \tau_{\leq 0}^{\text{CAlg}} B \simeq H\pi_0(B),$$

whose successive transition maps are structured square-zero extensions. For each fixed n , the composite

$$\tau_{\leq n}^{\text{CAlg}} B \longrightarrow H\pi_0(B)$$

is a finite composite of structured square-zero extensions; equivalently, its corresponding affine morphism is a finite spectral infinitesimal substitution.

Proof.

- (1) **Recover the inverse limit.** Postnikov completeness of connective spectra and creation of limits by the forgetful functor from commutative ring spectra give the inverse-limit equivalence.
- (2) **Identify the finite composite.** The second assertion follows from 3.2.1 by finite composition. $\square$

Remark 3.2.1 (Finite stages versus complete reconstruction). A finite infinitesimal substitution records only membership in the finite-composition closure of square-zero infinitesimal substitutions. On coordinate algebras, Postnikov reconstruction is an inverse limit of the finite structured square-zero stages. In this case the Postnikov functors supply the canonical finite-stage factorization.

Remark 3.2.2 (Postnikov reconstruction in the differentiated doctrine). For every n , the corresponding affine morphism of the finite composite

$$\tau_{\leq n}^{\text{CAlg}} B \longrightarrow H\pi_0(B)$$

is an object of the finite infinitesimal geometry constructed in Chapter 2, while

$$B \simeq \varprojlim_n \tau_{\leq n}^{\text{CAlg}} B$$

is obtained by passing from finite stages to their inverse limit. For each n , the finite stage fits into the compatible triangle

$$\begin{array}{ccc} \tau_{\leq n}^{\text{CAlg}} B & \longrightarrow & \tau_{\leq n-1}^{\text{CAlg}} B \\ & \searrow & \downarrow \\ & & H\pi_0(B). \end{array}$$

The infinitesimal site supplies the finite approximants and Postnikov completeness supplies the reconstruction.

Thus Postnikov height gives the first reconstruction prototype: finite infinitesimal stages are canonical approximants, and the inverse limit over all heights recovers the connective algebra. This controls homotopical height. Formal order at a fixed augmentation is a different coordinate, constructed next.

3.3 Formal order and the canonical adic filtration

Postnikov reconstruction varies homotopical height over a fixed discrete centre. Formal-order reconstruction instead fixes an augmentation

$$A \longrightarrow B$$

and resolves it by order of vanishing along B . The filtration is canonical: it is created by the left adjoint to filtration-zero augmentation and then completed by a symmetric-monoidal reflector.

The shifted cotangent complex supplies the linear generator of this filtration. The following dependency square states the calculation the section will establish:

$$\begin{array}{ccc} \text{augmentation } (A \rightarrow B) & \xrightarrow{\text{adic and completion}} & \widehat{\text{adic}}(A \rightarrow B) \\ \text{shifted cotangent} \downarrow & & \downarrow \text{Gr} \\ L_{B/A}[-1] & \xrightarrow{\text{free graded commutative algebra}} & \text{Sym}_B^{\text{gr}}(L_{B/A}[-1]\langle 1 \rangle). \end{array}$$

The square is organizational rather than internal to a single category. Its content is that the formal neighbourhood is functorially filtered and that its graded layers are freely generated by the same cotangent object that represented first-order variation in Chapter 2.

3.3.1 Filtered spectral algebra and the adic left adjoint

Definition 3.3.1 (Filtered and graded spectral objects). Let $\mathbb{N}_{\leq}$ denote the nonnegative integers as an ordered symmetric monoidal category under addition, and let $\mathbb{N}_{\text{disc}}$ denote the same commutative monoid regarded as a discrete symmetric monoidal category. Define

$$\text{Fil}(\text{Sp}) := \text{Fun}(\mathbb{N}_{\leq}^{\text{op}}, \text{Sp}), \quad \text{Gr}(\text{Sp}) := \text{Fun}(\mathbb{N}_{\text{disc}}, \text{Sp}),$$

with Day convolution. A filtered object is a tower

$$\cdots \longrightarrow F^2 M \longrightarrow F^1 M \longrightarrow F^0 M.$$

Its associated graded object is

$$\text{Gr}^n M := \text{cofib}(F^{n+1} M \rightarrow F^n M), \quad n \geq 0,$$

and these terms assemble into a symmetric monoidal functor

$$\text{Gr} : \text{Fil}(\text{Sp}) \longrightarrow \text{Gr}(\text{Sp}).$$

For a nonnegatively graded object N and $w \geq 0$, write

$$N\langle w \rangle$$

for the upward weight shift, with

$$(N\langle w \rangle)_k := N_{k-w},$$

where negative weights are zero. In particular, if an ungraded spectrum or module M is placed in weight zero, then $M\langle w \rangle$ is concentrated in weight w . For $r < n$, write

$$F^r M / F^n M := \text{cofib}(F^n M \rightarrow F^r M),$$

and put

$$F^\infty M := \varprojlim_n F^n M.$$

The filtration is **complete** if

$$F^\infty M \simeq 0.$$

Equivalently, for every $r \geq 0$, the canonical morphism

$$F^r M \longrightarrow \varprojlim_{n>r} F^r M / F^n M$$

is an equivalence. The indexing by $\mathbb{N}$ concerns filtration degree only. Connectivity hypotheses will be imposed explicitly when needed; otherwise the filtered terms range over arbitrary spectra.

Construction 3.3.1 (Completion of a filtered spectrum). Let

$$\text{const} : \text{Sp} \longrightarrow \text{Fil}(\text{Sp})$$

denote the constant-filtration functor, so that every transition map of $\text{const}(K)$ is the identity of K . The filtration-zero embedding of 3.3.2 is a different functor: it places an unfiltered object in filtration degree zero and vanishes in positive filtration degrees.

For a filtered spectrum $F^\bullet M$, the limit maps $F^\infty M \rightarrow F^r M$ assemble to a canonical morphism

$$\text{const}(F^\infty M) \longrightarrow F^\bullet M.$$

Define its **completion** by

$$\widehat{F^\bullet M} := \text{cofib}(\text{const}(F^\infty M) \longrightarrow F^\bullet M).$$

For every filtration degree r ,

$$\widehat{F^r M} \simeq \text{cofib}(F^\infty M \rightarrow F^r M) \simeq \varprojlim_{n>r} F^r M / F^n M.$$

The multiplicative and module completion reflectors induced by this localization are described in 3.3.1. Through 3.3.1, this is the complete filtered $\mathbb{E}_\infty$ -algebra construction used in [2, Section 2].

Proposition 3.3.1 (Universal property and associated graded of completion). *Completion is left adjoint to the inclusion of complete filtered spectra. It is idempotent and there are canonical equivalences*

$$\text{Gr } \widehat{F^\bullet M} \simeq \text{Gr } F^\bullet M.$$

The completion localization is compatible with Day convolution. Hence the subcategory of complete filtered spectra inherits the canonical localized symmetric monoidal structure for which completion is symmetric monoidal, and completion induces the corresponding reflectors on filtered commutative spectral algebras and on their module categories. On underlying filtered spectra these reflectors are computed by 3.3.1, and associated graded is unchanged.

Proof.

- (1) **Reflector property.** Let G be complete. The adjunction between constant filtrations and inverse limits gives

$$\text{Map}_{\text{Fil}(\text{Sp})}(\text{const}(K), G) \simeq \text{Map}_{\text{Sp}}\left(K, \varprojlim_r G^r\right) \simeq *$$

for every spectrum K . Applying $\text{Map}(-, G)$ to the defining cofiber sequence of 3.3.1 yields

$$\text{Map}(\widehat{F}, G) \simeq \text{Map}(F, G),$$

so completion has the required reflector universal property.

- (2) **Prove completeness.** Taking inverse limits in the defining cofiber sequence gives

$$\varprojlim_r \widehat{F^r} \simeq \text{cofib}(F^\infty M \xrightarrow{\text{id}} F^\infty M) \simeq 0.$$

Thus $\widehat{F}$ is complete.

- (3) **Compute associated graded.** The constant filtration has zero associated graded, and exactness of Gr gives

$$\text{Gr } \widehat{F} \simeq \text{Gr } F.$$

- (4) **Local equivalences.** A morphism $u : F \rightarrow G$ is inverted by completion precisely when $\mathrm{Gr}(u)$ is an equivalence. For the converse direction, put

$$H := \mathrm{cofib}(u).$$

If $\mathrm{Gr} H \simeq 0$, then every transition $H^{n+1} \rightarrow H^n$ is an equivalence. Hence H is a constant filtration and $\widehat{H} \simeq 0$. Exactness of completion then makes $\widehat{u}$ an equivalence.

- (5) **Monoidal localization.** Associated graded is symmetric monoidal. Tensor products therefore preserve completion equivalences. The symmetric-monoidal localization theorem [34, Definition 2.2.1.6 and Proposition 2.2.1.9] equips the complete filtered spectra with the localized symmetric monoidal structure and induces the corresponding localizations on commutative algebra objects and module categories. Their underlying filtered spectra are computed by 3.3.1.

□

Convention 3.3.1 (Comparison with integer-indexed filtrations). When comparing with literature using decreasing $\mathbb{Z}$ -indexed filtrations, an $\mathbb{N}$ -indexed filtration in this chapter is extended canonically by

$$F^r M := F^0 M \quad (r \leq 0).$$

This identifies our filtered objects with the full subcategory of decreasing integer-indexed filtrations which are constant in nonpositive filtration degrees. Under this identification, associated graded in nonnegative weights and completion agree with the corresponding integer-indexed constructions. In particular, the comparison with Antieau's filtered and complete filtered $\mathbb{E}_\infty$ -algebras uses this canonical extension.

Definition 3.3.2 (Weight-one filtered module). For a spectrum or module M , let

$$\mathrm{wt}_1(M)$$

denote the two-step descending filtered object

$$\cdots \longrightarrow 0 \longrightarrow M \xrightarrow{\mathrm{id}} M,$$

with M in filtration degrees one and zero. Equivalently,

$$F^0 \mathrm{wt}_1(M) = M, \quad F^1 \mathrm{wt}_1(M) = M, \quad F^n \mathrm{wt}_1(M) = 0 \quad (n \geq 2).$$

Its associated graded is concentrated in weight one:

$$\mathrm{Gr} \mathrm{wt}_1(M) \simeq M\langle 1 \rangle.$$

Convention 3.3.2 (Unfiltered bases in filtered and graded algebra). Whenever an unfiltered commutative ring spectrum R is used as a base in $\mathrm{Fil}(\mathrm{Sp})$ or $\mathrm{Gr}(\mathrm{Sp})$, it is embedded in filtration degree zero in the filtered case and in graded weight zero in the graded case. Thus in the filtered case

$$F^0 R = R, \quad F^n R = 0 \quad (n \geq 1),$$

and in the graded case the only nonzero graded term is weight zero. This convention applies, in particular, to expressions such as $\mathrm{CAlg}(\mathrm{Fil}(\mathrm{Sp}))_{R/}$.

Proposition 3.3.2 (Pointwise Postnikov truncation in filtered and graded spectral algebra). *Let*

$$\mathcal{C} = \text{Fil}(\text{Sp}) \quad \text{or} \quad \mathcal{C} = \text{Gr}(\text{Sp})$$

with its pointwise Postnikov t -structure. Day convolution is compatible with this t -structure: if $M \in \mathcal{C}_{\geq a}$ and $N \in \mathcal{C}_{\geq b}$, then

$$M \otimes N \in \mathcal{C}_{\geq a+b}.$$

Consequently the Postnikov truncation functors lift canonically to connective commutative algebra objects. For every pointwise connective commutative algebra C and every pointwise connective n -truncated commutative algebra E , there is a natural equivalence

$$\text{Map}_{\text{CAlg}(\mathcal{C})}(C, E) \simeq \text{Map}_{\text{CAlg}(\mathcal{C})}(\tau_{\leq n}^{\text{CAlg}} C, E).$$

Moreover every pointwise connective commutative algebra is recovered from its operadic Postnikov tower:

$$C \simeq \varprojlim_n \tau_{\leq n}^{\text{CAlg}} C.$$

Proof.

- (1) **Verify graded connectivity.** In the graded case the tensor-product connectivity estimate is termwise.
- (2) **Verify filtered connectivity.** In the filtered case each filtration term of the Day convolution is a colimit of tensor products of filtration terms, so the same estimate follows from the connective tensor-product theorem in spectra.
- (3) **Lift Postnikov truncations.** The general compatibility result for a monoidal t -structure therefore supplies Postnikov truncations on connective commutative algebra objects; see [34].
- (4) **Obtain the mapping equivalence.** Its universal property gives the displayed mapping equivalence.
- (5) **Recover the operadic tower.** Finally the forgetful functor from commutative algebra objects creates limits, while the pointwise Postnikov t -structure is left complete, so the operadic tower has limit C .

□

Fix a commutative ring spectrum R . For a filtered commutative R -algebra $F^\bullet C$, filtration degree zero and the weight-zero associated graded determine the augmentation

$$F^0 C \longrightarrow \text{Gr}^0 C.$$

Proposition 3.3.3 (Augmentation from filtered algebras). *The assignment*

$$\text{aug} : \text{CAlg}(\text{Fil}(\text{Sp}))_{R/} \longrightarrow \text{Fun}(\Delta^1, \text{CAlg}(\text{Sp})_{R/})$$

which sends a filtered algebra to $F^0 C \rightarrow \text{Gr}^0 C$ is accessible and preserves limits. The filtration-zero augmentation is represented on underlying spectra by the cofiber square

$$\begin{array}{ccc} F^1 C & \longrightarrow & F^0 C \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \text{Gr}^0 C. \end{array}$$

Hence it admits a left adjoint

$$\text{adic} \dashv \text{aug}.$$

Proof.

- (1) **Evaluate the filtration endpoints.** Evaluation in filtration degrees zero and one is accessible and preserves all limits.
- (2) **Identify the weight-zero graded term.** In the stable category of spectra,

$$\text{Gr}^0 C = \text{cofib}(F^1 C \rightarrow F^0 C)$$

is also a finite-limit construction.

- (3) **Preserve arbitrary limits.** Finite limits commute with arbitrary limits, and limits of commutative algebra objects are created in the underlying symmetric monoidal category. Thus both endpoints and their canonical comparison preserve limits.
- (4) **Apply the adjoint functor theorem.** The source and target are presentable, so the presentable adjoint functor theorem supplies the left adjoint.

□

Lemma 3.3.1 (The module-level augmentation filtration). *Let*

$$\text{aug}_{\text{mod}} : \text{Fil}(\text{Sp}) \longrightarrow \text{Fun}(\Delta^1, \text{Sp})$$

be the functor

$$F^\bullet M \longmapsto (F^0 M \longrightarrow \text{Gr}^0 M).$$

It admits a left adjoint. On an arrow $V \rightarrow V_0$ this left adjoint is the two-step filtered spectrum

$$\cdots \longrightarrow 0 \longrightarrow \text{fib}(V \rightarrow V_0) \longrightarrow V,$$

with V in filtration degree zero and the fibre in filtration degree one. In particular, its filtration-zero term is V and its weight-zero associated graded is V_0 .

Proof.

- (1) **Identify morphisms of fibre sequences.** For a filtered spectrum $F^\bullet N$, a morphism from the displayed two-step object to $F^\bullet N$ is the same as a morphism of the two fibre sequences

$$\begin{array}{ccccc} \text{fib}(V \rightarrow V_0) & \longrightarrow & V & \longrightarrow & V_0 \\ \downarrow & & \downarrow & & \downarrow \\ F^1 N & \longrightarrow & F^0 N & \longrightarrow & \text{Gr}^0 N. \end{array}$$

- (2) **Compute the mapping space.** In a stable ∞ -category, the space of such morphisms of fibre sequences is canonically the mapping space from the arrow $V \rightarrow V_0$ to $F^0 N \rightarrow \text{Gr}^0 N$.
- (3) **Conclude the adjunction.** This is the required adjunction.
- (4) **Identify the filtration terms.** The final assertion is the defining fibre sequence.

□

Proposition 3.3.4 (The adic construction recovers the input augmentation). *For every augmented arrow*

$$A \longrightarrow B$$

of commutative R -algebras, the filtered algebra $\mathrm{adic}(A \rightarrow B)$ has canonical identifications

$$F^0 \mathrm{adic}(A \rightarrow B) \simeq A, \quad \mathrm{Gr}^0 \mathrm{adic}(A \rightarrow B) \simeq B,$$

under which the canonical augmentation

$$F^0 \mathrm{adic}(A \rightarrow B) \longrightarrow \mathrm{Gr}^0 \mathrm{adic}(A \rightarrow B)$$

is the original morphism $A \rightarrow B$.

On underlying spectra, its first filtration quotient is displayed by

$$\begin{array}{ccc} F^1 \mathrm{adic}(A \rightarrow B) & \longrightarrow & A \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & B. \end{array}$$

Proof.

- (1) **Multiplicative left adjoint.** The algebraic left adjoint is the multiplicative extension of 3.3.1.
- (2) **Free augmented arrows.** Resolve the augmented arrow $A \rightarrow B$ by the simplicial resolution associated with free augmented commutative R -algebra arrows. On a free arrow

$$\mathrm{Sym}_R(V) \longrightarrow \mathrm{Sym}_R(V_0)$$

induced by a morphism of R -modules $V \rightarrow V_0$, the adjunction between free commutative algebras and underlying modules, together with 3.3.1, identifies its adic image with the free filtered commutative algebra generated by the two-step filtration

$$\mathrm{fib}(V \rightarrow V_0) \longrightarrow V.$$

Consequently its filtration-zero algebra is $\mathrm{Sym}_R(V)$, while its weight-zero associated graded algebra is $\mathrm{Sym}_R(V_0)$, and the comparison between them is the original free arrow.

- (3) **Passage to geometric realization.** The commutative-algebra monad preserves sifted colimits. Hence this free resolution is recovered by geometric realization, and evaluation at filtration zero, associated graded in weight zero, and the filtered free-algebra construction preserve the sifted colimits occurring in that realization. Passing to the realization therefore gives the two asserted identifications for an arbitrary augmented arrow. Their comparison is the adjunction unit and is identified with $A \rightarrow B$.

□

Definition 3.3.3 (Canonical adic filtration and completion). For an augmented arrow

$$A \longrightarrow B$$

of commutative R -algebras, define

$$\mathrm{adic}(A \rightarrow B)$$

to be its image under the left adjoint of 3.3.3. Its **completed canonical adic filtration** is the multiplicative completion

$$\widehat{\mathrm{adic}}(A \rightarrow B) := (\mathrm{adic}(\widehat{A \rightarrow B}))$$

of 3.3.1. On underlying filtered spectra, for every $r \geq 0$,

$$F^r \widehat{\mathrm{adic}}(A \rightarrow B) \simeq \varprojlim_{n > r} \mathrm{cofib}(F^n \mathrm{adic}(A \rightarrow B) \longrightarrow F^r \mathrm{adic}(A \rightarrow B)).$$

Write

$$A_B^\wedge := F^0 \widehat{\mathrm{adic}}(A \rightarrow B).$$

The adjunction unit of 3.3.4 followed by the completion unit gives a canonical morphism

$$A \longrightarrow A_B^\wedge.$$

3.3.2 Cotangent linearization of formal order

Definition 3.3.4 (Shifted cotangent complex of an augmentation). For an augmented arrow $A \rightarrow B$ of commutative R -algebras, its shifted cotangent complex is

$$L_{B/A}[-1].$$

If C is a filtered commutative algebra, regard $\mathrm{Gr}^0 C$ as concentrated in filtration degree zero. Its shifted cotangent complex relative to the filtered algebra C is

$$L_{\mathrm{Gr}^0 C/C}[-1]$$

in the filtered module category.

If C is a nonnegatively graded commutative algebra, write

$$C^{\mathrm{wt}=0}$$

for its weight-zero algebra, regarded as concentrated in weight zero. Its shifted cotangent complex relative to C is

$$L_{C^{\mathrm{wt}=0}/C}[-1]$$

in the graded module category.

Convention 3.3.3 (Almost-perfect filtered modules). If $F^\bullet A$ is a pointwise connective filtered commutative ring spectrum, an $F^\bullet A$ -module is called **almost perfect** when it is almost perfect in the stable module category $\mathrm{Mod}_{F^\bullet A}(\mathrm{Fil}(\mathrm{Sp}))$, equipped with the pointwise Postnikov t -structure. Perfectness and connectivity in filtered module categories are interpreted in the same intrinsic way.

Convention 3.3.4 (Almost finite presentation in filtered and graded spectral algebra). Let

$$\mathcal{C} = \text{Fil}(\text{Sp}) \quad \text{or} \quad \mathcal{C} = \text{Gr}(\text{Sp})$$

with its pointwise Postnikov t -structure, and let $A \rightarrow C$ be a morphism of pointwise connective commutative algebra objects. We say that C is **almost finitely presented over A** if, for every integer $n \geq 0$, the functor represented by C preserves filtered colimits on the full subcategory of pointwise connective n -truncated commutative A -algebras. This condition is a property of the morphism.

Remark 3.3.1 (Relation with formal integration). The filtered constructions in this section parallel the affine formal-integration machinery of Brantner–Magidson–Nuiten. The shifted cotangent complex used here is $L_{B/A}[-1]$; the adic filtration is left adjoint to filtration-zero augmentation; and the category of complete almost finitely presented augmentations is the domain of the shifted-cotangent–structured-square-zero comonadicity theorem. The main comparison results used below are [10, Remark 2.10, Lemma 2.12, Proposition 2.14, Lemma 2.16, Proposition 2.18, and Theorem 2.25]. The proofs are included here because the filtered connectivity and almost-perfectness estimates are also used in the later reconstruction and Artinian arguments.

Lemma 3.3.2 (Cotangent compatibility with grading, filtration zero, and adic formation). *Let $F^\bullet C$ be a filtered commutative algebra. There are canonical natural equivalences*

$$\text{Gr}(L_{\text{Gr}^0 C/F^\bullet C}[-1]) \simeq L_{\text{Gr}^0 C/\text{Gr} C}[-1],$$

$$F^0(L_{\text{Gr}^0 C/F^\bullet C}[-1]) \simeq L_{\text{Gr}^0 C/F^0 C}[-1],$$

and, for every augmented arrow $A \rightarrow B$,

$$L_{B/\text{adic}(A \rightarrow B)}[-1] \simeq \text{wt}_1(L_{B/A}[-1]).$$

The first equivalence is taken in graded modules over $\text{Gr} C$, and the second in $\text{Gr}^0 C$ -modules after evaluating the shifted relative cotangent complex in filtration zero.

The two comparison functors are summarized by

$$\text{Gr}(L_{\text{Gr}^0 C/F^\bullet C}[-1]) \xrightarrow{\simeq} L_{\text{Gr}^0 C/\text{Gr} C}[-1]$$

$$\begin{array}{c} F^0(L_{\text{Gr}^0 C/F^\bullet C}[-1]) \\ \simeq \downarrow \\ L_{\text{Gr}^0 C/F^0 C}[-1] \end{array}$$

where the horizontal and vertical arrows denote the canonical base-change comparisons.

Proof.

- (1) **Functoriality of cotangent complexes.** Associated graded and evaluation in filtration zero are symmetric monoidal, preserve colimits, and commute with free commutative algebra formation. Applying the universal derivation characterization of relative cotangent complexes therefore gives the first two equivalences.

- (2) **Relative shifted-cotangent arrow.** For the third, put

$$M := L_{B/A}[-1].$$

Consider the morphism of augmented arrows

$$(A \rightarrow B) \longrightarrow (B \rightarrow B).$$

Objectwise stabilization of the arrow category, together with the tangent equivalence of 2.2.1, identifies its relative shifted-cotangent object after restriction to the fixed target B with the B -module arrow

$$M \longrightarrow 0.$$

Indeed, a morphism from $M \rightarrow 0$ to a module arrow $N_0 \rightarrow N_1$ is a morphism $M \rightarrow N_0$ together with a specified nullhomotopy of its composite with $N_0 \rightarrow N_1$; by cotangent transitivity this is exactly the relative shifted-cotangent datum for deforming $A \rightarrow B$ while the target B is fixed. This is the arrow-category form of the ordinary relative cotangent classifier; compare [10, Definition 2.9 and Remark 2.10].

- (3) **Module-level augmentation adjunction.** Work in filtered B -modules. The module-level augmentation functor

$$\text{aug}_{\text{mod}} : \text{Fil}(\text{Mod}_B) \longrightarrow \text{Fun}(\Delta^1, \text{Mod}_B), \quad F^\bullet N \longmapsto (F^0 N \rightarrow \text{Gr}^0 N)$$

has the left adjoint computed in 3.3.1; in particular the arrow $M \rightarrow 0$ is sent to the two-step filtered module $\text{wt}_1(M)$.

- (4) **Tangent functor of augmentation.** The algebraic augmentation functor of 3.3.3 preserves split square-zero algebras. Indeed its two endpoint functors F^0 and Gr^0 are symmetric monoidal and carry a filtered split square-zero algebra to the corresponding split square-zero arrow. Hence the functor induced by aug on tangent module categories is exactly aug_{mod} .
- (5) **Yoneda identification.** Now let N be an arbitrary filtered B -module. Cotangent classification and the adjunction $\text{adic} \dashv \text{aug}$ identify maps from the shifted cotangent complex of $\text{adic}(A \rightarrow B)$ with maps from the relative shifted-cotangent arrow $M \rightarrow 0$ computed above into $\text{aug}_{\text{mod}} N$. Therefore

$$\begin{aligned} \text{Map}_{\text{Fil}(\text{Mod}_B)}(L_{B/\text{adic}(A \rightarrow B)}[-1], N) &\simeq \text{Map}_{\text{Fun}(\Delta^1, \text{Mod}_B)}(M \rightarrow 0, \text{aug}_{\text{mod}} N) \\ &\simeq \text{Map}_{\text{Fil}(\text{Mod}_B)}(\text{wt}_1(M), N). \end{aligned}$$

The equivalence is natural in every filtered B -module N , so Yoneda in the full filtered module category gives

$$L_{B/\text{adic}(A \rightarrow B)}[-1] \simeq \text{wt}_1(M).$$

The Yoneda argument takes place in the full filtered module category.

□

Theorem 3.3.1 (Associated graded of the canonical adic filtration). *For every augmented arrow $A \rightarrow B$ of commutative ring spectra there is a canonical equivalence of graded commutative B -algebras*

$$\widehat{\text{Gr adic}}(A \rightarrow B) \simeq \text{Sym}_B^{\text{gr}}(L_{B/A}[-1]\langle 1 \rangle).$$

Completion preserves associated graded.

Proof.

- (1) **Universal property.** Fix the source algebra A . By 3.3.4, $\mathrm{adic}(A \rightarrow B)$ has filtration-zero algebra A . Hence it is canonically a filtered commutative A -algebra. For a filtered commutative A -algebra $F^\bullet C$, the adjunction $\mathrm{adic} \dashv \mathrm{aug}$ gives

$$\begin{aligned} \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Fil}(\mathrm{Sp}))_{A/}}(\mathrm{adic}(A \rightarrow B), F^\bullet C) \\ \simeq \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})_{A/}}(B, \mathrm{Gr}^0 C). \end{aligned}$$

The source map from A is already fixed by the A -algebra structure. This is the universal property of the adic construction in [10, Section 2]; through 3.3.1, it also agrees with the relative $\mathbb{E}_\infty$ -Hodge filtration of [2, Section 2].

- (2) **Associated-graded calculation.** The associated-graded calculation [10, Lemma 2.12] gives

$$\mathrm{Gr} \, \mathrm{adic}(A \rightarrow B) \simeq \mathrm{Sym}_B^{\mathrm{gr}}(L_{B/A}[-1]\langle 1 \rangle).$$

Equivalently, this is the relative Hodge calculation of [2, Lemma 2.2]. Finally, 3.3.1 identifies the associated graded before and after completion:

$$\mathrm{Gr} \, \mathrm{adic}(A \rightarrow B) \xrightarrow{\sim} \widehat{\mathrm{Gr} \, \mathrm{adic}(A \rightarrow B)}.$$

Combining the two equivalences proves the theorem. □

Remark 3.3.2 (Interpretation of the associated graded). The theorem identifies the associated graded of the canonical adic filtration with the free graded commutative algebra on the shifted cotangent complex. The completed augmented algebra is then obtained by the completion reflector of 3.3.1.

The associated-graded theorem identifies the linear layers of formal order. Two detection results are still needed before those layers can reconstruct an augmentation: the canonical adic filtration must remain connective on the relevant domain, and an associated-graded equivalence between complete filtered objects must already be an equivalence before passing to graded pieces.

3.3.3 Completion and associated-graded detection

Lemma 3.3.3 (Connectivity of the canonical adic filtration). *Let $A \rightarrow B$ be a morphism of connective commutative ring spectra which is surjective on π_0 . Then every filtration term of $\mathrm{adic}(A \rightarrow B)$ is connective.*

Proof.

- (1) **Monadic resolution.** Connectivity is independent of the auxiliary base R , so we may work in the absolute arrow category. Resolve the arrow $A \rightarrow B$ by the monadic simplicial resolution for the objectwise free-forgetful adjunction

$$\mathrm{Fun}(\Delta^1, \mathrm{Sp}) \rightleftarrows \mathrm{Fun}(\Delta^1, \mathrm{CAlg}(\mathrm{Sp})).$$

Its simplicial terms are free algebra arrows

$$\mathrm{Sym}(V_q) \longrightarrow \mathrm{Sym}(V_{0,q}).$$

The underlying arrow $A \rightarrow B$ consists of connective spectra and is surjective on π_0 , equivalently its fibre is connective. The free commutative-algebra monad preserves this property. Indeed, if $V \rightarrow V_0$ is a morphism of connective spectra surjective on π_0 , then

$$\pi_0 \operatorname{Sym}(V) \simeq \operatorname{Sym}_{\mathbb{Z}}(\pi_0 V) \longrightarrow \operatorname{Sym}_{\mathbb{Z}}(\pi_0 V_0) \simeq \pi_0 \operatorname{Sym}(V_0)$$

is surjective. Iterating the monad shows that every $V_q \rightarrow V_{0,q}$ has connective source, target, and fibre.

- (2) **Connectivity of free filtered terms.** By 3.3.1, the module-level left adjoint sends $V_q \rightarrow V_{0,q}$ to the pointwise connective two-step filtration

$$\cdots \longrightarrow 0 \longrightarrow \operatorname{fib}(V_q \rightarrow V_{0,q}) \longrightarrow V_q.$$

For a free algebra arrow, the two free-forgetful adjunctions give a canonical equivalence

$$\begin{aligned} \operatorname{adic}(\operatorname{Sym}(V_q) \rightarrow \operatorname{Sym}(V_{0,q})) \\ \simeq \operatorname{Sym}^{\operatorname{Fil}}(\operatorname{adic}_{\operatorname{mod}}(V_q \rightarrow V_{0,q})). \end{aligned}$$

Indeed both sides represent, against a filtered commutative algebra $F^\bullet C$, maps from the arrow $V_q \rightarrow V_{0,q}$ to $F^0 C \rightarrow \operatorname{Gr}^0 C$ after passage to underlying spectra.

- (3) **Geometric realization.** Day convolution of pointwise connective filtered spectra is pointwise connective, since each filtration term of a tensor product is a colimit of tensor products of connective spectra. The free filtered commutative algebra on the displayed two-step filtration is therefore pointwise connective. Finally adic , being a left adjoint, preserves the geometric realization of the free simplicial resolution. Evaluation in every filtration degree preserves the sifted colimit underlying this realization, and geometric realizations of simplicial connective spectra are connective. Hence

$$F^n \operatorname{adic}(A \rightarrow B) \in \operatorname{Sp}_{\geq 0}$$

for every $n \geq 0$.

□

Remark 3.3.3 (Relation with the degree-zero augmentation ideal). For a connective augmentation surjective on π_0 , the canonical adic filtration records the full spectral augmentation, including its higher homotopy. The degree-zero ideal

$$I = \ker(\pi_0 A \rightarrow \pi_0 B)$$

enters after completion: on the almost finite presentation domain, 3.5.2 identifies canonical adic completion with derived I -completion. Thus the ideal-theoretic description concerns the completed object, while the individual filtration terms retain the full spectral augmentation data.

Lemma 3.3.4 (Complete filtered detection). *Let $\mathcal{C}$ be a stable ∞ -category admitting countable inverse limits, and let*

$$F^\bullet M \longrightarrow F^\bullet N$$

be a morphism of complete descending filtered objects in $\mathcal{C}$. If

$$\operatorname{Gr} M \xrightarrow{\sim} \operatorname{Gr} N,$$

then the filtered morphism is an equivalence. Equivalently,

$$F^r M \xrightarrow{\sim} F^r N$$

for every filtration degree $r \geq 0$.

Proof.

- (1) **Finite filtration quotients.** Fix $r \geq 0$. The shifted tails

$$F^{r+\bullet} M \quad \text{and} \quad F^{r+\bullet} N$$

are complete. The associated-graded hypothesis identifies every successive cofiber

$$\mathrm{cofib}(F^{k+1} M \rightarrow F^k M) \xrightarrow{\sim} \mathrm{cofib}(F^{k+1} N \rightarrow F^k N).$$

Induction on $n \geq 1$, using the cofiber sequences for the finite filtration quotients, gives

$$F^r M / F^{r+n} M \xrightarrow{\sim} F^r N / F^{r+n} N.$$

- (2) **Passage to the inverse limit.** Completeness identifies

$$\begin{array}{ccc} F^r M & \xrightarrow{\sim} & \varprojlim_{n>0} F^r M / F^{r+n} M \\ \downarrow & & \downarrow \sim \\ F^r N & \xrightarrow{\sim} & \varprojlim_{n>0} F^r N / F^{r+n} N. \end{array}$$

The right vertical map is an equivalence because countable inverse limits preserve equivalences. Hence $F^r M \rightarrow F^r N$ is an equivalence. Varying r proves the result. $\square$

Lemma 3.3.5 (Connectivity from a complete filtration with connective associated graded). *Let $F^\bullet C$ be a complete descending filtered spectrum such that every associated-graded term is connective. Then $F^0 C$ is connective.*

Proof.

- (1) **Define the finite quotients.** Put $C_n := F^0 C / F^{n+1} C$.
 (2) **Establish connectivity of the quotient tower.** Induction on n shows that every C_n is connective and that the transition maps are surjective on π_0 .
 (3) **Use completeness.** By completeness,

$$F^0 C \simeq \varprojlim_n C_n.$$

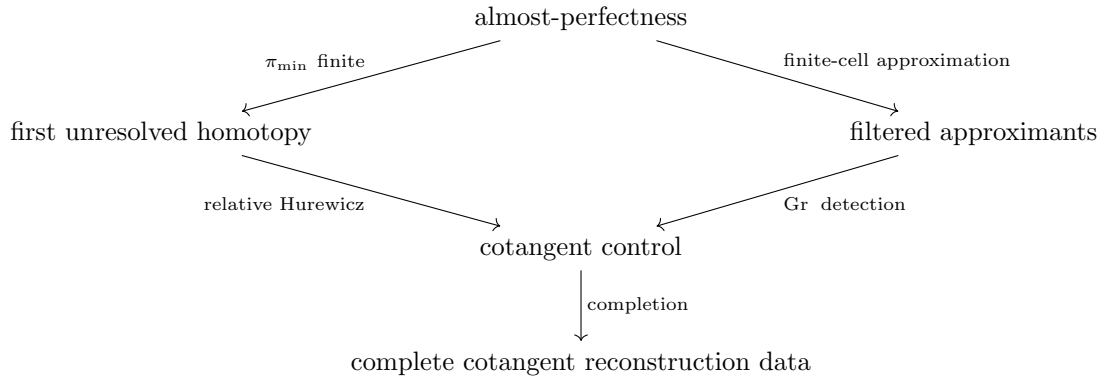
- (4) **Apply the Milnor exact sequence.** The Milnor exact sequence kills π_{-1} because the $\lim^1$ term vanishes for the surjective tower, and all lower negative homotopy groups vanish termwise. $\square$

We can now retain a compact interface from this section: every augmentation has a canonical filtered formal neighbourhood, its associated graded is cotangent-generated, and complete filtered objects are detected on their graded layers. What remains is to show that the finiteness and connectivity needed for comonadic reconstruction survive through this formal-order calculus.

3.4 Finiteness and convergence for fixed-centre reconstruction

The next task is therefore convergence rather than construction. The fixed-centre theorem will use augmentations $A \rightarrow B$ that are connective, finite enough for cotangent theory, and complete for the canonical formal-order filtration. This section establishes the estimates that make those conditions stable under the constructions used later.

Four mechanisms provide the control. Almost-perfectness governs linear coefficients, almost finite presentation governs algebra maps, finite-cell approximation transports those finiteness properties through filtered constructions, and relative Hurewicz comparison detects the first unresolved nonlinear homotopy in cotangent terms. Their dependency is summarized by



The lower-right object is the only output that must remain active: the completed adic filtration and its shifted cotangent complex stay finite, connective, and complete enough for Barr–Beck. The subsections establish the four links in the order in which they are used.

3.4.1 Finiteness notions and first unresolved homotopy

Definition 3.4.1 (Almost-perfect module). Let A be connective. An A -module M is **almost perfect** if it is bounded below and, for every integer n , its truncation $\tau_{\leq n} M$ is compact in the ∞ -category of n -truncated A -modules. Write

$$\mathrm{APerf}(A) \subseteq \mathrm{Mod}_A$$

for the full stable subcategory of almost-perfect modules.

Lemma 3.4.1 (Restriction transitivity for almost-perfect modules). *Let $A \rightarrow B$ be a morphism of connective commutative ring spectra. If B is almost perfect as an A -module, then restriction of scalars carries almost-perfect B -modules to almost-perfect A -modules:*

$$\mathrm{Res}_A^B(\mathrm{APerf}(B)) \subseteq \mathrm{APerf}(A).$$

Proof.

- (1) **Reduce to the connective case.** After a suspension we may take the B -module M to be connective.

- (2) **Resolve by finite free modules.** By the connective almost-perfect resolution theorem, M is the geometric realization of a simplicial B -module whose terms are finite free B -modules; see [34, Proposition 7.2.4.11(5)].
- (3) **Restrict the simplicial terms.** After restriction to A , each simplicial term is a finite direct sum of the almost-perfect A -module B .
- (4) **Use closure under realization.** The connective almost-perfect A -modules are closed under these geometric realizations by [34, Proposition 7.2.4.11(4)]. Hence the realization M , regarded as an A -module, is almost perfect.
- (5) **Undo the suspension.** Undoing the suspension gives the claim.

□

Lemma 3.4.2 (First unresolved homotopy of an almost-perfect module). *Let A be connective and let $M \in \mathrm{APerf}(A)$. If M is n -connective, then its first possible nonzero homotopy module*

$$\pi_n(M)$$

is finitely presented over $\pi_0(A)$. The conclusion holds for every connective commutative ring spectrum A .

Proof.

- (1) **Identify the first truncation.** Because M is n -connective,

$$\tau_{\leq n} M \simeq H\pi_n(M)[n].$$

- (2) **Use compactness.** Almost-perfectness says that this truncation is compact in the ∞ -category of n -truncated A -modules.
- (3) **Test on Eilenberg–Mac Lane diagrams.** Test compactness only on filtered diagrams of Eilenberg–Mac Lane modules $HN_\alpha[n]$. Taking connected components of the resulting mapping-space comparison gives

$$\mathrm{Hom}_{\pi_0(A)}\left(\pi_n(M), \operatorname{colim}_{\alpha} N_{\alpha}\right) \simeq \operatorname{colim}_{\alpha} \mathrm{Hom}_{\pi_0(A)}(\pi_n(M), N_{\alpha}).$$

- (4) **Read off finite presentation.** Thus $\pi_n(M)$ is finitely presented over $\pi_0(A)$.
- (5) **Record the filtered and graded variants.** The same argument applies verbatim to modules over connective graded or filtered commutative ring spectra with their pointwise Postnikov t -structures.

□

Definition 3.4.2 (Almost finite presentation). Let $A \rightarrow C$ be a morphism of connective commutative ring spectra. We say that C is **almost finitely presented over A** if, for every integer $n \geq 0$, the functor represented by C on n -truncated connective commutative A -algebras preserves filtered colimits.

Definition 3.4.3 (Coherent commutative ring spectrum). A connective commutative ring spectrum B is **coherent** if $\pi_0(B)$ is a coherent ordinary ring and every $\pi_n(B)$ is finitely presented over $\pi_0(B)$.

Definition 3.4.4 (Coherent module). For a coherent connective commutative ring spectrum B , write

$$\mathrm{Coh}(B) \subseteq \mathrm{APerf}(B)$$

for the full subcategory of eventually coconnective almost-perfect B -modules. A **connective coherent B -module** is an object of

$$\mathrm{Coh}(B) \cap (\mathrm{Mod}_B)_{\geq 0}.$$

Proposition 3.4.1 (Almost-perfect modules over a coherent commutative ring spectrum). *Let B be a coherent connective commutative ring spectrum. A B -module M is almost perfect if and only if it is bounded below and, for every integer n , the $\pi_0(B)$ -module $\pi_n(M)$ is finitely presented. Consequently*

$$\mathrm{Coh}(B)$$

is exactly the full subcategory of bounded B -modules whose homotopy modules are finitely presented over $\pi_0(B)$.

Proof.

- (1) **Invoke the coherent-module characterization.** This is the commutative $\mathbb{E}_\infty$ -specialization of [34, Proposition 7.2.4.17].
- (2) **Identify the coherent subcategory.** The final statement is the intersection with the eventually coconnective subcategory in 3.4.4.

□

Lemma 3.4.3 (Relative spectral Hurewicz in the first unresolved degree). *Let $A \rightarrow C$ be a morphism of connective commutative ring spectra inducing an isomorphism on π_0 , and suppose*

$$Q := \mathrm{cofib}(A \rightarrow C)$$

is n -connective for $n \geq 1$. Then the canonical relative-cofibre Hurewicz morphism

$$C \otimes_A Q \longrightarrow L_{C/A}$$

induces an isomorphism

$$\pi_n(Q) \xrightarrow{\cong} \pi_n L_{C/A}.$$

The comparison factors through extension of scalars:

$$Q \longrightarrow C \otimes_A Q \longrightarrow L_{C/A}.$$

Proof.

- (1) **Hurewicz comparison.** By the relative Hurewicz theorem of 2.4.3, the fibre of

$$C \otimes_A Q \longrightarrow L_{C/A}$$

is $2n$ -connective. Since $n \geq 1$, the induced morphism on the first possible nonzero homotopy group is therefore an isomorphism:

$$\pi_n(C \otimes_A Q) \xrightarrow{\cong} \pi_n L_{C/A}.$$

- (2) **Identify the first homotopy group.** It remains to identify the source. Because A , C , and Q are connective, the connective tensor-product spectral sequence applies. The assumption

$$\pi_0(A) \xrightarrow{\cong} \pi_0(C)$$

and n -connectivity of Q imply

$$\pi_n(C \otimes_A Q) \simeq \pi_0(C) \otimes_{\pi_0(A)} \pi_n(Q) \simeq \pi_n(Q).$$

Combining the two canonical identifications gives

$$\pi_n(Q) \xrightarrow{\cong} \pi_n L_{C/A}.$$

□

Almost-perfectness reduces the first unresolved homotopy to finite module data. To improve connectivity without losing that finiteness, we need an algebraic operation whose linear layer is prescribed while higher word-length layers become progressively more connective. Relation cells provide exactly that operation.

3.4.2 Relation cells and algebraic finiteness

Lemma 3.4.4 (Suspended free algebras are finite-cell). *Let C be connective. For every $n, r \geq 0$, the commutative C -algebra*

$$\mathrm{Sym}_C(C[n]^{\oplus r})$$

is finite-cell in the sense of Chapter 1.

Proof.

- (1) **Express the suspension as a pushout.** In the stable category Mod_C ,

$$C[1] \simeq 0 \amalg_C 0.$$

- (2) **Apply the free algebra functor.** The free commutative-algebra functor preserves colimits, so

$$\mathrm{Sym}_C(C[1]) \simeq C \amalg_{\mathrm{Sym}_C(C)} C.$$

- (3) **Recognize the suspended generator as finite-cell.** Thus the free algebra on a once-suspended generator lies in the finite-colimit closure of the degree-zero free generators used in the definition of finite cellularity.
- (4) **Iterate the construction.** Iterating suspension and taking finite sums proves the general case.

□

Lemma 3.4.5 (Finite-group homotopy orbits and spectral symmetric powers). *Let A be connective, or let $F^\bullet A$ be pointwise connective in $\mathrm{Fil}(\mathrm{Sp})$ or $\mathrm{Gr}(\mathrm{Sp})$.*

- (i) *If P is a connective finite cell module carrying an action of a finite group G , then P_{hG} is connective and almost perfect. More generally, if P is q -connective, then so is P_{hG} .*

(ii) If M is connective and almost perfect, then for every $r \geq 0$,

$$\mathrm{Sym}_A^r(M) \simeq (M^{\otimes_A r})_{h\Sigma_r}$$

is connective and almost perfect. The same assertion holds in the filtered and graded module categories with their pointwise t -structures.

Proof.

- (1) **Finite-group homotopy orbits.** For (i), compute homotopy orbits by the simplicial bar construction. In every simplicial degree its underlying module is a finite direct sum of copies of P , because G is finite. Every finite skeletal realization is therefore a finite cell module. The skeletal filtration has successive cofibres given by increasingly suspended latching cofibres; consequently the map from the s -skeleton to P_{hG} becomes arbitrarily connective as $s \rightarrow \infty$. The finite-cell approximation criterion for almost-perfect modules, and its filtered or graded analogue, proves almost-perfectness. Connectivity is preserved because homotopy orbits are colimits of connective objects.
- (2) **Spectral symmetric powers.** For (ii), fix a Postnikov degree d . Choose a connective finite cell approximation

$$P \longrightarrow M$$

whose cofibre is d -connective; after increasing d we may take the approximating module itself connective. Replacing the tensor factors one at a time shows that

$$P^{\otimes_A r} \longrightarrow M^{\otimes_A r}$$

has d -connective cofibre, since every successive cofibre contains the d -connective module $\mathrm{cofib}(P \rightarrow M)$ tensored with connective factors. Finite-group homotopy orbits preserve this connectivity. Thus

$$\mathrm{Sym}_A^r(P) \longrightarrow \mathrm{Sym}_A^r(M)$$

is d -connective. By (i), the source is almost perfect, hence admits a finite-cell approximation through degree d . Composing the two approximations and varying d proves that $\mathrm{Sym}_A^r(M)$ is almost perfect. The proof is pointwise in the filtered and graded cases.

□

Lemma 3.4.6 (Positive relation-cell attachment). *Let C be connective, let $n \geq 0$, let*

$$P \simeq C[n]^{\oplus r}$$

be finite free and connective, and let $\chi : P \rightarrow C$ be C -linear. Put

$$S := \mathrm{Sym}_C(P),$$

let $\epsilon_\chi, \epsilon_0 : S \rightrightarrows C$ be the maps classified by χ and zero, and define

$$\mathrm{RelCell}_C(\chi) := C_{\epsilon_\chi} \otimes_S C_{\epsilon_0}.$$

Then:

- (i) $\mathrm{RelCell}_C(\chi)$ is a finite-cell commutative C -algebra;

(ii) the two-sided bar construction has a natural exhaustive multiplicative filtration with

$$\mathrm{Gr}^k \mathrm{RelCell}_C(\chi) \simeq \mathrm{Sym}_C^k(P[1]);$$

(iii) $\mathrm{RelCell}_C(\chi)$ is almost perfect as a C -module.

Now suppose that $C \rightarrow D$ induces an isomorphism on π_0 , that

$$Q := \mathrm{cofib}(C \rightarrow D)$$

is $(n+1)$ -connective, and that

$$u : P \longrightarrow Q[-1]$$

is surjective on π_n . Let $\partial : Q[-1] \rightarrow C$ be the boundary of the cofibre sequence and define

$$\chi := \partial \circ u.$$

Then the canonical nullhomotopy of $P \rightarrow C \rightarrow D$ induces a map

$$\mathrm{RelCell}_C(\chi) \longrightarrow D$$

whose cofibre is $(n+2)$ -connective.

Proof.

- (1) **Finite-cell algebra.** By 3.4.4, $\mathrm{Sym}_C(P)$ is finite-cell. The pushout defining $\mathrm{RelCell}_C(\chi)$ is a finite colimit of finite-cell commutative C -algebras, which proves (i).
- (2) **Bar filtration.** Compute the pushout by the two-sided bar construction

$$\mathrm{Bar}_\bullet(C_{\epsilon_\chi}, S, C_{\epsilon_0}).$$

Filter its normalized realization by total symmetric degree in the occurrences of the generator P . Internal multiplication faces preserve total degree, the endpoint differential determined by χ lowers total degree, and the zero endpoint kills positive degree. Hence χ disappears on associated graded. The associated graded is therefore the zero-zero derived self-intersection

$$\begin{aligned} C \otimes_{\mathrm{Sym}_C(P)} C &\simeq \mathrm{Sym}_C(0) \amalg_{\mathrm{Sym}_C(P)} \mathrm{Sym}_C(0) \\ &\simeq \mathrm{Sym}_C(0 \amalg_P 0) \\ &\simeq \mathrm{Sym}_C(P[1]). \end{aligned}$$

Thus the weight- k layer is $\mathrm{Sym}_C^k(P[1])$, proving (ii).

- (3) **Almost-perfectness.** Since $P[1]$ is $(n+1)$ -connective, $\mathrm{Sym}_C^k(P[1])$ is $k(n+1)$ -connective. It is almost perfect by 3.4.5. Consequently every fixed Postnikov range sees only finitely many word-length layers. Finite extensions of almost-perfect modules are almost perfect, and the omitted tail can be made arbitrarily connective. This gives finite-cell approximations in every Postnikov range and proves (iii).
- (4) **Connectivity improvement.** For the final statement, the identity $\chi = \partial u$ supplies a canonical nullhomotopy of $P \rightarrow C \rightarrow D$, hence a morphism $\mathrm{RelCell}_C(\chi) \rightarrow D$. The induced morphism from the relative cofibre of $C \rightarrow \mathrm{RelCell}_C(\chi)$ to Q has linear layer

$$P[1] \xrightarrow{u[1]} Q,$$

which is surjective on π_{n+1} . Every nonlinear layer contains at least two copies of $P[1]$, and is therefore at least $2(n+1)$ -connective. Since $2(n+1) \geq n+2$, the cofiber of the map from the relative cofibre of $C \rightarrow \text{RelCell}_C(\chi)$ to Q is $(n+2)$ -connective. Octahedrality for

$$C \longrightarrow \text{RelCell}_C(\chi) \longrightarrow D$$

identifies this cofibre with $\text{cofib}(\text{RelCell}_C(\chi) \rightarrow D)$, proving the assertion. $\square$

Theorem 3.4.1 (Spectral criterion for almost finite presentation of a surjective augmentation). *Let $A \rightarrow B$ be a morphism of connective commutative ring spectra inducing a surjection on π_0 . The following conditions are equivalent:*

- (i) *B is almost finitely presented as a connective commutative A -algebra;*
- (ii) *$L_{B/A}$ is almost perfect and $\pi_0(B)$ is finitely presented as a $\pi_0(A)$ -algebra;*
- (iii) *B is almost perfect as an A -module.*

The criterion is valid for arbitrary connective commutative ring spectra satisfying the displayed hypotheses.

Proof.

- (1) **Cotangent criterion.** The equivalence of (i) and (ii) is the connective $\mathbb{E}_\infty$ -cotangent criterion for almost finite presentation; see [30, Theorem 2.12(2)]. The same criterion is recorded in [10, Proposition A.19(1)–(2)].
- (2) **From module finiteness to algebra finiteness.** The implication (iii) $\Rightarrow$ (i) follows from [10, Proposition A.19(3)]: almost-perfectness of B as an A -module implies almost finite presentation of the algebra map $A \rightarrow B$.
- (3) **From algebra finiteness to module finiteness.** For the reverse implication, assume (i). The standing hypothesis gives a surjection

$$\pi_0(A) \twoheadrightarrow \pi_0(B).$$

Then [10, Proposition A.19(4)] implies

$$B \in \text{APerf}(A).$$

Thus (i), (ii), and (iii) are equivalent. $\square$

Corollary 3.4.1 (Cotangent criterion for almost finite presentation). *Let $A \rightarrow C$ be a morphism of connective commutative ring spectra. Suppose $\pi_0(C)$ is finitely presented as a $\pi_0(A)$ -algebra and $L_{C/A}$ is almost perfect as a C -module. Then C is almost finitely presented over A .*

Proof.

- (1) **Apply the cotangent criterion.** Apply the connective $\mathbb{E}_\infty$ -cotangent criterion for almost finite presentation, [30, Theorem 2.12(2)].

- (2) **Verify its hypotheses.** Its hypotheses are precisely finite presentation of $\pi_0(C)$ over $\pi_0(A)$ together with almost-perfectness of $L_{C/A}$.
- (3) **Record the equivalent formulation.** The equivalent formulation in [10, Proposition A.19(1)–(2)] gives the same conclusion.

□

Lemma 3.4.7 (Finite presentation of the degree-zero augmentation kernel). *Let $A \rightarrow B$ be a morphism of connective commutative ring spectra which is surjective on π_0 . Suppose that B is coherent and almost perfect as an A -module. Put*

$$I := \ker(\pi_0(A) \rightarrow \pi_0(B)).$$

Then I is finitely presented as a $\pi_0(A)$ -module.

Proof.

- (1) **Almost-perfect fibre.** Put $K := \text{fib}(A \rightarrow B)$. Since the map is surjective on π_0 , the long exact homotopy sequence makes K connective. The almost-perfect subcategory is stable under finite limits and colimits; since A is perfect over itself and B is almost perfect over A , the fibre K is almost perfect. For a connective almost-perfect module, π_0 is finitely presented over $\pi_0(A)$. Thus $\pi_0(K)$ is finitely presented.
- (2) **Finite restriction of the first higher homotopy.** Apply 3.4.2 to the connective almost-perfect A -module B . It gives

$$\pi_0(B) \in \text{Mod}_{\pi_0(A)}^{\text{fp}}.$$

Coherence of B makes $\pi_1(B)$ finitely presented over $\pi_0(B)$. Choose a finite presentation of $\pi_1(B)$ by finite free $\pi_0(B)$ -modules. After restriction of scalars, the two finite free terms are finite direct sums of the finitely presented $\pi_0(A)$ -module $\pi_0(B)$. Hence $\pi_1(B)$, and in particular its image in any $\pi_0(A)$ -module, is finitely generated over $\pi_0(A)$.

- (3) **Degree-zero exact sequence.** The fibre sequence gives an exact sequence

$$\pi_1(B) \rightarrow \pi_0(K) \rightarrow \pi_0(A) \rightarrow \pi_0(B) \rightarrow 0.$$

Thus there is a canonical exact sequence

$$\text{im}(\pi_1(B) \rightarrow \pi_0(K)) \rightarrow \pi_0(K) \rightarrow I \rightarrow 0.$$

The middle module is finitely presented over $\pi_0(A)$ by the first step of the proof, while the image on the left is finitely generated by the preceding restriction-of-scalars argument. A quotient of a finitely presented module by a finitely generated submodule is finitely presented. Therefore I is finitely presented over $\pi_0(A)$.

□

Lemma 3.4.8 (The positive graded symmetric augmentation is almost finitely presented). *Let B be connective and let M be a connective almost-perfect B -module. Then the augmentation*

$$\text{Sym}_B^{\text{gr}}(M\langle 1 \rangle) \rightarrow B$$

is almost finitely presented in connective nonnegatively graded commutative B -algebras.

Proof.

- (1) **Fix the graded algebra and test object.** Put $C := \mathrm{Sym}_B^{\mathrm{gr}}(M\langle 1 \rangle)$. Let D be an n -truncated connective graded commutative C -algebra.
- (2) **Apply the free graded-algebra adjunction.** By the free graded-algebra adjunction, a morphism of augmented graded B -algebras from B to D over C is equivalently a nullhomotopy, compatible with the augmentation, of the image of the universal weight-one map

$$M \longrightarrow D_1.$$

- (3) **Express the mapping space as a finite limit.** Thus its mapping space is a finite-limit expression built from $\mathrm{Map}_{\mathrm{Mod}_B}(M, D_1)$.
- (4) **Use almost-perfectness and weight evaluation.** Evaluation in weight one preserves filtered colimits, and almost-perfectness of M makes mapping out of M commute with filtered colimits on every fixed truncated connective module category.
- (5) **Commute filtered colimits with finite limits.** Filtered colimits of spaces commute with finite limits, proving the claim.

□

Lemma 3.4.9 (The positive symmetric augmentation module is almost perfect). *Let B be connective and let M be a connective almost-perfect B -module. Put*

$$C := \mathrm{Sym}_B^{\mathrm{gr}}(M\langle 1 \rangle).$$

Then the weight-zero augmentation module B , concentrated in graded weight zero, is almost perfect as a graded C -module.

Proof.

- (1) **Resolve the augmentation module.** Resolve the augmentation module by the two-sided bar construction for $C \rightarrow B$.
- (2) **Filter by word length.** Filter its normalization by word length in the positive weight-one generator.
- (3) **Identify the homogeneous terms.** Every homogeneous term is assembled from finite tensor powers of $M[1]$ and finite-group homotopy orbits.
- (4) **Approximate each fixed word length.** For a fixed word length and a fixed Postnikov range, a sufficiently large finite skeleton of the corresponding finite-group free resolution gives a perfect approximation.
- (5) **Use increasing connectivity.** Because M is connective and occurs in positive weight, the connectivity of word length k tends to $+\infty$ with k .
- (6) **Bound the relevant word lengths.** Hence every fixed Postnikov range sees only finitely many word lengths.
- (7) **Assemble the perfect approximations.** Finite extensions of the resulting perfect approximations give a perfect graded C -module approximation to B through every Postnikov degree.

□

The preceding relation-cell calculus is unfiltered. To use it for the canonical adic filtration, we now transport the same approximation argument to filtered modules. Compact generators indexed by filtration degree make almost-perfectness accessible to finite filtered cells, while completion lets associated graded detect the result.

3.4.3 Filtered cell approximation and complete detection

Construction 3.4.1 (Compact filtered weight generators). For $w \geq 0$, let $\mathbb{S}\langle w \rangle \in \text{Fil}(\text{Sp})$ be the filtered sphere

$$F^n \mathbb{S}\langle w \rangle = \begin{cases} \mathbb{S}, & 0 \leq n \leq w, \\ 0, & n > w, \end{cases}$$

with identity transition maps in the nonzero range. Its associated graded is the sphere concentrated in weight w . For a filtered commutative ring spectrum $F^\bullet A$, put

$$F^\bullet A\langle w \rangle := F^\bullet A \otimes \mathbb{S}\langle w \rangle.$$

Then for every filtered A -module N there is a natural equivalence

$$\text{map}_{F^\bullet A}(F^\bullet A\langle w \rangle, N) \simeq F^w N,$$

and

$$\text{Gr}(F^\bullet A\langle w \rangle) \simeq \text{Gr}(A)\langle w \rangle.$$

The objects $F^\bullet A\langle w \rangle$, together with their suspensions, form a family of compact generators of the filtered module category. A **finite filtered cell module** is an object in the smallest full subcategory containing these suspended weight generators and closed under finite colimits. The perfect filtered modules form the thick subcategory generated by the same weight generators. Thus every finite filtered cell module is perfect, and every perfect filtered module is a retract of a finite filtered cell module.

Proof.

- (1) **Represent filtration evaluation.** The filtered sphere $\mathbb{S}\langle w \rangle$ represents evaluation in filtration degree w .
- (2) **Apply free-module adjunction.** Free-module adjunction gives the mapping formula.
- (3) **Prove compactness of the weight generators.** Evaluation preserves filtered colimits, so the weight generators are compact.
- (4) **Prove generation.** If all mapping spectra from all suspensions of all weight generators vanish, then every filtration term of N vanishes; hence they generate.
- (5) **Compute associated graded.** Associated graded is symmetric monoidal and carries $\mathbb{S}\langle w \rangle$ to the graded sphere in weight w , proving the associated-graded formula.
- (6) **Identify finite filtered cells as perfect.** Finite colimits of compact objects are compact, so every finite filtered cell module is compact and therefore perfect.

- (7) **Identify perfect modules as retracts.** Compact objects in a compactly generated stable category form the thick subcategory generated by any set of compact generators; equivalently, every perfect filtered module is a retract of a finite filtered cell module.

□

Lemma 3.4.10 (Finite-cell approximation criterion for filtered almost-perfect modules). *Let $F^\bullet A$ be pointwise connective and let N be a bounded-below filtered A -module. Then N is almost perfect if and only if, for every integer d , there exists a finite filtered cell module P_d and a map*

$$P_d \longrightarrow N$$

whose cofibre is d -connective in the pointwise Postnikov t -structure.

Proof.

- (1) **Inductive approximation.** After a suspension, assume that N is connective. Suppose first that N is almost perfect. Set

$$C_0 := N, \quad P_0 := 0.$$

Assume inductively that

$$P_d \longrightarrow N \longrightarrow C_d$$

is a cofiber sequence with P_d a finite filtered cell and C_d pointwise d -connective. Since finite filtered cells are perfect and almost-perfect modules are closed under finite cofibers, C_d is almost perfect. Therefore

$$\tau_{\leq d} C_d \simeq H\pi_d(C_d)[d]$$

is compact in the category of pointwise d -truncated filtered A -modules, so $\pi_d(C_d)$ is finitely presented in the heart.

- (2) **Kill the first unresolved homotopy.** For each $w \geq 0$, the free weight module $A\langle w \rangle$ represents evaluation in filtration degree w . Its heart object $\pi_0(A\langle w \rangle)$ is therefore projective, and these heart objects form a family of generators. Finite presentation of $\pi_d(C_d)$ gives weights $w_1, \dots, w_r$ and an epimorphism

$$\bigoplus_{i=1}^r \pi_0(A\langle w_i \rangle) \twoheadrightarrow \pi_d(C_d).$$

Choose representatives of these classes to obtain

$$Q_d := \bigoplus_{i=1}^r A\langle w_i \rangle[d] \longrightarrow C_d$$

surjective on pointwise π_d . Let C_{d+1} be its cofiber and put

$$P_{d+1} := \text{fib}(N \rightarrow C_{d+1}).$$

Then C_{d+1} is pointwise $(d+1)$ -connective, and the 3×3 lemma supplies a cofiber sequence

$$P_d \longrightarrow P_{d+1} \longrightarrow Q_d.$$

Thus P_{d+1} is again a finite filtered cell. Induction yields finite filtered-cell approximations with arbitrarily connective cofiber.

- (3) **Converse implication.** Conversely, suppose that such approximations exist. Fix n and choose $P \rightarrow N$ with $(n+1)$ -connective cofiber. Then

$$\tau_{\leq n}P \xrightarrow{\cong} \tau_{\leq n}N.$$

A finite filtered cell is compact, and mapping into a pointwise n -truncated module depends only on its n -truncation. Hence $\tau_{\leq n}N$ is compact in the pointwise n -truncated module category. Varying n proves that N is almost perfect. Desuspending gives the bounded-below case.

□

Lemma 3.4.11 (Restriction transitivity for filtered almost-perfect modules). *Let*

$$F^\bullet A \longrightarrow F^\bullet B$$

be a morphism of pointwise connective filtered commutative ring spectra. If $F^\bullet B$ is almost perfect as a filtered $F^\bullet A$ -module, then restriction of scalars carries almost-perfect filtered B -modules to almost-perfect filtered A -modules.

Proof.

- (1) **Filtered B -cell approximation.** After a suspension it is enough to treat a connective almost-perfect filtered B -module M . For every integer d , 3.4.10 supplies a finite filtered B -cell

$$P_d \longrightarrow M$$

with d -connective cofibre.

- (2) **Restriction to A .** Every filtered weight generator $B\langle w \rangle$, after restriction to A , is almost perfect. Indeed B is almost perfect over A , and tensoring with the finite filtered sphere $\mathbb{S}\langle w \rangle$ preserves finite-cell approximations and their pointwise connectivity. The almost-perfect subcategory is stable under finite limits, colimits, shifts, and retracts, so the restricted module underlying every finite filtered B -cell P_d is almost perfect over A .

- (3) **Finite A -cell approximation.** Apply 3.4.10 to the restricted module P_d . Choose a finite filtered A -cell

$$Q_d \longrightarrow P_d$$

with d -connective cofibre. The composite

$$Q_d \longrightarrow P_d \longrightarrow M$$

again has d -connective cofibre by octahedrality. Since this can be done for every d , the filtered finite-cell criterion makes M almost perfect over A .

□

Theorem 3.4.2 (Complete filtered almost-perfect detection). *Let $F^\bullet A$ be a complete pointwise connective filtered commutative ring spectrum, and let $F^\bullet N$ be a pointwise connective filtered A -module. Then the following are equivalent:*

- (i) $F^\bullet N$ is almost perfect in $\text{Mod}_{F^\bullet A}(\text{Fil}(\text{Sp}))$;
- (ii) $F^\bullet N$ is complete and $\text{Gr } N$ is almost perfect over $\text{Gr } A$.

Under these conditions, $F^0 N$ is almost perfect as an $F^0 A$ -module.

Proof.

- (1) **Associated graded.** Assume first that N is almost perfect. By 3.4.10, for every d choose a finite filtered cell module

$$P_d \longrightarrow N$$

with d -connective cofibre C_d . Associated graded sends each compact weight generator to the corresponding compact graded weight generator and is exact. Hence it sends finite filtered cells to finite graded cells, and the same approximations show that $\mathrm{Gr} N$ is almost perfect.

- (2) **Completeness.** Every weight generator $A\langle w \rangle$ is complete when A is complete: its descending tail is a finite filtration shift of the tail of A . Completeness is stable under finite fibres, cofibres, sums, and retracts, so every P_d is complete. Therefore

$$\varprojlim_n F^n N \simeq \varprojlim_n F^n C_d.$$

Each $F^n C_d$ is d -connective. The Milnor exact sequence makes the inverse limit $(d-1)$ -connective. Since d is arbitrary, the inverse limit vanishes and N is complete.

- (3) **Filtration zero.** Evaluation in filtration zero is exact and sends every weight generator $A\langle w \rangle$ to the free rank-one $F^0 A$ -module. Hence it sends finite filtered cell modules to perfect $F^0 A$ -modules and preserves the connectivity of the cofibre of each approximation. Thus $F^0 N$ is almost perfect over $F^0 A$.

- (4) **Inductive converse.** Conversely suppose that N is complete and $\mathrm{Gr} N$ is almost perfect. Set $C_0 = N$. Suppose inductively that C_d is complete, pointwise d -connective, and has almost-perfect associated graded. Then $\mathrm{Gr} C_d$ is d -connective, and 3.4.2 in the graded module category makes

$$\pi_d \mathrm{Gr} C_d$$

a finitely presented graded module. Choose finitely many homogeneous generators in weights $w_1, \dots, w_r$.

- (5) **Lift graded generators.** For each w_i , the cofibre sequence

$$F^{w_i+1} C_d \longrightarrow F^{w_i} C_d \longrightarrow \mathrm{Gr}^{w_i} C_d$$

and pointwise d -connectivity give a surjection on π_d . Choose lifts of the selected generators. They assemble to a morphism

$$\bigoplus_{i=1}^r F^\bullet A\langle w_i \rangle[d] \longrightarrow C_d.$$

Let C_{d+1} be its cofibre. Associated graded is exact, so the chosen classes kill the first possible homotopy group and $\mathrm{Gr} C_{d+1}$ is $(d+1)$ -connective. It remains almost perfect. The source is a finite filtered cell and is complete; hence C_{d+1} is complete.

- (6) **Recover pointwise connectivity.** Apply 3.3.5 to every shifted filtration tail of C_{d+1} . Since each associated-graded term is $(d+1)$ -connective, every filtration term of C_{d+1} is $(d+1)$ -connective. Iterating and splicing the finite cell attachments by octahedrality gives, through every Postnikov degree, a finite filtered-cell approximation to N . By 3.4.10, N is almost perfect.

□

Corollary 3.4.2 (Almost-perfectness from a complete filtration). *Let $F^\bullet A$ be complete and pointwise connective, and let $F^\bullet N$ be pointwise connective and complete. If $\mathrm{Gr} N$ is almost perfect over $\mathrm{Gr} A$, then $F^0 N$ is almost perfect over $F^0 A$.*

Proof.

- (1) **Apply complete filtered detection.** Apply 3.4.2.

□

Corollary 3.4.3 (Almost-perfect filtered modules are complete). *Let $F^\bullet A$ be a complete pointwise connective filtered commutative ring spectrum. Every pointwise connective almost-perfect filtered A -module is complete.*

Proof.

- (1) **Apply the forward detection implication.** Apply the forward implication of 3.4.2.

□

Complete filtered detection now controls objects once their graded layers are known. The last missing estimate is cotangent control: the nonlinear Hurewicz comparison must persist in filtered spectral algebra. The following results combine filtered relation cells, Postnikov approximation, and cotangent transitivity to supply it.

3.4.4 Filtered Hurewicz control and cotangent convergence

Lemma 3.4.12 (Filtered relative Hurewicz in the first unresolved degree). *Let $F^\bullet A \rightarrow F^\bullet C$ be a morphism of pointwise connective filtered commutative ring spectra. Suppose its relative cofibre Q is pointwise n -connective for some $n \geq 1$, and that the map is an isomorphism on pointwise π_0 . Then $L_{C/A}$ is pointwise n -connective and the canonical map*

$$F^\bullet C \otimes_{F^\bullet A} Q \longrightarrow L_{C/A}$$

induces an isomorphism on pointwise π_n . More generally, if $F^\bullet A \rightarrow F^\bullet C$ is pointwise surjective on π_0 , then $L_{C/A}$ is pointwise 1-connective.

Proof.

- (1) **Verify the Hurewicz hypotheses.** The stable presentable symmetric monoidal category

$$\mathrm{Fil}(\mathrm{Sp}) = \mathrm{Fun}(\mathbb{N}_{\leq}^{\mathrm{op}}, \mathrm{Sp})$$

with Day convolution and the pointwise Postnikov t -structure satisfies the hypotheses of the general Hurewicz theorem of [34, Construction 7.4.3.10 and Theorem 7.4.3.12]: Day convolution preserves colimits separately, its unit is pointwise connective, and tensor products of pointwise connective filtered spectra are pointwise connective.

- (2) **Apply the filtered Hurewicz theorem.** Applying that theorem inside $\mathrm{Fil}(\mathrm{Sp})$ gives a canonical Hurewicz morphism

$$F^\bullet C \otimes_{F^\bullet A} Q \longrightarrow L_{C/A}$$

whose fibre is pointwise $2n$ -connective when Q is pointwise n -connective.

- (3) **Read off the first homotopy group.** This implies pointwise n -connectivity of $L_{C/A}$ and the asserted isomorphism on the first possible homotopy object.
- (4) **Handle the surjective degree-zero case.** If the original morphism is pointwise surjective on π_0 , its relative cofibre is pointwise 1-connective, and the same theorem gives pointwise 1-connectivity of the relative cotangent complex.

□

Lemma 3.4.13 (Filtered positive relation-cell attachment). *Let $F^\bullet C$ be a pointwise connective filtered commutative ring spectrum, let $n \geq 0$, let P be a pointwise n -connective finite filtered cell C -module, and let $\chi : P \rightarrow C$ be C -linear. Put*

$$S := \mathrm{Sym}_C(P),$$

let $\epsilon_\chi, \epsilon_0 : S \rightrightarrows C$ be the maps classified by χ and zero, and define

$$\mathrm{RelCell}_C(\chi) := C_{\epsilon_\chi} \otimes_S C_{\epsilon_0}.$$

Then $\mathrm{RelCell}_C(\chi)$ is compact as a commutative filtered C -algebra, is almost perfect as a filtered C -module, and its relative cotangent complex is perfect. More precisely,

$$L_{\mathrm{RelCell}_C(\chi)/C} \simeq \mathrm{RelCell}_C(\chi) \otimes_C P[1].$$

The two-sided bar construction has an exhaustive multiplicative filtration whose word-length k associated-graded term is

$$\mathrm{Sym}_C^k(P[1]).$$

Now let $C \rightarrow D$ be a morphism of pointwise connective filtered commutative ring spectra, put

$$Q := \mathrm{cofib}(C \rightarrow D),$$

and suppose Q is pointwise $(n+1)$ -connective. If

$$u : P \longrightarrow Q[-1]$$

is surjective on the first possible homotopy object and $\partial : Q[-1] \rightarrow C$ is the boundary morphism, define

$$\chi := \partial \circ u.$$

Then the canonical nullhomotopy of $P \rightarrow C \rightarrow D$ constructs a morphism

$$\mathrm{RelCell}_C(\chi) \longrightarrow D$$

whose relative cofibre is pointwise $(n+2)$ -connective.

Proof.

- (1) **Compactness.** Because P is a finite filtered cell, it is compact and perfect as a filtered C -module. The forgetful functor from commutative filtered C -algebras preserves filtered colimits, so the free-algebra adjunction makes $\mathrm{Sym}_C(P)$ compact. Mapping out of the pushout $\mathrm{RelCell}_C(\chi)$ is a finite pullback of mapping spaces, and filtered colimits of spaces commute with finite limits; hence $\mathrm{RelCell}_C(\chi)$ is compact as a commutative filtered C -algebra.

- (2) **Relative cotangent complex.** Cotangent base change for the defining pushout and transitivity for

$$C \longrightarrow \mathrm{Sym}_C(P) \xrightarrow{\epsilon_0} C,$$

whose composite is the identity, give

$$L_{\mathrm{RelCell}_C(\chi)/C} \simeq \mathrm{RelCell}_C(\chi) \otimes_C P[1].$$

This module is perfect.

- (3) **Bar filtration.** For the underlying filtered module, compute the pushout by the two-sided bar construction and filter its normalized realization by total symmetric degree in the occurrences of P . Internal multiplication preserves this degree, the endpoint differential determined by χ lowers it, and the zero endpoint kills positive degree. Thus the associated graded is the zero-zero self-intersection

$$\begin{aligned} C \otimes_{\mathrm{Sym}_C(P)} C &\simeq \mathrm{Sym}_C(0) \amalg_{\mathrm{Sym}_C(P)} \mathrm{Sym}_C(0) \\ &\simeq \mathrm{Sym}_C(P[1]). \end{aligned}$$

The word-length k layer is therefore $\mathrm{Sym}_C^k(P[1])$. It is pointwise $k(n+1)$ -connective and almost perfect by 3.4.5. Hence every fixed Postnikov range sees only finitely many word lengths; finite extensions of their finite-cell approximations prove that $\mathrm{RelCell}_C(\chi)$ is almost perfect as a filtered C -module.

- (4) **Connectivity improvement.** For the final statement, $\chi = \partial u$ supplies a canonical nullhomotopy of $P \rightarrow C \rightarrow D$, and hence a map $\mathrm{RelCell}_C(\chi) \rightarrow D$. The linear word-length layer of the induced map on relative cofibres is

$$P[1] \longrightarrow Q,$$

which is surjective on the first possible homotopy object. Every nonlinear layer contains at least two copies of $P[1]$ and is therefore pointwise at least $2(n+1)$ -connective. Since $2(n+1) \geq n+2$, octahedrality for

$$C \longrightarrow \mathrm{RelCell}_C(\chi) \longrightarrow D$$

shows that the remaining relative cofibre is pointwise $(n+2)$ -connective.

□

Lemma 3.4.14 (Composition of compact spectral algebra morphisms). *Let $\mathcal{C}$ be Sp , $\mathrm{Fil}(\mathrm{Sp})$, or $\mathrm{Gr}(\mathrm{Sp})$, and let*

$$A \longrightarrow C \longrightarrow D$$

be morphisms of commutative algebra objects. If C is compact as a commutative A -algebra and D is compact as a commutative C -algebra, then D is compact as a commutative A -algebra.

Proof.

- (1) **Choose a filtered target diagram.** Let $\{E_\alpha\}$ be a filtered diagram of commutative A -algebras and let

$$\varphi : D \longrightarrow \mathrm{colim}_\alpha E_\alpha$$

be an A -algebra map.

- (2) **Restrict the algebra map to C .** Its restriction along $C \rightarrow D$ is a point of

$$\mathrm{Map}_{\mathrm{CAlg}(\mathcal{C})_{A/}} \left(C, \mathrm{colim}_{\alpha} E_{\alpha} \right).$$

- (3) **Represent the C -structure at a finite stage.** Compactness of C implies that this C -algebra structure is represented at some stage E_{α_0} .
- (4) **Pass to a cofinal diagram of C -algebras.** After restricting to the cofinal filtered subcategory of stages under α_0 , all objects carry the induced compatible C -algebra structure.
- (5) **Use compactness of D .** Compactness of D in $\mathrm{CAlg}(\mathcal{C})_{C/}$ then implies that φ , together with every higher homotopy between such maps, is represented at a finite stage.
- (6) **Identify the mapping-space comparison.** Therefore the canonical map

$$\mathrm{colim}_{\alpha} \mathrm{Map}_{\mathrm{CAlg}(\mathcal{C})_{A/}} (D, E_{\alpha}) \longrightarrow \mathrm{Map}_{\mathrm{CAlg}(\mathcal{C})_{A/}} \left(D, \mathrm{colim}_{\alpha} E_{\alpha} \right)$$

is an equivalence.

- (7) **Conclude compactness.** Thus D is compact as a commutative A -algebra. □

Lemma 3.4.15 (Filtered surjective almost finite presentation and cotangent criterion). *Let*

$$F^{\bullet}A \longrightarrow F^{\bullet}B$$

be a morphism of pointwise connective filtered commutative ring spectra which is surjective on π_0 in every filtration degree. Suppose that $F^{\bullet}B$ is almost perfect as a filtered $F^{\bullet}A$ -module. Then the morphism is almost finitely presented in the sense of 3.3.4, and its relative cotangent complex is an almost-perfect filtered $F^{\bullet}B$ -module. The shifted relative cotangent complex is pointwise connective.

Proof.

- (1) **Initial relative cofiber.** Write A and B for the two filtered algebras and put

$$Q_0 := \mathrm{cofib}(A \rightarrow B).$$

Pointwise surjectivity on π_0 makes Q_0 pointwise 1-connective. Since A is perfect over itself and B is almost perfect over A , the relative cofibre Q_0 is almost perfect as a filtered A -module.

- (2) **Inductive relation-cell construction.** We construct compact filtered A -algebras

$$C_d \longrightarrow B, \quad d \geq 0,$$

starting with $C_0 = A$, such that

$$Q_d := \mathrm{cofib}(C_d \rightarrow B)$$

is pointwise $(d+1)$ -connective and both C_d and Q_d are almost perfect as filtered A -modules. Suppose C_d has been constructed. Then $Q_d[-1]$ is pointwise d -connective and almost perfect over A . By 3.4.2 in the filtered module category, its first possible homotopy object is finitely

presented in the pointwise heart. The images of the weight generators $A\langle w \rangle$ generate that heart, so there are weights $w_1, \dots, w_r$ and a morphism

$$G_d := \bigoplus_{j=1}^r A\langle w_j \rangle[d] \longrightarrow Q_d[-1]$$

surjective on pointwise π_d .

- (3) **Compact algebra approximants.** Base change the finite filtered cell to C_d :

$$P_d^{\text{rel}} := C_d \otimes_A G_d.$$

The induced map

$$u_d : P_d^{\text{rel}} \longrightarrow Q_d[-1]$$

remains surjective on the first possible homotopy object. Let $\partial_d : Q_d[-1] \rightarrow C_d$ be the boundary and put

$$\chi_d := \partial_d u_d, \quad C_{d+1} := \text{RelCell}_{C_d}(\chi_d).$$

By 3.4.13, the canonical morphism $C_{d+1} \rightarrow B$ has relative cofibre Q_{d+1} pointwise $(d+2)$ -connective, and C_{d+1} is almost perfect as a filtered C_d -module. Since C_d is almost perfect over A , the filtered restriction-transitivity lemma 3.4.11 makes C_{d+1} almost perfect over A . The cofibre sequence

$$C_{d+1} \longrightarrow B \longrightarrow Q_{d+1}$$

then makes Q_{d+1} almost perfect over A .

- (4) **Almost finite presentation.** Each relative relation-cell morphism $C_d \rightarrow C_{d+1}$ is compact as a commutative filtered algebra by 3.4.13. Compact algebra morphisms are stable under composition by 3.4.14. Hence every C_d is compact as a filtered A -algebra. The relative cotangent complex of each relation-cell stage is perfect by 3.4.13, and cotangent transitivity propagates perfectness to

$$L_{C_d/A}$$

for every d .

- (5) **Cotangent transitivity.** Let E be a pointwise connective m -truncated filtered commutative A -algebra and choose $d \geq m$. Since Q_d is pointwise $(d+1)$ -connective, $C_d \rightarrow B$ induces an equivalence on operadic Postnikov truncations through degree m . By 3.3.2,

$$\begin{aligned} \text{Map}_{\text{CAlg}(\text{Fil}(\text{Sp}))_{A'}}(B, E) &\simeq \text{Map}_{\text{CAlg}(\text{Fil}(\text{Sp}))_{A'}}(\tau_{\leq m}^{\text{CAlg}} B, E) \\ &\simeq \text{Map}_{\text{CAlg}(\text{Fil}(\text{Sp}))_{A'}}(\tau_{\leq m}^{\text{CAlg}} C_d, E) \\ &\simeq \text{Map}_{\text{CAlg}(\text{Fil}(\text{Sp}))_{A'}}(C_d, E). \end{aligned}$$

The right-hand mapping functor preserves filtered colimits because C_d is compact. Since m was arbitrary, $A \rightarrow B$ is almost finitely presented in the sense of 3.3.4.

- (6) **Perfect cotangent approximants.** It remains to prove that the relative cotangent complex is almost perfect. For every $d \geq 1$, cotangent transitivity for

$$A \longrightarrow C_d \longrightarrow B$$

gives a cofibre sequence

$$B \otimes_{C_d} L_{C_d/A} \longrightarrow L_{B/A} \longrightarrow L_{B/C_d}.$$

Put

$$P_d^{\text{cot}} := B \otimes_{C_d} L_{C_d/A}.$$

Since $L_{C_d/A}$ is perfect, P_d^{cot} is a perfect filtered B -module.

- (7) **Hurewicz estimate.** For $d \geq 1$, the morphism $C_d \rightarrow B$ is pointwise an isomorphism on π_0 , and its relative cofibre Q_d is pointwise $(d+1)$ -connective. Hence 3.4.12 makes

$$L_{B/C_d}$$

pointwise $(d+1)$ -connective. Consequently, for every fixed integer $m \geq 0$ and every $d \geq \max\{m, 1\}$, the preceding transitivity sequence induces an equivalence

$$\tau_{\leq m} P_d^{\text{cot}} \xrightarrow{\simeq} \tau_{\leq m} L_{B/A}.$$

- (8) **Truncation compactness.** The module P_d^{cot} is perfect and therefore compact in the filtered B -module category. Its m -truncation is consequently compact in the full subcategory of pointwise m -truncated filtered B -modules: for every filtered diagram $\{E_\alpha\}$ in that subcategory,

$$\begin{aligned} \text{Map}\left(\tau_{\leq m} P_d^{\text{cot}}, \text{colim}_\alpha E_\alpha\right) &\simeq \text{Map}\left(P_d^{\text{cot}}, \text{colim}_\alpha E_\alpha\right) \\ &\simeq \text{colim}_\alpha \text{Map}(P_d^{\text{cot}}, E_\alpha) \\ &\simeq \text{colim}_\alpha \text{Map}(\tau_{\leq m} P_d^{\text{cot}}, E_\alpha). \end{aligned}$$

Therefore

$$\tau_{\leq m} L_{B/A}$$

is compact in the pointwise m -truncated filtered module category for every m .

- (9) **Connectivity of the shifted cotangent complex.** Finally, the pointwise-surjective case of 3.4.12 makes $L_{B/A}$ pointwise 1-connective. Thus $L_{B/A}$ is bounded below, and the preceding truncation-compactness statement proves directly that $L_{B/A}$ is almost perfect. Shifting by $[-1]$ shows that the relative shifted cotangent complex is pointwise connective.

□

Lemma 3.4.16 (Cotangent completeness for the canonical adic filtration). *Let $A \rightarrow B$ be connective and surjective on π_0 , and suppose*

$$M := L_{B/A}[-1]$$

is connective and almost perfect. Then the shifted cotangent complex of the completed canonical adic filtration is a connective almost-perfect complete filtered B -module, and completion induces an equivalence

$$L_{B/\text{adic}(A \rightarrow B)}[-1] \xrightarrow{\simeq} L_{B/\widehat{\text{adic}(A \rightarrow B)}}[-1].$$

Proof.

- (1) **Completed filtered algebra.** Put

$$F^\bullet C := \widehat{\text{adic}}(A \rightarrow B).$$

By 3.3.1,

$$\text{Gr } C \simeq \text{Sym}_B^{\text{gr}}(M\langle 1 \rangle).$$

The associated graded is connective. Since $F^\bullet C$ is complete, 3.3.5, applied to every shifted tail, shows that $F^\bullet C$ is pointwise connective. Its augmentation to the filtered algebra B concentrated in filtration degree zero is therefore pointwise surjective on π_0 .

- (2) **Almost-perfect augmentation module.** Regard B as a filtered C -module concentrated in filtration degree zero. It is complete, and its associated graded is the augmentation module over $\text{Sym}_B^{\text{gr}}(M\langle 1 \rangle)$. By 3.4.9, that graded module is almost perfect. The filtered detection theorem 3.4.2 therefore gives

$$B \in \text{APerf}(F^\bullet C).$$

Apply 3.4.15 to the filtered augmentation $F^\bullet C \rightarrow B$. Its shifted cotangent complex

$$L_{B/F^\bullet C}[-1]$$

is connective and almost perfect. Because the base filtered algebra is complete, 3.4.3 makes this shifted cotangent complex complete.

- (3) **Define the cotangent comparison.** Put

$$C_0^\bullet := \text{adic}(A \rightarrow B), \quad C_1^\bullet := \widehat{\text{adic}}(A \rightarrow B),$$

and let $q : C_0^\bullet \rightarrow C_1^\bullet$ be the completion morphism. The induced base-change map gives the canonical cotangent comparison

$$\kappa : L_{B/C_0^\bullet}[-1] \longrightarrow L_{B/C_1^\bullet}[-1].$$

- (4) **Apply naturality of graded cotangent compatibility.** Naturality of the first equivalence in 3.3.2 gives the commutative diagram

$$\begin{array}{ccc} \text{Gr } L_{B/C_0^\bullet}[-1] & \xrightarrow{\text{Gr}(\kappa)} & \text{Gr } L_{B/C_1^\bullet}[-1] \\ \simeq \downarrow & & \downarrow \simeq \\ L_{B/\text{Gr } C_0}[-1] & \longrightarrow & L_{B/\text{Gr } C_1}[-1]. \end{array}$$

- (5) **Identify the associated-graded comparison.** Completion preserves associated graded, so $\text{Gr}(q)$ is an equivalence. Cotangent functoriality therefore makes the lower horizontal morphism, and hence $\text{Gr}(\kappa)$, an equivalence.
- (6) **Identify the source before completion.** Before completion, 3.3.2 also identifies

$$L_{B/C_0^\bullet}[-1] \simeq \text{wt}_1(M),$$

which is complete.

- (7) **Use completeness of the target.** The target is complete by the preceding step.
- (8) **Apply complete filtered detection.** Thus κ is an associated-graded equivalence between complete filtered modules, and 3.3.4 makes it an equivalence.

□

This completes the technical package needed later: finite-cell approximation controls algebra and module finiteness, associated graded detects complete filtered objects, and filtered Hurewicz comparison keeps the shifted cotangent classifier connective and almost perfect after completion. We can now define the nonlinear fixed-centre domain on which formal-order reconstruction converges.

3.5 The complete fixed-centre reconstruction domain

The preceding estimates isolate the desired nonlinear domain. A complete almost finitely presented augmentation is a fixed point of canonical adic completion; on the connective, surjective, almost-finitely-presented sector, that fixed-point condition will be identified with derived completeness for the degree-zero augmentation ideal.

Two facts are needed before the comonadic theorem. First, canonical adic completion must be identified with its Amitsur and ideal-theoretic descriptions. Second, structured square-zero augmentations generated by connective almost-perfect cotangent data must already lie in the complete domain. Together these facts allow the cotangent and square-zero functors to restrict to one reconstruction category.

3.5.1 Canonical completeness and the complete almost finitely presented sector

Let $A \rightarrow B$ be a connective augmented arrow surjective on π_0 , and write

$$I := \ker(\pi_0(A) \rightarrow \pi_0(B)).$$

Derived I -completeness is used in the sense of [37, Sections 7.3.3–7.3.4]. It gives the intrinsic fixed-point description of the canonical adic completion.

Definition 3.5.1 (Complete almost finitely presented augmentation). Fix a morphism $R \rightarrow B$ of connective commutative ring spectra. Define

$$\mathrm{CAlg}(\mathrm{Sp})_{R//B}^{\mathrm{cn}, \mathrm{afp}, \wedge} \subseteq \mathrm{CAlg}(\mathrm{Sp})_{R//B}^{\mathrm{cn}}$$

to be the full subcategory spanned by augmented arrows $A \rightarrow B$ such that:

- (i) $\pi_0(A) \rightarrow \pi_0(B)$ is surjective;
- (ii) B is almost perfect as an A -module;
- (iii) the canonical adic completion morphism of 3.3.3 is an equivalence

$$A \xrightarrow{\sim} A_B^\wedge.$$

Objects of this full subcategory are **complete almost finitely presented augmentations** over R with centre B . Completeness is a fixed-point property of a functorially constructed completion.

The fixed-point condition is compatible with the augmentation:

$$\begin{array}{ccc} A & \xrightarrow{\quad} & A_B^\wedge \\ & \searrow & \downarrow \\ & & B. \end{array}$$

Lemma 3.5.1 (Nilpotent modules are derived complete). *Let A be a connective commutative ring spectrum and let $I \subseteq \pi_0(A)$ be a finitely generated ideal. If an A -module N is annihilated by a power I^m , then N is derived I -complete. The full subcategory of derived I -complete A -modules is stable and closed under limits.*

Proof.

- (1) **Use nilpotence on homotopy modules.** Every homotopy module $\pi_k(N)$ is annihilated by I^m .
- (2) **Identify ordinary adic completeness.** Its ordinary I -adic tower is therefore eventually constant, so it is I -adically complete and hence I -complete.
- (3) **Apply the spectral completion criterion.** The homotopy-group criterion for spectral derived completion, [37, Theorem 7.3.4.1], implies that N is derived I -complete.
- (4) **Record stability under limits.** Stability and closure under limits are the formal properties of the complete subcategory in the completion theory of [37, Section 7.3.3].

□

Lemma 3.5.2 (A complete filtration gives derived ideal completeness). *Let $F^\bullet C$ be a complete pointwise connective filtered commutative ring spectrum, put*

$$A := F^0 C, \quad B := \mathrm{Gr}^0 C, \quad I := \ker(\pi_0(A) \rightarrow \pi_0(B)),$$

and suppose I is finitely generated. If $F^\bullet M$ is a complete filtered C -module, then $F^0 M$ is derived I -complete. In particular, A is derived I -complete.

Proof.

- (1) **Express filtration zero as an inverse limit.** Completeness gives

$$F^0 M \simeq \varprojlim_n \mathrm{cofib}(F^n M \rightarrow F^0 M).$$

- (2) **Identify the augmentation ideal.** Since $F^1 C \rightarrow F^0 C \rightarrow B$ is a cofibre sequence of connective spectra, the ideal I is the image of $\pi_0 F^1 C \rightarrow \pi_0 F^0 C$.
- (3) **Use multiplicativity of the filtration.** Multiplicativity of the filtration sends a product of n lifts from $F^1 C$ into $F^n C$, and the action of $F^n C$ on $F^0 M$ lands in $F^n M$. Hence I^n acts trivially on

$$\mathrm{cofib}(F^n M \rightarrow F^0 M).$$

- (4) **Complete the finite quotients.** Each finite filtration quotient is therefore derived I -complete by 3.5.1.
- (5) **Pass completeness through the inverse limit.** The same lemma shows that the derived I -complete subcategory is closed under limits.
- (6) **Conclude derived completeness.** Taking the displayed inverse limit proves the claim.

□

Theorem 3.5.1 (Hodge–Adams interpretation of canonical adic completion). *Let $A \rightarrow B$ be a morphism of connective commutative ring spectra which is surjective on π_0 . Then there is a canonical equivalence*

$$A_B^\wedge \simeq \operatorname{Tot}_{[n] \in \Delta} B^{\otimes_A (n+1)}.$$

More generally, for every bounded-below A -module N , filtration zero of the completion of

$$N \otimes_A \operatorname{adic}(A \rightarrow B)$$

is canonically equivalent to the Amitsur totalization

$$\operatorname{Tot}_{[n] \in \Delta} (N \otimes_A B^{\otimes_A (n+1)}).$$

The comparison uses only connectivity and surjectivity on π_0 .

Proof.

- (1) **Identify with the Hodge filtration in the literature.** The universal property used in 3.3.1 identifies $\operatorname{adic}(A \rightarrow B)$ with the relative $\mathbb{E}_\infty$ -Hodge filtration attached to the augmentation $A \rightarrow B$. Through 3.3.1, the completion is therefore the complete relative Hodge filtration considered in [2, Construction 2.1 and Lemma 2.2].
- (2) **Apply the Hodge–Adams comparison.** For a connective morphism $A \rightarrow B$ that is surjective on π_0 , [2, Proposition 2.17] identifies the completed N -valued relative Hodge filtration with a complete filtration on the B -Adams completion

$$N_B^\wedge := \operatorname{Tot}_{[n] \in \Delta} (N \otimes_A B^{\otimes_A (n+1)})$$

for every bounded-below A -module N . Its filtration-zero term is therefore the displayed Amitsur totalization. Setting $N = A$ gives the canonical equivalence for $A_B^\wedge$.

□

Theorem 3.5.2 (Ideal-theoretic interpretation of canonical adic completion). *Let $A \rightarrow B$ be a morphism of connective commutative ring spectra which is surjective on π_0 and almost finitely presented. Put*

$$I := \ker(\pi_0(A) \rightarrow \pi_0(B)).$$

Then I is finitely generated and, for every bounded-below A -module N , there is a canonical equivalence

$$N_I^\wedge \simeq \operatorname{Tot}_{[n] \in \Delta} (N \otimes_A B^{\otimes_A (n+1)}).$$

In particular,

$$A_B^\wedge \simeq A_I^\wedge,$$

where $A_I^\wedge$ denotes derived I -completion in spectral algebraic geometry. Consequently

$$A \longrightarrow A_B^\wedge$$

is an equivalence if and only if A is derived I -complete.

Proof.

- (1) **Finite augmentation ideal.** Almost finite presentation and surjectivity on π_0 give

$$\pi_0(B) \simeq \pi_0(A)/I$$

with I finitely generated. Put

$$X := \mathrm{Spec}(A), \quad Z := \mathrm{Spec}(B).$$

Then $Z \rightarrow X$ is an almost finitely presented closed immersion with underlying closed support $V(I)$.

- (2) **Čech descent on the formal completion.** Let $\widehat{X}_Z$ be the formal completion of X along this support. Formal-completion descent for an almost finitely presented closed immersion identifies bounded-below quasi-coherent modules on $\widehat{X}_Z$ with the totalization over the Čech nerve of $Z \rightarrow X$; see [23, Corollary 3.1.6]. The n -simplices of that Čech nerve are

$$Z^{\times_X(n+1)} \simeq \mathrm{Spec}(B^{\otimes_A(n+1)}),$$

so the restriction of the quasi-coherent module associated with N has global sections

$$\mathrm{Tot}_{[n] \in \Delta} (N \otimes_A B^{\otimes_A(n+1)}).$$

The descent geometry can be displayed as

$$\cdots \rightrightarrows \mathrm{Spec}(B \otimes_A B) \rightrightarrows \mathrm{Spec}(B) \longrightarrow \widehat{X}_Z.$$

- (3) **Derived completion description.** Since I is finitely generated, the same formal completion is

$$\widehat{X}_Z \simeq \mathrm{Spf}(A, I).$$

The affine formal-completion theorem [37, Corollary 8.2.4.15] identifies quasi-coherent modules on $\mathrm{Spf}(A, I)$ with derived I -complete A -modules. Under this equivalence, the restriction of N corresponds to $N_I^\wedge$. Thus the two descriptions of the same formal restriction yield

$$N_I^\wedge \simeq \mathrm{Tot}_{[n] \in \Delta} (N \otimes_A B^{\otimes_A(n+1)}).$$

- (4) **Compare the two descriptions.** Taking $N = A$ and using 3.5.1 gives

$$A_B^\wedge \simeq A_I^\wedge.$$

The fixed points of derived I -completion are precisely the derived I -complete A -modules, which proves the final assertion. On the complete almost finitely presented theorem domain, this is the same completion criterion recorded in [10, Proposition 2.18].

□

Proposition 3.5.1 (Intrinsic characterization of complete almost finitely presented augmentations). *Let $A \rightarrow B$ be connective and surjective on π_0 , and put $I = \ker(\pi_0(A) \rightarrow \pi_0(B))$. The following are equivalent:*

- (i) $A \rightarrow B$ belongs to $\mathrm{CAlg}(\mathrm{Sp})_{R//B}^{\mathrm{cn}, \mathrm{afp}, \wedge}$;
- (ii) B is almost perfect over A and A is derived I -complete;
- (iii) $A \rightarrow B$ is almost finitely presented and A is derived I -complete.

Under these conditions I is finitely generated and

$$\widehat{\mathrm{Gr\,adic}}(A \rightarrow B) \simeq \mathrm{Sym}_B^{\mathrm{gr}}(L_{B/A}[-1]\langle 1 \rangle).$$

Proof.

- (1) **Compare the finiteness conditions.** The equivalence of (ii) and (iii) is 3.4.1.
- (2) **Deduce finite generation of the ideal.** Under either condition the augmentation is almost finitely presented, so I is finitely generated.
- (3) **Compare the two completions.** By 3.5.2,

$$A_B^\wedge \simeq A_I^\wedge.$$

- (4) **Identify the fixed-point condition.** Thus the fixed-point condition in (i) is exactly derived I -completeness, proving the equivalence of all three conditions.
- (5) **Record the associated graded.** The associated-graded formula is 3.3.1.

□

3.5.2 Structured square-zero finiteness and coherent refinement

Lemma 3.5.3 (Structured square-zero finiteness and coherent refinement). *Let $R \rightarrow B$ be a morphism of connective commutative ring spectra, let M be a connective almost-perfect B -module, and let*

$$B_\alpha \longrightarrow B$$

be the structured square-zero extension classified by

$$\alpha : L_{B/R}[-1] \longrightarrow M.$$

Then B is almost perfect as a B_α -module and $B_\alpha \rightarrow B$ is almost finitely presented. If B is coherent, then

$$I_\alpha := \ker(\pi_0(B_\alpha) \rightarrow \pi_0(B))$$

is finitely presented as a $\pi_0(B_\alpha)$ -module and B_α is a coherent connective commutative ring spectrum.

Proof.

- (1) **Almost finite presentation.** The spectral square-zero finiteness theorem [37, Corollary 5.2.2.5] says that a structured square-zero extension by a connective almost-perfect coefficient is almost finitely presented. Since the map is surjective on π_0 , 3.4.1 then gives

$$B \in \mathrm{APerf}(B_\alpha).$$

- (2) **Degree-zero kernel.** Assume now that B is coherent. Put

$$A_0 := \pi_0(B_\alpha), \quad B_0 := \pi_0(B).$$

The long exact sequence of the square-zero fibre sequence gives

$$\pi_1(B) \longrightarrow \pi_0(M) \longrightarrow I_\alpha \longrightarrow 0.$$

By 3.4.1, $\pi_1(B)$ and $\pi_0(M)$ are finitely presented over the coherent ring B_0 . The image of $\pi_1(B) \rightarrow \pi_0(M)$ is therefore finitely generated, hence finitely presented over B_0 ; its cokernel I_α is consequently finitely presented over B_0 .

The degree-zero kernel of a square-zero extension is square-zero, so $I_\alpha^2 = 0$ and

$$B_0 \simeq A_0/I_\alpha.$$

A finite B_0 -generating set of I_α is also a finite A_0 -generating set. Applying 3.7.1 to $A_0 \twoheadrightarrow B_0$ now makes I_α finitely presented over A_0 .

- (3) **Spectral coherence.** Apply the square-zero case of 3.7.2 to $A_0 \twoheadrightarrow B_0$. It makes A_0 coherent. Moreover every homotopy module of B and M is finitely presented over B_0 ; since $A_0 \twoheadrightarrow B_0$ has finitely generated kernel, these modules are finitely presented over A_0 by 3.7.1. The long exact homotopy sequence of

$$M \longrightarrow B_\alpha \longrightarrow B$$

then expresses each $\pi_n(B_\alpha)$ through kernels, cokernels, and extensions of finitely presented A_0 -modules. Coherence of A_0 keeps these modules finitely presented, proving spectral coherence. $\square$

Corollary 3.5.1 (Structured square-zero objects lie in the complete almost finitely presented sector). *Fix a morphism $R \rightarrow B$ of connective commutative ring spectra. Let M be connective and almost perfect, let*

$$\alpha : L_{B/R}[-1] \longrightarrow M,$$

and let $B_\alpha \rightarrow B$ be the corresponding structured square-zero extension over R . Then

$$B_\alpha \rightarrow B \in \mathrm{CAlg}(\mathrm{Sp})_{R/B}^{\mathrm{cn}, \mathrm{afp}, \wedge}.$$

Proof.

- (1) **Establish finiteness of the augmentation.** By 3.5.3, B is almost perfect over B_α , and the augmentation is almost finitely presented.
- (2) **Deduce finite generation of the ideal.** Hence its degree-zero augmentation ideal I_α is finitely generated.

- (3) **Use square-zero nilpotence.** The square-zero construction gives $I_\alpha^2 = 0$; hence I_α^2 annihilates the underlying B_α -module B_α .
- (4) **Apply derived completeness.** Therefore B_α is derived I_α -complete by 3.5.1.
- (5) **Invoke the intrinsic characterization.** Apply 3.5.1.

□

We now have the theorem domain and know that structured square-zero formation preserves it. Completeness comes from the canonical adic filtration, finiteness from the cotangent criterion, and the ideal-theoretic comparison makes the fixed-point condition intrinsic. These are exactly the inputs required for the fixed-centre cotangent–square-zero theorem.

3.6 Comonadic reconstruction from anchored shifted-cotangent data

Fix a connective centre B over R . The principal theorem compares two constructions: an augmentation $A \rightarrow B$ produces the anchored shifted cotangent object

$$L_{B/R}[-1] \longrightarrow L_{B/A}[-1],$$

whereas an anchored map $L_{B/R}[-1] \rightarrow M$ produces a structured square-zero augmentation of B . These assignments are adjoint. The proof of comonadicity has three stages: restrict the adjunction to the complete theorem domain, prove conservativity by associated-graded detection, and verify the split totalizations required by Barr–Beck through filtered reconstruction.

The adjunction unit already exhibits the resulting coalgebra structure:

$$A \xrightarrow{\eta_A} \mathrm{sqz}_{B/R} \cot_{B/R}(A)$$

$$\cot_{B/R}(A) \xrightarrow[\cot_{B/R}(\eta_A)]{\quad} \mathbb{C}_{B/R}^{\mathrm{sqz}} \cot_{B/R}(A).$$

The lower map is therefore generated by the adjunction rather than imposed independently. Comonadicity will show that this anchored cotangent coalgebra determines the complete augmentation. Coherence of B is unnecessary here; it enters only when the complete theory is restricted to bounded coherent Artinian tests.

3.6.1 Anchored shifted-cotangent data

Definition 3.6.1 (Anchored shifted-cotangent category). Fix a morphism of connective commutative ring spectra

$$R \longrightarrow B.$$

Define

$$\mathrm{AnchCot}_{B/R}^{\mathrm{sp}} := (\mathrm{Mod}_B)_{L_{B/R}[-1]}.$$

Thus an object is a pair (M, α) with

$$\alpha : L_{B/R}[-1] \longrightarrow M.$$

Let

$$\mathrm{AnchCot}_{B/R, \geq 0}^{\mathrm{aperf}}$$

be the full subcategory in which M is connective and almost perfect.

Definition 3.6.2 (Shifted-cotangent and structured square-zero functors). For an augmented R -algebra $A \rightarrow B$, define

$$\mathrm{cot}_{B/R}(A) := (L_{B/R}[-1] \rightarrow L_{B/A}[-1]),$$

where the anchor is induced by cotangent transitivity.

For $(M, \alpha) \in \mathrm{AnchCot}_{B/R}^{\mathrm{sp}}$, let

$$B \xrightarrow{d_\alpha} B \oplus M[1]$$

be the section classified by the degree-one derivation $L_{B/R} \rightarrow M[1]$. Define

$$B_\alpha := B \times_{B \oplus M[1]} B$$

so that

$$\begin{array}{ccc} B_\alpha & \longrightarrow & B \\ \downarrow & & \downarrow d_\alpha \\ B & \xrightarrow{d_0} & B \oplus M[1] \end{array}$$

is Cartesian, where d_0 is the zero derivation. Set

$$\mathrm{sqz}_{B/R}(M, \alpha) := (B_\alpha \rightarrow B).$$

This structured square-zero functor is defined in unrestricted commutative ring spectra; it restricts to connective arrows on connective coefficients by 2.7.1.

Proposition 3.6.1 (Shifted-cotangent–structured-square-zero adjunction). *There is a canonical adjunction*

$$\mathrm{cot}_{B/R} \dashv \mathrm{sqz}_{B/R}.$$

More explicitly,

$$\mathrm{Map}_{\mathrm{AnchCot}_{B/R}^{\mathrm{sp}}}(\mathrm{cot}_{B/R}(A), (M, \alpha)) \simeq \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})_{R//B}}(A, B_\alpha).$$

A point of the right-hand mapping space is equivalently a commutative lifting diagram

$$\begin{array}{ccc} A & \dashrightarrow & B_\alpha \\ \downarrow & & \downarrow \\ B & \xlongequal{\quad} & B. \end{array}$$

Proof.

- (1) **Interpret the right-hand mapping space.** A point of the right-hand mapping space is a lift of the augmentation $A \rightarrow B$ through the structured square-zero extension classified by $L_{B/R} \rightarrow M[1]$.

- (2) **Apply the square-zero lifting classifier.** By 2.11.1, such lifts are equivalent to maps

$$L_{B/A} \longrightarrow M[1]$$

whose restriction along $L_{B/R} \rightarrow L_{B/A}$ is the specified classifying term.

- (3) **Shift to anchored cotangent data.** Shifting by $[-1]$ identifies this exactly with a morphism from $\cot_{B/R}(A)$ to (M, α) in the undercategory of $L_{B/R}[-1]$.

□

3.6.2 Filtered reconstruction

Lemma 3.6.1 (Filtered reconstruction of a complete almost finitely presented augmentation). *Let $F^\bullet C$ be a complete filtered commutative R -algebra whose associated-graded terms are connective. Suppose its filtration-zero augmentation is*

$$A := F^0 C \longrightarrow B := \mathrm{Gr}^0 C$$

and that for a connective almost-perfect B -module M there is an equivalence

$$\mathrm{Gr} C \simeq \mathrm{Sym}_B^{\mathrm{gr}}(M\langle 1 \rangle).$$

Then $A \rightarrow B$ belongs to $\mathrm{CAlg}(\mathrm{Sp})_{R//B}^{\mathrm{cn}, \mathrm{afp}, \wedge}$. Moreover the filtered augmentation module B , concentrated in filtration degree zero, is almost perfect over $F^\bullet C$.

Proof.

- (1) **Connectivity.** Apply 3.3.5 to every shifted filtration tail of $F^\bullet C$. Every $F^n C$, in particular $A = F^0 C$, is connective. Since

$$F^1 C \longrightarrow F^0 C \longrightarrow B$$

is a cofibre sequence with connective first term, $\pi_0(A) \rightarrow \pi_0(B)$ is surjective.

- (2) **Almost-perfect augmentation module.** Regard B as the filtered C -module concentrated in filtration degree zero. It is complete. Its associated graded is the augmentation module over $\mathrm{Sym}_B^{\mathrm{gr}}(M\langle 1 \rangle)$, which is almost perfect by 3.4.9. Hence 3.4.2 makes B almost perfect as a filtered C -module and, after evaluation at filtration zero, gives

$$B \in \mathrm{APerf}(A).$$

The spectral criterion for almost finite presentation therefore makes the degree-zero augmentation ideal finitely generated.

- (3) **Derived completeness.** Finally apply 3.5.2 to the complete filtered algebra $F^\bullet C$, viewed as a module over itself. Its filtration-zero term A is derived complete for that augmentation ideal. The intrinsic characterization 3.5.1 now gives

$$A \rightarrow B \in \mathrm{CAlg}(\mathrm{Sp})_{R//B}^{\mathrm{cn}, \mathrm{afp}, \wedge}.$$

□

3.6.3 Comonadicity

Theorem 3.6.1 (Shifted-cotangent-structured-square-zero comonadicity). *Let $R \rightarrow B$ be a morphism of connective commutative ring spectra. The adjunction of 3.6.1 restricts to*

$$\mathrm{cot}_{B/R} : \mathrm{CAlg}(\mathrm{Sp})_{R//B}^{\mathrm{cn}, \mathrm{afp}, \wedge} \rightleftarrows \mathrm{AnchCot}_{B/R, \geq 0}^{\mathrm{aperf}} : \mathrm{sqz}_{B/R},$$

and the left adjoint $\mathrm{cot}_{B/R}$ is comonadic.

Proof.

- (1) **Restriction to the theorem domain.** The right adjoint lands in the displayed source category by 3.5.1. Conversely, if $A \rightarrow B$ is complete almost finitely presented, 3.5.1 makes the augmentation almost finitely presented. Hence $L_{B/A}$ is almost perfect, and surjectivity on π_0 makes $L_{B/A}[-1]$ connective. Thus the shifted-cotangent functor lands in $\mathrm{AnchCot}_{B/R, \geq 0}^{\mathrm{aperf}}$. This is the spectral form of the theorem domain appearing in [10, Theorem 2.25].
- (2) **Conservativity.** Suppose a morphism $A \rightarrow A'$ over B induces an equivalence

$$L_{B/A}[-1] \xrightarrow{\simeq} L_{B/A'}[-1]$$

of anchored shifted cotangent complexes. Functoriality of canonical adic formation gives a morphism between the completed filtrations. By 3.3.1, its associated graded is the free graded symmetric algebra on the displayed cotangent equivalence and is therefore an equivalence. Both filtered algebras are complete, so 3.3.4 makes the filtered morphism an equivalence. Taking filtration zero and using that both augmented arrows are fixed by canonical adic completion gives $A \simeq A'$.

- (3) **Split totalization data.** Let $A^\bullet$ be a cosimplicial object of $\mathrm{CAlg}(\mathrm{Sp})_{R//B}^{\mathrm{cn}, \mathrm{afp}, \wedge}$ which is $\mathrm{cot}_{B/R}$ -split in the Barr–Beck sense: there is an augmentation

$$(M, \alpha) \longrightarrow \mathrm{cot}_{B/R}(A^\bullet)$$

in $\mathrm{AnchCot}_{B/R, \geq 0}^{\mathrm{aperf}}$ making the resulting augmented cosimplicial object split. In particular,

$$(M, \alpha) \xrightarrow{\simeq} \mathrm{Tot}_{[n] \in \Delta} \mathrm{cot}_{B/R}(A^n),$$

and this totalization is absolute: it is preserved by every functor out of $\mathrm{AnchCot}_{B/R, \geq 0}^{\mathrm{aperf}}$. After forgetting the anchors, the underlying split cosimplicial B -module therefore has absolute totalization

$$M \xrightarrow{\simeq} \mathrm{Tot}_{[n] \in \Delta} L_{B/A^n}[-1].$$

- (4) **Totalization in filtered algebras.** Form

$$A^u := \mathrm{Tot}_{[n] \in \Delta} A^n$$

in the unrestricted category $\mathrm{CAlg}(\mathrm{Sp})_{R//B}$, and form, likewise in unrestricted filtered commutative spectra, the complete filtered algebra

$$F^\bullet C := \mathrm{Tot}_{[n] \in \Delta} \widehat{\mathrm{adic}}(A^n \rightarrow B).$$

Limits preserve complete filtered objects. Evaluation in filtration zero commutes with limits, and every A^n is fixed by completion, so

$$F^0 C \simeq A^u.$$

The totalization has been formed in unrestricted commutative ring spectra; filtered reconstruction will show that A^u is connective.

- (5) **Commute associated graded with totalization.** Associated graded commutes with this totalization. Indeed each graded piece is the cofibre of two filtration evaluations; filtration evaluation preserves limits, and in a stable category finite cofibres are finite limits and therefore commute with arbitrary limits.
- (6) **Compute the totalized associated graded.** Hence

$$\begin{aligned} \mathrm{Gr} C &\simeq \mathrm{Tot}_{[n] \in \Delta} \widehat{\mathrm{Gr} \mathrm{adic}(A^n \rightarrow B)} \\ &\simeq \mathrm{Tot}_{[n] \in \Delta} \mathrm{Sym}_B^{\mathrm{gr}}(L_{B/A^\bullet}[-1]\langle 1 \rangle). \end{aligned}$$

- (7) **Use splitness of the cotangent cosimplicial module.** Because the underlying augmented cosimplicial B -module $L_{B/A^\bullet}[-1]$ is split, its totalization is absolute.
- (8) **Identify the free graded algebra.** Therefore the graded free commutative-algebra functor preserves this totalization, giving

$$\begin{aligned} \mathrm{Gr} C &\simeq \mathrm{Sym}_B^{\mathrm{gr}} \left(\left(\mathrm{Tot}_{[n] \in \Delta} L_{B/A^n}[-1] \right) \langle 1 \rangle \right) \\ &\simeq \mathrm{Sym}_B^{\mathrm{gr}}(M\langle 1 \rangle). \end{aligned}$$

- (9) **Record finiteness of the totalized coefficient.** Since

$$(M, \alpha) \in \mathrm{AnchCot}_{B/R, \geq 0}^{\mathrm{aperf}},$$

the underlying B -module M is connective and almost perfect.

- (10) **Recover the connective totalization.** Lemma 3.6.1 now shows that $A^u = F^0 C$ is connective and that

$$A^u \rightarrow B$$

belongs to the complete almost finitely presented sector. Because the inclusion of connective commutative ring spectra is full, this unrestricted totalization therefore computes the required totalization in the source category. Write

$$A := A^u$$

from now on.

- (11) **Construct the shifted-cotangent comparison.** Write

$$C_n^\bullet := \widehat{\mathrm{adic}(A^n \rightarrow B)}.$$

The totalization cone defining $F^\bullet C$ has canonical projection morphisms

$$p_n : F^\bullet C \longrightarrow C_n^\bullet$$

of filtered commutative R -algebras, all compatible with the displayed augmentations to B . Cotangent functoriality in the base therefore gives natural morphisms

$$L_{B/F^\bullet C}[-1] \longrightarrow L_{B/C_n^\bullet}[-1].$$

Their cosimplicial compatibility is inherited from the totalization cone. By the universal property of the limit they assemble to the canonical comparison

$$\gamma : L_{B/F^\bullet C}[-1] \longrightarrow \operatorname{Tot}_{[n] \in \Delta} L_{B/C_n^\bullet}[-1].$$

- (12) **Apply associated graded to the comparison.** Apply associated graded to γ . Since associated graded commutes with the totalization, and since 3.3.2 is natural in the filtered algebra, $\operatorname{Gr}(\gamma)$ sits in the canonical comparison diagram

$$\begin{array}{ccc} \operatorname{Gr} L_{B/F^\bullet C}[-1] & \xrightarrow{\operatorname{Gr}(\gamma)} & \operatorname{Tot}_{[n] \in \Delta} \operatorname{Gr} L_{B/C_n^\bullet}[-1] \\ \simeq \downarrow & & \downarrow \simeq \\ L_{B/\operatorname{Gr} C}[-1] & \longrightarrow & \operatorname{Tot}_{[n] \in \Delta} L_{B/\operatorname{Gr} C_n}[-1]. \end{array}$$

- (13) **Define the stagewise cotangent modules.** Put

$$M_n := L_{B/A^n}[-1].$$

- (14) **Compute the associated-graded algebras.** By 3.3.1 and the calculation in the preceding step,

$$\operatorname{Gr} C_n \simeq \operatorname{Sym}_B^{\operatorname{gr}}(M_n\langle 1 \rangle), \quad \operatorname{Gr} C \simeq \operatorname{Sym}_B^{\operatorname{gr}}(M\langle 1 \rangle).$$

- (15) **Compute the graded relative cotangent complexes.** Cotangent transitivity for the zero augmentation of a free graded commutative B -algebra gives canonical equivalences

$$L_{B/\operatorname{Gr} C_n}[-1] \simeq M_n\langle 1 \rangle, \quad L_{B/\operatorname{Gr} C}[-1] \simeq M\langle 1 \rangle.$$

- (16) **Identify the lower comparison map.** Under these equivalences the lower horizontal morphism is exactly

$$M\langle 1 \rangle \longrightarrow \operatorname{Tot}_{[n] \in \Delta} M_n\langle 1 \rangle.$$

- (17) **Use split totalization.** This is an equivalence because the augmented cosimplicial module $M \rightarrow M_\bullet$ is split and hence has absolute totalization. Therefore $\operatorname{Gr}(\gamma)$ is an equivalence.

- (18) **Completeness of the comparison.** For every n , the shifted cotangent module $M_n = L_{B/A^n}[-1]$ is connective and almost perfect by the theorem-domain condition. Hence 3.4.16 makes

$$L_{B/C_n^\bullet}[-1]$$

a complete filtered B -module. Its totalization, the target of γ , is therefore complete because complete filtered modules are closed under limits.

For the source, 3.6.1 makes the augmentation module B almost perfect over $F^\bullet C$. The augmentation is pointwise surjective on π_0 , so 3.4.15 makes $L_{B/F^\bullet C}[-1]$ connective and almost perfect as a filtered B -module. The coefficient algebra B , concentrated in filtration degree zero, is complete; hence 3.4.3 makes the source complete. Since γ is an associated-graded equivalence between complete filtered modules, 3.3.4 makes γ an equivalence.

- (19) **Filtration-zero comparison.** Finally 3.3.2 identifies filtration zero of these shifted cotangent complexes in filtered modules with the ordinary shifted cotangent complexes. Thus the underlying B -modules satisfy

$$\mathrm{cot}_{B/R}(A) \xrightarrow{\cong} \mathrm{Tot}_{[n] \in \Delta} \mathrm{cot}_{B/R}(A^n).$$

Naturality of cotangent transitivity identifies this equivalence with the totalization of the canonical anchor maps from $L_{B/R}[-1]$. Hence the comparison is an equivalence in the undercategory

$$(\mathrm{Mod}_B)_{L_{B/R}[-1]/} = \mathrm{AnchCot}_{B/R}^{\mathrm{sp}},$$

and therefore preserves the anchor from $L_{B/R}[-1]$. Consequently, $\mathrm{cot}_{B/R}$ preserves every $\mathrm{cot}_{B/R}$ -split totalization required by the Barr–Beck–Lurie criterion.

- (20) **Barr–Beck conclusion.** Together with conservativity, the Barr–Beck–Lurie theorem gives comonadicity.

□

Definition 3.6.3 (Shifted-cotangent-structured-square-zero comonad). Under the hypotheses of 3.6.1, define

$$\mathbb{C}_{B/R}^{\mathrm{sqz}} := \mathrm{cot}_{B/R} \mathrm{sqz}_{B/R}$$

on $\mathrm{AnchCot}_{B/R, \geq 0}^{\mathrm{aperf}}$, with its canonical comonad structure induced by the adjunction.

Corollary 3.6.1 (Comonadic classification of complete formal augmentations). *The Barr–Beck comparison functor induces a canonical equivalence*

$$\mathrm{CAlg}(\mathrm{Sp})_{R//B}^{\mathrm{cn}, \mathrm{afp}, \wedge} \xrightarrow{\cong} \mathrm{Coalg}_{\mathbb{C}_{B/R}^{\mathrm{sqz}}}(\mathrm{AnchCot}_{B/R, \geq 0}^{\mathrm{aperf}}),$$

where $\mathrm{Coalg}_{\mathbb{C}_{B/R}^{\mathrm{sqz}}}(-)$ denotes the ∞ -category of coalgebras for the displayed comonad. Thus the Eilenberg–Moore category is the precise category of comonadic shifted-cotangent data used for reconstruction.

Proof.

- (1) **Invoke the comonadic comparison.** This is the comparison equivalence furnished by the comonadicity statement of 3.6.1.
- (2) **Identify the comonad.** Its comonad is $\mathbb{C}_{B/R}^{\mathrm{sqz}}$ by 3.6.3.

□

Corollary 3.6.2 (Canonical structured-square-zero cobar resolution). *Every*

$$A \rightarrow B \in \mathrm{CAlg}(\mathrm{Sp})_{R//B}^{\mathrm{cn}, \mathrm{afp}, \wedge}$$

admits a canonical augmented cosimplicial resolution

$$A \longrightarrow A^\bullet.$$

Its first terms are functorially of the form

$$A \longrightarrow \mathrm{sqz}_{B/R} \cot_{B/R}(A) \begin{array}{c} \longrightarrow \\ \longrightarrow \end{array} \mathrm{sqz}_{B/R} \mathbb{C}_{B/R}^{\mathrm{sqz}} \cot_{B/R}(A) \begin{array}{c} \longrightarrow \\ \longrightarrow \end{array} \cdots,$$

The resolution satisfies:

- (i) *every positive term is a constructed structured square-zero extension of B ;*
- (ii) *$\cot_{B/R}(A^\bullet)$ is split cosimplicial;*
- (iii) *the augmentation exhibits*

$$A \xrightarrow{\sim} \mathrm{Tot} A^\bullet.$$

The resolution is the cobar resolution functorially generated by the adjunction.

Proof.

- (1) **Invoke the canonical cobar resolution.** This is the canonical cobar resolution associated to the comonadic adjunction of 3.6.1.
- (2) **Identify its positive terms.** Positive terms lie in the image of the right adjoint $\mathrm{sqz}_{B/R}$,
- (3) **Identify the splitting codegeneracies.** the canonical extra codegeneracies split the shifted-cotangent image,
- (4) **Recover the totalization.** and comonadicity identifies the augmentation with the totalization.

□

Remark 3.6.1 (Comonadic interpretation). The comonadic adjunction reconstructs the complete almost finitely presented augmentation category from shifted relative cotangent data. Its cobar construction supplies the corresponding functorial cosimplicial resolution.

The fixed-centre problem is now solved on the complete almost-finitely-presented sector: the nonlinear augmentation is equivalent to its anchored shifted-cotangent object with the canonical comonadic coherence, and the cobar construction reconstructs it by a functorial cosimplicial square-zero resolution. The next task is to extract the bounded finite-test fragment needed for formal-moduli semantics.

3.7 Artinian finite tests inside the infinitesimal geometry

The complete fixed-centre category contains infinite formal neighbourhoods, whereas Schlessinger semantics is built from bounded tests. Spectral Artinian extensions provide this finite fragment by combining nilpotence on π_0 , almost finite presentation, and eventual coconnectivity of the relative fibre.

The section proves that these intrinsic conditions are equivalent to finite structured square-zero generation with coherent coefficients. The relation between the intrinsic and factorized descriptions is summarized by

$$\begin{array}{ccc}
 \text{intrinsic Artinian conditions} & \xrightarrow{\text{finite generation}} & \text{finite coherent square-zero factorization} \\
 \downarrow \text{complete formal object} & & \downarrow \text{Spec} \\
 \text{complete augmentation at } B & \xrightarrow[\text{affine realization}]{} & \text{finite infinitesimal test at } \text{Spec}(B).
 \end{array}$$

The finite factorization is a theorem and remains a proof witness. Affine realization, by contrast, produces a presentation-free object of the finite infinitesimal geometry already constructed in Chapter 2. Thus Artinianity adds bounded coherent control without replacing intrinsic infinitesimal geometry by a chosen presentation.

Remark 3.7.1 (Relation with the Artinian formal-integration category). The intrinsic Artinian conditions below agree with the fixed-centre formal-integration test category of Brantner–Magidson–Nuiten. Their finite coherent structured square-zero description is given in [10, Proposition 4.3 and Corollaries 4.4–4.5]. The argument below is written in the present spectral conventions so that the resulting factorizations can be compared directly with the finite infinitesimal substitutions of Chapter 2.

3.7.1 Intrinsic Artinian extensions

Definition 3.7.1 (Spectral Artinian extension). Fix a morphism of connective commutative ring spectra

$$R \longrightarrow B$$

with B coherent. A connective augmented R -algebra

$$A \longrightarrow B$$

is a **spectral Artinian extension** if:

- (i) $\pi_0(A) \rightarrow \pi_0(B)$ is surjective with nilpotent kernel;
- (ii) $A \rightarrow B$ is almost finitely presented;
- (iii) $\text{fib}(A \rightarrow B)$ is eventually coconnective.

Write

$$\text{Art}_{B/R}^{\text{sp}} \subseteq \text{CAlg}(\text{Sp})_{R//B}^{\text{cn}}$$

for the full subcategory on these arrows.

Lemma 3.7.1 (Finite presentation across a finitely generated quotient). *Let $R \twoheadrightarrow S = R/J$ be a surjection of ordinary commutative rings with J finitely generated.*

- (i) Every finitely presented S -module is finitely presented after restriction of scalars to R .
- (ii) If an R -module M is annihilated by J and finitely presented over R , then it is finitely presented over S .

Proof.

- (1) **Prove finite presentation after restriction.** For (i), start with a finite presentation over S , lift its finite matrix of relations to R , and adjoin finitely many relations expressing the action of a chosen finite generating set of J as zero on the finite set of module generators.
- (2) **Tensor the second presentation with the quotient.** For (ii), tensor a finite R -presentation of M with S .
- (3) **Identify the resulting cokernel.** Since J acts trivially on M , the resulting cokernel is still M , giving a finite S -presentation.

□

Lemma 3.7.2 (Coherence across a finite nilpotent extension). *Let $A_0 \twoheadrightarrow B_0$ be a surjection of ordinary commutative rings with nilpotent kernel I . Suppose B_0 is coherent and I is finitely presented as an A_0 -module. Then:*

- (i) A_0 is coherent;
- (ii) every power I^k is finitely presented as an A_0 -module;
- (iii) for every finitely presented A_0 -module M , each quotient

$$I^k M / I^{k+1} M$$

is finitely presented over B_0 .

Proof.

- (1) **Square-zero base case.** We first treat a square-zero extension

$$C \twoheadrightarrow D = C/J, \quad J^2 = 0,$$

with D coherent and J finitely presented over C . Since J is finitely generated, D is finitely presented over C , and because J is annihilated by itself the finite-presentation comparison of 3.7.1 makes J finitely presented over D .

- (2) **Choose generators of the ideal.** Let $K = (x_1, \dots, x_r) \subseteq C$, let

$$\varphi : C^{\oplus r} \twoheadrightarrow K$$

send the standard basis to the chosen generators, and reduce modulo J .

- (3) **Reduce the presentation modulo the square-zero ideal.** Write

$$\bar{\varphi} : D^{\oplus r} \twoheadrightarrow \bar{K}$$

for the resulting presentation of the image ideal $\bar{K} \subseteq D$.

- (4) **Choose lifts of the reduced relations.** Since D is coherent, the kernel

$$L := \ker(\bar{\varphi})$$

is finitely generated. Choose lifts $\tilde{\ell}_1, \dots, \tilde{\ell}_s \in C^{\oplus r}$ of a finite generating set of L .

- (5) **Prove finite generation of the intersection.** We first show that $K \cap J$ is finitely generated as a D -module. If $y \in K \cap J$, choose $v \in C^{\oplus r}$ with $\varphi(v) = y$. Its reduction $\bar{v}$ lies in L . Subtracting a C -linear combination of the $\tilde{\ell}_j$, we may arrange that the resulting vector lies in $J^{\oplus r}$. Consequently y is a D -linear combination of the finitely many elements $\varphi(\tilde{\ell}_j) \in J$ together with the image of $J^{\oplus r}$ under φ . The latter image is finitely generated because J is a finitely generated D -module.
- (6) **Promote the intersection to finite presentation.** Thus $K \cap J$ is finitely generated over D , hence finitely presented because D is coherent.
- (7) **Compare finite presentations over the extension.** The ideal $\bar{K}$ is finitely presented over D by coherence. The finite-presentation comparison 3.7.1 carries both $K \cap J$ and $\bar{K}$ to finitely presented C -modules.
- (8) **Conclude coherence of the square-zero extension.** Therefore the exact sequence

$$0 \longrightarrow K \cap J \longrightarrow K \longrightarrow \bar{K} \longrightarrow 0$$

makes K finitely presented over C . Every finitely generated ideal of C is therefore finitely presented, so C is coherent.

- (9) **Nilpotent induction.** Choose m with $I^m = 0$ and put

$$R_n := A_0/I^n, \quad 1 \leq n \leq m.$$

Then $R_1 = B_0$, $R_m = A_0$, and for $n \geq 2$ the kernel of

$$R_n \longrightarrow R_{n-1}$$

is

$$J_n := I^{n-1}/I^n,$$

which is square-zero. Suppose inductively that R_{n-1} is coherent. Because I is finitely presented over A_0 , base change gives a finitely presented R_{n-1} -module

$$I \otimes_{A_0} R_{n-1} \simeq I/I^n.$$

The submodule

$$J_n = I^{n-1}/I^n \subseteq I/I^n$$

is finitely generated: I is finitely generated, so the same is true of I^{n-1} . Coherence of R_{n-1} therefore makes J_n finitely presented over R_{n-1} . Since its R_n -action factors through $R_{n-1} = R_n/J_n$, the finite-presentation comparison of 3.7.1 makes J_n finitely presented over R_n . The square-zero step now shows that R_n is coherent. Induction gives coherence of every R_n , in particular A_0 .

- (10) **Establish finite presentation of the powers.** Every I^k is a finitely generated ideal in the coherent ring A_0 and is therefore finitely presented.

- (11) **Establish finite presentation of the associated quotients.** If M is finitely presented over A_0 , the submodules $I^k M$ and $I^{k+1} M$ are finitely generated submodules of the finitely presented module M , hence finitely presented by coherence. Their quotient is finitely presented over A_0 and is annihilated by I ; 3.7.1 makes it finitely presented over $B_0 = A_0/I$.

□

Nilpotence and boundedness alone do not yet give a finite structured presentation. The next step constructs one by separating the degree-zero nilpotent tower from the positive-degree Postnikov layers and then refining the resulting coefficients over the fixed centre.

3.7.2 Finite structured square-zero generation

Lemma 3.7.3 (Bounded refinement of a square-zero coefficient filtration). *Let*

$$C \longrightarrow D$$

be a structured square-zero extension of connective commutative ring spectra classified by a connective D -module M . Suppose M admits a finite filtration by D -submodules whose successive cofibres are $M_1, \dots, M_r$, and for every j there exists $n_j \geq 0$ such that

$$M_j \in (\text{Mod}_D)_{\geq n_j} \cap (\text{Mod}_D)_{\leq 2n_j}.$$

Then $C \rightarrow D$ admits a finite factorization by structured square-zero extensions whose successive coefficients are $M_1, \dots, M_r$. The factorization is functorial for the chosen filtered coefficient.

Proof.

- (1) **One filtration step.** It is enough to treat a cofibre sequence of connective D -modules

$$M' \longrightarrow M \longrightarrow M''$$

with M' in the stated small-extension range. Let

$$\eta : L_{D/A} \longrightarrow M[1]$$

classify $C \rightarrow D$ in some compatible relative algebra category. Compose with $M[1] \rightarrow M''[1]$ and let

$$C'' \longrightarrow D$$

be the corresponding structured square-zero extension. Functoriality of split square-zero formation and of the defining pullbacks produces a canonical morphism

$$C \longrightarrow C''$$

over D .

- (2) **Compare the two fibre sequences.** Compare the underlying fibre sequences

$$M \longrightarrow C \longrightarrow D, \quad M'' \longrightarrow C'' \longrightarrow D.$$

- (3) **Identify the relative fibre.** The 3×3 lemma in spectra identifies

$$\text{fib}(C \rightarrow C'') \simeq M'.$$

- (4) **Analyze the augmentation-ideal action.** We must also identify the module structure on this fibre. The augmentation ideal

$$\mathrm{fib}(C'' \rightarrow D) \simeq M''$$

acts on M' through multiplication inside the original square-zero fibre M .

- (5) **Use null multiplication.** Since the nonunital multiplication

$$M \otimes_C M \longrightarrow M$$

is coherently null, this ideal action is null.

- (6) **Identify the restricted module structure.** Hence the C'' -module structure on M' factors canonically through $C'' \rightarrow D$, so the fibre is precisely the restriction of the specified D -module M' . Its nonunital multiplication is likewise coherently null.

- (7) **Verify the small-extension range.** If

$$M' \in (\mathrm{Mod}_D)_{\geq n} \cap (\mathrm{Mod}_D)_{\leq 2n},$$

then $C \rightarrow C''$ is an n -small extension in the sense of [34, Theorem 7.4.1.26 and Corollary 7.4.1.27].

- (8) **Apply small-extension recognition.** The small-extension recognition theorem therefore identifies it with a structured square-zero extension by M' .
- (9) **Factor the relative class.** In a relative algebra category, the structural map from the base supplies the required nullhomotopy of the absolute class on restriction, and cotangent transitivity factors the class through the relative cotangent complex.
- (10) **Iterate the refinement.** Iterating this construction along the finite coefficient filtration gives the asserted factorization. In all applications below, the successive coefficients are Eilenberg–Mac Lane modules $HN[j]$, which lie in the required range with $n = j$.

□

Lemma 3.7.4 (Connective pullbacks over a degree-zero surjection). *Let*

$$C^u := A_0 \times_{A_{01}} A_1$$

be a pullback formed in unrestricted commutative ring spectra, where A_0, A_1, A_{01} are connective. If one of the maps

$$\pi_0(A_i) \longrightarrow \pi_0(A_{01})$$

is surjective, then C^u is connective. Hence it also computes the pullback in connective commutative ring spectra.

Proof.

- (1) **Create the underlying fibre sequence.** The forgetful functor from commutative ring spectra to spectra creates limits, so there is a fibre sequence

$$C^u \longrightarrow A_0 \oplus A_1 \longrightarrow A_{01}.$$

- (2) **Identify the possible negative homotopy.** All three displayed algebras are connective. Consequently the only possible negative homotopy group of C^u is

$$\pi_{-1}(C^u) \simeq \operatorname{coker}(\pi_0(A_0) \oplus \pi_0(A_1) \longrightarrow \pi_0(A_{01})).$$

- (3) **Use degree-zero surjectivity.** Surjectivity of either component kills this cokernel, and all lower negative homotopy groups vanish termwise.
- (4) **Conclude connectivity.** Thus C^u is connective.
- (5) **Recover the connective pullback.** The inclusion of connective commutative ring spectra is full, so the same object satisfies the pullback universal property in the connective subcategory.

□

Lemma 3.7.5 (Discrete square-zero quotients are structured). *Let $Q \twoheadrightarrow P$ be a surjection of ordinary commutative rings with square-zero kernel J . Then the induced morphism of connective commutative ring spectra*

$$HQ \longrightarrow HP$$

is a structured square-zero extension of connective commutative ring spectra with coefficient HJ . The same assertion holds in any compatible relative commutative-algebra category.

Proof.

- (1) **Identify the square-zero fibre.** The homotopy fibre of $HQ \rightarrow HP$ is the discrete HP -module HJ , and the multiplication on that fibre is null because $J^2 = 0$.
- (2) **Recognize a small extension.** Thus the arrow is a connective 0-small extension.
- (3) **Apply square-zero recognition.** The connective square-zero recognition theorem for commutative ring spectra identifies such an arrow with the structured extension classified by its canonical derivation

$$L_{HP} \longrightarrow HJ[1];$$

see [30, 34].

- (4) **Factor the relative class.** If the original square-zero quotient is equipped with a compatible base-algebra structure, the structural map from the base supplies the nullhomotopy of the absolute class after restriction; cotangent transitivity therefore factors the class uniquely through the relative cotangent complex.
- (5) **Conclude the relative statement.** This gives the relative statement.

□

Proposition 3.7.1 (Bounded nilpotent finite structured square-zero generation). *Let $A \rightarrow B$ be a morphism of connective commutative ring spectra such that $\pi_0(A) \rightarrow \pi_0(B)$ is surjective with nilpotent kernel and $\operatorname{fib}(A \rightarrow B)$ is eventually coconnective. Then $A \rightarrow B$ admits a finite factorization by structured square-zero extensions whose coefficients are restrictions of connective eventually coconnective B -modules. The hypotheses are exactly surjectivity on π_0 with nilpotent kernel and eventual coconnectivity of the relative fibre.*

Proof.

- (1) **Recall the reconstruction tower.** The general reconstruction tower by structured square-zero extensions for a map surjective on π_0 with nilpotent kernel is [33, Lemma 2.1.14]. We retain the explicit bounded argument here because the theorem additionally records finiteness under eventual coconnectivity and refines every coefficient to the restriction of a B -module.

- (2) **Separate degree zero.** Put

$$I := \ker(\pi_0(A) \rightarrow \pi_0(B)), \quad I^m = 0.$$

We first separate the degree-zero nilpotent part. Form, in unrestricted commutative ring spectra,

$$A^{(1)} := H\pi_0(A) \times_{H\pi_0(B)} B.$$

Since $H\pi_0(A) \rightarrow H\pi_0(B)$ is surjective on π_0 , 3.7.4 shows that $A^{(1)}$ is connective; hence the same pullback computes the limit in connective commutative ring spectra. The canonical map $A \rightarrow A^{(1)}$ is an isomorphism on π_0 .

- (3) **Define the degree-zero pullbacks.** For $1 \leq k \leq m$, put

$$D_k := H(\pi_0(A)/I^k), \quad \tilde{D}_k^u := D_k \times_{D_1} B$$

with the pullback formed first in unrestricted commutative ring spectra.

- (4) **Prove the pullbacks are connective.** The map $D_k \rightarrow D_1$ is surjective on π_0 , so 3.7.4 makes $\tilde{D}_k^u$ connective. It therefore computes the same pullback in connective commutative ring spectra; write

$$\tilde{D}_k := \tilde{D}_k^u.$$

- (5) **Identify the endpoints and transition pullbacks.** Then

$$\tilde{D}_1 \simeq B, \quad \tilde{D}_m \simeq A^{(1)},$$

and pullback pasting gives

$$\tilde{D}_{k+1} \simeq D_{k+1} \times_{D_k} \tilde{D}_k.$$

- (6) **Identify the discrete square-zero stages.** The map $D_{k+1} \rightarrow D_k$ is the discrete structured square-zero quotient with coefficient $H(I^k/I^{k+1})$ by 3.7.5.
- (7) **Recognize their algebraic pullbacks.** The displayed identity identifies

$$\tilde{D}_{k+1} \longrightarrow \tilde{D}_k$$

with its algebraic pullback along $\tilde{D}_k \rightarrow D_k$.

- (8) **Obtain the finite structured factorization.** Hence 2.7.2 gives

$$A^{(1)} = \tilde{D}_m \longrightarrow \cdots \longrightarrow \tilde{D}_1 = B$$

is a finite structured square-zero factorization.

- (9) **Identify the coefficients over the centre.** Since $I(I^k/I^{k+1}) = 0$, the coefficient action factors through $\pi_0(B) = \pi_0(A)/I$, and each coefficient is the restriction of the connective discrete B -module $H(I^k/I^{k+1})$.

- (10) **Begin the positive-degree tower.** It remains to factor $A \rightarrow A^{(1)}$. The fibre of $A^{(1)} \rightarrow B$ is HI , while the fibre of $A \rightarrow B$ is eventually coconnective. Fibre transitivity for

$$A \longrightarrow A^{(1)} \longrightarrow B$$

therefore shows that $\text{fib}(A \rightarrow A^{(1)})$ is eventually coconnective. Put

$$C_1 := \text{cofib}(A \rightarrow A^{(1)}).$$

The map is an isomorphism on π_0 , so C_1 is 1-connective; choose T such that it is T -coconnective.

- (11) **State the induction hypothesis.** Suppose inductively that maps

$$A \longrightarrow A^{(i)} \longrightarrow A^{(1)}$$

have been constructed with

$$C_i := \text{cofib}(A \rightarrow A^{(i)})$$

i -connective and T -coconnective.

- (12) **Construct the relative Hurewicz map.** The canonical relative cofibre–cotangent morphism is the composite

$$h_i : C_i \longrightarrow A^{(i)} \otimes_A C_i \longrightarrow L_{A^{(i)}/A}.$$

- (13) **Identify its first homotopy map.** Since $\pi_0(A) \simeq \pi_0(A^{(i)})$, the connective tensor-product spectral sequence makes the first arrow an isomorphism on π_i , and 3.4.3 makes the second arrow an isomorphism on π_i . Hence

$$\pi_i(h_i) : \pi_i C_i \xrightarrow{\cong} \pi_i L_{A^{(i)}/A}.$$

- (14) **Define the coefficient modules.** Put

$$N_i := \pi_i C_i, \quad M_i := HN_i[i-1].$$

- (15) **Restrict the coefficient action.** Regard M_i as an $A^{(i)}$ -module through the canonical truncation

$$A^{(i)} \longrightarrow H\pi_0(A^{(i)}) \simeq H\pi_0(A).$$

- (16) **Define the classifying map.** The relative cotangent complex is i -connective. Its first Postnikov projection followed by the inverse of $\pi_i(h_i)$ defines

$$\eta_i : L_{A^{(i)}/A} \longrightarrow H\pi_i L_{A^{(i)}/A}[i] \xrightarrow{H(\pi_i(h_i)^{-1})[i]} HN_i[i] = M_i[1].$$

- (17) **Compute the composite on first homotopy.** The composite

$$C_i \xrightarrow{h_i} L_{A^{(i)}/A} \xrightarrow{\eta_i} HN_i[i]$$

induces the identity on $N_i = \pi_i C_i$.

- (18) **Identify the Postnikov projection.** Because C_i is i -connective and $HN_i[i]$ is i -truncated, maps between them are detected on π_i ; the composite is therefore exactly the canonical Postnikov projection

$$C_i \longrightarrow H\pi_i(C_i)[i].$$

(19) **Construct the next structured square-zero extension.** Let $A^{(i+1)} \rightarrow A^{(i)}$ be the structured square-zero extension in commutative A -algebras classified by η_i .

(20) **Record its fibre sequence.** Its underlying fibre sequence is

$$M_i \longrightarrow A^{(i+1)} \longrightarrow A^{(i)}.$$

(21) **Apply octahedrality.** Octahedrality for $A \rightarrow A^{(i+1)} \rightarrow A^{(i)}$ gives a cofibre sequence

$$C_{i+1} \longrightarrow C_i \longrightarrow M_i[1].$$

(22) **Identify the boundary map.** By the square-zero boundary comparison 2.7.1, the final map is the composite

$$C_i \xrightarrow{h_i} L_{A^{(i)}/A} \xrightarrow{\eta_i} M_i[1],$$

which was just identified with the canonical Postnikov projection.

(23) **Identify the next Postnikov tail.** Therefore

$$C_{i+1} \simeq \text{fib}(C_i \rightarrow H\pi_i(C_i)[i]) \simeq \tau_{\geq i+1} C_i.$$

(24) **Update the connectivity bounds.** Thus the lower connectivity increases by one and the upper coconnective bound remains T .

(25) **Refine coefficients over the centre.** The $\pi_0(A)$ -module N_i has the finite I -adic filtration

$$N_i \supseteq IN_i \supseteq \cdots \supseteq I^m N_i = 0.$$

Its successive quotients are annihilated by I , so their action factors through $\pi_0(B)$. Consequently M_i has a finite filtration whose successive cofibres are restrictions of the connective, eventually coconnective B -modules

$$H(I^k N_i / I^{k+1} N_i)[i-1].$$

Each displayed quotient is concentrated in homotopy degree $i-1$, so it lies in the small-extension range required by 3.7.3. Applying that lemma refines the single stage $A^{(i+1)} \rightarrow A^{(i)}$ into finitely many structured square-zero extensions with coefficients restricted from those B -modules; the refinement preserves the composite and the Postnikov calculation above.

(26) **Termination.** For $i > T$, one has $C_i \simeq 0$, so $A \simeq A^{(i)}$. Concatenating the finitely many refined Postnikov-killing stages with the degree-zero nilpotent tower from $A^{(1)}$ to B gives the asserted finite structured square-zero factorization.

□

Proposition 3.7.2 (Finite structured square-zero generation of spectral Artinian extensions). *An augmented arrow $A \rightarrow B$ is spectral Artinian if and only if it admits a finite factorization*

$$A = A_m \xrightarrow{q_m} A_{m-1} \longrightarrow \cdots \longrightarrow A_1 \xrightarrow{q_1} A_0 = B.$$

in which every

$$q_i : A_i \longrightarrow A_{i-1}$$

is a structured square-zero extension whose coefficient is the restriction of a connective coherent B -module. Moreover, whenever $A \rightarrow B$ is spectral Artinian, A is coherent and

$$A \rightarrow B \in \text{CAlg}(\text{Sp})_{R//B}^{\text{cn}, \text{afp}, \wedge}.$$

Proof.

- (1) **Forward induction on the structured square-zero chain.** Suppose first that such a finite structured square-zero factorization is given. We prove inductively upward from $A_0 = B$ that every source A_i is coherent, that A_{i-1} is almost perfect over A_i , and that B is almost perfect over A_i . The assertions hold at $i = 0$. Assume them through A_{i-1} . The coefficient of $A_i \rightarrow A_{i-1}$ is the restriction of a connective coherent B -module M_i . Since B is almost perfect over A_{i-1} and M_i is almost perfect over B , 3.4.1 makes M_i almost perfect over A_{i-1} . Apply 3.5.3 with centre A_{i-1} . It makes A_{i-1} almost perfect over A_i , makes the stage almost finitely presented, and makes A_i coherent. 3.4.1 then makes B almost perfect over A_i .
- (2) **Verify the Artinian conditions.** Thus the composite $A \rightarrow B$ is almost finitely presented. Every stage has square-zero degree-zero kernel, so the kernel of the composite on π_0 is nilpotent. Every connective coherent B -module is eventually coconnective, and the relative fibre of a finite composite is obtained from the stage fibres by finitely many extensions; hence $\text{fib}(A \rightarrow B)$ is eventually coconnective. Therefore $A \rightarrow B$ is spectral Artinian. The nilpotent augmentation ideal makes A derived complete, while B is almost perfect over A ; 3.5.1 puts the arrow in the complete almost finitely presented sector. The induction already proved coherence of A .
- (3) **Reverse implication: initial finiteness.** Conversely let $A \rightarrow B$ be spectral Artinian and put

$$I := \ker(\pi_0(A) \rightarrow \pi_0(B)), \quad I^m = 0.$$

The spectral criterion for almost finite presentation gives

$$B \in \text{APerf}(A).$$

Since B is coherent, 3.4.7 makes I finitely presented as a $\pi_0(A)$ -module. Notice that the long exact sequence gives only a canonical quotient

$$\pi_0 \text{fib}(A \rightarrow B) \twoheadrightarrow I;$$

we never identify these two modules. By 3.7.2, the ordinary ring $\pi_0(A)$ is coherent, every I^k is finitely presented, and for every finitely presented $\pi_0(A)$ -module N the pieces

$$I^k N / I^{k+1} N$$

are finitely presented over $\pi_0(B)$. At this point the available coherence input concerns the ordinary ring $\pi_0(A)$, and the following construction uses precisely that input.

- (4) **Invoke bounded nilpotent factorization.** We now run the explicit bounded-nilpotent factorization of 3.7.1, keeping track of almost-perfectness so that its coefficients can be refined to coherent B -modules.
- (5) **Degree-zero coherent stages.** First form

$$A^{(1)} := H\pi_0(A) \times_{H\pi_0(B)} B$$

and the degree-zero tower

$$A^{(1)} = \tilde{D}_m \longrightarrow \cdots \longrightarrow \tilde{D}_1 = B, \quad \tilde{D}_k = H(\pi_0(A)/I^k) \times_{H\pi_0(B)} B.$$

Each displayed pullback may be formed in unrestricted commutative ring spectra: the map $H(\pi_0(A)/I^k) \rightarrow H\pi_0(B)$ is surjective on π_0 , so 3.7.4 shows that the result is connective and hence is also the pullback in the connective algebra category. Its stage coefficient is

$$H(I^k/I^{k+1}).$$

By 3.7.2, I^k/I^{k+1} is finitely presented over $\pi_0(B)$; since B is coherent, the corresponding Eilenberg–Mac Lane module is a connective coherent B -module. Thus the degree-zero tower already has the required coefficient class.

- (6) **Almost-perfectness of the degree-zero tower.** We also need its sources to be almost perfect over A . The target $B = \widetilde{D}_1$ is almost perfect over A . Every stage coefficient is almost perfect over B , hence is almost perfect over A by 3.4.1. The underlying fibre sequence of each structured square-zero stage then shows inductively that every $\widetilde{D}_k$, in particular $A^{(1)}$, is almost perfect over A . Therefore

$$C_1 := \text{cofib}(A \rightarrow A^{(1)})$$

is almost perfect over A , because A is perfect over itself. As in 3.7.1, C_1 is 1-connective and eventually coconnective; choose t such that it is t -coconnective.

- (7) **State the positive-degree induction hypothesis.** Inductively suppose that

$$A \longrightarrow A^{(i)} \longrightarrow A^{(1)}$$

has been constructed such that $A^{(i)}$ and

$$C_i := \text{cofib}(A \rightarrow A^{(i)})$$

are almost perfect over A , while C_i is i -connective and t -coconnective.

- (8) **Apply relative Hurewicz.** The relative Hurewicz construction of 3.7.1 gives an isomorphism

$$\pi_i C_i \xrightarrow{\cong} \pi_i L_{A^{(i)}/A}.$$

- (9) **Define the coefficient modules.** Put

$$N_i := \pi_i C_i, \quad M_i := HN_i[i-1].$$

- (10) **Restrict the coefficient action.** Regard M_i as an $A^{(i)}$ -module through $A^{(i)} \rightarrow H\pi_0(A^{(i)}) \simeq H\pi_0(A)$.

- (11) **Establish finite presentation of the coefficient.** Since C_i is i -connective and almost perfect, 3.4.2 makes N_i finitely presented over $\pi_0(A)$.

- (12) **Define the classifying term.** The first Postnikov projection of the relative cotangent complex and the inverse Hurewicz identification define the classifying term

$$\eta_i : L_{A^{(i)}/A} \longrightarrow HN_i[i] = M_i[1].$$

- (13) **Construct the next square-zero stage.** Let $A^{(i+1)} \rightarrow A^{(i)}$ be the corresponding structured square-zero extension over A .

- (14) **Identify the next Postnikov tail.** Exactly as in 3.7.1, octahedrality identifies

$$C_{i+1} := \text{cofib}(A \rightarrow A^{(i+1)}) \simeq \tau_{\geq i+1} C_i.$$

- (15) **Update the connectivity bound.** Thus connectivity increases by one and the fixed upper bound t remains.

- (16) **Defer almost-perfectness to the refinement.** Almost-perfectness will be established from the refined structured square-zero stage using the coherence information already available for the graded pieces.

- (17) **Refine the coefficients.** The finitely presented $\pi_0(A)$ -module N_i has the finite I -adic filtration

$$N_i \supseteq IN_i \supseteq \cdots \supseteq I^m N_i = 0.$$

By 3.7.2, every quotient

$$I^k N_i / I^{k+1} N_i$$

is finitely presented over $\pi_0(B)$. Therefore M_i has a finite filtration whose successive cofibres are the connective coherent B -modules

$$H(I^k N_i / I^{k+1} N_i)[i-1].$$

Each displayed quotient is concentrated in homotopy degree $i-1$, hence lies in the small-extension range of 3.7.3. Applying that lemma refines $A^{(i+1)} \rightarrow A^{(i)}$ into finitely many structured square-zero stages with coefficients restricted from those coherent B -modules.

- (18) **Propagate almost-perfectness.** Each refined coefficient is almost perfect over B , and B is almost perfect over A ; hence 3.4.1 makes each refined coefficient almost perfect over A . Starting from the inductive hypothesis that $A^{(i)}$ is almost perfect over A , the square-zero fibre sequences through the finite refinement show that $A^{(i+1)}$ is almost perfect over A . Since A is perfect over itself, its cofibre C_{i+1} is also almost perfect over A . This closes the induction using only the coherence input established for $\pi_0(A)$.
- (19) **Terminate the tower.** For $i > t$, one has $C_i \simeq 0$, so $A \simeq A^{(i)}$. Concatenating the finitely many refined Postnikov-killing stages with the degree-zero tower from $A^{(1)}$ to B gives a finite factorization of $A \rightarrow B$ by structured square-zero extensions whose coefficients are restrictions of connective coherent B -modules.
- (20) **Recover spectral coherence.** It remains to prove spectral coherence of A from the constructed finite structured square-zero chain. Start at the coherent target $A_0 = B$ and move backward through the finite coherent structured square-zero chain just constructed. Suppose A_{j-1} is coherent and B is almost perfect over A_{j-1} . The coefficient of $A_j \rightarrow A_{j-1}$ is the restriction of a connective coherent B -module, hence is almost perfect over A_{j-1} . Applying 3.5.3 with centre A_{j-1} proves that A_j is coherent and that A_{j-1} is almost perfect over A_j . 3.4.1 then makes B almost perfect over A_j . Induction reaches $A_m = A$ and proves that A is coherent.
- (21) **Derived completeness.** Finally the nilpotence of I makes A derived I -complete, while B is already almost perfect over A . Hence 3.5.1 gives

$$A \rightarrow B \in \text{CAlg}(\text{Sp})_{R//B}^{\text{cn}, \text{afp}, \wedge}.$$

□

Finite square-zero generation gives bounded coefficients. For Schlessinger excision we need the stronger relative statement that each stage can be refined by coefficients restricted from connective coherent B -modules, uniformly at the chosen centre.

3.7.3 Coherent refinement and relative factorization

Lemma 3.7.6 (Coherent refinement of an Artinian structured square-zero stage). *Let*

$$C \longrightarrow D$$

be a structured square-zero extension in $\mathrm{CAlg}(\mathrm{Sp})_{R//B}^{\mathrm{cn}}$, with coefficient $M \in (\mathrm{Mod}_D)_{\geq 0}$, and suppose that the augmented arrows $C \rightarrow B$ and $D \rightarrow B$ are spectral Artinian. Then M admits a finite filtration by D -submodules whose successive cofibres are restrictions of connective coherent B -modules. Consequently $C \rightarrow D$ admits a finite refinement by structured square-zero extensions with coefficients restricted from connective coherent B -modules.

Proof.

- (1) **Boundedness of the coefficient.** Because $C \rightarrow B$ and $D \rightarrow B$ are spectral Artinian, their relative fibres are eventually coconnective. Stable fibre calculus for the compatible maps to B identifies

$$M \simeq \mathrm{fib}(C \rightarrow D)$$

with the fibre of the induced map between those two relative fibres. Hence M is eventually coconnective; it is connective by the definition of the structured square-zero stage, so it is bounded.

- (2) **Almost-perfectness.** By 3.7.2, choose a finite coherent structured square-zero factorization

$$D = D_r \longrightarrow D_{r-1} \longrightarrow \cdots \longrightarrow D_0 = B.$$

The Artinian augmentation $C \rightarrow B$ satisfies $B \in \mathrm{APerf}(C)$. Every stage coefficient in the displayed factorization is almost perfect over B , hence is almost perfect over C by 3.4.1. Starting from $D_0 = B$, the square-zero fibre sequences therefore show inductively that

$$D \in \mathrm{APerf}(C).$$

The fibre sequence

$$M \longrightarrow C \longrightarrow D$$

then gives

$$M \in \mathrm{APerf}(C).$$

Both C and D are coherent by 3.7.2. Hence 3.4.1 makes every $\pi_j(M)$ finitely presented over $\pi_0(C)$.

- (3) **Coherence over the intermediate centre.** Put

$$J := \ker(\pi_0(C) \rightarrow \pi_0(D)).$$

The C -action on the square-zero coefficient factors through D , so $J\pi_j(M) = 0$. If a module annihilated by J is finitely presented over $\pi_0(C)$, tensoring a finite presentation with $\pi_0(D) = \pi_0(C)/J$ makes it finitely presented over $\pi_0(D)$. Thus every $\pi_j(M)$ is finitely presented over $\pi_0(D)$, and 3.4.1 gives

$$M \in \mathrm{Coh}(D) \cap (\mathrm{Mod}_D)_{\geq 0}.$$

- (4) **Define the centre ideal.** Now put

$$I := \ker(\pi_0(D) \rightarrow \pi_0(B)).$$

- (5) **Establish nilpotence and finite presentation.** This ideal is nilpotent. Since $D \rightarrow B$ is Artinian, 3.4.7 makes I finitely presented over $\pi_0(D)$.
- (6) **Control the I -adic quotients.** Apply 3.7.2. For each finitely presented module $N_j := \pi_j(M)$, the finite I -adic filtration

$$N_j \supseteq IN_j \supseteq \cdots \supseteq I^s N_j = 0$$

has successive quotients

$$I^k N_j / I^{k+1} N_j$$

finitely presented over $\pi_0(B)$.

- (7) **Bound the Postnikov layers.** Because M is connective and eventually coconnective, its canonical Postnikov tower contains only finitely many nonzero layers

$$HN_j[j].$$

- (8) **Refine each Postnikov layer.** For each such layer, pull the finite I -adic filtration of N_j back successively along the canonical Postnikov projection $\tau_{\leq j} M \rightarrow HN_j[j]$.
- (9) **Assemble the coefficient filtration.** Splicing these finite refinements over the finitely many Postnikov degrees gives a finite filtration of M by D -submodules whose successive cofibres are restrictions of the modules

$$H(I^k N_j / I^{k+1} N_j)[j].$$

- (10) **Identify coherent centre coefficients.** Each of these is a connective coherent B -module by 3.4.1.
- (11) **Verify the small-extension range.** Moreover it is concentrated in a single homotopy degree and therefore lies in the small-extension range of 3.7.3.
- (12) **Refine the structured extension.** Applying that lemma refines the original structured square-zero extension along this finite coefficient filtration.

□

Proposition 3.7.3 (Relative factorization for surjections on π_0 with nilpotent kernel). *Let*

$$A' \longrightarrow A$$

be a morphism in $\mathrm{CAlg}(\mathrm{Sp})_{R//B}^{\mathrm{cn}}$, and suppose that $A \rightarrow B$ is spectral Artinian. The following conditions are equivalent:

- (i) $A' \rightarrow B$ is spectral Artinian and

$$\pi_0(A') \longrightarrow \pi_0(A)$$

is surjective with nilpotent kernel;

(ii) there exists a finite factorization

$$A' = A_m \longrightarrow A_{m-1} \longrightarrow \cdots \longrightarrow A_0 = A$$

in $\mathrm{CAlg}(\mathrm{Sp})_{R//B}^{\mathrm{cn}}$ such that every $A_i \rightarrow A_{i-1}$ is a structured square-zero extension whose coefficient is the restriction of a connective coherent B -module.

Under these conditions every intermediate augmentation

$$A_i \longrightarrow B$$

is spectral Artinian.

Moreover, the full subcategory

$$\mathrm{Art}_{B/R}^{\mathrm{sp}}$$

is stable under pullback along morphisms that are surjective on π_0 with nilpotent kernel: if

$$\begin{array}{ccc} C & \longrightarrow & A_0 \\ \downarrow & & \downarrow \\ A_1 & \longrightarrow & A_{01} \end{array}$$

is a pullback in connective commutative R -algebras over B , all three augmentations

$$A_0 \rightarrow B, \quad A_1 \rightarrow B, \quad A_{01} \rightarrow B$$

are spectral Artinian, and $A_0 \rightarrow A_{01}$ is surjective on π_0 with nilpotent kernel, then $C \rightarrow B$ is spectral Artinian.

Proof.

- (1) **From Artinianity to relative module finiteness.** Because $A' \rightarrow B$ is spectral Artinian, 3.4.1 gives

$$B \in \mathrm{APerf}(A').$$

Choose a coherent structured square-zero factorization of the Artinian augmentation $A \rightarrow B$ supplied by 3.7.2:

$$A = C_r \longrightarrow C_{r-1} \longrightarrow \cdots \longrightarrow C_0 = B.$$

Every stage coefficient is the restriction of a connective coherent B -module, hence is almost perfect over B and therefore almost perfect over A' by 3.4.1. Starting from $C_0 = B \in \mathrm{APerf}(A')$, the square-zero fibre sequences show inductively that

$$C_j \in \mathrm{APerf}(A') \quad (0 \leq j \leq r).$$

In particular,

$$A \in \mathrm{APerf}(A').$$

- (2) **Artinianity over the intermediate centre.** The map $A' \rightarrow A$ is surjective on π_0 , so 3.4.1 now makes it almost finitely presented. Stable fibre calculus for the composable maps

$$A' \longrightarrow A \longrightarrow B$$

gives a fibre sequence

$$\mathrm{fib}(A' \rightarrow A) \longrightarrow \mathrm{fib}(A' \rightarrow B) \longrightarrow \mathrm{fib}(A \rightarrow B).$$

The last two terms are eventually coconnective because both augmentations to B are Artinian. Hence $\mathrm{fib}(A' \rightarrow A)$ is eventually coconnective. Together with the assumed nilpotence of the degree-zero kernel, this shows that

$$A' \longrightarrow A$$

is a spectral Artinian extension with centre A . The centre A is coherent by 3.7.2 applied to $A \rightarrow B$.

- (3) **Factor over the intermediate centre.** Apply 3.7.2 again, now to the Artinian augmentation $A' \rightarrow A$. We obtain a finite factorization

$$A' = D_s \longrightarrow D_{s-1} \longrightarrow \cdots \longrightarrow D_0 = A$$

by structured square-zero extensions whose coefficients are restrictions of connective coherent A -modules. The forward implication in 3.7.2, applied stage by stage, shows that every $D_j \rightarrow A$ is spectral Artinian. Since $A \rightarrow B$ is Artinian, each composite

$$D_j \longrightarrow A \longrightarrow B$$

is Artinian as well: almost finite presentation is stable under composition, nilpotent degree-zero kernels are stable under composition, and the relative fibre of a composite is an extension of the two relative fibres.

- (4) **Refine coefficients over B .** For every structured square-zero stage

$$D_j \longrightarrow D_{j-1},$$

both endpoint augmentations to B are therefore Artinian. Apply 3.7.6. It refines the stage, into finitely many structured square-zero extensions whose coefficients are restrictions of connective coherent B -modules. Concatenating these finite refinements gives the factorization required in (ii).

- (5) **Reverse induction.** Start with the Artinian augmentation $A_0 = A \rightarrow B$ and argue inductively. Suppose $A_{i-1} \rightarrow B$ is Artinian and

$$A_i \longrightarrow A_{i-1}$$

is a structured square-zero extension with coefficient the restriction of a connective coherent B -module M_i . The spectral criterion for almost finite presentation gives

$$B \in \mathrm{APerf}(A_{i-1}),$$

and restriction transitivity makes M_i almost perfect over A_{i-1} . Then 3.5.3 makes the structured square-zero stage almost finitely presented. Hence the composite $A_i \rightarrow B$ is almost finitely presented. Its degree-zero kernel is an extension of the nilpotent kernel of $\pi_0(A_{i-1}) \rightarrow \pi_0(B)$ by a square-zero kernel and is therefore nilpotent. Fibre transitivity expresses $\mathrm{fib}(A_i \rightarrow B)$ as a finite extension of the eventually coconnective fibre of $A_{i-1} \rightarrow B$ by the eventually coconnective coefficient M_i . Therefore $A_i \rightarrow B$ is spectral Artinian.

- (6) **Complete the reverse implication.** Induction proves that every intermediate augmentation, in particular $A' = A_m \rightarrow B$, is spectral Artinian. Since each map $\pi_0(A_i) \rightarrow \pi_0(A_{i-1})$ is surjective with square-zero kernel, their finite composite

$$\pi_0(A') \longrightarrow \pi_0(A)$$

is surjective with nilpotent kernel. This proves (i).

- (7) **Choose a factorization of the nilpotent-surjective leg.** Consider the Artinian morphism

$$A_0 \longrightarrow A_{01},$$

which is surjective on π_0 with nilpotent kernel. By the equivalence just proved, choose a finite factorization:

$$A_0 = D_m \longrightarrow D_{m-1} \longrightarrow \cdots \longrightarrow D_0 = A_{01}.$$

- (8) **Form the unrestricted pullbacks.** Form first, in unrestricted commutative ring spectra,

$$E_k^u := D_k \times_{A_{01}} A_1.$$

- (9) **Prove the pullbacks are connective.** The composite

$$D_k \longrightarrow A_{01}$$

is surjective on π_0 , since it is a composite of square-zero surjections. Hence 3.7.4 makes every E_k^u connective. It therefore computes the same pullback in connective commutative R -algebras over B ; write

$$E_k := E_k^u.$$

- (10) **Identify the endpoint pullbacks.** Then

$$E_0 \simeq A_1, \quad E_m \simeq C.$$

- (11) **Pull back the structured square-zero stages.** Algebraic pullback of structured square-zero extensions is structured by 2.7.2; the coefficient of each

$$E_k \longrightarrow E_{k-1}$$

is the restriction of the same connective coherent B -module occurring in $D_k \rightarrow D_{k-1}$.

- (12) **Propagate Artinianity.** Starting from the Artinian target $E_0 = A_1$, the one-stage argument above shows inductively that every $E_k \rightarrow B$ is Artinian. In particular $C = E_m \rightarrow B$ is spectral Artinian.

□

The coherent factorization theorem has two semantic consequences. Artinian augmentations lie inside the complete reconstruction domain, and affine realization identifies them with the bounded coherent fixed-centre part of the finite infinitesimal geometry.

3.7.4 Artinian tests as finite infinitesimal contexts

Corollary 3.7.1 (Artinian extensions are complete almost finitely presented). *There is a fully faithful inclusion*

$$\mathrm{Art}_{B/R}^{\mathrm{sp}} \hookrightarrow \mathrm{CAlg}(\mathrm{Sp})_{R/B}^{\mathrm{cn}, \mathrm{afp}, \wedge}.$$

Proof.

- (1) **Invoke the Artinian generation theorem.** This is the final assertion of 3.7.2. □

Proposition 3.7.4 (Artinian augmentations as coherent finite infinitesimal tests). *Let*

$$U = \mathrm{Spec}(B) \longrightarrow V = \mathrm{Spec}(R).$$

Affine realization defines a fully faithful functor

$$(\mathrm{Art}_{B/R}^{\mathrm{sp}})^{\mathrm{op}} \longrightarrow \mathrm{Inf}^{\mathrm{fin}}(U/V)$$

which sends an augmentation $A \rightarrow B$ to the relative infinitesimal test

$$\begin{array}{ccc} U & \xlongequal{\quad} & U \\ \downarrow & & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & V. \end{array}$$

Its essential image is the full subcategory of relative finite infinitesimal tests whose open part is all of U and whose affine coordinate augmentation is spectral Artinian. In particular, spectral Artinian tests form the bounded coherent fixed-centre fragment of the finite infinitesimal geometry constructed in Chapter 2.

Proof.

- (1) **Affine realization.** By 3.7.2, an Artinian augmentation admits a finite factorization

$$A = A_m \longrightarrow A_{m-1} \longrightarrow \cdots \longrightarrow A_0 = B$$

by structured square-zero extensions. Passing to affine geometry gives a finite composite

$$\mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A_1) \longrightarrow \cdots \longrightarrow \mathrm{Spec}(A)$$

of square-zero infinitesimal substitutions. Hence $\mathrm{Spec}(B) \rightarrow \mathrm{Spec}(A)$ is a finite spectral infinitesimal substitution by 2.8.3. The maps from R and to B give the displayed incidence diagram over $U \rightarrow V$.

- (2) **Full faithfulness.** Morphisms in $\mathrm{Art}_{B/R}^{\mathrm{sp}}$ are precisely commutative morphisms of augmented R -algebras over B . Under affine realization these are precisely the morphisms between the displayed relative infinitesimal tests whose open part is the identity of U . This proves full faithfulness. The description of the essential image is immediate from the construction and the intrinsic definition of spectral Artinian extension. The functor depends only on the augmented arrow. □

One final presentation is needed only for the Schlessinger argument. When the coefficient of a structured square-zero stage comes from the centre, the stage can be rewritten as a pullback against the split square-zero algebra at that centre.

3.7.5 Centre presentations for Schlessinger excision

Lemma 3.7.7 (Centre presentation of a relative structured square-zero stage). *Let*

$$C \longrightarrow D$$

be a structured square-zero extension in $\mathrm{CAlg}(\mathrm{Sp})_{R//B}^{\mathrm{cn}}$, and suppose its coefficient M is the restriction of a connective B -module. Then there is a canonical morphism over B

$$\sigma : D \longrightarrow B \oplus M[1]$$

such that

$$C \simeq D \times_{B \oplus M[1]} B,$$

where $B \rightarrow B \oplus M[1]$ is the zero section. The comparison is natural in the structured relative square-zero stage.

Proof.

- (1) **Choose the classifying derivation section.** Let

$$d_\delta : D \longrightarrow D \oplus M[1]$$

be the derivation section classifying $C \rightarrow D$.

- (2) **Use split square-zero base change.** Split square-zero formation for a coefficient restricted from B gives a Cartesian square

$$\begin{array}{ccc} D \oplus M[1] & \longrightarrow & B \oplus M[1] \\ \downarrow & & \downarrow \\ D & \longrightarrow & B. \end{array}$$

- (3) **Define the centre map.** Compose d_δ with the upper horizontal morphism to obtain $\sigma : D \rightarrow B \oplus M[1]$.
- (4) **Apply pullback pasting.** The defining square-zero pullback and the preceding Cartesian square then give, by pullback pasting,

$$\begin{aligned} C &= D \times_{D \oplus M[1]} D \\ &\simeq D \times_{B \oplus M[1]} B. \end{aligned}$$

- (5) **Record functoriality.** All comparisons are functorial consequences of split square-zero formation and pullback pasting.

□

We may now retain a compact description of the Artinian category: it is simultaneously a full subcategory of complete fixed-centre reconstruction and a bounded coherent subcategory of finite infinitesimal geometry. Its coherent square-zero factorizations reduce nilpotent pullbacks to elementary centre-based stages, which is the finite test calculus needed next.

3.8 Artinian formal-moduli problems and Schlessinger localization

The Artinian category is now a finite test theory at the fixed centre B . We seek space-valued functors that behave as nonlinear formal neighbourhoods. Rather than choosing additional coherence data, we impose two equations by localization: the centre is sent to a point, and every elementary square-zero pullback is sent to a homotopy pullback.

For a generator

$$A_\sigma = A \times_{B \oplus M[1]} B,$$

the algebraic pullback and the corresponding semantic condition are displayed together:

$$\begin{array}{ccc} A_\sigma & \longrightarrow & A \\ \downarrow & & \downarrow \sigma \\ B & \xrightarrow{d_0} & B \oplus M[1] \end{array} \qquad \begin{array}{ccc} F(A_\sigma) & \longrightarrow & F(A) \\ \downarrow & & \downarrow \\ F(B) & \longrightarrow & F(B \oplus M[1]). \end{array}$$

The left square constructs the elementary Artinian extension, and locality requires the right square to be Cartesian. Finite coherent factorization then promotes these elementary equations to the usual Schlessinger excision condition for arbitrary Artinian nilpotent pullbacks.

Definition 3.8.1 (Artinian functor category). Put

$$\mathcal{P}_{B/R}^{\text{Art}, \text{sp}} := \text{Fun}(\text{Art}_{B/R}^{\text{sp}}, \mathcal{S}).$$

For $A \rightarrow B$ in the Artinian category, write

$$y^A(C) := \text{Map}_{\text{Art}_{B/R}^{\text{sp}}}(A, C)$$

for the covariant Yoneda object. Thus

$$\text{Map}(y^A, F) \simeq F(A).$$

Lemma 3.8.1 (Schlessinger pullbacks remain Artinian). *Let $A \rightarrow B$ be spectral Artinian, let M be a connective coherent B -module, and let*

$$\sigma : A \longrightarrow B \oplus M[1]$$

be a morphism over B . Put

$$A_\sigma := A \times_{B \oplus M[1]} B,$$

where the second map is the zero section. Then $A_\sigma \rightarrow A$ is a structured square-zero extension with coefficient the restriction of M , and $A_\sigma \rightarrow B$ is spectral Artinian.

Proof.

- (1) **Form the unrestricted pullback.** Form A_σ first in unrestricted commutative ring spectra. The zero section

$$B \longrightarrow B \oplus M[1]$$

is an isomorphism on π_0 , so 3.7.4 makes A_σ connective. Hence this unrestricted pullback is also the pullback in connective commutative R -algebras over B .

- (2) **Identify the base-changed split square-zero square.** Regard M as an A -module through $A \rightarrow B$. Split square-zero formation gives a Cartesian square

$$\begin{array}{ccc} A \oplus M[1] & \longrightarrow & B \oplus M[1] \\ \downarrow & & \downarrow \\ A & \longrightarrow & B. \end{array}$$

- (3) **Lift the centre map.** Because σ lies over $A \rightarrow B$, it has a contractibly unique lift

$$\tilde{\sigma} : A \longrightarrow A \oplus M[1]$$

over A .

- (4) **Identify the derivation section.** This is the derivation section classified by the corresponding degree-one derivation of A with coefficient M .
- (5) **Apply pullback pasting.** Pullback pasting gives

$$A_\sigma \simeq A \times_{A \oplus M[1]} A.$$

- (6) **Identify the structured extension.** Thus $A_\sigma \rightarrow A$ is the structured square-zero extension classified by $\tilde{\sigma}$. Its coefficient is the restriction of the connective coherent B -module M .
- (7) **Artinianity.** Choose a finite coherent structured square-zero factorization of $A \rightarrow B$ using 3.7.2. Prepending the constructed stage $A_\sigma \rightarrow A$ gives such a factorization of $A_\sigma \rightarrow B$. The same proposition therefore makes $A_\sigma \rightarrow B$ spectral Artinian.

□

Definition 3.8.2 (Artinian Schlessinger generators). Let

$$W_{B/R}^{\text{Sch, Art}}$$

be the following set of morphisms in $\mathcal{P}_{B/R}^{\text{Art, sp}}$:

- (i) the centre generator

$$\emptyset \longrightarrow y^B;$$

- (ii) for every spectral Artinian $A \rightarrow B$, every connective coherent B -module M , and every morphism over B

$$\sigma : A \longrightarrow B \oplus M[1],$$

the Yoneda morphism

$$y^A \amalg_{y^{B \oplus M[1]}} y^B \longrightarrow y^{A_\sigma}, \quad A_\sigma = A \times_{B \oplus M[1]} B.$$

The corresponding algebra square is Cartesian:

$$\begin{array}{ccc} A_\sigma & \longrightarrow & A \\ \downarrow & & \downarrow \sigma \\ B & \xrightarrow{d_0} & B \oplus M[1]. \end{array}$$

The split square-zero augmentation $B \oplus M[1] \rightarrow B$ is Artinian: its map on π_0 is an isomorphism, its fibre $M[1]$ is eventually coconnective because M is coherent, and its augmentation is almost finitely presented by 3.5.3. The object $A_\sigma \rightarrow B$ is Artinian by 3.8.1.

Definition 3.8.3 (Spectral formal-moduli problem). Define

$$\mathrm{FMP}_{B/R}^{\mathrm{sp}} \subseteq \mathcal{P}_{B/R}^{\mathrm{Art}, \mathrm{sp}}$$

to be the full subcategory of $W_{B/R}^{\mathrm{Sch}, \mathrm{Art}}$ -local objects. Its objects are the **spectral formal-moduli problems** at the centre B over R , with B coherent.

Theorem 3.8.1 (Artinian Schlessinger reflector). *The inclusion*

$$\mathrm{FMP}_{B/R}^{\mathrm{sp}} \hookrightarrow \mathcal{P}_{B/R}^{\mathrm{Art}, \mathrm{sp}}$$

admits an accessible left adjoint

$$\mathcal{P}_{B/R}^{\mathrm{Art}, \mathrm{sp}} \begin{array}{c} \xrightarrow{L_{B/R}^{\mathrm{Sch}, \mathrm{Art}}} \\ \xleftarrow{\quad} \end{array} \mathrm{FMP}_{B/R}^{\mathrm{sp}}$$

A functor F is local precisely when

$$F(B) \simeq *$$

and, for every generator square,

$$F(A_\sigma) \xrightarrow{\simeq} F(A) \times_{F(B \oplus M[1])} F(B).$$

Proof.

- (1) **Choose a set of generators.** The Artinian category is essentially small in the fixed universe, so the Yoneda generators form a set.
- (2) **Use presentability of the functor category.** Its space-valued functor category is pre-sentable.
- (3) **Construct the accessible localization.** Accessible localization at the strongly saturated class generated by that set therefore exists by [35, Proposition 5.5.4.15].
- (4) **Apply Yoneda.** Yoneda gives

$$\mathrm{Map}(y^A, F) \simeq F(A).$$
- (5) **Identify the centre condition.** Locality for $\emptyset \rightarrow y^B$ is exactly $F(B) \simeq *$.
- (6) **Identify the square-zero condition.** Mapping the second generator into F turns its pushout into the displayed pullback of values.

□

Proposition 3.8.1 (Schlessinger excision follows from the generators). *Let*

$$F \in \mathrm{FMP}_{B/R}^{\mathrm{sp}}.$$

Consider a pullback square in $\text{Art}_{B/R}^{\text{sp}}$

$$\begin{array}{ccc} A & \longrightarrow & A_0 \\ \downarrow & & \downarrow \\ A_1 & \longrightarrow & A_{01}, \end{array} \quad A \simeq A_0 \times_{A_{01}} A_1,$$

and suppose that

$$A_0 \longrightarrow A_{01}$$

is surjective on π_0 with nilpotent kernel. Then the canonical map

$$F(A) \longrightarrow F(A_0) \times_{F(A_{01})} F(A_1)$$

is an equivalence.

Consequently, the generator formulation of 3.8.3 is equivalent to the ordinary Schlessinger requirement that Artinian pullback squares with one leg surjective on π_0 with nilpotent kernel be carried to pullback squares of spaces. The conclusion is independent of every finite structured square-zero factorization, coherent refinement, and centre presentation used inside the proof.

Proof.

- (1) **Factor the leg surjective on π_0 with nilpotent kernel.** By 3.7.3, the Artinian morphism

$$A_0 \longrightarrow A_{01},$$

which is surjective on π_0 with nilpotent kernel, admits a finite factorization

$$A_0 = D_m \longrightarrow D_{m-1} \longrightarrow \cdots \longrightarrow D_0 = A_{01},$$

where every

$$D_k \longrightarrow D_{k-1}$$

is a structured square-zero extension whose coefficient is the restriction of a connective coherent B -module M_k . Every $D_k \rightarrow B$ is Artinian.

- (2) **Pull back the factorization.** Pull this factorization back along

$$A_1 \longrightarrow A_{01}$$

by forming, first in unrestricted commutative ring spectra,

$$E_k^u := D_k \times_{A_{01}} A_1.$$

Every composite $D_k \rightarrow A_{01}$ is surjective on π_0 . Therefore 3.7.4 makes each E_k^u connective, so it also computes the pullback in the connective algebra category. Put

$$E_k := E_k^u.$$

Then

$$E_0 \simeq A_1, \quad E_m \simeq A.$$

By algebraic pullback stability of structured square-zero extensions, 2.7.2, every

$$E_k \longrightarrow E_{k-1}$$

is a structured square-zero extension with coefficient the restriction of M_k . The pullback clause of 3.7.3 makes every $E_k \rightarrow B$ spectral Artinian.

- (3) **Present each stage at the centre.** For each stage

$$D_k \longrightarrow D_{k-1},$$

the centre-presentation lemma 3.7.7 constructs a canonical map

$$\sigma_k : D_{k-1} \longrightarrow B \oplus M_k[1]$$

and a Cartesian identification

$$D_k \simeq D_{k-1} \times_{B \oplus M_k[1]} B.$$

After pullback to E_{k-1} , the same construction gives

$$E_k \simeq E_{k-1} \times_{B \oplus M_k[1]} B.$$

- (4) **Apply generator locality.** Since F is local for the Schlessinger generators,

$$F(D_k) \xrightarrow{\simeq} F(D_{k-1}) \times_{F(B \oplus M_k[1])} F(B)$$

and

$$F(E_k) \xrightarrow{\simeq} F(E_{k-1}) \times_{F(B \oplus M_k[1])} F(B).$$

- (5) **Combine the two pullback descriptions.** Combining these two pullback descriptions gives a canonical equivalence

$$F(E_k) \xrightarrow{\simeq} F(D_k) \times_{F(D_{k-1})} F(E_{k-1}).$$

- (6) **Identify the elementary pullback square.** Thus F carries every elementary pullback square

$$\begin{array}{ccc} E_k & \longrightarrow & D_k \\ \downarrow & & \downarrow \\ E_{k-1} & \longrightarrow & D_{k-1} \end{array}$$

to a pullback square of spaces.

- (7) **Paste the elementary squares.** Finite pasting now gives

$$F(E_m) \simeq F(D_m) \times_{F(D_0)} F(E_0).$$

Using

$$E_m \simeq A, \quad D_m = A_0, \quad D_0 = A_{01}, \quad E_0 = A_1,$$

we obtain

$$F(A) \simeq F(A_0) \times_{F(A_{01})} F(A_1).$$

- (8) **Converse implication.** Conversely, every Schlessinger generator square of 3.8.2 is itself an Artinian pullback square with one leg surjective on π_0 with nilpotent kernel: the zero section

$$B \longrightarrow B \oplus M[1]$$

is an isomorphism on π_0 . Hence preservation of all such Artinian pullbacks implies locality for the generator squares. Together with the centre condition $F(B) \simeq *$, this proves the asserted equivalence of the two formulations.

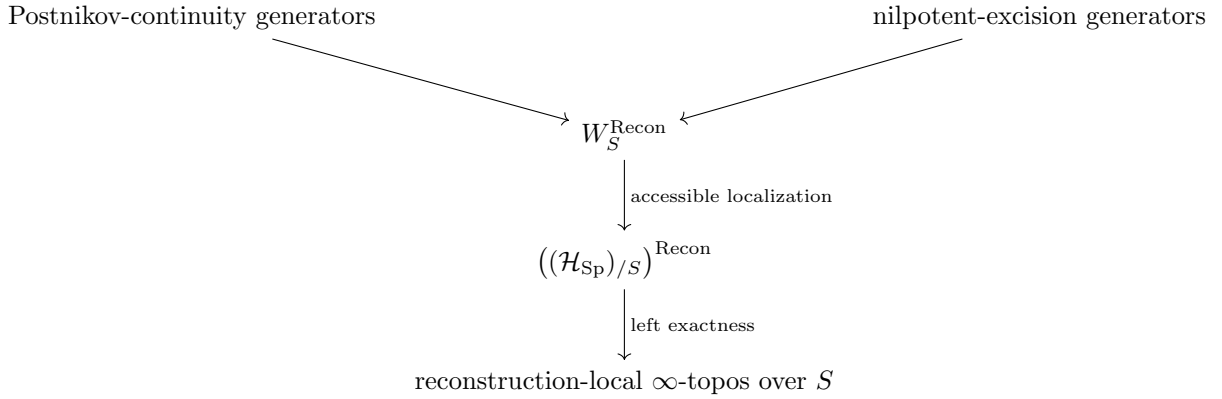
□

Thus spectral formal-moduli problems are exactly the Artinian functors satisfying the centre condition and elementary square-zero excision; finite coherent factorization upgrades those generators to full nilpotent excision. The centre can now be removed by retaining only the two reconstruction equations that make sense globally.

3.9 Centre-free reconstruction by continuity and nilpotent excision

The Artinian theory is centred and finitary. Over an arbitrary spectral Zariski base S , two of its reconstruction principles survive without choosing a centre, and we impose exactly those two principles in the slice over S .

The first is Postnikov continuity: values on a connective affine test are recovered from its finite Postnikov truncations. The second is nilpotent excision: algebraic pullbacks with one nilpotent-surjective leg are sent to homotopy pullbacks of mapping spaces. Accessible localization imposes these equations; left exactness is then the extra statement that makes the localized semantics compatible with finite-limit geometry. Their roles are summarized by



The reflector therefore gives only the candidate semantics. The substantive step is to show that its generating local equivalences are stable under pullback. That result implies base-change stability, finite-limit preservation, and the existence of centred leaves inside the localized semantics.

3.9.1 Reconstruction generators

Fix a base

$$S \in \mathcal{H}_{\text{Sp}}$$

and work in the slice

$$(\mathcal{H}_{\text{Sp}})_{/S}.$$

An **affine spectral test over S** is a morphism

$$h_A \longrightarrow S$$

with A connective; these are the tests on which the reconstruction equations are imposed.

Definition 3.9.1 (Representability by spectral schemes). A morphism

$$Y \longrightarrow S$$

in $\mathcal{H}_{\text{Sp}}$ is **representable by spectral schemes** if, for every affine spectral test $T \rightarrow S$, the fibre product

$$Y \times_S T$$

is a spectral scheme. This condition is a property of the morphism. In particular, every morphism of spectral schemes is representable by spectral schemes, by finite-limit closure of spectral schemes from Chapter 1.

Definition 3.9.2 (Global spectral reconstruction generators). Let W_S^{Recon} be the following set of morphisms in $(\mathcal{H}_{\text{Sp}})_S$.

- (i) For every affine spectral test $h_A \rightarrow S$, the **Postnikov-continuity generator**

$$c_A : \text{colim}_{n \geq 0} h_{\tau_{\leq n}^{\text{CAlg}} A} \longrightarrow h_A,$$

where every $h_{\tau_{\leq n}^{\text{CAlg}} A}$ maps to S through $h_A \rightarrow S$.

- (ii) Let

$$\begin{array}{ccc} C & \longrightarrow & A_0 \\ \downarrow & & \downarrow \\ A_1 & \longrightarrow & A_{01} \end{array} \quad C \simeq A_0 \times_{A_{01}} A_1$$

be a pullback of connective commutative ring spectra in which one $A_i \rightarrow A_{01}$ is surjective on π_0 with nilpotent kernel. For every affine spectral test $h_C \rightarrow S$, regard $h_{A_0}, h_{A_1}, h_{A_{01}}$ as objects over S through $h_C \rightarrow S$. The associated **nilpotent-excision generator** is

$$w_{\square} : h_{A_0} \amalg_{h_{A_{01}}} h_{A_1} \longrightarrow h_C.$$

The pullback C in (ii) is connective by 3.7.4.

Lemma 3.9.1 (Representable nilpotent excision). *Let*

$$C \simeq A_0 \times_{A_{01}} A_1$$

be a generator square of 3.9.2, equipped with its affine test $\text{Spec}(C) \rightarrow S$. If $Y \rightarrow S$ is representable by spectral schemes, then the canonical map

$$\text{Map}_S(\text{Spec } C, Y) \longrightarrow \text{Map}_S(\text{Spec } A_0, Y) \times_{\text{Map}_S(\text{Spec } A_{01}, Y)} \text{Map}_S(\text{Spec } A_1, Y)$$

is an equivalence.

Proof.

- (1) **Base change to the affine source.** Put

$$Y_C := Y \times_S \text{Spec}(C).$$

By representability, Y_C is a spectral scheme. Since every object in the generator square maps to S through $\text{Spec}(C)$, the mapping spaces in the statement identify with the corresponding mapping spaces over $\text{Spec}(C)$ into Y_C .

- (2) **Nilpotent geometry.** After interchanging the two indices if necessary, assume that

$$\pi_0(A_0) \longrightarrow \pi_0(A_{01})$$

is surjective with nilpotent kernel J . Then

$$\mathrm{Spec}(A_{01}) \longrightarrow \mathrm{Spec}(A_0)$$

is a nilpotent closed immersion and a homeomorphism on underlying topological spaces.

The pullback algebra $C = A_0 \times_{A_{01}} A_1$ is connective by 3.7.4, and its degree-zero homotopy ring is

$$\pi_0(C) \simeq \pi_0(A_0) \times_{\pi_0(A_{01})} \pi_0(A_1).$$

The projection $\pi_0(C) \rightarrow \pi_0(A_1)$ is surjective, and its kernel identifies with J through the first projection. In particular that kernel is nilpotent. Consequently

$$\mathrm{Spec}(A_1) \longrightarrow \mathrm{Spec}(C)$$

is also a nilpotent closed immersion and a homeomorphism on underlying topological spaces. This conclusion is obtained from the algebraic pullback; no geometric Cartesian identification of the displayed affine square is being asserted.

- (3) **Affine target.** If $Y_C = \mathrm{Spec}(D)$ is affine, the result is the algebraic pullback universal property:

$$\begin{aligned} \mathrm{Map}_C(\mathrm{Spec} C, \mathrm{Spec} D) &\simeq \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})_{C/}}(D, C) \\ &\simeq \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})_{C/}}(D, A_0) \times_{\mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})_{C/}}(D, A_{01})} \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})_{C/}}(D, A_1). \end{aligned}$$

- (4) **Choose affine local data.** For general Y_C , let a point of the right-hand side be represented by compatible maps

$$f_0 : \mathrm{Spec}(A_0) \longrightarrow Y_C, \quad f_1 : \mathrm{Spec}(A_1) \longrightarrow Y_C.$$

Choose a finite principal affine cover of $\mathrm{Spec}(C)$, equivalently of $\mathrm{Spec}(A_1)$, such that each restriction of f_1 lands in an affine open of Y_C . Because $\mathrm{Spec}(A_{01}) \rightarrow \mathrm{Spec}(A_0)$ is a nilpotent homeomorphism on support and f_0 agrees with f_1 after restriction to $\mathrm{Spec}(A_{01})$, the corresponding restrictions of f_0 land in the same affine opens.

- (5) **Čech refinement.** Refine each intersection of affine target opens by affine opens, and refine the corresponding principal source intersection by a finite principal cover subordinate to them, following the affine-open mapping-descent argument of Chapter 1. Repeat on higher intersections. On every affine member of the resulting Čech refinement, the preceding affine pullback calculation applies. Compatibility on further affine refinements is canonical because all local maps are restrictions of the original compatible pair.
- (6) **Mapping descent.** Mapping descent along the resulting effective epimorphic affine cover and its iterated intersections, 1.8.1, therefore reconstructs a unique map

$$\mathrm{Spec}(C) \longrightarrow Y_C$$

together with all higher homotopies, and identifies its restrictions with f_0 and f_1 . This proves the asserted equivalence of mapping spaces.

□

Theorem 3.9.1 (Global spectral reconstruction reflector). *Accessible localization at W_S^{Recon} exists and gives a reflective adjunction*

$$(\mathcal{H}_{\text{Sp}})_{/S} \begin{array}{c} \xrightarrow{L_S^{\text{Recon}}} \\ \xleftarrow{\iota_S} \end{array} ((\mathcal{H}_{\text{Sp}})_{/S})^{\text{Recon}}$$

An object $Y \rightarrow S$ is local if and only if:

(i) for every affine spectral test $h_A \rightarrow S$,

$$\text{Map}_S(h_A, Y) \xrightarrow{\sim} \varprojlim_n \text{Map}_S(h_{\tau_{\leq n}^{\text{CAlg}} A}, Y);$$

(ii) for every nilpotent-excision generator,

$$\text{Map}_S(h_C, Y) \xrightarrow{\sim} \text{Map}_S(h_{A_0}, Y) \times_{\text{Map}_S(h_{A_{01}}, Y)} \text{Map}_S(h_{A_1}, Y).$$

Every morphism $Y \rightarrow S$ representable by spectral schemes is L_S^{Recon} -local.

Proof.

- (1) **Verify the localization domain.** The spectral Zariski slice is presentable, and after fixing universes the family W_S^{Recon} is a set.
- (2) **Construct the accessible localization.** Accessible localization therefore exists by [35, Proposition 5.5.4.15].
- (3) **Identify the locality equations.** Yoneda identifies locality for the two classes of generators with the two displayed equations.
- (4) **Affine Postnikov continuity.** Let $Y \rightarrow S$ be representable by spectral schemes. Nilpotent excision is 3.9.1. For Postnikov continuity, fix an affine spectral test

$$T = \text{Spec}(A) \longrightarrow S$$

and put

$$Y_A := Y \times_S T.$$

This is a spectral scheme. The required comparison is

$$\text{Map}_A(\text{Spec } A, Y_A) \longrightarrow \varprojlim_n \text{Map}_A(\text{Spec } \tau_{\leq n}^{\text{CAlg}} A, Y_A).$$

If $Y_A = \text{Spec}(D)$ is affine, operadic Postnikov completeness gives

$$A \simeq \varprojlim_n \tau_{\leq n}^{\text{CAlg}} A,$$

and mapping out of the relative A -algebra D gives the desired equivalence.

- (5) **Affine-open reduction.** For a general spectral scheme Y_A , start with a compatible point of the right-hand limit. Its degree-zero member determines a finite principal cover of $\text{Spec } \pi_0(A)$ subordinate to affine opens of Y_A . The same elements of $\pi_0(A)$ determine compatible principal covers of every $\text{Spec}(\tau_{\leq n}^{\text{CAlg}} A)$ and of $\text{Spec}(A)$. Compatibility of the Postnikov family forces the restriction to each principal member to land in the same affine target.

- (6) **Čech descent.** Choose affine open refinements of the target intersections and finite principal refinements of the corresponding source intersections, and continue this construction on the higher Čech intersections. On every affine member of the resulting refinement the affine Postnikov calculation gives a unique compatible map from the corresponding principal open of $\mathrm{Spec}(A)$. Mapping descent along the effective epimorphic affine refinement and its iterated intersections glues the maps and all higher homotopies. Thus the relative mapping-space comparison is an equivalence for every affine test, and $Y \rightarrow S$ is local.

□

Definition 3.9.3 (Reconstruction-local objects). An object of

$$((\mathcal{H}_{\mathrm{Sp}})_{/S})^{\mathrm{Recon}}$$

is **reconstruction-local over S** if it is local for the reflector L_S^{Recon} .

Remark 3.9.1 (Recognition by nilcompleteness and infinitesimal cohesiveness). In spectral DAG terminology, the Postnikov-continuity condition above is Lurie’s **nilcompleteness**. Represented spectral Deligne–Mumford functors are cohesive and nilcomplete by [33, Proposition 2.1.7]. Moreover, for a nilcomplete functor admitting a cotangent complex, the pullback condition with one leg surjective on π_0 with nilpotent kernel is equivalent to infinitesimal cohesiveness by [33, Proposition 2.1.13].

These citations are recognition results only. The category $((\mathcal{H}_{\mathrm{Sp}})_{/S})^{\mathrm{Recon}}$ is defined by the explicit localization L_S^{Recon} on arbitrary spectral Zariski stacks; existence of a cotangent complex, a cohesive structure, or infinitesimal cohesiveness is not part of the definition; the definition remains the explicit localization above.

Proposition 3.9.1 (Base change of reconstruction-local objects). *Let*

$$p : S' \longrightarrow S$$

be a morphism of spectral Zariski stacks. Pullback carries reconstruction-local objects over S to reconstruction-local objects over S' :

$$p^* : ((\mathcal{H}_{\mathrm{Sp}})_{/S})^{\mathrm{Recon}} \longrightarrow ((\mathcal{H}_{\mathrm{Sp}})_{/S'})^{\mathrm{Recon}}.$$

This proposition records the stability of the localization equations under base change.

Proof.

- (1) **Slice mapping spaces.** Let $Y \rightarrow S$ be reconstruction-local and put

$$Y' := Y \times_S S'.$$

For every affine spectral test $T \rightarrow S'$, slice base change gives a natural equivalence

$$\mathrm{Map}_{S'}(T, Y') \simeq \mathrm{Map}_S(T, Y),$$

where the right-hand side regards T as an object over S through $S' \rightarrow S$.

- (2) **Postnikov continuity after base change.** Apply this equivalence to $T = \mathrm{Spec}(A)$ and to all of its Postnikov truncations. The Postnikov-continuity comparison for Y' is identified with the corresponding comparison for Y , and is therefore an equivalence.

- (3) **Nilpotent excision after base change.** Likewise, a nilpotent-excision pullback diagram of connective affine tests over S' , after composition with $S' \rightarrow S$, is the same algebraic pullback diagram regarded over S . The mapping-space comparison for Y' identifies with the corresponding comparison for Y , hence is an equivalence. Therefore Y' satisfies both defining locality equations and is reconstruction-local over S' .

□

Remark 3.9.2 (Reconstruction locality versus formal-moduli problems). The adjective **reconstruction-local** refers to the centre-free nonlinear localization of 3.9.1. The centred Artinian notion of 3.8.3 is recovered from it by the centred Artinian leaf construction below.

3.9.2 Left exactness of the reconstruction localization

Theorem 3.9.2 (Left exactness of spectral reconstruction localization). *The accessible localization*

$$L_S^{\text{Recon}}$$

is left exact. Consequently

$$((\mathcal{H}_{\text{Sp}})_{/S})^{\text{Recon}}$$

is an ∞ -topos, its fully faithful inclusion into $(\mathcal{H}_{\text{Sp}})_{/S}$ preserves all limits, and L_S^{Recon} preserves finite limits.

Proof.

- (1) **Localization criterion.** Let W be the strongly saturated class generated by W_S^{Recon} . The left-exactness criterion for a localization of an ∞ -topos, [35, Proposition 6.2.1.1], reduces the theorem to showing that W is stable under pullback. It is enough to prove that every generating morphism is universally a local equivalence.
- (2) **Pull back a nilpotent-excision generator.** First consider a nilpotent-excision generator

$$w_{\square} : h_{A_0} \amalg_{h_{A_{01}}} h_{A_1} \longrightarrow h_C, \quad C \simeq A_0 \times_{A_{01}} A_1,$$

and pull it back along an affine morphism $h_E \rightarrow h_C$. Put

$$E_i := E \otimes_C A_i, \quad E_{01} := E \otimes_C A_{01}.$$

- (3) **Record the underlying fibre sequence.** The defining pullback of algebras has an underlying fibre sequence

$$C \longrightarrow A_0 \oplus A_1 \longrightarrow A_{01}.$$

- (4) **Extend scalars to the affine pullback.** Extension of scalars from C to E is exact on module spectra, so it produces a fibre sequence

$$E \longrightarrow E_0 \oplus E_1 \longrightarrow E_{01}.$$

- (5) **Recover the algebra pullback.** Since limits of unrestricted commutative ring spectra are created on underlying spectra, the induced morphism

$$E \longrightarrow E_0 \times_{E_{01}} E_1$$

is an equivalence of commutative ring spectra.

- (6) **Base change the nilpotent-surjective degree-zero map.** Suppose, for definiteness, that $\pi_0(A_0) \rightarrow \pi_0(A_{01})$ is surjective with nilpotent kernel. Connectivity of the pushouts gives

$$\pi_0(E_i) \simeq \pi_0(E) \otimes_{\pi_0(C)} \pi_0(A_i).$$

Thus $\pi_0(E_0) \rightarrow \pi_0(E_{01})$ is again surjective and its kernel is a quotient of the scalar extension of the original nilpotent ideal; in particular it is nilpotent.

- (7) **Apply connective pullback recognition.** By 3.7.4, the displayed unrestricted pullback is connective.
- (8) **Conclude stability of the generator.** Therefore the affine pullback of $w_\square$ is again a nilpotent-excision generator.
- (9) **Pull back the Postnikov-continuity generator.** Now consider a Postnikov-continuity generator

$$c_A : \operatorname{colim}_n h_{\tau_{\leq n}^{\mathrm{CAlg}} A} \longrightarrow h_A$$

and an affine morphism $h_E \rightarrow h_A$, represented by a connective algebra map $A \rightarrow E$. Pullback in the spectral Zariski topos preserves colimits, so the pullback of c_A is

$$\operatorname{colim}_n h_{E_n} \longrightarrow h_E, \quad E_n := E \otimes_A \tau_{\leq n}^{\mathrm{CAlg}} A.$$

- (10) **Define the Postnikov fibre.** Let

$$K_n := \operatorname{fib}(A \rightarrow \tau_{\leq n}^{\mathrm{CAlg}} A).$$

- (11) **Record its connectivity.** Then K_n is $(n+1)$ -connective.
- (12) **Base change the fibre.** Exactness of extension of scalars gives

$$\operatorname{fib}(E \rightarrow E_n) \simeq E \otimes_A K_n,$$

which is again $(n+1)$ -connective because E is connective.

- (13) **Identify stable truncations.** Hence for every fixed d ,

$$\tau_{\leq d}^{\mathrm{CAlg}} E_n \xrightarrow{\simeq} \tau_{\leq d}^{\mathrm{CAlg}} E \quad (n \geq d).$$

- (14) **Evaluate a reconstruction-local object.** If Z is reconstruction-local, then

$$\begin{aligned} \varprojlim_n Z(E_n) &\simeq \varprojlim_n \varprojlim_d Z(\tau_{\leq d}^{\mathrm{CAlg}} E_n) \\ &\simeq \varprojlim_d \varprojlim_n Z(\tau_{\leq d}^{\mathrm{CAlg}} E_n) \\ &\simeq \varprojlim_d Z(\tau_{\leq d}^{\mathrm{CAlg}} E) \\ &\simeq Z(E). \end{aligned}$$

- (15) **Conclude local equivalence.** Thus every affine pullback of c_A is an L_S^{Recon} -equivalence.

- (16) **Pass to arbitrary pullbacks.** Finally let P be an arbitrary object over the target of a generator. Affine representables generate the corresponding slice of the spectral Zariski ∞ -topos under colimits. Write P as such a colimit. Pullback in an ∞ -topos preserves colimits, so the pullback of the generator along P is the colimit in the arrow category of its affine pullbacks. The localization is a left adjoint and therefore sends this colimit to a colimit of equivalences. Hence every generator is universally local. The class of morphisms all of whose pullbacks lie in W is strongly saturated and contains the generators, so it contains W . Therefore the local-equivalence class is pullback-stable, which proves left exactness. $\square$

3.9.3 The reconstruction localization and differential cohesion

Let

$$\iota_S : ((\mathcal{H}_{\text{Sp}})_{/S})^{\text{Recon}} \hookrightarrow (\mathcal{H}_{\text{Sp}})_{/S}$$

denote the fully faithful inclusion.

Definition 3.9.4 (Reconstruction localization endofunctor). Define

$$\text{Recon}_S := \iota_S L_S^{\text{Recon}} : (\mathcal{H}_{\text{Sp}})_{/S} \longrightarrow (\mathcal{H}_{\text{Sp}})_{/S}.$$

We call Recon_S the **reconstruction localization endofunctor** over S .

Proposition 3.9.2 (Idempotence of the reconstruction localization). *The endofunctor Recon_S is an idempotent left-exact localization endofunctor. Its fixed objects are precisely the reconstruction-local objects over S . The unit fits into the idempotence diagram*

$$Y \xrightarrow{\eta_Y} \text{Recon}_S Y \xrightarrow{\text{Recon}_S(\eta_Y)} \text{Recon}_S^2 Y,$$

where the second arrow is an equivalence. The first arrow is universal from Y to an object satisfying Postnikov continuity and nilpotent excision.

Proof.

- (1) **Use the reflective-localization identities.** Idempotence and the fixed-point characterization are the reflective-localization identities for the adjunction

$$L_S^{\text{Recon}} \dashv \iota_S.$$

- (2) **Apply left exactness.** Left exactness is 3.9.2.

- (3) **Apply the localization universal property.** The universal property is the defining universal property of localization at W_S^{Recon} , whose local objects are characterized by 3.9.1. $\square$

Remark 3.9.3 (Two left-exact localization constructions). The infinitesimal-shape modality

$$\mathfrak{S} = i_* i^*$$

of Chapter 2 acts on the infinitesimal Zariski topos and records the domain endpoint of an infinitesimal arrow. The reconstruction localization

$$\text{Recon}_S = \iota_S L_S^{\text{Recon}}$$

acts on the ordinary spectral Zariski slice over S and enforces Postnikov continuity together with nilpotent excision. The centred Artinian leaf construction below provides the comparison relevant to this chapter.

The centre-free nonlinear semantics is now available: reconstruction-local objects are the fixed points of a left-exact localization imposing Postnikov continuity and nilpotent excision. Because locality is stable under base change, we can choose an affine centre inside such an object and restrict to its bounded Artinian thickenings. This returns the global theory to the infinitesimal endpoint comparison of the preceding chapter.

3.10 Centred Artinian leaves and tangent recovery

Choose an affine centre

$$U = \text{Spec}(B) \longrightarrow V = \text{Spec}(R) \longrightarrow S$$

and a U -point $u : U \rightarrow Y$ of a reconstruction-local object. Restricting Y to coherent Artinian thickenings produces a centred formal-moduli problem; on each test, its value is the fibre at u of the endpoint restriction underlying differential cohesion.

The passage from nonlinear reconstruction to tangent semantics is summarized by

$$\begin{array}{ccc} \text{Artinian thickening } j_A : U \rightarrow T_A & \xrightarrow{\text{endpoint evaluation}} & \text{Map}_V(T_A, Y_V) \rightarrow \text{Map}_V(U, Y_V) \\ \downarrow \text{split square-zero restriction} & & \downarrow \text{fib}_u \\ \text{connective coefficient } M[n] & \xrightarrow{\text{centred lifting fibre}} & T_{Y/V, u}^n(M). \end{array}$$

The upper row retains the nonlinear finite infinitesimal test, while the lower row extracts the lifting fibre in a coefficient direction. Nilpotent excision turns these fibres into an Ω -spectrum and hence into a stable tangent functor.

3.10.1 Centred Artinian leaves

Definition 3.10.1 (Centred Artinian leaf). Let

$$V = \text{Spec}(R) \longrightarrow S, \quad U = \text{Spec}(B) \longrightarrow V$$

be affine spectral schemes with B coherent. Let $Y \rightarrow S$ be a spectral Zariski stack, and let

$$u : U \longrightarrow Y$$

be a chosen U -point over S . For $A \rightarrow B \in \text{Art}_{B/R}^{\text{sp}}$, define

$$\text{Leaf}_{U/V, u}^{\text{Art}}(Y)(A \rightarrow B) := \text{fib}_u(\text{Map}_S(\text{Spec } A, Y) \longrightarrow \text{Map}_S(U, Y)).$$

Equivalently, it is the upper-left corner of the Cartesian square

$$\begin{array}{ccc} \mathrm{Leaf}_{U/V,u}^{\mathrm{Art}}(Y)(A \rightarrow B) & \longrightarrow & \mathrm{Map}_S(\mathrm{Spec} A, Y) \\ \downarrow & & \downarrow \\ * & \xrightarrow{u} & \mathrm{Map}_S(U, Y). \end{array}$$

Proposition 3.10.1 (Centred Artinian leaves as differential-cohesive endpoint fibres). *Let $A \rightarrow B$ be spectral Artinian and let*

$$j_A : U = \mathrm{Spec}(B) \longrightarrow T_A = \mathrm{Spec}(A)$$

be the associated finite infinitesimal test of 3.7.4. Put

$$Y_V := Y \times_S V.$$

Under the endpoint formulas of Chapter 2, the restriction map

$$\mathrm{Map}_V(T_A, Y_V) \longrightarrow \mathrm{Map}_V(U, Y_V)$$

is the comparison obtained by evaluating

$$i_! Y_V \longrightarrow i_* Y_V$$

at the infinitesimal arrow j_A and then restricting to the fibre over the fixed structural maps to V . Consequently

$$\mathrm{Leaf}_{U/V,u}^{\mathrm{Art}}(Y)(A \rightarrow B)$$

is the homotopy fibre at the chosen point u of this differential-cohesive endpoint restriction.

Proof.

- (1) **Evaluate the endpoint Kan extensions.** For an infinitesimal arrow $j : U' \rightarrow T'$, the endpoint Kan-extension formulas of 2.10.1 give

$$(i_! F)(j) \simeq F(T'), \quad (i_* F)(j) \simeq F(U').$$

- (2) **Identify the endpoint comparison.** The natural comparison $i_! F \rightarrow i_* F$ is induced point-wise by restriction along $U' \rightarrow T'$.
- (3) **Specialize to the Artinian arrow.** Apply this with $F = Y_V$ and $j = j_A$.
- (4) **Pass to relative mapping spaces.** Taking the homotopy fibres of the two evaluation spaces over their prescribed maps to V gives the relative restriction

$$\mathrm{Map}_V(T_A, Y_V) \longrightarrow \mathrm{Map}_V(U, Y_V).$$

- (5) **Apply slice base change.** The slice base-change adjunction identifies these with the corresponding mapping spaces over S used in 3.10.1.
- (6) **Take the centred fibre.** Taking the further homotopy fibre at the chosen point u gives the asserted centred Artinian leaf.

□

Proposition 3.10.2 (Centred Artinian leaves are formal-moduli problems). *If $Y \rightarrow S$ is reconstruction-local, then*

$$\mathrm{Leaf}_{U/V,u}^{\mathrm{Art}}(Y) \in \mathrm{FMP}_{B/R}^{\mathrm{sp}}.$$

Proof.

- (1) **Centre condition.** At the identity augmentation $B \rightarrow B$, the defining restriction map is the identity of $\mathrm{Map}_S(U, Y)$, so its homotopy fibre over u is contractible.
- (2) **Structured square-zero excision.** Let

$$A_\sigma = A \times_{B \oplus M[1]} B$$

be a Schlessinger square with $A \rightarrow B$ Artinian and M connective coherent. The zero section

$$B \longrightarrow B \oplus M[1]$$

is an isomorphism on π_0 , hence this is a nilpotent-excision pullback. Reconstruction locality therefore gives

$$Y(A_\sigma) \simeq Y(A) \times_{Y(B \oplus M[1])} Y(B).$$

Taking the homotopy fibre of this equivalence over the compatible chosen point $u \in Y(B)$, and using pullback pasting, gives exactly the Schlessinger condition for $\mathrm{Leaf}_{U/V,u}^{\mathrm{Art}}(Y)$. Apply 3.8.3.

□

3.10.2 Tangent recovery from split square-zero lifting

The centred Artinian leaf already identifies nonlinear reconstruction with actual infinitesimal lifting at a fixed point. We now restrict those tests to split square-zero augmentations and stabilize in the coefficient variable.

For connective coefficients, $\mathbb{T}_{Y/V,u}^n(M)$ is literally the centred lifting space across $B \oplus M[n] \rightarrow B$. Nilpotent excision supplies both the loop equivalences and right exactness. The Spanier–Whitehead universal property then extends the resulting spectrum-valued functor from connective to eventually connective coefficients. No extension to all unbounded modules is claimed without an additional continuity theorem.

Actual split-square-zero lifting on connective coefficients

Definition 3.10.2 (Actual split-square-zero tangent levels). Let $Y \rightarrow S$ be reconstruction-local and let

$$V = \mathrm{Spec}(R) \longrightarrow S, \quad U = \mathrm{Spec}(B) \longrightarrow V$$

be a connective affine centre. Let

$$u : U \longrightarrow Y$$

be a chosen U -point over S . For a connective B -module M and $n \geq 0$, define

$$\mathbb{T}_{Y/V,u}^n(M) := \mathrm{fib}_u \left(\mathrm{Map}_V(\mathrm{Spec}(B \oplus M[n]), Y \times_S V) \longrightarrow \mathrm{Map}_V(U, Y \times_S V) \right).$$

The map is induced by the split square-zero augmentation $B \oplus M[n] \rightarrow B$. The split algebra section

$$B \longrightarrow B \oplus M[n]$$

induces a retraction

$$\mathrm{Spec}(B \oplus M[n]) \longrightarrow U,$$

and postcomposition of this retraction with the chosen point u defines a canonical basepoint of the displayed homotopy fibre. Thus $\mathbb{T}_{Y/V,u}^n(M)$ is canonically pointed, functorially in M .

Proposition 3.10.3 (Centred Artinian leaves recover the coherent split-square-zero tangent levels). *Let $Y \rightarrow S$ be reconstruction-local and let*

$$U = \mathrm{Spec}(B) \longrightarrow V = \mathrm{Spec}(R) \longrightarrow S$$

be an affine centre with B coherent. Let

$$u : U \longrightarrow Y$$

be a chosen U -point over S . For every connective coherent B -module M and every $n \geq 0$, the split square-zero augmentation

$$B \oplus M[n] \longrightarrow B$$

is a spectral Artinian extension, and there is a canonical equivalence

$$\mathrm{Leaf}_{U/V,u}^{\mathrm{Art}}(Y)(B \oplus M[n] \rightarrow B) \simeq \mathbb{T}_{Y/V,u}^n(M),$$

where the right-hand side is the split-square-zero lifting space defined in 3.10.2. Under these equivalences the loop maps induced by

$$B \oplus M[n] \simeq B \times_{B \oplus M[n+1]} B$$

agree. Hence stabilizing the centred Artinian leaf on connective coherent coefficients gives the restriction of the recovered tangent functor constructed below.

Proof.

- (1) **Artinianity of split square-zero augmentations.** Since M is coherent, it is connective, almost perfect, and eventually coconnective. The split square-zero augmentation is an isomorphism on π_0 for $n > 0$, and has square-zero degree-zero kernel for $n = 0$. Its fibre is $M[n]$, hence is eventually coconnective. Almost finite presentation follows from 3.5.3. Thus it is spectral Artinian.
- (2) **Identify the leaf with the lifting space.** The slice base-change adjunction gives natural equivalences

$$\mathrm{Map}_S(\mathrm{Spec}(B \oplus M[n]), Y) \simeq \mathrm{Map}_V(\mathrm{Spec}(B \oplus M[n]), Y \times_S V)$$

and likewise for U . Taking the homotopy fibre over the chosen point u identifies the defining value of the centred Artinian leaf with $\mathbb{T}_{Y/V,u}^n(M)$. Both loop maps are obtained by applying the same nilpotent-excision pullback identity for consecutive split square-zero algebras, so the equivalences are compatible with stabilization.

□

Lemma 3.10.1 (Retractive self-intersections are loop objects). *Let*

$$X \xrightarrow{s} E \xrightarrow{r} X, \quad r \circ s \simeq \text{id}_X,$$

be a retractive diagram of spaces and let $x \in X$. Give $X \times_E X$ the two maps to E induced by s . Then the homotopy fibre over x of the first projection

$$X \times_E X \longrightarrow X$$

is canonically equivalent to

$$\Omega_{s(x)} \text{fib}_x(r : E \rightarrow X).$$

The equivalence is natural in retractive diagrams.

Proof.

- (1) **Base change to the chosen point.** Base change the retractive diagram along the point $x : * \rightarrow X$.
- (2) **Reduce to a pointed space.** It reduces the statement to a pointed space $(F, *)$, where $F = \text{fib}_x(r)$, and to the homotopy fibre of the basepoint inclusion

$$* \longrightarrow F$$

over the basepoint.

- (3) **Identify the loop space.** That homotopy fibre is the based loop space ΩF .
- (4) **Record naturality.** Naturality is inherited from pullback.

□

Proposition 3.10.4 (Square-zero stabilization and right exactness). *For every connective B -module M , the canonical split square-zero pullback squares give natural equivalences*

$$\mathbb{T}_{Y/V,u}^n(M) \xrightarrow{\simeq} \Omega \mathbb{T}_{Y/V,u}^{n+1}(M), \quad n \geq 0.$$

These equivalences are compatible with identities, morphisms of coefficients, and iterated suspension. Hence the pointed functors

$$\mathbb{T}_{Y/V,u}^n : (\text{Mod}_B)_{\geq 0} \longrightarrow \mathcal{S}_*$$

form a canonical Ω -spectrum object in

$$\text{Fun}((\text{Mod}_B)_{\geq 0}, \mathcal{S}_*).$$

Equivalently, they determine a canonical functor

$$\mathbb{T}_{Y/V,u}^{\text{cn}} : (\text{Mod}_B)_{\geq 0} \longrightarrow \text{Sp}.$$

The resulting functor is zero-preserving and right exact. Explicitly, if

$$M' \longrightarrow M \longrightarrow M''$$

is a cofibre sequence of connective B -modules, then

$$\mathbb{T}_{Y/V,u}^{\text{cn}}(M') \longrightarrow \mathbb{T}_{Y/V,u}^{\text{cn}}(M) \longrightarrow \mathbb{T}_{Y/V,u}^{\text{cn}}(M'')$$

is a fibre and hence also a cofibre sequence of spectra.

Proof.

- (1) **Base change the reconstruction-local target.** By 3.9.1,

$$Y_V := Y \times_S V$$

is reconstruction-local over V .

- (2) **Form the coefficient pullback square.** For every connective B -module M and every $n \geq 0$, exactness in Mod_B gives the Cartesian square

$$\begin{array}{ccc} M[n] & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & M[n+1], \end{array}$$

since $M[n] \simeq \Omega M[n+1]$.

- (3) **Apply the split square-zero functor.** The split square-zero functor

$$\text{Mod}_B \longrightarrow \text{CAlg}(\text{Sp})_{B//B}, \quad N \longmapsto B \oplus N,$$

preserves limits by 2.2.1.

- (4) **Obtain the algebra pullback square.** Applying it to the displayed module square gives the canonical pullback square of connective commutative B -algebras

$$\begin{array}{ccc} B \oplus M[n] & \longrightarrow & B \\ \downarrow & & \downarrow d_0 \\ B & \xrightarrow{d_0} & B \oplus M[n+1]. \end{array}$$

- (5) **Identify the square-zero self-intersection.** Thus the identity

$$B \oplus M[n] \simeq B \times_{B \oplus M[n+1]} B$$

is a functorial square-zero pullback identity.

- (6) **Record compatibility under iteration.** The canonical comparisons for successive values of n are compatible under iterated pullback by limit functoriality and pullback pasting.
- (7) **Loop comparison.** The lower horizontal morphism is an isomorphism on π_0 for every $n \geq 0$, and hence is a nilpotent-excision leg. Since $Y \rightarrow S$ is reconstruction-local, evaluation on the displayed algebraic pullback gives a homotopy pullback square of mapping spaces. After passing to the slice over V and taking the homotopy fibre over the chosen point u , the retractive self-intersection lemma 3.10.1 gives a natural equivalence

$$\mathsf{T}_{Y/V,u}^n(M) \xrightarrow{\simeq} \Omega \mathsf{T}_{Y/V,u}^{n+1}(M).$$

- (8) **Spectrum coherence.** The split square-zero pullback construction is itself functorial, and the comparison between a two-step and an iterated three-step pullback is the canonical associativity comparison for finite limits. Thus the preceding natural equivalences satisfy the compatibility required for an Ω -spectrum object in

$$\mathrm{Fun}((\mathrm{Mod}_B)_{\geq 0}, \mathcal{S}_*).$$

Equivalently, they determine the functor

$$\mathsf{T}_{Y/V,u}^{\mathrm{cn}} : (\mathrm{Mod}_B)_{\geq 0} \longrightarrow \mathrm{Sp}.$$

The spectrum structure is induced by the functorial square-zero pullback identities.

- (9) **Coefficient cofiber square.** We next prove right exactness. Let

$$M' \longrightarrow M \longrightarrow M''$$

be a cofibre sequence of connective B -modules. For every $n \geq 0$, exactness in Mod_B gives

$$M'[n] \simeq \mathrm{fib}(M[n] \rightarrow M''[n]).$$

Apply the limit-preserving split square-zero functor of 2.2.1 to the corresponding Cartesian square

$$M'[n] \simeq M[n] \times_{M''[n]} 0.$$

This gives the canonical Cartesian square

$$\begin{array}{ccc} B \oplus M'[n] & \longrightarrow & B \\ \downarrow & & \downarrow d_0 \\ B \oplus M[n] & \longrightarrow & B \oplus M''[n]. \end{array}$$

- (10) **Nilpotent-excision hypothesis.** The lower horizontal morphism is surjective on π_0 . For $n > 0$ this is immediate because all three shifted coefficient modules have trivial degree-zero homotopy. For $n = 0$, the long exact sequence of the connective cofibre sequence gives

$$\pi_0(M) \twoheadrightarrow \pi_0(M'').$$

The kernel of the corresponding algebra map lies in the square-zero coefficient ideal, and is therefore square-zero. Thus in every degree the displayed square is a nilpotent-excision square.

- (11) **Right exactness.** Reconstruction locality gives a homotopy pullback

$$\begin{array}{ccc} Y(B \oplus M'[n]) & \longrightarrow & Y(B) \\ \downarrow & & \downarrow \\ Y(B \oplus M[n]) & \longrightarrow & Y(B \oplus M''[n]), \end{array}$$

where the relative mapping-space notation over V and S is suppressed. Taking the compatible homotopy fibres over u and using pullback pasting gives

$$\mathsf{T}_{Y/V,u}^n(M') \simeq \mathrm{fib}\left(\mathsf{T}_{Y/V,u}^n(M) \longrightarrow \mathsf{T}_{Y/V,u}^n(M'')\right).$$

These fibre identifications are natural in n and commute with the Ω -spectrum comparisons constructed above. Since limits in spectra are computed levelwise, they assemble to a fibre sequence

$$\mathsf{T}_{Y/V,u}^{\mathrm{cn}}(M') \longrightarrow \mathsf{T}_{Y/V,u}^{\mathrm{cn}}(M) \longrightarrow \mathsf{T}_{Y/V,u}^{\mathrm{cn}}(M'').$$

The target category Sp is stable, so the same sequence is a cofibre sequence.

(12) **Zero coefficient.** For the zero coefficient,

$$B \oplus 0 \simeq B,$$

and the defining restriction map is the identity. Hence

$$\mathsf{T}_{Y/V,u}^{\mathrm{cn}}(0) \simeq 0,$$

A zero-preserving functor from the connective prestable category $(\mathrm{Mod}_B)_{\geq 0}$ to a stable category which carries connective cofibre sequences to cofibre sequences preserves finite colimits. Therefore $\mathsf{T}_{Y/V,u}^{\mathrm{cn}}$ is right exact.

□

Proposition 3.10.5 (Postnikov continuity on connective coefficients). *For every connective B -module M , there is a canonical equivalence*

$$\mathsf{T}_{Y/V,u}^{\mathrm{cn}}(M) \xrightarrow{\simeq} \varprojlim_k \mathsf{T}_{Y/V,u}^{\mathrm{cn}}(\tau_{\leq k} M).$$

Proof.

(1) **Levelwise reduction.** It is enough to prove the assertion levelwise before stabilization. Fix $n \geq 0$. Reconstruction locality gives

$$Y(B \oplus M[n]) \simeq \varprojlim_d Y(\tau_{\leq d}^{\mathrm{CAlg}}(B \oplus M[n]))$$

and

$$Y(B) \simeq \varprojlim_d Y(\tau_{\leq d}^{\mathrm{CAlg}} B).$$

(2) **Stabilization of truncations.** For fixed d , use the canonical Postnikov truncation

$$M \longrightarrow \tau_{\leq k} M.$$

It induces a morphism of split square-zero algebras

$$B \oplus M[n] \longrightarrow B \oplus \tau_{\leq k} M[n].$$

Its underlying fibre is $\tau_{\geq k+1} M[n]$, hence is $(k+n+1)$ -connective. Therefore, once k is sufficiently large relative to d and n , it induces an equivalence

$$\tau_{\leq d}^{\mathrm{CAlg}}(B \oplus M[n]) \xrightarrow{\simeq} \tau_{\leq d}^{\mathrm{CAlg}}(B \oplus \tau_{\leq k} M[n]).$$

Thus, for every fixed d , the truncated square-zero evaluation stabilizes along the Postnikov tower of M .

- (3) **Define the truncated centred fibres.** For brevity let u_d denote the restriction of the chosen centre point u along

$$\mathrm{Spec}(\tau_{\leq d}^{\mathrm{CAlg}} B) \longrightarrow U$$

and put

$$\Phi_d(N) := \mathrm{fib}_{u_d} \left(Y(\tau_{\leq d}^{\mathrm{CAlg}}(B \oplus N[n])) \longrightarrow Y(\tau_{\leq d}^{\mathrm{CAlg}} B) \right).$$

- (4) **Use reconstruction locality.** Reconstruction locality gives

$$\mathrm{T}_{Y/V,u}^n(N) \simeq \varprojlim_d \Phi_d(N)$$

for $N = M$ and for every $N = \tau_{\leq k} M$.

- (5) **Use eventual constancy.** By the preceding stabilization calculation, for each fixed d the tower $\Phi_d(\tau_{\leq k} M)$ is eventually constant with value $\Phi_d(M)$.
- (6) **Interchange the inverse limits.** Therefore

$$\begin{aligned} \varprojlim_k \mathrm{T}_{Y/V,u}^n(\tau_{\leq k} M) &\simeq \varprojlim_k \varprojlim_d \Phi_d(\tau_{\leq k} M) \\ &\simeq \varprojlim_d \varprojlim_k \Phi_d(\tau_{\leq k} M) \\ &\simeq \varprojlim_d \Phi_d(M) \\ &\simeq \mathrm{T}_{Y/V,u}^n(M). \end{aligned}$$

- (7) **Identify the stated comparison.** Inverse limits commute because both are limits in spaces. Reversing the displayed canonical equivalence gives the comparison stated in the proposition.
- (8) **Promote the comparison to spectra.** The equivalences are natural in M and commute with the loop identifications of 3.10.4; they therefore assemble to the asserted equivalence of spectra.

□

Remark 3.10.1 (Accessibility is a separate finiteness property). Accessibility of the tangent functor requires a separate finiteness hypothesis on the nonlinear target. For example, an accessibility or local almost finite presentation condition can be used to prove that square-zero evaluation preserves the relevant filtered colimits.

The eventually connective stable envelope

Definition 3.10.3 (Eventually connective coefficient category). For a connective commutative ring spectrum B , define

$$\mathrm{Mod}_B^+ := \bigcup_{r \in \mathbb{Z}} (\mathrm{Mod}_B)_{\geq r} \subseteq \mathrm{Mod}_B.$$

Thus Mod_B^+ is the full stable subcategory of B -modules which are bounded below, with the lower bound allowed to depend on the object.

Proposition 3.10.6 (Spanier–Whitehead envelope of connective modules). *There is a canonical equivalence of stable ∞ -categories*

$$\mathrm{SW}((\mathrm{Mod}_B)_{\geq 0}) \xrightarrow{\sim} \mathrm{Mod}_B^+.$$

For every stable ∞ -category $\mathcal{D}$, restriction along the connective inclusion induces an equivalence

$$\mathrm{Fun}^{\mathrm{ex}}(\mathrm{Mod}_B^+, \mathcal{D}) \xrightarrow{\sim} \mathrm{Fun}^{\mathrm{rex}}((\mathrm{Mod}_B)_{\geq 0}, \mathcal{D}),$$

where rex denotes right-exact functors, equivalently zero-preserving finite-colimit-preserving functors.

Proof.

- (1) **Identify the stabilization.** The connective aisle $(\mathrm{Mod}_B)_{\geq 0}$ is a prestable ∞ -category: it is pointed, admits finite colimits, and its suspension functor is fully faithful. Its Spanier–Whitehead stabilization is obtained by formally inverting suspension. The inclusion

$$(\mathrm{Mod}_B)_{\geq 0} \hookrightarrow \mathrm{Mod}_B$$

therefore induces an exact functor from the stabilization to Mod_B . Its essential image consists precisely of the finite desuspensions of connective modules, namely the eventually connective modules. Full faithfulness follows from the stabilization mapping formula and the fact that suspension is already invertible in Mod_B . This proves the first equivalence.

- (2) **Universal property.** The second assertion is the universal property of the Spanier–Whitehead stabilization: for a pointed ∞ -category with finite colimits and a stable target, exact functors out of its stabilization are equivalent to right-exact functors out of the original category. See [34].

□

Theorem 3.10.1 (Eventually connective tangent functor). *Let $Y \rightarrow S$ be reconstruction-local and let*

$$U = \mathrm{Spec}(B) \longrightarrow V = \mathrm{Spec}(R) \longrightarrow S$$

be a connective affine centre. Let

$$u : U \rightarrow Y$$

be a chosen U -point over S . There is a canonical exact functor

$$\mathrm{T}_{Y/V,u}^{\mathrm{sp},+} : \mathrm{Mod}_B^+ \longrightarrow \mathrm{Sp}$$

whose restriction to connective modules is $\mathrm{T}_{Y/V,u}^{\mathrm{cn}}$. It is characterized, through a contractible space of choices, by this property.

For every connective coefficient M , its value is therefore the actual stabilized split-square-zero lifting spectrum of 3.10.2, and that connective restriction is Postnikov-continuous by 3.10.5.

Proof.

- (1) **Use right exactness on connective modules.** By 3.10.4,

$$\mathrm{T}_{Y/V,u}^{\mathrm{cn}} : (\mathrm{Mod}_B)_{\geq 0} \rightarrow \mathrm{Sp}$$

is right exact.

- (2) **Apply the Spanier–Whitehead universal property.** Apply the universal property of 3.10.6 with $\mathcal{D} = \text{Sp}$.
- (3) **Construct the exact extension.** It supplies a unique exact extension to Mod_B^+ , and the space of such extensions is contractible.
- (4) **Identify the remaining assertions.** The final statements are the defining restriction and 3.10.5.

□

Proposition 3.10.7 (Cotangent representation of the represented tangent functor). *For the represented target $U = \text{Spec}(B) \rightarrow V = \text{Spec}(R)$ with chosen U -point $\text{id}_U : U \rightarrow U$, there is a natural equivalence of exact functors on eventually connective modules*

$$\mathbb{T}_{U/V}^{\text{sp},+}(M) \simeq \text{map}_{\text{Mod}_B}(L_{B/R}, M), \quad M \in \text{Mod}_B^+.$$

The displayed equivalence concerns Mod_B^+ . The mapping-spectrum functor on the right extends canonically to all of Mod_B .

Proof.

- (1) **Identify the connective lifting spaces.** For connective M , the n -th space of the actual lifting spectrum is the space of R -algebra sections of

$$B \oplus M[n] \longrightarrow B$$

restricting to the identity.

- (2) **Apply cotangent classification.** Cotangent classification identifies it with

$$\Omega^\infty \text{map}_{\text{Mod}_B}(L_{B/R}, M[n]).$$

- (3) **Compare loops with suspension.** The loop identifications used in 3.10.4 agree with suspension in the mapping spectrum, so

$$\mathbb{T}_{U/V}^{\text{cn}}(M) \simeq \text{map}_{\text{Mod}_B}(L_{B/R}, M)$$

- (4) **Obtain the connective mapping-spectrum formula.** for connective M .
- (5) **Use exactness of the mapping-spectrum functor.** The mapping-spectrum functor on the right is exact on all B -modules.
- (6) **Restrict to eventually connective modules.** Restrict it to Mod_B^+ ; both sides are then exact extensions of the same right-exact functor on connective modules.
- (7) **Apply the stabilization universal property.** The Spanier–Whitehead universal property identifies them uniquely.

□

Definition 3.10.4 (Relative tangent functor of a centred map). Let $Y \rightarrow S$ be reconstruction-local and let

$$U = \operatorname{Spec}(B) \longrightarrow V = \operatorname{Spec}(R) \longrightarrow S$$

be a connective affine centre. Let

$$u : U \longrightarrow Y$$

be a chosen U -point over S . Postcomposition with u gives a natural morphism of exact functors on Mod_B^+

$$\operatorname{map}_{\operatorname{Mod}_B}(L_{B/R}, -)|_{\operatorname{Mod}_B^+} \longrightarrow \mathbb{T}_{Y/V, u}^{\operatorname{sp}, +}.$$

Define the **relative tangent functor of the centre** by

$$\mathbb{T}_{U/Y; V}^{\operatorname{sp}, +} := \operatorname{fib} \left(\operatorname{map}_{\operatorname{Mod}_B}(L_{B/R}, -)|_{\operatorname{Mod}_B^+} \longrightarrow \mathbb{T}_{Y/V, u}^{\operatorname{sp}, +} \right).$$

It is an exact functor $\operatorname{Mod}_B^+ \rightarrow \operatorname{Sp}$.

Remark 3.10.2 (Domain of the tangent functor). For a general reconstruction-local target, the construction above produces an exact tangent functor on the eventually connective category

$$\operatorname{Mod}_B^+.$$

An extension to all of Mod_B requires an additional continuity theorem, such as a right-left or pro-coherent continuity result. Represented targets already carry the full exact functor

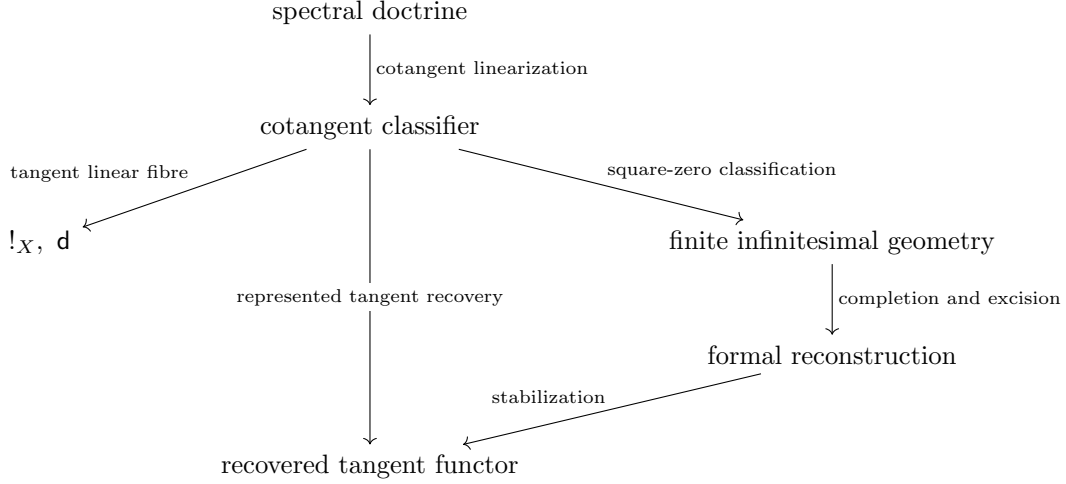
$$M \longmapsto \operatorname{map}_{\operatorname{Mod}_B}(L_{B/R}, M)$$

on Mod_B .

The reconstruction theory has now returned to stable coefficients. At a represented centre the tangent functor is represented by the cotangent complex; for a general reconstruction-local target it is an exact functor on eventually connective coefficients. With both nonlinear reconstruction and tangent recovery in hand, we can finally compare them with the differential-linear branch and with the stable boundary of reconstruction localization.

3.11 Comparison with the differentiated spectral doctrine

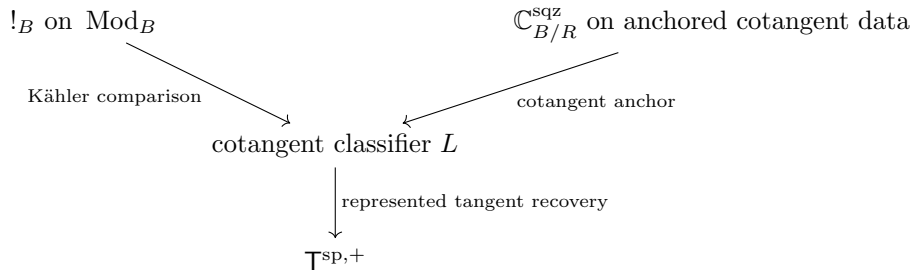
All objects needed for the deferred comparison have now been constructed. The following diagram can therefore be read from top to bottom without introducing future dependencies:



The left branch is the differential-linear structure internal to the stable coefficient fibre. The right branch is the nonlinear infinitesimal geometry used for reconstruction. They are not identified; their common interface is the cotangent classifier. The represented finite-projective case is where that interface becomes an explicit universal derivation.

There are three comparisons. First, the differential exponential and the square-zero reconstruction comonad act on different categories. Second, stabilization of the centre-free localization produces only the unit-boundary sector of the four-adjoint fracture formalism. Third, on represented finite-projective models, the differential-linear derivative, infinitesimal first-order variation, and recovered tangent functor meet at the universal cotangent derivation.

The cotangent-mediated comparison, and the absence of a direct comparison between the two comonads, is summarized by



The diagonal arrows are therefore mediated by the cotangent classifier; no natural transformation between the two comonads is asserted.

3.11.1 The differential exponential and the square-zero reconstruction comonad

Chapter 2 constructed, in every tangent fibre $\mathrm{Mod}_B \simeq \mathrm{QCoh}(\mathrm{Spec} B)$, the cofree cocommutative-coalgebra exponential

$$!_B : \mathrm{Mod}_B \longrightarrow \mathrm{Mod}_B$$

with its Seely comparisons, primitive infinitesimal coalgebras, codereliction, and deriving transformation. The present chapter has constructed a different canonical comonad,

$$\mathbb{C}_{B/R}^{\mathrm{sqz}} = \mathrm{cot}_{B/R} \mathrm{sqz}_{B/R},$$

on the anchored shifted-cotangent category

$$\mathrm{AnchCot}_{B/R, \geq 0}^{\mathrm{aperf}}.$$

Remark 3.11.1 (The two reconstruction-related comonads). The differential exponential

$$!_B : \mathrm{Mod}_B \longrightarrow \mathrm{Mod}_B$$

governs differential-linear use of coefficients in the tangent fibre. The comonad

$$\mathbb{C}_{B/R}^{\mathrm{sqz}} = \mathrm{cot}_{B/R} \mathrm{sqz}_{B/R}$$

acts on the anchored shifted-cotangent category $\mathrm{AnchCot}_{B/R, \geq 0}^{\mathrm{aperf}}$ and governs comonadic reconstruction of complete augmented algebras. Their comparison in this chapter is mediated by the cotangent complex and, on finite-projective models, by the universal derivation described in 3.11.1.

The fixed-centre comparison above is nonlinear and comonadic. The centre-free localization has a different stable shadow: after stabilization, its unit has an exact fibre. We record that boundary next and compare only its formal shape with the BNV fracture theorem.

3.11.2 Stable reconstruction defect and the BNV comparison

The left-exact reconstruction localization yields a canonical stable localization sequence. This is intentionally weaker than the BNV-type fracture theorem of Chapter 2: a reflective adjunction supplies one exact unit boundary after stabilization, whereas the remaining BNV sectors require the four-adjoint formalism constructed there.

Definition 3.11.1 (Stable reconstruction categories). Put

$$\mathcal{D}_{\mathrm{Sp}/S} := \mathrm{Sp}(((\mathcal{H}_{\mathrm{Sp}})_S)_*), \quad \mathcal{D}_{\mathrm{Sp}/S}^{\mathrm{Recon}} := \mathrm{Sp}(((\mathcal{H}_{\mathrm{Sp}})_S)^{\mathrm{Recon}})_*).$$

These are the stabilizations of the pointed ordinary and reconstruction-local spectral Zariski slices.

Theorem 3.11.1 (Stabilized reconstruction localization). *The adjunction*

$$L_S^{\mathrm{Recon}} \dashv \iota_S$$

stabilizes canonically to an exact reflective localization

$$\mathcal{D}_{\mathrm{Sp}/S} \begin{array}{c} \xrightarrow{L_S^{\mathrm{Recon}, \mathrm{st}}} \\ \xleftarrow{\iota_S^{\mathrm{st}}} \end{array} \mathcal{D}_{\mathrm{Sp}/S}^{\mathrm{Recon}}$$

with ι_S^{st} fully faithful. Consequently

$$\text{Recon}_S^{\text{st}} := \iota_S^{\text{st}} L_S^{\text{Recon, st}}$$

is an exact idempotent localization endofunctor of $\mathcal{D}_{\text{Sp}/S}$.

Proof.

- (1) **Use left exactness before stabilization.** By 3.9.2, L_S^{Recon} preserves finite limits as well as all colimits, and ι_S is a fully faithful right adjoint preserving limits.
- (2) **Pass the adjunction to pointed objects.** The induced adjunction on pointed objects therefore preserves the finite-limit structure used in stabilization.
- (3) **Stabilize the adjunction.** Functoriality of spectrum objects for finite-limit preserving functors, [34, Section 1.4.2], gives an adjunction of stable ∞ -categories

$$L_S^{\text{Recon, st}} \dashv \iota_S^{\text{st}}.$$

- (4) **Prove exactness of the stabilized functors.** The left adjoint is exact because it is induced by a finite-limit preserving functor on pointed objects, and the right adjoint is exact because it preserves finite limits between stable categories.
- (5) **Prove full faithfulness after stabilization.** Full faithfulness of ι_S^{st} follows from full faithfulness of ι_S .
- (6) **Apply the localization identities.** The reflective-localization identities then make $\text{Recon}_S^{\text{st}}$ exact and idempotent.

□

Definition 3.11.2 (Stable reconstruction defect). For $E \in \mathcal{D}_{\text{Sp}/S}$, define

$$\mathcal{K}_S^{\text{Recon}}(E) := \text{fib}(E \longrightarrow \text{Recon}_S^{\text{st}}(E)).$$

Let

$$\mathcal{K}_S^{\text{Recon}} := \ker(L_S^{\text{Recon, st}}) \subseteq \mathcal{D}_{\text{Sp}/S}.$$

We call $\mathcal{K}_S^{\text{Recon}}(E)$ the **stable reconstruction defect** of E .

Proposition 3.11.1 (Stable reconstruction defect triangle). *For every $E \in \mathcal{D}_{\text{Sp}/S}$ there is a canonical exact triangle*

$$\mathcal{K}_S^{\text{Recon}}(E) \longrightarrow E \longrightarrow \text{Recon}_S^{\text{st}}(E).$$

Moreover:

(i)

$$L_S^{\text{Recon, st}} \mathcal{K}_S^{\text{Recon}}(E) \simeq 0;$$

(ii) the functor

$$\mathcal{K}_S^{\text{Recon}} : \mathcal{D}_{\text{Sp}/S} \longrightarrow \mathcal{K}_S^{\text{Recon}}$$

is right adjoint to the fully faithful inclusion

$$\mathcal{K}_S^{\text{Recon}} \hookrightarrow \mathcal{D}_{\text{Sp}/S};$$

(iii) there is therefore a canonical stable localization sequence

$$\mathcal{K}_S^{\text{Recon}} \hookrightarrow \mathcal{D}_{\text{Sp}/S} \xrightarrow{L_S^{\text{Recon, st}}} \mathcal{D}_{\text{Sp}/S}^{\text{Recon}}.$$

The fibre construction is functorial in E .

Proof.

- (1) **Kernel property.** The exact triangle is the defining fibre sequence of 3.11.2. Apply the exact localization functor $L_S^{\text{Recon, st}}$. Since the localization unit becomes an equivalence after localization,

$$L_S^{\text{Recon, st}}(E) \xrightarrow{\simeq} L_S^{\text{Recon, st}} \text{Recon}_S^{\text{st}}(E),$$

its fibre vanishes. This proves (i).

- (2) **Right-adjoint property.** Let $K \in \mathcal{K}_S^{\text{Recon}}$. By the stabilized adjunction,

$$\begin{aligned} \text{map}_{\mathcal{D}_{\text{Sp}/S}}(K, \text{Recon}_S^{\text{st}}(E)) &\simeq \text{map}_{\mathcal{D}_{\text{Sp}/S}^{\text{Recon}}}(L_S^{\text{Recon, st}} K, L_S^{\text{Recon, st}} E) \\ &\simeq 0. \end{aligned}$$

Applying $\text{map}_{\mathcal{D}_{\text{Sp}/S}}(K, -)$ to the defining exact triangle therefore gives

$$\text{map}_{\mathcal{D}_{\text{Sp}/S}}(K, \mathcal{K}_S^{\text{Recon}}(E)) \xrightarrow{\simeq} \text{map}_{\mathcal{D}_{\text{Sp}/S}}(K, E).$$

Together with (i), this is precisely the right-adjoint universal property in (ii). The localization sequence in (iii) is the corresponding exact reflective localization.

□

Remark 3.11.2 (Comparison with the stable exact-adjoint fracture). The fibre

$$\mathcal{K}_S^{\text{Recon}}(E) = \text{fib}(E \rightarrow \iota_S^{\text{st}} L_S^{\text{Recon, st}} E)$$

has the unit-boundary form

$$\mathbf{A}(E) = \text{fib}(E \rightarrow jLE)$$

from the exact-adjoint fracture theorem of Chapter 2. The reconstruction localization supplies the reflective adjunction

$$L_S^{\text{Recon, st}} \dashv \iota_S^{\text{st}},$$

so the comparison identifies the unit-boundary sector of that fracture pattern. Recovering the secondary and cycle sectors and the characteristic morphism of the four-adjoint BNV formalism requires the additional right adjoint constructed in the preceding chapter.

The final comparison is the represented finite-projective case, where the differential-linear, infinitesimal, and reconstructed tangent constructions all admit the same explicit cotangent representative.

3.11.3 Common cotangent interface with differential-linear differentiation

Corollary 3.11.1 (Differential-linear differentiation and formal reconstruction meet at the cotangent classifier). *Let B be connective, let P be a finite projective B -module, and put*

$$A := \mathrm{Sym}_B(P^\vee).$$

Then

$$L_{A/B} \simeq A \otimes_B P^\vee,$$

and the represented tangent functor of the affine context

$$\mathrm{Spec}(A) \longrightarrow \mathrm{Spec}(B)$$

is

$$\mathsf{T}_{\mathrm{Spec}(A)/\mathrm{Spec}(B)}^{\mathrm{sp},+}(N) \simeq \mathrm{map}_{\mathrm{Mod}_A}(L_{A/B}, N)$$

for every $N \in \mathrm{Mod}_A^+$.

Under the differential-linear construction of Chapter 2, the deriving transformation

$$\mathsf{d}_P : !_B P \otimes_B P \longrightarrow !_B P$$

induces the convolution derivation

$$D : \mathcal{O}_!(P) \longrightarrow \mathcal{O}_!(P) \otimes_B P^\vee, \quad \mathcal{O}_!(P) := \underline{\mathrm{Hom}}_B(!_B P, B).$$

The comonad counit determines the canonical algebra morphism

$$j_P : A = \mathrm{Sym}_B(P^\vee) \longrightarrow \mathcal{O}_!(P),$$

and the differential-linear comparison theorem gives a canonical homotopy

$$D \circ j_P \simeq (j_P \otimes 1) \circ d_{\mathrm{univ}},$$

equivalently, the square

$$\begin{array}{ccc} A & \xrightarrow{j_P} & \mathcal{O}_!(P) \\ d_{\mathrm{univ}} \downarrow & & \downarrow D \\ A \otimes_B P^\vee & \xrightarrow{j_P \otimes 1} & \mathcal{O}_!(P) \otimes_B P^\vee \end{array}$$

commutes up to the canonical homotopy. Here

$$d_{\mathrm{univ}} : A \longrightarrow A \otimes_B P^\vee \simeq L_{A/B}$$

is the spectral universal derivation.

The canonical first infinitesimal neighbourhood of the diagonal constructed in Chapter 2 has first-order difference given by the same universal relative derivation. Hence, on this common represented finite-projective domain, differential-linear differentiation, differential-cohesive first-order variation, and tangent recovery from formal reconstruction meet at the cotangent classifier $L_{A/B}$, through the canonical comparisons displayed above.

Proof.

- (1) **Identify the cotangent complex.** The cotangent calculation

$$L_{A/B} \simeq A \otimes_B P^\vee$$

is 2.3.2.

- (2) **Identify the represented tangent functor.** The represented tangent identification is 3.10.7.

- (3) **Convolution derivative.** The cofree function algebra and the canonical map

$$j_P : \mathrm{Sym}_B(P^\vee) \longrightarrow \mathcal{O}_!(P)$$

are 2.6.2. The differential-linear deriving transformation induces the convolution derivation D by 2.6.8, and 2.6.3 gives the canonical commuting comparison

$$D \circ j_P \simeq (j_P \otimes 1) \circ d_{\mathrm{univ}}.$$

Thus the differential-linear derivative is compatible with the universal spectral Kähler derivative after passage along the canonical comparison j_P ; the comparison is therefore mediated by j_P .

- (4) **First infinitesimal neighbourhood.** Finally, 2.11.3 identifies the first-order difference of the two retractions of the canonical first infinitesimal neighbourhood with the same universal relative derivation. Together with cotangent representation of the recovered tangent functor, these comparisons establish the claimed common cotangent interface.

□

The comparison is now precise. Formal reconstruction neither identifies the differential-linear and nonlinear comonads nor reproduces the full BNV fracture. Instead, the cotangent classifier is the shared interface: it generates formal order, classifies square-zero variation, represents the recovered tangent functor on represented targets, and mediates the finite-projective Kähler comparison.

3.12 The formal reconstruction interface

The chapter exports five packages rather than one undifferentiated list of constructions: approximation and completion; fixed-centre comonadic reconstruction; Artinian and centre-free reconstruction semantics; centred tangent recovery; and comparison with the differentiated doctrine. The theorem below records the individual results that make up those packages so that later chapters can cite them precisely.

Theorem 3.12.1 (Formal reconstruction of the spectral doctrine). *The constructions of this chapter provide the following.*

- (i) **Postnikov reconstruction.** *Every connective commutative ring spectrum is recovered as*

$$B \simeq \varprojlim_n \tau_{\leq n}^{\mathrm{CAlg}} B,$$

and each positive Postnikov transition is a structured square-zero extension. The finite composite

$$\tau_{\leq n}^{\text{CAlg}} B \longrightarrow H\pi_0(B)$$

is therefore a finite composite of structured square-zero extensions; its corresponding affine morphism is a finite spectral infinitesimal substitution.

(ii) **Canonical adic filtration.** Filtration-zero augmentation admits a left adjoint

$$\text{adic} \dashv \text{aug},$$

and filtered completion is a symmetric-monoidal localization. Together they assign a functorial complete filtered algebra to every augmentation.

(iii) **Cotangent description of the associated graded.** For every augmentation $A \rightarrow B$,

$$\widehat{\text{Gr adic}}(A \rightarrow B) \simeq \text{Sym}_B^{\text{gr}}(L_{B/A}[-1](1)).$$

For connective maps surjective on π_0 , filtration zero of the completion is the Amitsur totalization. Under almost finite presentation it is derived completion for $I = \ker(\pi_0(A) \rightarrow \pi_0(B))$.

(iv) **Complete almost finitely presented augmentations.** On the domain of connective almost finitely presented augmentations that are surjective on π_0 , the fixed points of canonical adic completion are exactly the derived I -complete augmentations.

(v) **Comonadic shifted-cotangent-structured-square-zero reconstruction.** For a connective centre B over R , the adjunction

$$\text{CAlg}(\text{Sp})_{R//B}^{\text{cn,afp},\wedge} \xrightleftharpoons[\text{sqz}_{B/R}]{\text{cot}_{B/R}} \text{AnchCot}_{B/R,\geq 0}^{\text{aperf}}$$

is comonadic. Hence

$$\text{CAlg}(\text{Sp})_{R//B}^{\text{cn,afp},\wedge} \simeq \text{Coalg}_{\text{C}^{\text{sqz}}_{B/R}}(\text{AnchCot}_{B/R,\geq 0}^{\text{aperf}}).$$

The associated cobar resolution reconstructs each complete almost finitely presented augmentation from its comonadic shifted-cotangent data.

(vi) **Artinian infinitesimal tests.** At a centre B over R , with B coherent, spectral Artinian extensions are generated by finitely many structured square-zero stages with coefficients in connective coherent B -modules. Affine realization identifies them with the bounded coherent fixed-centre subcategory of the finite infinitesimal geometry of Chapter 2.

(vii) **Schlessinger formal-moduli problems.** The explicit localization of

$$\text{Fun}(\text{Art}_{B/R}^{\text{sp}}, \mathcal{S})$$

at the centre and square-zero excision maps exists and has as local objects the spectral formal-moduli problems $\text{FMP}_{B/R}^{\text{sp}}$.

- (viii) **Reconstruction localization over a base.** For every spectral Zariski stack S , the localization L_S^{Recon} generated by Postnikov continuity and nilpotent excision is left exact. Its essential image $((\mathcal{H}_{\text{Sp}})_{/S})^{\text{Recon}}$ is therefore an ∞ -topos, and

$$\text{Recon}_S = \iota_S L_S^{\text{Recon}}$$

is an idempotent left-exact localization endofunctor of $(\mathcal{H}_{\text{Sp}})_{/S}$.

- (ix) **Stable localization boundary.** Stabilization gives an exact reflective localization

$$\mathcal{D}_{\text{Sp}/S} \begin{array}{c} \xrightarrow{L_S^{\text{Recon, st}}} \\ \xleftarrow{\iota_S^{\text{st}}} \end{array} \mathcal{D}_{\text{Sp}/S}^{\text{Recon}}$$

and a functorial exact triangle

$$\mathcal{K}_S^{\text{Recon}}(E) \longrightarrow E \longrightarrow \text{Recon}_S^{\text{st}}(E).$$

The first term belongs to $\ker(L_S^{\text{Recon, st}})$ and is right adjoint to the inclusion of that kernel.

- (x) **Centred Artinian leaves.** At an affine centre with B coherent,

$$U = \text{Spec}(B) \longrightarrow V = \text{Spec}(R) \longrightarrow S,$$

and a chosen U -point $u : U \rightarrow Y$ of a reconstruction-local object $Y \rightarrow S$, the centred Artinian leaf is a spectral formal-moduli problem. Objectwise it is the centred fibre of the endpoint restriction associated with the corresponding finite infinitesimal arrow.

- (xi) **Eventually connective tangent functor.** Lifting along split square-zero augmentations on connective B -modules forms an Ω -spectrum and is right exact. Its Spanier–Whitehead extension is an exact functor

$$\mathbb{T}_{Y/V, u}^{\text{sp}, +} : \text{Mod}_B^+ \longrightarrow \text{Sp}.$$

For represented targets,

$$\mathbb{T}_{\text{Spec}(B)/\text{Spec}(R)}^{\text{sp}, +}(M) \simeq \text{map}_{\text{Mod}_B}(L_{B/R}, M).$$

- (xii) **Comparison with differential-linear differentiation.** For P finite projective and $A = \text{Sym}_B(P^\vee)$, the represented tangent functor, the first infinitesimal neighbourhood of the diagonal, and the differential-linear convolution derivative all meet at the universal derivation

$$d_{\text{univ}} : A \longrightarrow L_{A/B} \simeq A \otimes_B P^\vee.$$

The comparison with the convolution derivative is mediated by the canonical map

$$j_P : \text{Sym}_B(P^\vee) \longrightarrow \mathcal{O}_!(P).$$

Proof.

- (1) **Cite Postnikov reconstruction.** Item (i) is 3.2.1 together with 3.2.2.
- (2) **Cite filtered reconstruction.** Items (ii)–(iv) follow from 3.3.1, 3.3.1, 3.5.1, 3.5.2, and 3.5.1.

- (3) **Cite comonadic reconstruction.** Item (v) is 3.6.1 and its two corollaries 3.6.1 and 3.6.2.
- (4) **Cite Artinian reconstruction.** Item (vi) follows from 3.7.2 and 3.7.4.
- (5) **Cite Schlessinger reconstruction.** Item (vii) is 3.8.1 together with 3.8.1.
- (6) **Cite reconstruction localization.** Item (viii) follows from 3.9.1, 3.9.2, and 3.9.2.
- (7) **Cite the stable boundary.** Item (ix) is 3.11.1 and 3.11.1.
- (8) **Cite the centred leaf comparison.** Item (x) is 3.10.1 and 3.10.2.
- (9) **Cite tangent recovery.** Item (xi) is 3.10.1 together with 3.10.7.
- (10) **Cite the differential-linear comparison.** Item (xii) is 3.11.1.

□

3.13 Chapter summary

Two approximation coordinates now organize reconstruction. Postnikov height resolves a connective spectral algebra into canonical finite square-zero stages over its discrete centre and recovers it by inverse limit. Formal order resolves an augmentation through its canonical adic filtration, whose complete associated graded is the free graded commutative algebra on the shifted relative cotangent complex. These coordinates are different, but the cotangent classifier links formal order to the infinitesimal geometry of the preceding chapter.

At a fixed centre, that link becomes the principal reconstruction theorem. Complete almost finitely presented augmentations are comonadically recovered from the anchored shifted cotangent object

$$L_{B/R}[-1] \longrightarrow L_{B/A}[-1]$$

with coalgebra coherence generated by structured square-zero formation. Its bounded coherent fragment is the Artinian test category and yields Schlessinger formal-moduli semantics. Removing the centre leaves Postnikov continuity and nilpotent excision as the global reconstruction equations, whose left-exact localization defines the centre-free semantics.

Returning to a centre closes the loop. Split square-zero lifting stabilizes to an exact tangent functor on eventually connective coefficients, and represented targets recover the cotangent complex as its representing object. The differential-linear comonad, the square-zero reconstruction comonad, and the stable localization boundary remain distinct constructions; what unifies them is the cotangent classifier, which measures first-order variation, generates formal order, classifies square-zero reconstruction, and represents tangent behaviour.

Chapter 4

Spectral de Rham Reflection and Infinitesimal Descent

4.1 Orientation: from infinitesimal invariance to de Rham descent

The preceding chapters supplied two complementary pieces of structure. The differentiated spectral doctrine constructed structured square-zero extensions, finite infinitesimal substitutions, and their endpoint semantics; formal reconstruction then showed how suitably complete nonlinear geometry can be recovered from infinitesimal data. The present chapter asks a different question. We identify contexts that differ only by finite infinitesimal variation and ask what descent theory survives after that identification.

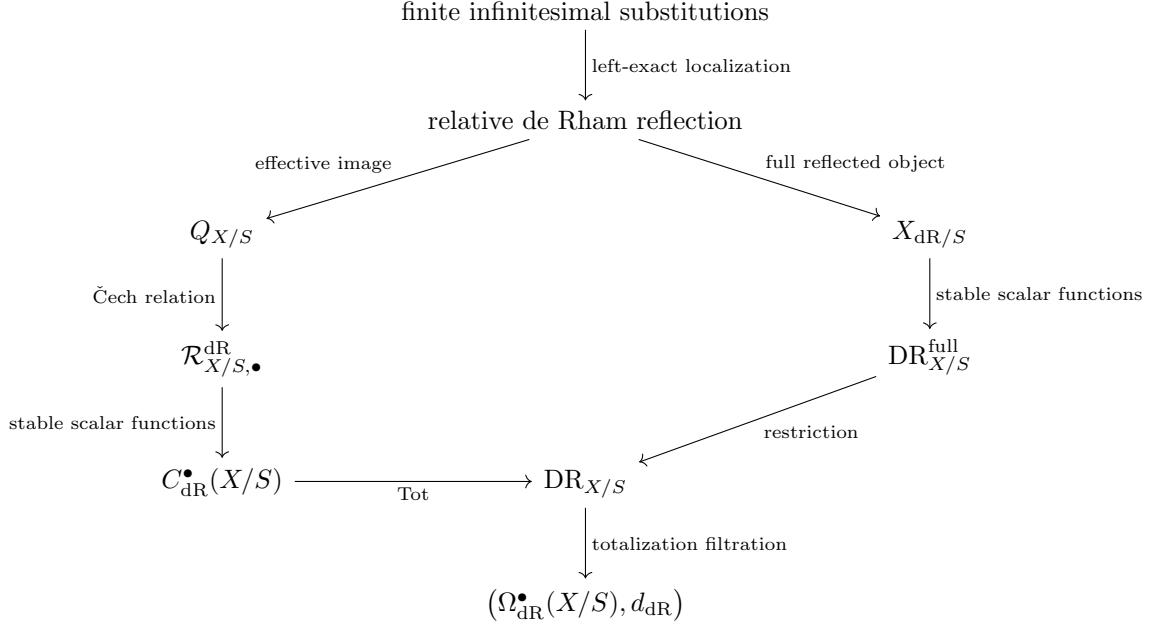
The principal result is a de Rham descent calculus attached to that identification. We construct the universal left-exact reflection that inverts finite infinitesimal variation, separate its universal target from the effective quotient actually presented by points of the original context, and show that stable scalar descent on the resulting effective relation produces intrinsic forms, a homotopy-coherent differential, products, closedness data, and primitives. Two later comparisons then measure different boundaries of this calculus: first-order restriction recovers the cotangent complex, whereas reduced-classical reflection adds the convergence needed to pass beyond finite infinitesimal stages.

The first step is therefore a localization problem. Affine square-zero substitutions generate a pullback-stable congruence, and the universal left-exact localization at that congruence is the spectral de Rham reflection. Its local objects are precisely the stacks that cannot distinguish a finite infinitesimal thickening from its centre. The following display records the eventual reflector; its local-equivalence class and relative form are constructed before the notation is used systematically:

$$L^{\mathrm{dR}} : \mathcal{H}_{\mathrm{Sp}} \rightleftarrows \mathcal{H}_{\mathrm{Sp}}^{\mathrm{dR}} : \iota^{\mathrm{dR}}.$$

A second issue appears as soon as we try to extract a descent object from the relative unit. The effective image of a localization unit can be strictly smaller than the whole reflected object. Its kernel-pair nerve therefore presents that effective image. We separate the universal and effective parts: the relative unit factors through an effective quotient $Q_{X/S}$, and the resulting kernel-pair groupoid presents $Q_{X/S}$. This distinction will remain active throughout the chapter. Stable scalar functions on the groupoid recover functions on $Q_{X/S}$; functions on the full reflected object are compared to them by restriction.

The dependency structure therefore branches into a universal object and an effective presentation; the effective branch supplies the descent calculus, while the universal branch remains available for comparison. The next diagram is only a route map: its symbols label objects introduced later, and no notation in it is used systematically before its construction.



The chapter follows this dependency order. We first construct the absolute and indexed de Rham reflectors and identify finite-infinitesimal invariance. We then factor the relative unit and prove that its effective relation is the Čech groupoid that supports descent. Only then do we build the stable scalar interface needed to take functions on that relation. Effective descent produces the cochain algebra and its totalization; stable Dold–Kan then exposes forms, the homotopy-coherent differential, products, closedness, and primitives. The final sections establish descent and comprehension and treat the cotangent and reduced-classical comparisons as two separate exits from the construction.

The conventions on universes, sheaves, effective epimorphisms, higher coherence, and notation from Chapter 1 remain in force. In particular, once the functorial source of a coherence has been identified, routine identity, composition, mate, and higher-pasting coherences are governed by the standing convention and will not be restated at every later occurrence.

4.2 From infinitesimal substitution to indexed de Rham reflection

The differentiated doctrine already supplies the arrows that should become invisible: underlying square-zero infinitesimal substitutions. To turn that test geometry into a modality, we must prove that the class generated by those arrows is stable under arbitrary pullback. That pullback stability is the substantive input: once it is established, accessible localization produces a left-exact reflector, and left exactness then makes the construction compatible with substitution.

This section therefore has three tasks. We first pass from affine square-zero generators to a pullback-stable strongly saturated congruence. We then construct the universal reflector and identify

its local objects by finite-infinitesimal invariance, including the endpoint comparison of the preceding chapter. Finally we relativize the construction over a base and prove coherent base change, thereby obtaining an indexed reflection rather than merely an ambient localization.

4.2.1 Affine square-zero generators and the pullback-stable congruence

The desired reflector will be left exact only if its local-equivalence class is stable under pullback. We therefore begin one level below the reflector itself. Smallness gives an accessible localization at the affine square-zero arrows, while affine density lets us test pullback stability on affine base changes. The point of the next lemma is to show that no additional pullback-generators are required.

By Convention 1.1.4, the category of $\mathbb{U}$ -small affine spectral schemes is $\mathbb{V}$ -small. Let Σ_{sqz} be the resulting $\mathbb{V}$ -small collection of morphisms of representables induced by the underlying square-zero infinitesimal substitutions of 2.8.2 between $\mathbb{U}$ -small affine spectral schemes. Thus a generator has the form

$$h_A \longrightarrow h_{A'}$$

for a spectral square-zero extension $A' \rightarrow A$.

For a map $h_V \rightarrow h_{A'}$, pullback is represented by the affine base-change square

$$\begin{array}{ccc} h_V \times_{h_{A'}} h_A & \longrightarrow & h_A \\ \downarrow & & \downarrow \\ h_V & \longrightarrow & h_{A'}. \end{array}$$

Lemma 4.2.1 (Affine density in slices). *Let $S \in \mathcal{H}_{\text{Sp}}$ and $Y \rightarrow S$ be an object of the slice. Then the canonical morphism*

$$\text{colim}_{(T,u) \in \text{AffPt}(Y)} h_T \longrightarrow Y$$

is an equivalence in $(\mathcal{H}_{\text{Sp}})_{/S}$.

Proof.

- (1) **Establish the next claim.** Regard Y through the fully faithful inclusion $\mathcal{H}_{\text{Sp}} \hookrightarrow \text{PSh}(\text{Aff}_{\text{Sp}})$.
- (2) **Apply Yoneda density.** Yoneda density gives, in the presheaf slice over the underlying presheaf of S ,

$$Y \simeq \text{colim}_{(T,u) \in \text{AffPt}(Y)} h_T.$$

- (3) **Record smallness of the indexing category.** The indexing category is $\mathbb{V}$ -small by Convention 1.1.4.
- (4) **Use colimit preservation by sheafification.** Spectral Zariski sheafification preserves colimits, every representable is already a spectral Zariski sheaf, and Y is already local.
- (5) **Use the stated construction.** Applying sheafification to the density colimit therefore gives the displayed equivalence in $(\mathcal{H}_{\text{Sp}})_{/S}$.
- (6) **Identify the affine-point indexing category.** The affine-point category is the canonical indexing object for this density presentation. Compare the Yoneda-density and sheaf-localization formalism of [35].

□

Definition 4.2.1 (De Rham local-equivalence class). Let W^{dR} be the smallest strongly saturated class of morphisms in $\mathcal{H}_{\text{Sp}}$ which contains Σ_{sqz} and is stable under arbitrary pullback.

Lemma 4.2.2 (Generated square-zero congruence). *The strongly saturated class $\text{Sat}(\Sigma_{\text{sqz}})$ is already stable under arbitrary pullback. Hence*

$$W^{\text{dR}} = \text{Sat}(\Sigma_{\text{sqz}}).$$

If $f : A \rightarrow B$ lies in W^{dR} , every iterated diagonal of f also lies in W^{dR} .

Proof.

(1) **Base-change every square-zero generator.** Put

$$W_0 := \text{Sat}(\Sigma_{\text{sqz}}),$$

and let L_Σ be the accessible localization generated by the small family Σ_{sqz} , whose existence follows from [35, Proposition 5.5.4.15]. Its local-equivalence class is W_0 .

Let $h_U \rightarrow h_T$ be a generator and let $Y \rightarrow h_T$ be arbitrary. Lemma 4.2.1 gives

$$Y \simeq \text{colim}_{(V,u) \in \text{AffPt}(Y)} h_V$$

in the slice over h_T . Colimits in an infinity-topos are universal, so

$$Y \times_{h_T} h_U \simeq \text{colim}_{(V,u)} (h_V \times_{h_T} h_U),$$

and the pullback arrow

$$Y \times_{h_T} h_U \longrightarrow Y$$

is the corresponding colimit in the arrow category. Every stage

$$h_V \times_{h_T} h_U \longrightarrow h_V$$

is represented by the affine base change $V \times_T U \rightarrow V$, hence is again a square-zero infinitesimal substitution by 2.8.1. Since L_Σ preserves colimits and inverts every stage map, it inverts their colimit. Thus every pullback of every generator belongs to W_0 .

(2) **Fix the indicated data.** Let

$$W_0^{\text{univ}} := \{f : X \rightarrow Y \mid \text{every pullback of } f \text{ belongs to } W_0\}.$$

(3) **Establish the indicated equivalence.** This class is stable under arbitrary pullback by definition and contains all equivalences.

(4) **Establish the next claim.** It is closed under retracts and small colimits in the arrow category: after any base change, retract diagrams remain retract diagrams and, by universality of colimits in an infinity-topos, arrow-category colimits become the corresponding colimits of the base-changed arrows.

(5) **Establish the next claim.** Strong saturation of W_0 then applies.

(6) **Establish the next claim.** We use the arrow-category characterization of strong saturation from [35, Section 5.5.4].

(7) **Establish the next claim.** It remains to verify three-for-two.

(8) **Fix the indicated data.** Let

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

be composable.

(9) **Treat the stated case.** If $f, g \in W_0^{\text{univ}}$, then every pullback of gf is a composite of pullbacks of f and g , so $gf \in W_0^{\text{univ}}$.

(10) **Treat the stated case.** If $f, gf \in W_0^{\text{univ}}$, let $Z' \rightarrow Z$ be arbitrary and put

$$Y' := Y \times_Z Z', \quad X' := X \times_Z Z'.$$

(11) **Use pullback stability.** Then $X' \rightarrow Z'$, being a pullback of gf , lies in W_0 , while $X' \rightarrow Y'$, being a pullback of f , lies in W_0 .

(12) **Apply two-out-of-three.** Two-out-of-three in W_0 gives $Y' \rightarrow Z' \in W_0$. Hence $g \in W_0^{\text{univ}}$.

(13) **Establish the next claim.** For the remaining cancellation case assume $g, gf \in W_0^{\text{univ}}$, and let $b : Y' \rightarrow Y$ be arbitrary.

(14) **Introduce the notation.** Put

$$\overline{Y} := Y \times_Z Y',$$

where $Y' \rightarrow Z$ is gb .

(15) **Use pullback stability.** The projection

$$p : \overline{Y} \longrightarrow Y'$$

is a pullback of g , hence belongs to W_0^{univ} .

(16) **Construct the indicated map.** The map b determines a canonical section

$$s : Y' \longrightarrow \overline{Y}, \quad ps \simeq \text{id}_{Y'}.$$

(17) **Use pullback stability.** Every pullback of s remains a section of the corresponding pullback of p .

(18) **Use the stated hypothesis.** Since every pullback of p lies in W_0 , two-out-of-three with the identity shows that every pullback of s lies in W_0 . Thus $s \in W_0^{\text{univ}}$.

(19) **Establish the next claim.** Now put

$$X'_Z := X \times_Z Y'.$$

(20) **Use pullback stability.** The map $X'_Z \rightarrow Y'$ is a pullback of gf , hence lies in W_0 .

(21) **Use pullback stability.** Moreover the canonical map

$$X \times_Y Y' \longrightarrow X'_Z$$

is the pullback of s along the morphism $X'_Z \rightarrow \bar{Y}$ induced by f .

(22) **Use the stated hypothesis.** Since $s \in W_0^{\text{univ}}$, this canonical map lies in W_0 .

(23) **Apply two-out-of-three.** Their composite is the base change of f along b , so two-out-of-three gives

$$X \times_Y Y' \longrightarrow Y' \in W_0.$$

(24) **Establish the next claim.** As b was arbitrary, $f \in W_0^{\text{univ}}$.

(25) **Conclude the stated claim.** Hence W_0^{univ} is strongly saturated.

(26) **Establish the next claim.** Step (1) shows $\Sigma_{\text{sqz}} \subseteq W_0^{\text{univ}}$. Therefore

$$W_0 = \text{Sat}(\Sigma_{\text{sqz}}) \subseteq W_0^{\text{univ}}.$$

(27) **Prove the converse implication.** Conversely, the identity base change of any $f \in W_0^{\text{univ}}$ shows $f \in W_0$.

(28) **Conclude the stated claim.** Thus

$$W_0 = W_0^{\text{univ}},$$

proving that W_0 is stable under arbitrary pullback.

(29) **Use the stated construction.** The defining minimality of W^{dR} now gives

$$W^{\text{dR}} = W_0.$$

(30) **Treat the stated case.** If $f : A \rightarrow B$ lies in W^{dR} , the projection

$$A \times_B A \longrightarrow A$$

is a pullback of f , hence belongs to W^{dR} .

(31) **Establish the next claim.** Its diagonal section

$$A \longrightarrow A \times_B A$$

composes with the projection to the identity.

(32) **Apply two-out-of-three.** Two-out-of-three therefore puts the diagonal in W^{dR} .

(33) **Establish the next claim.** Applying the same argument recursively proves the assertion for every iterated diagonal.

□

4.2.2 The universal spectral de Rham reflector

The generated congruence is now known to be pullback-stable. This is exactly the condition needed to promote accessible localization to a left-exact localization of the spectral Zariski topos. After constructing the reflector, we extend its affine generating property to global square-zero and finite infinitesimal substitutions and record the resulting mapping property.

Theorem 4.2.1 (Intrinsic spectral de Rham localization). *There is an accessible left-exact localization*

$$L^{\mathrm{dR}} : \mathcal{H}_{\mathrm{Sp}} \rightleftarrows \mathcal{H}_{\mathrm{Sp}}^{\mathrm{dR}} : \iota^{\mathrm{dR}}$$

displayed by the adjunction

$$\mathcal{H}_{\mathrm{Sp}} \begin{array}{c} \xrightarrow{L^{\mathrm{dR}}} \\ \xleftarrow{\iota^{\mathrm{dR}}} \end{array} \mathcal{H}_{\mathrm{Sp}}^{\mathrm{dR}}$$

with localization unit $\eta_X^{\mathrm{dR}} : X \rightarrow X_{\mathrm{dR}}$, whose local-equivalence class is W^{dR} . It is initial among accessible left-exact localizations of $\mathcal{H}_{\mathrm{Sp}}$ which invert every affine square-zero infinitesimal substitution.

Proof.

- (1) **Establish the locality claim.** The spectral Zariski topos is presentable, so localization at the $\mathbb{V}$ -small family Σ_{sqz} exists by the presentable localization theorem; see [35, Proposition 5.5.4.15].
- (2) **Establish the indicated equivalence.** Its local-equivalence class is the strongly saturated closure W^{dR} , which is pullback-stable by Lemma 4.2.2.
- (3) **Establish the locality claim.** The left-exactness criterion for a localization of an infinity-topos therefore makes the localization left exact; see [35, Proposition 6.2.1.1].
- (4) **Establish the locality claim.** Any accessible left-exact localization which inverts every affine square-zero infinitesimal substitution inverts every member of Σ_{sqz} .
- (5) **Establish the indicated equivalence.** Its local-equivalence class is strongly saturated and therefore contains

$$\mathrm{Sat}(\Sigma_{\mathrm{sqz}}) = W^{\mathrm{dR}}.$$

- (6) **Establish the next claim.** The universal factorization follows.

□

Definition 4.2.2 (Spectral de Rham reflection). Define the ambient idempotent left-exact endofunctor

$$\mathrm{dR} := \iota^{\mathrm{dR}} L^{\mathrm{dR}} : \mathcal{H}_{\mathrm{Sp}} \longrightarrow \mathcal{H}_{\mathrm{Sp}}.$$

For $X \in \mathcal{H}_{\mathrm{Sp}}$, put

$$X_{\mathrm{dR}} := \mathrm{dR}(X),$$

and write

$$\eta_X^{\mathrm{dR}} : X \longrightarrow X_{\mathrm{dR}}$$

for the localization unit.

Lemma 4.2.3 (Target descent for de Rham equivalences). *Let*

$$f : A \longrightarrow B$$

be a morphism in $\mathcal{H}_{\text{Sp}}$, let $e : B' \rightarrow B$ be an effective epimorphism, and put $A' := A \times_B B'$. If $A' \rightarrow B'$ is a de Rham equivalence, then $A \rightarrow B$ is a de Rham equivalence.

Proof.

- (1) **Fix the indicated data.** Let $B'_\bullet \rightarrow B$ be the Čech nerve of e .
- (2) **Use the stated hypothesis.** Since the local-equivalence class of the left-exact localization L^{dR} is pullback-stable, every map

$$A \times_B B'_n \longrightarrow B'_n$$

is a de Rham equivalence.

- (3) **Treat the stated case.** If F is de Rham-local, effective descent in the ambient infinity-topos gives

$$\text{Map}(B, F) \simeq \text{Tot}_n \text{Map}(B'_n, F)$$

and

$$\text{Map}(A, F) \simeq \text{Tot}_n \text{Map}(A \times_B B'_n, F).$$

- (4) **Pass to totalization.** The degreewise comparison is an equivalence, hence so is its totalization.
- (5) **Apply orthogonality.** Orthogonality to all de Rham-local objects therefore identifies $A \rightarrow B$ as a de Rham equivalence.

□

Proposition 4.2.1 (Global finite-infinitesimal arrows are de Rham equivalences). *Every square-zero infinitesimal substitution of spectral schemes is a de Rham equivalence. Consequently every finite spectral infinitesimal substitution of 2.8.3 is a de Rham equivalence.*

Proof.

- (1) **Fix the indicated data.** Let $j : U \rightarrow T$ be a global square-zero infinitesimal substitution.
- (2) **Construct the indicated map.** Choose an affine spectral Zariski covering morphism

$$e : \coprod_{\alpha} T_{\alpha} \longrightarrow T.$$

- (3) **Use the stated hypothesis.** Since j is affine, each pullback

$$U_{\alpha} := U \times_T T_{\alpha} \longrightarrow T_{\alpha}$$

is an affine square-zero infinitesimal substitution, hence belongs to Σ_{sqz} up to equivalence and is inverted by L^{dR} .

- (4) **Establish the indicated equivalence.** Their coproduct is therefore a de Rham equivalence.

- (5) **Apply the cited lemma.** Lemma 4.2.3 descends this conclusion along e , proving that j is a de Rham equivalence.
- (6) **Establish the next claim.** The finite-composite statement follows from strong saturation and 2.8.3.

□

Corollary 4.2.1 (Global universal property of de Rham localization). *The localization*

$$L^{\mathrm{dR}} : \mathcal{H}_{\mathrm{Sp}} \rightleftarrows \mathcal{H}_{\mathrm{Sp}}^{\mathrm{dR}} : \iota^{\mathrm{dR}}$$

is initial among accessible left-exact localizations of $\mathcal{H}_{\mathrm{Sp}}$ which invert every global square-zero infinitesimal substitution of spectral schemes.

Proof.

- (1) **Verify inversion of global square-zero arrows.** Proposition 4.2.1 shows that L^{dR} itself inverts every global square-zero infinitesimal substitution of spectral schemes.
- (2) **Prove the converse implication.** Conversely, any accessible left-exact localization which inverts every such global square-zero infinitesimal substitution in particular inverts every affine square-zero generator in Σ_{sqz} .
- (3) **Use the stated construction.** The initiality statement of Theorem 4.2.1 therefore supplies the required universal factorization.

□

Theorem 4.2.2 (de Rham locality criterion). *For $F \in \mathcal{H}_{\mathrm{Sp}}$, the following are equivalent.*

- 1. F is L^{dR} -local.
- 2. For every affine square-zero infinitesimal substitution $\mathrm{Spec} A \rightarrow \mathrm{Spec} A'$, restriction induces

$$F(\mathrm{Spec} A') \xrightarrow{\sim} F(\mathrm{Spec} A).$$

- 3. Restriction is an equivalence along every finite spectral infinitesimal substitution of 2.8.3.

Proof.

- (1) **Apply orthogonality.** Conditions (1) and (2) are the orthogonality characterization of local objects for the localization generated by Σ_{sqz} .
- (2) **Treat the stated case.** If F is local and $j : U \rightarrow T$ is any global finite spectral infinitesimal substitution, then Proposition 4.2.1 makes j a de Rham equivalence, so

$$\mathrm{Map}(T, F) \xrightarrow{\sim} \mathrm{Map}(U, F).$$

- (3) **Establish the next claim.** This proves (1) $\Rightarrow$ (3).
- (4) **Prove the converse implication.** Conversely, (3) contains all affine square-zero infinitesimal substitutions and hence implies (2).

□

Proposition 4.2.2 (Universal mapping property). *If Z is de Rham-local, composition with η_X^{dR} induces a natural equivalence*

$$\text{Map}_{\mathcal{H}_{\text{Sp}}}(X_{\text{dR}}, Z) \xrightarrow{\simeq} \text{Map}_{\mathcal{H}_{\text{Sp}}}(X, Z).$$

Proof.

- (1) **Construct the indicated map.** This is the mapping-space adjunction for $L^{\text{dR}} \dashv \iota^{\text{dR}}$.

□

4.2.3 Recognition by finite-infinitesimal invariance

The locality criterion can now be compared directly with the infinitesimal endpoint semantics of the differentiated doctrine. On an infinitesimal arrow the canonical endpoint comparison is nothing other than restriction from the thickening to its centre. Thus the previous chapter already contains a pointwise recognition test for the new reflector.

Let

$$i_! \dashv i^* \dashv i_* \dashv i^!$$

be the endpoint adjoint string of Chapter 2. The canonical comparison $i_! \rightarrow i_*$ gives

$$\gamma_F : i_! F \longrightarrow i_* F.$$

On a finite infinitesimal arrow $j : U \rightarrow T$, the endpoint formulas give

$$(i_! F)(j) \simeq F(T), \quad (i_* F)(j) \simeq F(U),$$

and $\gamma_F(j)$ is the restriction map $F(T) \rightarrow F(U)$.

Thus the endpoint comparison is read pointwise through

$$\begin{array}{ccc} (i_! F)(j) & \xrightarrow{\gamma_F(j)} & (i_* F)(j) \\ \simeq \downarrow & & \downarrow \simeq \\ F(T) & \longrightarrow & F(U). \end{array}$$

Remark 4.2.1 (Comparison with crystals). Finite-infinitesimal invariance is the spectral analogue of the invariance condition underlying the de Rham-prestack formulation of crystals; compare [18].

Theorem 4.2.3 (Finite-infinitesimal recognition of de Rham locality). *For $F \in \mathcal{H}_{\text{Sp}}$, the following are equivalent:*

1. F is de Rham-local;
2. F is invariant under every finite spectral infinitesimal substitution;
- 3.

$$\gamma_F : i_! F \longrightarrow i_* F$$

is an equivalence in the infinitesimal Zariski topos $\mathcal{H}_{\text{inf}}$.

Hence $\mathcal{H}_{\text{Sp}}^{\text{dR}}$ is the reflective full subcategory characterized by finite-infinitesimal invariance.

Proof.

- (1) **Identify the stated objects.** The pointwise endpoint formulas identify condition (3) with invariance along every affine finite infinitesimal substitution, since the infinitesimal Zariski topos of Chapter 2 is built from the affine finite-infinitesimal arrow category.
- (2) **Establish the locality claim.** Such invariance contains the affine square-zero generators and therefore implies de Rham locality by Theorem 4.2.2.
- (3) **Prove the converse implication.** Conversely, a de Rham-local object is invariant under every global finite infinitesimal substitution by the same theorem, hence in particular on the affine endpoint category.
- (4) **Establish the indicated equivalence.** This proves the equivalence of all three conditions.

□

Corollary 4.2.2 (De Rham locality and terminal formal étaleness). *An object F is de Rham-local if and only if its terminal morphism is formally étale with respect to the finite spectral infinitesimal substitutions of Chapter 2.*

The absolute reflector now has both its universal property and its intrinsic recognition criterion. To use it as part of the spectral doctrine, however, the construction must live in every slice and commute with substitution. The relative object to be justified in the next subsection is

$$X_{\text{dR}/S} := S \times_{S_{\text{dR}}} X_{\text{dR}}, \quad \eta_{X/S}^{\text{dR}} : X \longrightarrow X_{\text{dR}/S}.$$

The next step gives the indexed form of the same localization.

4.2.4 Relative reflection and substitution in context

Definition 4.2.3 (Relative de Rham object). For $X \rightarrow S$ in $\mathcal{H}_{\text{Sp}}$, define

$$X_{\text{dR}/S} := S \times_{S_{\text{dR}}} X_{\text{dR}}.$$

The induced map $\eta_{X/S}^{\text{dR}} : X \rightarrow X_{\text{dR}/S}$ is the relative de Rham unit.

Construction 4.2.1 (Slice adjunction). For fixed S , applying absolute de Rham reflection to structure maps gives

$$L_S^{\text{dR}} : (\mathcal{H}_{\text{Sp}})_{/S} \longrightarrow (\mathcal{H}_{\text{Sp}}^{\text{dR}})_{/S_{\text{dR}}}, \quad (X \rightarrow S) \longmapsto (X_{\text{dR}} \rightarrow S_{\text{dR}}).$$

Pullback along $\eta_S^{\text{dR}} : S \rightarrow S_{\text{dR}}$ gives

$$\iota_S^{\text{dR}} : (\mathcal{H}_{\text{Sp}}^{\text{dR}})_{/S_{\text{dR}}} \longrightarrow (\mathcal{H}_{\text{Sp}})_{/S}, \quad E \longmapsto S \times_{S_{\text{dR}}} E.$$

For $X \rightarrow S$ and local $E \rightarrow S_{\text{dR}}$, the pullback universal property and the absolute localization adjunction give a canonical equivalence

$$\text{Map}_S(X, \iota_S^{\text{dR}} E) \simeq \text{Map}_{S_{\text{dR}}}(X_{\text{dR}}, E).$$

Thus

$$L_S^{\text{dR}} \dashv \ell_S^{\text{dR}}.$$

The relative reflector sits over the absolute one in the diagram

$$\begin{array}{ccc} (\mathcal{H}_{\text{Sp}})_{/S} & \xrightarrow{L_S^{\text{dR}}} & (\mathcal{H}_{\text{Sp}}^{\text{dR}})_{/S_{\text{dR}}} \\ \downarrow & & \downarrow \\ \mathcal{H}_{\text{Sp}} & \xrightarrow{L^{\text{dR}}} & \mathcal{H}_{\text{Sp}}^{\text{dR}}. \end{array}$$

Lemma 4.2.4 (Local-map criterion). *Let $L : \mathcal{Y} \rightarrow \mathcal{Y}_L$ be a left-exact localization of an infinity-topos with local-equivalence class W . For a morphism $u : U \rightarrow X$, the following are equivalent:*

1. *u is right orthogonal to every morphism of W ;*
2. *the localization-unit square*

$$\begin{array}{ccc} U & \xrightarrow{\eta_U} & LU \\ u \downarrow & & \downarrow Lu \\ X & \xrightarrow{\eta_X} & LX \end{array}$$

is Cartesian.

Proof.

- (1) **Treat the stated case.** If the square is Cartesian, then u is a pullback of the map $Lu : LU \rightarrow LX$ between local objects.
- (2) **Apply orthogonality.** A map between local objects is right orthogonal to every local equivalence, and right orthogonality is preserved by pullback, so u is right orthogonal to W .
- (3) **Prove the converse implication.** Conversely, put

$$P := X \times_{LX} LU,$$

let $\alpha : U \rightarrow P$ be the comparison, and let $p : P \rightarrow X$ be the projection.

- (4) **Apply left exactness.** Left exactness gives

$$LP \simeq LX \times_{LLX} LLU \simeq LU,$$

so $L\alpha$ is an equivalence and hence $\alpha \in W$.

- (5) **Use pullback stability.** The map p is a pullback of Lu , so p is right orthogonal to W .
- (6) **Construct the indicated map.** Applying $\alpha \perp u$ to the square with upper map id_U and lower map p produces

$$s : P \longrightarrow U$$

with

$$s\alpha \simeq \text{id}_U, \quad us \simeq p.$$

- (7) **Establish the next claim.** Now apply $\alpha \perp p$.

- (8) **Use the stated construction.** The maps id_P and αs are two fillers of the same lifting problem, so contractibility of the filler space gives

$$\alpha s \simeq \mathrm{id}_P.$$

- (9) **Conclude the stated claim.** Thus α is an equivalence, which is precisely Cartesianity of the unit square. □

Theorem 4.2.4 (Slice de Rham localization). *The functor ι_S^{dR} of Construction 4.2.1 is fully faithful. Its composite with the left adjoint defines an accessible left-exact idempotent endofunctor*

$$\mathrm{dR}_S := \iota_S^{\mathrm{dR}} L_S^{\mathrm{dR}} : (\mathcal{H}_{\mathrm{Sp}})_S \longrightarrow (\mathcal{H}_{\mathrm{Sp}})_{/S},$$

with

$$\mathrm{dR}_S(X) \simeq X_{\mathrm{dR}/S}.$$

Its essential image, denoted

$$((\mathcal{H}_{\mathrm{Sp}})_{/S})^{\mathrm{dR}},$$

is canonically equivalent to

$$(\mathcal{H}_{\mathrm{Sp}}^{\mathrm{dR}})_{/S_{\mathrm{dR}}}.$$

An object $F \rightarrow S$ belongs to this essential image if and only if, for every affine square-zero infinitesimal substitution $U \rightarrow T$ over S ,

$$\mathrm{Map}_S(T, F) \xrightarrow{\simeq} \mathrm{Map}_S(U, F).$$

Equivalently, the same holds for every finite spectral infinitesimal substitution over S .

Proof.

- (1) **Use the stated construction.** For local $E \rightarrow S_{\mathrm{dR}}$, left exactness gives

$$\mathrm{dR}(S \times_{S_{\mathrm{dR}}} E) \simeq \mathrm{dR}(S) \times_{\mathrm{dR}(S_{\mathrm{dR}})} \mathrm{dR}(E) \simeq S_{\mathrm{dR}} \times_{S_{\mathrm{dR}}} E \simeq E.$$

- (2) **Conclude the stated claim.** Thus the counit $L_S^{\mathrm{dR}} \iota_S^{\mathrm{dR}} \rightarrow \mathrm{id}$ is an equivalence, and ι_S^{dR} is fully faithful.
- (3) **Establish the next claim.** The formula for dR_S follows from its definition.
- (4) **Use pullback stability.** Accessibility follows from accessibility of L^{dR} and pullback, and left exactness follows from left exactness of L^{dR} and finite limits in the slice.
- (5) **Identify the stated objects.** It remains to identify the local objects intrinsically.
- (6) **Fix the indicated data.** Let $F \rightarrow S$ be an object of the slice.
- (7) **Invoke the cited result.** By Lemma 4.2.4, F lies in the essential image of ι_S^{dR} if and only if its structure map $F \rightarrow S$ is right orthogonal to every absolute de Rham equivalence.

(8) **Use the stated hypothesis.** Since

$$W^{\mathrm{dR}} = \mathrm{Sat}(\Sigma_{\mathrm{sqz}}),$$

and, for a fixed right map, the class of left maps orthogonal to it is strongly saturated, this is equivalent to right orthogonality to every affine square-zero generator equipped with a structure map of its target to S .

(9) **Construct the indicated map.** Written in the slice, that condition is exactly

$$\mathrm{Map}_S(T, F) \xrightarrow{\simeq} \mathrm{Map}_S(U, F)$$

for every affine square-zero infinitesimal substitution $U \rightarrow T$ over S .

- (10) **Establish the next claim.** After the standing universe enlargement these tests form a small set, so they also exhibit L_S^{dR} as the corresponding set-generated localization of the presentable slice.
- (11) **Establish the next claim.** Finally, every global finite infinitesimal substitution over S has underlying arrow in W^{dR} by Proposition 4.2.1.
- (12) **Conclude the stated claim.** Hence every relative local object is invariant under all such arrows.
- (13) **Prove the converse implication.** Conversely the finite class contains the affine square-zero generators, so finite-infinitesimal invariance implies the displayed affine criterion.

□

Proposition 4.2.3 (Relative universal mapping property). *If $Z \rightarrow S$ is de Rham-local over S , composition with the relative unit induces*

$$\mathrm{Map}_S(X_{\mathrm{dR}/S}, Z) \xrightarrow{\simeq} \mathrm{Map}_S(X, Z).$$

The equivalence is natural in both variables.

Proof.

- (1) **Construct the indicated map.** This is the mapping-space adjunction for the reflective subcategory of Theorem 4.2.4.

□

Theorem 4.2.5 (Arbitrary base change). *For every Cartesian square*

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S, \end{array}$$

there is a canonical equivalence

$$X'_{\mathrm{dR}/S'} \xrightarrow{\simeq} X_{\mathrm{dR}/S} \times_S S'.$$

The comparison is the one supplied by finite-limit functoriality.

Proof.

- (1) **Apply left exactness.** Left exactness gives

$$\mathrm{dR}(X') \simeq \mathrm{dR}(X) \times_{\mathrm{dR}(S)} \mathrm{dR}(S').$$

- (2) **Use the stated construction.** Substituting this into the defining pullback of $X'_{\mathrm{dR}/S'}$ and cancelling the repeated pullback over $\mathrm{dR}(S')$ gives the result.
- (3) **Establish the next claim.** The coherence is the coherence of finite-limit functoriality.

□

Theorem 4.2.6 (Indexed spectral de Rham reflection). *The fibrewise reflectors*

$$\mathrm{dR}_S : (\mathcal{H}_{\mathrm{Sp}})_{/S} \longrightarrow (\mathcal{H}_{\mathrm{Sp}})_{/S}$$

assemble with substitution into an indexed accessible left-exact idempotent reflection on the spectral slice doctrine. More precisely, for every $c : S' \rightarrow S$ there is a canonical equivalence

$$c^* \mathrm{dR}_S \xrightarrow{\simeq} \mathrm{dR}_{S'} c^*,$$

These equivalences are represented by the commuting square of endofunctors

$$\begin{array}{ccc} (\mathcal{H}_{\mathrm{Sp}})_{/S} & \xrightarrow{\mathrm{dR}_S} & (\mathcal{H}_{\mathrm{Sp}})_{/S} \\ c^* \downarrow & \simeq & \downarrow c^* \\ (\mathcal{H}_{\mathrm{Sp}})_{/S'} & \xrightarrow{\mathrm{dR}_{S'}} & (\mathcal{H}_{\mathrm{Sp}})_{/S'}. \end{array}$$

with the pullback coherence fixed by the standing convention. The de Rham-local objects over S therefore form a reflective Cartesian subdoctrine of

$$S \longmapsto (\mathcal{H}_{\mathrm{Sp}})_{/S}.$$

For a judgment $X \rightarrow S$, the de Rham reflection rules are represented by

$$\begin{array}{ll} \text{formation} & : X \longmapsto X_{\mathrm{dR}/S}, \\ \text{introduction} & : \eta_{X/S}^{\mathrm{dR}} : X \rightarrow X_{\mathrm{dR}/S}, \\ \text{elimination} & : \mathrm{Map}_S(X_{\mathrm{dR}/S}, Z) \simeq \mathrm{Map}_S(X, Z) \quad (Z \text{ de Rham-local over } S), \\ \text{substitution} & : (X \times_S S')_{\mathrm{dR}/S'} \simeq X_{\mathrm{dR}/S} \times_S S'. \end{array}$$

Proof.

- (1) **Identify the stated objects.** For each fixed S , Theorem 4.2.4 constructs the accessible left-exact idempotent reflector and identifies its reflective full subcategory.
- (2) **Apply the cited proposition.** Proposition 4.2.3 supplies the displayed mapping property.
- (3) **Apply the cited theorem.** Theorem 4.2.5 supplies the substitution equivalence.

- (4) **Establish the locality claim.** These comparisons preserve the local essential images and assemble the fibrewise reflective subcategories into the asserted Cartesian subdoctrine.
- (5) **Establish the next claim.** Formation and introduction are given by the relative de Rham object and its unit.

□

We have therefore turned the finite infinitesimal test geometry into an indexed left-exact reflection. The reader may now retain two facts: de Rham-local objects are exactly the objects invariant under finite infinitesimal substitution, and relative reflection commutes with arbitrary base change. What is still missing is an effective presentation of the relative unit. That is a different problem, because a localization unit need not itself be an effective epimorphism.

4.3 Effective de Rham quotients and the infinitesimal relation

Relative reflection gives the universal map from X to a de Rham-local object. Effective descent is governed by the effective image of this unit: the unit can contain points of $X_{\text{dR}/S}$ beyond those generated effectively by points of X . We therefore factor the relative unit before forming its relation.

$$\begin{array}{ccc}
 X & \xrightarrow{e_{X/S}} Q_{X/S} & \xrightarrow{m_{X/S}} X_{\text{dR}/S} \\
 & \searrow e_{X/S} & \nearrow m_{X/S} \\
 & Q_{X/S} &
 \end{array}$$

The two factors play different roles. The monomorphism records how far the effective image is from the full reflected object, while the effective epimorphism is the map whose Čech nerve admits effective descent. This section first controls that factorization under base change and relates its right orthogonality to formal étaleness. It then identifies the kernel-pair nerve of $X \rightarrow Q_{X/S}$ with the simplicial relation already defined by fibre products over $X_{\text{dR}/S}$. Finally, de Rham neighbourhoods express the same relation as an infinitesimal closure of the diagonal.

The decisive comparison is the following pullback identity, proved below from the monomorphism $Q_{X/S} \hookrightarrow X_{\text{dR}/S}$:

$$\begin{array}{ccc}
 X \times_{Q_{X/S}} X & \xrightarrow{\simeq} & X \times_{X_{\text{dR}/S}} X \\
 \downarrow & & \downarrow \\
 X & \xlongequal{\quad} & X.
 \end{array}$$

It is this equivalence, rather than an identification $Q_{X/S} \simeq X_{\text{dR}/S}$, that makes the de Rham relation an effective groupoid in complete generality.

4.3.1 The effective image of the relative unit

Fix $X \rightarrow S$.

Definition 4.3.1 (Effective de Rham quotient). Factor the relative de Rham unit by the effective-epimorphism–monomorphism factorization in the slice [35]:

$$X \xrightarrow{e_{X/S}} Q_{X/S} \xrightarrow{m_{X/S}} X_{\mathrm{dR}/S}.$$

The object $Q_{X/S}$ is the *effective de Rham quotient* of X over S .

The factorization is a morphism in the slice:

$$\begin{array}{ccc} X & \xrightarrow{e_{X/S}} & Q_{X/S} \\ & \searrow \eta_{X/S}^{\mathrm{dR}} & \downarrow m_{X/S} \\ & & X_{\mathrm{dR}/S}. \end{array}$$

Proposition 4.3.1 (Functoriality and base change of the effective quotient). *The assignment $X \mapsto Q_{X/S}$ is functorial in the slice. For every $S' \rightarrow S$, with $X' = X \times_S S'$, there is a canonical equivalence*

$$Q_{X'/S'} \xrightarrow{\cong} Q_{X/S} \times_S S',$$

under arbitrary iterated base change.

Proof.

- (1) **Establish the next claim.** The effective-image factorization is the functorial factorization attached to the (effective epimorphism, monomorphism) factorization system.
- (2) **Apply naturality.** Naturality of the relative unit gives functoriality.
- (3) **Apply the cited theorem.** Theorem 4.2.5 identifies the base-changed target, and both effective epimorphisms and monomorphisms are pullback-stable.
- (4) **Use pullback stability.** The pullback of the factorization is therefore the effective image factorization of the base-changed unit.

□

Locality and lifting through the effective image

To use $Q_{X/S}$ in deformation-theoretic arguments, we need a criterion for its locality that can be checked after passing to an effective cover. The next lemmas isolate that criterion and identify the slice-local equivalences with the strongly saturated class generated by square-zero substitutions.

Lemma 4.3.1 (Target locality of de Rham equivalences over a base). *Let $f : A \rightarrow B$ be a morphism over S , let $e : B' \rightarrow B$ be an effective epimorphism, and put $A' = A \times_B B'$. If $A' \rightarrow B'$ is a de Rham equivalence over S , then $A \rightarrow B$ is a de Rham equivalence over S .*

Proof.

- (1) **Fix the indicated data.** Let $B'_\bullet \rightarrow B$ be the Čech nerve of e .

- (2) **Establish the indicated equivalence.** The local-equivalence class of the left-exact slice localization is pullback-stable, so every map

$$A \times_B B'_n \longrightarrow B'_n$$

is a de Rham equivalence over S .

- (3) **Use the stated construction.** For $F \rightarrow S$ de Rham-local over S , effective descent in the slice gives

$$\mathrm{Map}_S(B, F) \simeq \mathrm{Tot}_n \mathrm{Map}_S(B'_n, F)$$

and

$$\mathrm{Map}_S(A, F) \simeq \mathrm{Tot}_n \mathrm{Map}_S(A \times_B B'_n, F).$$

- (4) **Pass to totalization.** The degreewise comparison is an equivalence, hence so is its totalization.
- (5) **Apply orthogonality.** This is the orthogonality criterion for $A \rightarrow B$ to be a de Rham equivalence over S .

□

Lemma 4.3.2 (Relative image-locality criterion). *Let $m : Q \rightarrow Y$ be a monomorphism over S , with $Y \rightarrow S$ de Rham-local over S . Suppose that for every affine square-zero infinitesimal substitution $V \rightarrow T$ over S and every $q_0 : V \rightarrow Q$, the contractibly unique extension $a : T \rightarrow Y$ of mq_0 factors spectral-Zariski-locally on T through Q . Then $Q \rightarrow S$ is de Rham-local over S .*

Proof.

- (1) **Establish the next claim.** For $a : T \rightarrow Y$, write a_0 for its restriction to V and define

$$\Phi_T(a) := \mathrm{fib}_a(\mathrm{Map}_S(T, Q) \rightarrow \mathrm{Map}_S(T, Y)),$$

$$\Phi_V(a_0) := \mathrm{fib}_{a_0}(\mathrm{Map}_S(V, Q) \rightarrow \mathrm{Map}_S(V, Y)).$$

- (2) **Use the stated hypothesis.** Since m is a monomorphism, these fibres are (-1) -truncated.
- (3) **Establish the next claim.** A point of $\Phi_V(a_0)$ is precisely a factorization $q_0 : V \rightarrow Q$.
- (4) **Invoke the cited result.** By hypothesis the unique extension $a : T \rightarrow Y$ factors locally through Q .
- (5) **Identify the stated objects.** On overlaps two local factorizations have the same composite to Y , and the monomorphism property makes their identification space contractible.
- (6) **Establish the next claim.** They therefore descend to a global factorization.
- (7) **Conclude the stated claim.** Thus $\Phi_T(a) \rightarrow \Phi_V(a_0)$ is an equivalence for every a .
- (8) **Construct the indicated map.** Together with locality of Y , this makes

$$\begin{array}{ccc} \mathrm{Map}_S(T, Q) & \longrightarrow & \mathrm{Map}_S(T, Y) \\ \downarrow & & \downarrow \simeq \\ \mathrm{Map}_S(V, Q) & \longrightarrow & \mathrm{Map}_S(V, Y) \end{array}$$

- (9) **Establish the indicated equivalence.** Cartesian, so $\text{Map}_S(T, Q) \rightarrow \text{Map}_S(V, Q)$ is an equivalence.
- (10) **Apply the indicated result.** Apply Theorem 4.2.4.

□

Proposition 4.3.2 (De Rham equivalences over a base are square-zero generated). *For fixed S , the local-equivalence class of L_S^{dR} is the strongly saturated class generated by a small set of affine square-zero infinitesimal substitutions over S .*

Proof.

- (1) **Use the stated construction.** Localizing the presentable slice at those arrows gives precisely the local objects of Theorem 4.2.4.
- (2) **Establish the indicated equivalence.** Accessible reflective localizations with the same local objects are canonically equivalent, and the local-equivalence class of a set-generated localization is its strongly saturated closure.

□

Formal étale rigidity and smooth effectivity

The locality criterion now converts the formal lifting notions of the differentiated doctrine into statements about the relative de Rham unit. Formal étaleness gives Cartesianity and rigidity; formal smoothness gives the complementary existence statement that makes the unit effective.

Theorem 4.3.1 (Formal étale Cartesianity). *Let $u : U \rightarrow X$ be formally étale over S in the sense of Chapter 2. Then*

$$\begin{array}{ccc} U & \xrightarrow{u} & X \\ \eta_{U/S}^{\text{dR}} \downarrow & & \downarrow \eta_{X/S}^{\text{dR}} \\ U_{\text{dR}/S} & \longrightarrow & X_{\text{dR}/S} \end{array}$$

is Cartesian. Equivalently,

$$U \simeq X \times_{X_{\text{dR}/S}} U_{\text{dR}/S}.$$

Proof.

- (1) **Invoke the cited result.** By one-stage detection in Chapter 2, formal étaleness is right orthogonality to the affine square-zero substitutions.
- (2) **Construct the indicated map.** For a fixed right map, the class of left maps orthogonal to it is strongly saturated.
- (3) **Apply orthogonality.** Proposition 4.3.2 therefore gives right orthogonality to every de Rham equivalence over S .
- (4) **Apply the indicated result.** Apply Lemma 4.2.4 to the slice localization.

□

Theorem 4.3.2 (Formal étale rigidity). *If $X \rightarrow S$ is formally étale, then*

$$\eta_{X/S}^{\mathrm{dR}} : X \xrightarrow{\simeq} X_{\mathrm{dR}/S}.$$

Proof.

- (1) **Apply the indicated result.** Apply Theorem 4.3.1 to $X \rightarrow S$ regarded as a morphism in the slice over S .
- (2) **Use the stated hypothesis.** Since

$$S_{\mathrm{dR}/S} = S \times_{S_{\mathrm{dR}}} S_{\mathrm{dR}} \simeq S,$$

the resulting Cartesian square identifies

$$X \simeq S \times_{S_{\mathrm{dR}/S}} X_{\mathrm{dR}/S} \simeq X_{\mathrm{dR}/S}.$$

- (3) **Identify the stated objects.** Under this identification the comparison is the relative localization unit $\eta_{X/S}^{\mathrm{dR}}$.
- (4) **Establish the locality claim.** Equivalently, one may apply the locality criterion of Theorem 4.2.4.

□

Theorem 4.3.3 (Effectivity for formally smooth morphisms). *If $X \rightarrow S$ is formally smooth in the sense of Chapter 2, then*

$$\eta_{X/S}^{\mathrm{dR}} : X \longrightarrow X_{\mathrm{dR}/S}$$

is an effective epimorphism.

Proof.

- (1) **Establish the next claim.** Factor the unit as

$$X \xrightarrow{e} Q \xrightarrow{m} Y, \quad Y := X_{\mathrm{dR}/S},$$

with e an effective epimorphism and m a monomorphism.

- (2) **Establish the locality claim.** We first prove that $Q \rightarrow S$ is de Rham-local over S .
- (3) **Fix the indicated data.** Let $V \rightarrow T$ be an affine square-zero infinitesimal substitution over S , and let $q_0 : V \rightarrow Q$.
- (4) **Use the stated construction.** Pulling back e gives an effective epimorphism

$$V \times_Q X \longrightarrow V.$$

- (5) **Invoke the cited result.** By Convention 1.8.2, effective epimorphisms in the spectral Zariski topos are locally surjective on sections.
- (6) **Conclude the stated claim.** Hence q_0 lifts on a spectral Zariski covering sieve of V .

- (7) **Use the stated hypothesis.** Since V is affine, 1.8.5 and 1.8.7 refine this sieve to a finite affine spectral Zariski cover

$$\{V_\alpha \rightarrow V\}$$

on which q_0 lifts to X .

- (8) **Invoke the cited result.** By 2.8.3, pullback along $V \rightarrow T$ identifies the small spectral Zariski sites and their finite affine covering families.
- (9) **Conclude the stated claim.** Thus the chosen cover is transported, through a contractible space, to an affine spectral Zariski cover

$$\{T_\alpha \rightarrow T\}, \quad V_\alpha \simeq V \times_T T_\alpha.$$

- (10) **Establish the next claim.** Formal smoothness and one-stage detection then give, after a further spectral Zariski refinement, extensions $T_{\alpha\beta} \rightarrow X$.
- (11) **Establish the locality claim.** The object Y is local, so $mq_0 : V \rightarrow Y$ has a contractibly unique extension $a : T \rightarrow Y$.
- (12) **Identify the stated objects.** On each refined chart the composite through $X \rightarrow Y$ is another extension of the same closed-stage map, hence is canonically identified with a .
- (13) **Conclude the stated claim.** Thus a locally factors through Q .
- (14) **Apply the cited lemma.** Lemma 4.3.2 shows that Q is local.
- (15) **Apply the indicated result.** Apply the relative de Rham endofunctor to $X \xrightarrow{e} Q \xrightarrow{m} Y$.
- (16) **Use the stated hypothesis.** Since Q and Y are local, the image is

$$\mathrm{dR}_S(X) \xrightarrow{\mathrm{dR}_S(e)} Q \xrightarrow{m} Y.$$

- (17) **Establish the indicated equivalence.** The composite is the localization of the localization unit and is an equivalence.
- (18) **Conclude the stated claim.** Hence m admits the section

$$Y \xrightarrow{(m \mathrm{dR}_S(e))^{-1}} \mathrm{dR}_S(X) \xrightarrow{\mathrm{dR}_S(e)} Q.$$

- (19) **Establish the indicated equivalence.** A monomorphism which is a split epimorphism is an equivalence.
- (20) **Conclude the stated claim.** Therefore m is an equivalence and the original unit is the effective epimorphism e .

□

Corollary 4.3.1 (Effectivity domain). *If $\eta_{X/S}^{\mathrm{dR}}$ is an effective epimorphism—in particular if $X \rightarrow S$ is formally smooth—then*

$$Q_{X/S} \xrightarrow{\simeq} X_{\mathrm{dR}/S}.$$

Effectivity preservation and orthogonal factorization

The formally smooth case shows when one particular localization unit is effective. For descent we also need reflection itself to preserve effective epimorphisms. Once that is established, the de Rham equivalences and formally étale maps assemble into the orthogonal factorization system used throughout the remainder of the chapter.

Theorem 4.3.4 (Relative de Rham reflection preserves effective epimorphisms). *Fix S . If $e : A \rightarrow B$ is an effective epimorphism in $(\mathcal{H}_{\text{Sp}})_{/S}$, then*

$$A_{\text{dR}/S} \longrightarrow B_{\text{dR}/S}$$

is an effective epimorphism in the ambient slice. This assertion is compatible with arbitrary base change.

Proof.

- (1) **Construct the indicated map.** Factor the reflected map as

$$A_{\text{dR}/S} \xrightarrow{e'} I \xrightarrow{m} B_{\text{dR}/S},$$

with e' an effective epimorphism and m a monomorphism.

- (2) **Establish the locality claim.** We first show that I is local.
- (3) **Fix the indicated data.** Let $V \rightarrow T$ be an affine square-zero infinitesimal substitution and let $V \rightarrow I$ be a map.
- (4) **Use pullback stability.** The pullback of e' to V is an effective epimorphism.
- (5) **Invoke the cited result.** By Convention 1.8.2, the resulting section lifts on a spectral Zariski covering sieve of V .
- (6) **Establish the next claim.** The refinements of 1.8.5 and 1.8.7 refine that sieve to a finite affine spectral Zariski cover of V .
- (7) **Establish the indicated equivalence.** Transport this cover to T by the small-site equivalence of 2.8.3.
- (8) **Use the stated hypothesis.** Since $A_{\text{dR}/S}$ is local, the lifted maps extend across the corresponding square-zero charts.
- (9) **Construct the indicated map.** Their composites in $B_{\text{dR}/S}$ agree with the unique extension there; the monomorphism property of m then descends the local factorizations through I .
- (10) **Apply the cited lemma.** Lemma 4.3.2 makes I local.
- (11) **Introduce the notation.** Put
- $$J := B \times_{B_{\text{dR}/S}} I.$$
- (12) **Construct the indicated map.** The projection $j : J \rightarrow B$ is a monomorphism.

- (13) **Apply naturality.** Naturality of the relative unit and the factorization of the reflected map give a factorization

$$A \longrightarrow J \xrightarrow{j} B$$

of the original effective epimorphism $A \rightarrow B$.

- (14) **Establish the next claim.** In an infinity-topos the (effective epimorphism, monomorphism) factorization system is orthogonal.

- (15) **Apply orthogonality.** Applying this orthogonality to the square

$$\begin{array}{ccc} A & \longrightarrow & J \\ \downarrow & & \downarrow j \\ B & \xrightarrow{\text{id}_B} & B \end{array}$$

produces a section $B \rightarrow J$ of j .

- (16) **Establish the indicated equivalence.** A monomorphism with a section is an equivalence, so $J \simeq B$.
- (17) **Conclude the stated claim.** Hence the unit $B \rightarrow B_{\text{dR}/S}$ factors through I .
- (18) **Use the stated identification.** Because I is local, the universal property of the unit extends this factorization through a contractible space to a map

$$s : B_{\text{dR}/S} \longrightarrow I.$$

- (19) **Establish the next claim.** The composites ms and $\text{id}_{B_{\text{dR}/S}}$ have the same restriction along the unit $B \rightarrow B_{\text{dR}/S}$.
- (20) **Use the stated hypothesis.** Since the target is local, the unit mapping equivalence makes the space of such extensions contractible; hence $ms \simeq \text{id}$.
- (21) **Conclude the stated claim.** Thus m is a monomorphism with a section and therefore an equivalence.
- (22) **Construct the indicated map.** The reflected map is an effective epimorphism.
- (23) **Use pullback stability.** Base-change compatibility follows from Theorem 4.2.5 and pullback stability of effective epimorphisms.

□

Corollary 4.3.2 (Čech continuity of relative de Rham reflection). *If $A \rightarrow B$ is an effective epimorphism over S , with Čech nerve $A_\bullet \rightarrow B$, then*

$$(A_n)_{\text{dR}/S} \simeq \underbrace{A_{\text{dR}/S} \times_{B_{\text{dR}/S}} \cdots \times_{B_{\text{dR}/S}} A_{\text{dR}/S}}_{n+1 \text{ factors}}$$

compatibly with all simplicial operators, and

$$|(A_\bullet)_{\text{dR}/S}| \simeq B_{\text{dR}/S}.$$

Proof.

- (1) **Apply left exactness.** Left exactness gives the degree-wise identification, and Theorem 4.3.4 makes the reflected map an effective epimorphism.
- (2) **Establish the next claim.** The displayed simplicial object is therefore its Čech nerve.

□

Proposition 4.3.3 (Structured infinitesimal context extensions are de Rham equivalences). *Every structured infinitesimal context extension of spectral schemes constructed in Chapter 2 is a de Rham equivalence. The same holds for every finite composite and in every relative slice.*

Proof.

- (1) **Establish the next claim.** Choose an affine spectral Zariski cover of the target.
- (2) **Use pullback stability.** The pullback of the structured infinitesimal context extension to each affine chart is an affine square-zero infinitesimal substitution and is therefore inverted.
- (3) **Establish the indicated equivalence.** The coproduct of the affine charts is an effective epimorphism, so Lemma 4.3.1 detects the global arrow as a de Rham equivalence.
- (4) **Use the stated construction.** Closure under finite composition gives the remaining statement.

□

Theorem 4.3.5 (De Rham–formally étale factorization system). *For every morphism*

$$u : U \longrightarrow X$$

in $\mathcal{H}_{\text{Sp}}$, put

$$P^{\text{dR}}(u) := X \times_{X_{\text{dR}}} U_{\text{dR}}.$$

The defining pullback and the factorization may be displayed together as

$$\begin{array}{ccccc} U & \xrightarrow{\ell_u} & P^{\text{dR}}(u) & \longrightarrow & U_{\text{dR}} \\ & \searrow u & \downarrow r_u & & \downarrow \\ & & X & \xrightarrow{\eta_X^{\text{dR}}} & X_{\text{dR}} \end{array}$$

There is a canonical functorial factorization

$$U \xrightarrow{\ell_u} P^{\text{dR}}(u) \xrightarrow{r_u} X$$

in which ℓ_u is a de Rham equivalence and r_u is formally étale. The de Rham equivalences and formally étale morphisms form a pullback-stable orthogonal factorization system. The same factorization restricts in every slice over a fixed base S , where the left class is the class of de Rham equivalences over S and the right class is the class of morphisms right orthogonal to the finite spectral infinitesimal substitutions over S .

Proof.

- (1) **Apply naturality.** Naturality of the absolute de Rham unit gives the canonical comparison

$$U \longrightarrow X \times_{X_{\mathrm{dR}}} U_{\mathrm{dR}},$$

followed by the projection to X .

- (2) **Establish the next claim.** This is functorial in u .

- (3) **Apply left exactness.** Left exactness and idempotence give

$$\begin{aligned} \mathrm{dR}(P^{\mathrm{dR}}(u)) &\simeq X_{\mathrm{dR}} \times_{X_{\mathrm{dR}}} U_{\mathrm{dR}} \\ &\simeq U_{\mathrm{dR}}. \end{aligned}$$

- (4) **Establish the indicated equivalence.** Under this equivalence $\mathrm{dR}(\ell_u)$ is the identity of U_{dR} .

- (5) **Conclude the stated claim.** Hence ℓ_u is a de Rham equivalence.

- (6) **Identify the stated objects.** The unit-naturality square for r_u is, after the preceding identification, the defining Cartesian square

$$\begin{array}{ccc} P^{\mathrm{dR}}(u) & \longrightarrow & U_{\mathrm{dR}} \\ r_u \downarrow & & \downarrow \\ X & \xrightarrow{\eta_X^{\mathrm{dR}}} & X_{\mathrm{dR}}. \end{array}$$

- (7) **Apply the cited lemma.** Lemma 4.2.4 therefore makes r_u right orthogonal to every de Rham equivalence.

- (8) **Use the stated hypothesis.** Since the de Rham class is generated by the affine square-zero substitutions, one-stage detection from Chapter 2 identifies this right orthogonality with formal étaleness.

- (9) **Prove the converse implication.** Conversely, Theorem 4.3.1 identifies a formally étale morphism with a Cartesian de Rham-unit square, and Lemma 4.2.4 then makes it right orthogonal to every de Rham equivalence.

- (10) **Conclude the stated claim.** Thus the right class is precisely the formally étale class.

- (11) **Apply orthogonality.** The functorial factorization above and the orthogonality just proved give the asserted orthogonal factorization system.

- (12) **Establish the indicated equivalence.** Pullback stability follows from pullback stability of de Rham equivalences and from base-change stability of formal lifting.

- (13) **Use the stated construction.** Applying the same argument in a fixed slice, using Proposition 4.3.2, gives the relative statement.

□

4.3.2 The effective de Rham relation groupoid

The map $e_{X/S} : X \rightarrow Q_{X/S}$ is effective by construction, so its Čech nerve is the correct object for descent. The relation, however, was defined geometrically using fibre products over $X_{\mathrm{dR}/S}$. We must therefore show that these two simplicial objects agree. The monomorphism $Q_{X/S} \hookrightarrow X_{\mathrm{dR}/S}$ supplies exactly that comparison.

Definition 4.3.2 (De Rham relation groupoid). For $n \geq 0$, define

$$\mathcal{R}_{X/S,n}^{\mathrm{dR}} := \underbrace{X \times_{X_{\mathrm{dR}/S}} \cdots \times_{X_{\mathrm{dR}/S}} X}_{n+1 \text{ factors}}.$$

Deletion and repetition of factors define a simplicial object $\mathcal{R}_{X/S,\bullet}^{\mathrm{dR}}$ over S . Let

$$v_n : \mathcal{R}_{X/S,n}^{\mathrm{dR}} \longrightarrow X$$

denote the first-vertex projection. In degree one write

$$\mathcal{R}_{X/S}^{\mathrm{dR}} := \mathcal{R}_{X/S,1}^{\mathrm{dR}}, \quad \mathbf{s} := v_1 = \mathrm{pr}_1, \quad \mathbf{t} := \mathrm{pr}_2.$$

We call $\mathbf{s}$ the source and $\mathbf{t}$ the target.

In degree one the relation is displayed by

$$\begin{array}{ccc} & \mathcal{R}_{X/S}^{\mathrm{dR}} & \\ \mathbf{s} \swarrow & & \searrow \mathbf{t} \\ X & & X \\ e_{X/S} \searrow & & \swarrow e_{X/S} \\ & Q_{X/S} & \end{array} \quad e_{X/S}\mathbf{s} \simeq e_{X/S}\mathbf{t}.$$

Indeed, by definition of the fibre product over $X_{\mathrm{dR}/S}$,

$$\eta_{X/S}^{\mathrm{dR}}\mathbf{s} \simeq \eta_{X/S}^{\mathrm{dR}}\mathbf{t}.$$

Since $\eta_{X/S}^{\mathrm{dR}} = m_{X/S}e_{X/S}$ and $m_{X/S}$ is a monomorphism, its diagonal is an equivalence and cancellation along $m_{X/S}$ gives the displayed homotopy

$$e_{X/S}\mathbf{s} \simeq e_{X/S}\mathbf{t}.$$

Theorem 4.3.6 (Effective presentation). *There is a canonical equivalence of simplicial objects*

$$\mathcal{R}_{X/S,\bullet}^{\mathrm{dR}} \simeq \check{C}_{\bullet}(e_{X/S} : X \rightarrow Q_{X/S}),$$

and hence a canonical realization equivalence

$$|\mathcal{R}_{X/S,\bullet}^{\mathrm{dR}}| \simeq Q_{X/S}.$$

Proof.

- (1) **Use the stated hypothesis.** Since $m_{X/S} : Q_{X/S} \rightarrow X_{\mathrm{dR}/S}$ is a monomorphism,

$$Q_{X/S} \times_{X_{\mathrm{dR}/S}} Q_{X/S} \simeq Q_{X/S}.$$

- (2) **Use the stated construction.** Finite-limit pasting gives

$$X \times_{X_{\mathrm{dR}/S}} X \simeq X \times_{Q_{X/S}} X,$$

and the same argument in every degree gives the equivalence of nerves.

- (3) **Use the stated hypothesis.** Since $e_{X/S}$ is an effective epimorphism, its Čech nerve realizes to $Q_{X/S}$.

□

Proposition 4.3.4 (Effective groupoid structure). *The simplicial object $\mathcal{R}_{X/S, \bullet}^{\mathrm{dR}}$ is the effective equivalence-relation groupoid of $X \rightarrow Q_{X/S}$, as in the general effective descent theory of ∞ -topoi [35]. In particular it carries canonical unit, inversion, and composition maps*

$$X \longrightarrow \mathcal{R}_{X/S}^{\mathrm{dR}}, \quad \mathcal{R}_{X/S}^{\mathrm{dR}} \xrightarrow{\sim} \mathcal{R}_{X/S}^{\mathrm{dR}}, \quad \mathcal{R}_{X/S}^{\mathrm{dR}} \times_{\mathbf{t}, X, \mathbf{s}} \mathcal{R}_{X/S}^{\mathrm{dR}} \longrightarrow \mathcal{R}_{X/S}^{\mathrm{dR}},$$

with all coherences supplied by the simplicial identities.

In degree two, composition is encoded by the Segal diagram

$$\begin{array}{ccc} \mathcal{R}_{X/S, 2}^{\mathrm{dR}} & \xrightarrow[\sim]{(d_2, d_0)} & \mathcal{R}_{X/S}^{\mathrm{dR}} \times_X \mathcal{R}_{X/S}^{\mathrm{dR}} \\ d_1 \downarrow & & \downarrow \text{comp} \\ \mathcal{R}_{X/S}^{\mathrm{dR}} & \xlongequal{\quad} & \mathcal{R}_{X/S}^{\mathrm{dR}}. \end{array}$$

4.3.3 De Rham neighbourhoods as infinitesimal closures

The relation groupoid admits a second description that is useful for transport arguments. Each relation simplex is the de Rham closure of a diagonal inside a Cartesian power of X . Formulating this closure as a relative de Rham neighbourhood makes factorization through finite infinitesimal thickenings and formal étale invariance immediate consequences of the relative reflector.

Definition 4.3.3 (Relative de Rham neighbourhood). For $Z \rightarrow Y \rightarrow S$, define

$$\mathrm{Nbh}_{Z/S}^{\mathrm{dR}}(Y) := Y \times_{Y_{\mathrm{dR}/S}} Z_{\mathrm{dR}/S}.$$

It is an object over Y , equipped with the canonical map from Z .

Equivalently, it is defined by the Cartesian square

$$\begin{array}{ccc} \mathrm{Nbh}_{Z/S}^{\mathrm{dR}}(Y) & \longrightarrow & Z_{\mathrm{dR}/S} \\ \downarrow & & \downarrow \\ Y & \xrightarrow[\eta_{Y/S}^{\mathrm{dR}}]{} & Y_{\mathrm{dR}/S}. \end{array}$$

Proposition 4.3.5 (Base change). *For $S' \rightarrow S$, with $Y' = Y \times_S S'$ and $Z' = Z \times_S S'$,*

$$\mathrm{Nbh}_{Z'/S'}^{\mathrm{dR}}(Y') \xrightarrow{\cong} \mathrm{Nbh}_{Z/S}^{\mathrm{dR}}(Y) \times_S S'.$$

Proof.

- (1) **Establish the indicated equivalence.** Substitute the coherent relative base-change equivalences of Theorem 4.2.5 into the defining fibre product.

□

Proposition 4.3.6 (Factorization through a de Rham neighbourhood). *Suppose $Z \rightarrow T \rightarrow Y \rightarrow S$ and that $Z \rightarrow T$ is a de Rham equivalence over S . Then $T \rightarrow Y$ factors canonically through*

$$T \longrightarrow \mathrm{Nbh}_{Z/S}^{\mathrm{dR}}(Y) \longrightarrow Y.$$

The factorization is natural in the diagram and compatible with arbitrary base change.

Proof.

- (1) **Establish the indicated equivalence.** The hypothesis makes

$$L_S^{\mathrm{dR}}(Z) \longrightarrow L_S^{\mathrm{dR}}(T)$$

an equivalence.

- (2) **Identify the stated objects.** Precomposition with this equivalence identifies

$$\mathrm{Map}_S(T_{\mathrm{dR}/S}, Z_{\mathrm{dR}/S}) \xrightarrow{\cong} \mathrm{Map}_S(Z_{\mathrm{dR}/S}, Z_{\mathrm{dR}/S}),$$

and the identity of $Z_{\mathrm{dR}/S}$ therefore determines, through a contractible space, the inverse equivalence.

- (3) **Establish the next claim.** Compose the unit

$$T \longrightarrow T_{\mathrm{dR}/S}$$

with that canonically determined inverse to obtain $T \rightarrow Z_{\mathrm{dR}/S}$.

- (4) **Apply naturality.** Naturality of the localization unit identifies the two composites to $Y_{\mathrm{dR}/S}$, so the universal property of the fibre product produces the asserted factorization.
- (5) **Apply naturality.** Naturality and base-change compatibility follow from the same mapping-space construction and the coherent base-change equivalences for relative de Rham reflection.

□

Theorem 4.3.7 (De Rham relation simplices as diagonal neighbourhoods). *Let*

$$Y_n := X^{\times_S(n+1)}, \quad \Delta_n : X \rightarrow Y_n$$

be the diagonal. Then there is a canonical equivalence over Y_n

$$\mathcal{R}_{X/S,n}^{\mathrm{dR}} \xrightarrow{\cong} \mathrm{Nbh}_{X/S}^{\mathrm{dR}}(Y_n).$$

These equivalences are simplicially natural, functorial in X/S , and compatible with arbitrary base change.

Proof.

- (1) **Apply left exactness.** Left exactness gives

$$(Y_n)_{\mathrm{dR}/S} \simeq (X_{\mathrm{dR}/S})^{\times_S(n+1)}.$$

- (2) **Establish the next claim.** The relative de Rham image of Δ_n is the diagonal of $X_{\mathrm{dR}/S}$. Therefore

$$Y_n \times_{(Y_n)_{\mathrm{dR}/S}} X_{\mathrm{dR}/S} \simeq X \times_{X_{\mathrm{dR}/S}} \cdots \times_{X_{\mathrm{dR}/S}} X.$$

- (3) **Use the stated construction.** Deleting and repeating factors gives the simplicial compatibility. □

Theorem 4.3.8 (Formal étale invariance of de Rham neighbourhoods). *Let $u : U \rightarrow X$ be formally étale over S , and let $Z \rightarrow U$. Regard Z also as an object over X . Then there is a canonical equivalence over X*

$$\mathrm{Nbh}_{Z/S}^{\mathrm{dR}}(U) \xrightarrow{\simeq} \mathrm{Nbh}_{Z/S}^{\mathrm{dR}}(X),$$

where the left-hand object is viewed over X through $U \rightarrow X$. Under this equivalence the right-hand de Rham neighbourhood factors canonically through U . Moreover

$$Q_{U/S} \simeq Q_{X/S} \times_{X_{\mathrm{dR}/S}} U_{\mathrm{dR}/S}$$

and

$$\mathcal{R}_{U/S,n}^{\mathrm{dR}} \simeq U \times_{X,v_n} \mathcal{R}_{X/S,n}^{\mathrm{dR}} \simeq Q_{U/S} \times_{Q_{X/S}} \mathcal{R}_{X/S,n}^{\mathrm{dR}},$$

compatibly with the simplicial structure.

Proof.

- (1) **Use the stated construction.** Formal étale Cartesianity gives

$$U \simeq X \times_{X_{\mathrm{dR}/S}} U_{\mathrm{dR}/S}.$$

Hence

$$\begin{aligned} \mathrm{Nbh}_{Z/S}^{\mathrm{dR}}(U) &= U \times_{U_{\mathrm{dR}/S}} Z_{\mathrm{dR}/S} \\ &\simeq (X \times_{X_{\mathrm{dR}/S}} U_{\mathrm{dR}/S}) \times_{U_{\mathrm{dR}/S}} Z_{\mathrm{dR}/S} \\ &\simeq X \times_{X_{\mathrm{dR}/S}} Z_{\mathrm{dR}/S} = \mathrm{Nbh}_{Z/S}^{\mathrm{dR}}(X). \end{aligned}$$

- (2) **Use pullback stability.** The unit of U is the pullback of the unit of X along $U_{\mathrm{dR}/S} \rightarrow X_{\mathrm{dR}/S}$.
 (3) **Use pullback stability.** Pullback stability and uniqueness of effective-image factorization therefore give the formula for $Q_{U/S}$.
 (4) **Establish the next claim.** The de Rham relation formulas follow by taking the corresponding Čech nerves, or directly by finite-limit pasting. □

At this point the unstable geometry required for descent is complete. The relative reflector supplies the universal infinitesimal quotient, its effective image $Q_{X/S}$ supplies the object actually presented by generalized points, and $\mathcal{R}_{X/S,\bullet}^{\mathrm{dR}}$ is the effective relation groupoid presenting that image. The next task is therefore to construct a stable scalar-valued semantics on these already-defined objects.

4.4 Stable scalar semantics on de Rham geometry

The preceding section produced the effective groupoid on which descent must be computed. To extract a de Rham complex from it, we need stable scalar functions that can be pulled back, pushed forward by dependent product, and descended along effective epimorphisms. We reuse the stabilized slice fibres from formal reconstruction,

$$\mathcal{D}_{\mathrm{Sp}/S} = \mathrm{Sp}(((\mathcal{H}_{\mathrm{Sp}})_{/S})_*).$$

and establish only the interface used later: dependent products, Beck–Chevalley, internal function objects, effective descent, and a stable realization of quasi-coherent coefficients. The reader may keep the intended output in view throughout this section: for a map $a : T \rightarrow S$, stable scalar functions will be expressible both as $\Pi_a a^* \mathcal{O}_S^{\mathrm{add}}$ and as an internal function object, and both descriptions will commute with base change and effective descent.

Stable realization connects the two coefficient languages already present in the manuscript. Quasi-coherent modules remain the source of geometric coefficients, while parameterized spectra provide the ambient category in which functions on arbitrary spectral Zariski stacks can be formed. On an affine point these descriptions are compared by ordinary global sections:

$$\begin{array}{ccc} \mathrm{QCoh}(S) & \xrightarrow{\mathrm{Real}_S^{\mathrm{st}}} & \mathcal{D}_{\mathrm{Sp}/S} \\ a^* \downarrow & & \downarrow a^* \\ \mathrm{QCoh}(T) & \xrightarrow{\mathrm{Real}_T^{\mathrm{st}}} & \mathcal{D}_{\mathrm{Sp}/T} \\ \Gamma(T, -) \downarrow & & \downarrow \Gamma^{\mathrm{st}}(T, -) \\ \mathrm{Sp} & \xlongequal{\quad} & \mathrm{Sp}. \end{array}$$

The bottom comparison is what later identifies represented stable scalar functions with ordinary spectral functions and first-order scalar coefficients with cotangent coefficients.

4.4.1 Stable slices and dependent adjoints

The de Rham relation objects, effective quotients, and de Rham neighbourhoods live in $\mathcal{H}_{\mathrm{Sp}}$, so scalar functions on them must be formed in stabilized slices. We first establish the dependent-adjoint calculus for those slices. The affine-point presentation is then used only for the two jobs that require it: detecting stable function objects and realizing quasi-coherent coefficients.

Proposition 4.4.1 (Affine-point presentation of a spectral slice). *For $S \in \mathcal{H}_{\mathrm{Sp}}$, equip $\mathrm{AffPt}(S)$ with the topology induced by finite affine spectral Zariski covers on the affine source. Restriction along the affine-point functor induces a canonical equivalence*

$$(\mathcal{H}_{\mathrm{Sp}})_{/S} \xrightarrow{\cong} \mathrm{Sh}_{\tau_{\mathrm{Zar}}}(\mathrm{AffPt}(S)).$$

Under this equivalence an object $Y \rightarrow S$ is sent to the sheaf

$$(h_T \rightarrow S) \mapsto \mathrm{Map}_S(h_T, Y).$$

The statement applies to every $S \in \mathcal{H}_{\mathrm{Sp}}$.

Proof.

- (1) **Use the stated construction.** At the presheaf level, the category-of-elements construction gives the canonical equivalence

$$\mathrm{PSh}(\mathrm{Aff}_{\mathrm{Sp}})_{/S} \simeq \mathrm{PSh}(\mathrm{AffPt}(S)).$$

- (2) **Use pullback stability.** The spectral Zariski topology on the affine-point category is the pullback of the ambient topology: a covering family over $h_T \rightarrow S$ is precisely a finite affine spectral Zariski covering family of T , with the induced maps to S .
- (3) **Establish the locality claim.** Consequently the covering-sieve generators for the two sheaf localizations correspond.
- (4) **Use the stated construction.** Passing to the associated accessible left-exact localizations gives the displayed equivalence.
- (5) **Establish the next claim.** The pointwise formula is the Yoneda description of the slice presheaf.
- (6) **Use the stated construction.** Equivalently, Lemma 4.2.1 supplies the affine density needed to recover every object of the slice from these tests.

□

Construction 4.4.1 (Parameterized stable slice doctrine). For every $S \in \mathcal{H}_{\mathrm{Sp}}$, retain the stabilized slice category

$$\mathcal{D}_{\mathrm{Sp}/S} = \mathrm{Sp}(((\mathcal{H}_{\mathrm{Sp}})_{/S})_*)$$

and the notation of Chapter 3. Write $\mathbf{1}_S$ for its tensor unit. For an object $V \rightarrow S$, let

$$\Sigma_{+,S}^\infty V \in \mathcal{D}_{\mathrm{Sp}/S}$$

denote the suspension spectrum of the free pointed object V_+ in the slice. The cartesian symmetric monoidal structure on the slice stabilizes to the canonical presentably symmetric monoidal structure on $\mathcal{D}_{\mathrm{Sp}/S}$ [34]; in particular

$$\Sigma_{+,S}^\infty U \otimes \Sigma_{+,S}^\infty V \simeq \Sigma_{+,S}^\infty (U \times_S V).$$

As S varies, these already-fixed fibres and their pullback functors form the parameterized stable slice doctrine used below; see [34].

Theorem 4.4.1 (Stable dependent adjoints and base change). *For every morphism $f : T \rightarrow S$, slice pullback induces an exact strong symmetric monoidal functor*

$$f^* : \mathcal{D}_{\mathrm{Sp}/S} \longrightarrow \mathcal{D}_{\mathrm{Sp}/T}$$

with adjoints

$$\Sigma_f \dashv f^* \dashv \Pi_f.$$

For every Cartesian square

$$\begin{array}{ccc} T' & \xrightarrow{g'} & T \\ f' \downarrow & & \downarrow f \\ S' & \xrightarrow{g} & S, \end{array}$$

the pullback functors themselves form the square

$$\begin{array}{ccc} \mathcal{D}_{\mathrm{Sp}/S} & \xrightarrow{f^*} & \mathcal{D}_{\mathrm{Sp}/T} \\ g^* \downarrow & \simeq & \downarrow (g')^* \\ \mathcal{D}_{\mathrm{Sp}/S'} & \xrightarrow{(f')^*} & \mathcal{D}_{\mathrm{Sp}/T'}, \end{array}$$

and there are canonical Beck–Chevalley equivalences

$$g^* \Sigma_f \xrightarrow{\simeq} \Sigma_{f'}(g')^*, \quad g^* \Pi_f \xrightarrow{\simeq} \Pi_{f'}(g')^*.$$

Moreover there is a canonical projection equivalence

$$\Sigma_f(f^*E \otimes F) \xrightarrow{\simeq} E \otimes \Sigma_f F,$$

and Π_f carries the canonical lax symmetric monoidal structure right-adjoint to the strong symmetric monoidal structure on f^* . Their compatibility is determined by the slice pullback diagrams and the calculus of mates.

Proof.

- (1) **Use pullback stability.** Unstable slice pullback

$$f^* : (\mathcal{H}_{\mathrm{Sp}})_{/S} \longrightarrow (\mathcal{H}_{\mathrm{Sp}})_{/T}$$

preserves all limits and colimits because it has both the dependent sum and dependent product as adjoints.

- (2) **Use pullback stability.** It preserves finite limits in particular, so functoriality of spectrum objects induces the displayed exact stable pullback.
- (3) **Establish the next claim.** The induced functor is accessible and preserves all limits and colimits.
- (4) **Establish the next claim.** The presentable adjoint functor theorem therefore constructs both adjoints

$$\Sigma_f \dashv f^* \dashv \Pi_f.$$

- (5) **Use pullback stability.** The strong symmetric monoidal structure on stable pullback is the stabilization of the cartesian pullback structure; see [34].
- (6) **Fix the indicated data.** Let $v : V \rightarrow T$.
- (7) **Use the stated construction.** For every $E \in \mathcal{D}_{\mathrm{Sp}/S}$, the suspension-spectrum/zeroth-space adjunction in the two slices gives

$$\begin{aligned} \mathrm{map}_{\mathcal{D}_{\mathrm{Sp}/S}}(\Sigma_f \Sigma_{+,T}^\infty V, E) &\simeq \mathrm{map}_{\mathcal{D}_{\mathrm{Sp}/T}}(\Sigma_{+,T}^\infty V, f^* E) \\ &\simeq \mathrm{map}_{\mathcal{D}_{\mathrm{Sp}/S}}(\Sigma_{+,S}^\infty V, E), \end{aligned}$$

where on the last line V is regarded over S through $f \circ v$.

- (8) **Apply Yoneda density.** Yoneda therefore gives the canonical equivalence

$$\Sigma_f \Sigma_{+,T}^\infty V \simeq \Sigma_{+,S}^\infty V.$$

- (9) **Establish the next claim.** The canonical mate

$$g^* \Sigma_f \longrightarrow \Sigma_{f'}(g')^*$$

is a transformation between colimit-preserving exact functors.

- (10) **Establish the next claim.** Suspension spectra of objects of the slice, together with their shifts, generate the stabilization.

- (11) **Establish the next claim.** On a generator $\Sigma_{+,T}^\infty V$, the two sides are canonically

$$\Sigma_{+,S'}^\infty(V \times_S S') \quad \text{and} \quad \Sigma_{+,S'}^\infty(V \times_T T'),$$

which agree because the square is Cartesian.

- (12) **Conclude the stated claim.** Hence the mate is an equivalence.

- (13) **Apply the indicated result.** Apply the left Beck–Chevalley equivalence of the preceding step to the same Cartesian square with the horizontal and vertical roles exchanged.

- (14) **Use the stated construction.** This gives the canonical equivalence

$$f^* \Sigma_g \xrightarrow{\simeq} \Sigma_{g'}(f')^*.$$

- (15) **Identify the stated objects.** Passing to right adjoints identifies the right adjoint of the left-hand composite with $g^* \Pi_f$, and the right adjoint of the right-hand composite with $\Pi_{f'}(g')^*$.

- (16) **Use the stated construction.** Uniqueness of right adjoints therefore gives the canonical mate equivalence

$$g^* \Pi_f \xrightarrow{\simeq} \Pi_{f'}(g')^*.$$

- (17) **Establish the next claim.** This comparison is obtained entirely by transposition and the calculus of adjoint mates, so its functoriality is inherited from the left Beck–Chevalley equivalence.

- (18) **Establish the next claim.** Both sides of

$$\Sigma_f(f^* E \otimes F) \longrightarrow E \otimes \Sigma_f F$$

preserve colimits separately in E and F .

- (19) **Establish the next claim.** It therefore suffices to evaluate on

$$E = \Sigma_{+,S}^\infty U, \quad F = \Sigma_{+,T}^\infty V.$$

- (20) **Identify the stated objects.** Strong symmetric monoidality of suspension spectra and the calculation of Σ_f above identify both sides canonically with

$$\Sigma_{+,S}^\infty(U \times_S V).$$

- (21) **Conclude the stated claim.** Hence the projection morphism is an equivalence for all E, F .
- (22) **Use the stated hypothesis.** Since f^* is strong symmetric monoidal, the right adjoint Π_f receives its canonical lax symmetric monoidal structure.
- (23) **Use pullback stability.** All comparisons above are mates of the functorial pullback, product, and composition diagrams.

□

Corollary 4.4.1 (Parameterized stable slice hyperdoctrine recognition). *The parameterized stable slice doctrine*

$$S \longmapsto \mathcal{D}_{\mathrm{Sp}/S}$$

with reindexing f^ is, in particular, a stable presentably symmetric monoidal slice hyperdoctrine, in the sense of indexed categorical logic; compare [25]. Its dependent sum and dependent product are*

$$\Sigma_f \dashv f^* \dashv \Pi_f,$$

the Beck–Chevalley comparisons are equivalences, and Σ_f satisfies the projection formula. All substitution, dependent-adjoint, exchange, and projection coherences are those constructed from the slice adjunctions and their mates.

Proof.

- (1) **Apply the cited theorem.** Theorem 4.4.1 provides the adjoint string $\Sigma_f \dashv f^* \dashv \Pi_f$, the Beck–Chevalley equivalences, and the projection formula.
- (2) **Establish the next claim.** These data give the asserted stable presentably symmetric monoidal slice hyperdoctrine; compare the terminology of indexed categorical logic in [25].

□

4.4.2 Affine-point semantics and parameterized function objects

The dependent-adjoint calculus is now available abstractly. To turn it into a usable function theory, we identify stable slices on affine points. At this stage we need the resulting generating family, the behaviour of pullback, relative suspension spectra, internal function objects, and effective descent. The Postnikov and grouplike consequences of the same presentation will be recalled only when the additive delooping construction needs them.

Proposition 4.4.2 (Stable affine-point presentation). *For every $S \in \mathcal{H}_{\mathrm{Sp}}$, stabilization of Proposition 4.4.1 gives a canonical equivalence*

$$\mathcal{D}_{\mathrm{Sp}/S} \xrightarrow{\cong} \mathrm{Sh}_{\tau_{\mathrm{Zar}}}(\mathrm{AffPt}(S); \mathrm{Sp}).$$

Here

$$\mathrm{Sh}_{\tau_{\mathrm{Zar}}}(\mathrm{AffPt}(S); \mathrm{Sp})$$

denotes the stabilization of the spectral Zariski ∞ -topos $\mathrm{Sh}_{\tau_{\mathrm{Zar}}}(\mathrm{AffPt}(S))$. The notation uses the ambient spectral Zariski stack convention fixed in Chapter 1.

Under this equivalence, $E \in \mathcal{D}_{\mathrm{Sp}/S}$ is sent to the sheaf of spectra

$$(a : T \rightarrow S) \longmapsto \Gamma^{\mathrm{st}}(T, a^* E).$$

Consequently $\mathcal{D}_{\mathrm{Sp}/S}$ is generated under colimits by

$$\Sigma^n \Sigma_{+,S}^\infty T, \quad n \in \mathbb{Z},$$

where $T \rightarrow S$ ranges over affine spectral points, and a morphism in $\mathcal{D}_{\mathrm{Sp}/S}$ is an equivalence if and only if it induces equivalences on mapping spectra from all these generators.

Proof.

- (1) **Establish the next claim.** Spectrum objects are defined by finite-limit conditions.
- (2) **Use the stated hypothesis.** Since the equivalence of Proposition 4.4.1 is left exact, it induces an equivalence on stabilizations.
- (3) **Invoke the cited result.** By the notation fixed in the statement,

$$\mathrm{Sh}_{\tau_{\mathrm{Zar}}}(\mathrm{AffPt}(S); \mathrm{Sp}) := \mathrm{Sp}(\mathrm{Sh}_{\tau_{\mathrm{Zar}}}(\mathrm{AffPt}(S))_*).$$

- (4) **Conclude the stated claim.** Hence stabilization of the affine-point presentation gives

$$\mathrm{Sp}(((\mathcal{H}_{\mathrm{Sp}})_S)_*) \simeq \mathrm{Sh}_{\tau_{\mathrm{Zar}}}(\mathrm{AffPt}(S); \mathrm{Sp}).$$

- (5) **Establish the next claim.** Equivalently, the right-hand side is the category of spectrum objects satisfying spectral Zariski descent on the same affine-point site.
- (6) **Compute affine-point values.** The displayed value formula follows by applying the suspension-spectrum/zeroth-space adjunction in the slice over each affine point.
- (7) **Use the stated construction.** Affine density gives a set of colimit generators for the unstable slice.
- (8) **Establish the next claim.** Their relative suspension spectra and all integer shifts therefore generate the stabilization.
- (9) **Establish the next claim.** The final assertion is the generation criterion of a generating set in a presentable stable infinity-category.

□

Proposition 4.4.3 (Affine-point pullback). *Let $f : T \rightarrow S$. Under Proposition 4.4.2, stable pullback is restriction along*

$$\mathrm{AffPt}(T) \longrightarrow \mathrm{AffPt}(S), \quad (U \rightarrow T) \longmapsto (U \rightarrow T \rightarrow S).$$

Proof.

- (1) **Treat the stated case.** If $u : U \rightarrow T$ is affine and $E \in \mathcal{D}_{\mathrm{Sp}/S}$, the affine-point value of f^*E at u is the value of E at the composite $fu : U \rightarrow S$.
- (2) **Construct the indicated map.** This is the stabilized form of the Cartesian identity

$$\mathrm{Map}_T(U, Y \times_S T) \simeq \mathrm{Map}_S(U, Y),$$

natural in $Y \rightarrow S$.

- (3) **Conclude the stated claim.** Hence stable pullback is restriction on affine-point sheaves of spectra.

□

Construction 4.4.2 (Relative suspension spectra). Let $a : T \rightarrow S$. Write T_+ for the free pointed object on T in the slice over S , and define

$$\Sigma_{+,S}^\infty T := \Sigma_S^\infty T_+ \in \mathcal{D}_{\mathrm{Sp}/S}.$$

The dependent-sum description gives

$$\Sigma_{+,S}^\infty T \simeq \Sigma_a \mathbf{1}_T.$$

More generally, for $U \rightarrow T \rightarrow S$,

$$\Sigma_a \Sigma_{+,T}^\infty U \simeq \Sigma_{+,S}^\infty U.$$

Definition 4.4.1 (Stable sections). For $E \in \mathcal{D}_{\mathrm{Sp}/T}$, define

$$\Gamma^{\mathrm{st}}(T, E) := \mathrm{map}_{\mathcal{D}_{\mathrm{Sp}/T}}(\mathbf{1}_T, E).$$

Proposition 4.4.4 (Parameterized function object). *For $a : T \rightarrow S$ and $E \in \mathcal{D}_{\mathrm{Sp}/S}$, there is a canonical equivalence of objects of $\mathcal{D}_{\mathrm{Sp}/S}$*

$$\Pi_a a^* E \simeq \underline{\mathrm{Hom}}_{\mathcal{D}_{\mathrm{Sp}/S}}(\Sigma_{+,S}^\infty T, E).$$

Consequently

$$\mathrm{map}_{\mathcal{D}_{\mathrm{Sp}/S}}(\Sigma_{+,S}^\infty T, E) \simeq \Gamma^{\mathrm{st}}(T, a^* E).$$

Proof.

- (1) **Construct the indicated map.** For $B \in \mathcal{D}_{\mathrm{Sp}/S}$, the projection formula and the adjunction $\Sigma_a \dashv a^*$ give

$$\begin{aligned} \mathrm{map}_{\mathcal{D}_{\mathrm{Sp}/S}}(B, \underline{\mathrm{Hom}}(\Sigma_a \mathbf{1}_T, E)) &\simeq \mathrm{map}_{\mathcal{D}_{\mathrm{Sp}/S}}(B \otimes \Sigma_a \mathbf{1}_T, E) \\ &\simeq \mathrm{map}_{\mathcal{D}_{\mathrm{Sp}/T}}(a^* B, a^* E) \\ &\simeq \mathrm{map}_{\mathcal{D}_{\mathrm{Sp}/S}}(B, \Pi_a a^* E). \end{aligned}$$

- (2) **Apply Yoneda density.** Yoneda identifies the internal objects.

- (3) **Take the indicated specialization.** Taking $B = \mathbf{1}_S$ gives the second formula.

□

Proposition 4.4.5 (Closed base change for parameterized internal Homs). *Let $c : S' \rightarrow S$, and let $E, F \in \mathcal{D}_{\mathrm{Sp}/S}$. There is a canonical equivalence*

$$c^* \underline{\mathrm{Hom}}_{\mathcal{D}_{\mathrm{Sp}/S}}(E, F) \xrightarrow{\simeq} \underline{\mathrm{Hom}}_{\mathcal{D}_{\mathrm{Sp}/S'}}(c^* E, c^* F),$$

natural in c, E, F and compatible with the convolution algebra structures induced by cocommutative coalgebra structures on E and commutative algebra structures on F .

Proof.

- (1) **Fix the indicated data.** Let $B \in \mathcal{D}_{\mathrm{Sp}/S'}$.
- (2) **Use the stated construction.** Using $\Sigma_c \dashv c^*$, the internal-Hom adjunction, the projection formula of Theorem 4.4.1, and the same adjunction again gives

$$\begin{aligned}
 \mathrm{map}_{\mathcal{D}_{\mathrm{Sp}/S'}}(B, c^* \underline{\mathrm{Hom}}_{\mathcal{D}_{\mathrm{Sp}/S}}(E, F)) &\simeq \mathrm{map}_{\mathcal{D}_{\mathrm{Sp}/S}}(\Sigma_c B, \underline{\mathrm{Hom}}_{\mathcal{D}_{\mathrm{Sp}/S}}(E, F)) \\
 &\simeq \mathrm{map}_{\mathcal{D}_{\mathrm{Sp}/S}}(\Sigma_c B \otimes E, F) \\
 &\simeq \mathrm{map}_{\mathcal{D}_{\mathrm{Sp}/S}}(\Sigma_c(B \otimes c^* E), F) \\
 &\simeq \mathrm{map}_{\mathcal{D}_{\mathrm{Sp}/S'}}(B \otimes c^* E, c^* F) \\
 &\simeq \mathrm{map}_{\mathcal{D}_{\mathrm{Sp}/S'}}(B, \underline{\mathrm{Hom}}_{\mathcal{D}_{\mathrm{Sp}/S'}}(c^* E, c^* F)).
 \end{aligned}$$

- (3) **Apply Yoneda density.** Yoneda gives the displayed equivalence.
- (4) **Establish the next claim.** Every comparison in the calculation is obtained functorially from the displayed adjunctions and projection formula.
- (5) **Treat the stated case.** If E is cocommutative coalgebraic and F is commutative algebraic, convolution is obtained from the diagonal/comultiplication of E , the multiplication of F , and the closed symmetric monoidal structure.
- (6) **Use the stated hypothesis.** Since c^* is strong symmetric monoidal, it transports those structural maps to the corresponding maps of $c^* E$ and $c^* F$.
- (7) **Apply naturality.** Naturality of the preceding Yoneda comparison with respect to those maps identifies the two convolution multiplications and all higher commutative operations.

□

Proposition 4.4.6 (Multiplicative parameterized function object). *Let $a : T \rightarrow S$, and let $A \in \mathrm{CAlg}(\mathcal{D}_{\mathrm{Sp}/S})$. The diagonal $T \rightarrow T \times_S T$ and terminal map $T \rightarrow S$ give $\Sigma_{+,S}^\infty T$ its canonical cocommutative coalgebra structure. Endow*

$$\underline{\mathrm{Hom}}_{\mathcal{D}_{\mathrm{Sp}/S}}(\Sigma_{+,S}^\infty T, A)$$

with the resulting convolution commutative algebra structure. Then the equivalence of Proposition 4.4.4 refines canonically to an equivalence in $\mathrm{CAlg}(\mathcal{D}_{\mathrm{Sp}/S})$,

$$\Pi_a a^* A \xrightarrow{\simeq} \underline{\mathrm{Hom}}_{\mathcal{D}_{\mathrm{Sp}/S}}(\Sigma_{+,S}^\infty T, A).$$

The refinement is natural in a and A and is compatible with Cartesian base change.

Proof.

- (1) **Use the stated hypothesis.** Since a^* is strong symmetric monoidal, its right adjoint Π_a is canonically lax symmetric monoidal.
- (2) **Establish multiplicative compatibility.** The multiplication on $\Pi_a a^* A$ is therefore adjoint to

$$a^*(\Pi_a a^* A)^{\otimes 2} \longrightarrow a^* A^{\otimes 2} \longrightarrow a^* A,$$

where the first arrow is the tensor product of the two counits.

- (3) **Establish the next claim.** Under

$$\Sigma_a \mathbf{1}_T \simeq \Sigma_{+,S}^\infty T$$

and the projection formula, the mate of this composite is obtained by precomposing two functions with the diagonal

$$T \longrightarrow T \times_S T$$

and then multiplying their values in A .

- (4) **Establish multiplicative compatibility.** This is exactly convolution against the comultiplication

$$\Sigma_{+,S}^\infty T \longrightarrow \Sigma_{+,S}^\infty T \otimes \Sigma_{+,S}^\infty T.$$

- (5) **Identify the stated objects.** The same mate calculation for the k -fold diagonal $T \rightarrow T^{\times_S k}$ identifies every k -ary operation of the lax symmetric monoidal structure on Π_a with the corresponding convolution operation.
- (6) **Identify the stated objects.** The terminal map $T \rightarrow S$ identifies the units.
- (7) **Conclude the stated claim.** Thus the underlying equivalence of Proposition 4.4.4 is an equivalence of commutative algebras.
- (8) **Apply naturality.** Naturality and Cartesian base-change compatibility are inherited from the projection formula, Beck–Chevalley equivalences, and the functorial diagonal maps.

□

Proposition 4.4.7 (Effective descent for parameterized spectra). *If $e : T \rightarrow S$ is an effective epimorphism with Čech nerve $T_\bullet$, pullback induces an equivalence of symmetric monoidal presentable stable infinity-categories*

$$\mathcal{D}_{\mathrm{Sp}/S} \xrightarrow{\simeq} \mathrm{Tot}_{[n] \in \Delta} \mathcal{D}_{\mathrm{Sp}/T_n}.$$

Under this equivalence tensor units, tensor products, pullbacks, and commutative algebra objects are identified degreewise. Dependent adjoints and internal Homs are then identified by their universal properties and the Beck–Chevalley equivalences of Theorem 4.4.1. Pullback along e is conservative on parameterized spectra.

Proof.

- (1) **Apply effective descent.** Effective descent in the infinity-topos gives, as in [35],

$$(\mathcal{H}_{\mathrm{Sp}})_S \simeq \mathrm{Tot}_{[n] \in \Delta} (\mathcal{H}_{\mathrm{Sp}})_{T_n}.$$

- (2) **Take the indicated specialization.** Taking spectrum objects commutes with this limit because the spectrum-object conditions are finite-limit conditions.
- (3) **Identify the stated objects.** The fiberwise symmetric monoidal structure is functorial for pullback, so the resulting equivalence is symmetric monoidal and identifies tensor units, tensor products, pullbacks, and commutative algebra objects degreewise.

- (4) **Establish the indicated equivalence.** The dependent adjoints and internal Homs are characterized by their mapping universal properties and hence are transported by the same equivalence.
- (5) **Establish the indicated equivalence.** To prove conservativity, let $u : E \rightarrow F$ in $\mathcal{D}_{\mathrm{Sp}/S}$ become an equivalence after pullback to T .
- (6) **Use pullback stability.** Every structural map $T_n \rightarrow S$ of the Čech nerve factors through $T \rightarrow S$, so the pullback of u to every T_n is an equivalence.
- (7) **Pass to totalization.** Under the totalization equivalence the image of u is therefore degree-wise an equivalence, hence an equivalence.
- (8) **Conclude the stated claim.** Thus u itself is an equivalence.

□

4.4.3 Stable realization of quasi-coherent coefficients

The parameterized stable slice can now express functions on arbitrary geometric objects, but the cotangent comparison later requires a bridge from quasi-coherent coefficients to that stable slice. We construct this bridge by taking quasi-coherent global sections on every affine point and descending the resulting sheaf of spectra.

Construction 4.4.3 (Section stacks over spectral Zariski stacks). Let $S \in \mathcal{H}_{\mathrm{Sp}}$ and $\mathcal{F} \in \mathrm{QCoh}(S)$. Define

$$\mathrm{Sect}_S(\mathcal{F}) \longrightarrow S$$

by the affine-point formula: over $a : T \rightarrow S$, its fibre is

$$\mathrm{Map}_{\mathrm{QCoh}(T)}(\mathcal{O}_T, a^* \mathcal{F}).$$

It is a spectral Zariski stack over S , has a canonical zero section, and for every $f : T \rightarrow S$ there is a canonical Cartesian base-change equivalence

$$\mathrm{Sect}_S(\mathcal{F}) \times_S T \xrightarrow{\sim} \mathrm{Sect}_T(f^* \mathcal{F}).$$

More generally, for every $g : Y \rightarrow S$,

$$\mathrm{Map}_S(Y, \mathrm{Sect}_S(\mathcal{F})) \xrightarrow{\sim} \mathrm{Map}_{\mathrm{QCoh}(Y)}(\mathcal{O}_Y, g^* \mathcal{F}).$$

Proof.

- (1) **Establish the next claim.** The descent and base-change proof of 1.13.1 and 1.13.6 uses only that the base is a spectral Zariski sheaf and that quasi-coherent coefficients on affine tests satisfy Zariski descent.
- (2) **Construct the indicated map.** Those conditions hold for every $S \in \mathcal{H}_{\mathrm{Sp}}$, so the same Grothendieck construction and mapping-space totalization prove the first two claims.
- (3) **Establish the next claim.** For a general $Y \rightarrow S$, apply affine density to Y and use the affine-point limit definition of $\mathrm{QCoh}(Y)$.

- (4) **Construct the indicated map.** Both sides become the same limit of the affine mapping spaces

$$\mathrm{Map}_{\mathrm{QCoh}(U)}(\mathcal{O}_U, (g|_U)^*\mathcal{F}).$$

- (5) **Conclude the stated claim.** Thus the construction extends the section-stack interface of Chapter 1 to arbitrary bases in $\mathcal{H}_{\mathrm{Sp}}$.

□

Construction 4.4.4 (Stable realization of quasi-coherent coefficients). Let $S \in \mathcal{H}_{\mathrm{Sp}}$ and $\mathcal{F} \in \mathrm{QCoh}(S)$. On the affine-point site consider the presheaf of spectra

$$(a : T \rightarrow S) \mapsto \Gamma(T, a^*\mathcal{F}) = \mathrm{map}_{\mathrm{QCoh}(T)}(\mathcal{O}_T, a^*\mathcal{F}).$$

For a morphism of affine points $g : T' \rightarrow T$, restriction is the map on global sections induced by quasi-coherent pullback.

The construction is governed by the affine-point triangle

$$\begin{array}{ccc} T' & \xrightarrow{g} & T \\ & \searrow a' & \downarrow a \\ & & S, \end{array} \quad \Gamma(T, a^*\mathcal{F}) \longrightarrow \Gamma(T', (a')^*\mathcal{F}).$$

This presheaf is a spectral Zariski sheaf. Indeed, if $T_\bullet \rightarrow T$ is the Čech nerve of an affine spectral Zariski cover, quasi-coherent descent gives a symmetric monoidal equivalence

$$\mathrm{QCoh}(T) \simeq \mathrm{Tot}_{[n] \in \Delta} \mathrm{QCoh}(T_n),$$

under which $\mathcal{O}_T$ and $a^*\mathcal{F}$ correspond to their restrictions. Mapping spectra in a categorical limit are computed by the corresponding limit, so

$$\Gamma(T, a^*\mathcal{F}) \simeq \mathrm{Tot}_{[n] \in \Delta} \Gamma(T_n, a_n^*\mathcal{F}).$$

By Proposition 4.4.2, let

$$\mathrm{Real}_S^{\mathrm{st}}(\mathcal{F}) \in \mathcal{D}_{\mathrm{Sp}/S}$$

denote the canonical object corresponding to this sheaf of spectra.

Proposition 4.4.8 (Properties and monoidality of stable realization). *The construction*

$$\mathrm{Real}_S^{\mathrm{st}} : \mathrm{QCoh}(S) \longrightarrow \mathcal{D}_{\mathrm{Sp}/S}$$

is exact and canonically lax symmetric monoidal. Its affine-point evaluation is encoded by

$$\begin{array}{ccc} \mathrm{QCoh}(S) & \xrightarrow{\mathrm{Real}_S^{\mathrm{st}}} & \mathcal{D}_{\mathrm{Sp}/S} \\ a^* \downarrow & & \downarrow a^* \\ \mathrm{QCoh}(T) & \xrightarrow{\mathrm{Real}_T^{\mathrm{st}}} & \mathcal{D}_{\mathrm{Sp}/T}, \end{array}$$

which commutes up to the canonical base-change equivalence. More precisely:

1. for every $f : T \rightarrow S$, there is a coherent base-change equivalence

$$f^* \operatorname{Real}_S^{\operatorname{st}}(\mathcal{F}) \xrightarrow{\sim} \operatorname{Real}_T^{\operatorname{st}}(f^* \mathcal{F});$$

2. stable global sections recover quasi-coherent global sections,

$$\Gamma^{\operatorname{st}}(S, \operatorname{Real}_S^{\operatorname{st}}(\mathcal{F})) \xrightarrow{\sim} \Gamma(S, \mathcal{F});$$

3. there are canonical natural maps

$$\operatorname{Real}_S^{\operatorname{st}}(\mathcal{F}) \otimes \operatorname{Real}_S^{\operatorname{st}}(\mathcal{G}) \longrightarrow \operatorname{Real}_S^{\operatorname{st}}(\mathcal{F} \otimes \mathcal{G}), \quad \mathbf{1}_S \longrightarrow \operatorname{Real}_S^{\operatorname{st}}(\mathcal{O}_S),$$

which satisfy the symmetric-monoidal associativity, symmetry, and unit coherences. The base-change equivalences are compatible with this lax symmetric monoidal structure.

Proof.

- (1) **Establish the next claim.** Exactness may be checked on the stable affine-point presentation.
 (2) **Construct the indicated map.** For every affine point $a : T \rightarrow S$, the functor

$$\mathcal{F} \longmapsto \operatorname{map}_{\operatorname{QCoh}(T)}(\mathcal{O}_T, a^* \mathcal{F})$$

is exact: pullback is exact, and mapping spectra out of a fixed object preserve finite limits, hence finite colimits in a stable category.

- (3) **Invoke the cited result.** By Proposition 4.4.3, stable pullback is restriction of the affine-point sheaf of spectra.
 (4) **Conclude the stated claim.** Hence for an affine point $u : U \rightarrow T$,

$$\begin{aligned} \Gamma^{\operatorname{st}}(U, u^* f^* \operatorname{Real}_S^{\operatorname{st}}(\mathcal{F})) &\simeq \Gamma(U, (fu)^* \mathcal{F}) \\ &\simeq \Gamma(U, u^* f^* \mathcal{F}), \end{aligned}$$

which is exactly the affine-point value of $\operatorname{Real}_T^{\operatorname{st}}(f^* \mathcal{F})$.

- (5) **Apply naturality.** Naturality in u and the stable affine-point presentation give the claimed equivalence.
 (6) **Apply the indicated result.** Apply Lemma 4.2.1 to the terminal object $S \rightarrow S$ of the slice:

$$S \simeq \operatorname{colim}_{(T,a) \in \operatorname{AffPt}(S)} T.$$

- (7) **Establish the next claim.** The free-pointed functor on the slice and the relative suspension-spectrum functor are left adjoints, hence preserve this colimit.
 (8) **Use the stated hypothesis.** Since $\Sigma_{+,S}^\infty S \simeq \mathbf{1}_S$, we obtain canonically

$$\mathbf{1}_S \simeq \operatorname{colim}_{(T,a) \in \operatorname{AffPt}(S)} \Sigma_{+,S}^\infty T.$$

Therefore

$$\Gamma^{\operatorname{st}}(S, \operatorname{Real}_S^{\operatorname{st}}(\mathcal{F})) \simeq \lim_{(T,a) \in \operatorname{AffPt}(S)^{\operatorname{op}}} \Gamma(T, a^* \mathcal{F}).$$

- (9) **Identify the stated objects.** The affine-point limit definition of $\mathrm{QCoh}(S)$ and creation of mapping spectra in categorical limits identify the right side with $\Gamma(S, \mathcal{F})$.
- (10) **Use the stated construction.** Finally, on an affine point $a : T \rightarrow S$, tensoring two maps out of the tensor unit gives the canonical pairing

$$\begin{aligned} \mathrm{map}_{\mathrm{QCoh}(T)}(\mathcal{O}_T, a^* \mathcal{F}) \otimes \mathrm{map}_{\mathrm{QCoh}(T)}(\mathcal{O}_T, a^* \mathcal{G}) \\ \longrightarrow \mathrm{map}_{\mathrm{QCoh}(T)}(\mathcal{O}_T, a^*(\mathcal{F} \otimes \mathcal{G})), \end{aligned}$$

using $\mathcal{O}_T \otimes \mathcal{O}_T \simeq \mathcal{O}_T$.

- (11) **Use the stated construction.** The identity of $\mathcal{O}_T$ gives the unit map.
- (12) **Use pullback stability.** These pairings are natural under affine pullback because quasi-coherent pullback is strong symmetric monoidal, and their associativity, symmetry, and unit homotopies are precisely those of the symmetric monoidal coefficient categories.
- (13) **Construct the indicated map.** On the affine-point site they therefore define a map from the objectwise smash-product presheaf to the target sheaf; by the universal property of spectral Zariski sheafification this factors canonically through the tensor product in the sheaf category.
- (14) **Construct the indicated map.** The same construction for all arities and all operadic structure maps assembles to the displayed lax symmetric monoidal structure on $\mathrm{Real}_S^{\mathrm{st}}$.

□

Corollary 4.4.2 (Section-stack recovery from stable realization). *For every $n \geq 0$, there is a canonical equivalence over S*

$$\Omega^\infty(\mathrm{Real}_S^{\mathrm{st}}(\mathcal{F})[n]) \xrightarrow{\sim} \mathrm{Sect}_S(\mathcal{F}[n]).$$

Thus the shifted section stacks are the successive spaces of the spectrum object $\mathrm{Real}_S^{\mathrm{st}}(\mathcal{F})$.

Proof.

- (1) **Construct the indicated map.** On an affine point $a : T \rightarrow S$, both sides have fibre

$$\mathrm{Map}_{\mathrm{QCoh}(T)}(\mathcal{O}_T, a^* \mathcal{F}[n]).$$

- (2) **Establish the indicated equivalence.** The affine-point presentation and Yoneda give the equivalence, with the loop compatibilities inherited from suspension in the same mapping spectra.

□

4.4.4 The stable additive scalar

Definition 4.4.2 (Stable additive scalar). The tensor unit $\mathcal{O}_S \in \mathrm{QCoh}(S)$ carries its canonical commutative algebra structure. Applying the lax symmetric monoidal functor of Proposition 4.4.8 defines

$$\mathcal{O}_S^{\mathrm{add}} := \mathrm{Real}_S^{\mathrm{st}}(\mathcal{O}_S) \in \mathrm{CAlg}(\mathcal{D}_{\mathrm{Sp}/S}).$$

Equivalently, the scalar is obtained by applying stable realization to the tensor unit:

$$\mathcal{O}_S \xrightarrow{\text{Real}_S^{\text{st}}} \mathcal{O}_S^{\text{add}}.$$

On an affine point $a : T = \text{Spec } A \rightarrow S$, its value is canonically the E_∞ -ring spectrum

$$\Gamma(T, \mathcal{O}_T) \simeq A,$$

and its multiplication is the multiplication of this affine coordinate E_∞ -ring. The resulting coherences are inherited functorially from the lax symmetric monoidal realization.

Proposition 4.4.9 (Existence and base change of the stable additive scalar). *For every $f : T \rightarrow S$, there is a canonical equivalence of commutative algebra objects*

$$f^* \mathcal{O}_S^{\text{add}} \xrightarrow{\simeq} \mathcal{O}_T^{\text{add}},$$

with functoriality supplied by the standing coherence convention.

Proof.

- (1) **Invoke the cited result.** By Proposition 4.4.8 and pullback of the structure sheaf from Chapter 1,

$$\begin{aligned} f^* \mathcal{O}_S^{\text{add}} &= f^* \text{Real}_S^{\text{st}}(\mathcal{O}_S) \\ &\simeq \text{Real}_T^{\text{st}}(f^* \mathcal{O}_S) \\ &\simeq \text{Real}_T^{\text{st}}(\mathcal{O}_T) = \mathcal{O}_T^{\text{add}}. \end{aligned}$$

- (2) **Establish the indicated equivalence.** The base-change equivalence is compatible with the lax symmetric monoidal structure of Real^{st} , so it is multiplicative.
- (3) **Use pullback stability.** Coherence is inherited from functorial quasi-coherent pullback.

□

Proposition 4.4.10 (Recognition by the brave new affine line). *If S is represented by a spectral scheme, the grouplike commutative group object underlying $\mathcal{O}_S^{\text{add}}$ is canonically the additive group object underlying the brave new affine line*

$$\mathbb{A}_S^1 = \text{Spec}_S \text{Sym}_{\mathcal{O}_S}(\mathcal{O}_S).$$

Equivalently,

$$\Omega^\infty \mathcal{O}_S^{\text{add}} \xrightarrow{\simeq} (\mathbb{A}_S^1, +)$$

in the slice over S .

Proof.

- (1) **Identify the stated objects.** Corollary 4.4.2 identifies $\Omega^\infty \mathcal{O}_S^{\text{add}}$ with $\text{Sect}_S(\mathcal{O}_S)$.
- (2) **Identify the stated objects.** The structure sheaf is finite locally free of rank one, so 1.13.7 identifies this section stack with

$$\mathbf{V}_S(\mathcal{O}_S) \simeq \mathbb{L}_S(\mathcal{O}_S) = \mathbb{A}_S^1,$$

where the middle equivalence is the canonical self-duality of the rank-one free module.

- (3) **Identify the stated objects.** The affine-point identification is compatible with addition and multiplication because both operations are induced by those of the affine coordinate E_∞ -rings.

□

Proposition 4.4.11 (Stable scalar sections). *For every morphism $y : Y \rightarrow S$ in $\mathcal{H}_{\text{Sp}}$, there is a canonical natural equivalence of E_∞ -ring spectra*

$$\Gamma^{\text{st}}(Y, y^* \mathcal{O}_S^{\text{add}}) \xrightarrow{\sim} \Gamma(Y, \mathcal{O}_Y).$$

Proof.

- (1) **Apply the cited proposition.** Proposition 4.4.9 identifies

$$y^* \mathcal{O}_S^{\text{add}} \simeq \mathcal{O}_Y^{\text{add}} = \text{Real}_Y^{\text{st}}(\mathcal{O}_Y).$$

- (2) **Apply the cited proposition.** Proposition 4.4.8 gives the underlying spectrum equivalence.

- (3) **Use the stated construction.** Its lax symmetric monoidal construction is defined pointwise by the same multiplication of structure-sheaf sections that gives the E_∞ -ring structure on $\Gamma(Y, \mathcal{O}_Y)$; hence the equivalence is multiplicative.

□

The stable infrastructure now reduces to the promised interface. For every map $a : T \rightarrow S$, stable scalar functions may be written either as $\Pi_a a^* \mathcal{O}_S^{\text{add}}$ or as the corresponding internal function object; these descriptions are multiplicative, commute with base change, and satisfy effective descent. Nothing else from the stable-slice construction is needed to define de Rham cochains. The next section applies this interface to the effective relation and returns to the main argument.

4.5 De Rham cochains and effective infinitesimal descent

We now have both ingredients required for de Rham descent: the effective relation groupoid $\mathcal{R}_{X/S, \bullet}^{\text{dR}}$, and the stable additive scalar $\mathcal{O}_S^{\text{add}}$. The next construction is therefore forced. We take stable scalar functions degreewise on the relation and use effective descent along $e_{X/S} : X \rightarrow Q_{X/S}$ to identify their totalization.

The geometric and stable descriptions are related by a descent square:

$$\begin{array}{ccc} \mathcal{R}_{X/S, \bullet}^{\text{dR stable scalar functions}} & \xrightarrow{\quad} & \mathcal{C}_{\text{dR}}^{\bullet}(X/S) \\ \downarrow | - | & & \downarrow \text{Tot} \\ Q_{X/S} & \xrightarrow{\text{stable scalar functions}} & \text{DR}_{X/S}. \end{array}$$

The Čech–Amitsur comparison and effective descent identify the lower horizontal object with the totalization of the upper-right cosimplicial algebra. This is also the point at which the distinction from the full reflected object becomes algebraic: functions on $X_{\text{dR}/S}$ restrict to functions on $Q_{X/S}$, and the comparison is an equivalence precisely on the effectivity domain.

4.5.1 Stable scalar functions on the de Rham relation

We now apply the stable scalar interface degreeewise to the effective relation. For each simplex, dependent product along its structure map to S turns the pulled-back additive scalar into a scalar-function object on the base. The simplicial restriction maps then become the cofaces and codegeneracies of the desired cosimplicial algebra.

Let $a_n : \mathcal{R}_{X/S,n}^{\text{dR}} \rightarrow S$ be the structure map.

Each cochain degree is attached to the triangle

$$\begin{array}{ccc} \mathcal{R}_{X/S,n}^{\text{dR}} & \xrightarrow{a_n} & S \\ & \searrow v_n & \nearrow p \\ & X & \end{array}$$

whenever $p : X \rightarrow S$ denotes the structure morphism.

Definition 4.5.1 (Geometric de Rham cochains). Define

$$C_{\text{dR}}^n(X/S) := \Pi_{a_n} a_n^* \mathcal{O}_S^{\text{add}} \in \text{CAlg}(\mathcal{D}_{\text{Sp}/S}).$$

The simplicial operators of $\mathcal{R}_{X/S,\bullet}^{\text{dR}}$ induce a cosimplicial commutative algebra

$$C_{\text{dR}}^\bullet(X/S) : \Delta \longrightarrow \text{CAlg}(\mathcal{D}_{\text{Sp}/S}).$$

The passage from de Rham relation simplices to cochains is therefore

$$\mathcal{R}_{X/S,n}^{\text{dR}} \xrightarrow{\Sigma_{+,S}^\infty} \mathcal{R}_n^{\text{dR,st}}(X/S) \xrightarrow{\text{Hom}(-, \mathcal{O}_S^{\text{add}})} C_{\text{dR}}^n(X/S).$$

This de Rham relation-simplex formula is the definition; the Amitsur description below is a presentation theorem.

Proposition 4.5.1 (Cosimplicial structure). *If $\theta : [m] \rightarrow [n]$ is a morphism of Δ , with induced simplicial map $\theta^* : \mathcal{R}_{X/S,n}^{\text{dR}} \rightarrow \mathcal{R}_{X/S,m}^{\text{dR}}$, the cosimplicial structure map*

$$C_{\text{dR}}^m(X/S) \longrightarrow C_{\text{dR}}^n(X/S)$$

is the scalar restriction map obtained from the unit of $(\theta^)^* \dashv \Pi_{\theta^*}$ and composition of dependent products. These maps are morphisms of commutative algebras and satisfy the cosimplicial identities.*

Proof.

- (1) **Establish the next claim.** The triangle

$$\mathcal{R}_{X/S,n}^{\text{dR}} \xrightarrow{\theta^*} \mathcal{R}_{X/S,m}^{\text{dR}} \xrightarrow{a_m} S$$

satisfies $a_n = a_m \theta^*$.

- (2) **Use pullback stability.** Pullback is strong symmetric monoidal, dependent product is lax symmetric monoidal, and composition of dependent products is coherent.
- (3) **Use the stated construction.** The adjunction unit gives the stated restriction map.

(4) **Apply functoriality.** Functoriality of the units and mates gives the cosimplicial identities. $\square$

Construction 4.5.1 (Scalar restriction along a de Rham relation simplex). Let

$$r : \mathcal{R}_{X/S,n}^{\text{dR}} \longrightarrow \mathcal{R}_{X/S,m}^{\text{dR}}$$

be any simplicial structure map over S , so $a_n = a_m r$. Define the corresponding scalar-restriction morphism

$$\text{res}_r : C_{\text{dR}}^m(X/S) \longrightarrow C_{\text{dR}}^m(X/S)$$

to be the composite

$$\begin{aligned} \Pi_{a_m} a_m^* \mathcal{O}_S^{\text{add}} &\longrightarrow \Pi_{a_m} \Pi_r r^* a_m^* \mathcal{O}_S^{\text{add}} \\ &\simeq \Pi_{a_n} a_n^* \mathcal{O}_S^{\text{add}}, \end{aligned}$$

where the first arrow is induced by the unit $\text{id} \rightarrow \Pi_r r^*$. These are the cosimplicial structure maps of Proposition 4.5.1.

Convention 4.5.1 (Ordered degree-one faces and signs). In degree one write

$$\mathcal{R}_{X/S,1}^{\text{dR}} = X \times_{X_{\text{dR}/S}} X,$$

and order the two geometric faces by

$$\partial_0 = \mathbf{t} = \text{pr}_2, \quad \partial_1 = \mathbf{s} = \text{pr}_1.$$

The corresponding cofaces are therefore the typed scalar-restriction maps

$$d^0 = \text{res}_{\mathbf{t}}, \quad d^1 = \text{res}_{\mathbf{s}}.$$

For normalized cosimplicial cochains we use the cohomological convention

$$d_N^q = \sum_{i=0}^{q+1} (-1)^i d^i.$$

Thus the degree-zero alternating coface difference is

$$d^0 - d^1 = \text{res}_{\mathbf{t}} - \text{res}_{\mathbf{s}}.$$

The stable Dold–Kan comparison and the complete-filtered/coherent-cochain correspondence below are read with this convention.

Construction 4.5.2 (Suspension spectra of de Rham relation simplices). Put

$$\mathbf{R}_n^{\text{dR,st}}(X/S) := \Sigma_{+,S}^\infty \mathcal{R}_{X/S,n}^{\text{dR}} \in \mathcal{D}_{\text{Sp}/S}.$$

The de Rham relation groupoid makes $\mathbf{R}_\bullet^{\text{dR,st}}(X/S)$ a simplicial stable object. The diagonal of each de Rham relation degree induces its canonical cocommutative comultiplication.

Theorem 4.5.1 (Internal-function presentation). *For every $n \geq 0$, there is a canonical multiplicative equivalence*

$$C_{\text{dR}}^n(X/S) \xrightarrow{\simeq} \underline{\text{Hom}}_{\mathcal{D}_{\text{Sp}/S}}(\mathbf{R}_n^{\text{dR,st}}(X/S), \mathcal{O}_S^{\text{add}}).$$

The equivalences assemble into an equivalence of cosimplicial commutative algebras. Multiplication on the function object is convolution against the diagonal of $\mathbf{R}_n^{\text{dR,st}}(X/S)$.

Proof.

- (1) **Apply the cited proposition.** Proposition 4.4.6, applied to a_n and $A = \mathcal{O}_S^{\text{add}}$, gives the displayed equivalence directly in $\text{CAlg}(\mathcal{D}_{\text{Sp}/S})$.
- (2) **Establish multiplicative compatibility.** The convolution multiplication is induced by the diagonal of $\mathcal{R}_{X/S,n}^{\text{dR}}$.
- (3) **Apply naturality.** Naturality under simplicial operators follows from Beck–Chevalley, functoriality of the diagonals, and the mate identities.
- (4) **Identify the stated objects.** These compatibilities identify the degreewise equivalences as an equivalence of cosimplicial commutative algebras.

□

Remark 4.5.1 (Cochains as stable scalar functions on the de Rham relation). Under the internal-function presentation, a degree- n de Rham cochain is a stable scalar function on an n -simplex of the de Rham relation. The cofaces and codegeneracies are therefore restriction of scalar functions along the structural maps of that relation groupoid. The totalization filtration of the resulting descent object supplies the de Rham differential in the next section.

Corollary 4.5.1 (Mapping-spectrum formula). *For every n ,*

$$\Gamma^{\text{st}}(S, C_{\text{dR}}^n(X/S)) \simeq \text{map}_{\mathcal{D}_{\text{Sp}/S}}(\mathcal{R}_n^{\text{dR, st}}(X/S), \mathcal{O}_S^{\text{add}})$$

and equivalently

$$\Gamma^{\text{st}}(S, C_{\text{dR}}^n(X/S)) \simeq \Gamma^{\text{st}}(\mathcal{R}_{X/S,n}^{\text{dR}}, a_n^* \mathcal{O}_S^{\text{add}}).$$

Proposition 4.5.2 (Represented function-spectrum formula). *If $\mathcal{R}_{X/S,n}^{\text{dR}}$ is represented by a spectral scheme Y_n , then*

$$\Gamma^{\text{st}}(S, C_{\text{dR}}^n(X/S)) \simeq \Gamma(Y_n, \mathcal{O}_{Y_n})$$

canonically as E_∞ -ring spectra, compatibly with every represented face and degeneracy map.

Proof.

- (1) **Combine the preceding comparisons.** Combine the mapping-spectrum formula with Proposition 4.4.11.

□

4.5.2 Effective descent and the Čech–Amitsur comparison

The cosimplicial object has been defined geometrically. To prove that it really computes descent, we now use the effective epimorphism $e : X \rightarrow Q_{X/S}$. Its pullback–dependent-product adjunction produces both the descent comonad and the Amitsur monad; Beck–Chevalley identifies their iterates with the Čech relation degrees.

Put $Q := Q_{X/S}$, $e := e_{X/S} : X \rightarrow Q$, and let $p_Q : Q \rightarrow S$. The effective epimorphism e determines the adjunction

$$e^* : \mathcal{D}_{\text{Sp}/Q} \rightleftarrows \mathcal{D}_{\text{Sp}/X} : \Pi_e.$$

We display the adjunction together with its induced monad and comonad as

$$\begin{array}{ccc} \mathbb{T}_{X/S}^{\mathrm{dR}} & \xrightarrow{e^*} & \mathcal{D}_{\mathrm{Sp}/X}^{\mathbb{G}_{X/S}^{\mathrm{dR}}} \\ \mathcal{D}_{\mathrm{Sp}/Q}^{e^*} & \xleftarrow{\Pi_e} & \mathcal{D}_{\mathrm{Sp}/X}^{e^* \Pi_e} \end{array}$$

Two canonical endofunctors arise from this single adjunction. The monad $\Pi_e e^*$ on $\mathcal{D}_{\mathrm{Sp}/Q}$ produces the Amitsur cosimplicial object used for scalar functions, while the comonad $e^* \Pi_e$ on $\mathcal{D}_{\mathrm{Sp}/X}$ encodes the actual effective-descent coalgebra structure.

Definition 4.5.2 (Stable de Rham Amitsur monad and descent comonad). Define

$$\mathbb{T}_{X/S}^{\mathrm{dR}} := \Pi_e e^* : \mathcal{D}_{\mathrm{Sp}/Q} \longrightarrow \mathcal{D}_{\mathrm{Sp}/Q}$$

and

$$\mathbb{G}_{X/S}^{\mathrm{dR}} := e^* \Pi_e : \mathcal{D}_{\mathrm{Sp}/X} \longrightarrow \mathcal{D}_{\mathrm{Sp}/X}.$$

The adjunction $e^* \dashv \Pi_e$ supplies the monad structure on $\mathbb{T}_{X/S}^{\mathrm{dR}}$ and the comonad structure on $\mathbb{G}_{X/S}^{\mathrm{dR}}$. Since e^* is strong symmetric monoidal, Π_e is canonically lax symmetric monoidal, so the Amitsur monad acts functorially on commutative algebra objects. The adjunction $e^* \dashv \Pi_e$ supplies both structures functorially.

Theorem 4.5.2 (Comonadic effective descent). *Pullback along e is comonadic. More precisely, there is a canonical equivalence*

$$\mathcal{D}_{\mathrm{Sp}/Q} \xrightarrow{\simeq} \mathrm{Coalg}_{\mathbb{G}_{X/S}^{\mathrm{dR}}}(\mathcal{D}_{\mathrm{Sp}/X}),$$

under which an object over Q is recovered from its pullback to X together with the descent coalgebra structure induced by the Čech relation.

Proof.

(1) **Invoke the cited result.** By Proposition 4.4.7,

$$\mathcal{D}_{\mathrm{Sp}/Q} \xrightarrow{\simeq} \mathrm{Tot}_{[n] \in \Delta} \mathcal{D}_{\mathrm{Sp}/\check{C}_n(e)}.$$

(2) **Treat the stated case.** If a morphism becomes an equivalence after pullback along e , then every further pullback to every Čech degree is an equivalence.

(3) **Pass to totalization.** The totalization equivalence therefore makes the original morphism an equivalence.

(4) **Conclude the stated claim.** Hence e^* is conservative.

(5) **Establish the next claim.** The functor e^* has the left adjoint Σ_e , so it preserves all limits.

(6) **Establish the next claim.** In particular it preserves the totalizations required by the dual Barr–Beck theorem.

(7) **Use the stated construction.** Applying the comonadic form of Barr–Beck to $e^* \dashv \Pi_e$ gives the displayed equivalence; compare [34, Theorem 4.7.3.5].

□

Remark 4.5.2 (Logical reading of effective descent). The equivalence

$$\mathcal{D}_{\mathrm{Sp}/Q} \simeq \mathrm{Coalg}_{\mathbb{G}_{X/S}^{\mathrm{dR}}}(\mathcal{D}_{\mathrm{Sp}/X})$$

identifies a stable predicate on the effective quotient with a stable predicate on X equipped with the coherent transport data required along the de Rham relation. The associated Amitsur object is the corresponding cosimplicial form of that transport data, generated by the coalgebra structure of the adjunction.

Theorem 4.5.3 (Čech–Amitsur comparison). *Let $e_n : \mathcal{R}_{X/S,n}^{\mathrm{dR}} \rightarrow Q$ be the structural map of the Čech nerve of e . For every $A \in \mathcal{D}_{\mathrm{Sp}/Q}$, there are canonical equivalences*

$$(\mathbb{T}_{X/S}^{\mathrm{dR}})^{n+1}(A) \xrightarrow{\sim} \Pi_{e_n} e_n^* A$$

which assemble into an equivalence between the Amitsur cosimplicial object of the monad $\mathbb{T}_{X/S}^{\mathrm{dR}}$ and the stable Čech cosimplicial object. Degreeewise, the comparison has the form

$$(\mathbb{T}_{X/S}^{\mathrm{dR}})^{n+1}(A) \xrightarrow{\sim} \Pi_{e_n} e_n^* A.$$

If A is a commutative algebra, this is an equivalence of cosimplicial commutative algebras. The monad unit gives the cofaces and the monad multiplication gives the codegeneracies.

Proof.

- (1) **Apply the cited theorem.** Theorem 4.3.6 identifies $\mathcal{R}_{X/S,\bullet}^{\mathrm{dR}}$ with the Čech nerve of e .
- (2) **Establish the next claim.** For the first nontrivial iterate, right Beck–Chevalley for

$$\begin{array}{ccc} X \times_Q X & \xrightarrow{\mathrm{pr}_2} & X \\ \mathrm{pr}_1 \downarrow & & \downarrow e \\ X & \xrightarrow{e} & Q \end{array}$$

gives

$$e^* \Pi_e \simeq \Pi_{\mathrm{pr}_1} \mathrm{pr}_2^*.$$

- (3) **Use the stated construction.** Substitution into $(\Pi_e e^*)^2$ and composition of dependent products gives

$$(\mathbb{T}_{X/S}^{\mathrm{dR}})^2 A \simeq \Pi_{e_1} e_1^* A.$$

- (4) **Establish the next claim.** Iterating the same Cartesian comparison adds one Čech factor at each stage.
- (5) **Apply naturality.** Naturality of adjunction units, counits, and their mates identifies the cofaces and codegeneracies and supplies all cosimplicial coherences.

□

Corollary 4.5.2 (Čech–Amitsur totalization). *For every $A \in \mathcal{D}_{\mathrm{Sp}/Q}$, the canonical augmentation is a totalization equivalence*

$$A \xrightarrow{\sim} \mathrm{Tot} \, \mathrm{Am}_{\mathbb{T}_{X/S}^{\mathrm{dR}}}^\bullet(A).$$

Proof.

- (1) **Invoke the cited result.** By Theorem 4.5.3, the Amitsur object is the stable Čech object

$$[n] \mapsto \Pi_{e_n} e_n^* A.$$

- (2) **Apply the cited proposition.** Proposition 4.4.7 identifies its totalization with A .
 (3) **Pass to totalization.** This is the Čech totalization statement; the categorical descent theorem is the comonadic equivalence of Theorem 4.5.2.

□

Theorem 4.5.4 (Amitsur presentation of geometric de Rham cochains). *By Proposition 4.4.9,*

$$p_Q^* \mathcal{O}_S^{\text{add}} \xrightarrow{\sim} \mathcal{O}_Q^{\text{add}}$$

canonically in $\text{CAlg}(\mathcal{D}_{\text{Sp}/Q})$. There is a canonical equivalence of cosimplicial commutative algebras

$$C_{\text{dR}}^\bullet(X/S) \xrightarrow{\sim} \Pi_{p_Q} \text{Am}_{\mathbb{T}_{X/S}^{\text{dR}}}^\bullet(\mathcal{O}_Q^{\text{add}}).$$

Proof.

- (1) **Apply the cited theorem.** Theorem 4.5.3 gives

$$(\mathbb{T}_{X/S}^{\text{dR}})^{n+1}(\mathcal{O}_Q^{\text{add}}) \simeq \Pi_{e_n} e_n^* p_Q^* \mathcal{O}_S^{\text{add}}.$$

- (2) **Apply the indicated result.** Apply Π_{p_Q} and use $a_n = p_Q e_n$.
 (3) **Use pullback stability.** The strong symmetric monoidal structure on pullback and the canonical lax symmetric monoidal structure on dependent product make the comparison multiplicative in every degree, while the mate identities make it cosimplicial.

□

4.5.3 Effective and full de Rham algebras

Effective descent determines functions on $Q_{X/S}$, not automatically on the whole reflected object. We therefore name the two function algebras separately before comparing them. This keeps the effectivity distinction visible at the coefficient level.

Let

$$p_{\text{dR}} : X_{\text{dR}/S} \rightarrow S, \quad p_Q : Q_{X/S} \rightarrow S.$$

Definition 4.5.3 (Full and effective de Rham algebras). Define

$$\text{DR}_{X/S}^{\text{full}} := \Pi_{p_{\text{dR}}} p_{\text{dR}}^* \mathcal{O}_S^{\text{add}},$$

and define the effective de Rham algebra by

$$\text{DR}_{X/S} := \text{Tot } C_{\text{dR}}^\bullet(X/S).$$

Both are commutative algebra objects of $\mathcal{D}_{\text{Sp}/S}$.

Theorem 4.5.5 (Effective de Rham algebra). *There are canonical multiplicative equivalences*

$$\mathrm{DR}_{X/S} \simeq \Pi_{p_Q} p_Q^* \mathcal{O}_S^{\mathrm{add}}$$

and

$$\mathrm{DR}_{X/S} \simeq \underline{\mathrm{Hom}}_{\mathcal{D}_{\mathrm{Sp}/S}}(\Sigma_{+,S}^\infty Q_{X/S}, \mathcal{O}_S^{\mathrm{add}}).$$

Likewise

$$\mathrm{DR}_{X/S}^{\mathrm{full}} \simeq \underline{\mathrm{Hom}}_{\mathcal{D}_{\mathrm{Sp}/S}}(\Sigma_{+,S}^\infty X_{\mathrm{dR}/S}, \mathcal{O}_S^{\mathrm{add}}).$$

The monomorphism $m_{X/S} : Q_{X/S} \rightarrow X_{\mathrm{dR}/S}$ induces a canonical morphism of commutative algebras

$$\rho_{X/S} : \mathrm{DR}_{X/S}^{\mathrm{full}} \longrightarrow \mathrm{DR}_{X/S}.$$

It is induced contravariantly by the geometric inclusion

$$Q_{X/S} \xhookrightarrow{m_{X/S}} X_{\mathrm{dR}/S}$$

$$\mathrm{DR}_{X/S} \xleftarrow{\rho_{X/S}} \mathrm{DR}_{X/S}^{\mathrm{full}}.$$

The morphism $\rho_{X/S}$ is an equivalence whenever $m_{X/S}$ is an equivalence; in particular this holds on the formally smooth effectivity domain.

Proof.

- (1) **Construct the indicated map.** The de Rham relation groupoid is the Čech nerve of the effective epimorphism $e_{X/S} : X \rightarrow Q_{X/S}$.
- (2) **Use the stated construction.** Stable effective descent applied to $p_Q^* \mathcal{O}_S^{\mathrm{add}}$ gives

$$p_Q^* \mathcal{O}_S^{\mathrm{add}} \xrightarrow{\sim} \mathrm{Tot}_n \Pi_{e_n} e_n^* p_Q^* \mathcal{O}_S^{\mathrm{add}}.$$

- (3) **Apply the indicated result.** Apply Π_{p_Q} and use composition of dependent products to obtain the first formula.
- (4) **Apply the indicated result.** Apply Proposition 4.4.4 directly to $p_Q : Q_{X/S} \rightarrow S$ to obtain

$$\Pi_{p_Q} p_Q^* \mathcal{O}_S^{\mathrm{add}} \simeq \underline{\mathrm{Hom}}_{\mathcal{D}_{\mathrm{Sp}/S}}(\Sigma_{+,S}^\infty Q_{X/S}, \mathcal{O}_S^{\mathrm{add}}).$$

- (5) **Establish the next claim.** The full formula is the same proposition applied to p_{dR} .
- (6) **Use the stated construction.** Restriction along $m_{X/S}$ gives ρ , and its multiplicativity follows from naturality of the diagonal comultiplications.
- (7) **Treat the stated case.** If $m_{X/S}$ is an equivalence, so is the induced restriction map.

□

Proposition 4.5.3 (Augmentation to degree zero). *If $p : X \rightarrow S$, evaluation of the totalization cone in cosimplicial degree zero defines a canonical morphism*

$$\varepsilon_{\mathrm{dR}} : \mathrm{DR}_{X/S} \longrightarrow \Pi_p p^* \mathcal{O}_S^{\mathrm{add}}.$$

Under the effective-quotient description it is Π_{p_Q} applied to the monad unit

$$p_Q^* \mathcal{O}_S^{\mathrm{add}} \longrightarrow \Pi_e e^* p_Q^* \mathcal{O}_S^{\mathrm{add}}.$$

The effective descent problem has now been solved at the level of scalar functions. The total object $\mathrm{DR}_{X/S}$ is the algebra of stable scalar functions on $Q_{X/S}$, while $\mathrm{DR}_{X/S}^{\mathrm{full}}$ records functions on the full relative de Rham reflection. What remains is to resolve the totalization itself. Its partial totalizations and normalized layers will expose the successive coherence conditions of infinitesimal descent and thereby produce forms and the de Rham differential.

4.6 Totalization: forms, differential, products, and closedness

The previous section identified $\mathrm{DR}_{X/S}$ as the totalization of a cosimplicial stable scalar algebra. The remaining question is how the coherent descent conditions contained in that totalization should be read degree by degree. Stable Dold–Kan supplies the answer: normalization isolates the new datum appearing in each cosimplicial degree, while the partial-totalization tower records the successive extension conditions that assemble those data into a coherent descent object.

The two descriptions are the layer and tower views of the same object:

$$\begin{array}{ccc} C_{\mathrm{dR}}^\bullet(X/S) & \xrightarrow{\mathrm{Tot}} & \mathrm{DR}_{X/S} \\ \downarrow N^q & & \downarrow \mathrm{Gr}_{\mathrm{Tot}}^q[-] \\ \Omega_{\mathrm{dR}}^q(X/S) & \xrightarrow{\simeq} & \mathrm{Gr}_{\mathrm{Tot}}^q \mathrm{DR}_{X/S}[q]. \end{array}$$

The connecting maps between adjacent layers give the homotopy-coherent de Rham differential. Because the original cochains form a cosimplicial commutative algebra, the same construction also supplies products and the Leibniz rule. Keeping the full filtration stage rather than only its associated graded records closedness, exact-form lifts, and the higher coherences carried by a closed form.

4.6.1 Totalization layers and intrinsic de Rham forms

For each degree q , the partial totalization through degree $q - 1$ records all descent conditions already imposed below that degree. Its fibre inside the complete totalization measures the remaining coherence, while the normalized degree- q term is the next associated-graded layer. We now make this tower explicit.

For $q \geq 0$, let $\Delta_{< q} \subset \Delta$ be the full subcategory on $[0], \dots, [q - 1]$, and define

$$\mathrm{DR}_{X/S}^{< q} := \lim_{[n] \in \Delta_{< q}} C_{\mathrm{dR}}^n(X/S).$$

For $q = 0$, this is the zero object in the underlying stable category. Define the decreasing totalization filtration

$$F_{\mathrm{Tot}}^q \mathrm{DR}_{X/S} := \mathrm{fib}(\mathrm{DR}_{X/S} \longrightarrow \mathrm{DR}_{X/S}^{< q}).$$

The first stages form the inverse tower

$$\mathrm{DR}_{X/S} \longrightarrow \cdots \longrightarrow \mathrm{DR}_{X/S}^{<q+1} \longrightarrow \mathrm{DR}_{X/S}^{<q} \longrightarrow \cdots \longrightarrow \mathrm{DR}_{X/S}^{<1},$$

whose fibre tower is $F_{\mathrm{Tot}}^{\bullet} \mathrm{DR}_{X/S}$.

Definition 4.6.1 (Intrinsic de Rham forms). For $q \geq 0$, define $\Omega_{\mathrm{dR}}^q(X/S)$ to be the stable Dold–Kan normalization of the cosimplicial object $C_{\mathrm{dR}}^{\bullet}(X/S)$ in degree q . Equivalently, Ω_{dR}^q is the total fibre of the finite codegeneracy matching cube formed from all proper surjections $[q] \twoheadrightarrow [r]$.

Proposition 4.6.1 (Explicit normalization). *There is a canonical equivalence*

$$\Omega_{\mathrm{dR}}^q(X/S) \simeq \mathrm{tfib}\left(C_{\mathrm{dR}}^q(X/S) \longrightarrow \{C_{\mathrm{dR}}^r(X/S)\}_{[q] \twoheadrightarrow [r], r < q}\right).$$

In particular,

$$\Omega_{\mathrm{dR}}^0(X/S) = C_{\mathrm{dR}}^0(X/S)$$

and

$$\Omega_{\mathrm{dR}}^1(X/S) \simeq \mathrm{fib}(C_{\mathrm{dR}}^1(X/S) \rightarrow C_{\mathrm{dR}}^0(X/S)),$$

where the latter map is induced by the diagonal $X \rightarrow \mathcal{R}_{X/S,1}^{\mathrm{dR}}$. The degree-one normalization is displayed by the fibre square

$$\begin{array}{ccc} \Omega_{\mathrm{dR}}^1(X/S) & \longrightarrow & C_{\mathrm{dR}}^1(X/S) \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & C_{\mathrm{dR}}^0(X/S). \end{array}$$

Proof.

- (1) **Establish the next claim.** This is the matching-object description of stable Dold–Kan normalization; see [34].
- (2) **Identify the relevant fibre.** The matching diagram is finite, so its fibre is equivalently the total fibre of the codegeneracy cube.

□

Theorem 4.6.1 (Stable Dold–Kan layers and completeness). *For every $q \geq 0$, there is a canonical fibre/cofiber sequence*

$$F_{\mathrm{Tot}}^{q+1} \mathrm{DR}_{X/S} \longrightarrow F_{\mathrm{Tot}}^q \mathrm{DR}_{X/S} \longrightarrow \Omega_{\mathrm{dR}}^q(X/S)[-q].$$

Consequently

$$\mathrm{Gr}_{\mathrm{Tot}}^q \mathrm{DR}_{X/S} \simeq \Omega_{\mathrm{dR}}^q(X/S)[-q].$$

The filtration is already complete:

$$\varprojlim_q F_{\mathrm{Tot}}^q \mathrm{DR}_{X/S} \simeq 0,$$

and

$$\mathrm{DR}_{X/S} \xrightarrow{\sim} \varprojlim_q \mathrm{DR}_{X/S}^{<q}.$$

Proof.

- (1) **Invoke the cited result.** By stable Dold–Kan, equivalently the Bousfield–Kan partial-totalization tower of a cosimplicial object in a stable infinity-category, the successive fibres of the partial-totalization tower are the shifted normalized terms; see [34].
- (2) **Pass to totalization.** Passing to the fibre tower from the complete totalization gives the displayed sequence.
- (3) **Invoke the cited result.** By definition, the totalization is the limit of its finite partial totalizations.
- (4) **Take the indicated specialization.** Taking the inverse limit of the defining fibre sequences gives

$$\varprojlim_q F_{\text{Tot}}^q \text{DR}_{X/S} \simeq \text{fib} \left(\text{DR}_{X/S} \longrightarrow \varprojlim_q \text{DR}_{X/S}^{\leq q} \right) \simeq 0.$$

- (5) **Conclude the stated claim.** Thus the totalization filtration is complete. □

Extend the filtration by the constant value $\text{DR}_{X/S}$ in nonpositive filtration degrees. Since $\mathcal{D}_{\text{Sp}/S}$ is presentable and stable, it admits sequential limits. The complete-filtered/coherent-cochain correspondence of [3, Theorem 4.7 and Remark 4.9] therefore applies directly to the preceding complete filtration.

Lemma 4.6.1 (Stable Dold–Kan identification of totalization boundaries). *Let $C^\bullet$ be a cosimplicial object in a stable infinity-category, and use the cohomological normalization*

$$d_N^q = \sum_{i=0}^{q+1} (-1)^i d^i : N^q C \longrightarrow N^{q+1} C.$$

Put

$$\text{Tot}^{< q} C := \lim_{[n] \in \Delta_{< q}} C^n, \quad F^q := \text{fib}(\text{Tot } C \longrightarrow \text{Tot}^{< q} C).$$

Under the canonical stable Dold–Kan layer equivalences

$$\text{cofib}(F^{q+1} \rightarrow F^q) \xrightarrow{\simeq} N^q C[-q],$$

the shifted connecting morphism

$$N^q C \simeq \text{Gr}_F^q[q] \longrightarrow \text{Gr}_F^{q+1}[q+1] \simeq N^{q+1} C$$

is represented in the homotopy category by the normalized alternating coface differential d_N^q . The identification is natural in $C^\bullet$.

Proof.

- (1) **Identify the stated objects.** Stable Dold–Kan identifies a cosimplicial object in a stable ∞ -category, equivalently a simplicial object in the opposite stable ∞ -category, with the filtration obtained from its skeletal tower.
- (2) **Identify the stated objects.** Dualizing the skeletal filtration identifies the successive fibres of the partial-totalization tower with $N^q C[-q]$.

- (3) **Establish the next claim.** Under the comparison with the normalized complex in the homotopy category, the boundary is the cohomological differential

$$\sum_{i=0}^{q+1} (-1)^i d^i.$$

- (4) **Establish the next claim.** This is the compatibility of the stable Dold–Kan filtration differential with normalization; see [34, Section 1.2.4].
- (5) **Apply naturality.** Naturality follows from functoriality of the Dold–Kan construction. □

Theorem 4.6.2 (Homotopy-coherent de Rham differential). *The complete totalization filtration canonically determines a coherent nonnegative cochain object*

$$\Omega_{\mathrm{dR}}^0(X/S) \xrightarrow{d_{\mathrm{dR}}} \Omega_{\mathrm{dR}}^1(X/S) \xrightarrow{d_{\mathrm{dR}}} \Omega_{\mathrm{dR}}^2(X/S) \longrightarrow \cdots.$$

Its first two stages may be read together with the totalization layers as

$$\begin{array}{ccccc} F_{\mathrm{Tot}}^0 & \longleftarrow & F_{\mathrm{Tot}}^1 & \longleftarrow & F_{\mathrm{Tot}}^2 \\ \downarrow & & \downarrow & & \downarrow \\ \Omega_{\mathrm{dR}}^0 & \xrightarrow{d_{\mathrm{dR}}} & \Omega_{\mathrm{dR}}^1 & \xrightarrow{d_{\mathrm{dR}}} & \Omega_{\mathrm{dR}}^2, \end{array}$$

where the vertical arrows denote the associated layer maps with the shifts specified in Theorem 4.6.1. The construction retains specified nullhomotopies

$$d_{\mathrm{dR}}^{q+1} d_{\mathrm{dR}}^q \simeq 0$$

and the full higher coherence supplied by the complete filtration. With the cohomological normalization convention fixed above, the image of d_{dR}^q in the homotopy category is

$$d_{\mathrm{dR}}^q = \sum_{i=0}^{q+1} (-1)^i d^i : \Omega_{\mathrm{dR}}^q(X/S) \longrightarrow \Omega_{\mathrm{dR}}^{q+1}(X/S).$$

Proof.

- (1) **Establish the next claim.** Extend the complete decreasing totalization filtration by the constant value $\mathrm{DR}_{X/S}$ in nonpositive filtration degrees.
- (2) **Establish the next claim.** The complete-filtered/coherent-cochain correspondence of [3, Theorem 4.7 and Remark 4.9] sends this filtration to a coherent nonnegative cochain object whose degree- q term is

$$\mathrm{Gr}_{\mathrm{Tot}}^q \mathrm{DR}_{X/S}[q].$$

- (3) **Invoke the cited result.** By Theorem 4.6.1,

$$\mathrm{Gr}_{\mathrm{Tot}}^q \mathrm{DR}_{X/S}[q] \simeq \Omega_{\mathrm{dR}}^q(X/S).$$

- (4) **Conclude the stated claim.** Thus the terms of the resulting coherent cochain object are precisely the intrinsic de Rham forms.
- (5) **Construct the indicated map.** Under the complete-filtered/coherent-cochain correspondence, the first structure map from degree q to degree $q+1$ is the shifted connecting morphism between the adjacent filtration layers.
- (6) **Invoke the cited result.** By Lemma 4.6.1, its image in the homotopy category is exactly

$$\sum_{i=0}^{q+1} (-1)^i d^i.$$

- (7) **Construct the indicated map.** We denote the corresponding cochain morphism by d_{dR}^q .
- (8) **Record the higher cochain coherence.** The coherent-cochain object supplied by the equivalence contains, as part of the same functorial construction, the specified nullhomotopies

$$d_{\text{dR}}^{q+1} d_{\text{dR}}^q \simeq 0$$

and all higher compatibility among them.

□

Proposition 4.6.2 (Degree-zero de Rham differential). *With the ordered degree-one faces*

$$\partial_0 = \mathbf{t}, \quad \partial_1 = \mathbf{s},$$

the first differential

$$d_{\text{dR}}^0 : \Omega_{\text{dR}}^0(X/S) \longrightarrow \Omega_{\text{dR}}^1(X/S)$$

is induced by the scalar-restriction difference

$$\text{res}_{\mathbf{t}} - \text{res}_{\mathbf{s}} : C_{\text{dR}}^0(X/S) \longrightarrow C_{\text{dR}}^1(X/S).$$

Proof.

- (1) **Invoke the cited result.** By Lemma 4.6.1, the degree-zero homotopy-coherent differential is represented in the homotopy category by the normalized cohomological alternating coface sum

$$d_N^0 = d^0 - d^1.$$

- (2) **Establish the next claim.** Under the ordered degree-one face convention,

$$d^0 = \text{res}_{\mathbf{t}}, \quad d^1 = \text{res}_{\mathbf{s}},$$

so

$$d_N^0 = \text{res}_{\mathbf{t}} - \text{res}_{\mathbf{s}}.$$

- (3) **Establish the next claim.** The degree-one codegeneracy is induced by the diagonal

$$X \longrightarrow \mathcal{R}_{X/S,1}^{\text{dR}}.$$

- (4) **Establish the next claim.** Both projections t and s restrict along the diagonal to the identity of X .
- (5) **Establish the next claim.** Consequently the composite of

$$\text{res}_t - \text{res}_s$$

with the degree-one codegeneracy is canonically null.

- (6) **Identify the relevant fibre.** The stable fibre universal property therefore factors the alternating difference through

$$\Omega_{\text{dR}}^1(X/S) \simeq \text{fib}(C_{\text{dR}}^1(X/S) \longrightarrow C_{\text{dR}}^0(X/S)),$$

and that factor is d_{dR}^0 .

□

Corollary 4.6.1 (Associated-graded detection). *A morphism between two complete totalization filtrations of the form above is an equivalence if it is an equivalence on every normalized associated-graded term.*

Proof.

- (1) **Construct the indicated map.** The fibre of such a morphism is complete.
- (2) **Treat the stated case.** If all associated-graded terms vanish, every transition in the fibre filtration is an equivalence; completeness then forces the constant value to vanish.

□

Remark 4.6.1 (Higher-order information). The totalization filtration is defined from the de Rham relation itself. Its normalized terms can retain finite higher-order infinitesimal information beyond the first structured infinitesimal neighbourhood. Section 4.8 constructs the canonical first-order comparison with the cotangent complex.

Skeletal presentation

Let $\mathbf{R}_{\bullet}^{\text{dR}, \text{st}}(X/S)$ be the simplicial stable de Rham relation object constructed above. Define $\mathbf{R}_{\geq q}^{\text{dR}, \text{st}}(X/S)$ by the cofiber sequence

$$|\text{sk}_{q-1} \mathbf{R}_{\bullet}^{\text{dR}, \text{st}}(X/S)| \longrightarrow |\mathbf{R}_{\bullet}^{\text{dR}, \text{st}}(X/S)| \longrightarrow \mathbf{R}_{\geq q}^{\text{dR}, \text{st}}(X/S).$$

Theorem 4.6.3 (Skeletal totalization presentation). *There are canonical equivalences*

$$\begin{aligned} \text{DR}_{X/S} &\simeq \underline{\text{Hom}}(|\mathbf{R}_{\bullet}^{\text{dR}, \text{st}}(X/S)|, \mathcal{O}_S^{\text{add}}), \\ \text{DR}_{X/S}^{< q} &\simeq \underline{\text{Hom}}(|\text{sk}_{q-1} \mathbf{R}_{\bullet}^{\text{dR}, \text{st}}(X/S)|, \mathcal{O}_S^{\text{add}}), \end{aligned}$$

and

$$F_{\text{Tot}}^q \text{DR}_{X/S} \simeq \underline{\text{Hom}}(\mathbf{R}_{\geq q}^{\text{dR}, \text{st}}(X/S), \mathcal{O}_S^{\text{add}}).$$

They are natural in X/S and compatible with arbitrary base change.

Proof.

- (1) **Establish the next claim.** The degreewise internal-function theorem turns geometric realization in the simplicial source into totalization of the cosimplicial function object: internal Hom into the fixed scalar converts colimits in its first variable into limits.
- (2) **Pass to totalization.** Restricting to the $(q - 1)$ -skeleton gives the partial totalization, and applying internal Hom to the skeletal cofiber sequence gives the fibre sequence defining the filtration stage.
- (3) **Apply naturality.** Naturality in X/S follows degreewise from naturality of the de Rham relation nerve and scalar restriction. Let $c : S' \rightarrow S$.
- (4) **Identify the stated objects.** Relative de Rham base change identifies every relation degree with its pullback to S' , while Proposition 4.4.9 identifies the scalar coefficient.
- (5) **Establish multiplicative compatibility.** The functor c^* has both adjoints and therefore preserves all limits and colimits, so it commutes with geometric realizations, skeleta, and cofibres.
- (6) **Use the stated construction.** The internal-function terms are transported by the closed base-change equivalence of Proposition 4.4.5; its convolution compatibility gives the multiplicative comparison.
- (7) **Conclude the stated claim.** Thus all three displayed function-object formulas commute with arbitrary base change.

□

4.6.2 Multiplication and the normalized Leibniz rule

Every partial totalization and the complete totalization are limits of a cosimplicial commutative algebra and hence carry canonical E_∞ -algebra structures. The normalized terms carry the ordered Alexander–Whitney product used below.

Lemma 4.6.2 (Normalized Alexander–Whitney factorization). *Let $C^\bullet, D^\bullet, E^\bullet$ be cosimplicial objects in a stable symmetric monoidal infinity-category whose tensor product is exact separately in each variable, and let*

$$\mu^\bullet : C^\bullet \otimes D^\bullet \rightarrow E^\bullet$$

be a natural pairing. For $r, s \geq 0$, the consecutive-block map in degree $r + s$ canonically factors to an actual morphism of normalized objects

$$N^r C \otimes N^s D \longrightarrow N^{r+s} E.$$

In the homotopy category it is represented by

$$x \boxtimes_{\text{AW}} y = \mu^{r+s}(d^{r+s} \cdots d^{r+1} x, (d^0)^r y).$$

Proof.

- (1) **Identify the relevant fibre.** For $k \geq 0$, let $K_k(C)$ be the finite codegeneracy cube whose total fibre is $N^k C$.

- (2) **Establish the next claim.** The decomposition of the degeneracy directions

$$\{0, \dots, r+s-1\} = \{0, \dots, r-1\} \amalg \{r, \dots, r+s-1\}$$

produces, by the front and back consecutive-block inclusions, a morphism

$$K_r(C) \boxtimes K_s(D) \longrightarrow K_{r+s}(E).$$

- (3) **Apply naturality.** Naturality of μ supplies all cube compatibilities.
- (4) **Use the stated hypothesis.** Since tensor product is exact separately in each variable, taking the iterated total fibre of the source identifies it canonically with

$$N^r C \otimes N^s D.$$

- (5) **Take the indicated specialization.** Taking total fibres therefore gives the required normalized map.
- (6) **Establish the next claim.** The initial vertex is the displayed consecutive-block formula.

□

Theorem 4.6.4 (Ordered product and Leibniz rule). *For $r, s \geq 0$, the multiplication of $C_{\mathrm{dR}}^\bullet(X/S)$ induces canonical maps*

$$\smile_{\mathrm{AW}}: \Omega_{\mathrm{dR}}^r(X/S) \otimes \Omega_{\mathrm{dR}}^s(X/S) \longrightarrow \Omega_{\mathrm{dR}}^{r+s}(X/S).$$

Their images in the homotopy category are associative and unital, and

$$d_{\mathrm{dR}}(x \smile_{\mathrm{AW}} y) = d_{\mathrm{dR}}x \smile_{\mathrm{AW}} y + (-1)^r x \smile_{\mathrm{AW}} d_{\mathrm{dR}}y$$

there.

Proof.

- (1) **Apply the indicated result.** Apply Lemma 4.6.2 to the multiplication pairing.
- (2) **Establish multiplicative compatibility.** For three inputs, both parenthesizations are induced by multiplication along the same three consecutive vertex blocks, so associativity follows from the cosimplicial identities and associativity of multiplication.
- (3) **Establish the next claim.** The unit is the degree-zero unit.
- (4) **Establish the next claim.** For the differential, expand the alternating coface sum.
- (5) **Establish the next claim.** Cofaces before the break produce $dx \smile y$, those after the break produce $(-1)^r x \smile dy$, and the two contributions at the break agree by the ordinal identities and occur with opposite signs.
- (6) **Establish the next claim.** They cancel, giving the Leibniz formula.

□

Theorem 4.6.5 (Commutative algebra structures on totalizations). *Every $\mathrm{DR}_{X/S}^{<q}$ and $\mathrm{DR}_{X/S}$ carries its canonical commutative algebra structure, the transition maps form a tower in $\mathrm{CAlg}(\mathcal{D}_{\mathrm{Sp}/S})$, and*

$$\mathrm{DR}_{X/S} \xrightarrow{\sim} \varprojlim_q \mathrm{DR}_{X/S}^{<q}$$

is an equivalence of commutative algebras.

Proof.

- (1) **Establish multiplicative compatibility.** Limits of commutative algebra objects are created in the underlying stable category; see [34].
- (2) **Apply the indicated result.** Apply this to the finite and complete totalizations of $C_{\mathrm{dR}}^\bullet(X/S)$.

□

4.6.3 External products

Proposition 4.6.3 (Products of relative de Rham objects and effective images). *For $X, Y \rightarrow S$, there are canonical equivalences*

$$(X \times_S Y)_{\mathrm{dR}/S} \xrightarrow{\sim} X_{\mathrm{dR}/S} \times_S Y_{\mathrm{dR}/S},$$

$$Q_{X \times_S Y/S} \xrightarrow{\sim} Q_{X/S} \times_S Q_{Y/S},$$

and, for every $n \geq 0$,

$$\mathcal{R}_{X \times_S Y/S, n}^{\mathrm{dR}} \xrightarrow{\sim} \mathcal{R}_{X/S, n}^{\mathrm{dR}} \times_S \mathcal{R}_{Y/S, n}^{\mathrm{dR}}.$$

These equivalences are natural in both variables and compatible with iterated products and arbitrary base change.

Proof.

- (1) **Establish the next claim.** The endofunctor dR_S is left exact, so it preserves the binary product in the slice:

$$\mathrm{dR}_S(X \times_S Y) \simeq \mathrm{dR}_S(X) \times_S \mathrm{dR}_S(Y).$$
- (2) **Establish the indicated equivalence.** This is the first equivalence, and under it the unit of $X \times_S Y$ is the product of the two relative units.
- (3) **Introduce the notation.** Put $Q_X := Q_{X/S}$ and $Q_Y := Q_{Y/S}$, and write the two effective-image factorizations as

$$X \twoheadrightarrow Q_X \hookrightarrow X_{\mathrm{dR}/S}, \quad Y \twoheadrightarrow Q_Y \hookrightarrow Y_{\mathrm{dR}/S}.$$

- (4) **Construct the indicated map.** The product

$$X \times_S Y \longrightarrow Q_X \times_S Q_Y$$

is an effective epimorphism: it is the composite of

$$X \times_S Y \longrightarrow Q_X \times_S Y$$

and

$$Q_X \times_S Y \longrightarrow Q_X \times_S Q_Y,$$

each a pullback of one of the two effective epimorphisms.

- (5) **Construct the indicated map.** The product of the two monomorphisms is a monomorphism.
 (6) **Conclude the stated claim.** Thus

$$X \times_S Y \twoheadrightarrow Q_X \times_S Q_Y \hookrightarrow X_{\mathrm{dR}/S} \times_S Y_{\mathrm{dR}/S}$$

is the effective-epimorphism–monomorphism factorization of the product unit.

- (7) **Use the stated construction.** Uniqueness of the factorization gives

$$Q_{X \times_S Y/S} \simeq Q_X \times_S Q_Y.$$

- (8) **Establish the indicated equivalence.** Substitute the first equivalence into the iterated fibre-product definition of the relation nerve.
 (9) **Use the stated construction.** Finite-limit pasting gives, degreewise,

$$\mathcal{R}_{X \times_S Y/S, n}^{\mathrm{dR}} \simeq \mathcal{R}_{X/S, n}^{\mathrm{dR}} \times_S \mathcal{R}_{Y/S, n}^{\mathrm{dR}}.$$

- (10) **Apply naturality.** Naturality follows from finite products, pullback-stable effective-image factorization, and relative de Rham base change.

□

Proposition 4.6.4 (External products). *For $r, s \geq 0$, the scalar multiplication and ordered Alexander–Whitney construction induce natural products*

$$\Omega_{\mathrm{dR}}^r(X/S) \otimes \Omega_{\mathrm{dR}}^s(Y/S) \longrightarrow \Omega_{\mathrm{dR}}^{r+s}(X \times_S Y/S).$$

They are natural in both variables and under arbitrary base change. The normalized differential satisfies the signed Leibniz rule in the homotopy category.

Proof.

- (1) **Establish the next claim.** For each n , strong symmetric monoidality of relative suspension spectra and the product formula for de Rham relation degrees give

$$\mathcal{R}_n^{\mathrm{dR}, \mathrm{st}}(X \times_S Y/S) \simeq \mathcal{R}_n^{\mathrm{dR}, \mathrm{st}}(X/S) \otimes \mathcal{R}_n^{\mathrm{dR}, \mathrm{st}}(Y/S).$$

- (2) **Use the stated construction.** Evaluation followed by multiplication of the stable scalar therefore gives a canonical pairing

$$\begin{aligned} & \underline{\mathrm{Hom}}(\mathcal{R}_n^{\mathrm{dR}, \mathrm{st}}(X/S), \mathcal{O}_S^{\mathrm{add}}) \otimes \underline{\mathrm{Hom}}(\mathcal{R}_n^{\mathrm{dR}, \mathrm{st}}(Y/S), \mathcal{O}_S^{\mathrm{add}}) \\ & \longrightarrow \underline{\mathrm{Hom}}(\mathcal{R}_n^{\mathrm{dR}, \mathrm{st}}(X \times_S Y/S), \mathcal{O}_S^{\mathrm{add}}). \end{aligned}$$

- (3) **Invoke the cited result.** By Theorem 4.5.1, these maps form a natural pairing of cosimplicial de Rham cochain objects.
 (4) **Apply the indicated result.** Apply Theorem 4.6.4 to obtain the normalized product.
 (5) **Apply naturality.** Naturality of the de Rham relation product, scalar multiplication, and the consecutive-block construction gives functoriality and arbitrary base-change compatibility, while the signed Leibniz rule is the one already proved for normalized Alexander–Whitney products.

□

4.6.4 Closed forms, exact forms, and de Rham primitives

A normalized q -form records the q -th associated-graded layer, but closedness requires more: it is a lift of that form to the entire q -th stage of the complete totalization filtration. That lift retains a nullhomotopy of the next boundary together with all higher compatibility data. We package this retained coherence directly as the closed-form object.

Definition 4.6.2 (Closed de Rham forms). For $q \geq 0$ and $n \in \mathbb{Z}$, define the stable object

$$\Omega_{\mathrm{dR}}^{q, \mathrm{closed}}(X/S; n) := F_{\mathrm{Tot}}^q \mathrm{DR}_{X/S}[n + q].$$

The associated-graded quotient gives the underlying-form map

$$u_q : \Omega_{\mathrm{dR}}^{q, \mathrm{closed}}(X/S; n) \longrightarrow \Omega_{\mathrm{dR}}^q(X/S)[n].$$

Remark 4.6.2 (Closed forms and higher lift data). A global closed de Rham form is a point of

$$\Omega^\infty \Gamma^{\mathrm{st}}(S, \Omega_{\mathrm{dR}}^{q, \mathrm{closed}}(X/S; n)).$$

The complete filtration identifies such a point with an underlying de Rham q -form, a nullhomotopy of its boundary, and the higher compatibility data encoded by the associated coherent cochain object.

Definition 4.6.3 (Exact-form lift). For $q \geq 1$, the connecting morphism of

$$F_{\mathrm{Tot}}^q \mathrm{DR} \longrightarrow F_{\mathrm{Tot}}^{q-1} \mathrm{DR} \longrightarrow \Omega_{\mathrm{dR}}^{q-1}[1 - q]$$

defines, after shifting, a canonical morphism

$$d_{\mathrm{dR}}^{\mathrm{closed}} : \Omega_{\mathrm{dR}}^{q-1}(X/S)[n] \longrightarrow \Omega_{\mathrm{dR}}^{q, \mathrm{closed}}(X/S; n).$$

Proposition 4.6.5 (Underlying boundary of the exact-form lift). *The composite*

$$\Omega_{\mathrm{dR}}^{q-1}(X/S)[n] \xrightarrow{d_{\mathrm{dR}}^{\mathrm{closed}}} \Omega_{\mathrm{dR}}^{q, \mathrm{closed}}(X/S; n) \xrightarrow{u_q} \Omega_{\mathrm{dR}}^q(X/S)[n]$$

is the homotopy-coherent de Rham differential d_{dR}^{q-1} . The lift to the closed stage carries the canonical nullhomotopy of the next boundary and its higher coherences.

Proof.

- (1) **Construct the indicated map.** Under the complete-filtered/coherent-cochain correspondence, the shifted connecting morphism between adjacent filtration layers is precisely the cochain differential.
- (2) **Construct the indicated map.** The fact that the connecting map lands in the entire stage $F_{\mathrm{Tot}}^q[q+n]$, before projection to associated graded, is exactly the retained closed-form coherence of an exact form.

□

Definition 4.6.4 (De Rham primitive spaces). Let $q \geq 1$. Fix a global closed de Rham form

$$\omega \in \Omega^\infty \Gamma^{\mathrm{st}}(S, \Omega_{\mathrm{dR}}^{q, \mathrm{closed}}(X/S; n)).$$

Define

$$\mathrm{PrimDiff}_{\mathrm{dR}}^q(X/S; n) := \mathrm{fib}\left(\Omega_{\mathrm{dR}}^{q-1}(X/S)[n] \xrightarrow{d_{\mathrm{dR}}^{\mathrm{closed}}} \Omega_{\mathrm{dR}}^{q, \mathrm{closed}}(X/S; n)\right),$$

and define the space of de Rham primitives of ω by

$$\mathrm{PrimSpace}(\omega) := \mathrm{hofib}_\omega\left(\Omega^\infty \Gamma^{\mathrm{st}}(S, \Omega_{\mathrm{dR}}^{q-1}(X/S)[n]) \longrightarrow \Omega^\infty \Gamma^{\mathrm{st}}(S, \Omega_{\mathrm{dR}}^{q, \mathrm{closed}}(X/S; n))\right).$$

Theorem 4.6.6 (De Rham primitive torsor theorem). *If $\text{PrimSpace}(\omega)$ is nonempty, it is canonically a torsor for the grouplike E_∞ -space*

$$\Omega^\infty \Gamma^{\text{st}}(S, \text{PrimDiff}_{\text{dR}}^q(X/S; n)).$$

A chosen de Rham primitive identifies the torsor with this group object by translation. For the zero closed form, the zero de Rham primitive makes the identification canonical.

Proof.

- (1) **Construct the indicated map.** Stable sections preserve fibres because they are mapping spectra out of the tensor unit.
- (2) **Construct the indicated map.** The exact-form lift therefore induces a morphism of grouplike E_∞ -spaces whose fibre over the identity is the displayed infinite-loop space.
- (3) **Construct the indicated map.** Every nonempty homotopy fibre of a morphism of grouplike spaces is a torsor under the identity fibre.

□

Additive delooping from the skeletal quotient

The additive classifier uses one feature of the stable affine-point presentation that has not yet been needed: its canonical Postnikov structure and the recognition of connective spectrum objects as commutative group objects. We record that feature here, immediately before it becomes active.

Proposition 4.6.6 (Postnikov structure and grouplike recognition). *For every $S \in \mathcal{H}_{\text{Sp}}$, the stable slice $\mathcal{D}_{\text{Sp}/S}$ carries its canonical Postnikov t -structure. Every relative suspension spectrum $\Sigma_{+,S}^\infty U$ is connective, and $(\mathcal{D}_{\text{Sp}/S})_{\geq 0}$ is closed under small colimits and cofibres. Moreover, zeroth space induces a canonical equivalence*

$$(\mathcal{D}_{\text{Sp}/S})_{\geq 0} \xrightarrow{\simeq} \text{CGrp}((\mathcal{H}_{\text{Sp}})_S).$$

These statements take place in the ordinary spectral Zariski ∞ -topos.

Proof.

- (1) **Establish the next claim.** Stabilization of an infinity-topos carries the canonical accessible Postnikov t -structure.
- (2) **Establish the next claim.** Suspension spectra of pointed objects are connective, and the connective aisle is closed under small colimits and extensions, hence under cofibres; see [34, 37].
- (3) **Establish the indicated equivalence.** The equivalence between connective spectrum objects and grouplike commutative monoid objects is natural in finite-limit-preserving functors.
- (4) **Use the stated construction.** Applied to the spectral slice it gives

$$(\mathcal{D}_{\text{Sp}/S})_{\geq 0} \simeq \text{CGrp}((\mathcal{H}_{\text{Sp}})_S).$$

- (5) **Establish the next claim.** This is the claimed grouplike recognition.

□

The stable skeletal formula itself is the primary classifier:

$$\Gamma^{\text{st}}(S, \Omega_{\text{dR}}^{q, \text{closed}}(X/S; n)) \simeq \text{map}_{\mathcal{D}_{\text{Sp}/S}}(\mathbf{R}_{\geq q}^{\text{dR}, \text{st}}(X/S), \mathcal{O}_S^{\text{add}}[n+q]).$$

By Proposition 4.6.6, $\mathcal{D}_{\text{Sp}/S}$ carries its canonical Postnikov t -structure, every relative suspension spectrum $\Sigma_{+, S}^\infty T$ is connective, and the connective subcategory $(\mathcal{D}_{\text{Sp}/S})_{\geq 0}$ is closed under colimits and cofibres. Hence

$$|\text{sk}_{q-1} \mathbf{R}_{\bullet}^{\text{dR}, \text{st}}(X/S)| \quad \text{and} \quad |\mathbf{R}_{\bullet}^{\text{dR}, \text{st}}(X/S)|$$

are connective, and the defining cofiber sequence

$$|\text{sk}_{q-1} \mathbf{R}_{\bullet}^{\text{dR}, \text{st}}(X/S)| \longrightarrow |\mathbf{R}_{\bullet}^{\text{dR}, \text{st}}(X/S)| \longrightarrow \mathbf{R}_{\geq q}^{\text{dR}, \text{st}}(X/S)$$

shows that $\mathbf{R}_{\geq q}^{\text{dR}, \text{st}}(X/S)$ is connective.

Construction 4.6.1 (Additive delooping object). Define the commutative group object

$$\mathbf{R}_{X/S}^{\text{dR}, \langle q \rangle} := \Omega^\infty \mathbf{R}_{\geq q}^{\text{dR}, \text{st}}(X/S) \in \text{CGrp}((\mathcal{H}_{\text{Sp}})_{/S}).$$

This is the grouplike object corresponding to the connective spectrum $\mathbf{R}_{\geq q}^{\text{dR}, \text{st}}(X/S)$.

Theorem 4.6.7 (Additive-delooping classifier). *For every $q \geq 0$ and $n \in \mathbb{Z}$, there is a canonical equivalence of spaces*

$$\begin{aligned} & \Omega^\infty \Gamma^{\text{st}}(S, \Omega_{\text{dR}}^{q, \text{closed}}(X/S; n)) \\ & \simeq \text{Map}_{\text{CGrp}((\mathcal{H}_{\text{Sp}})_{/S})}(\mathbf{R}_{X/S}^{\text{dR}, \langle q \rangle}, \Omega^\infty(\mathcal{O}_S^{\text{add}}[n+q])). \end{aligned}$$

For $q = 0$, the stable source is

$$\mathbf{R}_{\geq 0}^{\text{dR}, \text{st}}(X/S) \simeq \Sigma_{+, S}^\infty Q_{X/S},$$

so the stable formula reduces to additive scalar functions on the effective quotient.

Proof.

- (1) **Establish the next claim.** The stable formula is Theorem 4.6.3.
- (2) **Invoke the cited result.** By the connectivity calculation above, the source $\mathbf{R}_{\geq q}^{\text{dR}, \text{st}}(X/S)$ belongs to $(\mathcal{D}_{\text{Sp}/S})_{\geq 0}$.
- (3) **Conclude the stated claim.** Hence, for every $E \in \mathcal{D}_{\text{Sp}/S}$, the canonical connective-cover map induces an equivalence

$$\text{Map}_{\mathcal{D}_{\text{Sp}/S}}(\mathbf{R}_{\geq q}^{\text{dR}, \text{st}}(X/S), \tau_{\geq 0} E) \xrightarrow{\simeq} \text{Map}_{\mathcal{D}_{\text{Sp}/S}}(\mathbf{R}_{\geq q}^{\text{dR}, \text{st}}(X/S), E),$$

because maps from a connective object to an object of $(\mathcal{D}_{\text{Sp}/S})_{\leq -1}$ are contractible.

- (4) **Apply the cited proposition.** Proposition 4.6.6 identifies $(\mathcal{D}_{\text{Sp}/S})_{\geq 0}$ with the commutative group objects in the slice.
- (5) **Apply the indicated result.** Apply the preceding mapping equivalence with

$$E = \mathcal{O}_S^{\text{add}}[n+q].$$

- (6) **Establish the indicated equivalence.** Under the grouplike-recognition equivalence, the connective spectrum $R_{\geq q}^{\text{dR}, \text{st}}(X/S)$ corresponds to the grouplike object

$$R_{X/S}^{\text{dR}, \langle q \rangle} = \Omega^\infty R_{\geq q}^{\text{dR}, \text{st}}(X/S),$$

and the connective cover of the target corresponds to the grouplike object

$$\Omega^\infty(\mathcal{O}_S^{\text{add}}[n+q]).$$

- (7) **Use the stated construction.** This gives the displayed mapping-space equivalence.

- (8) **Identify the degree-zero case.** For $q = 0$, the (-1) -skeleton is initial, so

$$R_{\geq 0}^{\text{dR}, \text{st}}(X/S) \simeq |R_{\bullet}^{\text{dR}, \text{st}}(X/S)| \simeq \Sigma_{+, S}^\infty Q_{X/S},$$

using $|\mathcal{R}_{X/S, \bullet}^{\text{dR}}| \simeq Q_{X/S}$ and preservation of geometric realizations by relative suspension spectra.

□

Remark 4.6.3 (Skeletal source in positive degree). For $q > 0$, the classifier is the additive delooping $R_{X/S}^{\text{dR}, \langle q \rangle}$ of the normalized skeletal quotient. The removal of lower-dimensional and degenerate simplices is therefore part of the positive-degree classification.

All coefficient objects of the de Rham calculus have now been constructed, so their structural behaviour can be stated without referring to objects that have not yet appeared. Contravariance in X comes from functoriality of the relative unit and effective-image factorization; substitution in the base comes from relative de Rham base change and right Beck–Chevalley. The basic geometric situation is the following Cartesian square:

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S, \end{array} \quad X'_{\text{dR}/S'} \simeq X_{\text{dR}/S} \times_S S'.$$

Because every later construction is obtained from the degreewise cochain diagram by limits, fibres, shifts, and the complete-filtered/coherent-cochain correspondence, the degreewise comparison propagates through the entire calculus.

4.6.5 Substitution and base change of the completed calculus

Contravariance in the geometric variable

Let $f : X \rightarrow Y$ be a morphism over S . Naturality of the relative de Rham unit gives

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \eta_{X/S}^{\text{dR}} \downarrow & & \downarrow \eta_{Y/S}^{\text{dR}} \\ X_{\text{dR}/S} & \longrightarrow & Y_{\text{dR}/S}. \end{array}$$

Functorial effective-image factorization gives $Q_{X/S} \rightarrow Q_{Y/S}$, and finite limits give a simplicial map

$$\mathcal{R}_{X/S, \bullet}^{\text{dR}} \longrightarrow \mathcal{R}_{Y/S, \bullet}^{\text{dR}}.$$

Scalar functions are contravariant, so these maps induce

$$f_{\text{dR}}^* : C_{\text{dR}}^\bullet(Y/S) \longrightarrow C_{\text{dR}}^\bullet(X/S)$$

as a morphism of cosimplicial commutative algebras.

Theorem 4.6.8 (Contravariant functoriality). *The preceding assignment is contravariant in X . The induced maps commute with the Amitsur presentation, full-to-effective comparison, partial and complete totalizations, normalized forms, homotopy-coherent differentials, closed stages, exact-form lifts, and Alexander–Whitney products. For composable $X \xrightarrow{f} Y \xrightarrow{g} Z$,*

$$(gf)_{\text{dR}}^* \simeq f_{\text{dR}}^* g_{\text{dR}}^*$$

by functorial image factorization, finite limits, and adjoint mates.

Proof.

- (1) **Establish the next claim.** The geometric terms are functorial constructions from the natural transformation $\eta_{-/S}^{\text{dR}}$, the functorial (effective epimorphism, monomorphism)-factorization, and finite limits.
- (2) **Construct the indicated map.** The cochain maps are the resulting scalar restriction maps.
- (3) **Construct the indicated map.** All subsequent objects are limits, fibres, shifts, or natural maps of the same diagrams, and the homotopy-coherent differential is functorial under exact maps of complete filtrations.

□

Arbitrary base change in the base

Theorem 4.6.9 (Parameterized base change). *Let $c : S' \rightarrow S$ and $X' = X \times_S S'$. There are coherent equivalences*

$$c^* C_{\text{dR}}^n(X/S) \xrightarrow{\sim} C_{\text{dR}}^n(X'/S')$$

for all n , assembling into an equivalence of cosimplicial commutative algebras. Consequently

$$c^* \text{DR}_{X/S} \simeq \text{DR}_{X'/S'}, \quad c^* \text{DR}_{X/S}^{< q} \simeq \text{DR}_{X'/S'}^{< q},$$

$$c^* \Omega_{\text{dR}}^q(X/S) \simeq \Omega_{\text{dR}}^q(X'/S'),$$

and

$$c^* \Omega_{\text{dR}}^{q, \text{closed}}(X/S; n) \simeq \Omega_{\text{dR}}^{q, \text{closed}}(X'/S'; n),$$

compatibly with the homotopy-coherent differential, exact-form lifts, and normalized Alexander–Whitney products. For morphisms of spectral schemes, the same statement holds for the first-order cotangent comparison.

Proof.

- (1) **Establish multiplicative compatibility.** The de Rham relation degrees commute with base change by Theorem 4.2.5 and finite-limit pasting:

$$\mathcal{R}_{X'/S',n}^{\text{dR}} \simeq \mathcal{R}_{X/S,n}^{\text{dR}} \times_S S'.$$

- (2) **Establish the next claim.** The scalar coefficient satisfies $c^* \mathcal{O}_S^{\text{add}} \simeq \mathcal{O}_{S'}^{\text{add}}$.
- (3) **Use the stated construction.** Right Beck–Chevalley from Theorem 4.4.1 therefore gives the degree-wise cochain equivalence.
- (4) **Use pullback stability.** The pullback functor has both adjoints between stable categories and hence preserves all limits and colimits; it consequently commutes with partial and complete totalizations, matching fibres, filtration fibres, shifts, geometric realizations, and skeletal quotients.
- (5) **Apply naturality.** Naturality gives compatibility with every structure map.
- (6) **Establish multiplicative compatibility.** The first structured infinitesimal neighbourhood and cotangent complex commute with Cartesian base change of spectral schemes by Chapter 2.

□

Construction 4.6.2 (Restriction on global section spectra). The coefficient equivalences above induce natural restriction morphisms

$$\Gamma^{\text{st}}(S, \Omega_{\text{dR}}^q(X/S)[n]) \longrightarrow \Gamma^{\text{st}}(S', \Omega_{\text{dR}}^q(X'/S')[n]),$$

and similarly for closed forms and de Rham algebras, by pulling a section along c^* and then using the base-change equivalence. These maps are the natural restriction morphisms on global section spectra.

The complete descent calculus is now stable under the two substitutions that matter later: contravariant change of the geometric object and Cartesian change of base. Its multiplication is likewise compatible with products. The remaining global issue is descent of these form objects with respect to covers of the geometric context. We address that next, together with the affine-descent step required for comprehension.

4.7 Formal étale descent and comprehension of de Rham forms

The intrinsic form coefficients are defined for every object of the spectral slice, and comprehension requires one further step: their stable-section semantics on affine tests must satisfy spectral Zariski descent. Establishing that descent requires an additional argument beyond base change. We therefore first control the de Rham relation under formally étale effective epimorphisms, the class that contains the finite Zariski covering morphisms used below.

Only two semantic levels will be distinguished in this section. The *stable-section semantics* is obtained directly from the intrinsic stable coefficient object. The *comprehended semantics* is its affine-descent extension represented by an object of the slice. Formally étale transport and finite-coproduct compatibility provide the descent bridge; effective descent reconstructs the cochain tower degree-wise; affine density then produces the representing comprehension objects.

The logic of the last step is expressed by the following square. The lower functor is the right Kan extension of the affine stable-section semantics:

$$\begin{array}{ccc} \mathrm{AffPt}(S)^{\mathrm{op}} & \xrightarrow{\mathfrak{F}\mathrm{orm}_{\mathrm{dR},S}^{q,\mathrm{stsec}}} & \mathcal{S} \\ j_S^{\mathrm{op}} \downarrow & & \parallel \\ ((\mathcal{H}_{\mathrm{Sp}})_S)^{\mathrm{op}} & \xrightarrow{\mathrm{Ran}_{j_S^{\mathrm{op}}}} & \mathcal{S}. \end{array}$$

The representing object obtained from the affine-point sheaf therefore comprehends exactly this affine-descended extension.

4.7.1 Formally étale transport

Theorem 4.3.8 gives the precise geometric invariance under a formally étale morphism $u : U \rightarrow X$:

$$\mathcal{R}_{U/S,n}^{\mathrm{dR}} \simeq U \times_{X,v_n} \mathcal{R}_{X/S,n}^{\mathrm{dR}}.$$

This Cartesian formula controls the geometric transport of the de Rham relation.

Proposition 4.7.1 (Degree-wise centred coefficient base change). *For $n \geq 0$, let $v_n : \mathcal{R}_{X/S,n}^{\mathrm{dR}} \rightarrow X$ be the first-vertex projection and put*

$$C_{\mathrm{dR}}^{n,\mathrm{ctr}}(X/S) := \Pi_{v_n} v_n^* \mathcal{O}_X^{\mathrm{add}} \in \mathcal{D}_{\mathrm{Sp}/X}.$$

If $u : U \rightarrow X$ is formally étale over S , then

$$u^* C_{\mathrm{dR}}^{n,\mathrm{ctr}}(X/S) \xrightarrow{\simeq} C_{\mathrm{dR}}^{n,\mathrm{ctr}}(U/S)$$

canonically in each degree.

Proof.

- (1) **Establish the next claim.** Use $\mathcal{R}_{U/S,n}^{\mathrm{dR}} \simeq U \times_X \mathcal{R}_{X/S,n}^{\mathrm{dR}}$ and right Beck–Chevalley for the Cartesian square defined by the first-vertex projection, together with $u^* \mathcal{O}_X^{\mathrm{add}} \simeq \mathcal{O}_U^{\mathrm{add}}$.

□

Remark 4.7.1 (First-vertex indexing). The objects $C_{\mathrm{dR}}^{n,\mathrm{ctr}}(X/S)$ are indexed by the distinguished first vertex. The face deleting that vertex changes the chosen centre, so the complete cosimplicial algebra is formed after dependent product to the base:

$$C_{\mathrm{dR}}^{\bullet}(X/S) \in \mathrm{CAlg}(\mathcal{D}_{\mathrm{Sp}/S}).$$

A formally étale map $U \rightarrow X$ induces the corresponding contravariant morphism of cosimplicial de Rham algebras.

Theorem 4.7.1 (Formal étale change of intermediate base). *Let $g : S' \rightarrow S$ be formally étale and let $V \rightarrow S'$. Regard V also as an object over S . Then*

$$V_{\mathrm{dR}/S'} \xrightarrow{\simeq} V_{\mathrm{dR}/S},$$

with the two relative units identified. Consequently

$$Q_{V/S'} \xrightarrow{\simeq} Q_{V/S}, \quad \mathcal{R}_{V/S', \bullet}^{\mathrm{dR}} \xrightarrow{\simeq} \mathcal{R}_{V/S, \bullet}^{\mathrm{dR}},$$

and for every $Z \rightarrow Y \rightarrow S'$,

$$\mathrm{Nbh}_{Z/S'}^{\mathrm{dR}}(Y) \xrightarrow{\simeq} \mathrm{Nbh}_{Z/S}^{\mathrm{dR}}(Y).$$

For coefficient objects the correctly typed comparison is

$$\Pi_g C_{\mathrm{dR}}^{\bullet}(V/S') \xrightarrow{\simeq} C_{\mathrm{dR}}^{\bullet}(V/S).$$

It induces corresponding equivalences

$$\Pi_g \mathrm{DR}_{V/S'} \simeq \mathrm{DR}_{V/S}, \quad \Pi_g \Omega_{\mathrm{dR}}^q(V/S') \simeq \Omega_{\mathrm{dR}}^q(V/S),$$

and similarly for partial totalizations, closed stages, and exact-form lifts. On stable global sections, the two intermediate-base constructions therefore agree canonically.

Proof.

(1) **Apply the indicated result.** Apply Theorem 4.3.1 to $g : S' \rightarrow S$ over the terminal object.

(2) **Use the stated construction.** It gives

$$S' \simeq S \times_{S_{\mathrm{dR}}} S'_{\mathrm{dR}}.$$

Hence

$$\begin{aligned} V_{\mathrm{dR}/S'} &= S' \times_{S'_{\mathrm{dR}}} V_{\mathrm{dR}} \\ &\simeq (S \times_{S_{\mathrm{dR}}} S'_{\mathrm{dR}}) \times_{S'_{\mathrm{dR}}} V_{\mathrm{dR}} \\ &\simeq S \times_{S_{\mathrm{dR}}} V_{\mathrm{dR}} = V_{\mathrm{dR}/S}. \end{aligned}$$

(3) **Establish the next claim.** The unit in both cases is induced by the same absolute unit $V \rightarrow V_{\mathrm{dR}}$.

(4) **Establish the next claim.** Uniqueness of effective-image factorization and finite-limit pasting give the quotient, de Rham relation, and de Rham neighbourhood formulas.

(5) **Introduce the notation.** Write

$$r_n : \mathcal{R}_{V/S', n}^{\mathrm{dR}} \longrightarrow S', \quad a_n := gr_n : \mathcal{R}_{V/S', n}^{\mathrm{dR}} \longrightarrow S.$$

Since $\mathcal{O}_{S'}^{\mathrm{add}} \simeq g^* \mathcal{O}_S^{\mathrm{add}}$,

$$\begin{aligned} \Pi_g C_{\mathrm{dR}}^n(V/S') &\simeq \Pi_g \Pi_{r_n} r_n^* \mathcal{O}_{S'}^{\mathrm{add}} \\ &\simeq \Pi_{a_n} a_n^* \mathcal{O}_S^{\mathrm{add}} \\ &= C_{\mathrm{dR}}^n(V/S). \end{aligned}$$

(6) **Identify the stated objects.** Composition of dependent products makes this identification cosimplicial and multiplicative.

(7) **Establish multiplicative compatibility.** The functor Π_g preserves all limits and is exact, so it commutes with totalization, normalization, and filtration fibres.

(8) **Establish the next claim.** Finally

$$\Gamma^{\mathrm{st}}(S, \Pi_g E) \simeq \Gamma^{\mathrm{st}}(S', E)$$

by adjunction. □

4.7.2 Finite disjoint coproducts

Finite affine Zariski covers enter the slice through a coproduct of their members. To use formal étale descent for such a covering map, we must know that maps into a finite coproduct decompose compatibly across an infinitesimal substitution. The next lemma transports the corresponding clopen decomposition from the centre to the thickening.

Lemma 4.7.1 (Finite clopen decompositions under infinitesimal substitution). *Let*

$$j : V \longrightarrow T$$

be a finite spectral infinitesimal substitution between affine spectral schemes, and let $\{F_i \rightarrow S\}_{i \in I}$ be a finite family of objects of the slice $(\mathcal{H}_{\text{Sp}})_{/S}$.

Every morphism

$$q : V \longrightarrow \coprod_{i \in I} F_i$$

over S canonically determines a finite disjoint open-and-closed decomposition

$$V \simeq \coprod_{i \in I} V_i$$

and morphisms

$$q_i : V_i \longrightarrow F_i.$$

Moreover this decomposition is the pullback, through a contractible space of choices, of a unique finite disjoint open-and-closed decomposition

$$T \simeq \coprod_{i \in I} T_i$$

such that

$$V_i \simeq V \times_T T_i.$$

Every T_i and V_i is affine, and

$$V_i \longrightarrow T_i$$

is again a finite spectral infinitesimal substitution. If j is a one-stage square-zero infinitesimal substitution, then every $V_i \rightarrow T_i$ is a one-stage square-zero infinitesimal substitution.

Proof.

(1) **Fix the indicated data.** Let

$$I_S := \coprod_{i \in I} S$$

be the finite constant object of the slice.

(2) **Construct the indicated map.** The structure maps $F_i \rightarrow S$ assemble to a canonical morphism

$$\coprod_{i \in I} F_i \longrightarrow I_S$$

which on the i -th summand lands in the i -th copy of S .

- (3) **Establish the next claim.** Compose with q to obtain

$$\bar{q} : V \longrightarrow I_S.$$

- (4) **Use the stated identification.** Because finite coproducts in an infinity-topos are disjoint and universal, pulling back the i -th summand $S \rightarrow I_S$ along $\bar{q}$ gives pairwise disjoint complemented subobjects

$$V_i \longrightarrow V$$

whose coproduct is V .

- (5) **Construct the indicated map.** The original map q restricts canonically to

$$q_i : V_i \longrightarrow F_i.$$

- (6) **Establish the next claim.** On the spectral Zariski site of the affine scheme V , a section of the finite constant sheaf I is equivalently a finite locally constant I -valued function, and hence equivalently a finite disjoint clopen decomposition of V .

- (7) **Conclude the stated claim.** Thus the subobjects V_i are represented by open-and-closed spectral subschemes.

- (8) **Use the stated hypothesis.** Since V is affine, each V_i is affine: its clopen support is represented by the spectral localization at the corresponding idempotent in $\pi_0\mathcal{O}(V)$.

- (9) **Invoke the cited result.** By 2.8.3, pullback along j induces an equivalence between the posets of open spectral subschemes of T and V , preserving finite intersections and covering families.

- (10) **Conclude the stated claim.** Hence each V_i is the pullback of a contractibly unique open spectral subscheme

$$T_i \longrightarrow T.$$

- (11) **Establish the indicated equivalence.** The equalities

$$V_i \cap V_k = \emptyset \quad (i \neq k)$$

and the covering relation

$$\coprod_i V_i \longrightarrow V$$

transport through this equivalence to

$$T_i \cap T_k = \emptyset \quad (i \neq k)$$

and to a covering family $\{T_i \rightarrow T\}$. Thus

$$T \simeq \coprod_i T_i.$$

- (12) **Establish the next claim.** Each T_i is open and its complement is the union of the remaining T_k , so it is open and closed.

- (13) **Use the stated hypothesis.** Since T is affine, every T_i is affine by the same idempotent-localization argument.
- (14) **Establish the next claim.** Finally,

$$V_i \simeq V \times_T T_i$$

by construction.

- (15) **Use the stated construction.** Base-change stability of finite infinitesimal substitutions gives that $V_i \rightarrow T_i$ is finite infinitesimal.
- (16) **Use the stated construction.** In the one-stage case, base-change stability of square-zero infinitesimal substitutions gives the final assertion.

□

Proposition 4.7.2 (Finite-coproduct compatibility). *For a finite disjoint family $\{X_i \rightarrow S\}$,*

$$\left(\coprod_i X_i \right)_{\mathrm{dR}/S} \simeq \coprod_i (X_i)_{\mathrm{dR}/S},$$

$$Q_{(\coprod_i X_i)/S} \simeq \coprod_i Q_{X_i/S},$$

and

$$\mathcal{R}_{(\coprod_i X_i)/S,n}^{\mathrm{dR}} \simeq \coprod_i \mathcal{R}_{X_i/S,n}^{\mathrm{dR}}.$$

de Rham scalar functions send these finite disjoint coproducts to finite products of coefficient objects, compatibly with normalization and totalization.

Proof.

- (1) **Prove locality of a finite coproduct.** We first prove that a finite disjoint coproduct of objects de Rham-local over S is again de Rham-local over S .
- (2) **Fix the indicated data.** Let

$$F := \coprod_i F_i \longrightarrow S$$

with every $F_i \rightarrow S$ de Rham-local over S .

- (3) **Fix the indicated data.** Let

$$j : V \longrightarrow T$$

be an affine square-zero infinitesimal substitution over S .

- (4) **Establish the indicated equivalence.** We show that

$$\mathrm{Map}_S(T, F) \longrightarrow \mathrm{Map}_S(V, F)$$

is an equivalence.

- (5) **Fix the indicated data.** Fix a point

$$q : V \longrightarrow F.$$

- (6) **Invoke the cited result.** By Lemma 4.7.1, q determines a finite disjoint open-and-closed decomposition

$$V \simeq \coprod_i V_i$$

together with maps

$$q_i : V_i \longrightarrow F_i.$$

- (7) **Establish the next claim.** The same lemma transports this decomposition through j to a contractibly unique finite disjoint open-and-closed decomposition

$$T \simeq \coprod_i T_i$$

such that

$$V_i \simeq V \times_T T_i.$$

- (8) **Establish the next claim.** Every T_i is affine, and each

$$V_i \longrightarrow T_i$$

is an affine square-zero infinitesimal substitution.

- (9) **Use the stated construction.** De Rham locality of F_i over S therefore gives a contractibly unique extension

$$\tilde{q}_i : T_i \longrightarrow F_i$$

of q_i .

- (10) **Construct the indicated map.** The maps $\tilde{q}_i$ assemble uniquely to an extension

$$\tilde{q} : T \simeq \coprod_i T_i \longrightarrow \coprod_i F_i = F.$$

- (11) **Establish the next claim.** The transported clopen decomposition is contractibly unique, and each component extension space is contractible.

- (12) **Conclude the stated claim.** Hence the homotopy fibre over q of

$$\mathrm{Map}_S(T, F) \longrightarrow \mathrm{Map}_S(V, F)$$

is contractible.

- (13) **Use the stated hypothesis.** Since this holds for every q , the restriction map is an equivalence.

- (14) **Invoke the cited result.** By Theorem 4.2.4, $F \rightarrow S$ is de Rham-local over S .

- (15) **Apply the indicated result.** Apply this to

$$F_i = (X_i)_{\mathrm{dR}/S}.$$

(16) **Use the stated construction.** The coproduct of the localization units gives

$$\coprod_i X_i \longrightarrow \coprod_i (X_i)_{\mathrm{dR}/S}.$$

(17) **Construct the indicated map.** For every $Z \rightarrow S$ de Rham-local over S , the induced mapping-space comparison is the finite product of the componentwise localization equivalences:

$$\begin{aligned} \mathrm{Map}_S \left(\coprod_i (X_i)_{\mathrm{dR}/S}, Z \right) &\simeq \prod_i \mathrm{Map}_S((X_i)_{\mathrm{dR}/S}, Z) \\ &\simeq \prod_i \mathrm{Map}_S(X_i, Z) \\ &\simeq \mathrm{Map}_S \left(\coprod_i X_i, Z \right). \end{aligned}$$

(18) **Conclude the stated claim.** Thus the coproduct map is a de Rham equivalence over S with target de Rham-local over S .

(19) **Establish the locality claim.** It therefore exhibits the relative de Rham localization:

$$\left(\coprod_i X_i \right)_{\mathrm{dR}/S} \simeq \coprod_i (X_i)_{\mathrm{dR}/S}.$$

(20) **Establish the next claim.** For each i , write the effective-image factorization as

$$X_i \xrightarrow{e_i} Q_{X_i/S} \xrightarrow{m_i} (X_i)_{\mathrm{dR}/S}.$$

(21) **Construct the indicated map.** Finite coproducts of effective epimorphisms are effective epimorphisms in an infinity-topos.

(22) **Invoke the cited result.** By disjointness and universality of finite coproducts, the coproduct of the monomorphisms m_i is again a monomorphism.

(23) **Conclude the stated claim.** Hence

$$\coprod_i X_i \longrightarrow \coprod_i Q_{X_i/S} \longrightarrow \coprod_i (X_i)_{\mathrm{dR}/S}$$

is the effective-epimorphism–monomorphism factorization of the relative de Rham unit of $\coprod_i X_i$.

(24) **Use the stated construction.** Functorial uniqueness of effective images therefore gives

$$Q_{(\coprod_i X_i)/S} \simeq \coprod_i Q_{X_i/S}.$$

- (25) **Compute the relation degrees.** Using the relative de Rham coproduct formula and disjointness of the summands, disjointness eliminates mixed components from the iterated fibre products. Therefore

$$\begin{aligned} \mathcal{R}_{(\coprod_i X_i)/S, n}^{\text{dR}} &= \underbrace{\left(\prod_i X_i \right) \times_{(\coprod_i X_i)_{\text{dR}/S}} \cdots \times_{(\coprod_i X_i)_{\text{dR}/S}} \left(\prod_i X_i \right)}_{n+1 \text{ factors}} \\ &\simeq \prod_i \underbrace{X_i \times_{(X_i)_{\text{dR}/S}} \cdots \times_{(X_i)_{\text{dR}/S}} X_i}_{n+1 \text{ factors}} \\ &= \prod_i \mathcal{R}_{X_i/S, n}^{\text{dR}}. \end{aligned}$$

- (26) **Establish the next claim.** Finally, dependent product of the stable additive scalar along a finite disjoint coproduct is the finite product of the corresponding dependent products.
- (27) **Conclude the stated claim.** Hence de Rham scalar functions carry the displayed de Rham relation decomposition to finite products of coefficient objects.
- (28) **Identify the stated objects.** Stable normalization, partial totalization, and complete totalization are limit constructions, so these identifications commute with normalization and totalization.

□

4.7.3 Effective formal étale descent

The preceding coproduct analysis shows that finite Zariski covering morphisms are formally étale. We can now combine formal étale transport with effective descent. The result reconstructs the effective quotient and every relation degree from the Čech nerve of the cover, and hence reconstructs the entire cochain tower.

Lemma 4.7.2 (Spectral Zariski covers are formally étale). *Every spectral Zariski open immersion is formally étale with respect to the finite spectral infinitesimal substitutions of Chapter 2. More generally, if $\{U_i \rightarrow X\}_{i \in I}$ is a finite spectral Zariski covering family, then the covering morphism*

$$\prod_{i \in I} U_i \longrightarrow X$$

is formally étale.

Proof.

- (1) **Invoke the cited result.** By 2.11.2, it is enough to test contractible lifting against affine one-stage square-zero substitutions $j : V \rightarrow T$.
- (2) **Fix the indicated data.** Let $U \hookrightarrow X$ be a spectral Zariski open immersion and consider such a lifting problem.
- (3) **Use pullback stability.** The inverse image

$$T \times_X U \longrightarrow T$$

is an open subobject whose pullback to V is all of V .

- (4) **Invoke the cited result.** By 2.8.1, pullback along j identifies the small spectral Zariski sites of T and V .
- (5) **Conclude the stated claim.** Hence the displayed open subobject is all of T .
- (6) **Establish the next claim.** The required factorization through U therefore exists, and its space is contractible because an open immersion is a monomorphism.
- (7) **Establish the next claim.** One-stage detection proves that $U \rightarrow X$ is formally étale.
- (8) **Fix the indicated data.** Let $\{U_i \rightarrow X\}_{i \in I}$ be a finite spectral Zariski covering family.
- (9) **Establish the next claim.** Again let $j : V \rightarrow T$ be an affine one-stage square-zero substitution.
- (10) **Construct the indicated map.** A map

$$V \longrightarrow \coprod_i U_i$$

determines, by Lemma 4.7.1, a finite disjoint open-and-closed decomposition

$$V \simeq \coprod_i V_i$$

together with maps $V_i \rightarrow U_i$.

- (11) **Establish the next claim.** The same lemma transports this through j to a contractibly unique decomposition
- $$T \simeq \coprod_i T_i, \quad V_i \simeq V \times_T T_i.$$
- (12) **Use pullback stability.** For each i , the inverse image of U_i in T_i is an open subobject whose pullback to V_i is all of V_i .
 - (13) **Establish the indicated equivalence.** The one-stage site equivalence therefore makes that inverse image all of T_i , so $T_i \rightarrow X$ factors through U_i .
 - (14) **Establish the next claim.** These factorizations assemble to a lift

$$T \longrightarrow \coprod_i U_i.$$

- (15) **Establish the next claim.** The transported clopen decomposition and the component factorizations have contractible spaces of choices, so the total filler space is contractible.
- (16) **Construct the indicated map.** One-stage detection proves that the covering morphism is formally étale.

□

Theorem 4.7.2 (Descent of effective de Rham quotients). *Let $u : U \rightarrow X$ be an effective epimorphism over S which is formally étale relative to X , and let $U_\bullet \rightarrow X$ be its Čech nerve. Then*

$$Q_{U/S} \rightarrow Q_{X/S}$$

is an effective epimorphism, the simplicial object $Q_{U_\bullet/S}$ is its Čech nerve, and

$$Q_{X/S} \simeq |Q_{U_\bullet/S}|.$$

For each de Rham relation degree n , the simplicial object in the cover degree

$$k \mapsto \mathcal{R}_{U_k/S,n}^{\text{dR}}$$

is the Čech nerve of the effective epimorphism

$$U \times_X \mathcal{R}_{X/S,n}^{\text{dR}} \rightarrow \mathcal{R}_{X/S,n}^{\text{dR}}.$$

Proof.

- (1) **Establish the next claim.** Formal-étale Cartesianity and Theorem 4.3.8 give

$$U \simeq X \times_{Q_{X/S}} Q_{U/S}.$$

- (2) **Construct the indicated map.** The composite $U \rightarrow X \rightarrow Q_{X/S}$ is an effective epimorphism and factors through $Q_{U/S} \rightarrow Q_{X/S}$.

- (3) **Construct the indicated map.** Factor the latter map as

$$Q_{U/S} \twoheadrightarrow I \hookrightarrow Q_{X/S}.$$

- (4) **Construct the indicated map.** Then the effective epimorphism $U \rightarrow Q_{X/S}$ factors through the monomorphism $I \hookrightarrow Q_{X/S}$.

- (5) **Apply orthogonality.** Orthogonality of effective epimorphisms and monomorphisms gives a section of this monomorphism, hence it is an equivalence.

- (6) **Conclude the stated claim.** Therefore $Q_{U/S} \rightarrow Q_{X/S}$ is an effective epimorphism.

- (7) **Identify the stated objects.** Applying the same Cartesian formula to the iterated fibre products U_k identifies

$$Q_{U_k/S} \simeq \underbrace{Q_{U/S} \times_{Q_{X/S}} \cdots \times_{Q_{X/S}} Q_{U/S}}_{k+1 \text{ factors}}.$$

- (8) **Establish the next claim.** This is the Čech nerve statement.

- (9) **Establish the next claim.** The de Rham relation formula follows from

$$\mathcal{R}_{U_k/S,n}^{\text{dR}} \simeq U_k \times_X \mathcal{R}_{X/S,n}^{\text{dR}}.$$

□

Theorem 4.7.3 (Effective descent for the intrinsic cochain tower). *Under the hypotheses of Theorem 4.7.2, for every de Rham relation degree n ,*

$$C_{\mathrm{dR}}^n(X/S) \xrightarrow{\simeq} \mathrm{Tot}_{[k] \in \Delta} C_{\mathrm{dR}}^n(U_k/S).$$

These equivalences assemble into an equivalence of cosimplicial commutative algebras. Consequently

$$\mathrm{DR}_{X/S} \simeq \mathrm{Tot}_k \mathrm{DR}_{U_k/S},$$

$$\Omega_{\mathrm{dR}}^q(X/S) \simeq \mathrm{Tot}_k \Omega_{\mathrm{dR}}^q(U_k/S),$$

and

$$F_{\mathrm{Tot}}^q \mathrm{DR}_{X/S} \simeq \mathrm{Tot}_k F_{\mathrm{Tot}}^q \mathrm{DR}_{U_k/S}.$$

The same holds for partial totalizations, closed de Rham forms, exact-form lifts, and the normalized Alexander–Whitney products.

Proof.

- (1) **Identify the stated objects.** For fixed n , Theorem 4.7.2 identifies $\mathcal{R}_{U_k/S, n}^{\mathrm{dR}}$ as the Čech nerve of an effective epimorphism over $\mathcal{R}_{X/S, n}^{\mathrm{dR}}$.
- (2) **Apply effective descent.** Effective descent for parameterized spectra applied to $a_n^* \mathcal{O}_S^{\mathrm{add}}$ gives, with

$$c_{k, n} : \mathcal{R}_{U_k/S, n}^{\mathrm{dR}} \longrightarrow \mathcal{R}_{X/S, n}^{\mathrm{dR}}$$

the cover-stage structure map,

$$a_n^* \mathcal{O}_S^{\mathrm{add}} \simeq \mathrm{Tot}_k \Pi_{c_{k, n}} c_{k, n}^* a_n^* \mathcal{O}_S^{\mathrm{add}}.$$

- (3) **Apply the indicated result.** Apply Π_{a_n} , which preserves limits, and compose dependent products.
- (4) **Use the stated construction.** This gives the degreewise cochain equivalence.
- (5) **Apply naturality.** Naturality in n makes it cosimplicial and multiplicative.
- (6) **Identify the relevant fibre.** All remaining constructions are obtained by limits, fibres, and shifts, which commute with the cover totalization.

□

Theorem 4.7.4 (Effective descent for full de Rham functions). *Under the same hypotheses,*

$$\mathrm{DR}_{X/S}^{\mathrm{full}} \xrightarrow{\simeq} \mathrm{Tot}_k \mathrm{DR}_{U_k/S}^{\mathrm{full}}$$

as commutative algebras, and the descended full-to-effective comparison is $\rho_{X/S}$.

Proof.

- (1) **Apply the cited theorem.** Theorem 4.3.4 makes $U_{\mathrm{dR}/S} \rightarrow X_{\mathrm{dR}/S}$ an effective epimorphism, while left exactness identifies the de Rham objects of the Čech stages with its Čech nerve.

- (2) **Apply the indicated result.** Apply effective descent for parameterized spectra to $p_{\mathrm{dR}}^* \mathcal{O}_S^{\mathrm{add}}$ and then apply $\Pi_{p_{\mathrm{dR}}}$.
- (3) **Apply naturality.** Naturality of restriction along $Q_{-/S} \hookrightarrow (-)_{\mathrm{dR}/S}$ gives compatibility with ρ .

□

4.7.4 Affine descent and comprehension

With effective formal étale descent available, the remaining argument is representability inside the slice. We first form the stable-section space-valued functors and verify Zariski descent on affine tests. The affine-point presentation then constructs the objects representing their comprehended semantics on arbitrary contexts.

Definition 4.7.1 (Stable-section form-space functors). For an arbitrary object $t : T \rightarrow S$, define

$$\mathfrak{Form}_{\mathrm{dR},S}^{q,\mathrm{stsec}}(T) := \Omega^\infty \Gamma^{\mathrm{st}}(S, \Omega_{\mathrm{dR}}^q(T/S)),$$

and

$$\mathfrak{Form}_{\mathrm{dR},S}^{q,\mathrm{closed},\mathrm{stsec}}(T) := \Omega^\infty \Gamma^{\mathrm{st}}(S, \Omega_{\mathrm{dR}}^{q,\mathrm{closed}}(T/S; 0)).$$

Contravariant functoriality in T makes these presheaves of spaces on $(\mathcal{H}_{\mathrm{Sp}})_{/S}$. Write

$$j_S : \mathrm{AffPt}(S) \longrightarrow (\mathcal{H}_{\mathrm{Sp}})_{/S}$$

for the affine-point inclusion.

Proposition 4.7.3 (Zariski descent of stable-section form spaces on affine tests). *The restrictions*

$$j_S^* \mathfrak{Form}_{\mathrm{dR},S}^{q,\mathrm{stsec}}, \quad j_S^* \mathfrak{Form}_{\mathrm{dR},S}^{q,\mathrm{closed},\mathrm{stsec}}$$

are spectral Zariski sheaves on $\mathrm{AffPt}(S)$. If $\{T_i \rightarrow T\}$ is a finite affine spectral Zariski cover with Čech nerve $T_\bullet$ formed in the slice, then

$$\mathfrak{Form}_{\mathrm{dR},S}^{q,\mathrm{stsec}}(T) \xrightarrow{\simeq} \mathrm{Tot}_{[k] \in \Delta} \mathfrak{Form}_{\mathrm{dR},S}^{q,\mathrm{stsec}}(T_k),$$

and similarly for closed forms.

Proof.

- (1) **Construct the indicated map.** The covering morphism

$$\coprod_i T_i \longrightarrow T$$

is an effective epimorphism and is formally étale by Lemma 4.7.2.

- (2) **Apply the indicated result.** Apply Theorem 4.7.3 to its Čech nerve.
- (3) **Establish the next claim.** This reconstructs the normalized and closed coefficient objects from the full Čech diagram, including its nonaffine iterated fibre products.

- (4) **Construct the indicated map.** Stable global sections are mapping spectra out of the tensor unit and therefore preserve limits, and Ω^∞ preserves limits.
- (5) **Establish the indicated equivalence.** The displayed equivalence follows.
- (6) **Establish the next claim.** On the affine-point site this is precisely the spectral Zariski sheaf condition.

□

Construction 4.7.1 (Affine-descent comprehension objects). By Proposition 4.4.1, the two affine-point sheaves of Proposition 4.7.3 determine canonical objects

$$\mathbf{Form}_{dR,S}^q, \quad \mathbf{Form}_{dR,S}^{q,\text{closed}} \in (\mathcal{H}_{\text{Sp}})_{/S}.$$

They are characterized on every affine point $U \rightarrow S$ by

$$\text{Map}_S(U, \mathbf{Form}_{dR,S}^q) \simeq \mathfrak{Form}_{dR,S}^{q,\text{stsec}}(U),$$

and

$$\text{Map}_S(U, \mathbf{Form}_{dR,S}^{q,\text{closed}}) \simeq \mathfrak{Form}_{dR,S}^{q,\text{closed},\text{stsec}}(U).$$

For an arbitrary $T \rightarrow S$, define the *comprehended de Rham form judgments* by

$$\mathfrak{Form}_{dR,S}^{q,\text{comp}}(T) := \text{Map}_S(T, \mathbf{Form}_{dR,S}^q),$$

and

$$\mathfrak{Form}_{dR,S}^{q,\text{closed},\text{comp}}(T) := \text{Map}_S(T, \mathbf{Form}_{dR,S}^{q,\text{closed}}).$$

Thus the comprehended semantics on arbitrary contexts is the affine-descent extension of the stable-section semantics on affine tests.

Proposition 4.7.4 (Right-Kan description of comprehended form judgments). *There are canonical equivalences of presheaves on the full slice*

$$\mathfrak{Form}_{dR,S}^{q,\text{comp}} \simeq \text{Ran}_{j_S^{\text{op}}} (j_S^* \mathfrak{Form}_{dR,S}^{q,\text{stsec}}),$$

and

$$\mathfrak{Form}_{dR,S}^{q,\text{closed},\text{comp}} \simeq \text{Ran}_{j_S^{\text{op}}} (j_S^* \mathfrak{Form}_{dR,S}^{q,\text{closed},\text{stsec}}).$$

Equivalently, for every $T \rightarrow S$,

$$\begin{aligned} \mathfrak{Form}_{dR,S}^{q,\text{comp}}(T) &\simeq \lim_{(U \rightarrow T) \in \text{AffPt}(T)^{\text{op}}} \mathfrak{Form}_{dR,S}^{q,\text{stsec}}(U), \\ \mathfrak{Form}_{dR,S}^{q,\text{closed},\text{comp}}(T) &\simeq \lim_{(U \rightarrow T) \in \text{AffPt}(T)^{\text{op}}} \mathfrak{Form}_{dR,S}^{q,\text{closed},\text{stsec}}(U). \end{aligned}$$

Proof.

- (1) **Apply the cited lemma.** Lemma 4.2.1 gives

$$T \simeq \text{colim}_{(U \rightarrow T) \in \text{AffPt}(T)} U$$

in the slice over S .

- (2) **Use the stated construction.** Mapping this colimit into either comprehension object gives the corresponding limit of affine mapping spaces.
- (3) **Identify the stated objects.** The affine characterizing equivalences of Construction 4.7.1 identify these mapping spaces with the stable-section affine form spaces.
- (4) **Establish the next claim.** The resulting limit is the pointwise formula for right Kan extension along j_S^{op} .

□

Construction 4.7.2 (Stable-section-to-comprehended comparison). For every $T \rightarrow S$, restriction of a stable-section form along every affine point $U \rightarrow T$ gives a compatible cone

$$\mathfrak{Form}_{\text{dR},S}^{q,\text{stsec}}(T) \longrightarrow \mathfrak{Form}_{\text{dR},S}^{q,\text{stsec}}(U).$$

Using Proposition 4.7.4, this cone determines a canonical natural transformation

$$\text{cmp}^q : \mathfrak{Form}_{\text{dR},S}^{q,\text{stsec}} \longrightarrow \mathfrak{Form}_{\text{dR},S}^{q,\text{comp}}.$$

The same construction gives

$$\text{cmp}^{q,\text{closed}} : \mathfrak{Form}_{\text{dR},S}^{q,\text{closed},\text{stsec}} \longrightarrow \mathfrak{Form}_{\text{dR},S}^{q,\text{closed},\text{comp}}.$$

If $T \rightarrow S$ is affine, the identity affine point $T \rightarrow T$ is terminal in $\text{AffPt}(T)$, so it is initial after passage to the opposite category. The preceding limit therefore evaluates at T , and

$$\text{cmp}_T^q, \quad \text{cmp}_T^{q,\text{closed}}$$

are equivalences.

Theorem 4.7.5 (Comprehension of affine-descended de Rham form judgments). *For every $q \geq 0$, the objects*

$$\mathbf{Form}_{\text{dR},S}^q, \quad \mathbf{Form}_{\text{dR},S}^{q,\text{closed}}$$

comprehend the affine-descended de Rham form judgments:

$$\text{Map}_S(T, \mathbf{Form}_{\text{dR},S}^q) \simeq \mathfrak{Form}_{\text{dR},S}^{q,\text{comp}}(T),$$

$$\text{Map}_S(T, \mathbf{Form}_{\text{dR},S}^{q,\text{closed}}) \simeq \mathfrak{Form}_{\text{dR},S}^{q,\text{closed},\text{comp}}(T)$$

for every $T \rightarrow S$, while on affine contexts these spaces are canonically the stable-section form spaces. Their identity sections are the universal terms for the comprehended judgments.

The underlying-form transformation induces

$$u_q : \mathbf{Form}_{\text{dR},S}^{q,\text{closed}} \longrightarrow \mathbf{Form}_{\text{dR},S}^q.$$

For $q \geq 1$, the exact-form lift induces

$$d_{\text{dR}}^{\text{closed}} : \mathbf{Form}_{\text{dR},S}^{q-1} \longrightarrow \mathbf{Form}_{\text{dR},S}^{q,\text{closed}}.$$

Their composite is the morphism of comprehension objects whose restriction to every affine test represents the intrinsic de Rham differential.

Proof.

- (1) **Use the stated construction.** Construction 4.7.1 gives the comprehension objects and their universal terms.
- (2) **Apply the cited proposition.** Proposition 4.7.4 and Construction 4.7.2 supply the right-Kan formula and the affine comparison.
- (3) **Construct the indicated map.** The underlying-form maps and exact-form lifts are natural in affine substitution, hence define morphisms of affine-point sheaves.
- (4) **Construct the indicated map.** Under Proposition 4.4.1, these natural transformations correspond to the displayed morphisms in the slice.
- (5) **Apply the cited proposition.** Proposition 4.6.5 identifies the composite on every affine test with the homotopy-coherent de Rham differential.
- (6) **Use the stated construction.** Affine descent gives the global statement.

□

We now have both semantic levels. On affine contexts the comprehended semantics agrees with the stable-section semantics; on arbitrary contexts it is the canonical affine-descent extension. The underlying-form and exact-form maps descend with the same construction. The remaining sections can therefore turn to comparison questions without reopening descent or representability.

4.8 First-order cotangent comparison

The de Rham calculus constructed above retains every finite infinitesimal stage visible to the generated congruence. To compare it with the differentiated spectral doctrine, we now restrict that nonlinear relation to the first structured infinitesimal neighbourhood of the diagonal. This comparison is deliberately one-way: first-order restriction extracts the cotangent coefficient, but the full degree-one de Rham normalization can contain higher-order infinitesimal classes that disappear under that restriction.

For a morphism of spectral schemes $p : X \rightarrow S$, the relevant geometric comparison begins with the first structured infinitesimal neighbourhood:

$$\begin{array}{ccc}
 X & \xleftarrow{i_1} & D_{X/S}^{(1)} \\
 & \searrow \Delta_{X/S} & \downarrow \\
 & & X \times_S X,
 \end{array}
 \quad \text{first-order functions} \longrightarrow \text{Real}_X^{\text{st}}(L_{X/S}).$$

The coefficient-level target of this restriction is the already constructed stable realization of the cotangent complex:

$$\Omega_{\text{dR}}^1(X/S) \longrightarrow \Pi_p \text{Real}_X^{\text{st}}(L_{X/S}),$$

This target becomes meaningful through the actual restriction map. Restriction of scalar functions along $D_{X/S}^{(1)} \rightarrow \mathcal{R}_{X/S,1}^{\text{dR}}$ fits into a fibrewise comparison:

$$\begin{array}{ccc}
 C_{\text{dR}}^{1,\text{ctr}}(X/S) & \longrightarrow & \mathcal{F}_{X/S}^{(1)} \\
 \sigma \downarrow & & \downarrow r_0 \\
 \mathcal{O}_X^{\text{add}} & \xlongequal{\quad} & \mathcal{O}_X^{\text{add}} \\
 \uparrow & & \uparrow \\
 \text{fib}(\sigma) & \longrightarrow & K_{X/S}^{(1)} \xrightarrow{\simeq} \text{Real}_X^{\text{st}}(L_{X/S}). \\
 & \searrow \lambda_{X/S}^1 &
 \end{array}$$

After dependent product to S , the bottom row becomes the first-order linearization of intrinsic one-forms, and the degree-zero differential maps to the stabilized universal derivation.

This section is restricted to spectral schemes so that $L_{X/S} \in \text{QCoh}(X)$ is available. The first structured infinitesimal neighbourhood, its base-change behaviour, and the identification of the difference of its two retractions with the universal relative derivation were constructed in Chapter 2. We use those results here as the first-order comparison interface. Write

$$X \xrightarrow{i_1} D_{X/S}^{(1)} \xrightarrow{\nu_1} X \times_S X$$

for that first structured infinitesimal neighbourhood, and

$$h_0 := \text{pr}_1 \nu_1, \quad h_1 := \text{pr}_2 \nu_1 : D_{X/S}^{(1)} \rightrightarrows X$$

for its two retractions.

Proposition 4.8.1 (First structured infinitesimal neighbourhood maps to the de Rham relation). *There is a canonical map over $X \times_S X$*

$$\phi_1 : D_{X/S}^{(1)} \longrightarrow \mathcal{R}_{X/S,1}^{\text{dR}}$$

whose composites with the source and target projections are h_0 and h_1 . It is canonical and compatible with arbitrary Cartesian base change of spectral schemes.

Proof.

- (1) **Establish the indicated equivalence.** The structured immersion $i_1 : X \rightarrow D_{X/S}^{(1)}$ is a square-zero infinitesimal substitution, hence a de Rham equivalence over S by Proposition 4.3.3.
- (2) **Apply the indicated result.** Apply Proposition 4.3.6 to

$$X \longrightarrow D_{X/S}^{(1)} \longrightarrow X \times_S X \longrightarrow S.$$

- (3) **Establish the next claim.** The resulting target de Rham neighbourhood is $\text{Nbh}_{X/S}^{\text{dR}}(X \times_S X)$, which is canonically $\mathcal{R}_{X/S,1}^{\text{dR}}$ by Theorem 4.3.7.

- (4) **Construct the indicated map.** The two projection identities follow from the defining fibre-product map.
- (5) **Apply naturality.** Naturality is inherited from the first-neighbourhood base-change theorem and the de Rham neighbourhood construction.

□

4.8.1 The first-order function coefficient and realized cotangent complex

Construction 4.8.1 (First-order function coefficient). Put

$$D := D_{X/S}^{(1)}.$$

By base change of the additive scalar,

$$h_0^* \mathcal{O}_X^{\text{add}} \simeq \mathcal{O}_D^{\text{add}}.$$

Form the first-order function object over the centre

$$\mathcal{F}_{X/S}^{(1)} := \Pi_{h_0} \mathcal{O}_D^{\text{add}} \simeq \Pi_{h_0} h_0^* \mathcal{O}_X^{\text{add}} \in \mathcal{D}_{\text{Sp}/X}.$$

Evaluation along the section $i_1 : X \rightarrow D$ gives

$$r_0 : \mathcal{F}_{X/S}^{(1)} \longrightarrow \mathcal{O}_X^{\text{add}},$$

while the unit of $h_0^* \dashv \Pi_{h_0}$ gives the constant-function section

$$s_0 : \mathcal{O}_X^{\text{add}} \longrightarrow \mathcal{F}_{X/S}^{(1)}.$$

Since $h_0 i_1 \simeq \text{id}_X$, the adjunction identities give

$$r_0 s_0 \simeq \text{id}.$$

Define the *first-order function coefficient*

$$K_{X/S}^{(1)} := \text{fib}(r_0).$$

Thus the splitting determined by s_0 gives a canonical stable equivalence

$$\mathcal{F}_{X/S}^{(1)} \simeq \mathcal{O}_X^{\text{add}} \oplus K_{X/S}^{(1)}.$$

Proposition 4.8.2 (First-order coefficient and the realized cotangent complex). *There is a canonical equivalence in $\mathcal{D}_{\text{Sp}/X}$*

$$\xi_{X/S} : K_{X/S}^{(1)} \xrightarrow{\simeq} \text{Real}_X^{\text{st}}(L_{X/S}),$$

compatible with Cartesian base change of spectral schemes. Consequently, for every affine spectral point $a : T \rightarrow X$,

$$\Gamma^{\text{st}}(T, a^* K_{X/S}^{(1)}) \xrightarrow{\simeq} \Gamma(T, a^* L_{X/S}),$$

and

$$\Gamma^{\text{st}}(S, \Pi_p K_{X/S}^{(1)}) \xrightarrow{\simeq} \Gamma(X, L_{X/S}).$$

Proof.

- (1) **Fix the indicated data.** Let $a : T \rightarrow X$ be an affine spectral point and form

$$\begin{array}{ccc} D_T := T \times_{X, h_0} D & \xrightarrow{\tilde{a}} & D \\ h_{0, T} \downarrow & & \downarrow h_0 \\ T & \xrightarrow{a} & X. \end{array}$$

- (2) **Establish the next claim.** Right Beck–Chevalley and base change of the additive scalar give

$$a^* \mathcal{F}_{X/S}^{(1)} \simeq \Pi_{h_{0, T}} \mathcal{O}_{D_T}^{\text{add}}.$$

- (3) **Establish the next claim.** The retraction identity $h_0 i_1 \simeq \text{id}_X$ determines a canonical centre

$$i_T : T \longrightarrow D_T, \quad i_T = (\text{id}_T, i_1 a).$$

- (4) **Establish the next claim.** The square

$$\begin{array}{ccc} T & \xrightarrow{i_T} & D_T \\ a \downarrow & & \downarrow \tilde{a} \\ X & \xrightarrow{i_1} & D \end{array}$$

is Cartesian.

- (5) **Establish the next claim.** Indeed,

$$\begin{aligned} D_T \times_D X &\simeq (T \times_X D) \times_D X \\ &\simeq T \times_X X \\ &\simeq T, \end{aligned}$$

where the middle fibre product uses $h_0 i_1 \simeq \text{id}_X$.

- (6) **Conclude the stated claim.** Thus this is precisely geometric Cartesian base change of the structured square-zero centre $i_1 : X \rightarrow D$.
- (7) **Invoke the cited result.** By 2.7.2, i_T is therefore canonically a structured square-zero extension with coefficient

$$a^* L_{X/S},$$

as supplied by the structured base-change construction.

- (8) **Use the stated hypothesis.** Since T is affine and i_T is a square-zero infinitesimal substitution, 2.9.2 makes D_T affine. Write

$$T = \text{Spec } A, \quad D_T = \text{Spec } A'.$$

- (9) **Construct the indicated map.** The opposite coordinate map

$$A' \longrightarrow A$$

is a structured spectral square-zero extension with coefficient $\Gamma(T, a^* L_{X/S})$.

- (10) **Invoke the cited result.** By 2.7.1, its homotopy fibre is canonically

$$\Gamma(T, a^* L_{X/S}).$$

- (11) **Establish the next claim.** Using the adjunction $h_{0,T}^* \dashv \Pi_{h_{0,T}}$ and Proposition 4.4.11,

$$\begin{aligned} \Gamma^{\text{st}}(T, a^* \mathcal{F}_{X/S}^{(1)}) &\simeq \Gamma^{\text{st}}(D_T, \mathcal{O}_{D_T}^{\text{add}}) \\ &\simeq A', \end{aligned}$$

while

$$\Gamma^{\text{st}}(T, \mathcal{O}_T^{\text{add}}) \simeq A.$$

- (12) **Identify the stated objects.** Under these identifications, the pulled-back evaluation map $a^* r_0$ is exactly the square-zero augmentation $A' \rightarrow A$.

- (13) **Identify the relevant fibre.** Stable sections preserve fibres, so

$$\begin{aligned} \Gamma^{\text{st}}(T, a^* K_{X/S}^{(1)}) &\simeq \text{fib}(A' \rightarrow A) \\ &\simeq \Gamma(T, a^* L_{X/S}). \end{aligned}$$

- (14) **Establish the next claim.** On the other hand, by the defining affine-point formula for stable realization,

$$\Gamma^{\text{st}}(T, a^* \text{Real}_X^{\text{st}}(L_{X/S})) \simeq \Gamma(T, a^* L_{X/S}).$$

- (15) **Construct the indicated map.** Every comparison above is functorial under morphisms of affine points.

- (16) **Conclude the stated claim.** Hence the stable affine-point presentation of Proposition 4.4.2 identifies the two sheaves of spectra and produces the canonical equivalence

$$\xi_{X/S} : K_{X/S}^{(1)} \xrightarrow{\simeq} \text{Real}_X^{\text{st}}(L_{X/S}).$$

- (17) **Establish the next claim.** Geometric base change of the first structured infinitesimal neighbourhood, cotangent base change, right Beck–Chevalley, and base change of Real^{st} show that $\xi_{X/S}$ is compatible with Cartesian base change of spectral schemes.

- (18) **Establish the next claim.** The affine-point formula is the first displayed consequence.

- (19) **Establish the next claim.** For the second, apply Π_p and stable global sections:

$$\begin{aligned} \Gamma^{\text{st}}(S, \Pi_p K_{X/S}^{(1)}) &\simeq \Gamma^{\text{st}}(X, K_{X/S}^{(1)}) \\ &\simeq \Gamma^{\text{st}}(X, \text{Real}_X^{\text{st}}(L_{X/S})) \\ &\simeq \Gamma(X, L_{X/S}), \end{aligned}$$

where the first equivalence is $p^* \dashv \Pi_p$ and the last is Proposition 4.4.8.

□

Construction 4.8.2 (Stabilized universal derivation over the base). The stabilized universal derivation over the base is obtained from the additive difference of the two first-neighbourhood retractions after dependent product to S .

Put

$$k := ph_0 \simeq ph_1 : D \longrightarrow S$$

and

$$\mathcal{E}_D := \Pi_k \mathcal{O}_D^{\text{add}} \in \mathcal{D}_{\text{Sp}/S}.$$

Composition of dependent products gives canonical equivalences

$$\mathcal{E}_D \simeq \Pi_p \Pi_{h_i} \mathcal{O}_D^{\text{add}}, \quad i = 0, 1.$$

For each i , the unit of $h_i^* \dashv \Pi_{h_i}$, followed by Π_p , gives

$$\bar{s}_i : \Pi_p \mathcal{O}_X^{\text{add}} \longrightarrow \mathcal{E}_D.$$

Restriction along $i_1 : X \rightarrow D$ gives

$$\bar{r} : \mathcal{E}_D \longrightarrow \Pi_p \mathcal{O}_X^{\text{add}}.$$

Since $h_i i_1 \simeq \text{id}_X$, functoriality of the adjunction units gives canonical homotopies

$$\bar{r} \bar{s}_i \simeq \text{id}_{\Pi_p \mathcal{O}_X^{\text{add}}}, \quad i = 0, 1.$$

Using the h_0 -description of $\mathcal{E}_D$, exactness of Π_p , and Proposition 4.8.2, we obtain canonical equivalences

$$\text{fib}(\bar{r}) \simeq \Pi_p K_{X/S}^{(1)} \xrightarrow{\sim} \Pi_p \text{Real}_X^{\text{st}}(L_{X/S}).$$

Therefore the additive difference

$$\bar{s}_1 - \bar{s}_0 : \Pi_p \mathcal{O}_X^{\text{add}} \longrightarrow \mathcal{E}_D$$

has canonically null composite with $\bar{r}$ and hence determines, through the stable fibre universal property, a canonical morphism

$$d_{X/S}^{\text{univ, st}} : \Pi_p \mathcal{O}_X^{\text{add}} \longrightarrow \Pi_p \text{Real}_X^{\text{st}}(L_{X/S}).$$

Proposition 4.8.3 (Affine identification of the stabilized derivation). *Suppose $p : X \rightarrow S$ is affine, with*

$$S = \text{Spec } R, \quad X = \text{Spec } A.$$

Then the map on stable global sections induced by $d_{X/S}^{\text{univ, st}}$ is the universal spectral derivation

$$d_{A/R} : A \longrightarrow L_{A/R}.$$

Thus, after restricting a morphism of spectral schemes to a compatible affine pair, the corresponding affine construction recovers the universal relative derivation.

Proof.

(1) **Construct the indicated map.** For the affine morphism

$$X = \text{Spec } A \longrightarrow S = \text{Spec } R,$$

- (2) **Identify the stated objects.** Chapter 2, 2.11.3, identifies the first structured infinitesimal neighbourhood after splitting by h_0 with a square-zero algebra

$$A' \simeq A \oplus L_{A/R}$$

for which the two algebra sections corresponding contravariantly to h_0 and h_1 are

$$\alpha_0(a) = (a, 0), \quad \alpha_1(a) = (a, d_{A/R}a).$$

- (3) **Identify the stated objects.** Under Proposition 4.4.11, $\bar{s}_0$ and $\bar{s}_1$ identify with these two sections.
- (4) **Construct the indicated map.** Their additive difference is

$$a \longmapsto (0, d_{A/R}a).$$

- (5) **Use the stated construction.** Projection to the evaluation fibre gives $d_{A/R}$, and Proposition 4.8.2 identifies that fibre with the stable realization of the cotangent coefficient.
- (6) **Conclude the stated claim.** Thus the resulting map on stable global sections is exactly the universal spectral derivation.

□

4.8.2 First-order restriction of intrinsic one-forms

Use the degree-one centred coefficient $C_{\mathrm{dR}}^{1,\mathrm{ctr}}(X/S)$ of Proposition 4.7.1; here $v_1 = \mathbf{s} : \mathcal{R}_{X/S,1}^{\mathrm{dR}} \rightarrow X$. The diagonal $X \rightarrow \mathcal{R}_{X/S,1}^{\mathrm{dR}}$ induces

$$\sigma : C_{\mathrm{dR}}^{1,\mathrm{ctr}}(X/S) \longrightarrow \mathcal{O}_X^{\mathrm{add}}.$$

Since the structure map of the de Rham relation is $p\mathbf{s}$,

$$C_{\mathrm{dR}}^1(X/S) \simeq \Pi_p C_{\mathrm{dR}}^{1,\mathrm{ctr}}(X/S), \quad C_{\mathrm{dR}}^0(X/S) \simeq \Pi_p \mathcal{O}_X^{\mathrm{add}}.$$

Because Π_p preserves fibres,

$$\Omega_{\mathrm{dR}}^1(X/S) \simeq \Pi_p \mathrm{fib}(\sigma).$$

Restriction along $\phi_1 : D_{X/S}^{(1)} \rightarrow \mathcal{R}_{X/S,1}^{\mathrm{dR}}$ gives a morphism

$$C_{\mathrm{dR}}^{1,\mathrm{ctr}}(X/S) \longrightarrow \mathcal{F}_{X/S}^{(1)}.$$

It intertwines σ with the centre evaluation $r_0 : \mathcal{F}_{X/S}^{(1)} \rightarrow \mathcal{O}_X^{\mathrm{add}}$, and hence induces

$$\mathrm{fib}(\sigma) \longrightarrow K_{X/S}^{(1)} \xrightarrow{\xi_{X/S}} \mathrm{Real}_X^{\mathrm{st}}(L_{X/S}).$$

Denote this composite by

$$\lambda_{X/S}^1 : \mathrm{fib}(\sigma) \longrightarrow \mathrm{Real}_X^{\mathrm{st}}(L_{X/S}).$$

Theorem 4.8.1 (First-order linearization). *Applying Π_p to $\lambda_{X/S}^1$ gives a canonical morphism*

$$\lambda_{X/S}^1 : \Omega_{\mathrm{dR}}^1(X/S) \longrightarrow \Pi_p \operatorname{Real}_X^{\mathrm{st}}(L_{X/S}).$$

It satisfies

$$\lambda_{X/S}^1 \circ d_{\mathrm{dR}}^0 \simeq d_{X/S}^{\mathrm{univ}, \mathrm{st}},$$

equivalently the triangle

$$\begin{array}{ccc} \Omega_{\mathrm{dR}}^0(X/S) & \xrightarrow{d_{\mathrm{dR}}^0} & \Omega_{\mathrm{dR}}^1(X/S) \\ & \searrow d_{X/S}^{\mathrm{univ}, \mathrm{st}} & \downarrow \lambda_{X/S}^1 \\ & & \Pi_p \operatorname{Real}_X^{\mathrm{st}}(L_{X/S}) \end{array}$$

commutes up to the canonical homotopy,

where the right-hand side is the base-valued stabilized universal derivation of Construction 4.8.2.

Proof.

- (1) **Establish the next claim.** Under the ordered face convention, the degree-zero boundary is the typed scalar-restriction difference

$$\operatorname{res}_{\mathbf{t}} - \operatorname{res}_{\mathbf{s}}.$$

- (2) **Establish the next claim.** Restriction along ϕ_1 replaces the target and source projections by h_1 and h_0 , respectively.
- (3) **Construct the indicated map.** After dependent product to S , these two restrictions are precisely the maps $\bar{s}_1$ and $\bar{s}_0$ into the common stable function object $\mathcal{E}_D$.
- (4) **Identify the relevant fibre.** Their difference factors through the evaluation fibre

$$\operatorname{fib}(\bar{r}) \simeq \Pi_p K_{X/S}^{(1)} \xrightarrow{\simeq} \Pi_p \operatorname{Real}_X^{\mathrm{st}}(L_{X/S}),$$

and by Construction 4.8.2 this factor is $d_{X/S}^{\mathrm{univ}, \mathrm{st}}$.

- (5) **Construct the indicated map.** The map on the same fibre induced by restriction of intrinsic one-forms is $\lambda_{X/S}^1$, giving the displayed identity.
- (6) **Apply the cited proposition.** Proposition 4.8.3 identifies the resulting map on compatible affine charts with the universal relative derivation.

□

Remark 4.8.1 (Higher-order kernel of first-order linearization). The morphism $\lambda_{X/S}^1$ records restriction of normalized degree-one de Rham functions to the first structured infinitesimal neighbourhood. Higher-order finite-infinitesimal functions can lie in its kernel, as the quadratic example below shows.

Proposition 4.8.4 (A quadratic class beyond first order). *Let*

$$S = \operatorname{Spec} H\mathbb{Q}, \quad X = \operatorname{Spec} H(\mathbb{Q}[t]).$$

Then the map

$$\pi_0 \Gamma^{\mathrm{st}}(S, \Omega_{\mathrm{dR}}^1(X/S)) \longrightarrow \pi_0 \Gamma(X, L_{X/S})$$

induced by $\lambda_{X/S}^1$ has nonzero kernel.

Proof.

- (1) **Fix the indicated data.** Let

$$t \in \pi_0 \Gamma^{\text{st}}(S, C_{\text{dR}}^0(X/S))$$

be the coordinate function.

- (2) **Introduce the notation.** Define its degree-one infinitesimal difference

$$\delta t := \text{res}_t(t) - \text{res}_s(t) \in \pi_0 \Gamma^{\text{st}}(S, C_{\text{dR}}^1(X/S)).$$

- (3) **Fix the indicated data.** Let

$$\sigma : C_{\text{dR}}^1(X/S) \longrightarrow C_{\text{dR}}^0(X/S)$$

be the degree-one codegeneracy induced by the diagonal.

- (4) **Establish the next claim.** The simplicial identities give canonical homotopies

$$\sigma \text{res}_t \simeq \text{id} \simeq \sigma \text{res}_s.$$

- (5) **Conclude the stated claim.** Hence $\sigma(\delta t)$ is canonically null.

- (6) **Use the stated construction.** Multiplication in the commutative algebra $C_{\text{dR}}^1(X/S)$ therefore gives

$$c_2 := (\delta t)^2$$

together with the induced canonical nullhomotopy

$$\sigma(c_2) \simeq 0.$$

- (7) **Invoke the cited result.** By the stable fibre description

$$\Omega_{\text{dR}}^1(X/S) \simeq \text{fib}(C_{\text{dR}}^1(X/S) \rightarrow C_{\text{dR}}^0(X/S)),$$

this pair determines a canonical class

$$\omega_2 \in \pi_0 \Gamma^{\text{st}}(S, \Omega_{\text{dR}}^1(X/S)).$$

- (8) **Conclude the stated claim.** Thus the normalization nullhomotopy is determined by the displayed fibre construction.

- (9) **Introduce the notation.** Put

$$Y = X \times_S X = \text{Spec } H(\mathbb{Q}[t_0, t_1]), \quad u = t_1 - t_0,$$

and let

$$T_2 := \text{Spec } H(\mathbb{Q}[t, u]/(u^3))$$

map to Y by

$$t_0 \longmapsto t, \quad t_1 \longmapsto t + u.$$

- (10) **Establish the next claim.** The centre $X \rightarrow T_2$ factors through

$$\mathbb{Q}[t, u]/(u^3) \longrightarrow \mathbb{Q}[t, u]/(u^2) \longrightarrow \mathbb{Q}[t],$$

whose successive kernels $(u^2)/(u^3)$ and $(u)/(u^2)$ are square-zero.

- (11) **Conclude the stated claim.** Hence $X \rightarrow T_2$ is a finite spectral infinitesimal substitution, and Proposition 4.3.6 supplies

$$T_2 \longrightarrow \mathcal{R}_{X/S,1}^{\text{dR}}.$$

- (12) **Establish the next claim.** Under the represented function-spectrum formula, δt restricts to u , so c_2 restricts to

$$u^2 \neq 0 \quad \text{in} \quad \mathbb{Q}[t, u]/(u^3).$$

- (13) **Conclude the stated claim.** Therefore the image of ω_2 in degree-one cochains is nonzero, and hence $\omega_2 \neq 0$.

- (14) **Establish the next claim.** Restrict δt along the canonical first structured infinitesimal neighbourhood

$$D_{X/S}^{(1)} \longrightarrow \mathcal{R}_{X/S,1}^{\text{dR}}.$$

- (15) **Establish the next claim.** Under the splitting by h_0 , the difference of the two retractions takes values in the square-zero coefficient $L_{X/S}$.

- (16) **Invoke the cited result.** By 2.7.1, the induced nonunitary multiplication on that coefficient fibre is coherently null.

- (17) **Establish the next claim.** Consequently the square

$$(\delta t)^2$$

restricts canonically to zero on $D_{X/S}^{(1)}$.

- (18) **Use the stated hypothesis.** Since $\lambda_{X/S}^1$ is induced precisely by restriction of normalized degree-one functions to this first structured infinitesimal neighbourhood, it follows that

$$\lambda_{X/S}^1(\omega_2) = 0.$$

- (19) **Conclude the stated claim.** Thus ω_2 is a nonzero element of the displayed kernel.

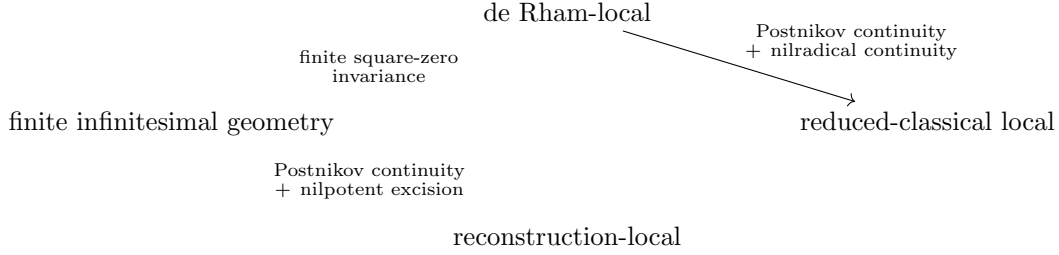
□

The cotangent comparison therefore has a precise scope. First-order restriction identifies the evaluation fibre with the realized cotangent complex and sends the intrinsic de Rham differential to the universal derivation. It does not collapse the finite-infinitesimal calculus to first order: the quadratic class above is detected by a second-order thickening and lies in the kernel of $\lambda_{X/S}^1$. The remaining comparison is of a different kind altogether. We now ask when finite-infinitesimal invariance, augmented by convergence, agrees with reduced-classical invariance.

4.9 Reduced-classical convergence of de Rham reflection

The first comparison restricted the de Rham relation to first order. The present comparison instead changes the localization. Finite-infinitesimal invariance already identifies every finite Postnikov or nilpotent stage. Reduced-classical reflection adds the convergence requirements for passage through an infinite Postnikov tower and an arbitrary nilradical.

This separates three locality conditions that are easy to conflate:



The lower and upper-right conditions encode different requirements: reconstruction uses nilpotent excision, whereas reduced-classical locality uses nilradical continuity after finite-infinitesimal invariance has already been imposed. The goal of this section is to construct the comparison $dR \rightarrow \text{Red}_{cl}$ and characterize exactly when it is invertible. For a connective commutative ring spectrum A , let r_{red} send

$$\text{Spec } A \mapsto \text{Spec } H((\pi_0 A)_{\text{red}}),$$

and define on affine tests

$$(\text{Red}_{cl} F)(\text{Spec } A) := F(\text{Spec } H((\pi_0 A)_{\text{red}})).$$

Lemma 4.9.1 (Reduced Zariski base change). *Let $A \rightarrow B$ represent an affine spectral Zariski open immersion. Then there is a canonical equivalence*

$$B \otimes_A H((\pi_0 A)_{\text{red}}) \xrightarrow{\sim} H((\pi_0 B)_{\text{red}}).$$

In particular, for a principal spectral localization $A \rightarrow A[f^{-1}]$,

$$A[f^{-1}] \otimes_A H((\pi_0 A)_{\text{red}}) \xrightarrow{\sim} H((\pi_0 A[f^{-1}])_{\text{red}}).$$

Reduced-classical reflection therefore carries every finite affine spectral Zariski Čech nerve to the Čech nerve of the corresponding reduced classical cover.

Proof.

(1) **Introduce the notation.** Put

$$R := \pi_0 A, \quad S := \pi_0 B.$$

(2) **Construct the indicated map.** The maps

$$A \longrightarrow B, \quad H(R_{\text{red}}) \longrightarrow H(S_{\text{red}})$$

induce a canonical B -algebra morphism

$$\theta : B \otimes_A H(R_{\text{red}}) \longrightarrow H(S_{\text{red}}).$$

- (3) **Establish the indicated equivalence.** We prove that θ is an equivalence after a finite affine Zariski covering family of $\mathrm{Spec} B$.
- (4) **Compute the principal-open case.** For $f \in R$, spectral localization is flat. The homotopy base-change formula for flat modules from 1.6.1 gives

$$\begin{aligned} A[f^{-1}] \otimes_A H(R_{\mathrm{red}}) &\simeq H(R_{\mathrm{red}}[\bar{f}^{-1}]) \\ &\simeq H((R[\bar{f}^{-1}])_{\mathrm{red}}) \\ &\simeq H((\pi_0 A[f^{-1}])_{\mathrm{red}}). \end{aligned}$$

Here the middle equivalence is the elementary identity that ordinary reduction commutes with localization.

- (5) **Invoke the cited result.** By 1.8.5 and 1.8.7, there are finitely many $f_i \in R$ such that

$$D_A^{\mathrm{sp}}(f_i) \longrightarrow \mathrm{Spec} A$$

factors through $\mathrm{Spec} B$, and the resulting family covers $\mathrm{Spec} B$.

- (6) **Use the stated hypothesis.** Since $\mathrm{Spec} B \rightarrow \mathrm{Spec} A$ is a monomorphism, the factorization implies

$$B \otimes_A A[f_i^{-1}] \xrightarrow{\simeq} A[f_i^{-1}].$$

- (7) **Conclude the stated claim.** Thus the morphisms $B \rightarrow A[f_i^{-1}]$ form a finite affine spectral Zariski cover of $\mathrm{Spec} B$.
- (8) **Establish the next claim.** Base-change θ along $B \rightarrow A[f_i^{-1}]$.
- (9) **Establish the next claim.** On the source, associativity and Step (2) give

$$\begin{aligned} (B \otimes_A H(R_{\mathrm{red}})) \otimes_B A[f_i^{-1}] &\simeq A[f_i^{-1}] \otimes_A H(R_{\mathrm{red}}) \\ &\simeq H((R[\bar{f}_i^{-1}])_{\mathrm{red}}). \end{aligned}$$

- (10) **Establish the next claim.** On degree zero the equality

$$S \otimes_R R[\bar{f}_i^{-1}] \simeq R[\bar{f}_i^{-1}]$$

follows from $B \otimes_A A[f_i^{-1}] \simeq A[f_i^{-1}]$.

- (11) **Use the stated hypothesis.** Since $B \rightarrow A[f_i^{-1}]$ is again a flat spectral localization, the flat homotopy base-change formula gives

$$\begin{aligned} H(S_{\mathrm{red}}) \otimes_B A[f_i^{-1}] &\simeq H(S_{\mathrm{red}} \otimes_S R[\bar{f}_i^{-1}]) \\ &\simeq H((S[\bar{f}_i^{-1}])_{\mathrm{red}}) \\ &\simeq H((R[\bar{f}_i^{-1}])_{\mathrm{red}}). \end{aligned}$$

- (12) **Identify the stated objects.** Under these identifications the base change of θ is the identity.
- (13) **Establish the indicated equivalence.** The covering family is jointly conservative on B -modules, so θ is an equivalence.

- (14) **Fix the indicated data.** Let $\{\mathrm{Spec}(B_i) \rightarrow \mathrm{Spec}(A)\}$ be a finite affine spectral Zariski covering family.
- (15) **Establish the next claim.** Every iterated intersection

$$B_{i_0} \otimes_A \cdots \otimes_A B_{i_n}$$

is again an affine spectral Zariski open over A .

- (16) **Identify the stated objects.** Applying the first assertion degreewise identifies its reduction with base change to $H(R_{\mathrm{red}})$.
- (17) **Identify the stated objects.** Associativity of relative tensor products then identifies these degreewise reductions with the iterated fibre products of the reduced classical opens.
- (18) **Establish the indicated equivalence.** The comparisons are natural in all face and degeneracy maps, so they assemble into the asserted equivalence of Čech nerves.

□

Proposition 4.9.1 (Reduced-classical localization). *The endofunctor $\mathrm{Red}_{\mathrm{cl}}$ preserves spectral Zariski sheaves and defines an accessible left-exact localization of $\mathcal{H}_{\mathrm{Sp}}$. Its local objects are precisely the F for which*

$$F(\mathrm{Spec} A) \xrightarrow{\sim} F(\mathrm{Spec} H((\pi_0 A)_{\mathrm{red}}))$$

for every connective A .

Proof.

- (1) **Fix the indicated data.** Let

$$\iota_{\mathrm{Zar}} : \mathcal{H}_{\mathrm{Sp}} \hookrightarrow \mathrm{PSh}(\mathrm{Aff}_{\mathrm{Sp}})$$

be the fully faithful inclusion and let r_{red}^* denote precomposition with reduced-classical reflection on affine spectral schemes.

- (2) **Establish the next claim.** At the presheaf level, r_{red}^* is accessible and preserves all limits.
- (3) **Apply the cited lemma.** Lemma 4.9.1 shows directly that it carries every finite affine spectral Zariski Čech nerve to the corresponding reduced classical Čech diagram.
- (4) **Use the stated hypothesis.** Since these finite affine covers generate the spectral Zariski topology of Chapter 1 and [37], $r_{\mathrm{red}}^* \iota_{\mathrm{Zar}}$ already lands in spectral Zariski sheaves.
- (5) **Conclude the stated claim.** Thus the sheaf endofunctor is canonically the composite

$$\mathrm{Red}_{\mathrm{cl}} \simeq a_{\mathrm{Zar}}^{\mathrm{sp}} r_{\mathrm{red}}^* \iota_{\mathrm{Zar}},$$

and hence is accessible.

- (6) **Use the stated identification.** Because the displayed presheaf precomposition preserves limits objectwise and the sheafification is left exact, $\mathrm{Red}_{\mathrm{cl}}$ preserves finite limits.

- (7) **Construct the indicated map.** The canonical map

$$A \longrightarrow H((\pi_0 A)_{\text{red}})$$

induces a coaugmentation $\eta^{\text{red}} : \text{id} \rightarrow \text{Red}_{\text{cl}}$.

- (8) **Establish the next claim.** The reduced-classical affine endofunctor is canonically idempotent, so both transformations

$$\text{Red}_{\text{cl}} \eta^{\text{red}}, \quad \eta^{\text{red}} \text{Red}_{\text{cl}} : \text{Red}_{\text{cl}} \longrightarrow \text{Red}_{\text{cl}}^2$$

are equivalences.

- (9) **Conclude the stated claim.** Therefore the coaugmented endofunctor is an accessible idempotent localization.
- (10) **Establish the indicated equivalence.** Its finite-limit preservation makes it left exact, and its fixed objects are exactly those satisfying the displayed affine-test equivalences.

□

Lemma 4.9.2 (Finite nilpotent quotients are de Rham-equivalent). *Let R be an ordinary commutative ring and $I \subseteq R$ nilpotent. Then*

$$HR \longrightarrow H(R/I)$$

is a finite composite of spectral square-zero extensions. Consequently every de Rham-local object evaluates $\text{Spec } HR$ and $\text{Spec } H(R/I)$ equivalently.

Proof.

- (1) **Establish the next claim.** This is the coordinate form of 2.8.5 from Chapter 2.
- (2) **Establish the next claim.** The proof of that proposition constructs the finite tower by powers of I , using 2.8.4 there to recognize each quotient stage as a spectral square-zero extension.
- (3) **Apply the cited theorem.** Theorem 4.2.2 then gives the displayed invariance.

□

4.9.1 From de Rham reflection to reduced-classical reflection

Reduced-classical reflection forgets both higher homotopy and nilpotent classical structure. The finite part of that forgetting is already invisible to de Rham-local objects. The following comparison therefore exists by the universal property of de Rham reflection.

Theorem 4.9.1 (Comparison with reduced-classical reflection). *Every de Rham generator is inverted by Red_{cl} . Hence the universal property of L^{dR} supplies a canonical transformation of left-exact ambient endofunctors*

$$\kappa : \text{dR} \longrightarrow \text{Red}_{\text{cl}}.$$

Thus the two localizations fit into

$$\text{dR} \xrightarrow{\kappa} \text{Red}_{\text{cl}}.$$

For $X \rightarrow S$, it induces

$$\kappa_{X/S} : X_{\mathrm{dR}/S} \longrightarrow S \times_{\mathrm{Red}_{\mathrm{cl}}(S)} \mathrm{Red}_{\mathrm{cl}}(X),$$

compatible with arbitrary base change.

The absolute comparison is compatible with the two localization units:

$$\begin{array}{ccc} X & \xrightarrow{\eta_X^{\mathrm{dR}}} & X_{\mathrm{dR}} \\ & \searrow & \downarrow \kappa_X \\ & & \mathrm{Red}_{\mathrm{cl}}(X). \end{array}$$

Proof.

- (1) **Fix the indicated data.** Let $A' \rightarrow A$ be a spectral square-zero extension, so the associated generator is

$$h_A \longrightarrow h_{A'}.$$

- (2) **Establish the next claim.** For a connective test algebra B , since the target $H((\pi_0 B)_{\mathrm{red}})$ is discrete, the 0-truncated adjunction of Chapter 1 gives

$$\begin{aligned} (\mathrm{Red}_{\mathrm{cl}} h_A)(\mathrm{Spec} B) &= \mathrm{Map}_{\mathrm{CAlg}(\mathrm{Sp})}(A, H((\pi_0 B)_{\mathrm{red}})) \\ &\simeq \mathrm{Hom}_{\mathrm{CRing}}(\pi_0 A, (\pi_0 B)_{\mathrm{red}}) \\ &\simeq \mathrm{Hom}_{\mathrm{CRing}}((\pi_0 A)_{\mathrm{red}}, (\pi_0 B)_{\mathrm{red}}). \end{aligned}$$

- (3) **Establish the next claim.** The same formula holds with A' in place of A .
- (4) **Establish the next claim.** A square-zero extension induces a square-zero quotient on π_0 , hence

$$(\pi_0 A')_{\mathrm{red}} \simeq (\pi_0 A)_{\mathrm{red}}.$$

- (5) **Conclude the stated claim.** Therefore $\mathrm{Red}_{\mathrm{cl}}$ sends the generator to a pointwise equivalence.
- (6) **Establish the next claim.** Initiality of Theorem 4.2.1 now constructs κ .
- (7) **Use pullback stability.** The relative comparison is obtained by applying the two left-exact constructions to the defining pullback of the relative de Rham object; its base-change coherence is inherited from finite-limit functoriality.

□

4.9.2 Finite collapse and the convergence gap

The comparison above is automatically an equivalence on tests for which the passage to the reduced classical ring is finite. This isolates the only remaining issue: convergence when the Postnikov tower or nilradical cannot be removed in finitely many square-zero stages.

Proposition 4.9.2 (Finite Postnikov and nilpotent collapse). *Let A be connective and eventually coconnective, and suppose the nilradical of $\pi_0 A$ is nilpotent. Then the coordinate morphism*

$$A \longrightarrow H((\pi_0 A)_{\text{red}})$$

is a finite composite of spectral square-zero extensions. Equivalently,

$$\text{Spec } H((\pi_0 A)_{\text{red}}) \longrightarrow \text{Spec } A$$

is a finite spectral infinitesimal substitution. Consequently every de Rham-local F satisfies

$$F(\text{Spec } A) \simeq F(\text{Spec } H((\pi_0 A)_{\text{red}})).$$

Proof.

- (1) **Invoke the cited result.** By 3.2.1 of Chapter 3, the multiplicative Postnikov tower has canonical structured square-zero transition maps.
- (2) **Establish the next claim.** Eventual coconnectivity makes the tower from A to $H\pi_0 A$ finite.
- (3) **Establish the next claim.** The nilpotent quotient from $\pi_0 A$ to its reduction is a finite composite of underlying spectral square-zero extensions by Lemma 4.9.2.
- (4) **Establish the next claim.** Concatenating these two finite chains proves the first assertion.
- (5) **Establish the next claim.** The canonical Postnikov stages and the finite nilpotent quotient tower together give the required finite composite.
- (6) **Establish the locality claim.** The de Rham-locality conclusion follows from Theorem 4.2.2.

□

4.9.3 Postnikov and nilradical convergence

We now formulate the two continuity conditions that close the finite-to- infinite gap. Their combination is exactly what distinguishes reduced-classical locality inside the de Rham-local sector.

Definition 4.9.1 (Nilradical continuity on affine tests). A de Rham-local object F is *nilradical-continuous on affine tests* if, for every ordinary ring R with nilradical N ,

$$\text{colim}_{\substack{J \subseteq N \\ J \text{ finitely generated}}} F(\text{Spec } H(R/J)) \xrightarrow{\simeq} F(\text{Spec } H(R/N)).$$

The transition maps are those induced by inclusion of finitely generated ideals.

Theorem 4.9.2 (Convergence criterion). *Let F be de Rham-local. The following are equivalent:*

1. F is Red_{cl} -local;
2. F is local for the Postnikov-continuity generators of 3.9.2 and is nilradical-continuous on affine tests.

Thus reduced-classical reflection is the further left-exact localization of the de Rham topos imposing Postnikov continuity together with nilradical continuity.

Proof.

- (1) **Fix the indicated data.** Let R be an ordinary ring with nilradical N , and let $J = (x_1, \dots, x_r) \subseteq N$ be finitely generated.
- (2) **Establish the next claim.** Choose $n_i \geq 1$ with $x_i^{n_i} = 0$.
- (3) **Establish the next claim.** Then

$$J^{n_1 + \dots + n_r} = 0,$$

so J is nilpotent. Hence

$$\sqrt{J} = N \quad \text{and} \quad (R/J)_{\text{red}} \simeq R/N.$$

- (4) **Treat the stated case.** If F is reduced-classical-local, then A , every $\tau_{\leq n}^{\text{CAlg}} A$, and $H((\pi_0 A)_{\text{red}})$ have the same reduced classical test.
- (5) **Conclude the stated claim.** Hence F is local for the Postnikov-continuity generators.
- (6) **Invoke the cited result.** By the preceding calculation, every finitely generated $J \subseteq N$ has

$$(R/J)_{\text{red}} \simeq R/N,$$

so the nilradical-continuity diagram is canonically constant up to equivalence.

- (7) **Prove the converse implication.** Conversely, by 3.2.2, every finite map

$$\tau_{\leq n}^{\text{CAlg}} A \longrightarrow H\pi_0 A$$

is a finite infinitesimal substitution.

- (8) **Use the stated construction.** De Rham locality therefore gives

$$F(\text{Spec } \tau_{\leq n}^{\text{CAlg}} A) \simeq F(\text{Spec } H\pi_0 A)$$

for every n .

- (9) **Establish the locality claim.** Locality for the Postnikov-continuity generators yields

$$F(\text{Spec } A) \simeq \varprojlim_n F(\text{Spec } \tau_{\leq n}^{\text{CAlg}} A) \simeq F(\text{Spec } H\pi_0 A).$$

- (10) **Establish the next claim.** For every finitely generated $J \subseteq N$, the first step shows that J is nilpotent.
- (11) **Apply the cited lemma.** Lemma 4.9.2 therefore gives

$$F(\text{Spec } HR) \simeq F(\text{Spec } H(R/J)).$$

- (12) **Establish the indicated equivalence.** The quotient maps $R \rightarrow R/J$ assemble these equivalences into a natural objectwise equivalence from the constant diagram at $F(\text{Spec } HR)$ to

$$J \longmapsto F(\text{Spec } H(R/J)).$$

- (13) **Establish the next claim.** The indexing poset of finitely generated ideals $J \subseteq N$ is filtered and has the initial object 0, so its nerve is contractible.
- (14) **Establish the next claim.** Its colimit is therefore canonically equivalent to $F(\mathrm{Spec} HR)$.
- (15) **Use the stated construction.** Nilradical continuity gives

$$F(\mathrm{Spec} HR) \simeq F(\mathrm{Spec} H(R/N)).$$

- (16) **Take the indicated specialization.** Taking $R = \pi_0 A$ and combining with the Postnikov comparison proves

$$F(\mathrm{Spec} A) \simeq F(\mathrm{Spec} H((\pi_0 A)_{\mathrm{red}})),$$

which is reduced-classical locality.

- (17) **Invoke the cited result.** By Theorem 4.9.1, every $\mathrm{Red}_{\mathrm{cl}}$ -local object is de Rham-local and

$$\mathrm{Red}_{\mathrm{cl}} \circ \mathrm{dR} \simeq \mathrm{Red}_{\mathrm{cl}}.$$

- (18) **Conclude the stated claim.** Therefore restriction of $\mathrm{Red}_{\mathrm{cl}}$ to the de Rham-local full subcategory is itself an accessible left-exact reflector, and the equivalence just proved identifies its fixed objects precisely by the two stated convergence properties.

□

Remark 4.9.1 (Comparison of the three left-exact reflections). The de Rham reflector, the reduced-classical reflector, and the reconstruction reflector of Chapter 3 encode three locality conditions. The functor dR inverts finite square-zero infinitesimal variation. On the de Rham-local subcategory, $\mathrm{Red}_{\mathrm{cl}}$ imposes the Postnikov and nilradical convergence conditions of Theorem 4.9.2. The reflector Recon_S is generated by Postnikov continuity together with nilpotent excision. These descriptions provide the comparison points used in the preceding results.

Corollary 4.9.1 (Invertibility criterion for the comparison). *For every $G \in \mathcal{H}_{\mathrm{Sp}}$,*

$$\kappa_G : \mathrm{dR}(G) \longrightarrow \mathrm{Red}_{\mathrm{cl}}(G)$$

is an equivalence if and only if $\mathrm{dR}(G)$ is local for the Postnikov-continuity generators of 3.9.2 and is nilradical-continuous on affine tests.

Proof.

- (1) **Establish the locality claim.** The functor $\mathrm{Red}_{\mathrm{cl}}$ is a further localization of the de Rham-local subcategory.
- (2) **Apply the indicated result.** Apply Theorem 4.9.2 to $\mathrm{dR}(G)$.

□

We can now state the comparison without collapsing distinct notions. De Rham reflection enforces invariance under finite infinitesimal variation. Reduced-classical reflection is the further localization obtained by adding Postnikov and nilradical continuity. Reconstruction, by contrast, combines Postnikov continuity with nilpotent excision. These three descriptions give the precise interfaces among the localizations and complete the comparison part of the chapter.

4.10 The spectral de Rham interface

All constructions named in the chapter are now active, so the global dependency diagrams can be read without introducing future notation. The first records how the effective relation feeds the stable descent calculus:

$$\begin{array}{ccc}
 X & \xrightarrow{\eta_{X/S}^{\text{dR}}} & X_{\text{dR}/S} & \rightsquigarrow & \mathcal{R}_{X/S, \bullet}^{\text{dR}} & \xrightarrow{\text{stable scalar functions}} & \mathcal{C}_{\text{dR}}^{\bullet}(X/S) \\
 \downarrow e_{X/S} & \nearrow m_{X/S} & & & & & \downarrow \text{Tot} \\
 Q_{X/S} & & & & (\Omega_{\text{dR}}^{\bullet}(X/S), d_{\text{dR}}) & \longleftarrow & \text{DR}_{X/S}.
 \end{array}$$

The second records the two comparison exits from that calculus: the reduced-classical comparison from the full reflection and the first-order cotangent comparison from intrinsic one-forms.

$$\begin{array}{ccc}
 X & \xrightarrow{\eta_{X/S}^{\text{dR}}} & X_{\text{dR}/S} & \xrightarrow{\kappa_{X/S}} & S \times_{\text{Red}_{\text{cl}}(S)} \text{Red}_{\text{cl}}(X) \\
 \downarrow e_{X/S} & & & & \\
 Q_{X/S} & \rightsquigarrow & \mathcal{R}_{X/S, \bullet}^{\text{dR}} & \longrightarrow & C_{\text{dR}}^{\bullet}(X/S) \\
 & & & & \downarrow \text{Tot} \\
 & & & & \text{DR}_{X/S} \\
 & & & & \rightsquigarrow \\
 & & & & \Omega_{\text{dR}}^1(X/S) \\
 & & & & \downarrow \lambda_{X/S}^1 \\
 & & & & \Pi_p \text{Real}_X^{\text{st}}(L_{X/S}).
 \end{array}$$

Theorem 4.10.1 (Structural summary of the spectral de Rham construction). *The constructions of this chapter assemble into the following sequence.*

- (i) *The underlying square-zero substitutions between affine spectral schemes generate a pullback-stable infinitesimal congruence W^{dR} on spectral contexts.*
- (ii) *The universal accessible left-exact localization at that congruence is the indexed de Rham reflection dR_S . Its local objects are characterized by finite-infinitesimal invariance, and its right orthogonal class consists of the formally étale morphisms.*
- (iii) *Effective image factorization of the relative de Rham unit produces the effective quotient $Q_{X/S}$, whose Čech nerve*

$$\mathcal{R}_{X/S, \bullet}^{\text{dR}},$$

is the de Rham relation groupoid generated on X .

- (iv) The stabilized slice categories of Chapter 3, with the dependent-adjoint structure established here, give the parameterized stable slice doctrine

$$S \longmapsto \mathcal{D}_{\mathrm{Sp}/S},$$

and stable realization interprets quasi-coherent coefficients inside that doctrine. The realized structure sheaf

$$\mathcal{O}_S^{\mathrm{add}}$$

is its distinguished stable additive scalar.

- (v) The cosimplicial object

$$C_{\mathrm{dR}}^\bullet(X/S)$$

consists of stable scalar functions on de Rham relation simplices. Comonadic effective descent identifies its totalization with invariant scalar functions on the effective quotient.

- (vi) The complete totalization filtration is the totalization filtration of that descent problem. Its normalized layers are the intrinsic de Rham forms, its connecting maps form the homotopy-coherent de Rham differential, and its higher stages classify higher closedness, exact-form lifts, and de Rham primitive torsors.

- (vii) Affine descent of the stable-section form-space functors constructs the internal comprehension objects

$$\mathbf{Form}_{\mathrm{dR},S}^q, \quad \mathbf{Form}_{\mathrm{dR},S}^{q,\mathrm{closed}},$$

together with the stable-section-to-comprehended transformations

$$\mathrm{cmp}^q, \quad \mathrm{cmp}^{q,\mathrm{closed}}.$$

These transformations are equivalences on affine contexts, and the displayed objects comprehend the corresponding affine-descended form judgments on arbitrary contexts.

- (viii) For a morphism $p : X \rightarrow S$ of spectral schemes, restriction to the canonical first structured infinitesimal neighbourhood gives the first-order coefficient

$$K_{X/S}^{(1)} = \mathrm{fib}(\mathcal{F}_{X/S}^{(1)} \longrightarrow \mathcal{O}_X^{\mathrm{add}})$$

and a canonical equivalence

$$K_{X/S}^{(1)} \xrightarrow{\cong} \mathrm{Real}_X^{\mathrm{st}}(L_{X/S}).$$

The resulting first-order linearization

$$\lambda_{X/S}^1 : \Omega_{\mathrm{dR}}^1(X/S) \longrightarrow \Pi_p \mathrm{Real}_X^{\mathrm{st}}(L_{X/S})$$

satisfies

$$\lambda_{X/S}^1 d_{\mathrm{dR}}^0 \simeq d_{X/S}^{\mathrm{univ},\mathrm{st}}.$$

- (ix) Reduced-classical reflection is a further left-exact localization of the de Rham-local sector characterized by Postnikov continuity and nilradical continuity. The comparison $\kappa : \mathrm{dR} \rightarrow \mathrm{Red}_{\mathrm{cl}}$ is invertible exactly on the corresponding convergence domain. This localization is distinct from the first-order cotangent comparison of the preceding item.

Proof.

- (1) **Collect localization, factorization, and the effective relation.** Items (i)–(iii) follow from Theorems 4.2.1, 4.2.6, 4.3.5, and 4.3.6, together with Theorem 4.2.3.
- (2) **Establish the next claim.** Item (iv) is Corollary 4.4.1 together with Construction 4.4.4 and Proposition 4.4.8.
- (3) **Establish the next claim.** Item (v) is Theorems 4.5.1, 4.5.2, and 4.5.5.
- (4) **Establish the next claim.** Item (vi) follows from Theorems 4.6.1 and 4.6.2, together with the subsequent closed-form and de Rham primitive constructions.
- (5) **Establish the next claim.** Item (vii) follows from Construction 4.7.1, Proposition 4.7.4, Construction 4.7.2, and Theorem 4.7.5.
- (6) **Establish the next claim.** Item (viii) follows from Proposition 4.8.2, Construction 4.8.2, and Theorem 4.8.1.
- (7) **Establish the next claim.** Item (ix) follows from Theorems 4.9.1 and 4.9.2.

□

The theorem above is the citation interface of the chapter. Later arguments can use the de Rham reflector, effective relation, stable scalar cochains, complete form filtration, comprehension objects, and the two comparison maps without reopening their constructions.

4.11 Chapter summary

The chapter now supplies a de Rham descent calculus for finite infinitesimal variation. The indexed left-exact reflector identifies precisely the contexts that are indistinguishable under finite infinitesimal substitution, and its right orthogonal class is the formally étale class. This is the localization interface used by every later construction in the chapter.

The relative unit contains two different pieces of information. Its target $X_{\mathrm{dR}/S}$ is the universal de Rham-local reflection, whereas its effective image $Q_{X/S}$ is the object actually presented by the generalized points of X . The de Rham relation is the effective equivalence-relation groupoid of $X \rightarrow Q_{X/S}$. On the formally smooth effectivity domain the distinction disappears, but the construction does not require that special case.

Stable scalar functions on this relation form a cosimplicial commutative algebra. Effective descent identifies its totalization with scalar functions on $Q_{X/S}$. The complete totalization filtration then exposes the successive descent layers: normalization gives the intrinsic de Rham forms, the connecting morphisms give the homotopy-coherent differential, and the remaining filtration data record multiplication, higher closedness, exact-form lifts, and primitive torsors. Formal étale descent and affine density turn the resulting affine stable-section semantics into internal comprehension objects on arbitrary contexts.

Two distinct comparison interfaces delimit what the resulting calculus remembers. Restriction to the first structured infinitesimal neighbourhood recovers the realized cotangent complex and the universal derivation, while the quadratic example shows that higher finite-infinitesimal information can survive outside that first-order image. Reduced-classical reflection tests a different boundary: it adds Postnikov and nilradical convergence to the de Rham-local sector. Later arguments may therefore use the cotangent comparison to extract first-order information and the reduced-classical comparison to test convergence, without conflating the two operations.

Chapter 5

Crystalline Jet Calculus and Formal Differential Equations

5.1 Orientation: from de Rham descent to crystalline jets

Chapter 4 turned finite infinitesimal variation into an effective descent relation while keeping that relation distinct from the universal de Rham reflection. The present chapter asks what this infinitesimal sameness does to objects varying over a context. For every relative context $X \rightarrow S$, the preceding chapter constructed the relative de Rham unit and its effective-image factorization

$$X \xrightarrow{e_{X/S}} Q_{X/S} \xrightarrow{m_{X/S}} X_{\mathrm{dR}/S},$$

together with the effective relation groupoid

$$\mathcal{R}_{X/S, \bullet}^{\mathrm{dR}} \simeq \check{C}_{\bullet}(X \xrightarrow{e_{X/S}} Q_{X/S}).$$

These two targets continue to play different roles. The universal object $X_{\mathrm{dR}/S}$ records the full reflection, whereas the effective image $Q_{X/S}$ is the quotient actually presented by points of X . Jets require effective transport, so this chapter works with the Čech groupoid of $X \rightarrow Q_{X/S}$; no second infinitesimal relation is introduced.

The existing infinitesimal neighbourhoods are already constructed; the remaining problem is to make their relation act on dependent objects. Pointwise de Rham neighbourhoods describe which generalized points are infinitesimally related. A differential-equation calculus must additionally transport an arbitrary object over X along those relations, compose that transport, and recognize when an equation is already stable under it. The dependent-adjoint triple of the effective quotient supplies exactly these operations:

$$\Sigma_{e_{X/S}} \dashv e_{X/S}^* \dashv \Pi_{e_{X/S}}.$$

Composing pullback with the two adjacent dependent adjoints produces the two endofunctors that will carry the calculus:

$$\mathrm{Disk}_{X/S}^{\mathrm{dR}, \infty} := e_{X/S}^* \Sigma_{e_{X/S}}, \quad J_{X/S}^{\infty} := e_{X/S}^* \Pi_{e_{X/S}}.$$

The left composite records infinitesimal extension along the relation and the right composite records sections over those infinitesimal variations. They are respectively the de Rham formal-disk monad

and the infinite-jet comonad, and the dependent-adjoint triple makes them adjoint:

$$\mathrm{Disk}_{X/S}^{\mathrm{dR},\infty} \dashv J_{X/S}^{\infty},$$

Mate formation then identifies disk actions with jet coactions. The main structural theorem adds the geometric content: effective Čech descent identifies either Eilenberg–Moore presentation with a single object over the quotient:

$$(\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}} \simeq \mathrm{Alg}_{\mathrm{Disk}^{\mathrm{dR},\infty}}((\mathcal{H}_{\mathrm{Sp}})_{/X}) \simeq \mathrm{Coalg}_{J^{\infty}}((\mathcal{H}_{\mathrm{Sp}})_{/X}).$$

This three-way identification is the chapter’s organizing result. It turns the same effective relation into a quotient-side descent object, a disk action, or a jet coaction; the later differential-equation calculus uses whichever presentation exposes the next operation most directly.

The route from the effective relation to these presentations is summarized by

$$\begin{array}{ccccc}
 & & \mathcal{R}_{X/S,\bullet}^{\mathrm{dR}} & & \\
 & \swarrow \text{left correspondence} & & \searrow \text{right correspondence} & \\
 \mathrm{Disk}_{X/S}^{\mathrm{dR},\infty} & & & & J_{X/S}^{\infty} \\
 \downarrow \text{algebras} & & & & \downarrow \text{coalgebras} \\
 \mathrm{Alg}_{\mathrm{Disk}^{\mathrm{dR},\infty}}((\mathcal{H}_{\mathrm{Sp}})_{/X}) & & & & \mathrm{Coalg}_{J^{\infty}}((\mathcal{H}_{\mathrm{Sp}})_{/X}) \\
 & \swarrow \sim & (\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}} & \searrow \sim & \\
 & & & &
 \end{array}$$

The display is a route map, not an additional layer of structure. Its lower equivalences are conclusions: the monad and comonad are constructed first, mate formation compares their Eilenberg–Moore objects, and effective descent then identifies both with the quotient slice.

Throughout the chapter, **crystalline transport** means the transport already encoded by the effective de Rham relation generated by

$$X \xrightarrow{e_{X/S}} Q_{X/S}.$$

It is therefore derived rather than independently chosen: a jet coaction transposes to relation transport, its coalgebra laws give the unit and cocycle, relation inversion gives the inverse, and effective descent identifies the whole structure with pullback from $Q_{X/S}$. The word *Cartan* will be used later for a different but related role: it describes an equation morphism that preserves this crystalline transport.

The construction sits at the intersection of two established descriptions of differential geometry. Jet-theoretic and synthetic approaches organize differential operators and formally integrable equations through infinite jets [38, 27, 1]; crystal and nonlinear D -geometric approaches organize infinitesimal compatibility by descent over de Rham objects [18, 5]. On the present spectral theorem domain, the square-zero-generated effective de Rham relation simultaneously produces the jet calculus over X and the quotient-side descent object over $Q_{X/S}$.

The chapter follows that identification in three phases. First, relative de Rham neighbourhoods are promoted from pointwise disks to functorial disk and jet endofunctors, and substitution is fixed

before any parameterized construction is used. Second, mate formation and effective descent identify disk actions, jet coactions, and crystalline transport; the resulting comonad then supplies holonomic sections, differential operators, generalized equations, solution objects, and Cartan prolongation. Third, base change, formally étale descent, and translation models expose the geometric content of the calculus. The final section keeps two boundaries separate: reduced-classical comparison is a convergence problem for a further localization, whereas finite order requires a new global filtration of the intrinsic relation.

The conventions on constructed coherence from the preceding chapters remain in force. In particular, once an adjunction, Beck–Chevalley comparison, descent limit, or relation groupoid has supplied the relevant higher structure, later prose uses that structure without reopening its coherence. Likewise, an *intrinsic formally integrable PDE* will mean an object already carrying effective de Rham descent, whereas a *generalized jet equation* will mean an ambient map into an infinite jet object that need not yet carry that descent. The intrinsic infinite jet functor J^∞ is constructed here; a global tower J^k is not used until a compatible finite-order globalization has actually been constructed.

5.2 Formal disks and the infinite jet adjunction

The effective relation is already available; this section turns it into an action on the whole slice over X . We first fix the dependent slice semantics and identify the pointwise formal disks. We then promote those disks to the formal-disk monad and infinite-jet comonad, identify their adjunction, and establish substitution. The reader should leave the section retaining the disk–jet adjunction and its fibrewise interpretation, not the intermediate bookkeeping used to construct them.

For orientation, the pointwise and functorial pictures fit into the following pullback architecture. For $a : T \rightarrow X$ and $E \rightarrow X$, the middle square is the formal disk over a ; the upper square pulls the coefficient object to that disk. Dependent product down the left column is the fibrewise operation that later computes $a^* J^\infty E$.

$$\begin{array}{ccc}
 \mathrm{ev}_a^* E & \longrightarrow & E \\
 \downarrow & & \downarrow \\
 T \times_{Q_{X/S}} X & \xrightarrow{\mathrm{ev}_a} & X \\
 \pi_a \downarrow & & \downarrow e_{X/S} \\
 T & \xrightarrow{e_{X/S} a} & Q_{X/S}
 \end{array}$$

5.2.1 Slice semantics and effective Čech descent

The effective quotient acts on coefficients through the slice semantics of the ambient spectral Zariski ∞ -topos. We fix those semantics once because every later jet comparison is inherited from the same dependent adjoints, Beck–Chevalley maps, and effective Čech descent.

Fix a morphism

$$p : X \longrightarrow S$$

in the spectral Zariski ∞ -topos.

For every map $f : U \rightarrow V$ in the ambient infinity-topos we use the dependent adjoint triple

$$\Sigma_f \dashv f^* \dashv \Pi_f.$$

Equivalently, the three functors fit into

$$(\mathcal{H}_{\mathrm{Sp}})_{/V} \begin{array}{c} \xleftarrow{\mathbb{M}_f^*} \\ \xleftarrow{\Sigma_f} \end{array} (\mathcal{H}_{\mathrm{Sp}})_{/U}$$

with the indicated adjunctions. Here Σ_f is composition with f , f^* is pullback, and Π_f is dependent product. These adjoints, together with their Beck–Chevalley transformations, are the only slice-theoretic input used below; see [36].

Theorem 5.2.1 (Slice descent and dependent adjoints). *The contravariant slice construction*

$$T \mapsto (\mathcal{H}_{\mathrm{Sp}})_{/T}$$

defines a functor

$$\mathcal{H}_{\mathrm{Sp}}^{\mathrm{op}} \longrightarrow \mathrm{Pr}^L$$

whose transition functor along $f : U \rightarrow V$ is pullback

$$f^* : (\mathcal{H}_{\mathrm{Sp}})_{/V} \rightarrow (\mathcal{H}_{\mathrm{Sp}})_{/U}.$$

For every such f , there is a dependent-adjoint triple

$$\Sigma_f \dashv f^* \dashv \Pi_f,$$

and the associated Beck–Chevalley transformations are compatible with Cartesian pasting.

Moreover, the slice construction carries colimits in $\mathcal{H}_{\mathrm{Sp}}$ to limits of infinity-categories. In particular, if

$$r : U \twoheadrightarrow T$$

is an effective epimorphism with Čech nerve $U_\bullet$, then restriction induces a canonical equivalence

$$(\mathcal{H}_{\mathrm{Sp}})_{/T} \xrightarrow{\cong} \mathrm{Tot}_{[n] \in \Delta} (\mathcal{H}_{\mathrm{Sp}})_{/U_n}.$$

All descent and transition maps used below are obtained from this contravariant slice construction.

Proof.

- (1) **Apply van Kampen slice descent.** Colimits in an infinity-topos are van Kampen, so for every small colimit presentation

$$T \simeq \mathrm{colim}_{i \in I} T_i$$

the canonical restriction functor is an equivalence

$$(\mathcal{H}_{\mathrm{Sp}})_{/T} \xrightarrow{\cong} \lim_{i \in I^{\mathrm{op}}} (\mathcal{H}_{\mathrm{Sp}})_{/T_i};$$

see [36, Theorem 6.1.3.9].

- (2) **Construct the dependent adjoints.** Universality of colimits makes each pullback functor f^* colimit-preserving, while local Cartesian closure gives the dependent adjoints

$$\Sigma_f \dashv f^* \dashv \Pi_f.$$

- (3) **Record Beck–Chevalley coherence.** Their Beck–Chevalley transformations and higher coherence are the canonical ones supplied by Cartesian pasting and mate formation.

- (4) **Present an effective epimorphism by its Čech nerve.** If $r : U \twoheadrightarrow T$ is effective, then

$$T \simeq \operatorname{colim}_{[n] \in \Delta^{\text{op}}} U_n.$$

- (5) **Identify effective Čech descent.** Applying the van Kampen equivalence gives

$$(\mathcal{H}_{\text{Sp}})_{/T} \simeq \lim_{[n] \in \Delta} (\mathcal{H}_{\text{Sp}})_{/U_n},$$

which is the asserted effective Čech descent equivalence. □

Let

$$\mathbf{u} : X \rightarrow \mathcal{R}_{X/S}^{\text{dR}}, \quad \text{inv} : \mathcal{R}_{X/S}^{\text{dR}} \xrightarrow{\simeq} \mathcal{R}_{X/S}^{\text{dR}}$$

denote the unit and inversion maps of the de Rham relation groupoid. Its first two nontrivial simplicial degrees are displayed by

$$\mathcal{R}_{X/S,2}^{\text{dR}} \begin{array}{c} \xrightarrow{d_0} \\ \xrightarrow{d_1} \\ \xrightarrow{d_2} \end{array} \mathcal{R}_{X/S}^{\text{dR}} \begin{array}{c} \xrightarrow{s} \\ \xrightarrow{t} \end{array} X.$$

The slice machinery is now fixed. The next step is geometric: identify what the effective relation looks like when centred at a generalized point.

5.2.2 Formal disks as pullbacks of the de Rham relation

Two points of X are related exactly when they have the same effective de Rham image. Pulling the source of that relation back along $a : T \rightarrow X$ therefore isolates the infinitesimal variations centred at a . The next theorem shows that this pullback is precisely the relative de Rham neighbourhood of the graph of a ; no additional notion of formal disk is introduced.

Let $a : T \rightarrow X$ be an object of $(\mathcal{H}_{\text{Sp}})_{/X}$, and let

$$\Gamma_a : T \longrightarrow T \times_S X$$

be its graph. We use the relative de Rham neighbourhood along Γ_a as the pointwise formal disk

$$\text{Nbh}_{T/S}^{\text{dR}}(T \times_S X)$$

taken along Γ_a . Write

$$\pi_a : \text{Nbh}_{T/S}^{\text{dR}}(T \times_S X) \rightarrow T, \quad \text{ev}_a : \text{Nbh}_{T/S}^{\text{dR}}(T \times_S X) \rightarrow X$$

for the maps induced by the two projections of $T \times_S X$.

Theorem 5.2.2 (Effective presentation of the de Rham formal disk). *There are canonical equivalences*

$$\mathrm{Nbh}_{T/S}^{\mathrm{dR}}(T \times_S X) \simeq T \times_{X_{\mathrm{dR}/S}} X \simeq T \times_{Q_{X/S}} X \simeq T \times_{X, \mathbf{s}} \mathcal{R}_{X/S}^{\mathrm{dR}}.$$

Under the final equivalence π_a is the pullback of $\mathbf{s}$, and ev_a is induced by $\mathbf{t}$. In particular π_a is an effective epimorphism. The relation presentation is summarized by the Cartesian square

$$\begin{array}{ccc} T \times_{X, \mathbf{s}} \mathcal{R}_{X/S}^{\mathrm{dR}} & \longrightarrow & \mathcal{R}_{X/S}^{\mathrm{dR}} \\ \pi_a \downarrow & & \downarrow \mathbf{s} \\ T & \xrightarrow{a} & X. \end{array}$$

Evaluation is obtained by following the upper horizontal map with $\mathbf{t}$.

Proof.

- (1) **Expand the relative de Rham neighbourhood.** By definition,

$$\mathrm{Nbh}_{T/S}^{\mathrm{dR}}(T \times_S X) = (T \times_S X) \times_{(T \times_S X)_{\mathrm{dR}/S}} T_{\mathrm{dR}/S}.$$

- (2) **Apply left exactness.** Left exactness of relative de Rham reflection gives

$$(T \times_S X)_{\mathrm{dR}/S} \simeq T_{\mathrm{dR}/S} \times_S X_{\mathrm{dR}/S}.$$

- (3) **Identify the reflected graph map.** Under this equivalence the map from $T_{\mathrm{dR}/S}$ is

$$(\mathrm{id}, a_{\mathrm{dR}/S}).$$

- (4) **Apply naturality and finite-limit pasting.** Naturality of the de Rham unit gives

$$a_{\mathrm{dR}/S} \eta_{T/S}^{\mathrm{dR}} \simeq \eta_{X/S}^{\mathrm{dR}} a,$$

so finite-limit pasting gives

$$\mathrm{Nbh}_{T/S}^{\mathrm{dR}}(T \times_S X) \simeq T \times_{X_{\mathrm{dR}/S}} X.$$

- (5) **Use the effective-image monomorphism.** Since

$$m_{X/S} : Q_{X/S} \longrightarrow X_{\mathrm{dR}/S}$$

is a monomorphism,

$$Q_{X/S} \times_{X_{\mathrm{dR}/S}} Q_{X/S} \simeq Q_{X/S}.$$

- (6) **Replace the reflected pullback by the effective quotient.** Both maps into $X_{\mathrm{dR}/S}$ factor through $Q_{X/S}$, hence

$$T \times_{X_{\mathrm{dR}/S}} X \simeq T \times_{Q_{X/S}} X.$$

- (7) **Identify the relation presentation.** Finally,

$$\mathcal{R}_{X/S}^{\mathrm{dR}} \simeq X \times_{Q_{X/S}} X$$

gives the relation presentation.

- (8) **Identify the structural maps and effectivity.** The description of π_a and ev_a follows from the two relation projections, and effectivity of π_a follows by base change from $e_{X/S} : X \rightarrow Q_{X/S}$.

□

Thus the pointwise formal disk has been identified intrinsically as the effective pullback $T \times_{Q_{X/S}} X$, with projection to the parameter and evaluation in X . What is still missing is functoriality in an arbitrary object over X .

5.2.3 Jet endofunctors from the effective quotient

The dependent-adjoint triple of $e_{X/S} : X \rightarrow Q_{X/S}$ packages these pointwise disks functorially. Pullback composed with the left adjoint gives the formal-disk endofunctor; pullback composed with the right adjoint gives the jet endofunctor. The remaining results in this subsection verify that these two endofunctors are exactly the left and right correspondence actions of the same Čech relation.

Endofunctors from the effective quotient

Definition 5.2.1 (De Rham formal-disk and jet endofunctors). Define endofunctors of $(\mathcal{H}_{\text{Sp}})_{/X}$ by

$$\text{Disk}_{X/S}^{\text{dR}, \infty} := e_{X/S}^* \Sigma_{e_{X/S}}, \quad J_{X/S}^{\infty} := e_{X/S}^* \Pi_{e_{X/S}}.$$

For $E \rightarrow X$, the object $J_{X/S}^{\infty} E \rightarrow X$ is the relative infinite jet object.

For a fixed relative context X/S , we suppress the subscript X/S from $\text{Disk}_{X/S}^{\text{dR}, \infty}$ and $J_{X/S}^{\infty}$ whenever the base is already active.

Theorem 5.2.3 (Effective-image formulas). *There are canonical equivalences*

$$\text{Disk}_{X/S}^{\text{dR}, \infty} \simeq (\eta_{X/S}^{\text{dR}})^* \Sigma_{\eta_{X/S}^{\text{dR}}}, \quad J_{X/S}^{\infty} \simeq (\eta_{X/S}^{\text{dR}})^* \Pi_{\eta_{X/S}^{\text{dR}}}.$$

These equivalences identify the monad and comonad structures constructed below. Hence the de Rham formal-disk and jet calculus depends only on the effective de Rham quotient $X \rightarrow Q_{X/S}$.

Proof.

- (1) **Record the effective-image factorization.** The effective-image factorization is the commutative triangle

$$\begin{array}{ccc} X & \xrightarrow{\eta_{X/S}^{\text{dR}}} & X_{\text{dR}/S} \\ & \searrow e_{X/S} & \nearrow m_{X/S} \\ & Q_{X/S} & \end{array}$$

$$\text{with } \eta_{X/S}^{\text{dR}} = m_{X/S} e_{X/S}.$$

- (2) **Contract the dependent sum along the monomorphism.** The monomorphism $m_{X/S} : Q_{X/S} \rightarrow X_{\text{dR}/S}$ has invertible diagonal, and hence

$$m_{X/S}^* \Sigma_{m_{X/S}} \simeq \text{id}_{(\mathcal{H}_{\text{Sp}})_{/Q_{X/S}}}.$$

(3) **State the dependent-product contraction.** Likewise

$$m_{X/S}^* \Pi_{m_{X/S}} \simeq \text{id}_{(\mathcal{H}_{\text{Sp}})_{/Q_{X/S}}}.$$

(4) **Compute the representing mapping spaces.** Indeed, for $Z, E \in (\mathcal{H}_{\text{Sp}})_{/Q_{X/S}}$,

$$\begin{aligned} \text{Map}_{Q_{X/S}}(Z, m_{X/S}^* \Pi_{m_{X/S}} E) &\simeq \text{Map}_{X_{\text{dR}/S}}(\Sigma_{m_{X/S}} Z, \Pi_{m_{X/S}} E) \\ &\simeq \text{Map}_{Q_{X/S}}(m_{X/S}^* \Sigma_{m_{X/S}} Z, E) \\ &\simeq \text{Map}_{Q_{X/S}}(Z, E). \end{aligned}$$

(5) **Apply Yoneda.** Yoneda gives the second equivalence.

(6) **Compose the dependent adjunctions.** Composition of dependent sums and products now gives

$$\begin{aligned} (\eta_{X/S}^{\text{dR}})^* \Sigma_{\eta_{X/S}^{\text{dR}}} &\simeq e_{X/S}^* m_{X/S}^* \Sigma_{m_{X/S}} \Sigma_{e_{X/S}} \\ &\simeq e_{X/S}^* \Sigma_{e_{X/S}}, \\ (\eta_{X/S}^{\text{dR}})^* \Pi_{\eta_{X/S}^{\text{dR}}} &\simeq e_{X/S}^* m_{X/S}^* \Pi_{m_{X/S}} \Pi_{e_{X/S}} \\ &\simeq e_{X/S}^* \Pi_{e_{X/S}}. \end{aligned}$$

(7) **Transport the monad and comonad structure.** Because the comparisons come from equivalences of the composing adjunctions, they identify the induced monad and comonad structures.

□

The effective-image calculation removes the universal reflected target from the active jet data: from this point through the jet construction, the map $e_{X/S} : X \rightarrow Q_{X/S}$ and its Čech relation are sufficient. The next formula rewrites the two endofunctors directly as left and right actions of that relation.

Proposition 5.2.1 (Relation-correspondence formulas). *There are canonical equivalences*

$$\text{Disk}_{X/S}^{\text{dR}, \infty} \simeq \Sigma_{\mathfrak{s}} \mathfrak{t}^*, \quad J_{X/S}^{\infty} \simeq \Pi_{\mathfrak{s}} \mathfrak{t}^*.$$

Transport along inversion also gives

$$\text{Disk}_{X/S}^{\text{dR}, \infty} \simeq \Sigma_{\mathfrak{t}} \mathfrak{s}^*.$$

Proof.

(1) **Identify the relation square as Cartesian.** The square

$$\begin{array}{ccc} \mathcal{R}_{X/S}^{\text{dR}} & \xrightarrow{\mathfrak{t}} & X \\ \mathfrak{s} \downarrow & & \downarrow e_{X/S} \\ X & \xrightarrow{e_{X/S}} & Q_{X/S} \end{array}$$

is Cartesian.

(2) **Apply Beck–Chevalley.** Left and right Beck–Chevalley give

$$e_{X/S}^* \Sigma_{e_{X/S}} \simeq \Sigma_{\mathbf{s}} \mathbf{t}^*, \quad e_{X/S}^* \Pi_{e_{X/S}} \simeq \Pi_{\mathbf{s}} \mathbf{t}^*.$$

(3) **Use relation inversion.** Relation inversion exchanges $\mathbf{s}$ and $\mathbf{t}$, yielding the final formula. □

The degree-one correspondence is the conceptual interface: $\text{Disk}^{\text{dR}, \infty}$ and J^∞ are the left and right dependent actions of the same relation. The iterated formulas below have one purpose: they verify that relation composition and units induce the monad/comonad structure coherently. The individual vertex notation need not be retained after that verification.

The simplicial relation and iterated correspondences

Proposition 5.2.2 (Iterated relation correspondences). *For every integer $n \geq 0$, let*

$$\mathbf{t}_n : \mathcal{R}_{X/S, n}^{\text{dR}} \rightarrow X$$

denote the last-vertex projection. Thus

$$\mathbf{t}_n(x_0, \dots, x_n) = x_n.$$

For $n = 0$,

$$v_0 = \mathbf{t}_0 = \text{id}_X,$$

and in degree one

$$v_1 = \mathbf{s}, \quad \mathbf{t}_1 = \mathbf{t}.$$

There are canonical equivalences

$$(\text{Disk}^{\text{dR}, \infty})^n \simeq \Sigma_{v_n} \mathbf{t}_n^*, \quad (J^\infty)^n \simeq \Pi_{v_n} \mathbf{t}_n^*,$$

compatible in n with the Cartesian pasting maps generated by insertion of the relation unit and composition of adjacent relation arrows.

Let

$$\text{rev}_n : \mathcal{R}_{X/S, n}^{\text{dR}} \xrightarrow{\simeq} \mathcal{R}_{X/S, n}^{\text{dR}}, \quad (x_0, \dots, x_n) \longmapsto (x_n, \dots, x_0)$$

be reversal. Then

$$v_n \text{rev}_n = \mathbf{t}_n, \quad \mathbf{t}_n \text{rev}_n = v_n.$$

Transport along rev_n identifies the right adjoint of $\Sigma_{v_n} \mathbf{t}_n^$ with $\Pi_{v_n} \mathbf{t}_n^*$. Hence*

$$(\text{Disk}^{\text{dR}, \infty})^n \dashv (J^\infty)^n$$

for every n . Under these adjunctions the simplicial structure maps inducing the monad structure on $\text{Disk}^{\text{dR}, \infty}$ are mates of the corresponding simplicial structure maps inducing the comonad structure on J^∞ . In particular, the unit and multiplication of $\text{Disk}^{\text{dR}, \infty}$ are mates of the counit and comultiplication of J^∞ , and the simplicial identities of $\mathcal{R}_{X/S, \bullet}^{\text{dR}}$ supply the remaining compatibility.

Proof.

- (1) **Check the initial degrees.** The assertion for $n = 0$ is tautological. For $n = 1$, the identities

$$v_1 = \mathbf{s}, \quad \mathbf{t}_1 = \mathbf{t}$$

reduce the statement to Proposition 5.2.1.

- (2) **Form the induction square.** Suppose the formulas have been constructed in degree n . Form the Cartesian square

$$\begin{array}{ccc} \mathcal{R}_{X/S, n+1}^{\mathrm{dR}} & \xrightarrow{\pi_{1\dots n+1}} & \mathcal{R}_{X/S, n}^{\mathrm{dR}} \\ \pi_{01} \downarrow & & \downarrow v_n \\ \mathcal{R}_{X/S, 1}^{\mathrm{dR}} & \xrightarrow{\mathbf{t}} & X, \end{array}$$

where

$$\pi_{01}(x_0, \dots, x_{n+1}) = (x_0, x_1), \quad \pi_{1\dots n+1}(x_0, \dots, x_{n+1}) = (x_1, \dots, x_{n+1}).$$

- (3) **Identify the first and last vertices.** By definition of the first and last vertices,

$$v_{n+1} = \mathbf{s} \pi_{01}, \quad \mathbf{t}_{n+1} = \mathbf{t}_n \pi_{1\dots n+1}.$$

- (4) **Apply left Beck–Chevalley.** Left Beck–Chevalley gives

$$\mathbf{t}^* \Sigma_{v_n} \simeq \Sigma_{\pi_{01}} \pi_{1\dots n+1}^*.$$

Therefore

$$\begin{aligned} \mathrm{Disk}^{\mathrm{dR}, \infty}(\mathrm{Disk}^{\mathrm{dR}, \infty})^n &\simeq \Sigma_{\mathbf{s}} \mathbf{t}^* \Sigma_{v_n} \mathbf{t}_n^* \\ &\simeq \Sigma_{\mathbf{s}} \Sigma_{\pi_{01}} \pi_{1\dots n+1}^* \mathbf{t}_n^* \\ &\simeq \Sigma_{v_{n+1}} \mathbf{t}_{n+1}^*. \end{aligned}$$

- (5) **Apply right Beck–Chevalley.** Right Beck–Chevalley similarly gives

$$\mathbf{t}^* \Pi_{v_n} \simeq \Pi_{\pi_{01}} \pi_{1\dots n+1}^*,$$

and hence

$$\begin{aligned} J^\infty(J^\infty)^n &\simeq \Pi_{\mathbf{s}} \mathbf{t}^* \Pi_{v_n} \mathbf{t}_n^* \\ &\simeq \Pi_{\mathbf{s}} \Pi_{\pi_{01}} \pi_{1\dots n+1}^* \mathbf{t}_n^* \\ &\simeq \Pi_{v_{n+1}} \mathbf{t}_{n+1}^*. \end{aligned}$$

- (6) **Complete the inductive correspondence.** This proves the iterated correspondence formulas.
- (7) **Identify the formal right adjoint.** The right adjoint of

$$\Sigma_{v_n} \mathbf{t}_n^*$$

is

$$\Pi_{\mathbf{t}_n} v_n^*.$$

- (8) **Record the reversal identities.** Reversal satisfies

$$v_n \operatorname{rev}_n = \mathbf{t}_n, \quad \mathbf{t}_n \operatorname{rev}_n = v_n,$$

and rev_n is an equivalence.

- (9) **Transport the right adjoint through reversal.** Composition of dependent products and transport along rev_n therefore give

$$\Pi_{\mathbf{t}_n} v_n^* \simeq \Pi_{v_n} \mathbf{t}_n^*.$$

Thus

$$(\operatorname{Disk}^{\operatorname{dR}, \infty})^n \dashv (J^\infty)^n.$$

- (10) **Match the degree-one structure maps.** In degree one, the relation unit $\mathbf{u} : X \rightarrow \mathcal{R}_{X/S,1}^{\operatorname{dR}}$ induces the monad unit by the dependent-sum unit and the comonad counit by the dependent-product counit. These transformations are mates.
- (11) **Match the degree-two structure maps.** In degree two, the middle face

$$d_1 : \mathcal{R}_{X/S,2}^{\operatorname{dR}} \rightarrow \mathcal{R}_{X/S,1}^{\operatorname{dR}}, \quad (x_0, x_1, x_2) \longmapsto (x_0, x_2),$$

induces the monad multiplication by contraction with the counit of

$$\Sigma_{d_1} \dashv d_1^*,$$

while it induces the comonad comultiplication by insertion of the unit of

$$d_1^* \dashv \Pi_{d_1}.$$

These transformations are mates under the displayed iterated adjunctions.

- (12) **Recover the higher coherences.** In higher degrees, all structural composites are obtained by deleting or repeating vertices of the same Čech simplex. Beck–Chevalley and mate formation are compatible with Cartesian pasting, so the two systems of transformations agree. The simplicial identities then supply the remaining monad–comonad compatibility. □

The iterated correspondence has now done its job: the simplicial operators of the Čech relation are identified with the structural maps of the disk monad and jet comonad. The next theorem extracts the pointwise consequence that will actually be used later.

Theorem 5.2.4 (Fibrewise interpretation of jets). *For $a : T \rightarrow X$ and $E \rightarrow X$, there is a canonical equivalence in the slice over T*

$$a^* J_{X/S}^\infty E \simeq \Pi_{\pi_a} \operatorname{ev}_a^* E.$$

Thus a generalized fibre of the jet object is the object of sections of E over the intrinsic de Rham formal disk centred at that generalized point.

Proof.

- (1) **Compute the pulled-back jet.** Pull back the formula

$$J^\infty \simeq \Pi_{\mathbf{s}} \mathbf{t}^*$$

along a . Proposition 5.2.2 identifies the pullback of $\mathbf{s}$ with π_a , and right Beck–Chevalley gives the result. □

Monad, comonad, and the disk–jet adjunction

The correspondence calculus already determines the structure maps geometrically. We now record the same structure in adjunction notation because later mate and Eilenberg–Moore arguments use the units and counits explicitly.

For the adjoint triple above, let

$$\eta^\Sigma : \mathrm{id}_{(\mathcal{H}_{\mathrm{Sp}})_{/X}} \longrightarrow e_{X/S}^* \Sigma_{e_{X/S}}, \quad \varepsilon^\Sigma : \Sigma_{e_{X/S}} e_{X/S}^* \longrightarrow \mathrm{id}_{(\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}}}$$

be the unit and counit of

$$\Sigma_{e_{X/S}} \dashv e_{X/S}^*,$$

and let

$$\eta^\Pi : \mathrm{id}_{(\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}}} \longrightarrow \Pi_{e_{X/S}} e_{X/S}^*, \quad \varepsilon^\Pi : e_{X/S}^* \Pi_{e_{X/S}} \longrightarrow \mathrm{id}_{(\mathcal{H}_{\mathrm{Sp}})_{/X}}$$

be the unit and counit of

$$e_{X/S}^* \dashv \Pi_{e_{X/S}}.$$

Proposition 5.2.3 (De Rham formal-disk monad). *The endofunctor $\mathrm{Disk}^{\mathrm{dR},\infty}$ carries the canonical monad structure*

$$(\mathrm{Disk}^{\mathrm{dR},\infty}, \eta^\Sigma, \mu), \quad \mu := e_{X/S}^* \varepsilon_{\Sigma_{e_{X/S}}}^\Sigma : \mathrm{Disk}^{\mathrm{dR},\infty} \mathrm{Disk}^{\mathrm{dR},\infty} \rightarrow \mathrm{Disk}^{\mathrm{dR},\infty}.$$

Under the relation formula, the unit is induced by $\mathbf{u} : X \rightarrow \mathcal{R}_{X/S}^{\mathrm{dR}}$ and the multiplication by

$$\mathrm{comp} : \mathcal{R}_{X/S}^{\mathrm{dR}} \times_{\mathbf{t}, X, \mathbf{s}} \mathcal{R}_{X/S}^{\mathrm{dR}} \rightarrow \mathcal{R}_{X/S}^{\mathrm{dR}}.$$

Proof.

- (1) **Identify the adjunction monad.** The monad is the monad of the adjunction $\Sigma_{e_{X/S}} \dashv e_{X/S}^*$.
- (2) **Translate the monad structure to the relation.** Under the Čech correspondence, its unit is the diagonal and its multiplication is relation composition.
- (3) **Verify the monad laws.** The monad laws are equivalently the unit and associativity identities of the relation groupoid.

□

Theorem 5.2.5 (Jet comonad). *The endofunctor J^∞ carries the canonical comonad structure*

$$(J^\infty, \varepsilon^\Pi, \delta^J), \quad \delta^J := e_{X/S}^* \eta_{\Pi_{e_{X/S}}}^\Pi : J^\infty \rightarrow J^\infty J^\infty.$$

Thus

$$\varepsilon_E^\Pi : J^\infty E \rightarrow E, \quad \delta_E^J : J^\infty E \rightarrow J^\infty J^\infty E$$

satisfy the counit and coassociativity laws supplied by the adjunction.

Theorem 5.2.6 (Adjoint-triple mate calculus). *The adjoint triple*

$$\Sigma_{e_{X/S}} \dashv e_{X/S}^* \dashv \Pi_{e_{X/S}}$$

canonically induces an adjunction

$$\mathrm{Disk}^{\mathrm{dR},\infty} = e_{X/S}^* \Sigma_{e_{X/S}} \dashv e_{X/S}^* \Pi_{e_{X/S}} = J^\infty.$$

For $Z, E \in (\mathcal{H}_{\mathrm{Sp}})_X$,

$$\begin{aligned} \mathrm{Map}_X(\mathrm{Disk}^{\mathrm{dR},\infty} Z, E) &= \mathrm{Map}_X(e_{X/S}^* \Sigma_{e_{X/S}} Z, E) \\ &\simeq \mathrm{Map}_{Q_{X/S}}(\Sigma_{e_{X/S}} Z, \Pi_{e_{X/S}} E) \\ &\simeq \mathrm{Map}_X(Z, e_{X/S}^* \Pi_{e_{X/S}} E) \\ &= \mathrm{Map}_X(Z, J^\infty E). \end{aligned}$$

Its unit is

$$\eta_Z^{\mathrm{adj}} : Z \xrightarrow{\eta_Z^\Sigma} e_{X/S}^* \Sigma_{e_{X/S}} Z \xrightarrow{e_{X/S}^* \eta_{\Sigma_{e_{X/S}} Z}^\Pi} e_{X/S}^* \Pi_{e_{X/S}} e_{X/S}^* \Sigma_{e_{X/S}} Z = J^\infty \mathrm{Disk}^{\mathrm{dR},\infty} Z,$$

and its counit is

$$\varepsilon_E^{\mathrm{adj}} : \mathrm{Disk}^{\mathrm{dR},\infty} J^\infty E = e_{X/S}^* \Sigma_{e_{X/S}} e_{X/S}^* \Pi_{e_{X/S}} E \xrightarrow{e_{X/S}^* \varepsilon_{\Pi_{e_{X/S}} E}^\Sigma} e_{X/S}^* \Pi_{e_{X/S}} E \xrightarrow{\varepsilon_E^\Pi} E.$$

Under mate correspondence for this composite adjunction, the monad unit

$$\eta^\Sigma : \mathrm{id} \rightarrow \mathrm{Disk}^{\mathrm{dR},\infty}$$

and the comonad counit

$$\varepsilon^\Pi : J^\infty \rightarrow \mathrm{id}$$

are mates. Likewise the monad multiplication

$$\mu : \mathrm{Disk}^{\mathrm{dR},\infty} \mathrm{Disk}^{\mathrm{dR},\infty} \rightarrow \mathrm{Disk}^{\mathrm{dR},\infty}$$

and the comonad comultiplication

$$\delta^J : J^\infty \rightarrow J^\infty J^\infty$$

are mates with respect to the induced adjunctions

$$(\mathrm{Disk}^{\mathrm{dR},\infty})^2 \dashv (J^\infty)^2 \quad \text{and} \quad \mathrm{Disk}^{\mathrm{dR},\infty} \dashv J^\infty.$$

These mate identifications are compatible with pasting.

Proof.

- (1) **Compose the two adjunctions.** The mapping-space equivalence is the composite of

$$\Sigma_{e_{X/S}} \dashv e_{X/S}^* \quad \text{and} \quad e_{X/S}^* \dashv \Pi_{e_{X/S}}.$$

The displayed formulas for η^{adj} and $\varepsilon^{\mathrm{adj}}$ are the unit and counit of the composite adjunction.

- (2) **Take homotopy-coherent mates.** In the ambient $(\infty, 2)$ -category of infinity-categories, mate formation identifies the relevant mapping spaces of natural transformations and is compatible with pasting. Applying that calculus to the unit and counit of $\Sigma_{e_{X/S}} \dashv e_{X/S}^* \dashv \Pi_{e_{X/S}}$ identifies η^Σ with ε^Π and $e_{X/S}^* \varepsilon_{\Sigma_{e_{X/S}}}^\Sigma$ with $e_{X/S}^* \eta_{\Pi_{e_{X/S}}}^\Pi$ under the induced adjunctions. The triangle identities supply the required comparisons. This is the adjunction calculus for infinity-categories; see [36, 34].

□

Corollary 5.2.1 (De Rham formal-disk–jet adjunction). *There is a canonical adjunction*

$$\mathrm{Disk}_{X/S}^{\mathrm{dR},\infty} \dashv J_{X/S}^{\infty}.$$

It is induced from the dependent-adjoint diagram

$$(\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}} \begin{array}{c} \xleftarrow{\Pi_{e_{X/S}}^*} \\ \xrightarrow{\Sigma_{e_{X/S}}} \end{array} (\mathcal{H}_{\mathrm{Sp}})_{/X}$$

by composing across the two adjunctions. Under the relation correspondence it may also be written

$$\begin{aligned} \mathrm{Map}_X(\mathrm{Disk}^{\mathrm{dR},\infty} Z, E) &\simeq \mathrm{Map}_{\mathcal{R}_{X/S}^{\mathrm{dR}}}(\mathfrak{t}^* Z, \mathfrak{s}^* E) \\ &\simeq \mathrm{Map}_{\mathcal{R}_{X/S}^{\mathrm{dR}}}(\mathfrak{s}^* Z, \mathfrak{t}^* E) \\ &\simeq \mathrm{Map}_X(Z, J^{\infty} E), \end{aligned}$$

where the middle equivalence is transport along inversion.

Proposition 5.2.4 (Maps from de Rham formal disks). *For $Z, E \in (\mathcal{H}_{\mathrm{Sp}})_{/X}$,*

$$\mathrm{Map}_X(Z, J^{\infty} E) \simeq \mathrm{Map}_X(\mathrm{Disk}^{\mathrm{dR},\infty} Z, E).$$

If $Z = a : T \rightarrow X$ is a generalized point, then under the inversion presentation

$$\mathrm{Disk}^{\mathrm{dR},\infty} \simeq \Sigma_{\mathfrak{t}} \mathfrak{s}^*$$

the object $\mathrm{Disk}^{\mathrm{dR},\infty}(a)$ is canonically the de Rham formal disk $\mathrm{Nbh}_{T/S}^{\mathrm{dR}}(T \times_S X)$ regarded over X by evaluation. Hence

$$\mathrm{Map}_X(a, J^{\infty} E) \simeq \mathrm{Map}_X(\mathrm{Nbh}_{T/S}^{\mathrm{dR}}(T \times_S X), E).$$

Proof.

(1) **Apply the disk–jet adjunction.** The first equivalence is the adjunction $\mathrm{Disk}^{\mathrm{dR},\infty} \dashv J^{\infty}$.

(2) **Use the inversion presentation.** For the second statement, the inversion form of 5.2.1 gives

$$\mathrm{Disk}^{\mathrm{dR},\infty}(a) \simeq \Sigma_{\mathfrak{t}} \mathfrak{s}^*(a).$$

(3) **Identify the generalized de Rham formal disk.** The pullback $\mathfrak{s}^*(a)$ is

$$T \times_{X,\mathfrak{s}} \mathcal{R}_{X/S}^{\mathrm{dR}} \simeq \mathrm{Nbh}_{T/S}^{\mathrm{dR}}(T \times_S X)$$

by 5.2.2, and postcomposition with $\mathfrak{t}$ is exactly the evaluation map of that disk.

□

The reader may now compress the jet construction to two facts:

$$\mathrm{Disk}_{X/S}^{\mathrm{dR},\infty} \dashv J_{X/S}^{\infty}, \quad a^* J^{\infty} E \simeq \Pi_{\pi_a} \mathrm{ev}_a^* E.$$

The remaining proposition records a closure property of the jet comonad needed for later pullback constructions; after that, substitution completes the section.

Proposition 5.2.5 (Limit preservation). *The functor J^∞ preserves all small limits. Consequently the forgetful functor*

$$U_J : \text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_{/X}) \rightarrow (\mathcal{H}_{\text{Sp}})_{/X}$$

creates all small limits.

Proof.

- (1) **Establish limit preservation.** Pullback preserves all limits and $\Pi_{e_{X/S}}$ is a right adjoint, hence $J^\infty = e_{X/S}^* \Pi_{e_{X/S}}$ preserves limits.
- (2) **Create limits in the coalgebra category.** The Eilenberg–Moore construction for a limit-preserving comonad [34] then gives the second assertion: form the limit of the underlying diagram and use $J^\infty(\lim E_i) \simeq \lim J^\infty E_i$ to assemble its contractibly unique coaction.

□

5.2.4 Substitution of the jet calculus

Because both jet endofunctors come from the same dependent-adjoint triple, substitution is the Beck–Chevalley comparison for base change of $e_{X/S}$. We establish that comparison now, before equations and solution objects use the calculus parametrically.

Theorem 5.2.7 (Base change). *Let $g : S' \rightarrow S$, put*

$$X' := X \times_S S',$$

and write

$$g_X : X' \rightarrow X, \quad g_Q : Q_{X'/S'} \rightarrow Q_{X/S}.$$

There is a canonical Cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g_X} & X \\ e_{X'/S'} \downarrow & & \downarrow e_{X/S} \\ Q_{X'/S'} & \xrightarrow{g_Q} & Q_{X/S}. \end{array}$$

Moreover,

$$Q_{X'/S'} \simeq Q_{X/S} \times_S S'.$$

Consequently there are canonical equivalences

$$g_X^* \text{Disk}_{X/S}^{\text{dR}, \infty} \simeq \text{Disk}_{X'/S'}^{\text{dR}, \infty} g_X^*, \quad g_X^* J_{X/S}^\infty \simeq J_{X'/S'}^\infty g_X^*.$$

They identify the monad and comonad structures and the adjunction $\text{Disk}^{\text{dR}, \infty} \dashv J^\infty$. The canonical disk coactions and Eilenberg–Moore comparison introduced below are transported by the same base-change equivalences.

Proof.

- (1) **Base-change the effective quotient.** Chapter 4 gives the coherent base-change equivalence

$$Q_{X'/S'} \simeq Q_{X/S} \times_S S',$$

and functoriality of the effective-image factorization identifies the displayed square as Cartesian.

(2) **Apply left Beck–Chevalley.** Left Beck–Chevalley gives

$$\begin{aligned} g_X^* \text{Disk}_{X/S}^{\text{dR}, \infty} &= g_X^* e_{X/S}^* \Sigma_{e_{X/S}} \\ &\simeq e_{X'/S'}^* g_Q^* \Sigma_{e_{X/S}} \\ &\simeq e_{X'/S'}^* \Sigma_{e_{X'/S'}} g_X^* = \text{Disk}_{X'/S'}^{\text{dR}, \infty} g_X^*. \end{aligned}$$

- (3) **Apply right Beck–Chevalley.** Right Beck–Chevalley gives the analogous calculation for J^∞ .
- (4) **Transport the structural maps by mate coherence.** These comparisons are mate transformations for a coherently base-changed adjoint triple, so they identify the units, counits, monad multiplication, comonad comultiplication, and composite-adjunction unit.
- (5) **Transport the later comparison structures.** Naturality of the later comparison coactions and mate equivalence then gives their base-change compatibility as well.
- (6) **Record higher Beck–Chevalley coherence.** All higher coherence is the higher Beck–Chevalley pasting coherence.

□

Arbitrary base change therefore transports the disk monad, the jet comonad, and their adjunction together. This completes the section’s interface: a generalized point sees sections over its intrinsic de Rham disk, while a change of base transports the entire construction by Beck–Chevalley. Later base-change statements for equations, prolongations, and solutions will be formal consequences of this result.

5.3 Crystalline transport and intrinsic formally integrable PDEs

The preceding section produced the disk–jet adjunction and its substitution law. The remaining question is geometric: what does an action or coaction on an object over X actually mean? This section answers that question once. Mate formation identifies disk actions with jet coactions, and effective descent identifies either one with an object over $Q_{X/S}$. Crystalline transport is then the degree-one relation-theoretic form of that same descent datum.

The section therefore proves one structural identification in progressively more concrete presentations. Once the quotient, algebra, coalgebra, and relation descriptions have been connected, later sections will use them without re-establishing their equivalence.

5.3.1 Homotopy-coherent mates and Eilenberg–Moore structures

The disk–jet adjunction already transposes individual structure maps. The next lemma is a book-keeping result: it upgrades that transposition to the full Eilenberg–Moore level, including structured morphisms. Its abstract notation is used only for this passage and need not remain active afterwards.

Lemma 5.3.1 (Adjoint monad–comonad duality). *Let*

$$T : \mathcal{C} \rightarrow \mathcal{C}$$

be a homotopy-coherent monad admitting a right adjoint

$$T \dashv J.$$

Transport the monad structure of T to J by taking right mates under the iterated adjunctions

$$T^n \dashv J^n.$$

Then J carries the resulting homotopy-coherent comonad structure, and mate formation induces a canonical equivalence

$$\mathfrak{M}_{T \dashv J} : \text{Alg}_T(\mathcal{C}) \xrightarrow{\sim} \text{Coalg}_J(\mathcal{C})$$

over $\mathcal{C}$. Thus, if

$$U_T : \text{Alg}_T(\mathcal{C}) \rightarrow \mathcal{C}, \quad U_J : \text{Coalg}_J(\mathcal{C}) \rightarrow \mathcal{C}$$

are the forgetful functors, there is a canonical equivalence

$$U_J \mathfrak{M}_{T \dashv J} \simeq U_T.$$

If

$$\eta^{T,J} : \text{id}_{\mathcal{C}} \rightarrow JT, \quad \varepsilon^{T,J} : TJ \rightarrow \text{id}_{\mathcal{C}}$$

are the unit and counit of $T \dashv J$, then the equivalence sends an action

$$s : TE \rightarrow E$$

to the coaction

$$s^\sharp := J(s) \eta_E^{T,J} : E \rightarrow JE,$$

and sends a coaction

$$\rho : E \rightarrow JE$$

to the action

$$\rho^\flat := \varepsilon_E^{T,J} T(\rho) : TE \rightarrow E.$$

These assignments are mutually inverse on structure spaces and preserve the underlying objects and morphisms of $\mathcal{C}$.

Proof.

- (1) **Form the iterated adjunctions.** In the ambient $(\infty, 2)$ -category of infinity-categories, the adjunction $T \dashv J$ induces adjunctions

$$T^n \dashv J^n$$

for every $n \geq 0$. Homotopy-coherent mate formation identifies the mapping spaces of natural transformations among these iterates and is compatible with whiskering and pasting; compare [36, 34].

- (2) **Construct the right-mate comonad.** Taking right mates of the unit, multiplication, and higher coherence maps of the monad T therefore produces the counit, comultiplication, and higher coherence maps of a comonad J .

- (3) **Identify algebra and coalgebra structure spaces.** The same mate equivalences identify, for every object E , the full space of T -algebra structures on E with the full space of J -coalgebra structures on E , and identify the corresponding structured mapping spaces. Hence they assemble to an equivalence

$$\mathrm{Alg}_T(\mathcal{C}) \xrightarrow{\simeq} \mathrm{Coalg}_J(\mathcal{C})$$

over $\mathcal{C}$.

- (4) **Recover the objectwise formulas.** The displayed formulas are the degree-one adjoint formulas for $T \dashv J$. Their composites are equivalent to the original structure maps by the triangle identities, so they describe the objectwise effect of the Eilenberg–Moore equivalence. $\square$

Theorem 5.3.1 (Mate equivalence of $\mathrm{Disk}^{\mathrm{dR},\infty}$ -algebras and J^∞ -coalgebras). *Mate formation for*

$$\mathrm{Disk}^{\mathrm{dR},\infty} \dashv J^\infty$$

extends to a canonical equivalence

$$\mathfrak{M} : \mathrm{Alg}_{\mathrm{Disk}^{\mathrm{dR},\infty}}((\mathcal{H}_{\mathrm{Sp}})_X) \xrightarrow{\simeq} \mathrm{Coalg}_{J^\infty}((\mathcal{H}_{\mathrm{Sp}})_X).$$

If

$$U_{\mathrm{Disk}} : \mathrm{Alg}_{\mathrm{Disk}^{\mathrm{dR},\infty}}((\mathcal{H}_{\mathrm{Sp}})_X) \longrightarrow (\mathcal{H}_{\mathrm{Sp}})_X, \quad U_J : \mathrm{Coalg}_{J^\infty}((\mathcal{H}_{\mathrm{Sp}})_X) \longrightarrow (\mathcal{H}_{\mathrm{Sp}})_X$$

are the two forgetful functors, then there is a canonical equivalence

$$U_J \mathfrak{M} \xrightarrow{\simeq} U_{\mathrm{Disk}}.$$

Thus $\mathfrak{M}$ is canonically an equivalence over $(\mathcal{H}_{\mathrm{Sp}})_X$.

Explicitly, if

$$s : \mathrm{Disk}^{\mathrm{dR},\infty} E \rightarrow E$$

is a $\mathrm{Disk}^{\mathrm{dR},\infty}$ -algebra action, its corresponding coaction is the adjoint

$$\rho_s := s^\sharp = J^\infty(s) \eta_E^{\mathrm{adj}} : E \longrightarrow J^\infty E.$$

Conversely, if

$$\rho : E \rightarrow J^\infty E$$

is a J^∞ -coalgebra coaction, its corresponding action is the adjoint

$$s_\rho := \rho^\flat = \varepsilon_E^{\mathrm{adj}} \mathrm{Disk}^{\mathrm{dR},\infty}(\rho) : \mathrm{Disk}^{\mathrm{dR},\infty} E \longrightarrow E.$$

These constructions are mutually inverse on the full structure spaces.

Under this correspondence the algebra unit law

$$s \eta_E^\Sigma \simeq \mathrm{id}_E$$

is the mate of the coalgebra counit law

$$\varepsilon_E^\Pi \rho_s \simeq \mathrm{id}_E,$$

and the algebra associativity law

$$s\mu_E \simeq s \operatorname{Disk}^{\mathrm{dR},\infty}(s)$$

is the mate of the coalgebra coassociativity law

$$\delta_E^J \rho_s \simeq J^\infty(\rho_s) \rho_s.$$

Likewise, if a structured morphism has underlying map

$$f : E \rightarrow E',$$

its mate has underlying map canonically identified with the same f ; preservation of the algebra actions is equivalent to preservation of the corresponding coalgebra coactions.

Proof.

- (1) **Identify the right-mate comonad.** By Proposition 5.2.2, the iterated adjunctions

$$(\operatorname{Disk}^{\mathrm{dR},\infty})^n \dashv (J^\infty)^n$$

identify the full monad structure of $\operatorname{Disk}^{\mathrm{dR},\infty}$ with the right-mate comonad structure of J^∞ . In particular, the right mates of

$$\eta^\Sigma : \operatorname{id} \rightarrow \operatorname{Disk}^{\mathrm{dR},\infty}, \quad \mu : (\operatorname{Disk}^{\mathrm{dR},\infty})^2 \rightarrow \operatorname{Disk}^{\mathrm{dR},\infty}$$

are precisely

$$\varepsilon^\Pi : J^\infty \rightarrow \operatorname{id}, \quad \delta^J : J^\infty \rightarrow (J^\infty)^2,$$

as identified by the simplicial relation calculus.

- (2) **Apply adjoint–monad duality.** Apply Lemma 5.3.1 to the adjunction

$$\operatorname{Disk}^{\mathrm{dR},\infty} \dashv J^\infty.$$

It produces a canonical equivalence

$$\mathfrak{M} : \operatorname{Alg}_{\operatorname{Disk}^{\mathrm{dR},\infty}}((\mathcal{H}_{\operatorname{Sp}})_{/X}) \xrightarrow{\simeq} \operatorname{Coalg}_{J^\infty}((\mathcal{H}_{\operatorname{Sp}})_{/X})$$

together with the canonical equivalence

$$U_J \mathfrak{M} \simeq U_{\operatorname{Disk}}.$$

Hence the comparison is canonically defined over $(\mathcal{H}_{\operatorname{Sp}})_{/X}$.

- (3) **Record the objectwise structure maps.** The objectwise structure-map formulas supplied by the lemma are

$$s^\sharp = J^\infty(s) \eta_E^{\operatorname{adj}}, \quad \rho^\flat = \varepsilon_E^{\operatorname{adj}} \operatorname{Disk}^{\mathrm{dR},\infty}(\rho).$$

- (4) **Verify the inverse structure maps.** They are mutually inverse by the triangle identities for

$$\operatorname{Disk}^{\mathrm{dR},\infty} \dashv J^\infty,$$

and the lemma identifies the full algebra and coalgebra structure spaces, with these degree-one maps as their objectwise components.

- (5) **Apply mate correspondence to the structural laws.** At the first two structural levels, the mate correspondence gives

$$s\eta_E^\Sigma \simeq \text{id}_E \quad \longleftrightarrow \quad \varepsilon_E^\Pi \rho_s \simeq \text{id}_E$$

and

$$s\mu_E \simeq s\text{Disk}^{\text{dR},\infty}(s) \quad \longleftrightarrow \quad \delta_E^J \rho_s \simeq J^\infty(\rho_s)\rho_s.$$

- (6) **Compare structured morphisms.** For an underlying morphism

$$f : E \rightarrow E',$$

the same mate correspondence carries the algebra-morphism compatibility

$$f s \simeq s' \text{Disk}^{\text{dR},\infty}(f)$$

to the coalgebra-morphism compatibility

$$J^\infty(f)\rho_s \simeq \rho_{s'}f,$$

and preserves the corresponding structured morphism data.

- (7) **Conclude the Eilenberg–Moore equivalence.** This proves the Eilenberg–Moore equivalence over $(\mathcal{H}_{\text{Sp}})_X$. □

Corollary 5.3.1 (Comparison actions and coactions are mates). *Let $F \rightarrow Q_{X/S}$, put $E = e_{X/S}^* F$, and define*

$$s_F := e_{X/S}^* \varepsilon_F^\Sigma : \text{Disk}^{\text{dR},\infty} E \rightarrow E, \quad \rho_F := e_{X/S}^* \eta_F^\Pi : E \rightarrow J^\infty E.$$

Then

$$\rho_F \simeq s_F^\sharp.$$

Consequently

$$\mathfrak{M}(E, s_F) \simeq (E, \rho_F)$$

canonically over E .

Proof.

- (1) **Expand the adjunct.** Using the formula for the unit η^{adj} of $\text{Disk}^{\text{dR},\infty} \dashv J^\infty$,

$$\begin{aligned} s_F^\sharp &= J^\infty(s_F)\eta_E^{\text{adj}} \\ &= e_{X/S}^* \Pi_{e_{X/S}}(e_{X/S}^* \varepsilon_F^\Sigma) e_{X/S}^* \eta_{\Sigma_{e_{X/S}} E}^\Pi \eta_E^\Sigma. \end{aligned}$$

- (2) **Use naturality of the unit.** Naturality of $\eta^\Pi : \text{id}_{(\mathcal{H}_{\text{Sp}})_{Q_{X/S}}} \rightarrow \Pi_{e_{X/S}} e_{X/S}^*$ with respect to $\varepsilon_F^\Sigma : \Sigma_{e_{X/S}} e_{X/S}^* F \rightarrow F$ gives

$$\Pi_{e_{X/S}} e_{X/S}^* (\varepsilon_F^\Sigma) \eta_{\Sigma_{e_{X/S}} e_{X/S}^* F}^\Pi \simeq \eta_F^\Pi \varepsilon_F^\Sigma.$$

- (3) **Apply the triangle identity.** Applying $e_{X/S}^*$ and composing with η_E^Σ therefore yields

$$\begin{aligned} s_F^\sharp &\simeq e_{X/S}^* \eta_F^\Pi e_{X/S}^* \varepsilon_F^\Sigma \eta_E^\Sigma \\ &\simeq e_{X/S}^* \eta_F^\Pi = \rho_F, \end{aligned}$$

where the second equivalence is the triangle identity for $\Sigma_{e_{X/S}} \dashv e_{X/S}^*$. □

The canonical jet coaction on a disk object

The Eilenberg–Moore equivalence supplies a distinguished jet coaction on every disk object. The following construction names it “free” in anticipation of the universal property proved later; until that theorem is reached, only the explicitly constructed comparison coaction is used.

Construction 5.3.1 (Free jet coalgebra on the disk object). For $Z \in (\mathcal{H}_{\text{Sp}})_{/X}$, define

$$\rho_Z^{\mathbb{F}} := e_{X/S}^* \eta_{\Sigma_{e_{X/S}} Z}^{\Pi} : \text{Disk}^{\text{dR}, \infty} Z \longrightarrow J^{\infty} \text{Disk}^{\text{dR}, \infty} Z,$$

and put

$$\mathbb{F}(Z) := (\text{Disk}^{\text{dR}, \infty} Z, \rho_Z^{\mathbb{F}}).$$

This is the canonical comparison J^{∞} -coalgebra on the object $e_{X/S}^* \Sigma_{e_{X/S}} Z$ supplied by the adjunction $e_{X/S}^* \dashv \Pi_{e_{X/S}}$. Equivalently, by Theorem 5.3.1, it is the coalgebra corresponding to the free $\text{Disk}^{\text{dR}, \infty}$ -algebra action

$$\mu_Z : \text{Disk}^{\text{dR}, \infty} \text{Disk}^{\text{dR}, \infty} Z \rightarrow \text{Disk}^{\text{dR}, \infty} Z.$$

Its coalgebra laws are supplied by the adjunction and mate calculus. Its free universal property is proved later in Theorem 5.4.1.

Proposition 5.3.1 (Unit factorization for the de Rham formal-disk–jet adjunction). *The unit $\eta_Z^{\text{adj}} : Z \rightarrow J^{\infty} \text{Disk}^{\text{dR}, \infty} Z$ satisfies*

$$\eta_Z^{\text{adj}} \simeq \rho_Z^{\mathbb{F}} \eta_Z^{\Sigma}.$$

Proof.

- (1) **Invoke the composite-adjunction unit formula.** The assertion is the unit formula in Theorem 5.2.6. □

Lemma 5.3.2 (Mate calculus). *Let $s : \text{Disk}^{\text{dR}, \infty} Z \rightarrow E$, and let*

$$s^{\sharp} : Z \rightarrow J^{\infty} E$$

be its adjunct. Then

$$\varepsilon_E^{\Pi} s^{\sharp} \simeq s \eta_Z^{\Sigma}.$$

For every $f : E \rightarrow F$, the adjunct of $f s$ is

$$J^{\infty}(f) s^{\sharp}.$$

Both identities are natural in the corresponding mapping spaces.

Proof.

- (1) **Write the adjunct through the adjunction unit.** Under the adjunction

$$\text{Disk}^{\text{dR}, \infty} \dashv J^{\infty},$$

the adjunct of

$$s : \text{Disk}^{\text{dR}, \infty} Z \rightarrow E$$

is canonically

$$s^\sharp \simeq J^\infty(s)\eta_Z^{\text{adj}},$$

where

$$\eta_Z^{\text{adj}} : Z \rightarrow J^\infty \text{Disk}^{\text{dR}, \infty} Z$$

is the unit of the formal-disk–jet adjunction.

- (2) **Apply naturality of the comonad counit.** Hence naturality of the comonad counit

$$\varepsilon^\Pi : J^\infty \rightarrow \text{id}_{(\mathcal{H}_{\text{Sp}})_/X}$$

gives

$$\begin{aligned} \varepsilon_E^\Pi s^\sharp &\simeq \varepsilon_E^\Pi J^\infty(s)\eta_Z^{\text{adj}} \\ &\simeq s \varepsilon_{\text{Disk}^{\text{dR}, \infty} Z}^\Pi \eta_Z^{\text{adj}}. \end{aligned}$$

- (3) **Insert the disk–jet unit factorization.** By Proposition 5.3.1,

$$\eta_Z^{\text{adj}} \simeq \rho_Z^\mathbb{F} \eta_Z^\Sigma.$$

- (4) **Use the comparison-coalgebra counit.** The map

$$\rho_Z^\mathbb{F} = e_{X/S}^* \eta_{\Sigma_{e_{X/S}} Z}^\Pi : \text{Disk}^{\text{dR}, \infty} Z \rightarrow J^\infty \text{Disk}^{\text{dR}, \infty} Z$$

is the canonical comparison coalgebra structure supplied by $e_{X/S}^* \dashv \Pi_{e_{X/S}}$, so its counit identity is

$$\varepsilon_{\text{Disk}^{\text{dR}, \infty} Z}^\Pi \rho_Z^\mathbb{F} \simeq \text{id}_{\text{Disk}^{\text{dR}, \infty} Z}.$$

- (5) **Conclude the first mate identity.** Therefore

$$\begin{aligned} \varepsilon_E^\Pi s^\sharp &\simeq s \varepsilon_{\text{Disk}^{\text{dR}, \infty} Z}^\Pi \rho_Z^\mathbb{F} \eta_Z^\Sigma \\ &\simeq s \eta_Z^\Sigma, \end{aligned}$$

which proves the first identity.

- (6) **Fix the target morphism.** Now let

$$f : E \rightarrow F.$$

- (7) **Apply target naturality.** Naturality of the mapping-space equivalence

$$\text{Map}_X(\text{Disk}^{\text{dR}, \infty} Z, -) \simeq \text{Map}_X(Z, J^\infty -)$$

in the target variable identifies the adjunct of

$$fs : \text{Disk}^{\text{dR}, \infty} Z \rightarrow F$$

with

$$J^\infty(f)s^\sharp : Z \rightarrow J^\infty F.$$

- (8) **Retain the higher homotopies.** Since these identities are obtained inside the natural equivalence of mapping spaces defining the adjunction, they are automatically compatible with all higher homotopies in those mapping spaces.

□

5.3.2 Crystalline transport and effective de Rham descent

The previous subsection established the formal equivalence between disk actions and jet coactions. We now supply its geometric meaning. Because $\mathcal{R}_{X/S, \bullet}^{\text{dR}}$ is already the Čech nerve of the effective quotient, effective descent identifies the Eilenberg–Moore categories with the quotient slice. Transposing a coaction then exposes the degree-one transport, while the coalgebra laws become exactly the unit and cocycle laws of the relation groupoid.

Proposition 5.3.2 (Effective Čech descent for the quotient). *Pullback along the effective epimorphism*

$$e_{X/S} : X \twoheadrightarrow Q_{X/S}$$

induces a canonical equivalence

$$(\mathcal{H}_{\text{Sp}})_{/Q_{X/S}} \xrightarrow{\cong} \text{Tot}_{[n] \in \Delta} (\mathcal{H}_{\text{Sp}})_{/\mathcal{R}_{X/S, n}^{\text{dR}}}.$$

Under this equivalence an object over $Q_{X/S}$ determines its degree-zero object over X , its degree-one crystalline transport, the Čech cocycle, and all higher simplicial coherence simultaneously.

Proof.

- (1) **Identify the effective Čech nerve.** Theorem 4.3.6 of Chapter 4 identifies $\mathcal{R}_{X/S, \bullet}^{\text{dR}}$ with the Čech nerve of $e_{X/S}$.
- (2) **Apply slice descent.** Apply Theorem 5.2.1.

□

Proposition 5.3.3 (Conservativity of effective pullback). *The pullback functor*

$$e_{X/S}^* : (\mathcal{H}_{\text{Sp}})_{/Q_{X/S}} \longrightarrow (\mathcal{H}_{\text{Sp}})_{/X}$$

is conservative.

Proof.

- (1) **Invoke effective descent for epimorphisms.** Effective epimorphisms are effective descent morphisms in an infinity-topos [36].
- (2) **Pull the morphism back to the Čech nerve.** If $f : F \rightarrow G$ becomes an equivalence after pullback to X , then every degree of its pullback to the Čech nerve $\mathcal{R}_{X/S, \bullet}^{\text{dR}}$ is an equivalence.
- (3) **Reconstruct the original morphism.** Under Proposition 5.3.2, the original morphism is the totalization of those degree-wise morphisms, hence is an equivalence.

□

Theorem 5.3.2 (Crystalline descent theorem). *The comparison functors*

$$K_{\Sigma} : (\mathcal{H}_{\text{Sp}})_{/Q_{X/S}} \longrightarrow \text{Alg}_{\text{Disk}^{\text{dR}}, \infty}((\mathcal{H}_{\text{Sp}})_{/X}),$$

$$F \longmapsto (e_{X/S}^* F, e_{X/S}^* \varepsilon_F^{\Sigma}),$$

and

$$K_{\Pi} : (\mathcal{H}_{\text{Sp}})_{/Q_{X/S}} \longrightarrow \text{Coalg}_{J^{\infty}}((\mathcal{H}_{\text{Sp}})_{/X}),$$

$$F \mapsto (e_{X/S}^* F, e_{X/S}^* \eta_F^\Pi),$$

are equivalences of infinity-categories.

Moreover, if

$$\mathfrak{M} : \mathrm{Alg}_{\mathrm{Disk}^{\mathrm{dR}, \infty}}((\mathcal{H}_{\mathrm{Sp}})_{/X}) \xrightarrow{\simeq} \mathrm{Coalg}_{J^\infty}((\mathcal{H}_{\mathrm{Sp}})_{/X})$$

is the mate equivalence of Theorem 5.3.1, then

$$\mathfrak{M}K_\Sigma \simeq K_\Pi$$

canonically over $(\mathcal{H}_{\mathrm{Sp}})_{/X}$. Hence there are canonical equivalences

$$(\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}} \simeq \mathrm{Alg}_{\mathrm{Disk}^{\mathrm{dR}, \infty}}((\mathcal{H}_{\mathrm{Sp}})_{/X}) \simeq \mathrm{Coalg}_{J^\infty}((\mathcal{H}_{\mathrm{Sp}})_{/X}).$$

Proof.

- (1) **Establish presentability.** The categories

$$(\mathcal{H}_{\mathrm{Sp}})_{/X}, \quad (\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}}$$

are presentable.

- (2) **Establish conservativity.** Pullback

$$e_{X/S}^* : (\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}} \longrightarrow (\mathcal{H}_{\mathrm{Sp}})_{/X}$$

is conservative by Proposition 5.3.3.

- (3) **Verify realization preservation.** It is also a left adjoint, since

$$e_{X/S}^* \dashv \Pi_{e_{X/S}},$$

and therefore preserves all geometric realizations required by the monadic Barr–Beck theorem.

- (4) **Identify the induced monad.** For the adjunction

$$\Sigma_{e_{X/S}} \dashv e_{X/S}^*,$$

the induced monad is

$$e_{X/S}^* \Sigma_{e_{X/S}} = \mathrm{Disk}^{\mathrm{dR}, \infty}.$$

- (5) **Apply monadic Barr–Beck.** Barr–Beck therefore gives

$$K_\Sigma : (\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}} \xrightarrow{\simeq} \mathrm{Alg}_{\mathrm{Disk}^{\mathrm{dR}, \infty}}((\mathcal{H}_{\mathrm{Sp}})_{/X});$$

see [34, Theorem 4.7.3.5].

- (6) **Fix the pulled-back object.** For $F \rightarrow Q_{X/S}$, put

$$E = e_{X/S}^* F.$$

- (7) **Identify the comparison structures.** The disk-algebra comparison structure is

$$s_F = e_{X/S}^* \varepsilon_F^\Sigma : \mathrm{Disk}^{\mathrm{dR}, \infty} E \rightarrow E,$$

whereas the jet-coalgebra comparison structure is

$$\rho_F = e_{X/S}^* \eta_F^\Pi : E \rightarrow J^\infty E.$$

(8) **Apply the mate comparison.** Corollary 5.3.1 identifies these maps as mates:

$$\rho_F \simeq s_F^\sharp.$$

(9) **Identify the comparison functors.** Consequently

$$\mathfrak{M}K_\Sigma \simeq K_\Pi$$

naturally over $(\mathcal{H}_{\text{Sp}})_{/X}$.

(10) **Conclude the coalgebraic presentation.** Since K_Σ and $\mathfrak{M}$ are equivalences, so is K_Π . Thus the quotient slice, disk-algebra category, and jet-coalgebra category are three presentations of the same effective de Rham descent category.

□

At this point the reader may replace the preceding constructions by one interface:

$\text{object over } Q_{X/S} \iff \text{formal-disk action} \iff \text{jet coaction} \iff \text{effective de Rham descent.}$
--

The results that follow unpack the degree-one geometric content of this equivalence as crystalline transport along $\mathcal{R}_{X/S}^{\text{dR}}$, so that unit, cocycle, inverse, and higher compatibility can be read directly from the same Čech groupoid.

Theorem 5.3.3 (Transport, algebra actions, and coalgebra coactions). *For $E \rightarrow X$, define the spaces*

$$\begin{aligned} \text{Alg}_{\text{Disk}^{\text{dR}}, \infty}(E) &:= \text{hofib}_E \left(\text{Alg}_{\text{Disk}^{\text{dR}}, \infty}((\mathcal{H}_{\text{Sp}})_{/X})^\simeq \longrightarrow (\mathcal{H}_{\text{Sp}})_{/X}^\simeq \right), \\ \text{Coalg}_{J^\infty}(E) &:= \text{hofib}_E \left(\text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_{/X})^\simeq \longrightarrow (\mathcal{H}_{\text{Sp}})_{/X}^\simeq \right), \end{aligned}$$

and let

$$\text{Desc}_{e_{X/S}}(E)$$

be the homotopy fibre over E of degree-zero evaluation from the maximal subgroupoid of

$$\text{Tot}_{[n] \in \Delta} (\mathcal{H}_{\text{Sp}})_{/\mathcal{R}_{X/S, n}^{\text{dR}}}.$$

Then there are canonical equivalences

$$\text{Alg}_{\text{Disk}^{\text{dR}}, \infty}(E) \simeq \text{Coalg}_{J^\infty}(E) \simeq \text{Desc}_{e_{X/S}}(E).$$

Thus a $\text{Disk}^{\text{dR}, \infty}$ -algebra structure, a jet coalgebra structure, and coherent de Rham-relation descent on the fixed underlying object are three presentations of the same structure space.

Proof.

- (1) **Take fibres of the mate equivalence.** The first equivalence is obtained by taking homotopy fibres of the mate equivalence over $(\mathcal{H}_{\text{Sp}})_{/X}$ in Theorem 5.3.1.
- (2) **Take fibres of the descent equivalence.** The second is obtained by combining Theorem 5.3.2 with Proposition 5.3.2 and again taking the homotopy fibre of the degree-zero underlying-object functor.

- (3) **Identify the fixed underlying object.** The homotopy fibres record the equivalence with the fixed object E at the level of maximal subgroupoids.

□

Construction 5.3.2 (Canonical crystalline transport). Let $F \rightarrow Q_{X/S}$, put $E := e_{X/S}^* F$, and use the comparison action s_F and coaction ρ_F of Corollary 5.3.1. On

$$\mathcal{R}_{X/S}^{\text{dR}} = X \times_{Q_{X/S}} X$$

the two composites $e_{X/S}\mathbf{s}$ and $e_{X/S}\mathbf{t}$ agree. Hence the two pullbacks of F are canonically identified, giving

$$\theta_F : \mathbf{s}^* E = (e_{X/S}\mathbf{s})^* F \xrightarrow{\sim} (e_{X/S}\mathbf{t})^* F = \mathbf{t}^* E.$$

This is the crystalline transport of F , displayed by

$$\begin{array}{ccc} \mathbf{s}^* E & \xrightarrow[\sim]{\theta_F} & \mathbf{t}^* E \\ \downarrow & & \downarrow \\ \mathcal{R}_{X/S}^{\text{dR}} & \xlongequal{\quad} & \mathcal{R}_{X/S}^{\text{dR}}. \end{array}$$

Under right Beck–Chevalley, the transpose of the comparison coaction ρ_F is precisely θ_F ; thus the descent presentation and the direct coalgebraic construction of Proposition 5.3.4 agree.

Proposition 5.3.4 (Crystalline transport extracted from a jet coaction). *Let*

$$\rho : E \longrightarrow J^\infty E$$

be a J^∞ -coalgebra structure. Under

$$J^\infty \simeq \Pi_{\mathbf{s}} \mathbf{t}^*,$$

its right-adjoint transpose is a canonical morphism

$$\theta_\rho : \mathbf{s}^* E \longrightarrow \mathbf{t}^* E.$$

The coalgebra laws canonically make this morphism an equivalence. Writing

$$\mathcal{R}_{X/S,2}^{\text{dR}} = X \times_{Q_{X/S}} X \times_{Q_{X/S}} X$$

and $p_{ij} : \mathcal{R}_{X/S,2}^{\text{dR}} \rightarrow \mathcal{R}_{X/S}^{\text{dR}}$ for the edge projections, one has

$$\mathbf{u}^* \theta_\rho \simeq \text{id}_E,$$

$$p_{12}^* \theta_\rho \circ p_{01}^* \theta_\rho \simeq p_{02}^* \theta_\rho,$$

and

$$\theta_\rho^{-1} \simeq \text{inv}^* \theta_\rho.$$

Thus the unit, cocycle, inverse, and remaining relation compatibility are derived from the coalgebra laws and the Čech-groupoid structure.

Proof.

- (1) **Transpose the jet coaction.** Under the right Beck–Chevalley equivalence

$$J^\infty E \simeq \Pi_{\mathfrak{s}\mathfrak{t}}^* E,$$

the map $\rho : E \rightarrow J^\infty E$ is adjoint to a uniquely determined map

$$\theta_\rho : \mathfrak{s}^* E \rightarrow \mathfrak{t}^* E.$$

- (2) **Recover the unit law.** The comonad counit

$$\varepsilon_E^\Pi : J^\infty E \rightarrow E$$

is induced, under the relation correspondence, by restriction along the relation unit $\mathfrak{u}$. Transposing the coalgebra counit law

$$\varepsilon_E^\Pi \rho \simeq \text{id}_E$$

therefore gives

$$\mathfrak{u}^* \theta_\rho \simeq \text{id}_E.$$

- (3) **Identify relation composition with monad multiplication.** By the formal-disk monad proposition, the monad multiplication

$$\mu : (\text{Disk}^{\text{dR}, \infty})^2 \rightarrow \text{Disk}^{\text{dR}, \infty}$$

is induced by composition in the de Rham relation groupoid.

- (4) **Identify the comonad mate.** By Theorem 5.2.6, its mate is the comultiplication

$$\delta^J : J^\infty \rightarrow J^\infty J^\infty.$$

- (5) **Transpose coassociativity to the relation.** Consequently, after applying right Beck–Chevalley twice, the two sides of the coalgebra coassociativity law

$$\delta_E^J \rho \simeq J^\infty(\rho) \rho$$

transpose respectively to

$$p_{02}^* \theta_\rho \quad \text{and} \quad p_{12}^* \theta_\rho \circ p_{01}^* \theta_\rho.$$

- (6) **Conclude the cocycle identity.** Hence

$$p_{12}^* \theta_\rho \circ p_{01}^* \theta_\rho \simeq p_{02}^* \theta_\rho.$$

- (7) **Define the first return simplex.** To derive invertibility, define

$$a : \mathcal{R}_{X/S}^{\text{dR}} \rightarrow \mathcal{R}_{X/S, 2}^{\text{dR}}, \quad a(x_0, x_1) := (x_0, x_1, x_0).$$

Then

$$p_{01} a = \text{id}_{\mathcal{R}_{X/S}^{\text{dR}}}, \quad p_{12} a = \text{inv}, \quad p_{02} a = \mathfrak{u} \mathfrak{s}.$$

- (8) **Derive a left inverse.** Pulling back the cocycle identity along a gives

$$\text{inv}^* \theta_\rho \circ \theta_\rho \simeq \mathfrak{s}^* \mathfrak{u}^* \theta_\rho \simeq \text{id}_{\mathfrak{s}^* E}.$$

(9) **Define the second return simplex.** Likewise, for

$$b : \mathcal{R}_{X/S}^{\text{dR}} \rightarrow \mathcal{R}_{X/S,2}^{\text{dR}}, \quad b(x_0, x_1) := (x_1, x_0, x_1),$$

one has

$$p_{01}b = \text{inv}, \quad p_{12}b = \text{id}_{\mathcal{R}_{X/S}^{\text{dR}}}, \quad p_{02}b = \mathbf{u} \mathbf{t},$$

(10) **Derive a right inverse.** and hence

$$\theta_\rho \circ \text{inv}^* \theta_\rho \simeq \mathbf{t}^* \mathbf{u}^* \theta_\rho \simeq \text{id}_{\mathbf{t}^* E}.$$

(11) **Identify inverse transport.** Thus θ_ρ is an equivalence with

$$\theta_\rho^{-1} \simeq \text{inv}^* \theta_\rho.$$

(12) **Recover higher simplicial coherence.** For the higher coherences, use Proposition 5.2.2:

$$(J^\infty)^n \simeq \Pi_{v_n} \mathbf{t}_n^*$$

identifies every iterated coalgebra structure map with transport over the n -simplex $\mathcal{R}_{X/S,n}^{\text{dR}}$. The coalgebra laws transpose to the face and degeneracy identities of the Čech nerve. The same simplicial object therefore supplies the remaining transport compatibility. $\square$

Proposition 5.3.5 (Crystalline transport extracted from a $\text{Disk}^{\text{dR},\infty}$ -algebra). *Let*

$$s : \text{Disk}^{\text{dR},\infty} E \longrightarrow E$$

be a $\text{Disk}^{\text{dR},\infty}$ -algebra structure. Under

$$\text{Disk}^{\text{dR},\infty} \simeq \Sigma_{\mathbf{s}} \mathbf{t}^*,$$

its left-adjoint transpose is a canonical equivalence

$$\tau_s : \mathbf{t}^* E \xrightarrow{\simeq} \mathbf{s}^* E.$$

Let

$$\rho := \mathfrak{M}(E, s) : E \longrightarrow J^\infty E$$

be the corresponding coalgebra structure of Theorem 5.3.1. Then

$$\tau_s \simeq \theta_\rho^{-1} \simeq \text{inv}^* \theta_\rho.$$

The algebra unit and associativity coherences are the reverse-oriented mates of the relation-unit and cocycle coherences.

Proof.

(1) **Identify the correspondence adjunction through inversion.** The adjunction

$$\text{Disk}^{\text{dR},\infty} \simeq \Sigma_{\mathbf{s}} \mathbf{t}^* \dashv J^\infty \simeq \Pi_{\mathbf{s}} \mathbf{t}^*$$

is obtained from the correspondence adjunction after transporting the right-adjoint presentation through relation inversion.

- (2) **Transpose the disk action.** Hence the mate of

$$s : \Sigma_s \mathfrak{t}^* E \rightarrow E$$

is obtained by transposing s to

$$\tau_s : \mathfrak{t}^* E \rightarrow \mathfrak{s}^* E$$

and then transporting along $\text{inv} : \mathcal{R}_{X/S}^{\text{dR}} \xrightarrow{\simeq} \mathcal{R}_{X/S}^{\text{dR}}$.

- (3) **Transpose the coalgebra coaction.** The corresponding coalgebra coaction $\rho = s^\sharp$ transposes to

$$\theta_\rho : \mathfrak{s}^* E \rightarrow \mathfrak{t}^* E.$$

- (4) **Identify inversion with inverse transport.** The two transposes are therefore exchanged by inversion. By Proposition 5.3.4,

$$\theta_\rho^{-1} \simeq \text{inv}^* \theta_\rho,$$

so

$$\tau_s \simeq \theta_\rho^{-1} \simeq \text{inv}^* \theta_\rho.$$

- (5) **Transport the higher coherences.** Compatibility of mates with pasting identifies the algebra unit and associativity coherences with the oppositely oriented relation unit and cocycle coherences, including all higher homotopies.

□

Proposition 5.3.6 (Degree-one transport and cocycle coherence). *For the canonical structures of Construction 5.3.2,*

$$\mathfrak{u}^* \theta_F \simeq \text{id}_E,$$

$$p_{12}^* \theta_F \circ p_{01}^* \theta_F \simeq p_{02}^* \theta_F,$$

and

$$\theta_F^{-1} \simeq \text{inv}^* \theta_F.$$

The remaining simplicial compatibility is inherited from the single object F on $Q_{X/S}$.

Proof.

- (1) **Apply the relation-transport theorem.** Apply Proposition 5.3.4 to ρ_F ; equivalently, use the degree-one and degree-two part of Proposition 5.3.2.

□

Proposition 5.3.7 (Explicit mate relation). *The canonical formal-disk action and jet coaction satisfy*

$$\varepsilon_E^\Pi \rho_F \simeq \text{id}_E, \quad \rho_F s_F \simeq J^\infty(s_F) \rho_E^\mathbb{R}.$$

Thus the formal-disk action and jet coaction are two adjoint presentations of the same effective descent structure.

Proof.

- (1) **Apply the counit identity.** The first equation is the triangle identity for $e_{X/S}^* \dashv \Pi_{e_{X/S}}$.

(2) **Use the coalgebra-morphism equation.** For the second, the action

$$s_F : (\mathrm{Disk}^{\mathrm{dR}, \infty} E, \mu_E) \longrightarrow (E, s_F)$$

is the canonical morphism from the free $\mathrm{Disk}^{\mathrm{dR}, \infty}$ -algebra to the comparison algebra. Under the equivalence over $(\mathcal{H}_{\mathrm{Sp}})_X$ of Theorem 5.3.2, the same underlying map is a morphism from the free jet coalgebra

$$(\mathrm{Disk}^{\mathrm{dR}, \infty} E, \rho_E^{\mathbb{F}})$$

to (E, ρ_F) . The coalgebra-morphism equation is precisely

$$\rho_F s_F \simeq J^\infty(s_F) \rho_E^{\mathbb{F}}.$$

□

The quotient, disk, jet, and relation descriptions have now been identified as four interfaces to one descent problem. We can therefore introduce the intrinsic category used later without adding another layer of structure.

5.3.3 Intrinsic formally integrable PDEs

We use the quotient slice for objects whose infinitesimal compatibility is already built in. The terminology must be kept distinct from the ambient equation calculus introduced in the next section: an intrinsic formally integrable PDE is already a descended object, whereas a generalized jet equation will merely be a map into an infinite jet object and may still fail to preserve crystalline transport.

Definition 5.3.1 (Intrinsic formally integrable PDE). Define

$$\mathrm{PDE}_{X/S}^{\mathrm{fi}} := (\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}}.$$

An object of this slice is an **intrinsic formally integrable spectral PDE**.

Here *formally integrable* means that the corresponding object over X carries coherent effective descent along the finite-infinitesimal de Rham relation

$$\mathcal{R}_{X/S, \bullet}^{\mathrm{dR}} \simeq \check{C}_\bullet(X \rightarrow Q_{X/S}).$$

Equivalently, it carries the associated formal-disk action, jet coaction, or crystalline transport. The terminology asserts this infinitesimal descent property; it does not assert analytic existence, regularity, or convergence of actual solutions.

By Theorem 5.3.2, the intrinsic category has the two equivalent presentations over X :

$$\mathrm{PDE}_{X/S}^{\mathrm{fi}} \simeq \mathrm{Alg}_{\mathrm{Disk}^{\mathrm{dR}, \infty}}((\mathcal{H}_{\mathrm{Sp}})_X) \simeq \mathrm{Coalg}_{J^\infty}((\mathcal{H}_{\mathrm{Sp}})_X).$$

Thus “intrinsic PDE”, “disk algebra”, and “jet coalgebra” describe the same already-integrable object through different interfaces. They should not be conflated with a generalized equation $E \rightarrow J^\infty Y$, which is an ambient presentation whose compatibility will be tested later by Cartan prolongation.

Remark 5.3.1 (Relation to crystals and nonlinear D -geometry). The slice over $Q_{X/S}$ is the nonlinear descent category associated to the de Rham relation groupoid. This is formally parallel to crystals over a derived de Rham prestack and to nonlinear D -geometry [18, 5]. When the reduced-classical comparison of Chapter 4 is invertible, this description agrees with the corresponding reduced-classical de Rham-prestack presentation.

Proposition 5.3.8 (Underlying object detects intrinsic morphisms). *Let $F, G \rightarrow Q_{X/S}$ be intrinsic formally integrable PDEs and put*

$$E := e_{X/S}^* F, \quad E' := e_{X/S}^* G.$$

Let

$$\theta_F : s^* E \xrightarrow{\sim} t^* E, \quad \theta_G : s^* E' \xrightarrow{\sim} t^* E'$$

be their canonical crystalline transports.

For a map

$$f : E \rightarrow E'$$

in $(\mathcal{H}_{\text{Sp}})_X$, the following are equivalent:

1. f descends to a map

$$F \rightarrow G$$

over $Q_{X/S}$;

2. f extends to a morphism between the corresponding effective Čech-descent objects;
3. f preserves crystalline transport together with its induced higher Čech coherence; in degree one this includes a specified homotopy filling the square

$$\begin{array}{ccc} s^* E & \xrightarrow{\theta_F} & t^* E \\ s^* f \downarrow & & \downarrow t^* f \\ s^* E' & \xrightarrow{\theta_G} & t^* E', \end{array}$$

compatible with the cocycle on $\mathcal{R}_{X/S,2}^{\text{dR}}$ and the remaining simplicial data;

4. f is a $\text{Disk}^{\text{dR},\infty}$ -algebra morphism;
5. f is a J^∞ -coalgebra morphism.

Thus preserving crystalline transport means preserving the effective-descent datum encoded by the degree-one square and its simplicial compatibilities.

Proof.

- (1) **Use the Čech totalization.** Proposition 5.3.2 gives a canonical equivalence

$$(\mathcal{H}_{\text{Sp}})_{/Q_{X/S}} \xrightarrow{\sim} \text{Tot}_{[n] \in \Delta} (\mathcal{H}_{\text{Sp}})_{/\mathcal{R}_{X/S,n}^{\text{dR}}}.$$

Under this equivalence, the objects F and G determine their underlying degree-zero objects E and E' , the degree-one transports θ_F and θ_G , their degree-two cocycle homotopies, and all higher simplicial coherence simultaneously.

- (2) **Characterize descended morphisms.** Consequently a map

$$f : E \rightarrow E'$$

descends to $Q_{X/S}$ exactly when it is the degree-zero component of a morphism between these two objects of the totalization. In degree one this requires the displayed transport square to commute up to the specified homotopy, and in higher degrees it requires precisely the compatibility of that homotopy with the cocycle and the remaining simplicial compatibility. This proves the equivalence of (1), (2), and (3) on the full mapping spaces.

- (3) **Pass to the monadic and comonadic presentations.** Theorem 5.3.2 gives equivalences of infinity-categories over $(\mathcal{H}_{\text{Sp}})_{/X}$,

$$(\mathcal{H}_{\text{Sp}})_{/Q_{X/S}} \simeq \text{Alg}_{\text{Disk}^{\text{dR},\infty}}((\mathcal{H}_{\text{Sp}})_{/X}) \simeq \text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_{/X}).$$

Under these equivalences a morphism over $Q_{X/S}$ is carried respectively to a $\text{Disk}^{\text{dR},\infty}$ -algebra morphism and a J^∞ -coalgebra morphism. Hence (1), (4), and (5) are equivalent as well.

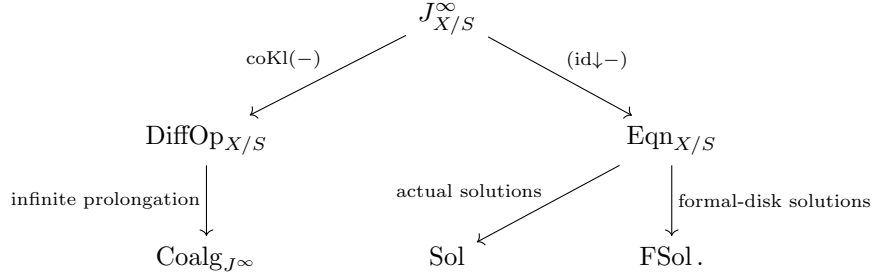
- (4) **Retain the full mapping-space coherence.** These equivalences hold on the complete mapping spaces and therefore retain the higher compatibilities carried by the totalization.

□

The intrinsic descent category is therefore fixed. Quotient-side language will be used when descent and limits are the issue; coalgebra language will be used when operators, equations, and prolongation are the issue.

5.4 Differential operators and generalized equations

We now use the jet comonad as a calculus rather than as a descent classifier. Its two principal uses have opposite variance. Maps out of a jet object give differential operators, while maps into a jet object give generalized equations. Holonomic sections test ordinary solutions of those equations; formal-disk families test infinitesimal compatibility. This variance split is the organizing principle of the section.



The diagram records roles rather than new functors: the constructions themselves are defined below.

5.4.1 Free and cofree jet coalgebras and holonomicity

The calculus uses two universal coalgebras. The disk object $\mathbb{F}(Z)$ carries the comparison coaction constructed above, and $\mathbb{K}(E)$ is cofree on an arbitrary object over X . Their adjunctions identify the terminal coalgebra and hence the holonomic prolongation of an ordinary section. The equation calculus uses two universal coalgebras: the disk coalgebra $\mathbb{F}(Z)$ introduced above and the cofree coalgebra on an arbitrary object over X . Their adjunctions also identify the canonical coalgebra on X , from which holonomic prolongation follows.

Construction 5.4.1 (Cofree jet coalgebras). For $E \in (\mathcal{H}_{\text{Sp}})_{/X}$, define

$$\mathbb{K}(E) := (J^\infty E, \delta_E^J).$$

The disk coalgebra $\mathbb{F}(Z)$ was constructed in Construction 5.3.1; $\mathbb{K}(E)$ is the cofree jet coalgebra.

Theorem 5.4.1 (Free–forgetful–cofree adjunctions). *Let*

$$U_J : \mathrm{Coalg}_{J^\infty}((\mathcal{H}_{\mathrm{Sp}})_{/X}) \rightarrow (\mathcal{H}_{\mathrm{Sp}})_{/X}$$

be the forgetful functor. There are canonical adjunctions

$$\mathbb{F} \dashv U_J \dashv \mathbb{K}.$$

They may be displayed as

$$(\mathcal{H}_{\mathrm{Sp}})_{/X} \begin{array}{c} \xrightarrow{\mathbb{F}} \\ \xleftarrow{U_J} \\ \xrightarrow{\mathbb{K}} \end{array} \mathrm{Coalg}_{J^\infty}((\mathcal{H}_{\mathrm{Sp}})_{/X})$$

with $\mathbb{F} \dashv U_J$ and $U_J \dashv \mathbb{K}$. In particular,

$$\mathrm{Map}_{\mathrm{Coalg}_{J^\infty}((\mathcal{H}_{\mathrm{Sp}})_{/X})}(\mathbb{F}(Z), C) \simeq \mathrm{Map}_X(Z, U_J C),$$

and

$$\mathrm{Map}_{\mathrm{Coalg}_{J^\infty}((\mathcal{H}_{\mathrm{Sp}})_{/X})}(C, \mathbb{K}(E)) \simeq \mathrm{Map}_X(U_J C, E).$$

Under effective descent, $\mathbb{F}(Z)$ corresponds to $\Sigma_{e_{X/S}} Z \rightarrow Q_{X/S}$, while $\mathbb{K}(E)$ corresponds to $\Pi_{e_{X/S}} E \rightarrow Q_{X/S}$.

Proof.

- (1) **Recall the Eilenberg–Moore adjunctions.** The Eilenberg–Moore construction for monads and comonads [34] gives the free and cofree adjunctions used here.
- (2) **Construct the free disk algebra.** The monad $\mathrm{Disk}^{\mathrm{dR}, \infty}$ has the canonical free-algebra functor

$$Z \mapsto (\mathrm{Disk}^{\mathrm{dR}, \infty} Z, \mu_Z)$$

left adjoint to the forgetful functor

$$\mathrm{Alg}_{\mathrm{Disk}^{\mathrm{dR}, \infty}}((\mathcal{H}_{\mathrm{Sp}})_{/X}) \rightarrow (\mathcal{H}_{\mathrm{Sp}})_{/X}.$$

- (3) **Compute its jet-coalgebra mate.** By Corollary 5.3.1 applied to the object $\Sigma_{e_{X/S}} Z \in (\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}}$, the mate of

$$\mu_Z = e_{X/S}^* \varepsilon_{\Sigma_{e_{X/S}} Z}^\Sigma$$

is

$$\rho_Z^\mathbb{F} = e_{X/S}^* \eta_{\Sigma_{e_{X/S}} Z}^\Pi.$$

- (4) **Identify the free jet coalgebra.** Therefore Theorem 5.3.1 carries the free $\mathrm{Disk}^{\mathrm{dR}, \infty}$ -algebra on Z to

$$\mathbb{F}(Z) = (\mathrm{Disk}^{\mathrm{dR}, \infty} Z, \rho_Z^\mathbb{F}).$$

- (5) **Transport the free–forgetful adjunction.** Since the mate equivalence is over $(\mathcal{H}_{\mathrm{Sp}})_{/X}$, it transports the free–forgetful adjunction to

$$\mathbb{F} \dashv U_J.$$

(6) **Construct the cofree coalgebra.** For the comonad J^∞ , the cofree coalgebra on E is

$$\mathbb{K}(E) = (J^\infty E, \delta_E^J),$$

with counit ε_E^Π , and it is right adjoint to U_J .

(7) **Assemble the adjoint triple.** This proves

$$\mathbb{F} \dashv U_J \dashv \mathbb{K}$$

and the displayed mapping-space equivalences.

(8) **Identify the descended adjoints.** Finally, under Theorem 5.3.2, the left and right adjoints of $e_{X/S}^*$ are $\Sigma_{e_{X/S}}$ and $\Pi_{e_{X/S}}$. The uniqueness of adjoints identifies the descended objects corresponding to $\mathbb{F}(Z)$ and $\mathbb{K}(E)$ with $\Sigma_{e_{X/S}} Z$ and $\Pi_{e_{X/S}} E$, respectively.

□

The theorem now justifies the terminology used earlier: $\mathbb{F}(Z)$ is the free jet coalgebra on Z , whereas $\mathbb{K}(E)$ is cofree on E . From this point onward the adjectives “free” and “cofree” refer to these proved universal properties.

Proposition 5.4.1 (Terminal jet coalgebra). *The terminal object*

$$\mathrm{id}_{Q_{X/S}} : Q_{X/S} \rightarrow Q_{X/S}$$

of $(\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}}$ pulls back to the terminal object

$$\mathrm{id}_X : X \rightarrow X$$

of $(\mathcal{H}_{\mathrm{Sp}})_{/X}$. Its comonadic comparison structure is a canonical coaction

$$\rho_X : X \longrightarrow J^\infty X,$$

and (X, ρ_X) is terminal in $\mathrm{Coalg}_{J^\infty}((\mathcal{H}_{\mathrm{Sp}})_{/X})$. The counit

$$\varepsilon_X^\Pi : J^\infty X \rightarrow X$$

is an equivalence with inverse ρ_X .

Proof.

(1) **Pull back the terminal descent object.** Under Theorem 5.3.2,

$$(\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}} \simeq \mathrm{Coalg}_{J^\infty}((\mathcal{H}_{\mathrm{Sp}})_{/X}),$$

and the underlying-object functor corresponds to $e_{X/S}^*$. The terminal object of $(\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}}$ therefore determines a terminal coalgebra whose underlying object is

$$e_{X/S}^*(\mathrm{id}_{Q_{X/S}}) = \mathrm{id}_X.$$

(2) **Apply the coalgebra counit equation.** The coalgebra counit equation gives

$$\varepsilon_X^\Pi \rho_X \simeq \mathrm{id}_X.$$

- (3) **Use terminality of the jet object.** By Proposition 5.2.5, $J^\infty X$ is terminal in $(\mathcal{H}_{\text{Sp}})_/X$.
- (4) **Identify the reverse composite.** Hence the reverse composite is the contractibly unique endomorphism of a terminal object and is equivalent to the identity.
- (5) **Conclude invertibility of the counit.** Thus $\rho_X \simeq (\varepsilon_X^\Pi)^{-1}$.

□

Definition 5.4.1 (Infinite prolongation of a section). For a section $\phi : X \rightarrow E$, define

$$j^\infty \phi := J^\infty(\phi)\rho_X : X \rightarrow J^\infty E.$$

Naturality of the counit gives

$$\varepsilon_E^\Pi j^\infty \phi \simeq \phi.$$

Definition 5.4.2 (Holonomic section). The terminology follows the infinite-jet formulation of holonomicity in geometric PDE theory [27]. A section $\sigma : X \rightarrow J^\infty E$ is **holonomic** if it is a jet-coalgebra morphism

$$(X, \rho_X) \longrightarrow \mathbb{K}(E).$$

Theorem 5.4.2 (Holonomic recognition). *There is a canonical equivalence*

$$\text{Map}_{\text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_/X)}((X, \rho_X), \mathbb{K}(E)) \simeq \text{Map}_X(X, E).$$

Under this equivalence ϕ corresponds to $j^\infty \phi$. The holonomic section fits into the commutative triangle

$$\begin{array}{ccc} X & \xrightarrow{\sigma} & J^\infty E \\ & \searrow \varepsilon_E^\Pi \sigma & \swarrow \varepsilon_E^\Pi \\ & E & \end{array}$$

and is holonomic precisely when $\sigma \simeq j^\infty(\varepsilon_E^\Pi \sigma)$.

Proof.

- (1) **Apply the cofree adjunction.** Apply the cofree adjunction $U_J \dashv \mathbb{K}$ to the terminal coalgebra.

□

5.4.2 Differential operators as co-Kleisli morphisms

The cofree adjunction turns the jet comonad into the co-Kleisli calculus of operators. A differential operator is therefore represented by a map $J^\infty E \rightarrow F$; comultiplication supplies composition, and infinite prolongation is simply the corresponding morphism between cofree coalgebras.

Definition 5.4.3 (Crystalline differential operator). Following the co-Kleisli formulation of infinite-jet operators used in categorical treatments of PDEs [38, 27], for $E, F \in (\mathcal{H}_{\text{Sp}})_/X$ a crystalline differential operator

$$D : E \rightsquigarrow F$$

is a morphism

$$\widehat{D} : J^\infty E \rightarrow F$$

in $(\mathcal{H}_{\text{Sp}})_{/X}$. Its action on a section $\phi : X \rightarrow E$ is

$$D(\phi) := \widehat{D} j^\infty \phi.$$

The action is represented by the composite

$$\begin{array}{ccc} X & \xrightarrow{j^\infty \phi} & J^\infty E \\ & \searrow D(\phi) & \downarrow \widehat{D} \\ & & F. \end{array}$$

Construction 5.4.2 (Differential-operator infinity-category). Define $\text{DiffOp}_{X/S}$ to have the same objects as $(\mathcal{H}_{\text{Sp}})_{/X}$ and mapping spaces

$$\text{Map}_{\text{DiffOp}_{X/S}}(E, F) := \text{Map}_{\text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_{/X})}(\mathbb{K}(E), \mathbb{K}(F)).$$

Equivalently, $\text{DiffOp}_{X/S}$ is the full infinity-subcategory of jet coalgebras spanned by the cofree objects, with each cofree object $\mathbb{K}(E)$ relabelled by E .

Proposition 5.4.2 (Co-Kleisli identification). *There are canonical equivalences*

$$\text{Map}_{\text{DiffOp}_{X/S}}(E, F) \simeq \text{Map}_X(J^\infty E, F).$$

Under these equivalences the identity is represented by ε_E^Π , and the composite of

$$\widehat{D}_1 : J^\infty E_1 \rightarrow E_2, \quad \widehat{D}_2 : J^\infty E_2 \rightarrow E_3$$

is represented by

$$\widehat{D}_2 J^\infty(\widehat{D}_1) \delta_{E_1}^J.$$

Hence

$$\text{DiffOp}_{X/S} \simeq \text{coKl}(J_{X/S}^\infty).$$

For two composable operators, the co-Kleisli composite is represented by

$$\begin{array}{ccccc} J^\infty E_1 & \xrightarrow{\delta_{E_1}^J} & J^\infty J^\infty E_1 & \xrightarrow{J^\infty(\widehat{D}_1)} & J^\infty E_2 \\ & \searrow \widehat{D}_2 \circ \text{coKl} \widehat{D}_1 & & & \downarrow \widehat{D}_2 \\ & & & & E_3. \end{array}$$

Proof.

- (1) **Apply the cofree adjunction.** Apply the cofree adjunction to maps between cofree coalgebras.
- (2) **Identify co-Kleisli composition.** Composition is composition in the coalgebra infinity-category, whose classifying map is the co-Kleisli composition formula.

- (3) **Inherit the higher coherence.** All higher associativity and unit coherence is therefore inherited from that infinity-category. □

Proposition 5.4.3 (Order-zero operators). *Every morphism $f : E \rightarrow F$ in $(\mathcal{H}_{\text{Sp}})_{/X}$ determines an order-zero operator*

$$\widehat{f}^{(0)} := f \varepsilon_E^\Pi : J^\infty E \rightarrow F.$$

These assignments form an identity-on-objects functor

$$\text{ord}_0 : (\mathcal{H}_{\text{Sp}})_{/X} \rightarrow \text{DiffOp}_{X/S}.$$

Full faithfulness of ord_0 requires additional hypotheses beyond the unrestricted slice.

Proof.

- (1) **Construct the order-zero functor.** The comonad counit defines the canonical identity-on-objects functor to the co-Kleisli infinity-category.
- (2) **Verify compatibility with composition.** Naturality of the counit and the comonad laws identify the co-Kleisli composite of two order-zero maps with the order-zero map of their ordinary composite. □

Definition 5.4.4 (Infinite prolongation of an operator). Let $D : E \rightsquigarrow F$ be a differential operator represented by

$$\widehat{D} : J^\infty E \rightarrow F.$$

Define its infinite prolongation by

$$\text{prol}_{\text{op}}^\infty(\widehat{D}) := J^\infty(\widehat{D})\delta_E^J : J^\infty E \rightarrow J^\infty F.$$

Theorem 5.4.3 (Cofree realization of differential operators). *Let $D : E \rightsquigarrow F$ be represented by*

$$\widehat{D} : J^\infty E \rightarrow F.$$

Then $\text{prol}_{\text{op}}^\infty(\widehat{D})$ is the contractibly unique cofree jet-coalgebra morphism

$$\mathbb{K}(E) \rightarrow \mathbb{K}(F)$$

classified by $\widehat{D}$. It satisfies

$$\varepsilon_F^\Pi \text{prol}_{\text{op}}^\infty(\widehat{D}) \simeq \widehat{D},$$

and infinite prolongation identifies $\text{DiffOp}_{X/S}$ fully faithfully with the full subcategory of cofree jet coalgebras.

Proof.

- (1) **Apply the cofree adjunction.** By the cofree adjunction $U_J \dashv \mathbb{K}$, the morphism classified by $\widehat{D}$ has underlying map $J^\infty(\widehat{D})\delta_E^J$; the displayed identity is the comonad counit law. □

Proposition 5.4.4 (Prolongation respects co-Kleisli composition). *For composable differential operators*

$$D_1 : E_1 \rightsquigarrow E_2, \quad D_2 : E_2 \rightsquigarrow E_3,$$

represented by

$$\widehat{D}_1 : J^\infty E_1 \rightarrow E_2, \quad \widehat{D}_2 : J^\infty E_2 \rightarrow E_3,$$

one has a canonical equivalence of jet-coalgebra morphisms

$$\mathrm{prol}_{\mathrm{op}}^\infty(\widehat{D}_2 \circ_{\mathrm{coKl}} \widehat{D}_1) \simeq \mathrm{prol}_{\mathrm{op}}^\infty(\widehat{D}_2) \circ \mathrm{prol}_{\mathrm{op}}^\infty(\widehat{D}_1).$$

For every section $\phi : X \rightarrow E_1$,

$$(D_2 \circ_{\mathrm{coKl}} D_1)(\phi) \simeq \widehat{D}_2 J^\infty(\widehat{D}_1) \delta_{E_1}^J j^\infty \phi.$$

Proof.

- (1) **Identify the classified composite.** Composition of the two cofree coalgebra morphisms is the cofree morphism whose classifying map is the co-Kleisli composite.
- (2) **Evaluate on a section.** The section formula is obtained by precomposition with $j^\infty \phi$.

□

Differential operators are thus maps *out of* jet objects, with infinite prolongation realizing their co-Kleisli composition coalgebraically. The second use of J^∞ reverses the variance: an equation is presented by a map *into* a jet object.

5.4.3 Generalized jet equations

Homotopy equalizers and derived zero loci of differential operators both produce morphisms into $J^\infty Y$. Rather than force such derived equations to be embedded subobjects, we retain the full morphism $E \rightarrow J^\infty Y$ and organize these presentations in a comma ∞ -category.

Construction 5.4.3 (Equation cut out by differential operators). Let

$$D_0, D_1 : Y \rightsquigarrow F$$

be differential operators represented by

$$\widehat{D}_0, \widehat{D}_1 : J^\infty Y \rightrightarrows F.$$

Define the derived equation object

$$E(D_0, D_1) := \mathrm{Eq}(\widehat{D}_0, \widehat{D}_1)$$

with its canonical map to $J^\infty Y$. This is the homotopy equalizer in $(\mathcal{H}_{\mathrm{Sp}})_/X$.

Construction 5.4.4 (Derived zero locus). Let $W \rightarrow X$ be equipped with a zero section

$$0 : X \rightarrow W,$$

and let $f : J^\infty Y \rightarrow W$. Define

$$Z(f) := J^\infty Y \times_W X.$$

It is characterized by the Cartesian square

$$\begin{array}{ccc} Z(f) & \longrightarrow & J^\infty Y \\ \downarrow & & \downarrow f \\ X & \xrightarrow{0} & W. \end{array}$$

The displayed homotopy fibre is the derived zero locus.

Definition 5.4.5 (Generalized and embedded jet equation). The formulation by morphisms into an infinite jet object is the derived analogue of generalized equations in geometric PDE theory [38, 27]. A **generalized jet equation** on Y relative to X/S is a morphism

$$i : E \longrightarrow J^\infty Y$$

in $(\mathcal{H}_{\text{Sp}})_{/X}$. It is **embedded** if i is a monomorphism.

Construction 5.4.5 (Infinity-category of generalized jet equations). Define

$$\text{Eqn}_{X/S} := (\text{id}_{(\mathcal{H}_{\text{Sp}})_{/X}} \downarrow J_{X/S}^\infty).$$

Thus an object is a triple

$$(Y, E, i), \quad i : E \longrightarrow J_{X/S}^\infty Y,$$

and a morphism

$$(Y, E, i) \longrightarrow (Y', E', i')$$

consists of maps

$$f : Y \rightarrow Y', \quad g : E \rightarrow E',$$

together with a commutative square

$$\begin{array}{ccc} E & \xrightarrow{i} & J^\infty Y \\ g \downarrow & & \downarrow J^\infty(f) \\ E' & \xrightarrow{i'} & J^\infty Y'. \end{array}$$

Equivalently,

$$\text{Eqn}_{X/S} \simeq \text{Fun}(\Delta^1, (\mathcal{H}_{\text{Sp}})_{/X}) \times_{(\mathcal{H}_{\text{Sp}})_{/X} \times (\mathcal{H}_{\text{Sp}})_{/X}} ((\mathcal{H}_{\text{Sp}})_{/X} \times (\mathcal{H}_{\text{Sp}})_{/X}),$$

where the first map is $(\text{ev}_0, \text{ev}_1)$ and the second sends (E, Y) to $(E, J^\infty Y)$. Hence the equation infinity-category is constructed by a finite limit in Cat_∞ .

For fixed Y , write

$$\text{Eqn}_{X/S}(Y) := \text{fib}_Y \left(\text{Eqn}_{X/S} \longrightarrow (\mathcal{H}_{\text{Sp}})_{/X} \right).$$

Equivalently,

$$\text{Eqn}_{X/S}(Y) \simeq ((\mathcal{H}_{\text{Sp}})_{/X})_{/J^\infty Y}.$$

Remark 5.4.1 (Generalized and embedded equations). A monomorphism in an infinity-topos is (-1) -truncated [36]. Derived zero loci and homotopy equalizers naturally carry higher homotopy data. Generalized jet equations retain this data by allowing arbitrary morphisms into a jet object, while embedded equations form the monomorphism-valued special case.

Proposition 5.4.5 (Pullback and simultaneous systems). *Let $f : Y \rightarrow Z$ and let $i : F \rightarrow J^\infty Z$ be a generalized jet equation. Its pullback to Y is*

$$J^\infty Y \times_{J^\infty Z} F \longrightarrow J^\infty Y.$$

If i is embedded, so is its pullback. More generally, any small family

$$\{E_\alpha \rightarrow J^\infty Y\}_{\alpha \in I}$$

has a simultaneous system given by its limit in $((\mathcal{H}_{\text{Sp}})_{/X})_{/J^\infty Y}$. For embedded members this is their intersection as subobjects.

Proof.

- (1) **Construct the pulled-back equation.** The first assertion is functoriality of J^∞ followed by pullback.
- (2) **Preserve embeddedness.** Monomorphisms are pullback-stable.
- (3) **Construct simultaneous systems.** Slices of an infinity-topos have all small limits, and limits of monomorphisms are monomorphisms.

□

Proposition 5.4.6 (Embedded predicates are internally classified). *If*

$$i : E \hookrightarrow J^\infty Y$$

is embedded, then it is classified by a morphism

$$\chi_i : J^\infty Y \longrightarrow \Omega_X^{\text{sub}}$$

to the subobject classifier of $(\mathcal{H}_{\text{Sp}})_{/X}$. Thus embedded equations are exactly proposition-valued differential predicates on the infinite jet object. Generalized jet equations retain the full morphism data into $J^\infty Y$.

Proof.

- (1) **Apply the subobject-classifier theorem.** The slice $(\mathcal{H}_{\text{Sp}})_{/X}$ is an infinity-topos, and its monomorphisms into a fixed object are classified by its subobject classifier [36].

□

The equation category separates the ambient variable Y from the equation object E while retaining derived intersections and higher homotopy. It still says nothing about whether E itself is transported coherently along the de Rham relation. That is the role of Cartan structure.

5.4.4 Cartan equations as coalgebra morphisms

The structured subclass consists of generalized equations whose source already carries a jet-coalgebra presentation and whose map into the ambient jet object preserves crystalline transport.

Definition 5.4.6 (Jet-coalgebra presentation). A **jet-coalgebra presentation** over X/S is a jet coalgebra

$$(E, \rho), \quad \rho : E \longrightarrow J^\infty E.$$

By Theorem 5.3.2, every such structure is equivalently the pullback presentation of an intrinsic formally integrable PDE

$$F \longrightarrow Q_{X/S}.$$

The term “Cartan” is reserved for this compatibility of an equation morphism with the already-defined crystalline transport. Thus *crystalline* describes descent along the effective de Rham relation, while *Cartan* describes preservation of that descent by an equation morphism. The cofree adjunction will recognize such morphisms as ordinary maps from the transported source to the ambient object.

Definition 5.4.7 (Cartan equation morphism). Let (E, ρ) be a jet coalgebra and let $Y \rightarrow X$. A map

$$y : E \rightarrow Y$$

constructs the Cartan equation morphism

$$i_y := J^\infty(y)\rho : E \rightarrow J^\infty Y.$$

The counit identity gives the commutative diagram

$$\begin{array}{ccccc} E & \xrightarrow{\rho} & J^\infty E & \xrightarrow{J^\infty(y)} & J^\infty Y \\ y \downarrow & & & & \downarrow \varepsilon_Y^\Pi \\ Y & \xlongequal{\quad\quad\quad} & & & Y. \end{array}$$

It is **embedded** if i_y is a monomorphism.

Theorem 5.4.4 (Cofree recognition of Cartan equation morphisms). *For a jet coalgebra (E, ρ) and $Y \in (\mathcal{H}_{\text{Sp}})_{/X}$, the cofree adjunction gives*

$$\text{Map}_{\text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_{/X})}((E, \rho), \mathbb{K}(Y)) \simeq \text{Map}_X(E, Y).$$

The coalgebra map classified by $y : E \rightarrow Y$ has underlying map

$$J^\infty(y)\rho : E \rightarrow J^\infty Y.$$

Conversely every coalgebra map

$$i : (E, \rho) \rightarrow \mathbb{K}(Y)$$

is canonically of this form with classifying map $y = \varepsilon_Y^\Pi i$.

Proof.

- (1) **Use the cofree mapping-space equivalence.** The mapping-space equivalence is the cofree adjunction $U_J \dashv \mathbb{K}$, as in the Eilenberg–Moore formalism [34].
- (2) **Recover the classifying map.** If i is a coalgebra map, then

$$\begin{aligned} J^\infty(\varepsilon_Y^\Pi i)\rho &\simeq J^\infty(\varepsilon_Y^\Pi)J^\infty(i)\rho \\ &\simeq J^\infty(\varepsilon_Y^\Pi)\delta_Y^J i \\ &\simeq i \end{aligned}$$

by the coalgebra-morphism identity and the comonad counit law.

□

Cartan equations are therefore the generalized equations whose sources already carry crystalline transport and whose ambient equation maps preserve it. The next question is different: how should such equations be tested against sections?

5.4.5 Actual and parameterized solutions

An actual solution compares an ordinary section of Y with its holonomic infinite jet and asks for a lift through the equation. The homotopy pullback below records the section, the lift, and the path identifying the two. Dependent products package the same condition uniformly in parameters over the base.

Definition 5.4.8 (Solution space). Let $i : E \rightarrow J^\infty Y$ be a generalized jet equation. Define its solution space by the homotopy pullback

$$\mathrm{Sol}(i) := \mathrm{Map}_X(X, Y) \times_{\mathrm{Map}_X(X, J^\infty Y)} \mathrm{Map}_X(X, E),$$

where the first map is $\phi \mapsto j^\infty \phi$ and the second is induced by i . Equivalently,

$$\begin{array}{ccc} \mathrm{Sol}(i) & \longrightarrow & \mathrm{Map}_X(X, E) \\ \downarrow & & \downarrow i_* \\ \mathrm{Map}_X(X, Y) & \xrightarrow{j^\infty} & \mathrm{Map}_X(X, J^\infty Y) \end{array}$$

is Cartesian. Thus a point of $\mathrm{Sol}(i)$ already consists of a section $\phi : X \rightarrow Y$, a map $a : X \rightarrow E$, and the path $ia \simeq j^\infty \phi$.

Construction 5.4.6 (Cartan structure on a parameterized copy of the base). Let $t : T \rightarrow S$, put

$$X_T := X \times_S T,$$

and regard X_T as an object of $(\mathcal{H}_{\mathrm{Sp}})_{/X}$ through the projection to X . The base-change equivalence

$$Q_{X_T/T} \simeq Q_{X/S} \times_S T$$

makes $Q_{X_T/T}$ an object over $Q_{X/S}$, and

$$X_T \simeq e_{X/S}^* Q_{X_T/T}.$$

The unit of $e_{X/S}^* \dashv \Pi_{e_{X/S}}$ constructs

$$\rho_T : X_T \longrightarrow J_{X/S}^\infty X_T.$$

For a T -parameterized section $\phi : X_T \rightarrow Y$, define

$$j_T^\infty \phi := J^\infty(\phi) \rho_T : X_T \rightarrow J^\infty Y.$$

Proposition 5.4.7 (Compatibility with base-changed prolongation). *Let $g_X : X_T \rightarrow X$ be the projection, put $Y_T := g_X^* Y$, and let $\tilde{\phi} : X_T \rightarrow Y_T$ correspond to $\phi : X_T \rightarrow Y$. Under the canonical equivalences*

$$\begin{aligned} \mathrm{Map}_X(X_T, J_{X/S}^\infty Y) &\simeq \mathrm{Map}_{X_T}(X_T, g_X^* J_{X/S}^\infty Y) \\ &\simeq \mathrm{Map}_{X_T}(X_T, J_{X_T/T}^\infty Y_T), \end{aligned}$$

the map $j_T^\infty \phi$ corresponds to

$$j^\infty \tilde{\phi} : X_T \rightarrow J_{X_T/T}^\infty Y_T.$$

Proof.

- (1) **Base-change the effective quotient.** Write $e_{X_T/T} : X_T \rightarrow Q_{X_T/T}$ for the base change of $e_{X/S}$. The base-change theorem of Chapter 4 gives

$$Q_{X_T/T} \simeq Q_{X/S} \times_S T$$

and therefore an equivalence of relation groupoids

$$\mathcal{R}_{X_T/T, \bullet}^{\text{dR}} \simeq \mathcal{R}_{X/S, \bullet}^{\text{dR}} \times_S T.$$

- (2) **Transpose the parameterized coaction.** Under

$$J_{X/S}^\infty \simeq \Pi_s \mathbf{t}^*,$$

the coaction

$$\rho_T : X_T \rightarrow J_{X/S}^\infty X_T$$

is adjoint to the canonical crystalline transport

$$\theta_T : \mathbf{s}^* X_T \xrightarrow{\simeq} \mathbf{t}^* X_T.$$

- (3) **Identify transport on the parameter.** Since $X_T = X \times_S T$, both sides identify with $\mathcal{R}_{X/S}^{\text{dR}} \times_S T$, and θ_T is the canonical identity on the parameter T .

- (4) **Compute the transpose of parameterized prolongation.** Consequently the relation transpose of

$$j_T^\infty \phi = J^\infty(\phi) \rho_T$$

is

$$\mathbf{s}^* X_T \xrightarrow{\theta_T} \mathbf{t}^* X_T \xrightarrow{\mathbf{t}^* \phi} \mathbf{t}^* Y.$$

- (5) **Compare the relation transposes.** Transporting this map through $\mathcal{R}_{X_T/T}^{\text{dR}} \simeq \mathcal{R}_{X/S}^{\text{dR}} \times_S T$ identifies it with

$$v_1^* X_T \longrightarrow \mathbf{t}^* Y_T$$

obtained by restricting $\tilde{\phi}$ to the evaluation vertex. By the relation formula for the base-changed jet comonad, this is precisely the transpose of

$$j^\infty \tilde{\phi} : X_T \rightarrow J_{X_T/T}^\infty Y_T.$$

- (6) **Apply right Beck–Chevalley.** Right Beck–Chevalley identifies $g_X^* J_{X/S}^\infty Y$ with $J_{X_T/T}^\infty Y_T$, so the two prolongations correspond as asserted.

□

Construction 5.4.7 (Parameterized prolongation map over the base). Since $p = p_Q e_{X/S}$, composition of dependent products gives

$$\Pi_p \simeq \Pi_{p_Q} \Pi_{e_{X/S}}.$$

Define

$$j_{/S}^\infty : \Pi_p Y \longrightarrow \Pi_p J^\infty Y$$

to be Π_{p_Q} applied to the unit

$$\Pi_{e_{X/S}} Y \longrightarrow \Pi_{e_{X/S}} e_{X/S}^* \Pi_{e_{X/S}} Y.$$

Definition 5.4.9 (Parameterized solution object). For $i : E \rightarrow J^\infty Y$, define

$$\mathbf{Sol}_{X/S}(i) := \Pi_p E \times_{\Pi_p J^\infty Y} \Pi_p Y$$

in the slice over S , where the lower map is $j_{/S}^\infty$. Thus

$$\begin{array}{ccc} \mathbf{Sol}_{X/S}(i) & \longrightarrow & \Pi_p E \\ \downarrow & & \downarrow \Pi_p(i) \\ \Pi_p Y & \xrightarrow{j_{/S}^\infty} & \Pi_p J^\infty Y \end{array}$$

is Cartesian in $(\mathcal{H}_{\text{Sp}})_{/S}$.

Theorem 5.4.5 (Parameterized solution object). *For every $T \rightarrow S$, there is a canonical equivalence*

$$\text{Map}_S(T, \mathbf{Sol}_{X/S}(i)) \simeq \text{Sol}(i_T),$$

where i_T is the base-changed equation over X_T/T . Thus $\mathbf{Sol}_{X/S}(i)$ represents parameterized solutions.

Proof.

- (1) **Apply dependent product.** Dependent product gives

$$\text{Map}_S(T, \Pi_p Z) \simeq \text{Map}_X(X_T, Z).$$

- (2) **Identify parameterized prolongation.** Proposition 5.4.7 identifies the T -point of $j_{/S}^\infty$ with ordinary base-changed jet prolongation.
- (3) **Identify the solution pullback.** Taking T -points of the defining pullback of $\mathbf{Sol}_{X/S}(i)$ therefore gives the homotopy pullback defining $\text{Sol}(i_T)$.

□

Actual solutions have now been represented both pointwise and in families, but they test only holonomic sections over X . Formal integrability requires a stronger infinitesimal test: the equation must also admit compatible families over every de Rham formal disk.

5.4.6 Formal-disk families and formal solutions

The disk monad supplies those test objects. A formal solution over a parameter Z consists of a lift from the disk object into the equation together with an ambient disk family whose canonical Cartan lift agrees with that equation map.

Definition 5.4.10 (Parameterized de Rham formal-disk family). For $Z, Y \in (\mathcal{H}_{\text{Sp}})_{/X}$, a Z -parameterized formal-disk family in Y is a morphism

$$\phi : \text{Disk}^{\text{dR}, \infty} Z \rightarrow Y.$$

Its ambient jet map is the constructed map

$$\sigma_\phi := J^\infty(\phi) \rho_Z^{\mathbb{F}} : \text{Disk}^{\text{dR}, \infty} Z \rightarrow J^\infty Y.$$

Theorem 5.4.6 (Constructed Cartan lift). *For every*

$$\phi : \text{Disk}^{\text{dR},\infty} Z \rightarrow Y,$$

the map σ_ϕ underlies the contractibly unique cofree jet-coalgebra morphism

$$\tilde{\sigma}_\phi : \mathbb{F}(Z) \rightarrow \mathbb{K}(Y)$$

classified by ϕ . Hence

$$\text{Map}_X(\text{Disk}^{\text{dR},\infty} Z, Y) \simeq \text{Map}_{\text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_X)}(\mathbb{F}(Z), \mathbb{K}(Y)).$$

The Cartan structure is therefore determined by ϕ .

Proof.

- (1) **Apply the cofree adjunction.** Apply the cofree adjunction to the free jet coalgebra.
- (2) **Identify the underlying Cartan lift.** The cofree morphism classified by ϕ has underlying map

$$J^\infty(\phi)\rho_Z^{\mathbb{F}}.$$

□

Definition 5.4.11 (Parameterized formal-solution space). For a generalized jet equation

$$i : E \rightarrow J^\infty Y,$$

define

$$\text{FSol}_i(Z) := \text{Map}_X(\text{Disk}^{\text{dR},\infty} Z, E) \times_{\text{Map}_X(\text{Disk}^{\text{dR},\infty} Z, J^\infty Y)} \text{Map}_X(\text{Disk}^{\text{dR},\infty} Z, Y),$$

where the second map sends ϕ to

$$J^\infty(\phi)\rho_Z^{\mathbb{F}}.$$

Equivalently, the defining square

$$\begin{array}{ccc} \text{FSol}_i(Z) & \xrightarrow{\quad\quad\quad} & \text{Map}_X(\text{Disk}^{\text{dR},\infty} Z, E) \\ \downarrow & & \downarrow i_* \\ \text{Map}_X(\text{Disk}^{\text{dR},\infty} Z, Y) & \xrightarrow[\phi \mapsto J^\infty(\phi)\rho_Z^{\mathbb{F}}]{} & \text{Map}_X(\text{Disk}^{\text{dR},\infty} Z, J^\infty Y) \end{array}$$

is Cartesian. Thus a point of $\text{FSol}_i(Z)$ contains maps

$$\ell : \text{Disk}^{\text{dR},\infty} Z \rightarrow E, \quad \phi : \text{Disk}^{\text{dR},\infty} Z \rightarrow Y,$$

together with the compatibility path

$$i\ell \simeq J^\infty(\phi)\rho_Z^{\mathbb{F}}.$$

Remark 5.4.2 (Literal de Rham formal-disk interpretation). If $Z = a : T \rightarrow X$ is a generalized point, then $\text{Disk}^{\text{dR},\infty}(a)$ is the relative de Rham neighbourhood $\text{Nbh}_{T/S}^{\text{dR}}(T \times_S X)$. Hence $\text{FSol}_i(a)$ is literally the homotopy-pullback space of maps from that de Rham formal disk into the equation and the ambient object.

Proposition 5.4.8 (Adjoint formula for a de Rham formal-disk family). *Let*

$$\sigma_\phi = J^\infty(\phi)\rho_Z^{\mathbb{F}} : \text{Disk}^{\text{dR},\infty} Z \rightarrow J^\infty Y,$$

and let

$$\sigma_\phi^\sharp : Z \rightarrow J^\infty J^\infty Y$$

be its adjunct under

$$\text{Disk}^{\text{dR},\infty} \dashv J^\infty.$$

Then

$$\sigma_\phi^\sharp \simeq \delta_Y^J \sigma_\phi \eta_Z^\Sigma.$$

Proof.

(1) **Use the coalgebra-morphism identity.** The cofree map

$$\tilde{\sigma}_\phi : \mathbb{F}(Z) \rightarrow \mathbb{K}(Y)$$

is a coalgebra morphism, so

$$J^\infty(\sigma_\phi)\rho_Z^{\mathbb{F}} \simeq \delta_Y^J \sigma_\phi.$$

(2) **Insert the adjunction-unit factorization.** Using

$$\eta_Z^{\text{adj}} \simeq \rho_Z^{\mathbb{F}} \eta_Z^\Sigma$$

from Proposition 5.3.1,

$$\begin{aligned} \sigma_\phi^\sharp &= J^\infty(\sigma_\phi)\eta_Z^{\text{adj}} \\ &\simeq J^\infty(\sigma_\phi)\rho_Z^{\mathbb{F}}\eta_Z^\Sigma \\ &\simeq \delta_Y^J \sigma_\phi \eta_Z^\Sigma. \end{aligned}$$

□

Proposition 5.4.9 (Adjoint form of a formal solution). *Let*

$$(\ell, \phi, h)$$

be a point of $\text{FSol}_i(Z)$, so

$$h : i\ell \simeq \sigma_\phi,$$

and let

$$\ell^\sharp : Z \rightarrow J^\infty E$$

be the adjunct of ℓ . The path h induces a path

$$J^\infty(i)\ell^\sharp \simeq \delta_Y^J i\ell\eta_Z^\Sigma,$$

and mate calculus gives

$$\varepsilon_E^\Pi \ell^\sharp \simeq \ell\eta_Z^\Sigma.$$

Proof.

(1) **Identify the adjunct of the equation composite.** The adjunct of $i\ell$ is

$$J^\infty(i)\ell^\sharp.$$

- (2) **Transport the formal-solution path.** Transport the path

$$i\ell \simeq \sigma_\phi$$

through the adjunction and apply Proposition 5.4.8.

- (3) **Apply mate calculus.** The final identity is Lemma 5.3.2.

□

5.5 Cartan prolongation and formal integrability

The preceding section separated four roles: operators are co-Kleisli maps, generalized equations are maps into jets, actual solutions test holonomic sections, and formal solutions test de Rham disk families. What remains is to force an arbitrary generalized equation to carry the crystalline compatibility detected by the last test.

Cartan prolongation supplies that universal repair. It is a pullback in jet coalgebras that retains the original ambient equation while imposing compatibility with the comonad. Its underlying object represents formal solutions, and the evaluation back to E measures whether any new compatibility was actually added. The section is organized around that single recognition map.

$$\begin{array}{ccc}
 & \text{Prol}^{\text{Cart}}(i) & \\
 p_1 \swarrow & & \searrow p_2 \\
 \mathbb{K}(E) & & \mathbb{K}(Y) \\
 \mathbb{K}(i) \searrow & & \swarrow \tilde{\delta}_Y^J \\
 & \mathbb{K}(J^\infty Y) &
 \end{array}$$

The pullback itself constructs the Cartan structure. The subsequent prolongation-completeness criterion asks exactly when its evaluation to E is already an equivalence; no separate notion of compatibility is introduced.

5.5.1 Cartan prolongation and its universal property

Fix a generalized jet equation

$$i : E \longrightarrow J^\infty Y.$$

Construction 5.5.1 (Infinite Cartan prolongation). The cofree functor gives

$$\mathbb{K}(i) : \mathbb{K}(E) \rightarrow \mathbb{K}(J^\infty Y).$$

There is also the cofree coalgebra map

$$\tilde{\delta}_Y^J : \mathbb{K}(Y) \rightarrow \mathbb{K}(J^\infty Y)$$

classified by $\text{id}_{J^\infty Y}$; its underlying map is δ_Y^J . Define

$$\text{Prol}^{\text{Cart}}(i) := \mathbb{K}(E) \times_{\mathbb{K}(J^\infty Y)} \mathbb{K}(Y)$$

in $\text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_X)$. Since the forgetful functor creates limits, its underlying object is

$$\text{Prol}(i) := J^\infty E \times_{J^\infty J^\infty Y} J^\infty Y.$$

Write

$$p_1 : \text{Prol}(i) \rightarrow J^\infty E, \quad p_2 : \text{Prol}(i) \rightarrow J^\infty Y$$

for the two projections. The construction is encoded by the Cartesian diagram

$$\begin{array}{ccc} \text{Prol}(i) & \xrightarrow{p_1} & J^\infty E \\ p_2 \downarrow & & \downarrow J^\infty(i) \\ J^\infty Y & \xrightarrow{\delta_Y^J} & J^\infty J^\infty Y. \end{array}$$

Theorem 5.5.1 (Canonical coalgebra on the prolongation). *The object $\text{Prol}(i)$ carries the canonical jet-coalgebra structure underlying $\text{Prol}^{\text{Cart}}(i)$. Write its coaction as*

$$\rho_{\text{Prol}(i)} : \text{Prol}(i) \longrightarrow J^\infty \text{Prol}(i).$$

Both projections p_1 and p_2 are coalgebra maps.

Proof.

- (1) **Create the pullback coalgebra.** Apply the limit-creation statement of Proposition 5.2.5 to the defining pullback of coalgebras.

□

Fix Y . Define

$$\text{CartEq}_{X/S}(Y) := \text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_X)/\mathbb{K}(Y).$$

Let

$$V_Y : \text{CartEq}_{X/S}(Y) \longrightarrow \text{Eqn}_{X/S}(Y)$$

be the underlying equation-morphism functor.

Theorem 5.5.2 (Cartan prolongation adjunction). *The construction above defines a functor*

$$\text{Prol}_Y^{\text{Cart}} : \text{Eqn}_{X/S}(Y) \rightarrow \text{CartEq}_{X/S}(Y)$$

and there is a canonical adjunction

$$V_Y \dashv \text{Prol}_Y^{\text{Cart}}.$$

The adjunction is displayed by

$$\text{CartEq}_{X/S}(Y) \begin{array}{c} \xrightarrow{V_Y} \\ \xleftarrow{\text{Prol}_Y^{\text{Cart}}} \end{array} \text{Eqn}_{X/S}(Y)$$

with V_Y left adjoint to Cartan prolongation. Equivalently, for $j : C \rightarrow \mathbb{K}(Y)$ in $\text{CartEq}_{X/S}(Y)$ and $i : E \rightarrow J^\infty Y$ in $\text{Eqn}_{X/S}(Y)$,

$$\text{Map}_{\text{CartEq}_{X/S}(Y)}(j, \text{Prol}_Y^{\text{Cart}}(i)) \simeq \text{Map}_{\text{Eqn}_{X/S}(Y)}(V_Y j, i).$$

Proof.

- (1) **Unpack maps into the Cartan pullback.** A map $j \rightarrow \text{Prol}_Y^{\text{Cart}}(i)$ is a coalgebra map

$$a : C \rightarrow \mathbb{K}(E)$$

whose composite with $\mathbb{K}(i)$ agrees with $\tilde{\delta}_Y^J j$.

- (2) **Apply the cofree adjunction.** The cofree adjunction identifies a with a map

$$\bar{a} : U_J C \rightarrow E.$$

- (3) **Compare the classifying maps.** The classifying map of $\mathbb{K}(i)a$ is $i\bar{a}$, while the classifying map of $\tilde{\delta}_Y^J j$ is

$$\varepsilon_{J^\infty Y}^\Pi \delta_Y^J U_J j \simeq U_J j.$$

- (4) **Identify the mapping-space adjunction.** Hence the compatibility is exactly the path

$$i\bar{a} \simeq U_J j$$

that defines a map from $V_Y j$ to i in the slice $\text{Eqn}_{X/S}(Y)$. The construction is natural on the full mapping spaces, giving the adjunction.

□

Theorem 5.5.3 (Universal formal-solution object). *For every $Z \in (\mathcal{H}_{\text{Sp}})_{/X}$, there is a natural equivalence*

$$\text{Map}_X(Z, \text{Prol}(i)) \simeq \text{FSol}_i(Z).$$

Thus the identity of $\text{Prol}(i)$ determines the universal formal-solution family.

Proof.

- (1) **Map the free coalgebra into the pullback.** Using $\mathbb{F} \dashv U_J$ and the pullback defining $\text{Prol}^{\text{Cart}}(i)$,

$$\begin{aligned} \text{Map}_X(Z, \text{Prol}(i)) &\simeq \text{Map}_{\text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_{/X})}(\mathbb{F}(Z), \text{Prol}^{\text{Cart}}(i)) \\ &\simeq \text{Map}_{\text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_{/X})}(\mathbb{F}(Z), \mathbb{K}(E)) \times_{\text{Map}_{\text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_{/X})}(\mathbb{F}(Z), \mathbb{K}(J^\infty Y))} \text{Map}_{\text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_{/X})}(\mathbb{F}(Z), \mathbb{K}(J^\infty Y)) \\ &\simeq \text{Map}_X(\text{Disk}^{\text{dR}, \infty} Z, E) \times_{\text{Map}_X(\text{Disk}^{\text{dR}, \infty} Z, J^\infty Y)} \text{Map}_X(\text{Disk}^{\text{dR}, \infty} Z, Y). \end{aligned}$$

- (2) **Identify the formal-solution space.** The right side is $\text{FSol}_i(Z)$.

□

Under the formal-disk–jet adjunction, the maps p_1 and p_2 are adjunct to maps

$$\ell_i^{\text{univ}} : \text{Disk}^{\text{dR}, \infty} \text{Prol}(i) \rightarrow E, \quad \phi_i^{\text{univ}} : \text{Disk}^{\text{dR}, \infty} \text{Prol}(i) \rightarrow Y.$$

The defining pullback path gives

$$i\ell_i^{\text{univ}} \simeq J^\infty(\phi_i^{\text{univ}})\rho_{\text{Prol}(i)}^\mathbb{F}.$$

This is the formal-solution family classified by $\text{id}_{\text{Prol}(i)}$.

Definition 5.5.1 (Prolongation evaluation map). Define

$$c_i : \text{Prol}(i) \rightarrow E$$

by

$$c_i := \varepsilon_E^\Pi p_1.$$

Equivalently,

$$\begin{array}{ccc} \text{Prol}(i) & \xrightarrow{p_1} & J^\infty E \\ & \searrow c_i & \swarrow \varepsilon_E^\Pi \\ & E & \end{array}$$

commutes.

Proposition 5.5.1 (Prolongation evaluation as the adjunction counit). *The prolongation evaluation map satisfies*

$$c_i \simeq \ell_i^{\text{univ}} \eta_{\text{Prol}(i)}^\Sigma, \quad i c_i \simeq p_2.$$

As a morphism in $\text{Eqn}_{X/S}(Y)$,

$$c_i : V_Y \text{Prol}_Y^{\text{Cart}}(i) \longrightarrow i$$

is the counit of $V_Y \dashv \text{Prol}_Y^{\text{Cart}}$.

Proof.

- (1) **Identify evaluation through adjunction.** The first identity is Lemma 5.3.2 applied to the adjunct pair $\ell_i^{\text{univ}} \dashv p_1$.
- (2) **Compute the ambient equation map.** Naturality of the comonad counit and the pullback relation $J^\infty(i)p_1 \simeq \delta_Y^J p_2$ give

$$\begin{aligned} i c_i &\simeq i \varepsilon_E^\Pi p_1 \\ &\simeq \varepsilon_{J^\infty Y}^\Pi J^\infty(i) p_1 \\ &\simeq \varepsilon_{J^\infty Y}^\Pi \delta_Y^J p_2 \\ &\simeq p_2. \end{aligned}$$

- (3) **Recognize the adjunction counit.** The mapping-space construction in the proof of Theorem 5.5.2 identifies this morphism with the adjunction counit.

□

Proposition 5.5.2 (Evaluation of formal solutions). *Under the natural equivalence*

$$\text{Map}_X(Z, \text{Prol}(i)) \simeq \text{FSol}_i(Z),$$

the map induced by c_i ,

$$\text{Map}_X(Z, \text{Prol}(i)) \rightarrow \text{Map}_X(Z, E),$$

sends a formal solution represented by $\ell : \text{Disk}^{\text{dR}, \infty} Z \rightarrow E$ to its value

$$\ell \eta_Z^\Sigma : Z \rightarrow E.$$

Proof.

- (1) **Identify the adjunct of the formal-solution lift.** If $f : Z \rightarrow \text{Prol}(i)$ represents the formal solution, then $p_1 f$ is the adjunct of ℓ .
- (2) **Compute prolongation evaluation.** Hence

$$c_i f = \varepsilon_E^\Pi p_1 f \simeq \ell \eta_Z^\Sigma$$

by Lemma 5.3.2.

□

Theorem 5.5.4 (Actual solutions as maps into Cartan prolongation). *For every generalized jet equation*

$$i : E \longrightarrow J^\infty Y,$$

there is a canonical natural equivalence

$$\text{Sol}(i) \xrightarrow{\simeq} \text{Map}_{\text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_{/X})}((X, \rho_X), \text{Prol}^{\text{Cart}}(i)).$$

Thus the Cartan prolongation represents actual solutions internally in the jet-coalgebra category. The representing correspondence is organized by

$$\begin{array}{ccc} (X, \rho_X) & \longrightarrow & \text{Prol}^{\text{Cart}}(i) \\ & \searrow & \downarrow \\ & & \mathbb{K}(Y), \end{array}$$

where the upper map is the coalgebraic form of a solution and the vertical map is the prolonged ambient equation.

Proof.

- (1) **Map the terminal coalgebra into the pullback.** Mapping from (X, ρ_X) into the defining pullback of $\text{Prol}^{\text{Cart}}(i)$ gives

$$\begin{aligned} & \text{Map}_{\text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_{/X})}((X, \rho_X), \text{Prol}^{\text{Cart}}(i)) \\ & \simeq \text{Map}_{\text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_{/X})}((X, \rho_X), \mathbb{K}(E)) \times_{\text{Map}_{\text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_{/X})}((X, \rho_X), \mathbb{K}(J^\infty Y))} \text{Map}_{\text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_{/X})}((X, \rho_X), \mathbb{K}(Y)) \end{aligned}$$

- (2) **Apply the cofree adjunction.** The cofree adjunction identifies the three mapping spaces with

$$\text{Map}_X(X, E), \quad \text{Map}_X(X, J^\infty Y), \quad \text{Map}_X(X, Y).$$

- (3) **Identify the two comparison maps.** The first comparison map is induced by i . The second sends $\phi : X \rightarrow Y$ to

$$J^\infty(\phi)\rho_X = j^\infty\phi.$$

- (4) **Recognize the solution pullback.** Hence the displayed pullback is exactly the homotopy pullback defining $\text{Sol}(i)$.

□

Corollary 5.5.1 (Actual solutions induce formal solutions). *Every point of $\text{Sol}(i)$ canonically determines a coalgebra map*

$$(X, \rho_X) \longrightarrow \text{Prol}^{\text{Cart}}(i)$$

and hence an underlying map

$$X \longrightarrow \text{Prol}(i),$$

equivalently an X -parameterized formal solution. The construction is natural in morphisms of generalized jet equations.

Proof.

- (1) **Forget the represented coalgebra structure.** Apply 5.5.4 and then forget the canonical coalgebra structure on $\text{Prol}^{\text{Cart}}(i)$.

□

Cartan prolongation has now produced both a universal Cartan equation and a single comparison back to the original equation. Formal integrability is the recognition problem for that comparison.

5.5.2 Prolongation-completeness and intrinsic formal integrability

We call an equation prolongation-complete when the evaluation $c_i : \text{Prol}(i) \rightarrow E$ is invertible. The results in this subsection identify that one condition in three useful forms: unique formal solvability from a value, a Cartesian comonad square for an already-Cartan equation, and a canonically determined jet-coalgebra structure on an arbitrary prolongation-complete equation.

Definition 5.5.2 (Prolongation-complete generalized jet equation). A generalized jet equation

$$i : E \rightarrow J^\infty Y$$

is **prolongation-complete** if

$$c_i : \text{Prol}(i) \xrightarrow{\sim} E.$$

Theorem 5.5.5 (Formal-solution recognition of prolongation-completeness). *The equation i is prolongation-complete if and only if, for every $Z \in (\mathcal{H}_{\text{Sp}})_{/X}$, evaluation is an equivalence*

$$\text{FSol}_i(Z) \xrightarrow{\sim} \text{Map}_X(Z, E).$$

Equivalently, every generalized jet equation point carries a coherently unique formal solution determined by its value. This criterion concerns formal infinitesimal solutions. Analytic existence of actual solutions belongs to a separate analytic problem.

Proof.

- (1) **Identify evaluation with mapping out of the counit.** By Theorem 5.5.3 and Proposition 5.5.2, the displayed map is $\text{Map}_X(Z, c_i)$.
- (2) **Apply Yoneda.** It is an equivalence for every Z exactly when c_i is an equivalence, by Yoneda.

□

Proposition 5.5.3 (Split prolongation evaluation for a Cartan equation morphism). *Let*

$$i : (E, \rho) \rightarrow \mathbb{K}(Y)$$

be a jet-coalgebra morphism. There is a canonical coalgebra map

$$h_i : (E, \rho) \rightarrow \text{Prol}^{\text{Cart}}(i)$$

whose composites with the two projections are ρ and i . It satisfies

$$c_i h_i \simeq \text{id}_E.$$

Thus the prolongation evaluation map of every Cartan equation morphism is canonically a split epimorphism.

Proof.

- (1) **Construct the split comparison.** The coalgebra-morphism identity

$$J^\infty(i)\rho \simeq \delta_Y^J i$$

constructs h_i by the pullback universal property of $\text{Prol}^{\text{Cart}}(i)$.

- (2) **Compute the prolongation evaluation.** Then

$$c_i h_i \simeq \varepsilon_E^\Pi \rho \simeq \text{id}_E.$$

□

Theorem 5.5.6 (Cartesian criterion). *For a Cartan equation morphism*

$$i : (E, \rho) \rightarrow \mathbb{K}(Y),$$

the following are equivalent:

1. *the underlying equation $i : E \rightarrow J^\infty Y$ is prolongation-complete;*
2. *the canonical comparison $h_i : E \rightarrow \text{Prol}(i)$ is an equivalence;*
3. *the Cartan square*

$$\begin{array}{ccc} E & \xrightarrow{\rho} & J^\infty E \\ i \downarrow & & \downarrow J^\infty(i) \\ J^\infty Y & \xrightarrow{\delta_Y^J} & J^\infty J^\infty Y \end{array}$$

is Cartesian.

Proof.

- (1) **Identify the prolongation pullback.** The object $\text{Prol}(i)$ is the pullback of $J^\infty(i)$ along δ_Y^J , and h_i is the comparison from E to that pullback.
- (2) **Compare the two equivalences.** Since $c_i h_i \simeq \text{id}_E$, the map c_i is an equivalence exactly when h_i is.

- (3) **Recognize the Cartesian square.** The latter is an equivalence exactly when the displayed square is Cartesian.

□

Theorem 5.5.7 (Tautological prolongation-completeness). *Let $\rho : E \rightarrow J^\infty E$ be any jet-coalgebra coaction. Regard ρ itself as a generalized jet equation on $Y = E$. Then its prolongation evaluation map is an equivalence. Hence every intrinsic formally integrable PDE has a canonical tautological prolongation-complete ambient equation.*

Proof.

- (1) **Form the tautological prolongation.** For $i = \rho$,

$$\text{Prol}(\rho) = J^\infty E \times_{J^\infty J^\infty E} J^\infty E,$$

where the two maps are $J^\infty(\rho)$ and δ_E^J . Write $p_1, p_2 : \text{Prol}(\rho) \rightrightarrows J^\infty E$ for the projections, so

$$J^\infty(\rho)p_1 \simeq \delta_E^J p_2.$$

- (2) **Identify the two projections.** Applying $J^\infty(\varepsilon_E^\Pi)$ and using $\varepsilon_E^\Pi \rho \simeq \text{id}_E$ and $J^\infty(\varepsilon_E^\Pi)\delta_E^J \simeq \text{id}$ gives

$$p_1 \simeq p_2.$$

- (3) **Apply the second counit identity.** Applying $\varepsilon_{J^\infty E}^\Pi$ instead gives

$$\rho \varepsilon_E^\Pi p_1 \simeq p_2 \simeq p_1.$$

- (4) **Construct the comparison maps.** The coalgebra law constructs

$$h = (\rho, \rho) : E \rightarrow \text{Prol}(\rho),$$

and the prolongation evaluation map is $c_\rho = \varepsilon_E^\Pi p_1$.

- (5) **Verify the inverse composites.** Hence

$$c_\rho h \simeq \text{id}_E,$$

while the two identities above give

$$h c_\rho \simeq \text{id}_{\text{Prol}(\rho)}.$$

- (6) **Conclude the equivalence.** Thus h and c_ρ are inverse equivalences.

□

Theorem 5.5.8 (Embedded Cartan equations are prolongation-complete). *If*

$$i : (E, \rho) \rightarrow \mathbb{K}(Y)$$

is a Cartan equation morphism whose underlying map $E \rightarrow J^\infty Y$ is a monomorphism, then $c_i : \text{Prol}(i) \rightarrow E$ is an equivalence.

Proof.

- (1) **Preserve the embedded equation.** The functor J^∞ preserves monomorphisms because it preserves limits. Hence $J^\infty(i)$ is a monomorphism, and its pullback

$$p_2 : \text{Prol}(i) \rightarrow J^\infty Y$$

is a monomorphism.

- (2) **Cancel along the monomorphism.** The split section h_i satisfies

$$p_2 h_i c_i \simeq i c_i \simeq p_2.$$

The monomorphism property of p_2 gives $h_i c_i \simeq \text{id}$, while $c_i h_i \simeq \text{id}$ was already constructed.

□

The preceding criteria detect when the prolongation evaluation is invertible. To turn that existence statement into a canonical intrinsic structure, one further point is needed: compatible Cartan lifts must have contractible moduli. The next lemma isolates exactly this categorical uniqueness step. Its abstract notation is used once, in the theorem that follows, and is not part of the later PDE interface.

Lemma 5.5.1 (Contractible lift over an invertible counit). *Let*

$$V : \mathcal{A} \rightleftarrows \mathcal{B} : P$$

be an adjunction $V \dashv P$, with V conservative, and let

$$\varepsilon^V : VP \rightarrow \text{id}_{\mathcal{B}}$$

be its counit. If

$$\varepsilon_b^V : VPb \xrightarrow{\sim} b$$

is an equivalence, then the homotopy fibre

$$\text{hofib}_b(V^\simeq : \mathcal{A}^\simeq \rightarrow \mathcal{B}^\simeq)$$

is contractible.

Proof.

- (1) **Choose the distinguished lift.** The pair

$$(Pb, \varepsilon_b^V)$$

defines a point of the fibre.

- (2) **Choose an arbitrary lift.** Let

$$(a, \varphi), \quad \varphi : Va \xrightarrow{\sim} b,$$

be any other point.

(3) **Apply the adjunction.** Adjunction identifies the space of maps

$$a \longrightarrow Pb$$

whose adjunct is φ with a contractible homotopy fibre of

$$\mathrm{Map}_{\mathcal{A}}(a, Pb) \xrightarrow{\simeq} \mathrm{Map}_{\mathcal{B}}(Va, b).$$

(4) **Construct the comparison lift.** Hence there is a contractibly unique map

$$\tilde{\varphi} : a \rightarrow Pb$$

such that

$$\varepsilon_b^V V(\tilde{\varphi}) \simeq \varphi.$$

(5) **Use conservativity.** Because both ε_b^V and φ are equivalences, $V(\tilde{\varphi})$ is an equivalence; conservativity of V therefore makes $\tilde{\varphi}$ an equivalence.

(6) **Compute the automorphism space.** The same adjunction calculation with $(a, \varphi) = (Pb, \varepsilon_b^V)$ makes the automorphism space of the distinguished point contractible.

(7) **Conclude contractibility.** Thus every point is connected to it through a contractible space of equivalences, and the entire homotopy fibre is contractible. □

Theorem 5.5.9 (Intrinsic structure of a prolongation-complete equation). *If*

$$i : E \rightarrow J^\infty Y$$

is prolongation-complete, then E carries a canonical J^∞ -coalgebra structure

$$\rho_i : E \rightarrow J^\infty E$$

for which

$$i : (E, \rho_i) \rightarrow \mathbb{K}(Y)$$

is a jet-coalgebra morphism.

Writing

$$p_1 : \mathrm{Prol}(i) \rightarrow J^\infty E$$

for the first projection of the Cartan prolongation, one has canonically

$$\rho_i \simeq p_1 c_i^{-1},$$

where c_i^{-1} denotes the contractibly unique inverse of

$$c_i : \mathrm{Prol}(i) \xrightarrow{\simeq} E.$$

The space of compatible Cartan structures on the fixed ambient equation i is the homotopy fibre

$$\mathcal{S}_i := \mathrm{hofib}_i \left(V_Y^\simeq : \mathrm{CartEq}_{X/S}(Y)^\simeq \longrightarrow \mathrm{Eqn}_{X/S}(Y)^\simeq \right).$$

A point consists of a Cartan equation together with a specified equivalence of its underlying ambient equation with i . The space $\mathcal{S}_i$ is contractible. Consequently E descends canonically, through effective infinitesimal descent, to an object of

$$\mathrm{PDE}_{X/S}^{\mathrm{fi}}.$$

Proof.

- (1) **Invoke the canonical prolongation coalgebra.** The Cartan prolongation $\text{Prol}^{\text{Cart}}(i)$ carries its canonical coalgebra structure by Theorem 5.5.1.
- (2) **Use prolongation-completeness.** Its underlying object is $\text{Prol}(i)$, and prolongation-completeness says that

$$c_i : \text{Prol}(i) \xrightarrow{\simeq} E$$

is an equivalence.

- (3) **Transport the coaction across the evaluation equivalence.** Let

$$\rho_{\text{Prol}(i)} : \text{Prol}(i) \rightarrow J^\infty \text{Prol}(i)$$

be the prolongation coaction. Transporting this coaction across c_i constructs

$$\rho_i : E \xrightarrow{c_i^{-1}} \text{Prol}(i) \xrightarrow{\rho_{\text{Prol}(i)}} J^\infty \text{Prol}(i) \xrightarrow{J^\infty(c_i)} J^\infty E.$$

- (4) **Record the contractible inverse choice.** The space of inverses of an equivalence is contractible.
- (5) **Introduce the first projection.** To compute the transported coaction, use the first projection

$$p_1 : \text{Prol}(i) \rightarrow J^\infty E.$$

- (6) **Use the coalgebra-morphism identity.** Since

$$p_1 : \text{Prol}^{\text{Cart}}(i) \rightarrow \mathbb{K}(E)$$

is a coalgebra morphism,

$$J^\infty(p_1)\rho_{\text{Prol}(i)} \simeq \delta_E^J p_1.$$

- (7) **Apply the comonad counit.** Using

$$c_i = \varepsilon_E^\Pi p_1$$

and the comonad counit identity gives

$$\begin{aligned} J^\infty(c_i)\rho_{\text{Prol}(i)} &= J^\infty(\varepsilon_E^\Pi)J^\infty(p_1)\rho_{\text{Prol}(i)} \\ &\simeq J^\infty(\varepsilon_E^\Pi)\delta_E^J p_1 \\ &\simeq p_1. \end{aligned}$$

- (8) **Identify the transported coaction.** After transport across c_i , this is precisely

$$\rho_i \simeq p_1 c_i^{-1}.$$

- (9) **Identify the second projection.** The second projection

$$p_2 : \text{Prol}^{\text{Cart}}(i) \rightarrow \mathbb{K}(Y)$$

is also a coalgebra morphism, and Proposition 5.5.1 gives

$$p_2 \simeq i c_i.$$

- (10) **Transport the ambient equation.** Transporting p_2 across

$$\mathrm{Prol}^{\mathrm{Cart}}(i) \xrightarrow{\simeq} (E, \rho_i)$$

therefore makes

$$i : (E, \rho_i) \rightarrow \mathbb{K}(Y)$$

a coalgebra morphism.

- (11) **Identify the Cartan prolongation adjunction.** It remains to prove the asserted contractible uniqueness. By Theorem 5.5.2,

$$V_Y \dashv \mathrm{Prol}_Y^{\mathrm{Cart}},$$

and Proposition 5.5.1 identifies the counit at i with

$$c_i : V_Y \mathrm{Prol}_Y^{\mathrm{Cart}}(i) \rightarrow i.$$

- (12) **Establish conservativity.** The functor V_Y is conservative: a morphism in $\mathrm{CartEq}_{X/S}(Y)$ whose underlying equation morphism is an equivalence has underlying map over X an equivalence, and the forgetful functor from J^∞ -coalgebras reflects equivalences.
- (13) **Apply the contractible-fibre lemma.** Since c_i is an equivalence, apply Lemma 5.5.1 to

$$V_Y \dashv \mathrm{Prol}_Y^{\mathrm{Cart}}.$$

It gives

$$\mathcal{S}_i \simeq *.$$

- (14) **Interpret contractible uniqueness.** Thus the compatible coalgebra structure is canonical in the precise sense that its moduli space is contractible.
- (15) **Identify effective jet descent.** Finally, Theorem 5.3.2 identifies

$$\mathrm{Coalg}_{J^\infty}((\mathcal{H}_{\mathrm{Sp}})_{/X}) \simeq (\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}}.$$

- (16) **Descend the canonical coalgebra.** Hence the canonical coalgebra (E, ρ_i) determines the stated intrinsic formally integrable PDE.

□

Corollary 5.5.2 (Cartan factorization of a prolongation-complete equation). *Let $i : E \rightarrow J^\infty Y$ be prolongation-complete and put*

$$y_i := \varepsilon_Y^\Pi i : E \rightarrow Y.$$

Then

$$i \simeq J^\infty(y_i)\rho_i.$$

Proof.

- (1) **Use the intrinsic Cartan structure.** Theorem 5.5.9 makes i a coalgebra morphism into $\mathbb{K}(Y)$.
- (2) **Apply Cartan recognition.** Apply Theorem 5.4.4.

□

Theorem 5.5.10 (Quotient-side criterion for prolongation-completeness). *Let $F \rightarrow Q_{X/S}$ be an intrinsic formally integrable PDE, put*

$$E = e_{X/S}^* F,$$

and let

$$y : E \rightarrow Y$$

be a map with associated Cartan equation

$$i_y : E \rightarrow J^\infty Y.$$

Put

$$G := \Pi_{e_{X/S}} Y$$

and let

$$\bar{y} : F \rightarrow G$$

be the adjunct of y .

Under effective descent, the Cartan prolongation of i_y corresponds to

$$\Pi_{e_{X/S}} e_{X/S}^* F \times_{\Pi_{e_{X/S}} e_{X/S}^* G} G.$$

The canonical Cartan comparison corresponds to the map

$$F \longrightarrow \Pi_{e_{X/S}} e_{X/S}^* F \times_{\Pi_{e_{X/S}} e_{X/S}^* G} G$$

whose two components are

$$\eta_F^\Pi : F \rightarrow \Pi_{e_{X/S}} e_{X/S}^* F \quad \text{and} \quad \bar{y} : F \rightarrow G.$$

The compatibility required to define this map is the naturality equivalence

$$\Pi_{e_{X/S}} e_{X/S}^* (\bar{y}) \eta_F^\Pi \simeq \eta_G^\Pi \bar{y}$$

for the monad unit

$$\eta^\Pi : \text{id}_{(\mathcal{H}_{\text{Sp}})/Q_{X/S}} \longrightarrow \Pi_{e_{X/S}} e_{X/S}^*.$$

Consequently i_y is prolongation-complete if and only if

$$\begin{array}{ccc} F & \xrightarrow{\eta_F^\Pi} & \Pi_{e_{X/S}} e_{X/S}^* F \\ \bar{y} \downarrow & & \downarrow \Pi_{e_{X/S}} e_{X/S}^* (\bar{y}) \\ G & \xrightarrow{\eta_G^\Pi} & \Pi_{e_{X/S}} e_{X/S}^* G \end{array}$$

is Cartesian.

Proof.

- (1) **Compute the quotient-side adjunct.** The adjunct of

$$y : e_{X/S}^* F \rightarrow Y$$

under

$$e_{X/S}^* \dashv \Pi_{e_{X/S}}$$

is

$$\bar{y} : F \xrightarrow{\eta_F^\Pi} \Pi_{e_{X/S}} e_{X/S}^* F \xrightarrow{\Pi_{e_{X/S}}(y)} \Pi_{e_{X/S}} Y = G.$$

- (2) **Identify the canonical coaction.** The canonical coaction on

$$E = e_{X/S}^* F$$

is

$$\rho_F = e_{X/S}^* \eta_F^\Pi.$$

- (3) **Identify the Cartan equation as a pullback.** Hence

$$\begin{aligned} i_y &= J^\infty(y) \rho_F \\ &= e_{X/S}^* \Pi_{e_{X/S}}(y) e_{X/S}^* \eta_F^\Pi \\ &= e_{X/S}^* (\Pi_{e_{X/S}}(y) \eta_F^\Pi) \\ &= e_{X/S}^* \bar{y}. \end{aligned}$$

- (4) **Translate the cofree objects.** Under Theorem 5.3.2, the cofree coalgebra on

$$E = e_{X/S}^* F$$

corresponds to

$$\Pi_{e_{X/S}} e_{X/S}^* F.$$

Likewise,

$$\mathbb{K}(Y)$$

corresponds to

$$G = \Pi_{e_{X/S}} Y,$$

and

$$\mathbb{K}(J^\infty Y)$$

corresponds to

$$\Pi_{e_{X/S}} e_{X/S}^* \Pi_{e_{X/S}} Y = \Pi_{e_{X/S}} e_{X/S}^* G.$$

- (5) **Translate the first cofree morphism.** Since

$$i_y = e_{X/S}^* \bar{y},$$

the cofree coalgebra morphism

$$\mathbb{K}(i_y) : \mathbb{K}(E) \rightarrow \mathbb{K}(J^\infty Y)$$

corresponds under effective descent to

$$\Pi_{e_{X/S}} e_{X/S}^* (\bar{y}) : \Pi_{e_{X/S}} e_{X/S}^* F \longrightarrow \Pi_{e_{X/S}} e_{X/S}^* G.$$

- (6) **Translate the second cofree morphism.** The cofree coalgebra map

$$\widetilde{\delta}_Y^J : \mathbb{K}(Y) \rightarrow \mathbb{K}(J^\infty Y)$$

classified by

$$\mathrm{id}_{J^\infty Y}$$

corresponds to the monad unit

$$\eta_G^\Pi : G \longrightarrow \Pi_{e_{X/S}} e_{X/S}^* G.$$

- (7) **Preserve the Cartan pullback.** The effective-descent equivalence preserves limits. Therefore the Cartan prolongation

$$\mathrm{Prol}^{\mathrm{Cart}}(i_y) = \mathbb{K}(E) \times_{\mathbb{K}(J^\infty Y)} \mathbb{K}(Y)$$

corresponds to

$$\Pi_{e_{X/S}} e_{X/S}^* F \times_{\Pi_{e_{X/S}} e_{X/S}^* G} G.$$

- (8) **Apply naturality of the monad unit.** Naturality of the monad unit gives the canonical equivalence

$$\Pi_{e_{X/S}} e_{X/S}^* (\bar{y}) \eta_F^\Pi \simeq \eta_G^\Pi \bar{y}.$$

- (9) **Construct the quotient-side comparison.** Thus the pair

$$(\eta_F^\Pi, \bar{y})$$

defines a canonical comparison map

$$F \longrightarrow \Pi_{e_{X/S}} e_{X/S}^* F \times_{\Pi_{e_{X/S}} e_{X/S}^* G} G.$$

- (10) **Identify the pulled-back comparison.** After applying the conservative functor

$$e_{X/S}^*,$$

this comparison is exactly the canonical Cartan map

$$h_{i_y} : E \longrightarrow \mathrm{Prol}(i_y)$$

of Proposition 5.5.3.

- (11) **Compare the Cartan section and evaluation.** Since

$$c_{i_y} h_{i_y} \simeq \mathrm{id}_E,$$

the map

$$h_{i_y}$$

is an equivalence if and only if

$$c_{i_y}$$

is an equivalence.

(12) **Recognize the quotient-side Cartesian criterion.** On the quotient side, the comparison

$$F \longrightarrow \Pi_{e_{X/S}} e_{X/S}^* F \times_{\Pi_{e_{X/S}} e_{X/S}^* G} G$$

is an equivalence exactly when the square

$$\begin{array}{ccc} F & \xrightarrow{\eta_F^\Pi} & \Pi_{e_{X/S}} e_{X/S}^* F \\ \bar{y} \downarrow & & \downarrow \Pi_{e_{X/S}} e_{X/S}^*(\bar{y}) \\ G & \xrightarrow{\eta_G^\Pi} & \Pi_{e_{X/S}} e_{X/S}^* G \end{array}$$

is Cartesian.

(13) **Transport the criterion back to the Cartan equation.** Conservativity of $e_{X/S}^*$ therefore identifies this condition with prolongation-completeness of i_y .

□

Corollary 5.5.3 (Recognition for embedded ambient equations). *Let*

$$i : E \hookrightarrow J^\infty Y$$

be an embedded generalized jet equation. The following are equivalent:

1. *i is prolongation-complete;*
2. *i lies in the essential image of*

$$V_Y : \text{CartEq}_{X/S}(Y) \longrightarrow \text{Eqn}_{X/S}(Y);$$

equivalently, there exist an object $F \rightarrow Q_{X/S}$, a map

$$y : e_{X/S}^* F \rightarrow Y,$$

and a specified equivalence in $\text{Eqn}_{X/S}(Y)$ from the associated Cartan equation

$$J^\infty(y)\rho_F : e_{X/S}^* F \rightarrow J^\infty Y$$

to i ;

3. *E carries a jet-coalgebra structure*

$$\rho : E \rightarrow J^\infty E$$

for which

$$i : (E, \rho) \rightarrow \mathbb{K}(Y)$$

is a coalgebra morphism.

When these conditions hold, the space of compatible Cartan structures on the fixed ambient equation i is contractible.

Proof.

- (1) **Construct the intrinsic Cartan structure.** Prolongation-completeness gives (2) and (3) by Theorem 5.5.9.
- (2) **Translate the Cartan structure through descent.** Theorem 5.3.2 identifies a jet coalgebra with an object over $Q_{X/S}$, while Theorem 5.4.4 identifies a coalgebra morphism into $\mathbb{K}(Y)$ with its classifying map $E \rightarrow Y$.
- (3) **Identify the equivalent intrinsic presentations.** This proves (2) $\Leftrightarrow$ (3) at the level of the corresponding infinity-categories.
- (4) **Recover the converse.** Conversely, a compatible coalgebra structure makes i an embedded Cartan equation morphism, hence prolongation-complete by the embedded Cartan theorem above.
- (5) **Recover contractible uniqueness.** Contractible uniqueness is the homotopy-fibre statement of Theorem 5.5.9 and follows from prolongation-completeness.

□

Corollary 5.5.4 (Cartan prolongation is intrinsically formally integrable). *For every generalized jet equation $i : E \rightarrow J^\infty Y$, its Cartan prolongation $(\text{Prol}(i), \rho_{\text{Prol}(i)})$ descends canonically to an object of $\text{PDE}_{X/S}^{\text{fi}}$. If i is embedded, the prolonged equation*

$$p_2 : \text{Prol}(i) \rightarrow J^\infty Y$$

is embedded and prolongation-complete.

Proof.

- (1) **Descend the prolongation coalgebra.** Every jet coalgebra descends by Theorem 5.3.2.
- (2) **Treat the embedded case.** If i is a monomorphism, then $J^\infty(i)$ is a monomorphism and its pullback p_2 is a monomorphism. It is therefore an embedded Cartan equation morphism and hence prolongation-complete.

□

Prolongation-completeness is therefore the bridge between the two uses of the word “equation” in the chapter. An ambient equation $E \rightarrow J^\infty Y$ becomes intrinsic precisely when its prolongation adds no new underlying object; then its compatible jet-coalgebra structure has contractible moduli and descends to the effective quotient.

5.5.3 Intrinsic solutions and equation calculus

Once an equation is prolongation-complete, its Cartan prolongation may be transported back to the original source. Actual solutions can then be read directly as morphisms from the terminal jet coalgebra, while the quotient-side presentation makes limits and colimits of intrinsic PDEs transparent.

Theorem 5.5.11 (Solutions intrinsically). *If $i : E \rightarrow J^\infty Y$ is prolongation-complete and (E, ρ_i) is the constructed jet-coalgebra presentation, then*

$$\text{Sol}(i) \simeq \text{Map}_{\text{Coalg}_{J^\infty}((\mathcal{H}_{\text{Sp}})_{/X})}((X, \rho_X), (E, \rho_i)).$$

Thus actual solutions are precisely morphisms from the terminal jet coalgebra to the descended equation presentation.

Proof.

- (1) **Represent solutions by Cartan prolongation.** For every equation, 5.5.4 gives

$$\mathrm{Sol}(i) \simeq \mathrm{Map}_{\mathrm{Coalg}_{J^\infty}((\mathcal{H}_{\mathrm{Sp}})_{/X})}((X, \rho_X), \mathrm{Prol}^{\mathrm{Cart}}(i)).$$

- (2) **Transport across prolongation evaluation.** If i is prolongation-complete, 5.5.9 transports the prolongation coalgebra across the equivalence

$$c_i : \mathrm{Prol}(i) \xrightarrow{\simeq} E.$$

Postcomposition with the resulting coalgebra equivalence gives the displayed mapping-space equivalence.

□

Theorem 5.5.12 (The PDE infinity-topos). *The category*

$$\mathrm{PDE}_{X/S}^{\mathrm{fi}} = (\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}}$$

is an infinity-topos and admits all small limits and colimits. Under either the monadic or comonadic presentation, the underlying-object functor to $(\mathcal{H}_{\mathrm{Sp}})_{/X}$ preserves all small limits and colimits. Consequently the underlying object over X of an intrinsic limit or colimit is canonically the corresponding limit or colimit of the underlying objects over X .

Proof.

- (1) **Use the slice-topos theorem.** A slice of an infinity-topos is again an infinity-topos [36].
- (2) **Identify the underlying-object functor.** Under Theorem 5.3.2, the underlying-object functor is identified with

$$e_{X/S}^* : (\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}} \rightarrow (\mathcal{H}_{\mathrm{Sp}})_{/X}.$$

- (3) **Use the dependent adjoints.** Since

$$\Sigma_{e_{X/S}} \dashv e_{X/S}^* \dashv \Pi_{e_{X/S}},$$

this pullback functor is both a left and a right adjoint and hence preserves all small colimits and limits.

- (4) **Transport preservation through the presentations.** Transport through the monadic or comonadic equivalence proves the statement.

□

Corollary 5.5.5 (Simultaneous formally integrable systems). *Products, fibre products, homotopy equalizers, arbitrary simultaneous systems, coproducts, and all other small colimits of intrinsic formally integrable PDEs are again intrinsic formally integrable PDEs and are computed on their underlying objects over X .*

Corollary 5.5.6 (Prolonged homotopy equalizers). *Let*

$$D_0, D_1 : E \rightsquigarrow F$$

be differential operators represented by

$$\widehat{D}_0, \widehat{D}_1 : J^\infty E \rightrightarrows F,$$

and write

$$i_0 : E_0 := \mathrm{Eq}(\widehat{D}_0, \widehat{D}_1) \longrightarrow J^\infty E$$

for the derived equation. Its Cartan prolongation satisfies

$$\mathrm{Prol}(i_0) \simeq \mathrm{Eq}(\mathrm{prol}_{\mathrm{op}}^\infty(\widehat{D}_0), \mathrm{prol}_{\mathrm{op}}^\infty(\widehat{D}_1)),$$

where the equalizer on the right is formed in $(\mathcal{H}_{\mathrm{Sp}})_{/X}$ over $J^\infty E$.

Proof.

- (1) **Apply the jet functor to the equalizer.** Since J^∞ preserves limits,

$$J^\infty E_0 \simeq \mathrm{Eq}(J^\infty(\widehat{D}_0), J^\infty(\widehat{D}_1))$$

inside $J^\infty J^\infty E$.

- (2) **Pull back along comultiplication.** Pulling this equalizer back along $\delta_E^J : J^\infty E \rightarrow J^\infty J^\infty E$ gives the equalizer of

$$J^\infty(\widehat{D}_0)\delta_E^J, \quad J^\infty(\widehat{D}_1)\delta_E^J,$$

which are the two prolonged operators.

□

Proposition 5.5.4 (Intrinsic morphisms preserve Cartan structure). *Let $F, G \rightarrow Q_{X/S}$ be intrinsic formally integrable PDEs, with underlying objects E, E' , and let $f : F \rightarrow G$ be a morphism over $Q_{X/S}$. Then*

$$e_{X/S}^* f : E \rightarrow E'$$

is simultaneously a $\mathrm{Disk}^{\mathrm{dR}, \infty}$ -algebra morphism and a J^∞ -coalgebra morphism. Consequently it preserves formal-disk actions, crystalline transport, and every formal-solution family obtained by precomposition.

Proof.

- (1) **Apply the mapping-space descent equivalence.** The mapping-space statement of Theorem 5.3.2 gives the claim.

□

The equation calculus is now internal: every generalized equation admits a Cartan prolongation, and a prolongation-complete equation already carries the resulting intrinsic descent structure. The remaining task is geometric functoriality.

5.6 Substitution, locality, and geometric models

So far the calculus has been developed over one relative context. Geometry requires two further checks. Arbitrary base change must transport the construction between bases, and effective formally étale descent must reconstruct it from local contexts. After these functoriality statements are fixed, group objects and affine-space charts serve as concrete translation models of the same intrinsic relation.

5.6.1 Base change of prolongations and solution objects

The substitution theorem already transports the jet functors. We now apply it to the finite-limit and dependent-product constructions defining prolongations and solution objects.

Corollary 5.6.1 (Base change of Cartan prolongations). *Let*

$$i : E \rightarrow J_{X/S}^\infty Y$$

be a generalized jet equation and let

$$i' : E' := g_X^* E \longrightarrow J_{X'/S'}^\infty (g_X^* Y)$$

be its pullback. Then there are canonical equivalences

$$g_X^* (\text{Prol}(i)) \simeq \text{Prol}(i'), \quad g_X^* (c_i) \simeq c_{i'}.$$

They identify the canonical prolongation coalgebras, both projections, and the universal formal-solution family. Consequently prolongation-completeness is preserved by arbitrary base change.

Proof.

- (1) **Express Cartan prolongation as a pullback.** Cartan prolongation is the finite pullback

$$\text{Prol}(i) = J^\infty E \times_{J^\infty J^\infty Y} J^\infty Y.$$

- (2) **Base-change the prolongation pullback.** Pullback along g_X preserves this limit, and Theorem 5.2.7 identifies every jet term and every comultiplication map with the corresponding object after base change.
- (3) **Base-change the evaluation map.** The prolongation evaluation map is $\varepsilon_E^\Pi p_1$, so compatibility with the comonad counit gives the second equivalence.

□

Corollary 5.6.2 (Base change of solution objects). *For a generalized jet equation i , the parameterized solution object satisfies*

$$\text{Sol}_{X'/S'}(i') \simeq \text{Sol}_{X/S}(i) \times_S S'.$$

The same base-change comparison holds for the formal-solution representing object $\text{Prol}(i)$.

Proof.

- (1) **Base-change the parameterized solution pullback.** The solution object is a finite pullback of dependent products and the prolongation map.

- (2) **Apply Beck–Chevalley.** Apply right Beck–Chevalley and Theorem 5.2.7.
- (3) **Base-change the formal-solution object.** The formal-solution statement is the preceding corollary.

□

Cartan prolongation, its evaluation map, and both solution constructions therefore inherit the substitution law of the jet comonad. Locality asks for the corresponding reconstruction statement from a cover.

5.6.2 Formally étale invariance and effective descent

Formally étale maps are the relevant local changes of context because the preceding de Rham chapter proves that they preserve the effective relation itself. The results below first restrict the calculus along one such map and then reconstruct it from an effective formally étale Čech nerve.

Theorem 5.6.1 (Formally étale restriction). *Let $u : U \rightarrow X$ be formally étale over S . The Cartesian effective-quotient square of Theorem 4.3.8 in Chapter 4 induces canonical equivalences*

$$u^* \text{Disk}_{X/S}^{\text{dR}, \infty} \simeq \text{Disk}_{U/S}^{\text{dR}, \infty} u^*, \quad u^* J_{X/S}^\infty \simeq J_{U/S}^\infty u^*,$$

compatibly with the monad, comonad, formal-disk–jet adjunction, disk coalgebras, and mate correspondence.

Proof.

- (1) **Apply Beck–Chevalley to the quotient square.** Apply left and right Beck–Chevalley to the Cartesian effective-quotient square of Theorem 4.3.8.
- (2) **Transport the structural maps by mates.** Compatibility with the structural maps is the mate coherence of the base-changed adjoint triple.

□

Theorem 5.6.2 (Effective formally étale descent of the jet calculus). *Let*

$$u : U \twoheadrightarrow X$$

be an effective epimorphism over S which is formally étale over S , and let $U_\bullet \rightarrow X$ be its Čech nerve. Write

$$u_n : U_n \rightarrow X$$

for the structure maps.

Then restriction induces a canonical equivalence

$$\text{PDE}_{X/S}^{\text{fi}} \xrightarrow{\simeq} \text{Tot}_{[n] \in \Delta} \text{PDE}_{U_n/S}^{\text{fi}}.$$

Moreover, for every $E \rightarrow X$, there are canonical reconstruction equivalences

$$\text{Disk}_{X/S}^{\text{dR}, \infty} E \simeq \text{Tot}_{[n] \in \Delta} \Pi_{u_n} \text{Disk}_{U_n/S}^{\text{dR}, \infty} (u_n^* E),$$

and

$$J_{X/S}^\infty E \simeq \text{Tot}_{[n] \in \Delta} \Pi_{u_n} J_{U_n/S}^\infty (u_n^* E).$$

These equivalences reconstruct the monad and comonad structure, the formal-disk–jet adjunction, the canonical disk coalgebras, and the mate comparison from their restrictions to the Čech stages.

Proof.

- (1) **Identify the quotient Čech nerve.** By the effective formally étale quotient-descent theorem of Chapter 4, the simplicial effective quotient $Q_{U_\bullet/S}$ is the Čech nerve of the effective morphism

$$Q_{U/S} \longrightarrow Q_{X/S}.$$

- (2) **Apply effective descent to the quotient slices.** Effective descent for slices therefore gives

$$(\mathcal{H}_{\text{Sp}})_{/Q_{X/S}} \simeq \text{Tot}_{[n] \in \Delta} (\mathcal{H}_{\text{Sp}})_{/Q_{U_n/S}}.$$

- (3) **Identify the intrinsic PDE totalization.** Using the definition

$$\text{PDE}_{-/S}^{\text{fi}} = (\mathcal{H}_{\text{Sp}})_{/Q_{-/S}}$$

gives the first assertion.

- (4) **Restrict the jet calculus.** For every n , Theorem 5.6.1 gives coherent equivalences

$$u_n^* \text{Disk}_{X/S}^{\text{dR}, \infty} \simeq \text{Disk}_{U_n/S}^{\text{dR}, \infty} u_n^*,$$

and

$$u_n^* J_{X/S}^\infty \simeq J_{U_n/S}^\infty u_n^*,$$

compatible with the corresponding adjunction and structural maps.

- (5) **State the slice reconstruction formula.** Effective Čech descent for the cover u gives, for every $K \in (\mathcal{H}_{\text{Sp}})_{/X}$,

$$K \simeq \text{Tot}_{[n] \in \Delta} \Pi_{u_n} u_n^* K.$$

- (6) **Verify the reconstruction formula on mapping spaces.** Indeed, for every $W \rightarrow X$,

$$\begin{aligned} \text{Map}_X \left(W, \text{Tot}_{[n] \in \Delta} \Pi_{u_n} u_n^* K \right) &\simeq \text{Tot}_{[n] \in \Delta} \text{Map}_{U_n} (u_n^* W, u_n^* K) \\ &\simeq \text{Map}_X (W, K), \end{aligned}$$

and Yoneda gives the reconstruction equivalence.

- (7) **Apply reconstruction to the disk and jet objects.** Apply this formula to

$$K = \text{Disk}_{X/S}^{\text{dR}, \infty} E$$

and to

$$K = J_{X/S}^\infty E,$$

then use the formally étale restriction equivalences from the preceding step. This gives the two displayed formulas.

- (8) **Reconstruct the structural maps.** The units, counits, multiplication, comultiplication, adjunction maps, disk coalgebras, and mate transformations are natural transformations between the functors just reconstructed. Effective descent is fully faithful on mapping spaces, so each such transformation and its higher compatibility are uniquely reconstructed from their degreewise restrictions.

□

Corollary 5.6.3 (Descent of equations and Cartan prolongation). *Under the hypotheses of Theorem 5.6.2, restriction induces a canonical equivalence*

$$\mathrm{Eqn}_{X/S} \xrightarrow{\simeq} \mathrm{Tot}_{[n] \in \Delta} \mathrm{Eqn}_{U_n/S}.$$

For fixed $Y \in (\mathcal{H}_{\mathrm{Sp}})_{/X}$, this restricts to

$$((\mathcal{H}_{\mathrm{Sp}})_{/X})_{/J_{X/S}^\infty Y} \xrightarrow{\simeq} \mathrm{Tot}_{[n] \in \Delta} ((\mathcal{H}_{\mathrm{Sp}})_{/U_n})_{/J_{U_n/S}^\infty (u_n^* Y)}.$$

If

$$i : E \rightarrow J_{X/S}^\infty Y$$

is a generalized jet equation and i_n its restriction to U_n , then

$$u_n^* \mathrm{Prol}(i) \simeq \mathrm{Prol}(i_n)$$

for every n , compatibly with the canonical prolongation coalgebras and evaluation maps. Consequently

$$\mathrm{Prol}(i) \simeq \mathrm{Tot}_{[n] \in \Delta} \Pi_{u_n} \mathrm{Prol}(i_n).$$

Proof.

- (1) **Express the equation category as a finite limit.** Construction 5.4.5 expresses $\mathrm{Eqn}_{X/S}$ as a finite limit in Cat_∞ formed from the slice $(\mathcal{H}_{\mathrm{Sp}})_{/X}$ and the endofunctor $J_{X/S}^\infty$.
- (2) **Reconstruct the finite-limit ingredients.** Effective descent reconstructs the slice, while Theorem 5.6.2 reconstructs the jet functor and its coherent restriction maps.
- (3) **Descend the equation category.** Since $\mathrm{Fun}(\Delta^1, -)$, products, fibres, and pullbacks preserve limits of infinity-categories, the equation category descends as asserted.
- (4) **Express Cartan prolongation as a pullback.** Cartan prolongation is the finite pullback

$$\mathrm{Prol}(i) = J^\infty E \times_{J^\infty J^\infty Y} J^\infty Y.$$

- (5) **Restrict Cartan prolongation.** Restriction preserves this pullback and identifies every jet term and the comultiplication with its corresponding restriction. Hence

$$u_n^* \mathrm{Prol}(i) \simeq \mathrm{Prol}(i_n).$$

- (6) **Reconstruct the global prolongation.** The final formula follows from effective reconstruction in the slice:

$$K \simeq \mathrm{Tot}_{[n] \in \Delta} \Pi_{u_n} u_n^* K$$

applied to $K = \mathrm{Prol}(i)$.

□

Corollary 5.6.4 (Descent of solution objects). *Under the same hypotheses, let*

$$i : E \rightarrow J_{X/S}^\infty Y$$

and let i_n denote its restriction to U_n . Then the parameterized solution object satisfies

$$\mathbf{Sol}_{X/S}(i) \simeq \operatorname{Tot}_{[n] \in \Delta} \mathbf{Sol}_{U_n/S}(i_n)$$

in $(\mathcal{H}_{\mathrm{Sp}})_{/S}$. The formal-solution functor is reconstructed from the corresponding degreewise Cartan prolongations through

$$\mathrm{Prol}(i) \simeq \operatorname{Tot}_{[n] \in \Delta} \Pi_{u_n} \mathrm{Prol}(i_n).$$

Proof.

- (1) **Define the Čech-stage structural maps.** Put

$$p_n := p_{u_n} : U_n \rightarrow S.$$

- (2) **Reconstruct dependent products.** For every $Z \rightarrow X$, effective reconstruction in the slice and preservation of limits by Π_p give

$$\begin{aligned} \Pi_p Z &\simeq \Pi_p \operatorname{Tot}_{[n] \in \Delta} \Pi_{u_n} u_n^* Z \\ &\simeq \operatorname{Tot}_{[n] \in \Delta} \Pi_{p_n} u_n^* Z. \end{aligned}$$

- (3) **Apply the reconstruction formula to the solution data.** Apply this to

$$Z = E, \quad Z = Y, \quad Z = J^\infty Y.$$

- (4) **Compare the parameterized prolongation maps.** The formally étale jet comparison identifies the restrictions of the parameterized prolongation map

$$j_{/S}^\infty : \Pi_p Y \rightarrow \Pi_p J^\infty Y$$

with the corresponding maps on the Čech stages.

- (5) **Reconstruct the solution pullback.** Since the solution object is the finite pullback

$$\mathbf{Sol}_{X/S}(i) = \Pi_p E \times_{\Pi_p J^\infty Y} \Pi_p Y,$$

the asserted totalization formula follows.

- (6) **Reconstruct the formal-solution object.** The formal-solution statement is Corollary 5.6.3. □

Corollary 5.6.5 (Locality of prolongation-completeness). *Under the hypotheses of Theorem 5.6.2, a generalized jet equation*

$$i : E \rightarrow J_{X/S}^\infty Y$$

is prolongation-complete if and only if its restriction

$$u^* i$$

to U is prolongation-complete.

Proof.

- (1) **Identify the restricted evaluation map.** By Corollary 5.6.3, restriction identifies the prolongation evaluation map

$$c_i : \text{Prol}(i) \rightarrow E$$

with

$$c_{u^*i} : \text{Prol}(u^*i) \rightarrow u^*E.$$

- (2) **Use conservative pullback.** Pullback along the effective epimorphism u is conservative. Hence c_i is an equivalence if and only if c_{u^*i} is an equivalence.

□

Remark 5.6.1 (Later Nisnevich specialization). If a later geometric theorem identifies the relevant spectral Nisnevich covers with effective formally étale covers, the preceding theorem and its corollaries yield Nisnevich descent for jets, equations, solution objects, Cartan prolongations, prolongation-completeness, and the intrinsic PDE infinity-topos. Nisnevich geometry therefore enters as a later localization theorem for this calculus.

Effective formally étale descent now reconstructs the jet calculus and detects prolongation-completeness locally. These results establish locality without choosing coordinates. Coordinates enter only as models of the already-constructed relation.

5.6.3 Formal translation models

When the relative context is a commutative group object, subtraction turns the abstract relation into a familiar action groupoid: two related points differ by a unique infinitesimal element in the formal neighbourhood of the identity. This translation description is a checksum for the intrinsic construction, not a replacement for it.

Proposition 5.6.1 (Formal translation in degree one). *Let $G \rightarrow S$ be a commutative group object in the slice and let*

$$0 : S \rightarrow G$$

be its zero section. The zero section determines the relative de Rham neighbourhood

$$\text{Nbh}_{S/S}^{\text{dR}}(G) \simeq G \times_{G_{\text{dR}/S}} S,$$

where $S \rightarrow G_{\text{dR}/S}$ is the de Rham image of the zero section. The zero section also induces $S \rightarrow Q_{G/S}$, and there is a canonical equivalence

$$\text{Nbh}_{S/S}^{\text{dR}}(G) \simeq G \times_{Q_{G/S}} S.$$

Moreover $\text{Nbh}_{S/S}^{\text{dR}}(G)$ carries the canonical commutative group-object structure over S induced as the kernel of the group morphism

$$\eta_{G/S}^{\text{dR}} : G \rightarrow G_{\text{dR}/S}.$$

Thus this relative de Rham neighbourhood uses the same effective de Rham quotient as the intrinsic jet calculus. There is a canonical equivalence

$$G \times_{G_{\text{dR}/S}} G \xrightarrow{\simeq} G \times_S \text{Nbh}_{S/S}^{\text{dR}}(G)$$

which in internal additive notation sends

$$(g_0, g_1) \mapsto (g_0, g_1 - g_0)$$

and whose inverse sends $(g, \xi) \mapsto (g, g + \xi)$. The source projection becomes pr_G and evaluation becomes the formal translation action

$$\alpha_G : G \times_S \text{Nbh}_{S/S}^{\text{dR}}(G) \rightarrow G, \quad (g, \xi) \mapsto g + \xi.$$

The degree-one relation therefore becomes the action-groupoid square

$$\begin{array}{ccc} G \times_S \text{Nbh}_{S/S}^{\text{dR}}(G) & \xrightarrow{\alpha_G} & G \\ \text{pr}_G \downarrow & & \downarrow e_{G/S} \\ G & \xrightarrow{e_{G/S}} & Q_{G/S}, \end{array}$$

and hence

$$\text{Disk}_{G/S}^{\text{dR}, \infty} \simeq \Sigma_{\text{pr}_G} \alpha_G^*, \quad J_{G/S}^\infty \simeq \Pi_{\text{pr}_G} \alpha_G^*.$$

Proof.

- (1) **Preserve commutative group objects under reflection.** Relative de Rham reflection is left exact, hence preserves commutative group objects, and $\eta_{G/S}^{\text{dR}} : G \rightarrow G_{\text{dR}/S}$ is a group morphism.
- (2) **Identify the relative neighbourhood as the kernel.** Thus $\text{Nbh}_{S/S}^{\text{dR}}(G)$ is its kernel.
- (3) **Induce the kernel group structure.** Kernels of morphisms of commutative group objects are finite-limit constructions, so the pullback defining $\text{Nbh}_{S/S}^{\text{dR}}(G)$ inherits its commutative group structure from G . In particular the expressions $\xi_1 + \xi_2$ and $-\xi$ used below are operations of this constructed commutative group object.
- (4) **Pass from the de Rham object to the effective quotient.** Write

$$G \xrightarrow{e_{G/S}} Q_{G/S} \xrightarrow{m_{G/S}} G_{\text{dR}/S}$$

for the effective-image factorization. Both the map $G \rightarrow G_{\text{dR}/S}$ and the de Rham image of the zero section factor through $Q_{G/S}$. Since $m_{G/S}$ is a monomorphism,

$$Q_{G/S} \times_{G_{\text{dR}/S}} Q_{G/S} \simeq Q_{G/S},$$

and finite-limit pasting gives

$$G \times_{G_{\text{dR}/S}} S \simeq G \times_{Q_{G/S}} S.$$

- (5) **Construct the translation equivalence.** Equality of the de Rham images of g_0 and g_1 is equivalent to vanishing of the de Rham image of $g_1 - g_0$. Addition and subtraction in the group object therefore construct the two displayed maps, and the group laws show that they are inverse equivalences.
- (6) **Transport the jet formulas.** The jet formulas are Proposition 5.2.1 transported through this equivalence.

□

Theorem 5.6.3 (Formal-translation nerve with explicit simplicial operators). *For every $n \geq 0$, there is a canonical equivalence*

$$\Phi_{G,n} : \mathcal{R}_{G/S,n}^{\text{dR}} \xrightarrow{\simeq} G \times_S (\text{Nbh}_{S/S}^{\text{dR}}(G))^{\times_{S^n}}$$

given internally by

$$\Phi_{G,n}(g_0, \dots, g_n) = (g_0, \xi_1, \dots, \xi_n), \quad \xi_j = g_j - g_{j-1},$$

with inverse

$$(g, \xi_1, \dots, \xi_n) \mapsto (g, g + \xi_1, g + \xi_1 + \xi_2, \dots, g + \xi_1 + \dots + \xi_n).$$

Under these coordinates the face maps are

$$\begin{aligned} d_0(g, \xi_1, \dots, \xi_n) &= (g + \xi_1, \xi_2, \dots, \xi_n), \\ d_i(g, \xi_1, \dots, \xi_n) &= (g, \xi_1, \dots, \xi_i + \xi_{i+1}, \dots, \xi_n), \quad 0 < i < n, \\ d_n(g, \xi_1, \dots, \xi_n) &= (g, \xi_1, \dots, \xi_{n-1}), \end{aligned}$$

and degeneracies insert the zero element of $\text{Nbh}_{S/S}^{\text{dR}}(G)$. In low degrees the nerve is

$$\begin{array}{ccc} G \times_S K \times_S K & \begin{array}{c} \xrightarrow{d_0} \\ \xrightarrow{d_1} \\ \xrightarrow{d_2} \end{array} & G \times_S K \begin{array}{c} \xrightarrow{\text{pr}_G} \\ \xrightarrow{\alpha_G} \end{array} G, \end{array}$$

where $K = \text{Nbh}_{S/S}^{\text{dR}}(G)$. Hence the entire relation groupoid is the action-groupoid nerve of formal translation.

Proof.

- (1) **Place successive differences in the kernel.** Every vertex of a relation simplex has the same image in $G_{\text{dR}/S}$, so

$$\eta_{G/S}^{\text{dR}}(\xi_j) = \eta_{G/S}^{\text{dR}}(g_j) - \eta_{G/S}^{\text{dR}}(g_{j-1}) = 0.$$

- (2) **Factor the successive differences through the kernel.** Thus every difference factors canonically through $\text{Nbh}_{S/S}^{\text{dR}}(G)$.
- (3) **Construct the inverse coordinates.** Successive partial sums construct the inverse.
- (4) **Compute the simplicial operators.** Deleting the first vertex translates the source point by ξ_1 ; deleting an interior vertex adds the adjacent differences; deleting the last vertex forgets the final difference; repeating a vertex inserts zero. These are precisely the displayed simplicial operators.

□

Remark 5.6.2 (Comparison with synthetic formal disks). The translation formula is the intrinsic spectral analogue of the formal translation models used in synthetic differential geometry [28, 27]. The infinitesimal translation object is $\text{Nbh}_{S/S}^{\text{dR}}(G)$, the fibre of the square-zero-generated relative de Rham unit at the identity.

Relative linear spaces as translation models

Proposition 5.6.2 (Canonical additive group structure on a relative linear space). *Let $\mathcal{E}$ be a finite locally free quasi-coherent module on the represented spectral base S . The relative linear space $\mathbb{L}_S(\mathcal{E})$ carries a canonical commutative group-object structure over S .*

On the coordinate algebra

$$\mathrm{Sym}_{\mathcal{O}_S}(\mathcal{E}),$$

the corresponding Hopf-algebra structure is induced by the maps of quasi-coherent modules

$$\mathcal{E} \xrightarrow{\Delta_{\mathcal{E}}} \mathcal{E} \oplus \mathcal{E}, \quad v \mapsto (v, v),$$

$$\mathcal{E} \longrightarrow 0,$$

and

$$-\mathrm{id}_{\mathcal{E}} : \mathcal{E} \longrightarrow \mathcal{E}.$$

Under the canonical equivalence

$$\mathrm{Sym}(\mathcal{E} \oplus \mathcal{E}) \simeq \mathrm{Sym}(\mathcal{E}) \otimes_{\mathcal{O}_S} \mathrm{Sym}(\mathcal{E}),$$

the resulting comultiplication is characterized on generators by

$$v \mapsto v \otimes 1 + 1 \otimes v.$$

The counit sends every generator to zero and the antipode sends v to $-v$.

This group-object construction is contravariantly functorial in $\mathcal{E}$: a morphism

$$\varphi : \mathcal{E} \rightarrow \mathcal{E}'$$

induces a canonical homomorphism of commutative group objects

$$\mathbb{L}_S(\mathcal{E}') \longrightarrow \mathbb{L}_S(\mathcal{E}).$$

It is also compatible with arbitrary base change on S .

Proof.

- (1) **Identify the symmetric-monoidal free-algebra comparison.** The free commutative-algebra functor

$$\mathrm{Sym}_{\mathcal{O}_S} : \mathrm{QCoh}(S) \longrightarrow \mathrm{CAlg}(\mathrm{QCoh}(S))$$

is symmetric monoidal with respect to direct sum in the source and tensor product in the target [34], giving

$$\mathrm{Sym}(\mathcal{E} \oplus \mathcal{E}) \simeq \mathrm{Sym}(\mathcal{E}) \otimes_{\mathcal{O}_S} \mathrm{Sym}(\mathcal{E}).$$

- (2) **Construct the Hopf comultiplication.** Applying the free-algebra functor to the diagonal

$$\Delta_{\mathcal{E}} : \mathcal{E} \rightarrow \mathcal{E} \oplus \mathcal{E}$$

therefore produces a comultiplication

$$\mathrm{Sym}(\mathcal{E}) \longrightarrow \mathrm{Sym}(\mathcal{E}) \otimes_{\mathcal{O}_S} \mathrm{Sym}(\mathcal{E}).$$

- (3) **Identify the comultiplication on generators.** By the free universal property it is characterized on the generating module $\mathcal{E}$ by

$$v \longmapsto v \otimes 1 + 1 \otimes v.$$

- (4) **Construct the remaining Hopf structure maps.** Likewise, the zero map

$$\mathcal{E} \rightarrow 0$$

induces the augmentation

$$\mathrm{Sym}(\mathcal{E}) \longrightarrow \mathcal{O}_S,$$

and the negation map

$$-\mathrm{id}_{\mathcal{E}} : \mathcal{E} \rightarrow \mathcal{E}$$

induces the antipode

$$\mathrm{Sym}(\mathcal{E}) \longrightarrow \mathrm{Sym}(\mathcal{E}).$$

- (5) **Verify the Hopf identities.** The commutative Hopf identities reduce, by the free-algebra universal property, to the corresponding identities for diagonal, zero, and negation in the additive commutative group object $\mathcal{E}$ of $\mathrm{QCoh}(S)$.
- (6) **Pass to relative spectrum.** Passing contravariantly through relative spectrum gives multiplication, zero section, and inversion on

$$\mathbb{L}_S(\mathcal{E}).$$

Thus $\mathbb{L}_S(\mathcal{E})$ is canonically a commutative group object over S .

- (7) **Establish functoriality.** Now let

$$\varphi : \mathcal{E} \rightarrow \mathcal{E}'$$

be a morphism of finite locally free quasi-coherent modules. Functoriality of the free commutative-algebra construction gives

$$\mathrm{Sym}(\varphi) : \mathrm{Sym}(\mathcal{E}) \longrightarrow \mathrm{Sym}(\mathcal{E}').$$

The naturality of diagonal, zero, and negation makes this a morphism of the displayed commutative Hopf algebras. Applying relative spectrum reverses its variance and therefore gives a homomorphism of commutative group objects

$$\mathbb{L}_S(\mathcal{E}') \longrightarrow \mathbb{L}_S(\mathcal{E}).$$

Hence the construction is contravariantly functorial in $\mathcal{E}$.

- (8) **Establish base change.** Finally, pullback of quasi-coherent modules preserves direct sums, diagonals, zero maps, negation, and the free commutative-algebra construction, while relative spectrum commutes with base change. Therefore the entire commutative-group-object structure and its contravariant functoriality are compatible with arbitrary base change on S .

□

Proposition 5.6.3 (Linear translation presentation). *Let $\mathcal{E}$ be a finite locally free quasi-coherent module on S . For every $n \geq 0$,*

$$\mathcal{R}_{\mathbb{L}_S(\mathcal{E})/S, n}^{\text{dR}} \simeq \mathbb{L}_S(\mathcal{E}) \times_S (\text{Nbh}_{S/S}^{\text{dR}}(\mathbb{L}_S(\mathcal{E})))^{\times_{S^n}},$$

compatibly with the full simplicial structure. The source map is projection to $\mathbb{L}_S(\mathcal{E})$, evaluation is formal addition, and

$$\text{Disk}_{\mathbb{L}_S(\mathcal{E})/S}^{\text{dR}, \infty} \simeq \Sigma_{\text{pr}_L} \alpha_{\mathbb{L}_S(\mathcal{E})}^*, \quad J_{\mathbb{L}_S(\mathcal{E})/S}^{\infty} \simeq \Pi_{\text{pr}_L} \alpha_{\mathbb{L}_S(\mathcal{E})}^*.$$

Writing

$$\pi_{\mathcal{E}} : \mathbb{L}_S(\mathcal{E}) \rightarrow S$$

for the structure morphism, the represented free-algebra cotangent calculation gives

$$L_{\mathbb{L}_S(\mathcal{E})/S} \simeq \pi_{\mathcal{E}}^* \mathcal{E}.$$

Thus the first-order linearization has the expected coefficient, while the full relation retains all finite infinitesimal information.

Proof.

- (1) **Apply the translation nerve.** Theorem 5.6.3 applies to the additive group object $\mathbb{L}_S(\mathcal{E})$. The jet formulas follow by transport of the relation correspondence.
- (2) **Recover the cotangent comparison.** The cotangent formula is the relative free commutative-algebra calculation established in the differentiated spectral doctrine. It supplies the first-order comparison with the structured neighbourhood, while the de Rham relation supplies the jet comonad.

□

The translation nerve therefore gives a concrete checksum: source is projection, target is formal translation, and higher simplices record successive infinitesimal increments. Relative linear spaces recover the expected cotangent coefficient at first order. A formally étale chart transports this same model to a represented local context.

5.6.4 Represented formally étale local models

A formally étale chart from a represented smooth context into affine space pulls back the affine translation relation independently of any group-object structure on the context. The following proposition is conditional only on the existence of such a chart. Later geometric results may supply charts on larger theorem domains; they are not inputs to the intrinsic jet construction.

Proposition 5.6.4 (Formally étale translation action and nerve). *Suppose a later geometric theorem supplies a formally étale morphism*

$$q : U \rightarrow \mathbb{A}_V^n$$

over a represented base V . Using the relative de Rham neighbourhood

$$\text{Nbh}_{V/V}^{\text{dR}}(\mathbb{A}_V^n) \simeq \mathbb{A}_V^n \times_{(\mathbb{A}_V^n)_{\text{dR}/V}} V$$

of the zero section, there is a canonical action

$$\alpha_U : U \times_V \text{Nbh}_{V/V}^{\text{dR}}(\mathbb{A}_V^n) \longrightarrow U$$

constructed from the evaluation map of the intrinsic relation of U/V . It satisfies the canonically commutative square

$$\begin{array}{ccc} U \times_V \mathrm{Nbh}_{V/V}^{\mathrm{dR}}(\mathbb{A}_V^n) & \xrightarrow{\alpha_U} & U \\ q \times \mathrm{id} \downarrow & & \downarrow q \\ \mathbb{A}_V^n \times_V \mathrm{Nbh}_{V/V}^{\mathrm{dR}}(\mathbb{A}_V^n) & \xrightarrow{\alpha_{\mathbb{A}_V^n}} & \mathbb{A}_V^n. \end{array}$$

Its unit and associativity equivalences are derived from the unit and composition of the de Rham relation groupoid:

$$\begin{aligned} \alpha_U(u, 0) &\simeq u, \\ \alpha_U(\alpha_U(u, \xi_1), \xi_2) &\simeq \alpha_U(u, \xi_1 + \xi_2). \end{aligned}$$

Relation inversion gives

$$\alpha_U(\alpha_U(u, \xi), -\xi) \simeq u.$$

For every $k \geq 0$, there is a canonical equivalence

$$\mathcal{R}_{U/V, k}^{\mathrm{dR}} \simeq U \times_V (\mathrm{Nbh}_{V/V}^{\mathrm{dR}}(\mathbb{A}_V^n))^{\times_V k}$$

compatible with the full simplicial structure. In these coordinates,

$$\begin{aligned} d_0(u, \xi_1, \dots, \xi_k) &= (\alpha_U(u, \xi_1), \xi_2, \dots, \xi_k), \\ d_i(u, \xi_1, \dots, \xi_k) &= (u, \xi_1, \dots, \xi_i + \xi_{i+1}, \dots, \xi_k), \quad 0 < i < k, \\ d_k(u, \xi_1, \dots, \xi_k) &= (u, \xi_1, \dots, \xi_{k-1}), \end{aligned}$$

and degeneracies insert 0. Consequently

$$\mathrm{Disk}_{U/V}^{\mathrm{dR}, \infty} \simeq \Sigma_{\mathrm{pr}_U} \alpha_U^*, \quad J_{U/V}^\infty \simeq \Pi_{\mathrm{pr}_U} \alpha_U^*.$$

Proof.

- (1) **Apply formally étale invariance.** Formal-étale invariance of the relation gives

$$\mathcal{R}_{U/V, k}^{\mathrm{dR}} \simeq U \times_{\mathbb{A}_V^n, v_k} \mathcal{R}_{\mathbb{A}_V^n/V, k}^{\mathrm{dR}}$$

simplicially in k .

- (2) **Insert the affine translation nerve.** By Theorem 5.6.3,

$$\mathcal{R}_{\mathbb{A}_V^n/V, k}^{\mathrm{dR}} \simeq \mathbb{A}_V^n \times_V (\mathrm{Nbh}_{V/V}^{\mathrm{dR}}(\mathbb{A}_V^n))^{\times_V k}.$$

- (3) **Pull back the translation nerve.** Pullback along $q : U \rightarrow \mathbb{A}_V^n$ therefore gives

$$\mathcal{R}_{U/V, k}^{\mathrm{dR}} \simeq U \times_V (\mathrm{Nbh}_{V/V}^{\mathrm{dR}}(\mathbb{A}_V^n))^{\times_V k}$$

for every k .

- (4) **Identify the source map in degree one.** In degree one, under

$$\mathcal{R}_{U/V}^{\mathrm{dR}} \simeq U \times_V \mathrm{Nbh}_{V/V}^{\mathrm{dR}}(\mathbb{A}_V^n),$$

the source map becomes the projection

$$\mathrm{pr}_U : U \times_V \mathrm{Nbh}_{V/V}^{\mathrm{dR}}(\mathbb{A}_V^n) \rightarrow U.$$

- (5) **Construct the induced translation action.** Transport the evaluation map

$$\mathbf{t} : \mathcal{R}_{U/V}^{\mathrm{dR}} \rightarrow U$$

through this equivalence and define the resulting map to be

$$\alpha_U : U \times_V \mathrm{Nbhd}_{V/V}^{\mathrm{dR}}(\mathbb{A}_V^n) \rightarrow U.$$

- (6) **Compare with affine translation.** The formal-étale relation square is Cartesian and intertwines evaluation, so

$$q\alpha_U \simeq \alpha_{\mathbb{A}_V^n}(q \times \mathrm{id}_{\mathrm{Nbhd}_{V/V}^{\mathrm{dR}}(\mathbb{A}_V^n)}).$$

- (7) **Identify the intrinsic action.** Thus α_U is the action induced by the intrinsic relation.

- (8) **Derive the unit law.** The relation unit $U \rightarrow \mathcal{R}_{U/V}^{\mathrm{dR}}$ corresponds to the zero section

$$u \mapsto (u, 0),$$

so the groupoid unit law gives

$$\alpha_U(u, 0) \simeq u.$$

- (9) **Derive the composition law.** In degree two,

$$\mathcal{R}_{U/V,2}^{\mathrm{dR}} \simeq U \times_V \mathrm{Nbhd}_{V/V}^{\mathrm{dR}}(\mathbb{A}_V^n) \times_V \mathrm{Nbhd}_{V/V}^{\mathrm{dR}}(\mathbb{A}_V^n).$$

Under this identification relation composition is

$$(u, \xi_1, \xi_2) \mapsto (u, \xi_1 + \xi_2),$$

while the successive evaluations are

$$u \mapsto \alpha_U(u, \xi_1) \mapsto \alpha_U(\alpha_U(u, \xi_1), \xi_2).$$

The groupoid composition identity therefore gives

$$\alpha_U(\alpha_U(u, \xi_1), \xi_2) \simeq \alpha_U(u, \xi_1 + \xi_2).$$

- (10) **Derive the inverse law.** Relation inversion corresponds to

$$(u, \xi) \mapsto (\alpha_U(u, \xi), -\xi),$$

and the inverse law gives

$$\alpha_U(\alpha_U(u, \xi), -\xi) \simeq u.$$

- (11) **Recover all simplicial operators.** Deleting and repeating vertices yields the displayed face and degeneracy maps in every degree.

- (12) **Read off the jet functors.** Finally, under the degree-one relation equivalence the source and evaluation maps are

$$\mathrm{pr}_U, \quad \alpha_U,$$

respectively. Proposition 5.2.1 therefore gives the formulas for $\mathrm{Disk}_{U/V}^{\mathrm{dR},\infty}$ and $J_{U/V}^\infty$.

□

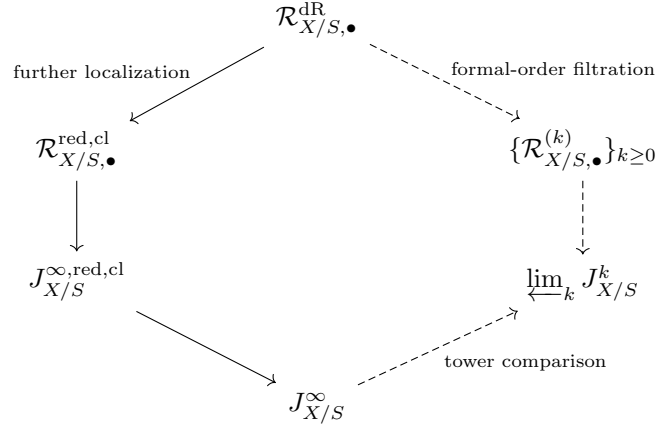
Remark 5.6.3 (Tubular comparison on represented smooth models). On the represented smooth theorem domain, a coordinate-free tubular theorem can compare the de Rham relation with a vector bundle constructed from the cotangent complex. Such a theorem supplies a geometric model for the relation and a first-order cotangent comparison. The jet comonad continues to arise from the de Rham relation, so higher finite-infinitesimal information remains visible in the resulting model.

Represented coordinates therefore model rather than define the jet calculus. With locality and the translation checksum in place, only two comparison questions remain, and they must not be conflated.

5.7 Comparison boundaries of the jet calculus

The first comparison changes the infinitesimal localization: when does the reduced-classical relation produce the same infinite jets as the intrinsic de Rham relation? The second keeps the intrinsic relation fixed but asks for a functorial filtration by formal order. The first is therefore a convergence problem. The second requires a new globalization construction.

The following fork records that distinction. Its right-hand branch is dashed because no global finite-order relation stages have yet been constructed.



5.7.1 Reduced-classical comparison

The reduced-classical boundary is obtained by applying a further localization to the target of the de Rham unit. The comparison of comonads is therefore controlled by contraction of the intermediate dependent adjunction, not by a new definition of jets.

The comparison $\kappa_{X/S}$ of Chapter 4 determines the reduced-classical relation simplicial object

$$\mathcal{R}_{X/S, \bullet}^{\text{red, cl}} := \tilde{C}_{\bullet} \left(X \xrightarrow{\kappa_{X/S} \eta_{X/S}^{\text{dR}}} S \times_{\text{Red}_{\text{cl}}(S)} \text{Red}_{\text{cl}}(X) \right)$$

and the reduced-classical jet comonad

$$J_{X/S}^{\infty, \text{red, cl}} := (\kappa_{X/S} \eta_{X/S}^{\text{dR}})^* \Pi_{\kappa_{X/S} \eta_{X/S}^{\text{dR}}}.$$

Remark 5.7.1 (Reduced-classical comparison comonad). The comonad $J_{X/S}^{\infty, \text{red}, \text{cl}}$ is attached to the composite $\kappa_{X/S} \eta_{X/S}^{\text{dR}}$ and serves as the reduced-classical comparison object. On the theorem domain where $\kappa_{X/S}$ is an equivalence, it agrees with the intrinsic jet comonad.

Lemma 5.7.1 (Contraction of a composite base-change comonad). *Let*

$$U \xrightarrow{f} V \xrightarrow{g} W$$

be composable morphisms in an infinity-topos. Put

$$J_f := f^* \Pi_f, \quad J_{gf} := (gf)^* \Pi_{gf}.$$

Using composition of dependent pullbacks and products, there is a canonical identification

$$J_{gf} \simeq f^* g^* \Pi_g \Pi_f.$$

The counit of

$$g^* \dashv \Pi_g$$

therefore induces a canonical natural transformation

$$\vartheta_{g,f} : J_{gf} \simeq f^* g^* \Pi_g \Pi_f \xrightarrow{f^* \varepsilon_{\Pi_f}^g} f^* \Pi_f = J_f.$$

This transformation is a morphism of comonads, is compatible with further composition, and is an equivalence whenever g is an equivalence.

Proof.

- (1) **Identify the two adjunction comonads.** The comonad J_f is induced by the adjunction

$$f^* \dashv \Pi_f,$$

whereas J_{gf} is induced by the composite adjunction

$$f^* g^* \dashv \Pi_g \Pi_f.$$

- (2) **Contract the inserted adjunction.** The displayed transformation contracts the inserted adjunction $g^* \dashv \Pi_g$ by its counit.
- (3) **Verify comonad compatibility.** Compatibility with the two comonad counits and comultiplications is therefore the standard compatibility of units, counits, and mates under composition of adjunctions. The same pasting formalism gives compatibility with further composition.
- (4) **Treat the equivalence case.** If g is an equivalence, then g^* and Π_g are inverse equivalences and the counit

$$g^* \Pi_g \rightarrow \text{id}$$

is invertible. Hence $\vartheta_{g,f}$ is an equivalence of comonads.

□

Theorem 5.7.1 (Canonical reduced-classical comparison of jets). *The map $\kappa_{X/S}$ induces a simplicial map*

$$\mathcal{R}_{X/S, \bullet}^{\text{dR}} \longrightarrow \mathcal{R}_{X/S, \bullet}^{\text{red, cl}}$$

which is the identity in degree zero, and a canonical natural transformation of comonads

$$J_{X/S}^{\infty, \text{red, cl}} \longrightarrow J_{X/S}^{\infty}.$$

If $\kappa_{X/S}$ is an equivalence, this comparison is an equivalence of comonads. In degree one the relation comparison sits in

$$\begin{array}{ccc} \mathcal{R}_{X/S}^{\text{dR}} & \longrightarrow & \mathcal{R}_{X/S}^{\text{red, cl}} \\ \downarrow \scriptstyle t & & \downarrow \scriptstyle s \\ X & \xlongequal{\quad} & X. \end{array}$$

Proof.

- (1) **Construct the simplicial comparison.** The comparison is encoded by the commutative triangle

$$\begin{array}{ccc} X & \xrightarrow{\kappa_{X/S} \eta_{X/S}^{\text{dR}}} & S \times_{\text{Red}_{\text{cl}}(S)} \text{Red}_{\text{cl}}(X) \\ & \searrow \scriptstyle \eta_{X/S}^{\text{dR}} & \nearrow \scriptstyle \kappa_{X/S} \\ & X_{\text{dR}/S} & \end{array}$$

and finite-limit functoriality gives the induced simplicial relation map.

- (2) **Apply the composite-comonad lemma.** Apply Lemma 5.7.1 with

$$f = \eta_{X/S}^{\text{dR}}, \quad g = \kappa_{X/S}.$$

- (3) **Construct the comparison comonad morphism.** Since

$$\Pi_{\kappa_{X/S} \eta_{X/S}^{\text{dR}}} \simeq \Pi_{\kappa_{X/S}} \Pi_{\eta_{X/S}^{\text{dR}}},$$

the resulting comonad morphism is

$$\begin{aligned} J_{X/S}^{\infty, \text{red, cl}} &= (\kappa_{X/S} \eta_{X/S}^{\text{dR}})^* \Pi_{\kappa_{X/S} \eta_{X/S}^{\text{dR}}} \\ &\simeq (\eta_{X/S}^{\text{dR}})^* \kappa_{X/S}^* \Pi_{\kappa_{X/S}} \Pi_{\eta_{X/S}^{\text{dR}}} \\ &\longrightarrow (\eta_{X/S}^{\text{dR}})^* \Pi_{\eta_{X/S}^{\text{dR}}} \\ &\simeq J_{X/S}^{\infty}, \end{aligned}$$

where the displayed arrow is induced by the counit of

$$\kappa_{X/S}^* \dashv \Pi_{\kappa_{X/S}}.$$

- (4) **Verify comonad compatibility.** Lemma 5.7.1 supplies compatibility with counit and comultiplication.

- (5) **Treat the equivalence case.** If $\kappa_{X/S}$ is an equivalence, the same lemma makes the comparison an equivalence of comonads.

□

Corollary 5.7.1 (Convergence criterion for the reduced-classical jet comparison). *If the absolute comparison maps*

$$\kappa_X : X_{\mathrm{dR}} \longrightarrow \mathrm{Red}_{\mathrm{cl}}(X), \quad \kappa_S : S_{\mathrm{dR}} \longrightarrow \mathrm{Red}_{\mathrm{cl}}(S)$$

are equivalences, then the relative comparison

$$\kappa_{X/S} : X_{\mathrm{dR}/S} \rightarrow S \times_{\mathrm{Red}_{\mathrm{cl}}(S)} \mathrm{Red}_{\mathrm{cl}}(X)$$

and hence

$$J_{X/S}^{\infty, \mathrm{red}, \mathrm{cl}} \xrightarrow{\cong} J_{X/S}^{\infty}$$

are equivalences.

Moreover, the convergence theorem of Chapter 4 identifies invertibility of κ_X with Postnikov continuity and nilradical continuity of the de Rham-local object X_{dR} on affine tests, and similarly for S .

Proof.

- (1) **Use the further localization.** Chapter 4 proves that $\mathrm{Red}_{\mathrm{cl}}$ is the further left-exact localization of the de Rham-local sector characterized by Postnikov continuity and nilradical continuity, and that

$$\mathrm{Red}_{\mathrm{cl}} \circ \mathrm{dR} \simeq \mathrm{Red}_{\mathrm{cl}}.$$

- (2) **Apply the convergence criterion to the de Rham object.** Apply Theorem 4.9.2 to the de Rham-local object

$$X_{\mathrm{dR}}.$$

It is $\mathrm{Red}_{\mathrm{cl}}$ -local if and only if it is local for the Postnikov-continuity generators and nilradical-continuous on affine tests.

- (3) **Identify the absolute comparison map.** The unit of the further localization is

$$X_{\mathrm{dR}} \longrightarrow \mathrm{Red}_{\mathrm{cl}}(X_{\mathrm{dR}}) \simeq \mathrm{Red}_{\mathrm{cl}}(X),$$

which is precisely κ_X . Hence κ_X is an equivalence exactly under those two continuity conditions.

- (4) **Apply the argument to the base.** The same argument applies to S .
- (5) **Pass to the relative comparison.** When both absolute comparisons are equivalences, substitute them into

$$X_{\mathrm{dR}/S} = S \times_{S_{\mathrm{dR}}} X_{\mathrm{dR}}.$$

Finite-limit functoriality gives

$$\kappa_{X/S} : X_{\mathrm{dR}/S} \xrightarrow{\cong} S \times_{\mathrm{Red}_{\mathrm{cl}}(S)} \mathrm{Red}_{\mathrm{cl}}(X).$$

Theorem 5.7.1 then gives

$$J_{X/S}^{\infty, \mathrm{red}, \mathrm{cl}} \xrightarrow{\cong} J_{X/S}^{\infty}.$$

□

Remark 5.7.2 (Comparison with crystals and nonlinear D -geometry). On the theorem domain where intrinsic de Rham reflection agrees with reduced-classical reflection, the equivalence

$$\mathrm{PDE}_{X/S}^{\mathrm{fi}} \simeq (\mathcal{H}_{\mathrm{Sp}})_{/Q_{X/S}}$$

is formally parallel to the description of crystals by descent over a de Rham prestack and to nonlinear D -geometry [18, 5]. The comparison applies on the stated theorem domain and identifies the two constructed architectures there. The intrinsic de Rham reflector is generated by the **square-zero infinitesimal substitutions**; no coefficient module, classifying derivation, or square-zero presentation is retained by the reflector.

Remark 5.7.3 (Comparison with reduced synthetic jets). On classical smooth input the preceding theorem gives the comparison with a reduced de Rham-groupoid jet construction on its convergence domain. The interaction of reduction with projective limits in synthetic smooth jet geometry is studied in [22].

Reduced-classical comparison is thus a convergence question about a further localization of an already constructed infinite-jet comonad. Finite order is a different problem: it asks for internal stages of the same intrinsic relation.

5.7.2 What is required for finite-order jets

Only J^∞ has been constructed globally. A global finite-jet construction requires a functorial, base-change-coherent filtration of the de Rham relation itself. Chosen finite factorizations encode presentation data through their lengths and intermediate square-zero stages. The reconstruction chapter supplies the intrinsic affine invariant—the canonical formal-order filtration—whose coherent extension to the unrestricted simplicial relation requires an additional theorem.

A finite-order refinement would therefore consist of a tower

$$J^0 E \longleftarrow J^1 E \longleftarrow J^2 E \longleftarrow \dots$$

together with a comparison of J^∞ with its inverse limit. The desired global relationship would have the schematic form

$$\begin{array}{c} J^\infty E \\ \text{comparison} \downarrow \text{dashed} \\ \dots \longrightarrow J^2 E \longrightarrow J^1 E \longrightarrow J^0 E. \end{array}$$

The constructions already available determine what that missing theorem must do.

First, Chapter 2 defines finite spectral infinitesimal substitutions as the equivalence-closed class generated under finite composition by square-zero substitutions. Membership in that class is intrinsic, but a chosen factorization and its length are not retained structure.

Second, Chapter 3 constructs a genuine **formal-order filtration** on every augmented spectral algebra from the adjunction

$$\mathrm{adic} \dashv \mathrm{aug}.$$

This filtration supplies an intrinsic affine notion of order. A finite jet tower would require a base-change-coherent globalization of these affine stages to the full de Rham relation and, compatibly, to its simplicial structure. Only after that construction would endofunctors J^k be available on the unrestricted slice.

Proposition 5.7.1 (Canonical formal order on an affine diagonal). *Let $R \rightarrow A$ be a morphism of connective commutative ring spectra and let*

$$\mu_A : A \otimes_R A \longrightarrow A$$

be multiplication, the coordinate augmentation of the affine diagonal. The formal-order reconstruction of Chapter 3 supplies the canonical completed adic filtration

$$\widehat{\mathrm{adic}}(A \otimes_R A \rightarrow A).$$

The cotangent comparison is displayed by

$$L_{A/(A \otimes_R A)}[-1] \longrightarrow L_{(A \otimes_R A)/R} \otimes_{A \otimes_R A} A \longrightarrow L_{A/R},$$

and in particular there is a canonical equivalence

$$L_{A/(A \otimes_R A)}[-1] \xrightarrow{\simeq} L_{A/R}.$$

Hence there is a canonical equivalence of graded commutative A -algebras

$$\mathrm{Gr} \widehat{\mathrm{adic}}(A \otimes_R A \rightarrow A) \simeq \mathrm{Sym}_A^{\mathrm{gr}}(L_{A/R}\langle 1 \rangle).$$

Proof.

- (1) **Apply cotangent transitivity.** Apply cotangent transitivity [34, 37] to

$$R \longrightarrow A \otimes_R A \xrightarrow{\mu_A} A.$$

It gives a cofibre sequence

$$L_{(A \otimes_R A)/R} \otimes_{A \otimes_R A} A \longrightarrow L_{A/R} \longrightarrow L_{A/(A \otimes_R A)}.$$

- (2) **Identify the first cotangent term.** Cotangent base change for the two tensor factors identifies the first term with

$$L_{A/R} \oplus L_{A/R},$$

and the map induced by multiplication is addition

$$+ : L_{A/R} \oplus L_{A/R} \longrightarrow L_{A/R}.$$

- (3) **Compute the fibre of addition.** Its fibre is canonically $L_{A/R}$, via

$$x \longmapsto (x, -x).$$

- (4) **Identify the shifted diagonal cotangent complex.** Stability therefore identifies its cofibre with $L_{A/R}[1]$, proving

$$L_{A/(A \otimes_R A)}[-1] \simeq L_{A/R}.$$

- (5) **Apply the associated-graded theorem.** Now apply the associated-graded theorem for the canonical adic filtration, 3.3.1, to the augmentation

$$A \otimes_R A \rightarrow A.$$

It gives

$$\begin{aligned} \mathrm{Gr} \widehat{\mathrm{adic}}(A \otimes_R A \rightarrow A) &\simeq \mathrm{Sym}_A^{\mathrm{gr}}(L_{A/(A \otimes_R A)}[-1]\langle 1 \rangle) \\ &\simeq \mathrm{Sym}_A^{\mathrm{gr}}(L_{A/R}\langle 1 \rangle). \end{aligned}$$

□

Remark 5.7.4 (Globalization required for finite-order jets). The preceding proposition supplies formal-order stages for a represented affine diagonal augmentation. A global finite-order jet theory requires a base-change coherent extension of those stages to the topos-level relation

$$\mathcal{R}_{X/S,1}^{\text{dR}} = X \times_{Q_{X/S}} X,$$

together with a compatible simplicial filtration of $\mathcal{R}_{X/S,\bullet}^{\text{dR}}$ and endofunctors

$$J_{X/S}^k : (\mathcal{H}_{\text{Sp}})_{/X} \rightarrow (\mathcal{H}_{\text{Sp}})_{/X}$$

whose inverse limit compares with $J_{X/S}^\infty$. The complete totalization filtration of Chapter 4 remains the descent filtration on de Rham cochains; its normalized layers supply the intrinsic de Rham forms.

Remark 5.7.5 (Finite dependence requires a constructed invariant). Let

$$D : E \rightsquigarrow F$$

be a differential operator represented by

$$\widehat{D} : J^\infty E \rightarrow F.$$

The present construction supplies a canonical order-zero class through ordinary morphisms

$$E \rightarrow F,$$

represented in the co-Kleisli category by

$$J^\infty E \xrightarrow{\varepsilon_E^\Pi} E \rightarrow F.$$

Beyond order zero, a numerical order function on arbitrary differential operators requires a functorial invariant that measures finite dependence, for example a filtration stage, a compactness condition, or a factorization through a constructed finite-order jet object

$$J^k E.$$

The finite-infinitesimal calculus by itself records membership in the equivalence-closed finite-composition class generated by square-zero substitutions; a chosen factorization length remains presentation data rather than an intrinsic order. The canonical adic filtration of Chapter 3 supplies an intrinsic affine formal-order invariant, and a numerical jet order on the unrestricted slice becomes available after that filtration has been globalized to a functorial tower J^k .

The boundary is therefore exact: affine diagonals carry canonical formal order with cotangent-controlled associated graded, while the global simplicial filtration needed for J^k remains a further construction. The infinite-jet calculus is independent of that refinement, and numerical order becomes intrinsic after the filtration has been globalized.

5.8 The crystalline jet interface

The chapter now has a reusable interface. The theorem below is intentionally a citation index: it collects the precise results that later chapters may invoke. The retrospective summary that follows has a different job and will compress these results to the conceptual structure that should remain active.

Theorem 5.8.1 (Crystalline jet calculus). *For every $X \rightarrow S$, the effective de Rham quotient*

$$X \xrightarrow{e_{X/S}} Q_{X/S}$$

and its de Rham relation groupoid organize the jet calculus through the commutative architecture

$$\begin{array}{ccccc}
 & & \mathcal{R}_{X/S, \bullet}^{\text{dR}} & & \\
 & \swarrow & & \searrow & \\
 \text{Disk}_{X/S}^{\text{dR}, \infty} & \xrightarrow{\quad \dashv \quad} & J_{X/S}^{\infty} & & \\
 & \nwarrow & & \nearrow & \\
 & & (\mathcal{H}_{\text{Sp}})_{/Q_{X/S}} & & \\
 & & \downarrow & & \\
 & & \text{PDE}_{X/S}^{\text{fi}} & &
 \end{array}$$

and canonically determine:

1. relative de Rham neighbourhoods serving as de Rham formal disks

$$\text{Nbh}_{T/S}^{\text{dR}}(T \times_S X) \simeq T \times_{Q_{X/S}} X$$

for every generalized point $a : T \rightarrow X$;

2. the adjoint triple

$$\Sigma_{e_{X/S}} \dashv e_{X/S}^* \dashv \Pi_{e_{X/S}}$$

and hence the monad/comonad adjunction

$$\text{Disk}_{X/S}^{\text{dR}, \infty} = e_{X/S}^* \Sigma_{e_{X/S}} \dashv J_{X/S}^{\infty} = e_{X/S}^* \Pi_{e_{X/S}};$$

3. the mate equivalence

$$\text{Alg}_{\text{Disk}^{\text{dR}, \infty}}((\mathcal{H}_{\text{Sp}})_{/X}) \xrightarrow{\simeq} \text{Coalg}_{J^{\infty}}((\mathcal{H}_{\text{Sp}})_{/X}),$$

together with the effective-descent recognition

$$\text{PDE}_{X/S}^{\text{fi}} = (\mathcal{H}_{\text{Sp}})_{/Q_{X/S}} \simeq \text{Alg}_{\text{Disk}^{\text{dR}, \infty}}((\mathcal{H}_{\text{Sp}})_{/X}) \simeq \text{Coalg}_{J^{\infty}}((\mathcal{H}_{\text{Sp}})_{/X});$$

4. crystalline transport as the relation transpose of the jet coaction, with its unit, inverse, cocycle, and simplicial compatibility induced by the de Rham relation groupoid;

5. the co-Kleisli differential-operator calculus and infinite prolongation;
6. the comma infinity-category

$$\mathrm{Eqn}_{X/S} = (\mathrm{id}_{(\mathcal{H}_{\mathrm{Sp}})_X} \downarrow J_{X/S}^\infty)$$

of generalized jet equations, together with derived equations, actual and parameterized solution objects, formal-solution spaces, Cartan prolongation, the coalgebraic representation of actual solutions, and prolongation-completeness;

7. arbitrary base change and effective formally étale descent of the jet calculus and equation infinity-category, together with the conditional reduced-classical comonad comparison;
8. for a relative commutative group object, the formal-translation model and full translation nerve, and for a formally étale affine-space chart, the induced translation action constructed from relation evaluation.

On represented affine diagonal augmentations, the canonical adic construction of Chapter 3 supplies genuine formal order with associated graded

$$\mathrm{Sym}_A^{\mathrm{gr}}(L_{A/R}\langle 1 \rangle),$$

and a global finite-order refinement would further supply a tower J^k and a comparison

$$J^\infty \longrightarrow \varprojlim_k J^k.$$

Such a comparison belongs to the additional globalization problem described above.

Proof.

- (1) **Record the formal-disk theorem.** The formal-disk statement is 5.2.2.
- (2) **Record the disk-jet adjunction data.** The effective-image formulas, jet comonad, and adjoint-triple mate structure are 5.2.3, 5.2.5, and 5.2.6.
- (3) **Record the Eilenberg–Moore and descent recognition.** The Eilenberg–Moore mate equivalence is 5.3.1, while effective Čech descent and Barr–Beck recognition are 5.3.2 and 5.3.2.
- (4) **Record crystalline transport.** Crystalline transport, its cocycle, and its inverse are constructed in 5.3.4.
- (5) **Record holonomic and Cartan recognition.** Holonomic and Cartan recognition are 5.4.2 and 5.4.4.
- (6) **Record the equation infinity-category.** The equation infinity-category is Construction 5.4.5.
- (7) **Record the prolongation and solution representation.** Cartan prolongation, formal-solution representability, and actual-solution representation are 5.5.2, 5.5.3, and 5.5.4.
- (8) **Record the prolongation-completeness criteria.** Prolongation-completeness is controlled by 5.5.5, 5.5.6, 5.5.7, and 5.5.9.

- (9) **Record the functoriality and reduced-comparison results.** Base change, formally étale restriction, effective descent of the jet calculus, equation descent, and the reduced comparison are 5.2.7, 5.6.1, 5.6.2, 5.6.3, and 5.7.1.
- (10) **Record translation and affine formal-order comparisons.** The group-object translation specialization is 5.6.3; the induced translation action on a formally étale affine-space chart is 5.6.4; and the affine formal-order comparison is 5.7.1.

□

5.8.1 Chapter summary

The chapter has converted one effective infinitesimal relation into three usable forms. Over X , it acts as the formal-disk monad and infinite-jet comonad; over $Q_{X/S}$, the same data are a single descended object. Crystalline transport is the relation-theoretic expression of that identification, not an extra field attached to an object.

The jet presentation then supplies the differential-equation calculus. Differential operators are co-Kleisli maps out of infinite jets, while generalized equations are maps into them. Cartan prolongation is the universal passage from an arbitrary ambient equation to one carrying compatible crystalline transport. Its evaluation $c_i : \text{Prol}(i) \rightarrow E$ is the recognition map: when it is invertible, the compatible Cartan structure has contractible moduli and the equation is already an intrinsic formally integrable PDE.

Two limits of the construction should remain separate. Reduced-classical comparison asks whether a further infinitesimal localization changes the already-defined J^∞ , and is controlled by convergence. Finite order asks for new structure on the intrinsic relation: a global simplicial formal-order filtration. The affine adic filtration supplies the local invariant, but no global J^k -tower is asserted or used before that globalization is constructed.

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